

Sherif Hanafy Mahmoud
Editor

Patient Assessment in Clinical Pharmacy

A Comprehensive Guide
Second Edition

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 Springer

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To my wonderful wife, Sally, whose love, patience, and continuous support make every endeavor possible.

To my precious kids, Basant, Omar, and Ali, you remind me that the most meaningful legacies are not written in books, but in the lives we touch

And to all pharmacists, may this work serve as a reminder of the profound impact you have in advancing patient care, and an inspiration to keep pushing the boundaries of what our profession can achieve to make a meaningful difference in people's lives.

Foreword

In recent decades, the profession of pharmacy has undergone a significant transformation. In addition to traditional roles of medication dispensing and compounding, pharmacists today are increasingly recognized as patient care providers. The evolution of pharmacy practice has been shaped by increased healthcare demands, the growing complexity and cost of medication therapies, and the need for accessible high-quality healthcare services. As a result, the scope of pharmacy practice has expanded in many jurisdictions—bringing with it both new responsibilities and new opportunities for meaningful impact.

At the core of patient care lies a fundamental clinical competency: patient assessment. Effective patient assessment enables pharmacists to make informed, evidence-based decisions that go far beyond selection of drug therapy. It is an essential step in the provision of patient-centered care and enables identification of medication-related problems, evaluation of therapeutic effectiveness, monitoring of adverse effects—all in the context of the individual patient's needs, conditions, and preferences.

Patient Assessment in Clinical Pharmacy is not only a timely resource, but also a critical tool in equipping current and future pharmacists for modern pharmacy practice. This textbook offers a systematic and practical guide to the essential components of patient evaluation, from history-taking and physical assessment to the interpretation of clinical data with practical tools and clinical pearls for patient care.

The 2nd edition of *Patient Assessment in Clinical Pharmacy* is divided into four parts:

Part I consists of chapters focused on an introduction to the patient care process, principles of patient assessment and physical assessment for pharmacists, with new chapters focused on assessment of complex patients and clinical decision-making.

Part II focuses on symptom assessment, with a range of common symptoms that patients present with in everyday practice. These chapters cover possible underlying etiology and offer guidance for comprehensive assessment that can identify patients who may need urgent medical attention, conditions that can be self-managed, and those that may require prescription therapy or further follow-up.

Part III is composed of chapters related to assessment of chronic illnesses ranging from diabetes mellitus and hypertension to asthma and depression. Pharmacists are integral to assessment and management of chronic conditions either as part of interprofessional teams, or in some cases independently.

Part IV highlights specialized or tailored assessments that may be needed depending on the patient population (e.g., older adults, pediatrics) and practice setting (e.g., acute or critical care). From understanding vital signs and interpreting laboratory data to recognizing signs and symptoms across a range of conditions, the content is both comprehensive and accessible.

This textbook is composed of chapters written by pharmacy practitioners and academics in Alberta and other parts of Canada and is intended to support pharmacy students and pharmacists around the world. In an era where pharmacists are increasingly involved in collaborative healthcare teams, primary care services, and even prescribing roles in some jurisdictions, these competencies are more relevant than ever. This resource serves not only as a source of knowledge but also as a catalyst for advancing pharmacy practice in a variety of settings.

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Christine Hughes

Preface

Pharmacy is in the midst of a remarkable transformation. Pharmacists are now recognized as essential providers of patient-centered care. Every day, across community pharmacies, hospitals, clinics, and specialized practice settings, pharmacists are assessing patients, identifying problems, preventing harm, and shaping therapeutic decisions that improve lives. We are not only medication experts; we are advocates, educators, and integral members of the healthcare team.

Patient assessment is the cornerstone of our contribution to healthcare. It provides the foundation upon which care plans are built, therapies are optimized, and outcomes are achieved. Strong assessment skills allow pharmacists to move beyond medications alone and into the realm of comprehensive, compassionate care.

When I first set out to write this book, there were limited resources that addressed patient assessment from the unique perspective of clinical pharmacy. The first edition sought to fill that gap, offering a structured framework for developing these critical skills. The positive reception of that edition reinforced my belief in the importance of this work. This second edition expands and deepens that vision, reflecting both the growing scope of pharmacy practice and the increasing complexity of the patients we serve.

Over the years, I have had the privilege of working with pharmacists and students who continually inspire me with their curiosity, resilience, and commitment to patients. Their questions, insights, and passion for advancing pharmacy practice have shaped not only this book but also my own growth as a clinician, educator, and researcher. Updating this edition has been both a professional and personal journey; one that reminded me why I first entered this field: to make a meaningful difference in people's lives.

The book remains organized into four parts: an introduction to the principles of patient assessment and the patient care process; a practical guide to evaluating common symptoms; a structured approach to assessing patients with chronic illnesses; and finally, specialized topics that highlight the breadth of pharmacy practice, from pharmacokinetics and critical illness to the unique considerations in pediatrics and geriatrics. In this second edition, I am pleased to introduce two new Chaps. 4 and 5: "Assessment of Complex Patients" and "Clinical Decision-Making for Pharmacists." These additions provide pharmacists and pharmacy students with additional tools essential for their day-to-day practice.

My hope is that this book not only equips pharmacists and pharmacy students with practical tools but also inspires them to embrace the responsibility, and privilege, of providing care that truly makes a difference. The future of pharmacy will be defined not simply by the medicines we dispense but by the lives we touch and the trust we earn through our clinical expertise.

May this resource serve as both a guide and a reminder of the profound role pharmacists play in advancing health and transforming patient care.

Edmonton, AB, Canada

Sherif Hanafy Mahmoud

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Part I

Introduction

Introduction to the Patient Care Process

1

Theresa L. Charrois

Chapter Objectives

1. Define and understand the role of the patient care process in providing care.
2. Describe the components of comprehensive patient history taking.
3. Apply a process to patient assessment that can be used in a variety of different patient care settings.
4. Apply a process to assess a patient for drug-related problems.
5. Outline the components of patients' comprehensive care plans.
6. Develop appropriate documentation of patient care.

Background

Pharmacists play an important role in patient-centered care. Being the most accessible of healthcare team members, pharmacists' role as healthcare practitioners is evolving as they are taking an active part in primary patient care. With the continuously expanding pharmacists' scope of practice, there is a need for a framework for pharmacists to follow a consistent approach to providing patient-centered care. Research has shown that pharmacists directly involved in the care of patients improve health outcomes; therefore, it is important for pharmacists to have a fundamental process in place to provide patient-centered care [1–4].

The patient care process provides such a framework and is central to our identity as pharmacists. It is what defines our role as professionals. It includes our scientific knowledge of medications, our clinical knowledge, and our interactions with the patient. The patient care process is a continuous and

dynamic mechanism to provide patient care (Fig. 1.1) and includes the essential components that can be adapted to suit various practice settings:

Step 1: Patient assessment

Step 2: Care plan development and implementation

Step 3: Monitoring and follow-up

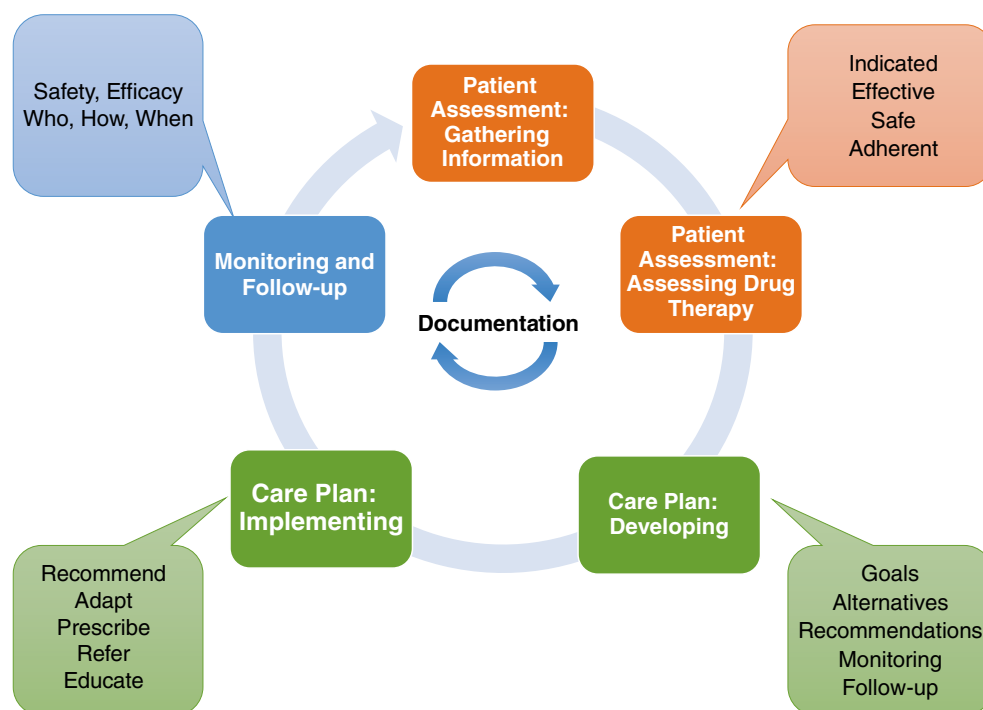
Assessment of a patient includes a complete history and an assessment of current medications (step 1). Developing a care plan including all drug-related problems, along with goals and implementation of recommendations, is the next step of the process (step 2). Finally, appropriate monitoring and follow-up helps to ensure that goals are being met and that safety is being monitored (step 3). This framework is consistent for complete pharmacist assessment of a patient as well as for targeted assessment that may be specific to a medical condition. Patient care is then documented to ensure that it is communicated to all other members of the healthcare team.

Assessment

The key purpose of an assessment in pharmacy practice is to determine if a patient's drug therapy needs are being met [5]. Assessment can occur during various types of patient encounters, for example, at prescription drop off and pick up, during routine refills, or during over-the-counter (OTC) medication counseling. Often, pharmacists think assessment can only happen in a private room during a thorough patient interview; however, it should be occurring at any stage of the medication use process. Specific elements of patient assessment will change based on the situation, i.e., an assessment during a refill will differ from an assessment with a new prescription, but the same overall process can be applied regardless of the scenario or setting. Within the patient care process, there are two primary components of assessment: (1) assessment of the patient, including the patient interview, and (2)

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Fig. 1.1 The patient care process



assessment of current drug therapy to determine if there are any drug-related problems.

Assessment of the Patient: Patient Interview and History Taking

Prior to conducting an interview, the pharmacist should establish a relationship with the patient to ensure trust building and clear definition of the patient's goals. This is followed by a structured patient interview. In order to assess a patient, the pharmacist will need to collect relevant information through multiple sources such as patient interviews, electronic medical records (EMRs) (prescription records, lab values, diagnostic tests, etc.), and physical examination. Complete and relevant patient history taking is essential for proper patient assessment. By collating this information, the pharmacist can develop a patient database suited for their particular setting. The information collected and inputted into the database can then be used to assess the patient and develop a care plan.

As part of the patient interview, the pharmacist needs to determine the following: (1) the patient's reason for seeking care, (2) their current symptoms (if presenting with a specific complaint), (3) medical history, and (4) medication history.

In determining the patient's reason for seeking care, it is imperative that the patient's perspective is considered. The patient's goals need to be the priority and can be negotiated with the pharmacist's goals as a care provider. Incorporating the patient's goals into the pharmacist's assessment can cre-

ate a sense of trust and shared values for moving forward in a treatment plan.

Symptom assessment can be done in a variety of ways, but the key pieces of information that need to be collected remain the same: the region/location, what helps or worsens the symptom, the severity, and the temporality of symptom onset. Using appropriate questions to get at each of these elements, the pharmacist can have a better and more complete understanding of the patient's primary symptom concern. Readers are referred to Chap. 2 for an in-depth discussion on symptom assessment.

A complete and thorough medical history, including current and ongoing medical issues, resolved medical issues, and surgical history, is a vital part of the process. In certain cases, a pharmacist may need to use physical assessment skills (Chap. 3) to evaluate current conditions, symptoms, and even the safety and efficacy of their medications. Whether or not physical assessment is required as part of patient assessment depends on the presenting complaint, reason for seeking care, and the setting where care is being provided.

Clearly, a complete medication history is our true domain as pharmacists, but pharmacists have to ensure that they capture the many facets of medication use. Specifically, medication history needs to include current and past medications, allergies, adverse reactions, immunizations, and patient adherence. Often, patients do not consider natural health products or over-the-counter medications as part of their medication list, and those products should be specifically interrogated. Other factors that may affect drug therapy

should also be considered, such as social history (tobacco, alcohol, and recreational drug use) and relevant dietary and physical activity information. Information such as social history may not always be relevant; therefore, judgment should be exercised before asking relevant questions. In addition, these questions should be asked in a nonjudgmental way, to again ensure the development of trust between the patient and pharmacist [6].

At the end of the interview, the pharmacist should be able to determine what is going on with the patient (in terms of the presenting complaint and symptoms) and the patient's primary concerns. From this point, the pharmacist can specifically assess the patient's medications. Table 1.1 provides a summary of the elements of comprehensive patient history.

Assessing Drug Therapy

After patient information is collected, the pharmacist must then determine if the patient's current drug therapy is appropriate.

The four primary questions a pharmacist should consider for each medication are the following:

- Is this medication indicated?
- Is this medication effective?
- Is this medication safe?
- Is the patient being adherent to this medication?

These can be remembered using the acronym IESA: indicated, effective, safe, and adherent. In addition, the pharmacist should consider if there are medical conditions that are not currently treated but may require drug therapy, as a lack of drug therapy for a condition is also a drug-related problem. When assessing indication, there should be a clear reason why a patient is on each medication. In addition, you should determine if the medication they are on for a condition is the most optimal therapy based on the relevant guidelines for practice, comorbidities, and outcomes.

For effectiveness, it makes the most sense to determine what the important patient and clinical outcomes are for a condition and if these are being met. This relates back to optimizing therapy. Considerations such as dose increases or additional therapy being added should be taken into account if drug therapy is not effective.

In terms of assessing safety, the main issue we should consider are the adverse effects from the medications. This includes a general overview of common adverse effects of a drug as well as those rare adverse effects that could be harmful. Asking the patient the question "Are you having any side effects?" is not always helpful because patients may not link symptoms to drugs. Questions should be tailored to the specific drug. In addition, medication safety can be assessed by reviewing appropriate lab work that was collected during the patient assessment. Drug interactions are also part of a complete assessment of safety parameters; this includes determining if the drug interaction is potentially relevant to this patient and the possible severity of the interaction if the interacting medications are continued.

Medication history also includes an assessment of adherence. When assessing drug adherence, pharmacists should ask the patient in a relevant time period, how often they forget their medications, i.e., *how many pills have you missed in the last week?* This is a relatively nonjudgmental approach to assessing adherence, and judgment when asking about adherence could lead to false answers. When a patient expresses that they are sometimes nonadherent, the pharmacist needs to assess what factors may be leading to this nonadherence in order to help develop strategies to overcome it, such as education or reminder systems. Nonadherence can be due to multiple factors and can be purposeful or unintended. An assessment of the root cause of the nonadherence will help in decision-making about the possible ways to improve it. Adherence may also be medication-specific, so

Table 1.1 Summary of the elements of comprehensive patient history

Elements of a patient's history	Details
History of present illness (HPI)	Assessment of presenting symptoms/complaint SCHOLARE (see Chap. 2)
Medical history; past medical history (PMH)	Current and previous medical conditions Hospitalizations, surgeries
Medication history	Current medications (including oral, topical, parenteral, eye, ear, nose medications): indication, dosage, duration (link to current medical conditions), adherence Previous medications Nonprescription medications: including complementary and alternative medications, vitamins, minerals, over-the-counter medications Immunizations
Allergies	Include reaction (date, onset, symptoms, management)
Social history (SH)	Lifestyle considerations: diet, exercise, living conditions Substance use: caffeine, alcohol, tobacco, recreational drug use
Family history (FH)	First-degree relatives Focus on conditions with familial linkages such as cardiovascular disease, diabetes, and cancer.
Laboratory	Complete blood count, electrolytes, renal function (including estimated creatinine clearance), liver function, microbiology, etc.
Review of systems (ROS)	Head to toe assessment Integumentary, head/neuro, eyes/ears/nose, neck, chest/lungs, cardiovascular, gastrointestinal, urinary, hepatic, renal, reproductive, musculoskeletal, endocrine

SCHOLARE Symptoms, Characteristics, History, Onset, Location, Aggravating factors, Remitting factors and Explanatory Model

assessing adherence globally may not always provide an accurate depiction of actual medication use; questions may need to be targeted to the individual drugs. Both adherence information and assessing drug safety may actually be uncovered during the initial patient interview. Further details regarding adherence are discussed in Chap. 2.

After working through the four medication assessment questions (indicated, effective, safe, adherent), the easiest way of approaching the next step of drug therapy assessment is to link medical conditions to medications. An example is shown in Table 1.2. By linking medical conditions to medications, pharmacists can identify if all conditions are being adequately treated or if all medications have a relevant indication. In addition, monitoring information can be included to assess efficacy and safety. This is a good first step in assessing the appropriateness of a person’s drug therapy.

Table 1.2 shows the scenario of IESA question application to a patient. In this example, hydrochlorothiazide is indicated for the patient’s hypertension and its effectiveness can be gauged by assessing blood pressure (BP). Safety may need to be further explored through other questioning on adverse effects, such as dizziness. Adherence is also indicated in the assessment. It is clear from this basic format that there are no drugs currently being used for seasonal allergies, but, perhaps, as determined from the initial patient assessment, it was unnecessary as the patient preferred to take nothing. In addition, there may be no indication the omeprazole, and that may need reassessment.

From this assessment of indicated, effective, safe, and adherence, you can develop a list of potential and actual drug-related problems. It is the identification of these drug-related problems that leads the pharmacist to the next step of the patient care process, which is developing a care plan.

Table 1.2 Example of assessing drug therapy appropriateness

Medical conditions	Medications	Other information to assess efficacy and safety
Hypertension	Hydrochlorothiazide 25 mg daily	CrCl = 75 mL/min BP 126/72 No reports of dizziness, falls Misses dose once/month (information can be used to assess efficacy and safety)
Seasonal allergies	(Not currently treated: potential DRP)	Only in spring
(No current indication: potential DRP)	Omeprazole	Previously had <i>Helicobacter pylori</i> (1 year ago) No current symptoms of reflux

CrCl creatinine clearance, DRP drug-related problem, BP blood pressure

Care Plans

The purpose of a care plan is for a pharmacist to document their assessment of a patient, along with a plan for resolving and monitoring medical conditions and medications. A care plan can take many forms, and a variety of templates are available to assist in this process. Table 1.3 provides one example for a template of a care plan. A care plan should not be kept separate from other patient care records, as it is a documented plan for care that other pharmacists on the team or other healthcare professionals in the pharmacist’s setting may want to refer to as well.

The primary aspects that should be included in the care plan, after the patient’s database is created, are the assessed drug-related problems and for each problem, the *goals, alternatives, and recommendations*. *Goals* are meant to be specific to a patient and not broad and overarching. For example, when considering blood pressure treatment goals, not only should a specific target blood pressure value based on the patient’s comorbidities and current guidelines be included but also any parameters specific to that patient, such as minimizing postural hypotension in a patient who is already concerned about falling, should be considered. The patient should be asked what their goal is with therapy to help construct goals that make sense to their lifestyle. Not all care plans require *alternatives* for management; however, it is a clear way to follow someone’s thought process as to why

Table 1.3 Example of a care plan template

Medical Conditions and MED-Related Needs: List and prioritize each medical condition first, followed by any DRPs identified for a given condition. Although some medical conditions may not have a DRP, a care plan is still necessary for ongoing patient monitoring XXX
Goals of Therapy: For each medical condition and/or DRP, state the desired goals of therapy/timeframe Goals: Cure, prevent, slow/stop progression, reduce/eliminate symptoms, normalize a lab value <i>Consider realistic goals determined through patient discussion. Goals of therapy are measurable or observable parameters that are used to evaluate the efficacy and safety of therapy</i> XXX
Alternatives: Compare relevant drug and nondrug therapies that will produce the desired goals. List the <i>pros</i> and <i>cons</i> of each therapy as well as the rationale for each being included <i>Consider: Indication ● Efficacy ● Safety ● Adherence ● Cost/coverage</i> XXX
Recommendations/plan: In collaboration with the patient and other healthcare providers, select the best alternative and implement the plan. Provide a rationale for the chosen plan relative to the other alternatives considered <i>Consider: Drugs: correct drug, formulation, route, dose, frequency, schedule, duration, medication management. Nondrugs: nondrug measures, education, patient referral</i> XXX

DRP drug-related problem

a particular medication was chosen. By listing relevant and appropriate alternatives, as well as considering the pros and cons of each, a pharmacist can be transparent with their decision-making process of determining optimal therapy. From this list of alternatives, a *recommendation* should then be determined. This may mean a need for drug therapy, additional drug therapy, or stopping drug therapy. Lifestyle parameters, such as diet and exercise, can be included in this section as well. Education needs for the patient regarding medical conditions and medications should also be addressed. The recommendations should be succinct but clear, with a plan for who is taking care of each part of the plan.

Monitoring and Follow-Up

Monitoring plans should be developed for each and every drug-related problem. Depending on the situation, a monitoring plan that encompasses all the included drug therapy problems and recommendations should be developed, ensuring that there is no overlap. For example, in a combined monitoring plan, you could include all required follow-up lab testing in one place. This helps ensure consistency in follow-up parameters and not sending the patient at multiple time points for blood tests or for multiple visits to care providers.

For monitoring, specific parameters for both efficacy and safety should be indicated. Efficacy parameters are directly related to the goals of therapy and the outcomes that are targeted. Safety parameters are generally related to the side effects of the medications. Parameters, for both efficacy and safety, can include signs and symptoms as well as laboratory parameters.

In addition, it is important to indicate who is responsible for the appropriate follow-up. In jurisdictions where pharmacists can order lab tests, they may be the ones to follow-up on lab results, however, this should be clearly indicated in the monitoring plan. In addition to noting the respective person responsible for follow-up, a timeframe for when each parameter is monitored needs to be determined. Follow-up visits with patients can be used for a variety of reasons: assessment of meeting goals and outcomes, effectiveness of drug therapy, safety of drug therapy, and also the assessment of any new drug-related problems (using the parameters of indicated, effective, safe, and adherent). Therefore, follow-up visits should be considered a vital component in the care provision of patients.

An example of a complete monitoring and follow-up plan is included in Table 1.4. This example is a monitoring plan developed based on the patient's medical conditions.

Table 1.4 Example of a monitoring and follow-up plan

Medical condition	Parameter	Frequency	Rationale
Hypertension (treated by ramipril 5-mg po BID)	Efficacy: BP <120/80	Daily (by patient: home BP monitor); every 3 months (by family doctor); at refills (by pharmacist)	Target based on patient's comorbidities
	Safety: Potassium, SCr, cough	1 Week after initiation of ramipril; prn afterward	
Diabetes (treated by metformin 750-mg po BID)	Efficacy: A1c <7%	q3 months (by pharmacist)	Target A1C as per guidelines
	Fasting blood glucose pre-meals 4–7 mmol/L postprandial blood glucose 5–8 mmol/L	3× per week (by patient)	
	Safety: GI disturbances such as diarrhea	At refills (by pharmacist)	

BID twice daily, *BP* blood pressure, *GI* gastrointestinal, *prn* when needed, *SCr* serum creatinine

Documentation

A care plan is meant to be used as a tool specifically for pharmacists, whereas documentation is meant to be a legal document of care provided to a patient and to inform other healthcare providers what care was provided to the patient. Usually, a care plan is significantly longer and more thorough, whereas a documentation note is shorter and more succinct. Documentation is essential owing to the following reasons:

- To help maintain patient safety
- To comply with legal requirements
- To avoid duplication of work
- To facilitate communication
- To comply with the standard of practice in different jurisdictions
- To facilitate quality assurance

Documentation can take many forms, such as a quick note in dispensing software or a longer consult letter that is related to only one particular concern. Notes can be structured (such

as DAP (data, assessment, and plan), SOAP (subjective, objective, assessment, plan)) or unstructured such as those focused on a complete medication history or only on one specific problem (e.g., assessment of drug levels). Only care that has been documented can be assumed to have been provided: If you did not document the care, it did not happen [5]. A practitioner's documentation will be highly dependent on where they practice, requirements for that place of practice, and how care is communicated between practitioners.

Structured Documentation

Using a structured format to help guide documentation, a pharmacist can ensure that the essential information is included. Two common formats are used in practice: a DAP note and an SOAP note. A DAP note includes data, assessment, and plan. Other types of structured documentation can include preprinted forms, such as medication reconciliation or allergy assessment forms, often used in care facilities and hospitals.

The difference between a DAP note and an SOAP note is that in an SOAP note, the data is further separated into subjective and objective data. Data includes a succinct summary of the information collected during the patient interview that is relevant to the problem being discussed. Only data related to completed patient assessment should be included; including too much data that is not directly related to the purpose of the note can make the DAP note unreasonably long. Long notes run the risk of not being read by other members of the care team. Other elements to include in the data section are the patient's goals and preferences.

The assessment component of the note is the pharmacist's assessment and determination of what the drug-related issue is and the rationale behind the assessment. This includes the pharmacist's professional interpretation of the data presented. The rationale should be clear so that other healthcare providers can understand the assessment and, consequently, the plan.

The plan includes specific recommendations for the problem as well as follow-up and monitoring. Recommendations or prescribing decisions should include the specifics of dose, route, and duration. Recommendations can also include nonmedication-focused interventions such as lifestyle factors and patient education. Follow-up should include who is responsible for what elements of follow-up, so work is not

being duplicated. In addition, it should include the timeframe for monitoring and follow-up. Two examples of DAP notes are provided in Boxes 1.1 and 1.2.

Box 1.1 Structured Documentation Example 1:

Community Pharmacy

August 1, 2025 9:30 am: Rx#1234567 New prescription for hypertension

Data

- Mr. Y received a new prescription for hydrochlorothiazide 25 mg daily
- Average BP over last three pharmacy visits: 150/82 (May 2025); 153/79 (June 2025); and 157/90 (July 2025)
- SCr 85 $\mu\text{mol/L}$, calculated CrCl 95 mL/min, K+ 4.2 mmol/L (labs conducted yesterday)
- No other medications or medical conditions
- Patient expresses concern about starting a regular medication

Assessment

- Hydrochlorothiazide is indicated as a first-line treatment for hypertension, and dose the appears appropriate. Target for patient is BP <140/90

Plan

- The pharmacist to check SCr, K+ in 1 week
The pharmacist to assess for safety in 1 month at next refill: dizziness, postural hypotension
- The pharmacist to assess for efficacy at next refill: BP measurement in pharmacy
- The pharmacist counseled the patient on how to take medication and common adverse effects; discussed patient's concerns about starting medication; and offered suggestions to ensure adherence
- The patient to monitor BP at home daily for the first week, then a few times a week until next refill The patient will call the pharmacist if any concerns

Abbreviations: Rx#, prescription number; BP, blood pressure; SCr, serum creatinine; K+, potassium

Box 1.2 Structured Documentation Example 2: Hospitalized Patient

Pharmacist's note RE: Vancomycin

D: 47 F (70 kg, 176 cm) admitted to the hospital for traumatic brain injury 6 days ago. She was started on vancomycin 3 days ago for MRSA hospital acquired pneumonia

Vitals (today): $T_{\max}$ 37.2 (was 38.7); BP 138/75; HR 75; RR 13; O_2 sat 95% currently on room air (was on 4L O_2 3 days ago)

Labs: SCr 75 $\mu\text{mol/L}$ (stable); est. CrCl ~87 ml/min; BUN 7 mmol/L; WBC $9 \leftarrow 13 \leftarrow 19 \times 10^9/L$; Neut. $6 \leftarrow 10 \leftarrow 14 \times 10^9/L$

Microbiology: Sputum culture (4 days ago): 3+ MRSA sensitive to vancomycin

Chest X-ray (3 days ago): LLL pneumonia

Current antibiotics: Vancomycin 1 g iv q8h started 3 days ago (dose times 00:30, 08:30, 16:30)

Vancomycin level drawn today at 00:02 was 17 mg/L (before the sixth dose)

A: Patient is appropriately treated with vancomycin given the MRSA pneumonia. Vancomycin level was drawn appropriately before the sixth dose and was within target for MRSA pneumonia (10–20 mg/L) and most likely reflects steady state given patient's weight and renal function. The patient is clinically improving (afebrile with less oxygen requirements; improving WBC and neut.)

P: Continue vancomycin at the current dose 1 g iv q8h for a total of 14 days. Team to monitor SCr and CBC at least 2x per week as long as the patient is on vancomycin. Pharmacist to follow up and order vancomycin levels as appropriate.

Abbreviations: A, assessment; BP, blood pressure; BUN, blood urea nitrogen; D, data; F, female HR, heart rate; P, plan; LLL, left lower lobe; MRSA, methicillin-resistant *Staphylococcus aureus*; RR, respiratory rate; SCr, serum creatinine; $T_{\max}$, maximum temperature in 24 h; WBC, white blood cell

Unstructured Documentation

There are situations in which a full DAP note is not required, such as a note to clarify a previous order, a follow-up note, allergy assessment, and renal dosage adjustment. A pharmacist should exercise their professional judgment about when an unstructured note is appropriate and, when in doubt, should err on the side of a structured note to ensure completeness of documentation.

Some key elements that should be considered in all forms of documentation are that documentation should:

- Be done in a timely manner: Ideally during or immediately after care is provided
- Be concise: Notes should be short and to the point
- Be complete: Assumptions should not be made about missing information
- Avoid dangerous abbreviations: E.g., use “daily” instead of “qd”
- Use a professional tone and avoid terms that are inappropriate, unnecessary, etc. and avoid judgments or blame for errors

Conclusions

Utilizing a structured process to approach patient care ensures that pharmacists are both thorough and complete and that all actual and potential drug-related problems are identified. Assessment starts with interviewing the patient and creating a database, which includes all relevant details of the patient's medical and medication history. From there, the pharmacist can assess drug therapy by considering the four following parameters: indicated, effective, safe, and adherent. After this initial assessment, a care plan can be developed (including recommendations) and implemented with appropriate follow-up and monitoring. Documentation is essential, so practitioners can be accountable for the care provided. These are the essential elements of the patient care process, which is the heart of the care that pharmacists provide.

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Principles of Patient Assessment

2

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Chapter Objectives

1. Describe the role of patient assessment in pharmacy practice.
2. Describe the steps of symptom assessment.
3. Demonstrate an understanding of chronic disease assessment at both the initial presentation and follow-up.
4. Apply the principles of patient assessment to allergies, adverse reactions, and drug interaction assessment.

Background

The pharmacist's role as a healthcare practitioner is evolving as they are taking a more active part in primary patient care. Clinical services are now becoming the forefront of pharmacy practice as pharmacists are helping patients manage their medications and diseases, providing patient education, and, in some provinces, prescribing and adapting medications. Pharmacists can be accessed in a variety of practice settings—for example, in community pharmacies, hospitals, and specialized practices. This wide range of practice settings gives pharmacists the opportunity to be involved in patient care for a variety of patients, from those who are acutely ill to those in long-term care. Pharmacists' involvement in patient care can make a significant difference to clinical outcomes and patient care and can decrease medication-related adverse events.

Pharmacists are the healthcare professionals most accessible to the public. Patients frequently approach pharmacists with their health-related questions. Many of these questions center around patients with acute minor illnesses who request

a pharmacist to help them select an over-the-counter (OTC) product. Furthermore, frequent visits to the pharmacy for medication refills provide an opportunity for pharmacists to have regular follow-ups with patients and to be involved in chronic disease management. Finally, as medication experts, pharmacists play a significant role in ensuring safe medication use for each patient throughout his/her treatment course. All of these roles require pharmacists to be familiar with and capable of providing patient assessment.

Patient assessment is the process of methodically collecting information while also using clinical judgment and therapeutic knowledge to identify a patient's actual and potential drug-related problems. The data collected can be objective, such as lab tests or diagnostic imaging, or subjective, such as those obtained from the patient. Patient assessment is a skill that requires an organized process and knowledge of the presenting symptom or disease. Pharmacists are required to be able to identify red flags and important patient-specific characteristics throughout the assessment. With this information, the pharmacist can formulate a plan and, if appropriate, provide patient education, a drug or nondrug recommendation, or a referral to another healthcare practitioner. Although patient assessment is a clinical task, it also requires strong communication skills, as the patient is a major source of information. The ability to connect with the patient and conduct a patient-centered interview is another facet of patient assessment that cannot be overlooked.

There are different types of patient assessments due to the wide range of symptoms, diseases, and medications that can be encountered by pharmacists. A symptom assessment could be conducted when a patient presents to the pharmacy requesting a recommendation for a minor ailment, such as a cough. Patients with chronic diseases, such as diabetes, could require a chronic disease assessment at both their initial diagnosis and at regular follow-up visits. In addition, patient assessment involves an approach to assessing adverse reactions, allergies, and drug–drug interactions. Patient assessments are part of the patient care that pharmacists pro-

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vide and play a significant role in the provision of clinical services.

This chapter provides an overview of patient assessment. Patient assessments specific to a symptom or disease can be found in the respective chapter.

Connection for Effective Patient Assessment

Patient assessment relies on gathering not only a patient's history, medications, physical assessments, and laboratory values but also their lived experiences with the disease and medications [1, 2]. These are often referred to as illness and medication experience, respectively, and are valuable in identifying the causes of a patient's medication-related problems [3]. A patient-centered interviewing approach, *Smith's Patient-Centered Interview: An Evidence-Based Approach*, was created to gather lived experiences alongside traditional biomedical information [4]. This model recognizes the complexity of collecting a patient's perspectives alongside specific clinical data. More than 35 years of research indicates that patient-centered interviewing increases clinician job satisfaction, elicits increased patient information, reduces malpractice suits, and improves both adherence and patient outcomes, including blood pressure (BP) and diabetic control [4].

Patient-centered interviewing has three main stages [4]. First, clinicians are encouraged to connect with the patient and set an agenda, followed by patient-centered interviewing to explore the patient's illness and medication experiences and the impact on their lives. Pharmacists would invite patients and encourage them to share the personal and emotional contexts of their story, respond to emotions, and summarize the information to check for accuracy. Pharmacists could explicitly transition to the middle stage with clinical-centered interviewing where the pharmacist would obtain a medical and medication history followed by a physical exam as required. In the end stage, the pharmacist could share information and make a plan.

The quality of patient assessment and patient-centered interviewing depends on the pharmacist and patient connection. A high-quality physician–patient relationship can improve patient health, including measures such as blood pressure, pain scores, and quality of life [5]. While comparable research is not available for pharmacists, patients have requested that pharmacists should recognize them as individuals. Although not all patients feel a need for an ongoing relationship with their pharmacist [6–8], most want to feel a connection and to be treated with respect in all pharmacist encounters [9]. Establishment of a patient connection, whether it leads to a relationship or not, relies on the pharmacist's frame of mind as well as strong communication skills.

Mindful Practice

Pharmacists have been called on to enhance their clinical skills and reduce medication errors by cultivating a mindful practice [10, 11]. The qualities of a mindful practice include observing one's thoughts and judgments, maintaining curiosity, and acting with compassion [12]. Specifically, in pharmacy practice, Shoemaker suggests that pharmacists should listen to the patient's story, acknowledge their individuality, and be curious about them [13]. Mindfulness can be as simple as taking a single deep breath to focus one's self before meeting with a patient and can extend to formal mindfulness practice. A practical and evidence-based approach for professionals has been outlined in *Search Inside Yourself* [14]. In the health context, limited research suggests that mindfulness training may reduce physician burnout [6] and increase attentiveness [15].

Communication Skills

Patient connections are facilitated by strong pharmacist communication skills, which can be learnt through practice [9]. A connection can be established early in the encounter when pharmacists introduce themselves and take a moment for small talk. These routine greeting rituals recognize a patient as an individual and may lead to insights into a patient's goals and preferences. Pharmacists should use a patient's name, as this shows respect and recognition that again fosters a connection. While these ideas are self-evident, their importance is easy to overlook in a fast-paced practice.

Patients often perceive pharmacists, whether in a community pharmacy, hospital, or clinic, as busy clinicians who may not have time for them. Three specific skills can help pharmacists efficiently create the necessary space for connections: acknowledgment of time, private space, and agenda setting. First, pharmacists can pause and reassure a patient that there is time, despite the fact that the environment may appear busy. Otherwise, patients may choose to save their concerns for another time. Second, pharmacists can offer a private area to talk (if appropriate). Many patients are not aware that pharmacists have space; however, patients are quite aware of the people standing behind or sitting near them who could overhear their conversation. A pharmacist could take the lead by saying “Why don't you come with me to a private area, so we will not be interrupted?” Third, pharmacists could set an agenda even for brief conversations to identify patient questions as well as prevent important questions at the end. Pharmacists can simply ask the patient what they would like to talk about and then follow-up with “what else” until all topics are raised. The pharmacist and patient can then agree on what can be discussed at present and what

can be scheduled in the future. Patients are often reluctant to raise all concerns, and encouragement helps prioritize issues.

Symptom Assessment (Utilize the SCHOLARE Acronym)

A symptom assessment is a process that a pharmacist uses to collect information regarding a symptom that a patient is experiencing, such as a cough, a sore throat, or a headache. Pharmacists conduct numerous symptom assessments during a single shift. Patients commonly seek pharmacist advice on how to treat their minor ailments and are looking for recommendations on an over-the-counter (OTC) product. Before a pharmacist delves into collecting information on a symptom, it is important to obtain a brief medical and medication history from the patient. This information can affect the questions asked during the assessment and the end recommendation. Throughout the process of the symptom assessment, the pharmacist should be aware of patient-specific characteristics that could affect their advice or plan. Such patient-specific characteristics include age, comorbidities, pregnancy or breastfeeding status, and medication history. Patient-specific characteristics are also important when the pharmacist is assessing for the presence of red flags. A red flag is a symptom or patient characteristic that alerts the pharmacist that there could be a more severe underlying problem, and, often, referral to another healthcare practitioner is required. Some examples of red flags would be the presence of a new-onset headache at 50 years or older or blood in the stool with diarrhea. If a pharmacist identifies a red flag, it is important to inform the patient of the next step, which is often seeking additional care. The pharmacist should also advise the patient how urgent the need to seek the attention of another healthcare practitioner is, such as the need to make an appointment with his/her family physician or the need to immediately proceed to the emergency department. For patient-specific characteristics and red flags that are relevant to a particular symptom, please refer to the respective chapter related to that topic.

Conducting a thorough symptom assessment is important to get the full history of the symptom, recognize red flags, and make a final recommendation to the patient. Although the goal is to conduct a comprehensive assessment, this can be lengthy and is not always feasible concerning the pharmacy workload or the patient's time. Possible time constraints are why it is important to have a developed process when conducting an assessment, which allows you to gather information in an organized and efficient manner. An organized process also helps pharmacists know that they have not missed important information during the assessment or, on the other hand, are repeating questions that will bring forth already known information.

An assessment should begin with an introduction so that the patient knows that they are speaking to a pharmacist. The next step is to determine the patient's chief complaint and his/her reason for seeking care. It is essential to know the chief complaint before gathering the patient's history as it may help the pharmacist pick up on key parts of a patient's history or patient characteristics when conducting the rest of the assessment. For example, if the patient has constipation and you begin to conduct a patient history and he/she is taking opioids for chronic pain then you will be primed to ask specific questions about this possible relationship during symptom assessment.

Many acronyms have been developed to help healthcare practitioners with the assessment process. Some examples of acronyms include LQQOSMA (Location, Quantity, Quality, Onset, Setting, Modifying and Aggravating factors), SOCRATES (Site, Onset, Character, Radiation, Association, Time Course, Exacerbating and Relieving Factors and Severity), and SCHOLAR or SCHOLARE. In this book, we will be using SCHOLAR/E. SCHOLARE stands for Symptoms, Characteristics, History, Onset, Location, Aggravating factors, Remitting factors and Explanatory model. Each category in this acronym has a purpose, and methodically going through each step will help the pharmacist collect a full history. The first letter is S, for *symptoms*, which is an opportunity to ask about the chief complaint and inquire about any other symptoms the patient may be experiencing. The next letter, C, for *characteristics*, focuses on the patient describing the symptom and the pharmacist learning about the quality and severity of the presenting symptom. This part of the assessment is also an opportunity to ask closed-ended questions about the presence or absence of specific characteristics that could also be present. Asking these particular questions can be helpful when determining if any red flags are present. Next is the *history* section. The goal is to determine how long the patient has been experiencing the symptom and if the patient has experienced this ailment before. A thorough history can be helpful in identifying any red flags regarding the length of time the symptom has been present or its frequency. Also, if the patient has had the same symptom before, it can be helpful to ask about any treatments tried in the past and if they were successful or not. *Onset* asks about when the symptom started and details about this time. This includes what the patient was doing at the time of presentation or if the patient has had any recent changes in his/her life. Determining the onset can help elicit the most likely cause, which could assist in deciding on symptom management. *Location* seeks to understand where the symptom is located in the body. It may be unnecessary to ask about the location if it is obvious, such as with a sore throat or an earache. If the patient is complaining of pain; however, it is important to know the location as well as if the pain is radiating. *Aggravating* and *remitting factors* explore

what makes the symptom worse or better and also if the patient has tried any treatment at this point and if that has been successful. Lastly, the *explanatory model* examines how individuals perceive and interpret their symptoms. Examples of questions in each category can be found in Table 2.1. It is important to note that not all questions will be relevant to every symptom assessment. For specific symptom assessment questions, see chapters on the related topic.

After completion of this process, the pharmacist should have a clear picture of the presenting symptom. Further questioning may be required to gather more specific information if needed. Additional information gathering such as laboratory test results may be needed for proper assessment. Once the assessment is complete, a quick summary of the data should be presented to the patient. A summary allows the pharmacist to check for accuracy, can help summarize the information, and may be helpful in recognizing additional information that is needed. Finally, ask the patient if there is anything else they would like to add that may have been missed. After the symptom assessment is complete, the pharmacist will be able to create a plan. The plan will be specific to the patient and the symptom and may include a recommendation of pharmacological treatment, a nonpharmacological treatment, or a referral to another healthcare practitioner. The recommendation of a drug could be either an OTC product or prescription medication based on the province the pharmacist practices in and its respective scope of practice.

Table 2.1 Examples of SCHOLARE questions

Part of assessment	Examples of questions
Symptoms	<i>What is your main symptom?</i> <i>Are you experiencing any other symptoms?</i>
Characteristics	<i>Describe the symptom</i> <i>On a scale of 1–10, how severe is your symptom?</i> <i>How often is this symptom present?</i>
History	<i>How long have you had this?</i> <i>Have you experienced this in the past?</i>
Onset	<i>When did it start?</i> <i>What were you doing when it started?</i> <i>Was it a gradual or abrupt onset?</i>
Location	<i>Describe the location of the symptom</i> <i>Is there any radiation from this location?</i>
Aggravating factors	<i>What makes it worse?</i>
Remitting factors	<i>What makes it better?</i> <i>Have you tried anything to treat the symptom?</i>
Explanatory model	<i>What do you think might be the cause of your symptoms?</i> <i>What do you worry about most regarding your illness?</i> <i>What kind of treatment do you think would be most helpful?</i>

SCHOLARE Symptoms, Characteristics, History, Onset, Location, Aggravating factors, Remitting factors and Explanatory Model

Chronic Disease Assessment

Initial Assessment (Utilize the IESA Acronym)

Pharmacists play a significant role in the initial assessment of patients who have been recently diagnosed with a chronic disease because, most often, this diagnosis accompanies the initiation of one or more new medications. A new medication requires a thorough assessment by the pharmacist whether at a community pharmacy or in a hospital. This assessment should begin with the formulation of a complete patient history. The history should include patient demographics, medical history, medication history, social history, and allergies. Although the focus of this section is chronic disease assessment, the following overview of an initial assessment can also be applied to transient illnesses. Patients with acute illnesses, such as an infection or anemia, are also initiated on new medications and require an initial assessment by a pharmacist as well. The pharmacist has a responsibility to ensure that the medication is indicated, effective, and safe and that the patient can adhere to the treatment. An acronym that can be used to remember these four parameters of assessment is *IESA*. For a summary of the IESA steps, refer to Table 2.2.

Indication

After a new medication is prescribed, the pharmacist should initially assess if the new drug is indicated. It is important to determine the reason this new medication was prescribed to the patient. Based on the medication prescribed, an assumption of the disease could be made, but it is also impor-

Table 2.2 Summary of the steps of a chronic disease initial assessment (IESA)

Indication	Determine the reason this new medication was prescribed to the patient Assess if the drug is indicated and if medication therapy is currently warranted
Effective	Assess if the drug prescribed is an optimal treatment choice Determine if the dose of medication is appropriate given the chronic disease Consider patient-specific characteristics that may influence drug efficacy (e.g., age, comorbidities, pregnancy status) Create a plan to monitor for medication efficacy
Safety	Assess if the dose and frequency of the medication prescribed are appropriate for the indication Determine if a medication is safe for the individual patient given the patient-specific characteristics (e.g., age, comorbidities, allergies, pregnancy status) Consider the potential for drug interactions Create a plan to address any safety concerns if present Create a plan for continued monitoring of safety
Adherence	The patient is the main source of information on this topic The pharmacist's role is to help brainstorm options to support patient adherence

tant to ask the patient if he/she knows the indication. Asking the patient may present you with an unexpected indication for the medication, as many drugs are used in multiple diseases. After confirmation of the diagnosis, an assessment of the appropriateness of the drug should be made given the chronic disease. It is often helpful to obtain a history of the present illness, as a more detailed history of what led to the diagnosis can be helpful in the initial assessment as well as during follow-up. The pharmacist can then assess if the medication is indicated to treat this disease and also if medication therapy is currently warranted.

Effective

After confirming the indication, the next step is to assess if the medication has the potential to be effective. The drug needs to be an appropriate choice for the patient's illness. The pharmacist should assess if the drug prescribed is an optimal treatment choice, which can be determined by evaluating if the medication is a first-line therapy option. Many chronic diseases have published clinical guidelines that provide evidence-based recommendations on first-line therapies for the disease. There are cases in which the patient for various reasons is intentionally not prescribed a first-line therapy, and this should be assessed on a case-by-case basis. If the medication is an appropriate option for the chronic disease, the pharmacist can progress to checking the prescribed dose. For a drug to be effective, usually, the dose of the medication should be in a particular dosing range. Keep in mind, however, that the recommended dose of a medication can differ depending on the type and severity of the illness. Also, patient-specific characteristics can impact the dose of the medication, and this should be taken into consideration (discussed below). Finally, prescribers may choose to titrate a medication up to the recommended dose to prevent adverse effects at the beginning of therapy; therefore, the original prescribed dose may be lower than the target. At this point, the pharmacist should also consider how efficacy will be monitored after therapy begins. The monitoring plan may include subjective information from the patient, laboratory tests, diagnostic imaging, or a physical examination. The pharmacist should create a plan to monitor for medication efficacy and inquire if there is a follow-up scheduled with the prescriber. If, during this part of the initial assessment, the pharmacist judges that the medication and/or its dosage regimen is unlikely to be an effective option, then the next step may be to adapt the prescription or contact the prescriber.

Safety

An essential part of the initial assessment is to assess the safety of the newly initiated medication. Ensuring a drug is safe for patients is the responsibility and one of the main roles of the pharmacist. The first step is to assess if the dose

and frequency of the medication prescribed are appropriate for the indication, as previously discussed. The next step is to determine if a drug is safe for the individual patient given the patient-specific characteristics. Such characteristics include the patient's age, comorbidities, allergies, and medication history. Certain medications need to be used with caution or are contraindicated in specific disease states or when used concomitantly with other drugs, and this can be a safety concern. An example of an interaction with a disease is the use of bupropion in patients with a seizure disorder as it can lower the seizure threshold and therefore is not preferable in those patients. The patient's age is important as it can affect the dose or dosing frequency of the medication; for example, a young child will often require a different dose than an adult. Patient comorbidities can also affect the dose, dosing frequency, and medication choice. For example, patients with impaired kidney function who have been prescribed a drug that is renally eliminated may require dose adjustments. Allergies can be a significant safety concern, and the pharmacist needs to know if a patient has an allergy before dispensing a new medication. In addition, an assessment of the compatibility of the new drug with the patient's other medications should be conducted. Drug interactions can occur among many drugs and can result in ineffective therapy and adverse effects. Furthermore, other patient factors that should be considered are physical characteristics (weight, height, etc.), previous medication intolerances, and the medication's side effect profile. If the pharmacist is concerned that the drug may be unsafe for the patient, this should be addressed with a plan. This plan may include prescribing or recommending an alternative medication, adjusting the dose or dosing frequency, or contacting the prescriber. Another part of the safety assessment is formulating a monitoring plan. This plan may include future lab tests, such as serum creatinine (SCr) or liver chemistry tests. The plan should indicate when the tests need to be conducted and the health-care practitioner who will be responsible for ordering the test and interpreting the results. A more detailed overview of an allergy, adverse effects, and drug interaction assessment are provided later in this chapter.

The chronic disease initial assessment is usually quite comprehensive due to the patient beginning therapy on new long-term medication. Many patients with chronic conditions are also on multiple other drugs and often have other comorbidities, which adds an extra layer of complexity to the pharmacist assessment. This interaction should end with a plan for follow-up and an agreement with the patient on the means of communication (e.g., phone, pharmacy visit) and the approximate time to follow-up (e.g., 2 weeks). For example, a patient starting on an antihypertensive could have a planned follow-up in 2 weeks, so the pharmacist can assess for initial efficacy and the development of adverse reactions and address any patient concerns or questions.

Adherence

Adherence is an important part of the initial assessment that is often overlooked. If a patient is unable to adhere to medication therapy, then it is unlikely to be effective and might also put the patient at harm. Even before the initiation of therapy, there can be indicators that the patient may have challenges with adherence for a variety of reasons:

- The frequency of medication doses (e.g., three times a day)
- Busy lifestyle
- Low health literacy
- Language barriers
- Problems with dexterity or ability to self-administer medications
- Memory problems or dementia
- Lack of motivation
- Low self-efficacy
- Lack of understanding of the chronic condition
- Insufficient or limited medication coverage by the patient's insurance

Patients are the main source of information on this topic, as they are the ones who have to fit the new medications into their schedule. If there is concern regarding nonadherence, the pharmacist's role is to help brainstorm options to support the patient. Adherence can be intentional or nonintentional, and factors that could contribute to either should be part of the assessment. An important consideration when assessing adherence concerns is the frequency of medication doses. For example, if a patient with a busy schedule is starting on a medication that is dosed three times a day, this may be an indicator that adhering to this dosing schedule may present a challenge. Other considerations could be the patient's health literacy, possible language barriers, and the ability of the patient to self-administer the medication. Medication adherence can also be due to a lack of understanding of the disease and the role medication plays in its management. All of these potential reasons would require the creation of a plan to help circumvent the possibility of nonadherence. The plan may include, for example, taking the time to write out patient-specific detailed instructions, the use of a translator, blister packing medications, or collaboration with other healthcare professionals to arrange assistance in medication administration.

Patient education is important for increasing patients' understanding of their disease, the benefit of the medication, and the potential harms if the disease is untreated. Another common reason for nonadherence is the medication cost. Pharmacists should consider the cost of the medication to the patient and if the patient will be able to afford to continue to pay for the drug. If this is a concern, options may include generic substitution, switching to a less expensive alterna-

tive, or looking into compassionate drug coverage options. Motivation is another key factor to adherence, and it can fluctuate throughout treatment. The diagnosis of a chronic disease and the initiation of a new medication mean that the patient will be asked to make changes in his/her life. This could include changes to a patient's lifestyle, such as exercise or diet, as well as the addition of a new medication into his/her schedule. A patient's motivation for a change can be an indicator of future adherence to therapy. Techniques to increase motivation could include motivational interviewing [16], patient education, and perhaps breaking down lifestyle changes into small gradual steps. Self-efficacy can also be a factor in patient adherence; if a patient does not believe he/she will succeed in making lifestyle changes, this may cause a defeated attitude and lack of motivation from the beginning. Similar techniques to motivational interviewing can be used to help increase self-efficacy.

Follow-Up Assessment (Utilize the 2CAs Acronym)

After a patient's initial diagnosis of a chronic disease and the pharmacist's initial assessment, follow-up assessments are conducted. Patients with chronic conditions require continued assessment throughout their disease course. They are often on long-term therapy, and, during this time, there can be changes in the disease, medications, the patient's lifestyle, and the patient's attitude toward his/her condition. Follow-up assessments allow the pharmacist to evaluate the patients' therapy, provide further education, and discuss patient concerns, questions, and thoughts. Follow-up assessments are conducted at regular intervals that usually coincide with the patient's visits to the pharmacy for medication refills. Follow-up can also occur in the hospital setting for a patient with a previously diagnosed disease or in a clinic with regularly scheduled follow-ups irrespective of the medication refill schedule. More frequent assessments usually occur soon after the initial diagnosis as patients often have more questions at this time and a pharmacist can address patients' concerns early on. Also, many medication adverse reactions occur soon after initiation of therapy, and these reactions should be addressed as quickly as possible. A prompt assessment is required because of the risk to patient safety, and adverse reactions can cause decreased medication adherence. Regardless of when the follow-up assessment happens in the trajectory of the disease, it should focus on four main topics. These areas of focus are disease control, complications, adverse reactions to the medications, and adherence to the treatment plan; this can be remembered by the two Cs and two As of a follow-up assessment (the *2CAs acronym*). For a summary of the steps of a follow-up assessment, refer to Table 2.3.

Table 2.3 Summary of the steps of a chronic disease follow-up assessment (2CAs acronym)

Disease control	Assess to determine if the medication is currently effective at treating the specific chronic disease Can include both objective (e.g., lab tests) and subjective data collected from the patient
Disease complications	Often, an indication that the current therapy is not effective or potentially that the disease is progressing Requires a reevaluation of the current treatment and potentially a change in medication, dose, or addition of a new medication
Adverse reactions to the drug	Assess early on and throughout the entire course of treatment May require a change to the medication, the dose, or a referral to another healthcare practitioner if severe
Adherence	Nonadherence can lead to disease complications and a lack of disease control Pharmacists should be nonjudgmental during this discussion with patients Work with the patient to discover his/her barriers to adherence and discuss possible solutions

Disease Control

An assessment of disease control should occur at each follow-up as it provides evidence of the efficacy of the current pharmacological and nonpharmacological therapies. This part of the assessment can include both objective and subjective data collected from the patient. Objective information could include lab tests (e.g., hemoglobin A1C for diabetic patients), diagnostic imaging, or the occurrence of an event (e.g., stroke, seizure). Collecting information from the patient is often a helpful means to facilitate assessment of disease control. Patients can provide information on the presence of disease symptoms, home test readings if applicable (e.g., blood pressure), and their thoughts on the control of their condition. Depending on the disease, this part of the assessment may be more focused on objective or subjective data. For example, to assess the disease control of a patient with hyperlipidemia, the pharmacist would be looking almost solely at lab test results. This example can be contrasted to a patient with chronic pain where the pharmacist would rely on the patient's account of his/her pain control. If the disease is not adequately controlled, then changes in therapy may need to occur, such as a medication change, change in dose, or addition of another medication.

Disease Complications

Complications of the disease should also be assessed at each follow-up. These can include the occurrence of an event (e.g., myocardial infarction, chronic obstructive pulmonary

disease (COPD) exacerbation) or a worsening of symptoms. Disease complications are often an indication that the current therapy is not effective or potentially that the disease is progressing. Identifying red flags is important for assessing the need for referral to another healthcare practitioner. An example would be the presence of a diabetic foot ulcer in a diabetic patient.

Follow-up assessments give pharmacists the opportunity to continue to assess a patient's medication therapy and disease control. If there are no concerns at the time of assessment, this assessment can often be brief with no changes required. If there are concerns or complications, the pharmacist can address this by formulating a plan to either modify medication treatment, refer the patient to another healthcare professional, or provide patient education.

Adverse Reactions to the Medication

Adverse drug reactions (ADRs) are often assessed early on but also need to be continued to be evaluated throughout the entire course of treatment. Some medications' adverse reactions do not occur for months to years after the initiation of therapy. Pharmacists play an essential role in monitoring and managing adverse medication reactions. Adverse reactions can range from mild to very severe and need to be addressed as they are a safety concern and can cause decreased adherence to therapy. If a patient is experiencing an adverse reaction, a medication or dose change may be required, and, if the reaction is severe, a referral to another healthcare practitioner for treatment may be needed. A pharmacist should be aware of the red flags that would prompt a referral to another healthcare professional. An example would include a patient on a nonsteroidal anti-inflammatory drug (NSAID) that presents with blood in the stool.

Adherence

As discussed earlier, pharmacists play a vital role in the assessment of a patient's adherence to treatment. A pharmacist can assess a patient's adherence to both pharmacological treatments and nonpharmacological therapy, such as lifestyle changes. Adherence is important to assess, as nonadherence can lead to disease complications and a lack of disease control. It is important to note that adherence can change throughout treatment, which is why it should be assessed at each follow-up visit. Adherence can be intentional and non-intentional, but pharmacists should be nonjudgmental and noncritical during discussions with patients. Pharmacists should work with the patient to discover his/her barriers to adherence and discuss possible solutions that the patient believes may help, such as alarms, dosettes, etc.

Allergy Assessment

A patient's allergies should always be kept on the patient's healthcare or pharmacy file. When a pharmacist encounters a new patient, they must ensure that a full list of the patient's allergies is collected. Also, when a patient is prescribed a new medication, the pharmacist should always check that the patient's allergy information is up to date so that an allergy assessment can be conducted. An allergy assessment is used to determine a patient's "true" allergies and their accompanying reactions. With the knowledge of the patient's drug allergies, the pharmacist can ensure the safety of the drug for the patient. An allergy assessment prevents not only morbidity and mortality but also the unnecessary use of less appropriate therapeutic alternatives. Adverse reactions or intolerances to drugs can often be mislabeled as allergies. Determining if a patient has a "true" drug allergy prevents a prescriber from avoiding certain drugs that may be better therapeutic options for the patient. Allergies usually present as one or all of the following symptoms: urticaria, skin eruptions, bronchospasm, angioedema, and/or anaphylaxis. The presence of these symptoms is evidence that the patient has an allergy to the drug. Intolerances that are labeled as an allergy are often gastrointestinal-related such as stomach upset, nausea and vomiting, or diarrhea. Other adverse reactions that are reported as allergies include dizziness, drowsiness, or delirium.

When a patient starts a new medication, the pharmacist should either have or compile a list of the patient's allergies and the reaction he/she experienced. Just writing down a list of allergies without determining the symptoms the patient had is a common cause for intolerances being mislabeled as allergies. The pharmacist should also ask when in the patient's life he/she last experienced the allergic reaction. The time period is important because there are allergic reactions that some patients may outgrow as they become adults, such as a penicillin allergy. Also, the pharmacist should establish a timeline of the symptoms, including the date the drug was dosed, the length of therapy, and the date the allergic reaction occurred. Determining a timeline helps the pharmacist evaluate if the allergy was likely due to the drug or other causes. Finally, the pharmacist should ask if the patient has ever had the drug allergy rechallenged. If the patient did receive the drug again and there was no allergic reaction, then the drug is likely safe for the patient to receive again.

If it is determined that the patient had a previous allergy to a drug, then the next step is to create a plan on how to manage this potential drug-related problem. If the reaction was mild, such as a minor skin rash, then the best option may be to rechallenge the allergy. If the patient is willing to rechallenge, then the pharmacist should provide education on the signs and symptoms of an allergic reaction and follow

up with the patient in a few days to ensure no reaction has occurred. The patient may not be willing to rechallenge the allergy due to fear of a reaction, and a therapeutic equivalent may need to be prescribed. If the previous allergic reaction was severe, such as angioedema, anaphylaxis, or Stevens–Johnson syndrome, then choosing an alternative drug would be the safest option. There are cases, however, where the drug that has been prescribed is a far superior therapeutic option or even the only option. In these cases, a risk-versus-benefit analysis can be conducted, and, if the benefit of the drug outweighs the risk of the allergy, then there are desensitization protocols that can be ordered under the supervision of a physician.

Another aspect of an allergy assessment is cross-sensitivity, which is the concept that an allergy to one drug predisposes an individual to have an allergy to another drug with a similar chemical structure. There are reports of cross-sensitivity reactions between drugs, but the risk is often quite low. For example, the risk of a patient with a penicillin allergy experiencing an allergic reaction with a cephalosporin is <2% [17]. The pharmacist should evaluate the risk for cross-sensitivity with the use of evidence and clinical judgment. Often, the risk of cross-sensitivity is low, and the pharmacist can dispense the drug as prescribed and provide patient education on the signs and symptoms of an allergic reaction. If the past drug allergy was severe and the cross-sensitivity between the other drug is non-negligible, then a risk-versus-benefit analysis should be conducted.

If a patient is currently experiencing an allergic reaction at the pharmacy or home, then the first step is to assess the severity of the reaction. If the reaction is self-manageable, such as urticaria or a mild skin rash, then the pharmacist could suggest an OTC product like diphenhydramine. Referral to another healthcare practitioner may be required if the reaction is severe, such as if the patient is having trouble breathing. The pharmacist should inform the patient of the level of urgency that is needed in seeking additional care, and, if the allergy is life-threatening, emergency services should be called. At this time or during the patient's next visit to the pharmacy, the pharmacist should discuss the cause of the reaction with the patient. Asking the patient about any recent new medications or changes to medications can help determine if the reaction was drug-related. Any drug allergies need to be recorded in the patient's file along with the type of reaction that the patient experienced. The pharmacist should also inform patients to alert their other healthcare providers. If a patient has an allergic reaction to a medication and has not completed the course of therapy, a decision needs to be made to either continue or discontinue the medication. The decision can depend on the severity of the allergy, the patient's wishes, and the length of treatment remaining. If a replacement drug is required, then the

pharmacist should discuss with the prescriber alternative therapeutic options, preferably not in the same class of drugs.

The following are some examples of common drug allergies. These drugs should alert pharmacists to be especially diligent in their allergy assessment:

- Acetylsalicylic acid
- NSAIDs
- Antiseizure medications
- Sulfonamides
- Beta-lactams
- Opioids

Of note, other drugs not listed can also lead to allergic reactions as well as fillers in medications, such as lactose. Educating patients about the signs and symptoms of an allergic reaction may increase their ability to recognize a reaction and seek treatment quicker.

Adverse Drug Reaction Assessment

An adverse drug reaction (ADR) is an unwanted response to a drug that occurs at standard therapeutic doses. As drug experts, pharmacists play an important role in preventing ADRs and in creating recommendations and plans to help circumvent an ADR if it arises. In order to avoid potential ADRs, a crucial step a pharmacist can take is to conduct a thorough patient history, symptom assessment, and/or chronic disease assessment. Knowledge of these factors allows the pharmacist to identify potential drug-related problems before the patient even begins therapy and also to adapt the treatment based on any concerns. Patient-specific characteristics such as age and weight can impact the dose of the drug or can predispose a patient to an ADR. For example, elderly patients prescribed a sedative, such as zopiclone, may be predisposed to experience an ADR such as falls. When a patient begins a new medication, the pharmacist should always educate the patient on the common ADRs and potential severe ADRs. A monitoring plan should be set, e.g., what laboratory tests to be ordered and how often and who will follow up with the test results. At each follow-up visit, patients always need to be asked if they have noticed any ADR; this can include open- or closed-ended questions asking about a specific common ADR of a drug. Any ADR experienced by the patient should always be recorded on his/her file with the name of the drug and the specific reaction.

If a patient presents to the pharmacy during a follow-up or comes to discuss a possible ADR, an assessment is required to determine if it was drug-induced. There are several factors that determine if the symptoms the patient is experiencing are likely to be ADRs:

- **Temporality:** After the pharmacist has a clear idea of the presenting reaction, he/she can begin to determine if a drug is the cause. A pharmacist may suspect an ADR if the patient had recently started on a new medication or the dose of the medication had been recently changed. For a drug to be the cause, there has to be a timeline that is consistent with the presentation of the reaction. On the other hand, if the patient experiences the symptoms before drug initiation, the patient's newly started medication is not likely the cause. It is important to remember, however, that an ADR can occur months to years after a medication was initiated.
- The reported reaction fits with the drug's possible ADRs: Another consideration is to check if the reaction or pattern of the reaction fits with the literature and what has been reported previously.
- The reported reaction is biologically plausible based on the drug's mechanism of action.
- The reported reaction disappears or reverses after medication discontinuation.
- There is no alternative explanation of the patient's reported adverse reaction.

After this assessment, the pharmacist can establish the probability of the reaction being drug-related. Table 2.4 depicts the Naranjo adverse drug reaction probability scale, a tool to check the probability if the reaction is drug-induced [18]. Based on the scores calculated from the scale, the pharmacist may label it as definite, probable, possible, or unlikely. If the symptom or reaction is deemed to be not drug-induced, an assessment should be made to help determine the cause as well as a plan for treatment or referral.

ADRs can be dose-related or not dose-related. They fall under any of the following categories:

- Allergic reactions (not dose-related)
- Idiosyncratic (unpredictable reactions that are not dose-related)
- Drug toxicity (generally dose-related; occurs with drug overdose)
- Side effects (could be acute or chronic; dose or not dose-related)
- Drug withdrawal symptoms
- Drug interactions (see the next section)

Once an ADR has been confirmed, pharmacists can create a management plan. The plan may consist of continuing the drug and providing patient education if the reaction is mild or is known to be transient and the patient is willing to continue therapy. Some ADRs can also be resolved with patient education, such as if the ADR is likely caused by how the patient is administering the medication. Some medications require the

Table 2.4 Naranjo adverse drug reaction probability scale

Question	Yes	No	Unknown	Score
1. Are there previous conclusive reports on this reaction?	+1	0	0	
2. Did the adverse event appear after the suspected drug was administered?	+2	-1	0	
3. Did the adverse event improve when the drug was discontinued or a specific antagonist was administered?	+1	0	0	
4. Did the adverse event reappear when the drug was readministered?	+2	-1	0	
5. Are there alternative causes that could on their own have caused the reaction?	-1	+2	0	
6. Did the reaction reappear when a placebo was given?	-1	+1	0	
7. Was the drug detected in blood or other fluids in concentrations known to be toxic?	+1	0	0	
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0	
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	
Total score				
<i>Total score</i>				<i>Interpretation</i>
≥9				Definite
5–8				Probable
1–4				Possible
≤0				Unlikely

Adapted with permission from John Wiley and Sons. Naranjo CA et al. [18]

patient to take the drug with food to avoid gastrointestinal discomfort. Assessing if the patient is taking the drug with food could be the key to finding an easy solution to the ADR. Other plans may include dose reduction or suggesting an alternative therapy. If the drug causing the ADR is clinically required and there are no therapeutic alternatives, then a benefit-vs.-risk analysis should be performed to determine if continuing the drug has a more significant benefit to the patient than the harm caused by the drug. Regardless of the recommendation, the plan should be documented in the patient's file for future reference. In the future, if the same or similar drug is prescribed to the patient, the ADR could be rechallenged depending on the severity of the original ADR and how necessary the drug is and the availability of therapeutic alternatives and with shared decision-making with the patient.

Drug Interaction Assessment

Drug–Drug Interactions

A drug–drug interaction occurs when a drug affects the pharmacokinetic or pharmacodynamic profile of another drug. Pharmacists play a key role in the identification and manage-

ment of drug–drug interactions. For a pharmacist to be able to assess a drug interaction, a complete patient medication history is needed that includes prescription and over-the-counter medications as well as any supplements or vitamins. A drug interaction can occur via multiple mechanisms. One common type of drug interaction is when one drug alters the metabolism of another drug by inhibiting or inducing an enzyme (e.g., cytochrome P450 (CYP450)). Drugs can also affect other pharmacokinetic properties such as absorption, distribution, and excretion from the body. Moreover, pharmacodynamic drug–drug interactions are possible. Concomitant administration of drugs might lead to synergistic, antagonistic, or de novo effects. Regardless of the mechanism, drug interactions can lead to detrimental effects such as an increased risk of adverse events or therapy failure. Pharmacists play an important role in being able to assess their patient's medications for potential drug interactions to prevent the significant clinical impact an interaction can have. It is difficult for a pharmacist to be familiar with all the possible interactions due to the existence of so many drugs and drug combinations; however, it is important for pharmacists to be educated on the common drugs' interactions that are seen in their practice site. Also, to help assist the pharmacist, most pharmacies have medication management software programs that can identify potential drug interactions and alert the pharmacist who can then make an assessment.

Drug interactions can range from not clinically significant to severe; a pharmacist needs to be able to identify where an interaction falls on this scale to be able to make an appropriate plan. Generally, interactions that are not clinically significant do not require an intervention. A method for managing some interactions can be the separation of the administration times of the two medications. Many of these interactions are due to a drug affecting the absorption of another due to altering the environment of the gastrointestinal tract or through binding or chelation. An example is the need to separate levothyroxine administration from calcium- or iron-containing products by 4 h due to decreased absorption if administered together. It is paramount that a pharmacist can recognize interactions and make an appropriate plan, especially as some drug interactions can be severe and even fatal. A plan for severe interactions may include changing a medication to a safe alternative or separating the administration of the drug by a period of time. An example of a clinically significant interaction is the dangerous hypotensive effect when sildenafil and isosorbide mononitrate are combined. The potential for this interaction makes it imperative that patients separate the two medications by at least 24 h. Some severe interactions can involve multiple drugs, and the addition of another drug can exponentially increase the risk of an adverse event. An example of this is the potentially fatal combination of multiple drugs that prolong the corrected QT interval (QTc) interval, such as ondansetron, citalopram, and quetiapine. If the patient must be on a combination of these

drugs, often, a baseline electrocardiogram (ECG) is warranted. After a pharmacist has identified a potential drug interaction, an appropriate intervention should follow. An intervention could include patient education, a medication substitution or dose change, a test for a patient's baseline status (e.g., ECG), therapeutic drug monitoring, or contacting the prescriber. The pharmacist should also consider patient-specific characteristics when deciding on an intervention. Older age, polypharmacy, and baseline factors such as renal or hepatic function can influence the likelihood of a patient experiencing an interaction.

Ideally, pharmacists can prevent drug interactions from occurring; however, not all drug interactions can be averted. Consequently, pharmacists will be faced with situations in which a patient is presenting with a drug interaction that has already occurred. Pharmacists should consider a drug interaction as a potential cause of an adverse event if the patient is taking more than one medication. An examination of the patient's medication list and inquiring about any nonprescription medication and supplements will allow the pharmacist to either rule out an interaction or prompt a more thorough investigation. If a drug interaction is believed to be the cause, steps should be taken to resolve this drug-related problem. A referral will be needed if the interaction is severe, and the patient may require immediate medical attention.

There are a vast number of drugs that can be implicated in drug–drug interactions. Table 2.5 depicts some examples of classes of medications or supplements that are known to cause drug interactions. Depending on the pharmacist's practice site, some of these medications may be seen frequently or rarely if at all. Pharmacists should familiarize themselves with the classes of drugs that are frequent culprits of drug interactions at their practice sites. Note that not all drugs listed under each medication class may cause drug interactions and that each medication class may not produce an interaction to the same degree.

Drug–Food Interactions

A drug–food interaction assessment is often approached differently than drug–drug or drug–disease interactions. Pharmacists can identify a potential drug–drug or drug–disease interaction when the medication is initially prescribed.

Table 2.5 Classes of medications that are major perpetrators of drug–drug interactions

Anticoagulants	Fluoroquinolone antibacterials
Antidepressants	HMG-CoA reductase inhibitors
Antiseizure medications	Macrolide antibacterials
Antineoplastic agents	Protease inhibitors
Antirejection agents	St. John's wort
Antituberculosis agents	

HMG-CoA 3-hydroxy-3-methylglutaryl-coenzyme A

Pharmacists can create a plan to change the drug, dose, or duration of the therapy based on this initial assessment because they have access to the patient's medication and medical history. On the other hand, pharmacists are unable to conduct a similar assessment with drug–food interactions because a patient's complete diet history is not available. At medication initiation, the pharmacist can ask the patient questions about his/her diet and meal schedule. However, most of the pharmacists' role will be to provide education to the patient on potential drug–food interactions and how to mitigate this risk. For example, risedronate needs to be taken 30 min before food intake. Drug–food interactions can range from mild to severe depending on the type of interaction, the drug, and the quantity of the specific food that is ingested. Drug–food interactions are predominantly due to the effect of food on the absorption of the drug and can include both solid food and drinks (e.g., juice and alcohol). Some drugs have better absorption with food in the stomach, such as amoxicillin/clavulanate, and are recommended to be taken with meals. Other drugs are absorbed less when taken with food and should be taken on an empty stomach, such as bisphosphonates. Food can also affect the metabolism of drugs and can act as an inducer or inhibitor of enzymes. One of the most documented examples is grapefruit and its inhibition of the enzyme CYP3A4. Inhibition of CYP3A4 decreases the metabolism of drugs metabolized by CYP3A4, such as simvastatin, tacrolimus, and felodipine. Food can also enhance the effects or side effects of some drugs, such as the increased sedation with alcohol and benzodiazepines, or counteract the effect of the drug, such as with vitamin K and warfarin. An example of a clinically significant drug–food interaction is the combination of metronidazole and alcohol. This combination can cause a disulfiram-like reaction, which produces symptoms ranging from nausea to tachycardia and hypotension. Examples like metronidazole and alcohol show the importance of pharmacist counseling and patient education when a patient is initiated on medication with the possibility of a drug–food interaction.

Some examples of food and drinks that are commonly implicated in drug–food interactions can be found in Table 2.6. This list is not exhaustive but provides a few examples of drugs that interact with a particular food or drink.

Drug–Disease Interactions

Pharmacists also need to be aware of potential drug–disease interactions. A drug–disease interaction is an interaction that occurs between the drug and one or more of the patient's comorbidities. It can lead to an increased or decreased effect of the drug or can aggravate the disease or its complications or lead to adverse drug effects. A complete medical and medication history is required for proper

Table 2.6 Examples of food–drug interactions

Food or drink	Drugs and mechanism of interaction
<i>Alcohol</i> (many alcoholic beverages contain tyramine; see foods that contain tyramine)	<i>Antiseizure medications</i> (e.g., carbamazepine), <i>benzodiazepines</i> , <i>narcotic analgesics</i> (e.g., morphine), <i>sedatives</i> (e.g., zopiclone): Synergistic effect leading to increased risk of adverse effects, such as CNS depression <i>Isotretinoin</i> : May enhance the adverse effects of the drug, such as an increased risk of elevated triglyceride concentrations <i>Metronidazole</i> : May enhance the adverse effects of alcohol and can cause a disulfiram-like reaction
<i>Caffeine</i>	<i>Ciprofloxacin</i> , <i>fluvoxamine</i> : Can increase the serum concentration of caffeine <i>Lithium</i> : Decreased serum concentration of the drug, which may diminish the therapeutic effect of the drug <i>Clozapine</i> : Increased serum concentration of the drug, leading to possible toxicity <i>Theophylline</i> : Synergistic effect, leading to increased risk of adverse effects
<i>Foods high in calcium</i> (e.g., milk)	<i>Allopurinol</i> , <i>bisphosphonates</i> , <i>cefuroxime</i> , <i>dabigatran</i> , <i>fosinopril</i> , <i>levothyroxine</i> , <i>quinolone antibacterials</i> , <i>tetracycline antibacterials</i> : May decrease the absorption and serum concentration of the drug, which may diminish the therapeutic effect
<i>Foods high in potassium</i> (e.g., bananas)	<i>ARBs</i> , <i>ACEIs</i> , <i>potassium-sparing diuretics</i> : May enhance the hyperkalemic effect, which can cause adverse effects such as muscular weakness, fatigue, arrhythmias, and bradycardia
<i>Foods high in tyramine</i> (e.g., strong or aged cheese)	<i>MAOIs</i> : May cause a hypertensive crisis
<i>Foods high in vitamin K</i> (e.g., dark, leafy green vegetables)	<i>Warfarin</i> : May diminish the therapeutic anticoagulant effect
<i>Grapefruit</i> (including juice)	<i>Amiodarone</i> , <i>clopidogrel</i> : May decrease the serum concentration of the drug, which may diminish the therapeutic effect <i>Antirejection agents</i> (e.g., <i>sirolimus</i>), <i>DHP calcium channel blockers</i> , <i>HMG-CoA reductase inhibitors</i> : May increase the serum concentration of the drug, leading to possible toxicity

ARBs angiotensin II receptor blockers, *ACEIs* angiotensin-converting enzyme inhibitors, *MAOIs* monoamine oxidase inhibitors, *CNS* central nervous system, *DHP* dihydropyridine, *HMG-CoA* 3-hydroxy-3-methylglutaryl-coenzyme A

drug–disease interaction assessment. If there is a potential for a drug–disease interaction, a plan should be formulated, which may include changing the medication or the dose or

Table 2.7 Examples of drug–disease interactions

Drug	Disease
Anticholinergics	<i>Constipation</i> : Due to slowed gastrointestinal motility and decreased secretions from the intestinal tract <i>Dementia</i> : Has been associated with greater cognitive decline <i>Glaucoma</i> : Can narrow the drainage angle of the eye and prevent eye fluid from properly exiting the eye, resulting in high eye pressure <i>BPH</i> : Reduces bladder muscle contractions and can aggravate BPH symptoms
NSAIDs	<i>Hypertension</i> : Can elevate blood pressure <i>Peptic ulcer disease</i> : Lowers prostaglandin levels and damages the gastroduodenal mucosa <i>Chronic kidney disease</i> : Lowers prostaglandin levels, which causes decreased blood flow and oxygen supply to the kidneys
Bupropion	<i>Seizures</i> : Can lower the seizure threshold
Benzodiazepines	<i>Dementia</i> : Increased risk of confusion and delirium <i>Falls</i> : Can cause sedation and daytime drowsiness; the risk is greatest in elderly patients
Beta-blockers (specifically non-cardioselective beta-blockers)	<i>Asthma</i> : Can cause bronchoconstriction and worsening of asthma symptoms
Non-DHP calcium channel blockers	<i>Heart failure</i> : Weakens the force of muscular contractions (have a negative inotropic effect) and can exacerbate heart failure symptoms and worsen the condition

NSAIDs nonsteroidal anti-inflammatory drugs, *BPH* benign prostatic hyperplasia, *DHP* dihydropyridine

careful monitoring of the disease. An example of such interactions that may be encountered in practice is renal insufficiency. Renal impairment can significantly impact the elimination of renally excreted drugs. A renal dosage adjustment or medication avoidance is often recommended in patients with decreased kidney function. In addition, some drugs have nephrotoxic potential and should be used with caution. Patients with impaired renal function and taking renally eliminated drugs or those on nephrotoxic drugs should have their kidney function assessed regularly throughout therapy.

Table 2.7 depicts examples of some common drug–disease interactions. This table is not exhaustive but is meant to provide a few examples of drug–disease interactions that a pharmacist could come across in practice. It is important to note that OTC drugs could be potentially implicated in drug–

disease interactions. Counseling is important, and pharmacists should ask patients to seek advice if they are starting any OTC medication or herbals. An initial chronic disease assessment can provide an opportunity to educate the patient on OTC medications that he/she should avoid.

Conclusions

Pharmacists expanding scopes of practice and emerging clinical roles allow pharmacists to conduct patient assessments daily for a variety of symptoms and acute and chronic diseases. Pharmacists can individualize and adapt the patient assessment process to their specific practice needs. They need to make sure that each medication is indicated, effective, and safe for every patient and that the patient can adhere to the treatment. Patient assessments conducted by pharmacists make a difference in patient safety and clinical outcomes and can improve patient-centered care. The following chapters in this book provide a detailed approach to patient assessments that focus on individual symptoms or chronic diseases, followed by specialized topics in patient assessment.

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Physical Assessment for Pharmacists

3

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Chapter Objectives

1. Outline the basic steps involved in physical assessment (PA).
2. Complete a general survey and review of systems for a specific patient.
3. Describe how PA can be used in pharmacy practice to enhance pharmaceutical care.
4. List available comprehensive guides to physical assessment so that one can begin to develop new PA skills as they evolve in their practice.

Background

Physical assessment (PA) is used by healthcare professionals to gather important information about their patients. For example, physicians and nurse practitioners (NPs) use PA as part of their comprehensive assessments and for diagnosis and management plans. Nurses on the hospital ward conduct physical assessments every shift to monitor the health status of acutely ill patients. Pharmacy practice can be enhanced

with the use of PA as well. It is useful for the initial and follow-up assessment of patients and in monitoring response to therapy and adverse drug reactions. In addition, PA aids in identifying red flag signs and symptoms that prompt referral to other healthcare practitioners or to the emergency department. Each practice area will lend itself to a different set of skills, and each pharmacist can determine for themselves which skills will be useful in their own practice. Compared to other professions, pharmacists have a unique perspective and scope of practice. They generally do not perform a comprehensive physical examination [1]. For this reason, this chapter will focus on specific skills and activities that can be used to identify drug therapy problems and to gather information that a pharmacist can use in the provision of pharmaceutical care. Resources such as *Bates' Guide to Physical Examination and History Taking* or *Patient Assessment in Pharmacy Practice* [1, 2] are available as comprehensive guides to the abundance of PA skills and activities that have been established in the medical, nursing, and other fields. Since physical assessment involves many specific terms that may not be familiar to all pharmacists, a glossary of terms is found in Table 3.1.

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Table 3.1 Glossary of terms

Glossary term	Meaning
Accessory muscles	Muscles in the shoulders and chest that are not used in normal breathing but are used by patients who are in respiratory distress or having trouble breathing
Adventitious	Abnormal, as in an abnormal lung sound
Anaphylaxis	Severe, IgE-mediated allergic reaction that involves bronchoconstriction, wheezing, itching/hives, facial/throat/tongue swelling, low blood pressure, and gastrointestinal symptoms like vomiting, diarrhea, and abdominal pain. It usually starts within minutes to hours of exposure to the allergen
Anterior	The front side of the body (toward the stomach)
Apical heart rate	Measured at the bottom (apex) of the heart by auscultation of the area where the fifth intercostal space and the midclavicular line intersect
Bronchial breath sounds	Louder, harsher sounds that are heard when the large bronchi are auscultated; also heard when a patient has pneumonia and consolidated lungs
Bronchovesicular breath sounds	Normal sounds in the mid-chest area or between the scapula, reflecting air moving through the mid-sized airways. They will be between the pitch of bronchial and vesicular sounds
Conjunctivitis	Inflammation of the conjunctiva in the eyes
Cyanosis	Blue-tinged skin caused by lack of oxygen
Delirium	Acute disturbance in state of mind and mental processes, which is characterized by incoherent thoughts and speech, restlessness, and illusions. Often related to fever, intoxication, or other physical disorders
Dentition	The teeth and how they are arranged in the mouth
Dermatome	An area of the skin that is innervated by a single spinal nerve
Encephalopathy	Altered function of the brain from damage, disease, or other malfunctions, usually presents as altered mental state
Exudate	A combination of fluid, cells, and pus that has been deposited in tissue after escaping from intact blood vessels. It usually occurs due to inflammation
Gastroenteritis	Inflammation of the stomach and intestines due to a multitude of causes. Symptoms are usually nausea, vomiting, and diarrhea
Hives	Itchy, raised patches of the skin that can be reddish or pale. Caused by an IgE-mediated reaction to allergens or other stressors and will disappear after the trigger is removed
Hypoxia	Lack of oxygen
Jaundice	Yellow-colored skin or sclera of the eyes due to buildup of bilirubin. Often associated with liver damage
Lymphedema	Lymph fluid accumulating in tissues, usually the legs, and producing swelling
Mottled	Skin that is blotchy or irregularly colored. It is often due to inadequate perfusion or cold
Myxedema	Synonymous with severe hypothyroidism but also describes the skin changes that occur with hypothyroidism including swelling and thickening of the tongue and mucous membranes
Neuralgia	Nerve pain or painful spasms; often described as burning, shooting, aching, and following a certain nerve
Neuritis	Inflammation of a nerve and tends to feel like burning pain
Nystagmus	Flickering, spasmodic, involuntary movement of the eyes
Posterior	Toward the back of the body
Pruritus	Itching of the skin
Rhinorrhea	Watery, mucous discharge from the nose
Sternal angle	The angle formed by the manubrium and the sternum, which provides an anatomical landmark. This landmark provides an “outer edge” of the thoracic plane and is where the jugular venous pressure is measured against
Sympathomimetic	Producing an effect similar to that of the sympathetic nervous system. Usually associated with increased heart rate and blood pressure, pupil dilation, sweating, and piloerection
Syncopal (syncope)	Temporary loss of consciousness due to lack of blood supply to the brain with paleness, sweating, nausea, and blurry vision
Toxidrome	A collection of symptoms that occur when a patient is exposed to toxic levels of a certain type of substance. They help rapidly rule in or out a possible causal substance in an unknown intoxication
Turgor (skin)	Elasticity of the skin and ability to return to normal shape after it is stretched
Vesicle	Raised, fluid-filled lesion on the skin, usually smaller than 0.5 cm across (if larger than 0.5 cm, it is called a bulla)
Vesicular breath sounds	Normal breath sounds that are soft and low-pitched. They have a rustling quality and are louder at the bases of the lungs because they reflect air movement in the terminal bronchioles and alveoli
Wernicke’s encephalopathy	The combination of altered mental status, abnormal gait, and nystagmus caused by thiamine deficiency. Usually associated with alcohol use disorder or malnutrition
Wheals	Well-defined areas of edema in the skin that are usually itchy and reddened

IgE immunoglobulin E

Physical Assessment Introduction

Physical assessment (PA) is the evaluation of objective anatomical findings gathered through four distinct activities, *inspection*, *palpation*, *percussion*, and *auscultation* [3], which are performed in the stated order. However, not all of the above activities are needed each time a physical assessment is done [1]. The objective information gathered through PA is considered in combination with the subjective information gathered from the patient's health history. Together with the general survey, the health history and PA comprise the overall health assessment of a given patient [1].

Inspection involves the use of vision, hearing, and smell to examine patients or the specific area that is being assessed. It is usually used in conducting a general survey (see below), and it yields important information that should be deliberately noted as one examines the patient [1].

Palpation uses the hands to feel for abnormalities on the surface or deeper if required. One should start with light palpation and move deeper as needed, for example, to feel the size of internal organs or to feel for tenderness. For example, palpation is used to measure a pulse, assess peripheral edema, or feel for palpable lymph nodes [1].

Percussion involves lightly striking the body surface with the purpose of producing a sound. For example, to percuss the lungs, one would place their index or middle finger on the patient's skin and then lightly strike that finger with the other hand. The sound that is produced provides clues about the nature of the underlying tissue. For example, a louder sound is produced by the air-filled lungs, in comparison to the solid biceps muscle [1]. This technique could be used to identify an area of consolidation in a patient with suspected pneumonia, where dullness may be heard.

Auscultation is the act of listening to the sounds created by the body, usually using a stethoscope. It can be performed on the heart, lungs, blood vessels, and viscera. Chest auscultation yields important information. Lung and heart sounds can provide important clues about serious illnesses. Examples of abnormalities that can be heard via auscultation include heart murmurs, extra heart sounds, lung crackles, decreased air entry, carotid bruit, apical heart rate (HR), etc. [1]. These abnormalities often signal a need for medical attention.

General Survey

Note the patient's physical appearance, behavior, and mobility. The general survey can tell us a great deal and is easy to perform. It can be performed while casually speaking with the patient. The following are the key pieces of information to be gathered during the general survey [4]:

- *Appearance*: Overall appearance, hygiene, grooming, attire, skin color, presence of lesions, height, and weight
- *Behavior*: Facial expressions, level of consciousness, orientation, speech, and demeanor
- *Mobility*: Posture, range of motion, use of mobility aids, and gait

Vital Signs

Vital signs provide vital information. Generally, the term "vital signs" includes blood pressure (BP), heart rate, respiratory rate (RR), and temperature. Many practice settings also include oxygen saturation. Every pharmacist should be able to interpret a patient's vital signs as reported by other healthcare professionals. Ideally, pharmacists should be able to perform vital sign measurements themselves as well. Table 3.2 shows the normal ranges in adults for the vital sign parameters. Vital signs for children, infants, and neonates should be looked up in references specific to these age groups, and further information about vital sign assessment in pediatric patients is found in Chap. 30.

Blood Pressure

Blood pressure (BP) measurement is a particularly important assessment and one that is prone to inaccuracy due to external factors (see Chap. 15 "Hypertension Assessment" for more details). Proper BP measurement is the first step to managing hypertension as well as BP-related adverse effects from medications. Since some patients will not monitor their BP independently, pharmacists should try to offer measurements to all patients whose BP could have therapeutic relevance. See Table 3.3 for the recommended techniques for BP measurements [5].

Table 3.2 Normal ranges for vital sign parameters in adults [1]

Vital sign parameter	Normal range
Blood pressure	SBP: ~120–140 mmHg DBP: ~80–90 mmHg
Heart rate	~60–100 bpm
Temperature (varies by the measurement method)	~36.1 °C–37.2 °C (fever generally considered to be >38 °C)
Respiratory rate	~12–20 bpm
Oxygen saturation	~95–100% considered normal <90% generally considered low For patients with COPD requiring oxygen therapy, the desired range is usually 88–92%

COPD chronic obstructive pulmonary disease, *DBP* diastolic blood pressure, *SBP* systolic blood pressure

Table 3.3 Recommended techniques for BP measurements^a

Recommended technique for automated office blood pressure (AOBP)	
1.	Measurements should be taken with a validated sphygmomanometer known to be accurate
2.	Choose a cuff with an appropriate bladder size matched to the size of the arm. Select the cuff size as recommended by its manufacturer
3.	Place the cuff so that the lower edge is 3 cm above the elbow crease and the bladder is centered over the brachial artery. There is no rest period needed before measurement. The arm should be bare and supported with the BP cuff at the heart level, as a lower position will result in an erroneously higher SBP and DBP. There should be no talking, and the patient's legs should not be crossed
4.	When using automated office oscillometric devices, the patient should be seated in a quiet room (no specified period of rest), with the device set to take measures at 1- or 2-min intervals. The first measurement is taken by a health professional to verify the cuff position and validity of the measurement. The patient is left alone after the first measurement, while the device automatically takes subsequent readings
5.	Record the average BP as displayed on the electronic device as well as the arm used and whether the patient was supine, sitting, or standing. Record the heart rate
Recommended technique for office blood pressure measurement (OBPM)	
1.	Measurements should be taken with a sphygmomanometer known to be accurate. A validated electronic device should be used. If not available, a recently calibrated aneroid device can be used. Aneroid devices or mercury columns need to be clearly visible at the eye level
2.	Choose a cuff with an appropriate bladder size matched to the size of the arm. For measurements taken by auscultation, bladder width should be close to 40% of the arm circumference and bladder length should cover 80–100% of the arm circumference. When using an automated device, select the cuff size as recommended by its manufacturer
3.	Place the cuff so that the lower edge is 3 cm above the elbow crease and the bladder is centered over the brachial artery. The patient should be resting comfortably for 5 min in the seated position with back support. The arm should be bare and supported with the BP cuff at the heart level, as a lower position will result in an erroneously higher SBP and DBP. There should be no talking, and the patient's legs should not be crossed. The first reading should be discarded and the latter two averaged. BP should also be assessed after 2 min of standing (with the arm supported) and at times when patients report symptoms suggestive of postural hypotension. Supine BP measurements may also be helpful in the assessment of elderly and diabetic patients
4.	When using automated office oscillometric devices such as the BpTRU (VSM MedTech Ltd., Vancouver, Canada), the patient should be seated in a quiet room (no specified period of rest). With the device set to take measures at 1- or 2-min intervals, the first measurement is taken by a health professional to verify the cuff position and validity of the measurement. The patient is left alone after the first measurement while the device automatically takes subsequent readings. The BpTRU automatically discards the first measure and averages the next five measures
5.	For auscultation, at least three measurements should be taken in the same arm with the patient in the same position. The first reading should be discarded and the latter two averaged
6.	Increase the pressure rapidly to 30 mmHg above the level at which the radial pulse is extinguished (to exclude the possibility of a systolic auscultatory gap)
7.	Place the bell or diaphragm of the stethoscope gently and steadily over the brachial artery
8.	Open the control valve so that the rate of deflation of the cuff is approximately 2 mmHg per heartbeat. A cuff deflation rate of 2 mmHg per beat is necessary for accurate systolic and diastolic estimation
9.	Read the systolic level, the first appearance of a clear tapping sound (phase I Korotkoff), and the diastolic level, the point at which the sounds disappear (phase V Korotkoff). If Korotkoff sounds persist as the level approaches 0 mmHg, then the point of muffling of the sound is used (phase IV) to indicate the diastolic pressure. Leaving the cuff partially inflated for too long will fill the venous system and make the sounds difficult to hear. To avoid venous congestion, it is recommended that at least 1 min should elapse between readings
10.	Record the BP to the closest 2 mmHg on the manometer (or 1 mmHg on electronic devices) as well as the arm used and whether the patient was supine, sitting, or standing. Avoid digit preference by not rounding up or down. Record the heart rate. Seated BP is used to determine and monitor treatment decisions. Standing BP is used to examine for postural hypotension, if present, which may modify the treatment
11.	In the case of arrhythmia, additional readings with auscultation may be required to estimate the average systolic and diastolic pressure. Isolated extra beats should be ignored. Note the rhythm and pulse rate
12.	BP should be taken in both arms on at least one visit, and, if one arm has a consistently higher pressure, that arm should be subsequently used for BP measurement and interpretation
Recommended technique for home blood pressure measurement (HBPM)	
1.	Measurements should be taken with a validated electronic device
2.	Choose a cuff with an appropriate bladder size matched to the size of the arm. Bladder width should be close to 40% of the arm circumference, and bladder length should cover 80–100% of the arm circumference. Select the cuff size as recommended by its manufacturer
3.	The cuff should be applied to the nondominant arm unless the SBP difference between arms is >10 mmHg, in which case the arm with the highest value obtained should be used
4.	The patient should be resting comfortably for 5 min in the seated position with back support
5.	The arm should be bare and supported with the BP cuff at the heart level
6.	Measurement should be performed before breakfast and 2 h after dinner, before taking medication
7.	No caffeine or tobacco in the hour and no exercise 30 min preceding the measurement
8.	Duplicate measurement should be taken in the morning and in the evening for 7 days (i.e., 28 measurements in total)
9.	Average the results excluding the first day's readings

(continued)

Table 3.3 (continued)

Recommended technique for ambulatory blood pressure monitoring (ABPM)	
1. The appropriately sized cuff should be applied to the nondominant arm unless the SBP difference between arms is >10 mm Hg, in which case the arm with the highest value obtained should be used	
2. The device should be set to record for a duration of at least 24 h with the measurement frequency set at 20–30-min intervals	
3. A patient-reported diary to define daytime (awake), nighttime (sleep), activities, symptoms, and medication administration is useful for study interpretation	
4. Daytime and nighttime should preferentially be defined using the patient’s diary. Alternatively, predefined thresholds can be used (e.g., 8 AM to 10 PM for awake and 10 PM to 8 AM for nighttime)	
5. The ambulatory BP monitoring report should include all of the individual BP readings (both numerically and graphically), the percentage of successful readings, the averages for each time frame (daytime, nighttime, 24 h), and the “dipping” percentage (the percentage the average BP changed from daytime to nighttime)	
6. Criteria for a successful ambulatory BP monitoring study are:	
a. At least 70% of the readings are successful	
b. At least 20 daytime readings and 7 nighttime readings are successful	

BP blood pressure, DBP diastolic BP, SBP systolic BP. Unless otherwise mentioned, steps apply to measurement by auscultation and oscillometry using an upper arm cuff

^aReprinted from *Canadian Journal of Cardiology*, 36, Rabi et al. [5], Copyright 2020, with permission from Elsevier

Pharmacists play a key role in managing hypertension. By helping patients monitor and interpret their BP, pharmacists help mitigate risks and maximize benefits of drug therapy. Research has shown that pharmacist prescribing for hypertension management leads to clinically and statistically significant reductions in BP [6]. Pharmacists are also well-positioned in the healthcare system to detect BP-related drug therapy problems. For example, heart failure (HF) patients are usually on multiple drugs that affect their BP, such as beta-blockers and diuretics. They often have BP in the lower range, but these “BP” medications are titrated up in order to achieve evidence-based doses that provide maximal reductions in both morbidity and mortality. Pharmacists can help HF patients get the most from these medications by assessing each scenario when patients have questions about their BP and medications. If a patient is tolerating a dose and is not symptomatic, then a lower BP may not be problematic. Conversely, for patients who are experiencing adverse effects while their medications are up-titrated, the pharmacist may be the first healthcare provider to know about it through follow-up assessment. Further assessment of hypertension is found in Chap. 15.

Heart Rate

Heart rate (HR) is another important parameter. For patients with HR outside the normal range of 60–100 beats per minute (bpm), one should ask the questions “What is causing the abnormality?” and “Is this problematic?” For example, a person who is very fit may have a resting HR < 50–60 bpm due to cardiovascular adaptations from exercise. This is “normal” for them and not concerning. For a patient who is taking a beta-blocker, bradycardia (HR < 60 bpm) may warrant dose reduction, depending on whether they are symptomatic,

how low their HR is, and the indication for beta-blocker therapy. For patients who report feeling heart palpitations, or those who are tachycardic (e.g., >100 bpm [1]), one should manually check the patient’s pulse to see if it feels irregular. An irregular pulse suggests arrhythmias, such as atrial fibrillation (AF). AF is a common arrhythmia that is described as irregularly irregular. If undiagnosed and untreated, AF puts the patient at risk for stroke and other complications. For patients with an irregular pulse, an electrocardiogram (ECG) is indicated to rule out AF or other arrhythmias. From an infectious disease perspective, tachycardia is one of the signs of sepsis and is cause for concern if any other symptoms are present (RR >20 bpm, SBP < 90 mmHg, temperature > 38 °C, white blood cells (WBCs) >12 × 10⁹/L, etc.) [7].

Temperature

Temperature is easy to measure and can be done in any setting. Single-use thermometers are available for purchase and can be used to measure a patient’s temperature when applicable. Figure 3.1 shows a commercially available single-use thermometer. The steps for measuring oral temperature with a single-use thermometer are as follows:

1. Wash your hands.
2. Carefully open the package.
3. Insert the thermometer under the patient’s tongue, into one of the posterior sublingual pockets. The dots can be facing up or down. Have the patient close their mouth.
4. Wait for 60 s.
5. Remove the thermometer and wait for 10 s before reading.
6. Read the temperature by determining the last blue dot on the matrix. Record the temperature.

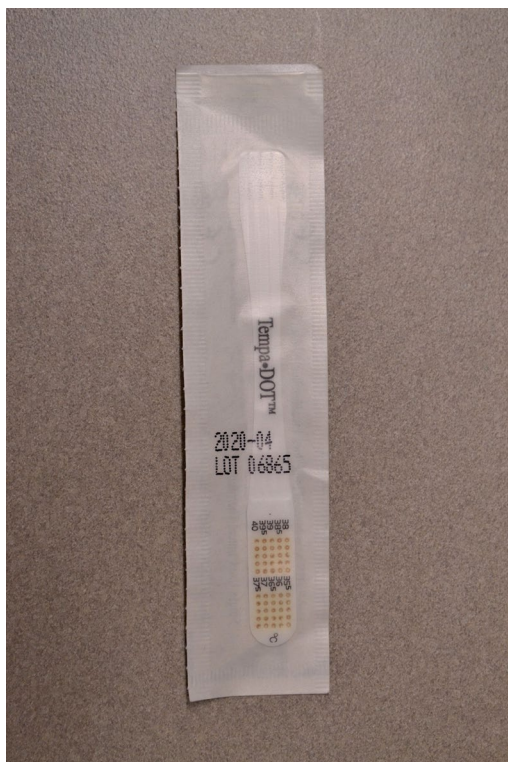


Fig. 3.1 An example of a commercially available single-use thermometer

Table 3.4 Temperature measurement by different routes [1]

Temperature route	Normal temperature in adults (°C)	Comments
Oral	37	Accurate and convenient
Axillary (armpit)	36.5	In adults, this route is used only when the oral route is not possible
Rectal	37.5	Preferred route if the oral route is not possible (e.g., intubation, facial surgery) Most accurate way to measure core body temperature
Tympanic (ear)	37.5	Noninvasive, quick, and efficient Infrared sensor detects flow of blood through the eardrum
Temporal artery (forehead)	37.5	Measures infrared heat in the temporal artery Next to the rectal route, this is the best estimation of core temperature

In addition to the oral route, temperature can be measured by the axillary, rectal, tympanic, or temporal artery route. Table 3.4 depicts the normal temperature ranges and considerations for the various routes.

Respiratory Rate

Respiratory rate (RR) is ascertained visually by noting the rising of a patient's chest to signify a breath, counting the number of breaths in 30 s, and then multiplying by 2. Alternatively, one may choose to count the number of breaths for the whole minute if the patient is breathing irregularly. Since a person's breathing may be altered if they become aware of it, it is best not to tell the patient when assessing their respiratory rate [1]. An elevated RR can be a sign of respiratory distress, such as in a severe acute exacerbation of chronic obstructive pulmonary disease (AECOPD), asthma exacerbation, or pneumonia. A low RR is also worrisome, as the patient may not be getting adequate gas exchange. They may become hypoxic or may develop respiratory acidosis if they are unable to exhale enough carbon dioxide. RR is also an important parameter when monitoring for opioid toxicity. Opioids suppress the body's respiratory drive when taken in excess, an effect that can be fatal. Any patient with a low or high RR needs to be referred, possibly even to the emergency room (ER).

Review of Systems

In general, the review of systems is a systematic approach to identifying abnormalities or issues that may not have been identified by the patient. Each body system is reviewed, often in a head-to-toe sequence. A review of systems can be completed using focused questioning, PA, and information gathered from other health professionals.

Neurological/Psychiatric

A pharmacist should be familiar with performing and interpreting specific aspects of the neurological exam if it is relevant to their practice. For example, when reflexes are reported, the designations 3+ and 4+ are more exaggerated than normal [1]. This would be relevant when assessing someone for the possibility of serotonin syndrome, since hyperreflexia is one of the possible signs. Moreover, pharmacists should be looking for signs of neurological concerns such as abnormal movements, abnormal gait, or signs of spasticity as seen in some patients post-stroke. These may help assess how well a patient's treatment is working or if there are any adverse effects occurring from medications. One important adverse effect to look for in patients who are using antipsychotics is tardive dyskinesia, which tends to present as repetitive odd movements of the face, mouth, tongue, and head [1].

Level of Consciousness

The general survey provides a good understanding of the patient's level of consciousness (LOC). A decreased level of consciousness, especially a change from baseline, warrants referral to the emergency room or to a primary care provider if the decrease is mild. For pharmacists working in the hospital setting, it is also helpful to be familiar with the Glasgow Coma Scale (GCS). For example, nurses may report GCS as a vital sign in the emergency room and critical care areas. Level of consciousness is relevant to medication therapy, as medications that suppress the central nervous system (CNS) may not be appropriate if the patient has a decreased LOC. Additionally, medications could be causing the decreased LOC, and this must be screened for. Patients with a decreased LOC may also not be able to swallow medications, and consideration for alternate routes of administration is required if this is the case. Nurses are a great resource to determine whether the oral route is appropriate for a given patient, as they are the individuals who administer medications, and the ability to take oral medications is part of their assessment. Table 3.5 shows the rubric for GCS scoring, which is based on the best response the patient gives in each category [8].

The numerical scores assigned during GCS assessment are typically added together to give a total out of 15, with a range of 3–15. A score of 15 is “normal,” and lower scores indicate an altered LOC. Information about how to determine a patient's GCS is available from a variety of resources [9].

Orientation

All pharmacists should be able to check if a patient is oriented to a person, place, and time. This should be included in the general survey. Often, this is reported in patient charts as “AO x (number),” which indicates if the patient is alert and

how many elements of orientation a patient is able to correctly identify. Simple questions can be used, such as “What is your name?,” “Where are we?,” and “What is today's date?” It is especially important to get the patient to state what year it is, as some patients may be able to answer all the other questions correctly and not know what year it is [1]. Assessing orientation helps determine if a patient is delirious. In addition to acute confusion, symptoms of delirium include inattention, hallucinations, behavioral changes, etc. Acute confusion or other delirium symptoms warrant a referral, as delirium is a medical emergency. The underlying cause of delirium must be ascertained and corrected, if possible. In addition to the need for a referral, detecting confusion is helpful if we are trying to gather information from the patient directly, since we may be unable to do so if the patient is significantly confused. Delirium is also an important side effect to look out for from many medications, including benzodiazepines, anticholinergic medications, opioids, or corticosteroids. Monitoring a patient's level of orientation/confusion is also highly relevant for the management of hepatic encephalopathy. This is an important efficacy parameter that helps guide therapy.

Psychiatric

Throughout the physical exam and health history interview, the pharmacist should observe the patient's affect, level of cooperation, etc. Other things to look out for would include suicidal ideation, anxiety, hallucinations, delusions, pressured speech, tangential speech, etc. [1]. Consider one's baseline if they have a history of psychiatric illness. Do not hesitate to ask about specific symptoms if you are concerned about a patient, and have a plan to manage or refer patients who are experiencing psychiatric symptoms. The *Handbook of Clinical Rating Scales and Assessment in Psychiatry and Mental Health* is an excellent resource for mental health assessment [10].

Table 3.5 Glasgow Coma Scale (GCS) [8]

Glasgow Coma Scale component	Best response (with the corresponding numerical score)
Eye opening	Spontaneous—4
	To voice—3
	To pain—2
	None—1
Verbal response	Oriented—5
	Confused—4
	Words—3
	Sounds—2
	None—1
Best motor response	Obeying commands—6
	Localizing—5
	Normal flexion (withdrawal)—4
	Abnormal flexion—3
	Extension—2
	None—1

Peripheral Neuropathies

Neuropathies can include a wide variety of conditions that can damage either single or multiple nerves in one area. They tend to cause symptoms such as tingling, numbness, pain, and weakness in the affected areas and can be very bothersome to patients. Patients who present with symptoms that suggest neuropathies should be referred to a physician for further assessment, but there are some assessments that pharmacists can perform if trained. Although a neuropathy itself cannot be seen from inspection, many tests can be done to determine if nerve function is intact in an area. These tests assess the patient's ability to feel pain, light touch, temperature, and vibration [1]. Two common tests are the monofilament test and tuning fork test, which help determine the specific deficiencies the patient is experiencing [11]. Often, patients who have long-standing or poorly controlled diabe-

tes will develop peripheral neuropathy in their lower limbs. For these patients, daily foot checks can help catch complications early. Information on how to perform foot checks for patients can be found on the Diabetes Canada website: <https://www.diabetes.ca/>.

Head, Eyes, Ears, Nose, and Throat (HEENT)

Visually inspect the head, eyes, ears, nose, and throat (HEENT) for abnormalities. In addition, lightly palpate structures on the head, eyes, ears, and nose if it is relevant to your assessment. Examples of notable findings include:

- A dry, flaky, scalp with reddened areas may indicate dandruff, psoriasis, or dermatitis.
- Dry mucous membranes in the mouth, which could be a sign of dehydration
- Poor dentition, which is a risk factor for many systemic illnesses and poor health in general
- White patches on the tongue or throughout the mouth can indicate oral candidiasis (“thrush”).
- Bulges or swelling in the throat that may indicate a goiter, which can be a sign of hypo- or hyperthyroidism.
- Xanthomas are deposits of cholesterol in the skin of patients who have very high cholesterol levels and indicate a need for a referral.
- Tender/swollen cervical lymph nodes can often be felt in patients with pharyngitis.

Eyes

Patients with eye concerns often present to pharmacists in community or other primary care settings for recommendations. By becoming familiar with simple methods of assessing the eyes, pharmacists can improve their patient recommendations for minor conditions and are better poised to detect signs of more serious ailments.

When the eyes are irritated, they may appear to be red, with or without discharge. Red eyes may be due to a variety of problems including dry eyes or conjunctivitis. Conjunctivitis may be infectious (viral or bacterial) or noninfectious (allergic, contact lens-related). Patients who have copious discharge likely have infection present and should be seen by a physician or optometrist for appropriate care. For individuals who present with allergic symptoms, including itching and exposure to a known allergen, there are both prescription and nonprescription treatments available. In terms of other infections, patients will often ask for treatment for styes, which are infections of the oil glands that surround the eyelid margin. They will look like bumps right on the margin of the eyelid and may be red and very itchy. The available combination of polymyxin B and gramicidin

(Polysporin Eye Drops®) may be effective in some patients, but, generally, it is preferred that they see a family doctor or optometrist to confirm diagnosis. There is also a relatively high rate of antimicrobial resistance to this product.

Protruding eyeballs, exophthalmos, is a sign of hyperthyroidism. This finding may identify a need for additional therapy if the patient is untreated or may indicate that the dose is too low if they are on treatment already. Serum thyroid-stimulating hormone (TSH), among other more specialized blood tests in conjunction with the clinical picture, can help interpret this finding.

While assessing the eyes, pay attention to the irises and pupils. First, inspect pupil size. Normal pupils should be equally sized and round, with an appropriate size based on the brightness of the room. Pupils can also change size in response to medications or other substances. Patients who are experiencing opioid toxicity will have pinpoint pupils (miosis). Sympathomimetic drugs often dilate the pupils (mydriasis), which can help identify causative agents in the setting of an unknown toxidrome. Dilatation of pupils may also be a sign of serotonin syndrome, which pharmacists can help identify based on a patient’s medication profile.

A pharmacist can also assess eye movement, looking for involuntary oscillating eye movements (nystagmus). Figure 3.2 describes the technique used for assessing eye movements. Nystagmus is relevant to pharmacy practice as it can be a sign of phenytoin toxicity. Also, for patients who present with alcohol use disorder, the presence of nystagmus

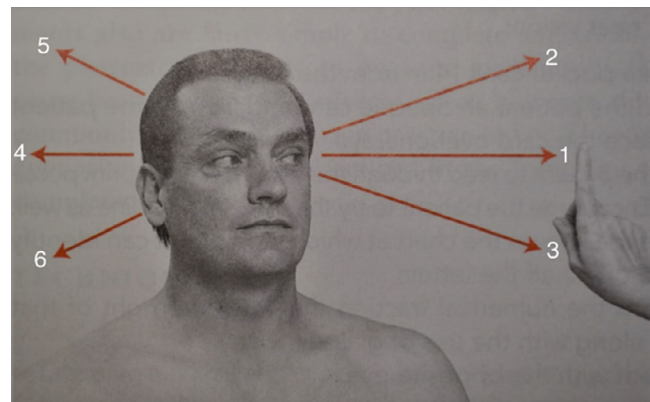


Fig. 3.2 The six cardinal positions for assessing eye movements. Follow these steps to perform the assessment: 1. Instruct the patient to follow the movement of your finger with their eyes and not to move their head. 2. Hold your finger ~12–18 inches from the patient’s head, starting in the middle. Move your finger to the right side. Watch for a fluttering movement or “beats” of nystagmus with each new position of your finger. 3. From position 1 (right side, midline), move your finger up while staying to the right. This is position 2. 4. From position 2, move your finger down, staying to the right side, to position 3. 5. From position 3, move your finger to the left side, midline, to position 4. 6. From position 4, move your finger upward, staying to the left, to position 5. 7. From position 5, move your finger downward, staying to the left, to position 6. (Reprinted with permission from Jones [1])

may signal the presence of Wernicke's encephalopathy (WE). The classic triad of symptoms is encephalopathy (disorientation, impaired memory, etc.), gait ataxia, and oculomotor dysfunction. Most patients with WE may have only one or two signs, which makes it harder to diagnose [12]. WE is believed to be underrecognized, and pharmacists can help by seeking out this need for additional therapy.

Rarely, patients can present with what looks like rings around the irises. This may occur in patients with extremely high cholesterol and diseases associated with too much copper (e.g., Wilson's disease) as well as other rare diseases. If it has not been addressed before, this is a sign that the patient needs to be referred.

Pharyngitis

Pharyngitis is another common complaint that is presented to pharmacists. Patients are often concerned that they have group A streptococcal (GAS) infections or "strep throat." With the increase in popularity of point-of-care testing (POCT) for GAS, pharmacists can both display antimicrobial stewardship and, for those who do test positive, expedite access to proper treatment. In patients complaining of a sore throat, use of a light and tongue depressor will help visualize the oropharynx (throat) and allow one to inspect for redness, inflammation, and tonsillar edema or exudate.

Palpate the neck area, looking for enlarged or tender lymph nodes. These are usually a sign of infection. While viral pharyngitis is more commonly associated with enlarged posterior cervical lymph nodes, bacterial pharyngitis affects the anterior cervical nodes [2]. However, this is not a guaranteed way to differentiate between the two.

As with all infections, antimicrobial stewardship is an essential component of the pharmacist's role in pharyngitis. For patients with pharyngitis, history and physical examination findings are very helpful to differentiate between a viral and bacterial source. In general, the presence of upper respiratory tract infection (URTI) viral symptoms such as cough, conjunctivitis, gastroenteritis, or rhinorrhea indicates a viral infection. Patients with any symptoms of viral infection should not undergo testing such as rapid antigen detection testing (RADT) for group A streptococcal (GAS) pharyngitis. Careful selection of patients for microbiological testing is an essential step in avoiding unnecessary antibiotic therapy. Since many people are asymptomatic carriers of GAS, a swab in a patient who has viral symptoms could easily lead to a false positive result. Signs and symptoms of GAS pharyngitis include acute-onset sore throat plus the factors included in the Centor criteria [13, 14]. These criteria also provide a threshold score above which RADT is indicated and below which GAS pharyngitis is unlikely. The modified Centor criteria include age and are presented in Table 3.6.

When ≥ 3 criteria are present, RADT is recommended. Less than three criteria indicates that GAS pharyngitis is

Table 3.6 The modified Centor criteria for probability of group A streptococcal pharyngitis [13, 14]

Criterion	Score		
Age	3–14 years (+1)	15–44 years (0)	≥ 45 years (–1)
Tonsillar swelling or exudate	Yes (+1)	No (0)	
Tender or swollen anterior cervical lymph nodes	Yes (+1)	No (0)	
Temperature $> 38^{\circ}\text{C}$	Yes (+1)	No (0)	
Absence of cough	Yes (+1)	No (0)	

unlikely [13]. For patients with symptomatic, confirmed GAS pharyngitis (RADT + follow-up culture), antibiotics are usually prescribed. Although GAS pharyngitis is usually self-limited (~8–10 days), antibiotics reduce the severity and duration of symptoms. They also reduce transmission and complications such as peritonsillar abscess and acute rheumatic fever [15].

Dermatological

Inspection and palpation are the main modes of PA of the skin. Since much of the information is collected from inspection, it should be performed in a well-lit room with either natural light or penlight to determine how raised a lesion is. Fluorescent light tends to make lesions appear flat [1]. When performing palpation, gloves are important both for patient comfort and practitioner safety. Patients may present with a rash, wart, or other specific dermatological concerns. One can inspect the area, looking closely for distinguishing features. Careful, focused history taking will play an important role in dermatological assessment as well. Included with skin assessment is assessing hair and nails. Changes to the nails can be a sign of autoimmune disorders, such as psoriasis, or infection if splinter hemorrhages are present. Nails that are very thick, lifting up from the nail bed, or are discolored can be a sign of a fungal infection. These are often found in older patients and on the toenails.

One rare but serious condition that pharmacists must be diligent in screening patients for is anaphylaxis, as this condition is a potential reaction to vaccines and other medications. Patients may ask their pharmacist when early anaphylaxis symptoms are present either right after receiving an injection or by phone after taking a new medication. Signs and symptoms include hives/wheals, pruritus, flushing, hypotension, difficulty breathing, nausea, vomiting, headache, and dizziness [1].

Another condition that may be encountered by pharmacists is shingles (herpes zoster). Shingles typically starts as neuritis/pain over a certain region and usually progresses to a rash over a specific dermatome, often with the development of vesicles [1]. Early recognition is key, as early treatment

(<72 h from onset of symptoms) speeds up resolution of rash, vesicles, and pain as well as reduces transmission to others. It may also reduce residual pain from postherpetic neuralgia [16, 17].

When assessing a patient, the pharmacist should take note of the patient's skin color. For example, the skin or sclera may have a yellow hue, which may be caused by a buildup of bilirubin due to liver dysfunction. This is known as jaundice. Bluish or purplish discoloration of the lips or fingers, also known as cyanosis, is a sign of hypoxia. Pale skin can indicate underlying anemia or could also occur in a person who is about to have a syncopal episode. Mottled skin, which is a blotchy purplish pattern, can be a sign of several conditions including shock. Alteration in skin color may also be due to adverse effects of medications. One of the most striking is "blue man syndrome" caused by amiodarone. This will appear as a bluish discoloration in sun-exposed areas in patients who have used the drug for a long time or at high doses. Oral contraceptives can also be associated with discoloration in the form of hyperpigmented patches called chloasma. Both types of discolorations can improve on their own after the drug is withdrawn, but it may take months. They are usually harmless but can be distressing to patients as they often occur on the face.

Decreased skin turgor, which is a sign of dehydration, is detected by pulling up on the skin. If the skin remains elevated in a tent-like formation, this is decreased skin turgor. Other physical symptoms that can be observed with dehydration include dry mucous membranes and eyes. Dehydration requires referral for further evaluation.

The temperature of the skin can be felt using the back of the hand. Notice if it is warm, cool, dry, or clammy. This may be considered with other symptoms the patient is experienc-

ing in determining a diagnosis. For example, if a patient has a red, swollen, hot area of skin, they may be experiencing an infection in that area.

For further assessment, we refer the reader to Chap. 13. It discusses dermatological assessment in more detail.

Respiratory System

Assessment of the respiratory system yields important information. Pharmacists should be familiar with general landmarks used to describe respiratory findings, as can be found in Figs. 3.3, 3.4, and 3.5. Aside from determining the patient's respiratory rate, the pharmacist can also use inspection to assess for use of accessory muscles during respiration. This can be seen as muscles in the neck contract with each breath, lifting the ribs and sternum to allow a larger volume of air to enter the lungs [1]. If not associated with exercise, accessory muscle use can be a sign of respiratory distress. Tripod breathing, when a patient leans forward and puts their hands on their knees, is another sign that can indicate respiratory distress and is often seen in patients with emphysema. Barrel chest is another abnormality that can be visualized. It is caused by overinflation of the lungs, and can be seen in patients with chronic obstructive pulmonary disease (COPD), or old age [1]. These patients will have a chest that is at least as wide from back to front as it is from left to right.

Auscultation of the lungs is a useful skill. Ideally, the lungs should be auscultated from the anterior and posterior aspects. Have the patient breathe slowly and deeply through their mouth. Using the diaphragm of the stethoscope (larger side of the round listening surface), auscultate each lung field, moving from upper to lower fields. Figure 3.6 depicts

Fig. 3.3 Cardiac and respiratory landmarks for physical assessment: bony chest skeleton. (Reprinted from *Textbook of Physical Diagnoses*, 7th ed., Swartz MH, The chest, 315–342.e1, Copyright 2025, with permission from Elsevier)

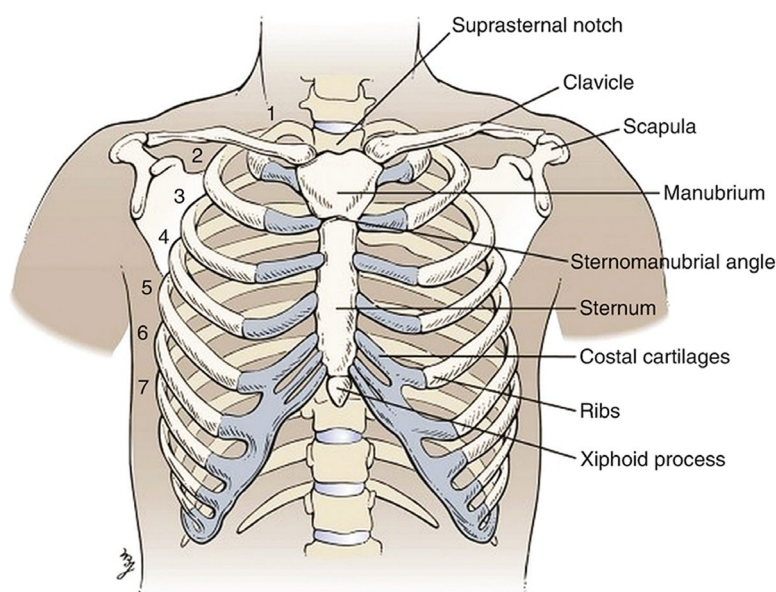


Fig. 3.4 Cardiac and respiratory landmarks for physical assessment: thoracic cage landmarks. (a) Topographic landmarks of the anterior thorax. (b) Landmarks on the lateral view. (Reprinted from *Textbook of Physical Diagnoses*, 7th ed., Swartz MH, The chest, 315–342.e1, Copyright 2025, with permission from Elsevier)

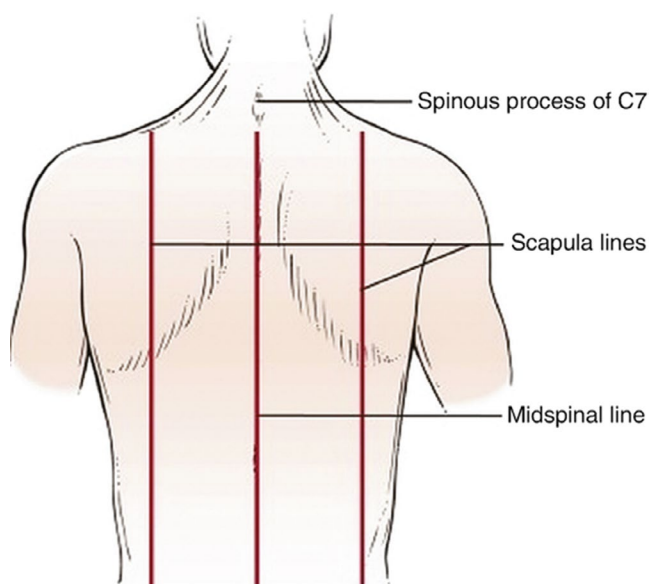
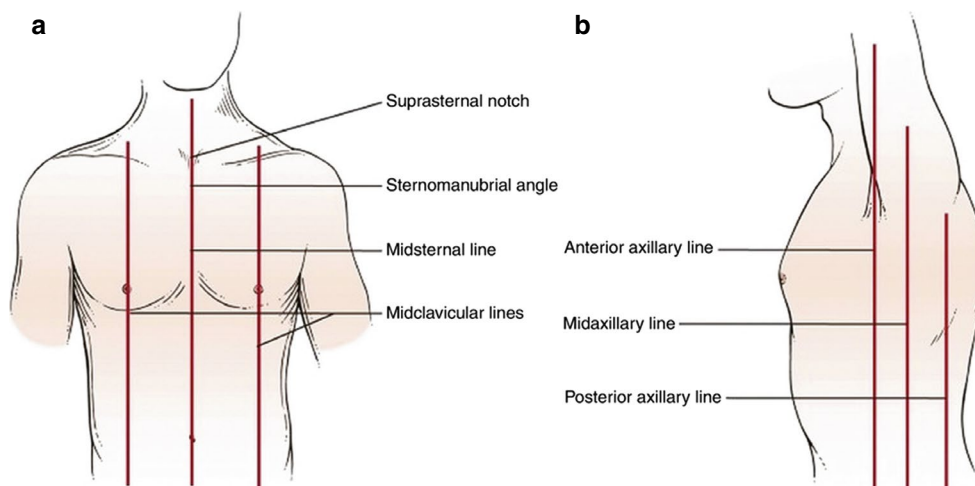


Fig. 3.5 Cardiac and respiratory landmarks for physical assessment: topographic landmarks of the posterior thorax. (Reprinted from *Textbook of Physical Diagnoses*, 7th ed., Swartz MH, The chest, 315–342.e1, Copyright 2025, with permission from Elsevier)

correct stethoscope positioning on the posterior thorax, to ensure that each lobe is listened to. Listen to a full breath cycle at each position before moving to the next. One should be familiar with the normal breath sounds prior to looking for abnormal (adventitious) sounds. Depending on where one places their stethoscope, they will hear bronchovesicular, vesicular, or bronchial breath sounds. Each has a characteristic sound, owing to the underlying structures (main bronchi, bronchioles, lobes, and trachea). The Internet hosts many educational websites (e.g., www.easyauscultation.com), which provide audio samples of both normal and adventitious breath sounds. Table 3.7 depicts examples of adventitious breath sounds.

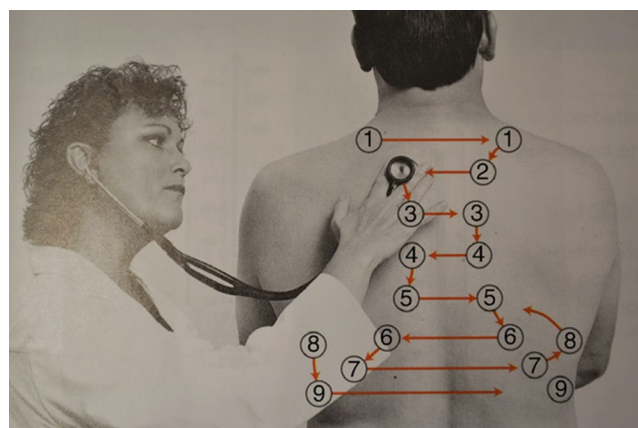


Fig. 3.6 Auscultation points for breath sounds. (Reprinted with permission from Jones [1])

Patients with respiratory conditions will often have spirometry testing done to determine the degree of airflow limitation they have. Patients with COPD should have this done during diagnosis as well as whenever an exacerbation is suspected. Pharmacists should be aware that a post-bronchodilator forced expiratory volume over 1 second (FEV1) over forced vital capacity (FVC) ratio (FEV1/FVC) value <0.7 is supportive of a COPD diagnosis. This means that after the patient is given a bronchodilator, they still have airflow limitation present [18]. Patients will have different degrees of airflow limitation (Table 3.8), which contributes to their classification between mild and severe disease in combination with other symptoms and degree of functional limitation. See Chap. 18 for more detailed information about COPD assessment.

In asthma, many patients use a peak flowmeter, which assesses how quickly they can blow air out of their lungs. An example of one available peak flowmeter is shown in Fig. 3.7. Patients should know what their peak flow is when they are

Table 3.7 Adventitious lung sounds [1, 2]

Adventitious sound	Characteristics of sound	Clinical correlation
Crackles (fine or coarse)	Intermittent, nonmusical, and brief Fine crackles are soft, high-pitched, and very brief Coarse crackles are louder, lower-pitched, and longer	Caused by fluid, mucus, or pus in the airway Fine crackles can be heard in pulmonary edema (HF) and pulmonary fibrosis Coarse crackles can be heard in pneumonia, COPD, lung abscess, and TB
Rhonchi	Low-pitched, rattling lung sound with snoring quality. Mostly heard with expiration. Usually clears after coughing	Caused by obstruction or secretions in the larger airway Can be heard in COPD, pneumonia, bronchitis, etc.
Wheezes	High-pitched, long, musical sound Can be heard with expiration or inspiration	Caused by airway narrowing Can be heard in asthma, COPD, and bronchitis
Friction rub	Grating or creaking sound that can usually be localized on the chest wall and can come and go	Inflamed pleura rubbing together is the cause, with friction arising from lack of lubricating fluid Any condition associated with pleural irritation can cause this, such as pneumonia
Stridor	High-pitched, musical sound. Can often be heard without a stethoscope	Caused by upper airway obstruction Can be heard in patients upon extubation

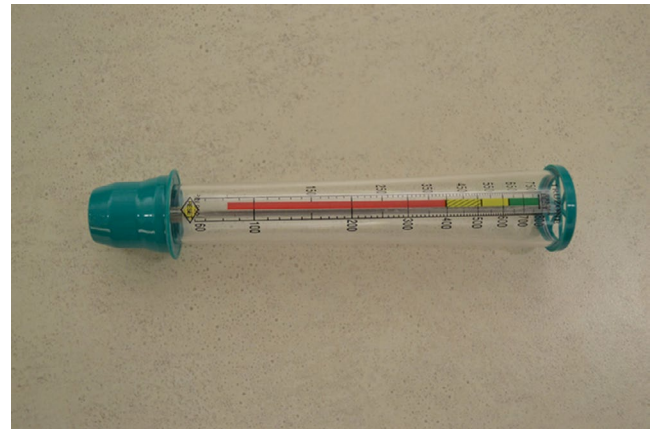
COPD chronic obstructive pulmonary disease, *TB* tuberculosis

Table 3.8 Classification of COPD airflow limitation based on post-bronchodilator FEV1 [18]

Degree of limitation	GOLD class	FEV1 measure in comparison to the expected value
Mild	1	≥80%
Moderate	2	50%–<80%
Severe	3	30%–<50%
Very severe	4	<30%

COPD chronic obstructive pulmonary disease, *FEV1* forced expiratory volume over 1 second, *GOLD* Global Initiative for Chronic Obstructive Lung Disease

not experiencing any symptoms, as a decrease can show a worsening in asthma control. From a pharmacy perspective, patients who have asthma and use a peak flowmeter could have their normal value recorded while completing a care plan in order to monitor disease progression. See Chap. 17 for more detailed information about asthma assessment.

**Fig. 3.7** An example of a peak flowmeter

Cardiovascular/Peripheral Vascular System

As discussed above, BP and HR are measured during the assessment of vital signs. Both of these values are important and are taken into consideration alongside the cardiovascular exam. During the general survey, the pharmacist should take note of how the patient is mobilizing and whether they experience any dyspnea with movement or at rest. This is especially relevant to patients with HF. If a patient reports chest pain, this requires referral for further assessment. Auscultation of the heart will reveal an abundance of information about the patient's cardiovascular system; however, most pharmacists are not trained to perform this. Some pharmacists who are in specialty practice, such as heart function clinics, may be trained to perform more advanced cardiac assessments. One example of how auscultation can be used is in patients with atrial fibrillation; the apical heart rate may be more accurate than measuring the peripheral pulse, since not every heartbeat will be transmitted peripherally. To illustrate, the peripheral pulse may be 80 bpm, but the apical rate could be 105 bpm, with the extra beats lost in transmission. The apical rate could help guide therapy if one is adjusting the patient's rate control to a resting target of <100 bpm. Heart sounds are also important to consider. All pharmacists should be aware that S1 and S2 sounds (corresponding to the closing of the heart valves) are normal, and the presence of additional sounds can indicate pathologies such as heart failure. Cardiac murmurs are also detected via auscultation. There is a grading system based on their qualities and intensities, from I to VI, i.e., faint to very loud [1]. A new cardiac murmur, taken in context with the patient's clinical picture, could be a sign of infective endocarditis, for example. This finding could affect antibiotic choice or dose, depending on the situation. Being aware of these findings and their implications when they are reported will allow pharmacists to optimize patient care whether or not they are able to perform the assessments themselves.

Assessment of the patient's QT interval is also relevant in patients who are on certain medications or have certain medical conditions. It will be reported if a patient has an electrocardiogram (ECG) done, but pharmacists should also be aware of when it should be requested in patients who have never had an ECG. Most pharmacists are not trained to interpret an ECG tracing, but it is important to be familiar with what a normal interval is as well as factors that are known to prolong it. A normal rate-corrected QT interval (QTc) is <430 ms for men and <450 ms for women. Many medications have an impact on the QT interval, including some anti-infectives, antidepressants, antiarrhythmic drugs, and drugs of abuse. Factors such as electrolyte imbalance, age, sex and cardiovascular disease add to the risk [1]. Patients with a prolonged QTc interval are at a higher risk of an arrhythmia known as Torsades de pointes (TdP), which can cause sudden cardiac death. Therefore, when the QTc interval exceeds 450 ms in males and 470 ms in females, it is imperative to address any modifiable factors including electrolyte correction and changing medications. There are many resources available to determine a patient's risk of QT prolongation, including www.crediblemeds.org, which will allow you to consider multiple risk factors together.

Circulation can be assessed by palpating peripheral pulses. One should note if the pulse is normal, diminished, or absent. For example, a person with peripheral arterial disease (PAD) may have diminished peripheral pulses and a person who is experiencing shock may have absent pulses [19]. Figure 3.8 shows the various sites for palpating peripheral pulses.

Volume Status

The patient's volume status is a key consideration for pharmacists, whether the patient is euvoletic, hypovolemic, or hypervolemic. For example, patients with hypovolemia are at risk of acute kidney injury, and they may accumulate any medication that is renally cleared. They are also at a higher risk of toxicity from nephrotoxic drugs. For patients who take diuretics, fluid status is essential for appropriate dosing, to ensure maximal efficacy and minimal toxicity. Volume status is also central to determining the etiology for patients with hyponatremia, a condition that frequently has implications for drug therapy. There are several PA skills that can help determine a patient's level of hydration. Inspection is used to examine the mucous membranes and skin turgor as discussed previously. Additionally, one would inspect the lower extremities for the presence of peripheral edema. Palpation is used to confirm the presence of edema and to determine whether it is pitting or not. If edema is present at the ankle level, one would inspect and palpate up the leg to see how high the edema goes. This is an important finding to note, as it is an indicator of the degree of fluid overload. Pitting edema is graded on a numeric scale, from 1+ to 4+,

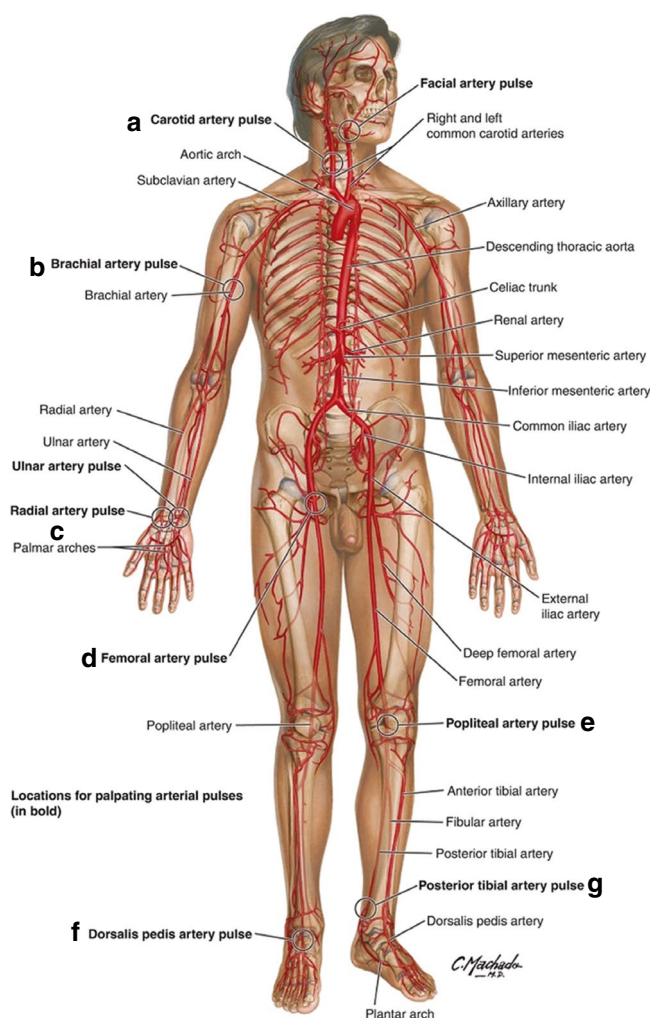


Fig. 3.8 Palpation of arterial pulse., (a) Carotid, (b) brachial, (c) radial, (d) femoral, (e) popliteal, (f) dorsalis pedis, and (g) posterior tibial. (Reprinted from *Atlas of Human Anatomy*, 7th ed., Netter FH, Chapter 1: Introduction, Plates 4, 11–10, Copyright 2025, with permission from Elsevier)

with 1+ being least severe. If the edema is nonpitting, it is usually due to either lymphedema or myxedema. One should also note whether the edema is bilateral and symmetrical. Unilateral edema could be due to deep vein thrombosis or venous insufficiency, for example [19].

Of note, peripheral edema is a common side effect of calcium channel blockers (CCBs). For example, up to 15% of non-HF patients taking amlodipine experience peripheral edema [20]. The risk is higher in women than in men and is also higher with dihydropyridine vs. non-dihydropyridine CCBs. The effect is likely dose-dependent and tends to build up gradually. Venodilators can counteract CCB-induced edema. Angiotensin-converting enzyme (ACE) inhibitors are the most studied class for this indication [21].

Another important aspect of volume status assessment is jugular venous pressure (JVP), as this is a marker of right

atrial pressure as well as central venous pressure. This can be a challenging skill to master, in part because the visibility of veins is variable [1]. However, even if one is not proficient at estimating JVP, it can still be useful to know how to interpret the results. The normal range of JVP is ~3–4 cm above the sternal angle. Values <3–4 cm would suggest hypovolemia, and values >4 cm indicate hypervolemia [1, 2]. JVP assessment is frequently performed by physicians and nurse practitioners (NPs) during care for patients with HF and other medical conditions (see Chap. 16 “Heart Failure Assessment”).

Genitourinary (GU), Renal, and Gastrointestinal (GI) Systems

For patients with genitourinary, renal, or gastrointestinal complaints, focused health history taking will be the main mode of information gathering. A structured approach to questioning, such as SCHOLARE (symptoms, characteristics, history, onset, location, aggravating factors, remitting factors, and explanatory model), will help ensure comprehensiveness. Review of laboratory findings plays an important role as well. Assessments of the liver and kidneys are explored in Chaps. 25 and 26, respectively.

Liver function should be considered, as many drugs are metabolized by the liver. Advanced liver disease puts patients at a high risk of adverse effects from medications, such as bleeding or encephalopathy/delirium. The Child–Pugh classification is a validated scoring system, which considers both lab values and physical findings in order to categorize patients based on their degree of liver dysfunction. The categories are A, B, and C, with A being milder dysfunction and C being severe liver disease. One of the physical findings that can often be seen is ascites, as patients’ abdomens can be very large with fluid accumulation. Child–Pugh scores can be used to determine whether dose reduction of medications is needed, as this is the parameter that is often used for drug dosing adjustment in liver failure. Online Child–Pugh calculators are readily available [22, 23]. There are other clinical scoring systems, which are more commonly used for prognostication, such as the MELD (model for end-stage liver disease) score.

Assessing renal function relies on a patient’s laboratory values, urine output, and medical history, rather than physical assessment. However, patients who have renal concerns such as stones or infections will often have pain in their lower back that varies in intensity depending on the condition. It is rare to be able to palpate the kidneys well, so this is not often done and other methods of investigation such as ultrasound are preferred [1].

Musculoskeletal (MSK)

The musculoskeletal (MSK) system has countless possible ailments. The pharmacist is well-positioned to screen patients and determine who is appropriate for self-care and who should be referred. If the patient is in distress, has decreased mobility, or may require diagnostic imaging, they should be referred to their primary care provider or the emergency room. Depending on one’s practice setting and expertise, a pharmacist may use inspection and palpation to assess injuries, including range of motion assessment. For example, pharmacists working in specialty rheumatology clinics will likely use PA and health history taking to complete an in-depth assessment of their patients. However, even those in general practice can inspect joints for signs of swelling and deformities that may indicate rheumatoid arthritis, such as swan neck and Boutonniere’s deformities in the fingers. Osteoarthritis is another common complaint of patients, but there are few physical signs and often patient assessment will rely on history taking or diagnostic imaging. Within the same differential diagnosis as different types of arthritides can be gout. In most patients, this will present as a single red, inflamed, acutely painful joint. It is often the first metatarsophalangeal joint of the foot. Patients who present with this picture should be referred to their physician for further assessment and diagnosis.

Adverse effects of medications can manifest as MSK symptoms as well. One important area that pharmacists should be familiar with is statin-induced muscle symptoms. These are a common complaint among patients taking statins. Figure 3.9 presents an assessment tool which may be used to help determine if muscle symptoms are due to the statin [24].

Patients who have osteoporosis with damage to the spine will often have abnormal curvature of their spine, called kyphosis, as a result of vertebral compression fractures. Upon inspection, they will have a hunched back in the shoulder area. They can also have a shortened distance between their rib cage and pelvis. If a patient cannot stand up straight against a wall with their head and shoulders on the wall, it can be a sign of kyphosis. Vertebral compression fractures can also be screened for by simply asking how tall they were when they were 20 years old and how tall they are now. Losing >2.5 inches from age 20 or >2 inches in a year is a sign that they need further assessment [25]. Osteoporotic patients may also have fractures from falls that are not expected to be damaging. These are often hip or forearm fractures caused by trying to break a fall. In patients who present after these types of fractures, a risk assessment can be done to decide if they need further screening or treatment for osteoporosis.

Statin-Associated Muscle Symptom Clinical Index (SAMS-CI)

Instructions:

- Use with patients who have had muscle symptoms that were **new** or **increased** after starting a statin regimen.
- A **statin regimen** includes any statin at any dose or frequency, including a statin the patient has used previously, at the same or a different dose.
- **Muscle symptoms** may include aches, cramps, heaviness, discomfort, weakness, or stiffness.
- Interpret overall score in light of **other possible causes** of the muscle symptoms, such as:

Recent physical exertion	Hypothyroidism	Concurrent illness
Changes in exercise patterns	Drug interaction with statin	Underlying muscle disease
- See reverse for Frequently Asked Questions

How many statin regimens has the patient had that involved new or increased muscle symptoms?

One

Two or more

Complete the questions on the left side of this page.

Complete the questions on the right side of this page.

Regarding this statin regimen:

A. Location and pattern of muscle symptoms

(if more than one category applies, record the highest number.)

Enter score:

Symmetric, hip flexors or thighs	3
Symmetric, calves	2
Symmetric, proximal upper extremity	2
Asymmetric, intermittent, or not specific to any area	1

B. Timing of muscle symptom onset in relation to starting statin regimen

<4 weeks	3
4–12 weeks	2
>12 weeks	1

C. Timing of muscle symptom improvement after withdrawal of statin (if patient is still taking statin, stop regimen and monitor symptoms.)

<2 weeks	2
2–4 weeks	1
No improvement after 4 weeks	0

Rechallenge the patient with a statin regimen, (even if same statin compound or regimen as above) **then complete final question:**

D. Timing of recurrence of similar muscle symptoms in relation to starting second regimen

<4 weeks	3
4–12 weeks	1
>12 weeks or similar symptoms did not reoccur	0

Total:

All four scores above must be entered before totaling

Regarding the statin regimen *before* the most recent regimen:

A. Location and pattern of muscle symptoms

(if more than one category applies, record the highest number.)

Enter score:

Symmetric, hip flexors or thighs	3
Symmetric, calves	2
Symmetric, proximal upper extremity	2
Asymmetric, intermittent, or not specific to any area	1

B. Timing of muscle symptom onset in relation to starting statin regimen

<4 weeks	3
4–12 weeks	2
>12 weeks	1

C. Timing of muscle symptom improvement after withdrawal of statin

<2 weeks	2
2–4 weeks	1
No improvement after 4 weeks	0

Regarding the *most recent* statin regimen: (even if same statin compound as above)

D. Timing of recurrence of similar muscle symptoms in relation to starting regimen

<4 weeks	3
4–12 weeks	1
>12 weeks or similar symptoms did not reoccur	0

Total:

All four scores above must be entered before totaling

Interpretation

Total score:

Likelihood that the patient's muscle symptoms are due to statin use:

2–6

Unlikely

7–8

Possible

9–11

Probable

10 Oct 2016, Based on Rosenson et al. An assessment by the Statin Muscle Safety Task Force: 2014 update, *J Clin Lipidol*. 2014 May-Jun;8(3 Suppl):S58–71.

Fig. 3.9 Statin-associated muscle symptom assessment. (Reprinted from Springer Nature under the terms of Creative Commons CC BY license from Rosenson et al. [24])

Lab Work and Microbiology

The pharmacist should review pertinent laboratory values and microbiology when required, as this can have a significant impact on drug therapy. Each unique clinical scenario will dictate what lab work is needed. The following are some of the tests that are frequently applicable to pharmacy practice:

- Blood work:
 - Complete blood count (CBC) with or without differential
 - Serum electrolytes (“lytes”)
 - Urea (blood urea nitrogen (BUN))
 - Creatinine (SCr)
 - “Liver chemistry tests” (alanine aminotransferase (ALT), aspartate aminotransferase (AST), bilirubin (total and conjugated), international normalized ratio (INR), alkaline phosphatase, serum albumin)
 - Thyroid-stimulating hormone (TSH)
 - Creatine kinase (CK)
 - Troponin
 - Lactate
- Drug levels for therapeutic drug monitoring (e.g., phenytoin, valproic acid, vancomycin, gentamicin, digoxin, etc.)
- Urinalysis, arterial blood gas (ABG), venous blood gas (VBG)
- Microbiology: Blood, urine, skin and soft tissue, sputum cultures, intraoperative bone/joint/hardware cultures, cerebrospinal fluid (CSF), etc.

Further discussion of laboratory tests is beyond the scope of this chapter. Like all PA and assessment skills, one’s ability to interpret lab values is developed through practice and repetition. Resources are available to provide background information about lab tests and their interpretation [26].

Red Flags (Referral Needed)

Screening for red flags during physical assessment is a broad but important topic. A comprehensive list of all red flags is not possible. Referral is needed any time a patient is in distress, whether that be psychological, respiratory, cardiovascular, etc. Another general principle is that signs and symptoms involving blood or bleeding are usually red flags. Any patient that you feel requires more treatment or assessment than you are competent to perform should be referred as well, whether it is to another pharmacist with different skills or to another healthcare provider. Examples of red flags identified during PA can include:

- *CNS*: Decreased LOC, acute confusion, delirium, sudden weakness, seizures, suicidal ideation, severe depression, signs or symptoms of mania, new gait disturbance
- *Eyes*: Vision changes, pain, feeling of a foreign body, light sensitivity, double vision, exophthalmos, nystagmus, irregular pupils (fixed, large, or small)
- *Respiratory*: Dyspnea, difficulty breathing, abnormal findings on lung auscultation (e.g., wheeze, crackles, rhonchi, or stridor), worsened asthma or chronic obstructive pulmonary disease (COPD) symptoms, persistent cough
- *Cardiovascular*: Symptomatic or significant hypotension, marked hypertension, tachycardia or bradycardia, chest pain, arrhythmia or “palpitations,” new or worsened heart failure symptoms, significant dehydration
- *MSK*: Distressing injury, poorly controlled rheumatoid arthritis, mobility issues, requirement for diagnostic imaging
- *GI*: Severe constipation, diarrhea, vomiting, hematemesis, melena stools, bright red blood per rectum (BRBPR), pencil-like stools
- *GU*: Urinary tract infection symptoms, difficulty urinating, decreased urine output, hematuria
- *Infectious*: Signs or symptoms of sepsis, prolonged or high fever, flank pain or tenderness, suspected untreated infection
- *Dermatology*: Jaundice, cyanosis, signs and symptoms of anaphylaxis, rash with a fever especially in children

Case

The following case example demonstrates the use of physical assessment findings in patient assessment.

Patient JB is a 75-year-old male admitted to the medicine ward for acute exacerbation of COPD. He presented with accessory muscle use, hypoxia, and dyspnea. Today is postadmission day 3, and you are seeing JB for the first time on a Monday afternoon, as you missed seeing him on team rounds. Your team is gone for the afternoon, so you are seeing him by yourself. You decide to review JB closely as he seems unwell.

Chief Complaint

“I can’t breathe very well.”

History of the Present Illness

He has had COPD for 10 years and was only hospitalized once, 9 years ago. He currently uses fluticasone/salmeterol (Advair®) 500 mg Diskus 1 puff twice a day (bid), tiotropium (Spiriva®) Respimat 2 puffs daily, and salbutamol (Ventolin®) 2 puffs four times a day (qid) + as needed (prn)

(uses about 14 puffs per week). According to community pharmacy, Spiriva and Ventolin are filled regularly every 3 months, but Advair is filled sporadically, with the last 30-day supply filled 53 days ago.

PMHx

HF (× 2 years), smoker (1 pack/day × 40 years), hypertension (× 10 years)

Current Medications

Perindopril 8 mg by mouth (po) daily
 Bisoprolol 5 mg po daily
 Furosemide 40 mg intravenous (IV) bid (autostopped this morning because it was only ordered for 3 days)
 Fluticasone /salmeterol (Advair®) 500 mg Diskus 1 puff bid
 Tiotropium (Spiriva Respimat®) 2 puffs daily
 Salbutamol 2 puffs qid + prn
 Ipratropium 2 puffs qid + prn
 Prednisone 40 mg po daily × 5 days

Vital Signs (Taken This Morning)

BP 110/80 mmHg, HR 70 bpm and regular, RR 25, O₂ saturation 90% on 4L, temperature 38.2 °C, GCS 15

Physical Exam Findings and Review of Systems (Your Own)

General survey: The patient is pleasant and appears at stated age; he is thin. Speech seems labored. He is alert and oriented x3 (to person, place, and time). He is bed-bound, and the head of the bed is elevated.

HEENT: Pupils equal, round, reactive to light, pursed lip breathing seen, no cyanosis observed

Dermatological: Not performed

Respiratory: Barrel chest, appears to be breathing hard, accessory muscle use visualized. Lungs have diffuse expiratory wheeze, rhonchi, and coarse crackles at the bases with dullness to percussion. Sputum is copious and green-colored.

Cardiovascular/peripheral vascular: Radial pulse regular, JVP 3 cm, 2+ pitting edema to calves bilaterally

Abdomen: Not performed

GI/GU: Not performed

MSK: Not performed

History Questions

You complete a medication history and discover that JB ran out of his Advair at home as it was not covered by his insurance plan. His dyspnea is worse than baseline; he is producing more sputum than normal, and it has changed from yellow to green. No hemoptysis. He feels some dizziness and is short of breath. He also tells you that his shoes are not fitting as well as they used to, and they feel tight. He reports

Table 3.9 Problem list and drug therapy problems for JB

Condition	Control/adherence	Drug therapy problems
AECOPD	Nonadherent to Advair because of cost Has signs of bacterial infection Multiple types of inhalers, assess technique	Reassessment of inhaler insurance coverage/cost Additional therapy with antibiotics is needed
Heart failure	Fine crackles at the bases present Pitting edema to the calves Bisoprolol dose is not optimized (10 mg daily is the target)	Requires continuation of diuretic therapy and reassessment of home dose Reassessment of the route of the diuretic (IV vs. po)
Dizziness	Blood pressure is 110/80, but does not have home values for comparison Assess for orthostasis	May require reassessment of antihypertensive therapy
Shortness of breath (SOB)	Could be due to any of the following: AECOPD, not using Advair, HF exacerbation	Need to clarify timeline when SOB worsened with patient

AECOPD acute exacerbation of chronic obstructive pulmonary disease, *HF* heart failure, *IV* intravenous, *po* by mouth, *SOB* shortness of breath

sleeping with two pillows because he “coughs a lot when he just uses one.”

Problem list and drug therapy problems are summarized in Table 3.9.

You decide to page the resident on call to discuss the drug therapy problems you have identified. Before calling, you prioritize the problems and come up with specific suggestions including drug, dose, and route.

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Assessment of Complex Patients in Pharmacy Practice

4

Sara Housh and Sherif Hanafy Mahmoud

Chapter Objectives

1. Define a complex patient, including the factors contributing to their complexity.
2. Review the implications of patient complexity for both the individual and healthcare system.
3. Identify tools and strategies for recognizing and assessing complex patients in clinical practice.
4. Examine the role of pharmacists in the care of complex patients across multiple healthcare settings.
5. Define deprescribing, and apply the principles when managing complex patients.

Background

The scope of practice and practice environments for pharmacists are expanding as healthcare adapts to the growing needs of the population. Positioned across various clinical settings, including community pharmacies and hospital units, pharmacists are exposed to a diverse range of patients and scenarios. They may assist patients with concerns related to minor ailments like allergies, address acute conditions such as urinary tract infections (UTIs), or provide ongoing management for chronic conditions like diabetes and osteoarthritis. Some patients have more than one chronic condition at a time, also referred to as comorbidities. When a patient has two or more chronic conditions, they may be considered complex. Complex patients often utilize high numbers of healthcare resources, have increased chances of mental health disorders, and overall experience lower quality of life [1]. These patients often present with unique healthcare needs and an increased need for interdisciplinary collaboration. It has been noted that patient care improves when phar-

macists are incorporated into the patient's healthcare team [2]. Patient complexity is often compounded by biopsychosocial factors, including lack of financial or residential security, or mental health conditions [3]. It is pertinent to identify when a patient is complex so that drug–disease interactions can be considered when optimizing their care. It is also relevant to understand whether a patient is complex when considering follow-up and monitoring plans for ongoing management. This chapter includes the definition of a complex patient, describes some of the implications of complexity, and explores the role of the pharmacist in providing complex patient care.

Who Is a Complex Patient?

It is agreed by many researchers that a clear definition of a “complex patient” is absent from the current literature [3]. Defining a complex patient is necessary for the continual improvement of the healthcare system, as it facilitates informed decision-making, enhances care coordination, and ensures consistent monitoring and follow-up [4]. In the literature, complexity is often interchangeable with the terms “multimorbidity” and “comorbidity” [3]. Generally, the term “complexity” denotes patients who require extra care and consideration in all aspects of the patient care process. Several domains are noted as contributing to complexity: demographics, personality traits, communication barriers, socioeconomic status, physical and mental health, and the environment [3]. This list is not exhaustive but provides a comprehensive view of the patient. Research has shown that the holistic model of illness can predict findings not explained by a traditional biomedical model [5]. An example from a review by Wade in 2011 shows that being in debt—a psychological stressor—is associated with lower back pain, despite an absence of any radiological abnormalities of the spine [5].

In this chapter, a “complex patient” refers to a person with two or more chronic conditions that may be influencing the care of their other conditions [6]. There may be compound-

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ing biopsychosocial factors that impact the patient state [6]. “Polypharmacy,” defined by the use of five or more concurrent medications, is another aspect of complexity that can be chiefly managed by pharmacists, who are trained to identify and solve drug therapy problems [7].

It is also important to make a distinction between *complex* and *challenging* patients, as confounding factors from the various biological, medical, socioeconomic, and cultural realms can be more convoluting for some patients more than others [4]. For example, take two patients with the same three conditions: diabetes, hypertension, and chronic obstructive pulmonary disease (COPD). Patient A lives with their spouse who actively aids in their care and also has an insurance plan that covers their medications fully. In contrast, patient B lives alone and lacks insurance, which further complicates their care. Despite both patients being considered complex based on their medical diagnoses, it can be concluded that patient B would be a more challenging complex case for their healthcare team because of these social factors.

As the most accessible healthcare professionals, pharmacists are in a unique position to assess patients for complexity as they help continually manage their care. Identifying complex patients aids healthcare teams in better prioritizing patient issues and promoting shared decision-making so patients can reach their healthcare goals [6]. Clinical pharmacy services have been shown to improve a variety of metrics regarding patient care, such as reaching blood pressure or sugar targets [7]. In fact, research has demonstrated that involving the skills of a clinical pharmacist in patient care can double the number of patients who reach their health goals [7]. The role of the pharmacist as a pharmacotherapy expert means that they should be addressed as key players in the healthcare team for management of chronic and complex conditions.

Currently, the most frequently cited tool for assessment of patient complexity is the INTERMED tool [8]. This clinician-led assessment organizes patient factors into four main domains: biological, social, psychological, and healthcare needs [9]. The four domains are assessed by time, history, current state, and prognosis.

Complex patients are commonly identified when they receive acute care, which is why the INTERMED tool was developed and validated for the hospital setting [10]. While it is true that complex patients typically present with higher rates of hospitalization than noncomplex patients and thus can be commonly found in hospitals, their care extends outside acute care settings. The INTERMED tool is holistic and has been deemed reliable, but, due to its length and complexity, it is not suitable for use in community pharmacy practice. Primary care providers, pharmacists included, oversee the patient’s care in the long term and can help find drug-related

problems in outpatient settings [6]. Continuous community monitoring allows pharmacists to identify patients with complications and interactions arising from chronic conditions that might otherwise go unrecognized. Therefore, it is crucial for pharmacists to be able to recognize complex patients in practice [11].

Implications of Complexity

The concept of a complex patient is not novel, and their population is increasing as time goes on. Approximately two-thirds of people over 65 are affected by more than one medical condition, and it is estimated that over 25% of patients over the age of 65 are considered complex within the Western healthcare system [12].

The number of patients with chronic illnesses has been steadily increasing over the past 20 years, and this trend is expected to continue. A study conducted at the University of Toronto in Ontario, Canada, projected that by 2040, one in four adults over the age of 30 will be living with a chronic condition, compared to one in eight in 2002 [13]. As a result, the prevalence of multimorbidity will undoubtedly rise, increasing the burden of chronic illness on the healthcare system. Currently, multimorbidity accounts for 78% of all primary health consultations [14]. As populations age, the complexity of care increases. An international study by Skou et al. in 2022 found that 30% of individuals aged 45–64 had multimorbidities, with the proportion rising to 65% for those aged 65–84 and to 82% for those aged ≥85 [14]. Utilization of healthcare services is largely driven by complex and multimorbid patients, as evidenced by hospital readmission and mortality rates. One study found that hospital mortality rates were 3.9% for patients without comorbidities but increased to 16.6% for those with multimorbidities [15]. In terms of readmission, the same study showed that the likelihood of readmission within 1 month was 4.8% for those without comorbidities, rising to 11.6% for those with comorbidities [15]. Considering readmission 1 year after discharge, the risk significantly increases with age [15]. Individuals without comorbidities had a 15.3% chance of readmission after 1 year, while those with comorbidities had a 34.3% chance. The risk of readmission increased by 34% for those aged 64–79 and by 53% for those over 79 [15]. Beyond the costs to the healthcare system, complex patients also experience higher personal financial burdens and strain on their support systems. They are also two to three times more likely to develop depression as compared to the general population [14]. These statistics highlight that being able to screen for complexity as a pharmacist becomes increasingly pertinent, especially in an aging demographic, where the number of citizens with multimorbidities is increasing with time [7].

How Can We Identify a Complex Patient?

Increased biopsychosocial complexity is related to an increased risk of negative health outcomes, including multimorbidity, longer disease duration, reduced clinical improvement, and a higher risk of mortality [16]. Consequently, identification of these patients is a critical component to providing optimal healthcare. The INTERMED tool is commonly used and cited for use in identifying complex patients [17]. It is a semi-structured clinician-led interview with 20 questions and takes at least 30 min to complete [18]. This tool has been frequently cited as the reference standard for complexity assessment [17], but its use is limited due to the fact that it is clinician-administered and its administration requires training [16]. To circumvent this issue, and aid in complexity assessment in a greater number of settings, several self-assessment tools have been developed using INTERMED as a basis, including the INTERMED Self-Assessment (IMSA) questionnaire and IMSA for the Elderly (IMESA) [10]. IMSA targets adults of all ages, while IMESA focuses on those aged ≥ 65 years [19]. Both IMSA and IMESA are under 20 questions long, covering the same domains as the INTERMED tool, and take approximately 10–20 min to complete [18]. While the scores do not correlate exactly with that of the INTERMED, evidence shows that when contrasted with the INTERMED tool, the IMSA questionnaire was able to adequately assess whether or not the patient required a case manager [10], who would then be responsible for connecting the patient with additional healthcare supports. Generally, both IMSA and IMESA are able to accurately indicate when a patient is likely to use health services at an increased frequency [18], which is a strong component of complex patient care.

With regard to hospital readmission rates, there are tools that can predict the probability of being hospitalized in a given period, with the most validated of these being the Probability of Repeated Admission (Pra) instrument. Pra is targeted at the elderly and looks at the probability of readmission in the next 4 years using eight questions that cover caregiver availability, age, diabetes, and number of physician visits in the past year, among others [18].

While these are the main tools currently used to assess complexity in the healthcare system, there are several others looking at different social determinants of health, including the Community Assessment Risk Screen (CARS) and even a Predicted Insurance Expenditures (Pie) tool [18].

There is currently no tool for complex patient screening that has been curated for use in community pharmacy practice. Such a tool may be developed on the basis of its predecessors and should consider the social determinants of health [20], as pharmacists are the most accessible healthcare providers and have great potential to identify complex patients in everyday practice, especially when conducting follow-up on chronic conditions.

What Is the Role of the Pharmacist in Complex Patient Management?

The responsibilities of a pharmacist go far beyond dispensing, and they are involved in a multitude of patient activities including reconciling medication, detecting drug interactions, monitoring of laboratory tests, extending prescriptions, and promoting patient education [21]. Studies have shown that integrating pharmacists into ambulatory clinics improves chronic disease management and optimizes use of medications [2]. When patients are given new medication orders, or medication-related changes, it is the role of the pharmacist to document these changes [22]. When given the opportunity, pharmacist recommendations on medication management can be implemented in collaborative practice to promote faster patient recovery and more efficient medication management [21]. Beyond medication reconciliation, and detecting drug interactions, pharmacists in Alberta, Canada, can personally order laboratory tests to monitor drug therapy and even prescribe, adapt, and extend prescriptions. When appropriate, pharmacists may advocate for consideration of deprescribing if it proves beneficial to the patient [22].

The capabilities of the pharmacist are demonstrated to their full potential in complex patient management. At each point of the patient care process, beginning with initial assessment and ending with monitoring and follow-up, the potential to improve health outcomes increases when a pharmacist is made an integral part of the patient care team [22]. As medication experts, pharmacists are trained to identify drug interactions, determine the indication, efficacy, and safety of medications, and simplify regimes when possible [23]. Much of the challenge in these cases arises from the fact that most complex patients are those with polypharmacy with a variety of drugs prescribed to them, each with their unique mechanisms, side effects, and interactions [24]. While complex patients often experience higher rates of hospitalization [3] and are thus likely to see pharmacists in hospitals, community pharmacists see chronic outpatients on a frequent basis and thus are in the ideal position to optimize medication regimens for the improvement of health outcomes [11].

The role of the pharmacist in an interprofessional care team helps improve chronic disease management and prevent administration errors [23]. Research has demonstrated that patients have preferences for healthcare sites that can provide them with interprofessional healthcare teams that include doctors, pharmacists, nurses, and social workers [21]. In hospital-based multidisciplinary teams, doctors frequently follow up on pharmacist recommendations regarding medication management [21]. Upon discharge, pharmacists can work with nurses and general practitioners to develop care plans that facilitate a smooth transition from

hospital to home [22]. Optimal use of interprofessional care teams results in a reduction of the patient's use of healthcare services. When pharmacists contribute to chronic care management, results indicate that there are improvements in vital measurements such as outpatient hypertension control [25]. One study focusing on team-based care determined that pharmacist intervention in hypertension management led to an overall decrease of 10 mmHg in systolic blood pressure [25]. Furthermore, data demonstrated that when community pharmacists provide advice on pain management, a decrease in the mean pain score for patients was observed [21]. Another study suggests that pharmacists are the most appropriate providers for managing complex patients with diabetes, particularly when endocrinologists are scarce [24]. As patients with diabetes are frequently deemed complex due to comorbidities and intensive insulin regimens, the management of the medication regimen is often delegated to pharmacists, suggesting that they are best-suited for reviewing polypharmacy cases [24]. Complex patients with diabetes are one such example of where pharmacist contribution can improve patient outcomes, but, as a profession, pharmacists' ability to improve patient outcomes in conditions beyond diabetes, including gastrointestinal (GI) disorders, high blood pressure, and respiratory conditions such as asthma, has also been demonstrated [25]. Pharmacists also have positive impacts on areas of specialized care, such as palliative care patients, through continuing medication management [26].

The social aspects of a patient's history are relevant in determining complexity and are included in multiple assessment tools [5]. *Social determinants of health* are defined as "the conditions in which people are born, grow, work, live, and age" and significantly affect the health and quality of life of patients [20].

Since these social determinants are important in assessing complexity, pharmacists should be educated to identify them. This ensures that they can address these domains at the patient level during their daily practice. There are various ways to address these issues at the patient, practice, and community levels [20]. The five main domains are healthcare access and quality, economic stability, education access and quality, social and community context, and neighborhood and built environment [20]. As demonstrated in Table 4.1, there are currently suggested strategies to manage each social determinant of health [20].

The primary method of addressing these issues is through social screening and referral processes, with tools like IMSA being potentially helpful [16]. At the patient level, strategies such as medication affordability, adherence interventions, and patient education are key [20]. Collecting social history during the patient care process is critical for assessing the impact of social factors, creating a reference plan, and deter-

Table 4.1 Pharmacist strategies to address the social determinants of health

Social determinants of health	Pharmacist strategies
Healthcare access and quality	Culturally sensitive patient education Medication affordability Medication adherence Interventions Primary prevention strategies Interdisciplinary collaboration Political advocacy
Economic stability	Medication affordability Political advocacy
Social and community context	Social needs screening and referral Community engagement Community needs assessment and planning Political advocacy
Neighborhood and built environment	Community engagement Community needs assessment and planning Political advocacy
Education access and quality	Community engagement Political advocacy

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mining how access to necessary services will be followed up [20].

Deprescribing in the Context of a Pharmacist's Role

Being medication experts also indicates that pharmacists know when to simplify medication regimens via deprescribing and are also capable of making helpful recommendations to primary care physicians if they feel a medication is indicated. As many complex patients are on multiple medications, ensuring that medication is discontinued when no longer needed is important for avoiding unnecessary drug interactions and other harmful effects [27]. Medication deprescribing is defined as the process of stopping or reducing medications that are no longer needed [27]. Complex medication regimens have been linked to increased chances of administration errors and poor health outcomes [28], which emphasizes the importance of deprescribing when appropriate. Specific examples of deprescribing include recommendations from pharmacists to halt usage of opioids or nonsteroidal anti-inflammatory drug (NSAID) analgesics in the elderly, who can suffer from increased adverse effects to these drug classes compared to younger patients [29]. One study investigating the frequency of safe deprescribing showed that when a pharmacist conducted medication reviews of elderly polypharmacy patients over a period of 3 months, >81% of patients in the study had at least one medication discontinued [30]. Evidence thus shows that

deprescribing is an important part of the medication optimization process [27].

Deprescribing can also play a role when identifying *potentially inappropriate medications* (PIMs). Such medications are more likely to be a part of a chronic medication regimen, particularly in the elderly [2]. Criteria exist to aid in identifying medications that may not be appropriate for a patient, such as the Beers, START (Screening Tool to Alert doctors to Right Treatment), and STOPP (Screening Tool of Older Person's Prescriptions) criteria [2]. One study aimed to identify potentially inappropriate medications in the elderly with chronic conditions and comorbidities [2]. The study classified potentially inappropriate medications if there was a greater risk of adverse effects as compared to benefits, duplication of drugs or drug classes, using a drug more frequently than necessary, or if there was an absence of a beneficial drug [2]. The study concluded that while using STOPP/START criteria helped pharmacists identify 65% of all PIMs in the enrolled population, 35% of PIMs were identified via clinician judgment alone [2]. Of this 35%, most patients were prescribers of intense polypharmacy, identified as taking more than 10 medications [2]. This study highlights the benefits and comprehensive knowledge of pharmacists in improving patient medication safety.

Medication simplification is another way by which pharmacists can improve the health outcomes of complex patients, by adjusting administration times, altering medication routes, and switching to combination formulations where possible [28].

Given the significant role pharmacists play in patient care, pharmacy school curricula should integrate the principles of deprescribing across various disease states [31].

Telehealth

Complex patients, due both to the number of comorbidities and polypharmacy, require more frequent monitoring and follow-up appointments as compared to noncomplex patients. In order to meet patient care needs, primary care networks trial different methods of offering continuing care. One example that has been increasing in prevalence, following the COVID-19 pandemic of 2020, is the concept of virtual, or telehealth, where patients access their healthcare teams from their own homes [26, 32]. Telehealth encompasses all communication technologies that are used to provide patient care from a distance and is used across many healthcare disciplines [33]. Rural communities lacking in healthcare providers also benefit greatly from the availability of such services [34]. Telehealth has been cited as a useful solution to maintaining patient monitoring and follow-up [33]. It has shown to promote high levels of patient satisfaction and acceptance, and integration of pharmacists into

existing virtual care networks has been proven to help detect potentially inappropriate medications and successfully implement deprescribing measures [26]. In their study in Maryland, Green et al. (2024) studied the effects of having pharmacists monitoring polypharmacy dementia patient medications in a telehealth model [30]. The patients were given options of video or phone calls, where the pharmacist revised both the benefits and harms of the patients' current medications. To measure the effects of the monitoring, the researchers used the *Family Caregiver Medication Administration Hassles Scale* (FCMAHS) scoring. The study found that the scores decreased by 3.3 points after 3 months of pharmacist monitoring, indicating a reduction in medication burden [30]. Another example of pharmacists benefiting patients through telehealth services is found in the specialty of pain management pharmacists [33]. Telehealth services enable pharmacists to review patient cases on a more regular basis and to improve access to appropriate pain medication management [33]. Clinical evidence has demonstrated that incorporation of pain management pharmacists reduces overall pain intensity, improves physical functioning, and reduces overall costs to the healthcare system [35]. The format of telehealth appointments involving counseling often takes 20–30 min, whereas follow-up appointments may be shorter at 6–10 min. Due to their shorter nature, follow-up appointments can be conducted on a monthly basis, and physicians have reported that pharmacists were thus able to fill gaps in laboratory monitoring and supportive care in between specialist appointments [34].

Initial Assessment of a Complex Patient

Initially, assessment of a complex patient begins in the same way as it does for any other patient. The assessment process must always involve the following initial steps:

1. Identifying the patient's chief complaint (*why is the patient currently seeking care?*)
2. Assessing the patient's medical history, including assessment of their present illness
3. Assessing the patient's medication regimen.

The general approach to patient assessment is summarized in Fig. 4.1.

During assessment of a patient's medical history, it may become clear that a patient is complex. The first clue to complexity would be the presence of two or more chronic conditions. Figure 4.2 summarizes one way a pharmacist might approach a scenario where this is the case.

However, determining complexity is multifaceted and cannot be completely summarized in a flowchart. This may be best illustrated by an example, as referred to in Box 4.1.

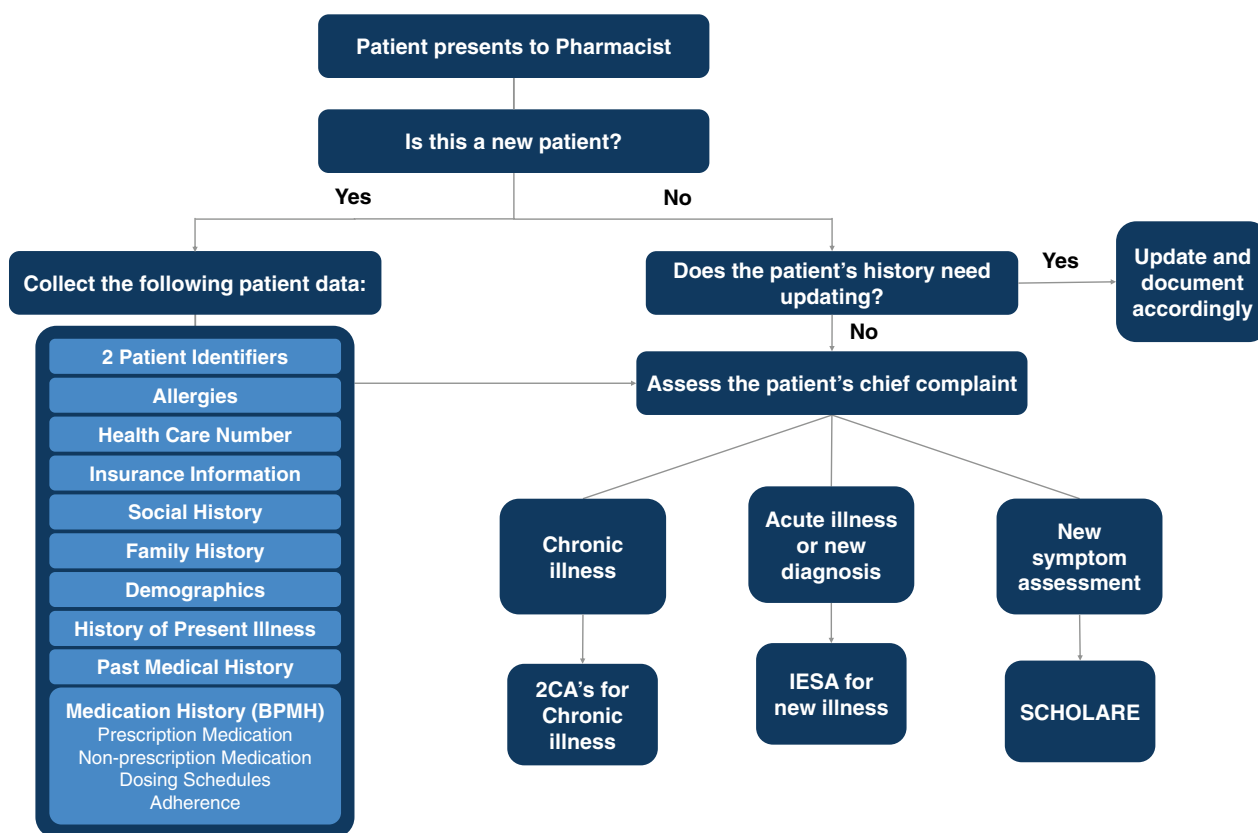


Fig. 4.1 Flowchart summarizing the general approach to initial patient assessment, including assessment tools BPMH (best possible medication history), 2CAs, IESA, and SCHOLARE. Refer to *Chapter 2 Principles of Patient Assessment* for expansion on these tools. 2CAs control of the disease, complications, adverse reactions to the medica-

tions, and adherence to the treatment plan, IESA indication, effectiveness, safety, and adherence, SCHOLARE symptoms, characteristics, history, onset, location, aggravating factors, and remitting factors and explanatory model

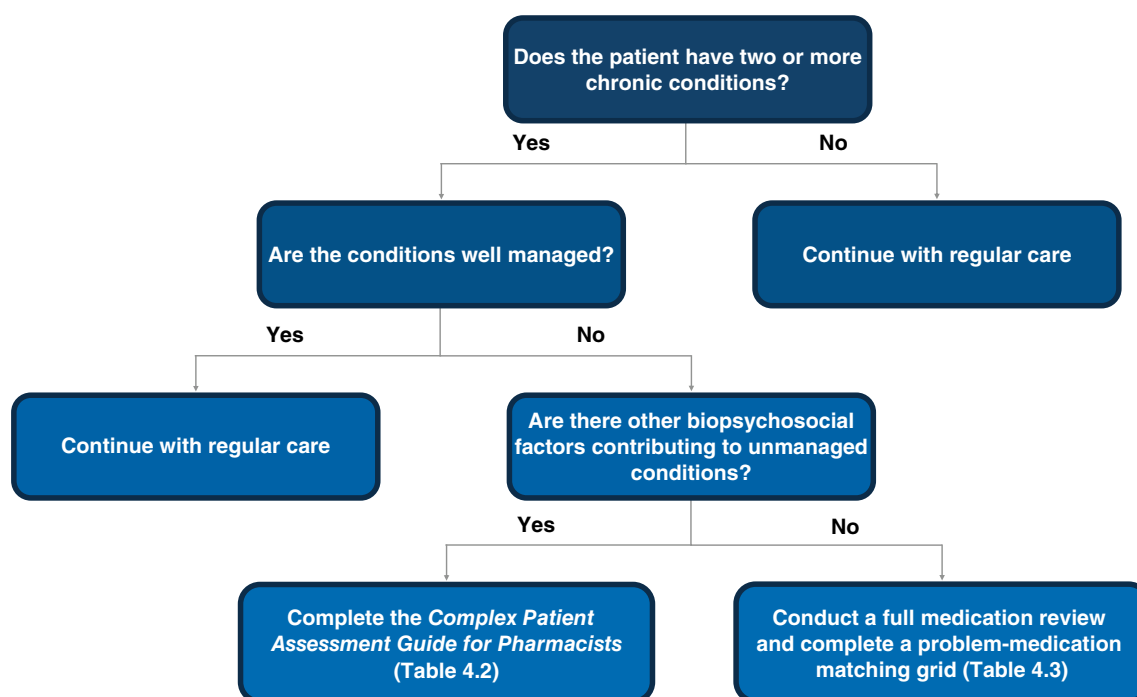


Fig. 4.2 Flowchart summarizing a pharmacist's approach to assessment of complex patients. Refer to Table 4.2 for *Complex Patient Assessment Guide for Pharmacists* and Table 4.3 for a sample template of the problem-medication matching grid

Box 4.1 Case 1 Example*The Case*

You are the pharmacist at an ambulatory medicine clinic. You have been asked by one of the physicians to conduct a medication review for Hassan who is here for an annual checkup. Continue reading to learn more about Hassan's case.

Chief Complaint

Hassan is a 72-year-old male who is here for an annual checkup. He appears to be a bit tired and says that he has been feeling fatigued for the past few weeks. He also says his knee pain has been restricting his mobility, and that saddens him as he can no longer walk as much as he would like to.

Allergies

None.

Past Medical History

Condition	Duration
Hypertension	12 years
Hypothyroidism	5 years
Type 2 diabetes mellitus	3 years
GERD	5 years
Osteoarthritis in both knees	5 years

GERD gastroesophageal reflux disease

Social History

Hassan is a retired accountant and has no medical insurance. He is married and lives with his wife in a house. They have a son and a daughter who visit them on a biweekly basis. There is no history of illicit drugs, alcohol, or tobacco usage. His caffeine intake consists of one cup of coffee per day. He is not very active as his mobility is limited due to his osteoarthritis, but he keeps a healthy diet to aid in management of his diabetes.

Medications

Medication	Dosage	Frequency
Amlodipine	5 mg	Daily
Hydrochlorothiazide	12.5 mg	Daily
Levothyroxine	75 mcg	Daily
Metformin	850 mg	tid (three times a day)
Pantoprazole	40 mg	Daily
Atorvastatin	40 mg	qhs (every night at bedtime)
Ibuprofen	300 mg	2 tabs po q6h

Hassan reports good adherence to his medication regimen, and his wife has confirmed this.

Vital Signs

Measurement	Value
Average home BP	148/95 mmHg
Clinic BP	162/110 mmHg
Heart rate (HR)	85 bpm
Respiratory rate (RR)	18 breaths/min
Temperature	37.3 °C
Weight	95 kg
Height	173 cm

Review of Systems

System	Findings
CNS	Unremarkable, except for feeling tired and fatigued often
EENT, skin, GI, GU, MSK	Unremarkable
CVS	No complaints of chest pain/angina
Resp	Unremarkable; clear chest; no problems with breathing

CNS central nervous system, *EENT* eye, ear, nose, and throat, *GI* gastrointestinal, *GU* genitourinary, *MSK* musculoskeletal, *CVS* cardiovascular, *Resp* respiratory

Labs

Lab test	Value	Reference range
HGB	111 ↓	120–160 g/L
RBC	4.3	4.3–6 × 10 ⁹ /L
HCT	0.37 ↓	0.41–0.52
MCV	70 ↓	80–100 fL
MCHC	319 ↓	320–360 g/L
RDW	14.1	<15.6%
WBC	5.6	4–11 × 10 ⁹ /L
PLT	320	140–450 × 10 ⁹ /L
SCr	66	45–125 μmol/L
Bili	17	<20 μmol/L
AST	25	<40 U/L
ALT	22	<50 U/L
ALP	38	30–130 U/L
Na	136	135–145 mmol/L
K	3.9	3.5–5.0 mmol/L
Glu (fasting)	9 ↑	3.3–6.0 mmol/L
Hb A1C	12.0↑	4–5.8%
TSH	2.56	0.2–4.00 mU/L
INR	1.1	0.9–1.2

HGB hemoglobin, *RBC* red blood cell, *HCT* hematocrit, *MCV* mean corpuscular volume, *MCHC* mean corpuscular hemoglobin concentration, *RDW* red cell distribution width, *WBC* white blood cell, *PLT* platelet, *SCr* serum creatinine, *AST* aspartate aminotransferase, *ALT* alanine transaminase, *ALP* alkaline phosphatase, *TSH* thyroid-stimulating hormone, *INR* international normalized ratio

Is Hassan a Complex Patient?

Box 4.1 contains several tables showing Hassan's medical and social history. After reviewing the patient data, a few preliminary questions can be asked before proceeding to more formal methods of identifying a patient as complex. Referring to Fig. 4.2 as a guide, the first question that indicates complexity is the presence of two or more chronic conditions. If the patient's medical history indicates multiple conditions, some of which are unmanaged, this may prompt the completion of the *Complex Patient Assessment Guide for Pharmacists* as seen in Table 4.2.

The questions outlined in Table 4.2 serve as a guide in deciding if a patient is complex. In Hassan's case, the questions could be answered as follows:

Category: Biological Chronicity

- *Does the patient have 2 or more chronic conditions? Are any of them unmanaged?*

Table 4.2 A series of categories and correlated questions derived from the IMSA questionnaire to assist pharmacists in determining if a patient is complex [10]

Assessment category	Question
Biological chronicity	<i>Does the patient have 2 or more chronic conditions? Are any of them unmanaged?</i>
	<i>Is the patient taking 5 or more medications?</i>
Symptom severity and therapeutic challenges	<i>Are the patient's daily activities restricted by their current health status?</i>
	<i>Has the patient been experiencing repeated hospital admissions?</i>
	<i>Has the patient been showing signs of increased healthcare utilization? (e.g. seeing primary care providers, pharmacists, or nurse practitioners more often than usual)</i>
	<i>Has the patient been experiencing psychological problems, such as anxiety or depression (e.g. feeling tense, anxious, down or sad)?</i>
Biopsychosocial factors	<i>Does the patient have an established support system, whether that be family, friends or health aids?</i>
	<i>Is the patient struggling with residential instability, or are they in need of home care, or home modifications?</i>
	<i>Is the patient struggling to adhere to their medication regimen?</i>
	<i>Is the patient struggling to access healthcare when they need it?</i>
	<i>Who are the healthcare providers currently involved in the patients' care?</i>
Healthcare team and collaborative care	<i>Are there multiple specialists involved?</i>
	<i>Is there consistent communication between members of the healthcare team? (Ask the patient if their healthcare team ever works together to manage their diagnoses)</i>

IMSA INTERMED Self-Assessment

Yes, Hassan has several conditions, some of which appear to be uncontrolled, including diabetes, hypertension, and osteoarthritis.

- *Is the patient taking 5 or more medications?*

Yes, Hassan is on six prescription medications.

Category: Symptom Severity and Therapeutic Challenges

- *Are the patient's daily activities restricted by their current health status?*

Hassan has noted that his mobility is limited by his worsening osteoarthritis.

- *Has the patient been experiencing repeated hospital admissions?*

No. Hassan has not been hospitalized recently.

- *Has the patient been showing signs of increased health-care utilization?*

Other than this annual checkup, Hassan has not seen a healthcare provider recently.

Category: Biopsychosocial Factors

- *Has the patient been experiencing psychological problems, such as anxiety or depression (examples: feeling tense, anxious, down or sad)?*

Hassan has indicated that his limited mobility is causing a depressive mood, which could get worse if left unaddressed.

- *Does the patient have an established support system, whether that be family, friends or health aides?*

Hassan is well-supported by his wife and children.

- *Is the patient struggling with residential instability, or are they in need of home care, or home modifications?*

No. Hassan lives at home with his wife, and she is able to help him with his daily routine.

- *Is the patient struggling to adhere to their medication regimen?*

No. Both Hassan and his wife confirm he is very adherent to his prescribed medications.

- *Is the patient struggling to access healthcare when they need it?*

Hassan has no health insurance and finds medication costs stressful. This could limit his access to healthcare.

After answering these questions, it is clear that Hassan demonstrates several signs of complexity. Once a patient is noted as being complex, appropriate considerations can be made toward their management and care. One major example of how care differs in complex patients is the consideration of drug–disease interactions. In summary, a drug–disease interaction is defined as situations where medications used to

treat one chronic disease may cause negative implications of the patient’s other comorbidities [36]. Complex patients are more likely to experience drug–disease interactions than noncomplex patients [36]. In Hassan’s case, for example, he is taking ibuprofen every few hours to manage his osteoarthritis. The medication is not working adequately, which is something that should be addressed, but, additionally, this medication is an NSAID, which can cause gastrointestinal irritation. Hassan is experiencing gastroesophageal reflux disease (GERD) and is taking pantoprazole to manage it. Reviewing Hassan’s medications indicates that the medication he is taking to manage his osteoarthritis (ibuprofen) may be worsening his GERD, an example of a drug–disease interaction.

When considering medication simplification or deprescribing for complex patients, resources such as the Beers criteria are helpful tools. The Beers criteria are a list of medications that may be potentially inappropriate in older adults and aim to help healthcare practitioners improve medication safety [37]. Hassan’s current medication regimen has two medications that are referenced in the Beers criteria: pantoprazole and ibuprofen. Hassan’s use of ibuprofen should be assessed, and alternative medication could be recommended, such as a topical diclofenac cream. If Hassan’s GERD symptoms are resolved after deprescribing ibuprofen, then pantoprazole may be deprescribed as well. These changes would reduce Hassan’s pill burden while hopefully leading to better management of his osteoarthritis and GERD.

Infodumps and Problem–Medication Matching Grids

Complex patient care can sometimes involve what is referred to in this textbook as an *infodump*, where a lot of information about the patient is provided to the healthcare practitioner at once. Large volumes of information can quickly become overwhelming and convolute the patient’s care. When faced with an infodump, deciding what information is relevant to the patient’s care is necessary for an efficient assessment. For example, if a patient needs treatment for a UTI, and has recently had their bloodwork done, but they also have osteoporosis, the information from their recent bone mineral density scan may be available alongside blood culture results in the electronic medical record. It is up to the clinician to determine that information is not necessary when providing drug therapy to treat the patient’s UTI. Additionally, when viewing large amounts of information, particularly labs and

Table 4.3 A sample template for a problem–medication matching grid

Problem list	Current medications	Drug-related problems
Problem 1	Medication A	Write notes, if applicable
Problem 2	Medication B	Write notes, if applicable
Problem 3	Medication C	Write notes, if applicable
	Medication D	Write notes, if applicable

vital signs, it is important to *determine what you need to look for* before delving into the available data. It is also important to consider trends over numbers at face value. For example, when treating the previous patient’s UTI, it may be relevant to monitor kidney function when deciding what antimicrobial agent to administer. Serum creatinine levels are one indicator of kidney function, and normal levels are typically between 65 and 119 $\mu\text{mol/L}$ in males. A high serum creatinine level of 125 mmol/L , for example, would be alarming in someone whose levels have been stable at 65 mmol/L for an extended period of time; however, these levels may be considered a positive sign in someone who historically has had elevated levels of 135 mmol/L . A helpful mnemonic for remembering this rule is “*the trend is your friend*,” and it can be applied when looking at a variety of healthcare markers including but not limited to blood pressure readings, blood sugar, and electrolyte and ferritin levels.

One method of managing instances of an *infodump* is to use helpful organizational tools. One such tool is the *problem–medication matching grid*, which can be used to sort a patient’s current issues and medications. This can help ensure that medication complications and interactions are not missed as well as flagging the presence of drug–disease interactions. Table 4.3 depicts an example of a template for a medication grid, which can be extended based on the patient’s conditions and medication regimen.

The medication grid is one helpful way to make connections between drug, disease, and other safety concerns in the complex patient’s healthcare regimen. Such tools work for all patients but are especially helpful when patients subscribe to intense polypharmacy and could be taking ≥ 10 medications [2]. After organizing the patient’s medications and conditions, and completing an assessment of IESA (indication, efficacy, safety, and adherence) of the medications, a color-coding system can be implemented to help summarize necessary changes. *Red* indicates a medication that should be discontinued, *yellow* indicates a medication that can be continued with changes, and *green* indicates a medication that can be continued as is. This will be illustrated with another patient case described in Box 4.2.

Box 4.2 Case 2 Example*The Case*

You are the pharmacist providing consultative coverage in the internal medicine ward. You have been told that Victoria and her husband requested to speak with a pharmacist to review Victoria's current medications.

Chief Complaint

Victoria is a 62-year-old female patient who was admitted to the hospital 6 months ago following a fall from a ladder on a concrete surface. Prior to her admission to the hospital, Victoria had no previous medical history and was not taking any medications. The fall resulted in a T3–4 fracture with complete spinal cord injury. Victoria became paraplegic and experienced complications in the hospital, which have been provided to you on this list:

Victoria's Hospital Stay Complications

- Upper GI bleed
- Infected sacral ulcer with bone involvement
- Severe pain associated with the sacral ulcer
- Severe neuropathic pain
- Chronic constipation
- Skin rash to imipenem
- Spastic muscles
- Oral thrush since starting the antibiotics

After checking Victoria's medical administration record in the hospital software, you compile a list of medications she has taken while in the hospital.

Victoria's Medication List

- Lansoprazole 30 mg by mouth (po) twice a day (bid)
- Magnesium hydroxide as needed (prn)
- Imipenem 500 mg intravenous (IV) Q6h (finish in 7 days)
- Metronidazole 500 mg IV Q12h
- Acetaminophen 975 mg Q6h ATC
- Hydromorphone 0.5–1 mg subcutaneously (SC) q4h prn
- Gabapentin 100 mg po three times a day (tid)
- Pregabalin 50 mg po tid
- Bisacodyl 1 suppository daily
- Docusate/lactulose/PEG 100 mg prn
- Diphenhydramine 50 mg po pre-imipenem
- Baclofen 5 mg po bid and 10 mg at bedtime (qhs)
- Nystatin 4 ml swish and swallow (SWSP) four times a day (qid)

- Lorazepam 1 mg sublingual (SL) Q6h prn
- Dalteparin 5000 IU SC daily
- Quetiapine 50 mg po qhs
- Zopiclone 7.5 mg qhs prn

After looking at your lists, you conclude that Victoria is *complex* because she has multiple ongoing conditions at once and is subscribed to extreme *polypharmacy* as her current list of medications exceeds ten agents. This would appear to be a large amount of information, an example of an *infodump*.

In order to help assess Victoria's medication regimen and determine what changes need to be made, you decide to organize Victoria's conditions and match them with the medications using a *problem–medication matching grid*, which is shown in Table 4.4.

After organizing the table, you have some questions, which you place in the drug-related problems column. For example, why is Victoria taking two antibiotics to treat the infected sacral ulcer? After inquiring with the infectious disease team, they confirm that metronidazole is no longer needed and should be discontinued. You decide to indicate this medication will be discontinued by highlighting it in red in the table. Additionally, you noticed that Victoria is taking both pregabalin and gabapentin to treat her neuropathic pain. As these medications are from the same class, it is not necessary or potentially safe to administer both concurrently. You decide to discontinue pregabalin, highlighting it in red. There is another issue with her neuropathic pain medications: gabapentin is administered at a subtherapeutic dose, so you highlight this medication in yellow and make a note to the physician that the dosing should be increased to 300 mg by mouth (po) thrice a day (tid). There are more changes that need to be made to Victoria's regimen, but these examples demonstrate the principles that need to be applied when considering complex patients.

Conclusions

Identification of complex patients in practice is a skill that all healthcare providers must develop. Complex patient care is multifaceted, and the patient is best served by a well-informed interdisciplinary healthcare team. Pharmacists are an integral part of the healthcare team and are involved in patient care at all stages, including acute and chronic disease management. As pharmacists are accessible and knowledgeable healthcare providers, they must be able to identify com-

Table 4.4 Example of using a medication matching grid to summarize a large quantity of information given as an *infodump*

Problem List	Medications	Drug Related Problems
1. Upper GI bleed	Lansoprazole 30 mg PO BID	-
	Magnesium hydroxide PRN	
2. Infected sacral ulcer with bone involvement	Imipenem 500 mg IV Q6h (finish in 7d)	Why are there two antibiotics being used to treat this condition?
	Metronidazole	
3. Severe pain associated with the ulcer	Acetaminophen 975 mg Q6h ATC	-
	Hydromorphone PRN	
4. Severe neuropathic pain	Gabapentin 100 mg PO TID	Two medications from the same class may not be needed. The gabapentin dose also seems too low for therapeutic efficacy.
	Pregabalin 50 mg po TID	
5. Constipation	Bisacodyl 1 suppository daily	-
	Docusate/lactulose/PEG PRN	
6. Skin rash to imipenem	Diphenhydramine 50 mg pre-imipenem	-
7. BPH in/out catheter problems	Tamsulosin CR 0.4 mg PO daily	-
8. Spastic muscles	Baclofen 5 mg PO BID and 10 mg QHS	-
9. Oral thrush since started antibiotics	Nystatin 4 ml SWSP QID	-
10. Anxiety	Lorazepam 1 mg SL Q6h PRN	Lorazepam is not a good choice for long term anxiety management, but this medication could continue to work until the patient is discharged.
11. DVT prophylaxis	Dalteparin 5000 IU SC daily	The healthcare team should reassess the need for DVT prophylaxis, as the patient has already been taking Dalteparin for 6 months.
12. Sleep	Quetiapine 50 mg PO QHS	Daily use of antipsychotics for sleep aids is not an effective management strategy for insomnia and should be stopped. Zopiclone can continue as needed.
	Zopiclone 7.5 mg QHS PRN	

GI gastrointestinal, *BPH* benign prostatic hyperplasia, *DVT* deep vein thrombosis, *po* by mouth, *bid* twice a day, *prn* as needed, *IV* intravenous, *tid* three times a day, *qhs* every night at bedtime, *SWSP* swish and swallow, *SL* sublingual, *SC* subcutaneously

plex patients in practice, by increasing their familiarity with social determinants of health and the impact of biopsychosocial factors on a patient's health. Understanding a complex patient case requires making connections between drug–disease interactions, monitoring for drug interactions, and deprescribing when appropriate, among other responsibilities. In the future, development of a standardized complex patient assessment tool that could be easily administered across all pharmacy practice settings would benefit the profession.

Clinical Pearls

- Identifying a complex patient is critical for proper management of their care.
 - The first indicator of complexity is the presence of two or more chronic conditions.
 - Pharmacists should be aware of the various domains impacting patient complexity, including biological, social, psychological, and healthcare factors.
 - The role of the pharmacist in complex patient care is multifaceted and includes roles such as detecting drug interactions, monitoring of laboratory tests, extending prescriptions, and promoting patient education.
- An *infodump* is an instance in which a lot of information is provided at once. There are different ways to manage an infodump to manage care efficiently.
 - It is important that the pharmacist be selective when treating the issue at hand and reviewing relevant history, labs, and medications.
 - Tools such as the *problem–medication matching grid* can be utilized to help sort information from an infodump.

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Clinical Decision-Making for Pharmacists

5

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Chapter Objectives

1. Describe clinical decision-making and critical thinking in pharmacy practice.
2. Describe cognitive biases and barriers to clinical decision-making.
3. Describe tools toward clinical decision-making in pharmacy practice.

Background

Clinical decision-making and critical thinking are required essential skills for pharmacists regardless of the setting they practice in. Pharmacists have a large role to play in managing patients and have sufficient knowledge and skills to help make decisions in the best interests of their patients. Pharmacists must be comfortable and have enough confidence in order to care for the patients because the patients should feel comfortable knowing that they are in good hands and that their needs will be taken care of appropriately. Once obtained their pharmacy degree, some pharmacists who start working may feel overwhelmed and unconfident in their abilities to make a decision. There may be some hesitancy at first to be proactive and communicate effectively with other healthcare professionals. On the other hand, some pharmacists may graduate feeling very confident and think their decision is the only correct one. They may not consider other alternatives and may overlook important details. In both cases, harm to the patient can occur. This chapter will focus on definitions of clinical decision-making and critical thinking, clinical decision-making in pharmacy and medicine, and the various components that should be considered when making a decision. At the end of this chapter, the pharmacist

should gain an appreciation of the complexity and intricacies of clinical decision-making and critical thinking.

What Is Clinical Decision-Making?

To put it simply, clinical decision-making is a process that involves *gathering and interpreting* information, *evaluating* options, and *deciding* how the problem will be resolved based on the information that is given. However, it is actually a complex process that requires the clinician to consider all the components and evidence in order to formulate an appropriate action. There are several other definitions of clinical decision-making, which may include choosing among alternatives, problem-solving, using logic or reason, or intuition. A framework for clinical decision-making has been proposed by Tiffen et al., which includes four sequential steps in the process: data gathering (e.g., medical history and physical exam), data interpretation, data evaluation, and making the final decision. The purpose of this model is to show that clinical decision-making is not a linear process, but rather, a fluid and evolving one. It takes into account the experience and knowledge of the decision-maker as well as the characteristics of the situation since these factors can affect how one makes a clinical decision [1]. Although this model was designed for nurse practitioners and is specific to diagnosing patients, the concepts depicted in the framework can still be applicable to most healthcare professionals. Here is an example of how pharmacists could utilize this framework in practice:

1. In a community pharmacy setting, a patient may come in asking for a recommendation for an over-the-counter product for acid reflux. The pharmacist *gathers information* about the acid reflux and the patient's medical history, which may include how often it occurs, location of pain, when it started, other symptoms such as nausea or vomiting, current medications, medical conditions, allergies, etc.

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2. After that, the pharmacist *interprets this information* and determines how to proceed from here. If any red flags are present, the pharmacist may refer the patient to another practitioner.
3. If there are not any red flags and the pharmacist determines that the acid reflux can be managed, alternatives such as non-pharmacological treatment, antacids, H₂-receptor antagonists, or proton pump inhibitors can be considered. A discussion is facilitated with the patient about their goals of therapy and their preferences. The pharmacist will then *evaluate the available options* and consider the patient's characteristics.
4. Finally, the pharmacist *makes a decision* to recommend a treatment and establishes a monitoring plan that is specific to the patient. Follow-up and documentation may also be done.

While following these steps can serve as a guide in making clinical decisions, the thought processes that go into making a decision are essential. To execute these steps, one must be able to have effective critical thinking skills to make appropriate decisions and try to avoid cognitive bias.

Critical Thinking

In order to make a decision, critical thinking is needed. Critical thinking has several definitions, and there are many ways to frame this definition. One could describe critical thinking as an ability or process that pulls together, assesses, synthesizes, analyzes, and uses information to establish a therapeutic plan. Another way of defining it includes the characteristics of the person rather than their abilities, which include paying careful attention, keeping an open mind, and reflecting on one's own actions. Additionally, there are definitions that combine the abilities and characteristics of the individual that include analyzing and assessing the information using an open mind [2]. Critical thinking is a combination of attitude, knowledge, and thinking skills that lay the foundation for problem solving, clinical reasoning, and clinical decision-making. According to Persky et al., there are six core skills that collectively constitute critical thinking: interpretation, analysis, evaluation, inference, explanation, and self-regulation. These skills are defined in Table 5.1 [3]. Critical thinking is a skill that is incorporated into the pharmacy school curricula, and it takes time to develop, as it is a cognitive process and involves several steps. In pharmacy school, students are taught how to approach a problem, the steps in evaluating the problem, and the possible solutions to resolve the problem. As they cultivate their skills over time with increased practice, the process will eventually become second nature. A prospective cohort study done by Cone et al., in which a Health Sciences Reasoning Test (HSRT)

Table 5.1 Core skills in critical thinking [3]

Attitude	Willingness to learn and plan, demonstrate flexibility and persistence, confidence, be curious, be open-minded, and attentive
Knowledge	For pharmacists, the requisite knowledge from school and the knowledge gained from experience
Thinking skills (knowledge of the critical thinking process)	
Interpretation	Showing one's understanding of the information and being able to explain it to others
Analysis	Examining the different elements and putting them together to find the meaning
Evaluation	Judging if the information given is reliable and unbiased
Inference	Developing a conclusion by using and recognizing the elements
Explanation	Making one's statement of the information clear enough for others to understand
Self-regulation	Monitoring and managing one's own thoughts, emotions, and actions

was administered to pharmacy students at the University of New Mexico to see if the students' critical thinking skills improved in the second year after exposure to a curriculum that focused on critical thinking. The HSRT has been validated as a tool to assess critical thinking. It was found that there was a significant improvement in critical thinking skills after exposure to this type of curriculum, showing that critical thinking skills are learned [4].

Shared Decision-Making

An enormous part of clinical decision-making includes involving the patient as much as possible. In shared decision-making, healthcare professionals inform the patient about the risks and benefits of the treatment and work with the patient to create a plan to manage their conditions. Evidence may be incorporated along with the patient's beliefs and preferences in the decision-making process. This leads to *focusing on outcomes that are important to the patient*. The main point to note about shared decision-making is that patients are not the sole-decision-makers here, and it is not about the clinician doing all the decision-making [5]. There needs to be effective communication between the patient and healthcare professional, an understanding of the patient's beliefs, values, and concerns, and enough information should be provided so that the patient can feel like they are in control of their health. When facilitating shared decision-making, it is important to keep in mind the patient's characteristics that may impact the interaction with the clinician, such as disabilities, language barriers, and sight or hearing problems. Patients should be encouraged to ask questions about their medical conditions and medications and should be asked open-ended questions to get detailed

responses. When involving the patient in decision-making, note that their perspective may be different, and the patient can decide if they want the treatment or not, even if you disagree with them [6].

Cognitive Biases

Cognitive biases can impact clinical decision-making and can lead to medical errors and poorer outcomes for patients. Cognitive bias, also known as heuristics, can be defined as “*cognitive short-cuts*” that clinicians make in order to make a final decision. To make clinical decision-making easier and more efficient, a “rule of thumb” or “cognitive guideline” can be used for clinical scenarios subconsciously. However, this can lead to diagnostic errors and poorer management of patient care [7]. There are several types of cognitive biases that exist in medicine and the most commonly known biases will be talked about in this chapter, and each of these is defined in Table 5.2 [8]. A systematic review found that anchoring and framing effects and information biases were

the most commonly reported cognitive biases in physicians when making clinical decisions. For diagnostic errors, overconfidence, the anchoring effect, and information or availability bias were the most implicated. Anchoring, information bias, overconfidence, premature closure, representativeness, and confirmation bias were associated with therapeutic or management error [9]. Many of these cognitive biases can be applied to pharmacists as well. For example, in confirmation bias, a patient comes in with a cough and may think they have a cold and the pharmacist will look for evidence to support that they have a cough but an asthma flare-up may be more likely. Another example with search satisfying bias would be a pharmacist answering a question from the patient that involves a literature search. The pharmacist may end up using the first article they found and stop searching for alternative answers to the question that could be different. Having an awareness of these cognitive biases can help ensure that pharmacists are making appropriate decisions with their patients and that there will be optimal management of the patient’s conditions.

Table 5.2 Cognitive biases commonly seen in practice [8]

Bias	Definition	Example
Aggregate bias	Aggregated data that is used to make clinical practice guidelines, but it does not apply to certain patients. There is an assumption that the patient is atypical and does not fit the criteria	Certain clinical practice guidelines may advise not to order an x-ray for patients, but the physician thinks that the patient does need it because they believe that the patient is the exception
Anchoring bias	Placing emphasis on salient features too early in the initial consultation, and even with new information coming to light, not changing the original impression. This type of bias is similar to confirmation bias	An elderly female patient comes in with chest pain, and due to the patient’s age, emphasis may be placed on angina even if the patient mentions symptoms aligning more with heartburn
Availability bias	This type of bias can be influenced by recent experience of the disease or by lack of experience with the disease	In the emergency department, three patients were diagnosed with acute kidney injury; therefore, this diagnosis comes into mind when the next patient comes in with similar symptoms Conversely, if a certain disease has not been seen for a long time, it can lead to underdiagnosis of the disease
Bandwagon effect	The tendency of an individual adopting a certain attitude or behavior because everyone else is doing it, even if there is no rationale in doing so	Because all the pharmacists from the same pharmacy tend to recommend a product for a certain minor ailment, a new pharmacist may do the same In hospitals, some physicians choose one DVT prophylaxis over another such as UFH over LMWH, because UFH is assumed to have a lower bleeding risk even though there is no sufficient evidence of this A community pharmacy may receive many prescriptions for a cream that has PLO gel as the base, but the pharmacy does not carry that base, and instead, all the pharmacists adapt it to Versapro base
Commission bias	This may happen in over-confident clinicians where there is an obligation to keep with the ethical principle of beneficence, so there is a tendency toward action instead of inaction. This is less common than omission bias	Prescribing a medication that aids with sleep, but the patient may not need it
Confirmation bias	Looking for evidence that supports the diagnosis instead of looking for evidence for another diagnosis. This type of bias is similar to anchoring bias	A patient comes in with a cough and may think they have a cold, and the pharmacist will look for evidence to support that they have a cough, but an asthma flare-up may be more likely

(continued)

Table 5.2 (continued)

Bias	Definition	Example
Dunning-Kruger effect [10]	The tendency for individuals with a low level of knowledge or experience to overestimate their abilities. Figure 5.1 provides an illustration of this type of bias	A new graduate pharmacist starts working at a pharmacy, thinking they know everything
Framing effect	Decision-making is influenced by how the problem is being framed.	A treatment may be framed as 70% effective or 30% ineffective and this could affect the prescriber's decision to prescribe the treatment
Gambler's fallacy	A belief that an event that occurred frequently will not occur as frequently in the future.	A series of patients in the emergency department were diagnosed with acute coronary syndrome and the physician has the belief that the sequence will not continue after this
Hindsight bias	Upon learning of the outcome, the tendency to think of past events as predictable and not having a realistic reflection of what actually happened. This leads to underestimation or overestimation of the abilities of the decision-maker. This type of bias may negatively impact learning from one's own mistakes	"I knew it the whole time" or "I never saw it coming"
Information bias	A belief that the more evidence that is collected, the better it will be to make a decision. This can lead to finding irrelevant information that may not have any value	When doing a literature search for a clinical question, one may come across several articles that may not have valuable information but may still use that information for their answer
Omission bias	The opposite of commission bias, where there is a tendency for inaction instead of action to prevent harm (ethical principle of non-maleficence); however, harm can still occur	A patient comes into the pharmacy with a prescription for an antibiotic (e.g., cephalexin to be taken once daily), but the dose or dosing interval should be adjusted. The pharmacist may have a fear of adapting the dose because they may be worried about causing harm to the patient. However, not doing anything in this case can lead to worsening infection or unwanted adverse effects
Overconfidence bias	Believing that we know more than we actually do. There is too much confidence in our abilities, leading to a disregard for important information and more reliance on intuition and hunches rather than evidence	An experienced clinician may feel very confident in diagnosing certain conditions and, as a result, often does not do a thorough medical history with the patient A pharmacist doing a medication review may feel very confident knowing about their patient's medications, that they do not ask questions about their over-the-counter medications
Representativeness restraint (pattern recognition)	The tendency to look for patterns or a prototype of the disease. This can result in missing patients who show atypical symptoms	Atypical angina symptoms can be present in the elderly, females and diabetics, but they are missed
Sutton's slip	Going for the obvious	One diagnosis may be obvious, but other possible diagnoses are not considered
Search satisfying	Calling off the search once information is found	A pharmacist answers a question from the patient that involves a literature search. The pharmacist may end up using the first article they found and stop searching for alternative answers to the question that could be different A physician arrives at a diagnosis but does not search for other possible diagnoses

DVT deep vein thrombosis; *LMWH* low-molecular-weight heparin; *UFH* unfractionated heparin

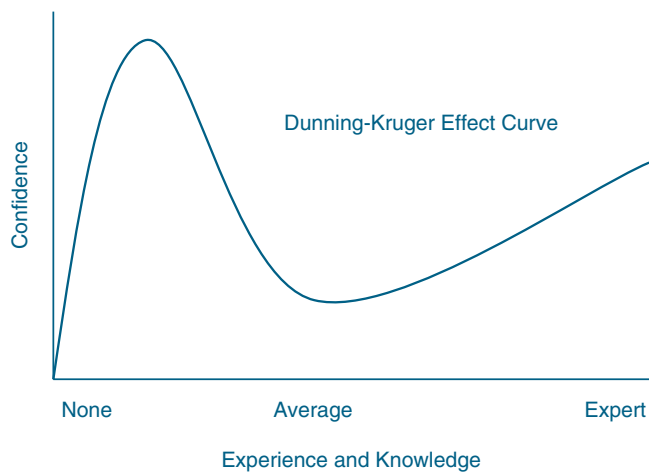


Fig. 5.1 A healthcare professional with none to minimal experience may have overconfidence when they start practicing. Over time as they gain knowledge and experience, they start to have self-doubt and confidence lowers. This is where they might make changes to their processes in decision-making. As they learn more, their confidence comes back, but they are not overconfident as they once were

Clinical Decision-Making in Pharmacy

In pharmacy practice, pharmacists are always making decisions, whether it is checking a prescription or adjusting a medication based on a drug-therapy problem, and each of these decisions has a process to it. The ways in which pharmacists make decisions are highly dependent on multiple factors, which will be explored in this section.

Role of Pharmacists in Clinical Decision-Making

In Canada, pharmacists have an extended scope of practice where they are more involved in the clinical decision-making process. Despite the differences in scope of practice between provinces and territories, pharmacists can still make decisions to maximize benefit and minimize harm for their patients. The new scope of practice makes it easier for collaboration between other healthcare professionals and puts more responsibility and accountability on the profession. In most of the provinces, pharmacists can extend or adapt prescriptions, prescribe in an emergency, or initiate a prescription (e.g., the province of Alberta) [11]. Historically, pharmacy practice was more technical and objective where dispensing and distribution practices were more important. Ensuring the right medication is given to the right patient is the main role of the pharmacist; however, there is now more emphasis on pharmacists to take on clinical responsibilities. Pharmacy technicians are able to final check prescriptions (provided that clinical verification was already done by the

pharmacist), which gives the pharmacist more time to carry out other clinical activities such as immunizations, medication reviews, care plans, and prescribing activities [12]. For immunizations, it is up to the pharmacist's discretion if the patient would be an appropriate candidate, and the pharmacist can choose not to give a vaccine if they believe it is not appropriate. For medication reviews and care plans, the pharmacist may notice some drug-therapy problems and decide how they want to resolve them. For prescribing, the pharmacist determines that the patient cannot see another provider in a timely manner; therefore, they may prescribe or do an extension/adaptation of a medication. Other decisions pharmacists can make in practice are choosing to recommend a treatment or a dose adjustment of a medication to a patient's family physician, collaborating with other health care providers, recommending an over-the-counter medication for a minor ailment, refer the patient, or do nothing. Before making a decision, pharmacists should be doing an assessment of the issue first to determine if any intervention is needed, even if they decide to do nothing. Pharmacists are the most accessible health care professionals and with an expanded scope of practice, they are equipped with the knowledge and skills to make appropriate clinical decisions.

Uncertainty of Clinical Decision-Making in Pharmacy

Despite the advanced role of pharmacists, there is still reluctance in making clinical decisions. *Professional abstinence* has been defined as "consciously choosing not to provide the full scope of patient care activities" by Biggs et al. [13]. Professional abstinence has a huge impact on pharmacy practice, especially pharmacy students and new graduate pharmacists. When preceptors model this type of behavior to students completing their practicum placements, it can be discouraging to students who were otherwise taught the concepts of patient-centered care [14]. When there are situations that are in the grey area, there is a lack of confidence in making these types of decisions. Pharmacists come across situations with uncertainty every day and should have enough confidence to face them and resolve them. There may be times when there are not many options to choose from; therefore, decisions may be more difficult to make. A few reasons for this reluctance could be the curriculum that is learned in school, changing scope of practice, fear of compromising relationships with other health care professionals and patients, fear of taking responsibility for the outcome, lack of knowledge, or lack of experience. A qualitative study by Noble et al. found that new graduate pharmacists from Australia had a difficult transition period from being a student to a pharmacist due to a lack of workplace experience

and budding professional identities in university. The development of the interns' professional identities depended greatly on the workplace environment and found that there was a bigger emphasis on dispensing and distribution rather than clinical services. This study shows that what students learn in school is not always going to translate into real-life practice [15]. Another qualitative study done in Ontario by Gregory et al. found that the most common behaviors that pharmacists were doing to avoid making independent decisions were educating the patient or building a relationship with the patient, getting advice (e.g., from a regulatory body), or deferring responsibility to another health care provider (e.g., physician) and strict rule-following. There was a fear of the pharmacists potentially losing their license, a fear of losing business opportunities for owners, and a personal need to be liked [12]. It should be noted that pharmacists do have a positive impact on patient care when they are involved in decision-making. There are numerous studies that show an increase in achieving patient outcomes when pharmacists are involved. An example of this is a prospective study done that was designed to see the impact of pharmacist-led shared decision-making in managing atrial fibrillation. There was an overall greater satisfaction with anticoagulant therapy from patients and increased adherence to therapy. This shows how much pharmacists are valued in managing patient care, and that they should be practicing to the full scope because it leads to better outcomes for patients [16].

Factors and Barriers That Affect Clinical Decision-Making in Pharmacy

There are a multitude of factors and barriers that contribute to pharmacists making clinical decisions. Some of these can be contextual, social, intrinsic to the pharmacist, or be dependent on the patient [17, 18].

Contextual Factors Where a pharmacist practices can determine the kinds of clinical decisions they will make, and this is dependent on the scope of practice in that area. For example, some provinces allow pharmacists to prescribe in an emergency, but others do not. Some provinces allow pharmacists to make therapeutic substitutions but others do not [11]. Depending on where the pharmacist practices, they may be limited or have more options in treating patients, which can affect how they make clinical decisions. Pharmacists may also be specialized in managing certain diseases. They could be working in a specialty clinic (e.g., anticoagulation clinic) or be working in a hospital unit that provides specialized care for certain patients (e.g., pediatrics). Pharmacists in these areas are likely to be more comfortable in making clinical decisions because they have extensive knowledge and have more experience with these

kinds of diseases. Workflow in the pharmacy, especially in the community setting, can be another barrier in clinical decision-making. A busy pharmacy can lead to stress and fatigue, which can hinder the pharmacist from taking the time to carry out clinical activities. Important information could be missed in the interactions, which can negatively impact patient care.

Social Factors Relationships with other prescribers may affect pharmacists' decision-making. There may be some physicians in the area who are uncomfortable with pharmacists doing prescribing activities for their patients, which can contribute to the fear of damaging relationships with the physicians. This can lead to pharmacists not making decisions for their patients. On the other hand, there are physicians who are comfortable with pharmacists prescribing and may even ask the pharmacist for a recommendation for drug therapy. This may be due in part of having a good working relationship with the prescriber. In practice, it is usually required to have good collaborative relationships with a patient's prescriber before taking part in prescribing activities. Being on a multidisciplinary team can often affect how a pharmacist may approach in making a decision. In this setting, pharmacists may have to think about other healthcare professionals' treatment of the patient and take that into consideration. It is also important that pharmacists are aware of what decisions are being made by other healthcare professionals as that can impact decision-making. For example, a patient has a non-functioning gastrointestinal tract (GI) after a bowel resection and is unable to take anything by mouth. In consultation with a dietician, the physician decided that the patient was a candidate for total parenteral nutrition (TPN). This can affect how the patient's medications will be taken. Since the patient does not have a functioning GI tract, they are unable to absorb many medications. The pharmacist will have to figure out dosing adjustments since the bioavailability of some medications will be different when given intravenously.

Intrinsic Factors The comfort level of a pharmacist also has an impact on clinical decision-making. There may be diseases that are not commonly seen and diseases that were not taught in school. If there is low to no experience in managing these diseases, there is a higher level of discomfort associated with making decisions. A pharmacist may mitigate this through self-directed learning by educating themselves on the disease and trying to get familiar with the management of the disease. To maintain licensure as a pharmacist in Canada, pharmacists must complete a certain number of continuing education credits to ensure they are competent and have the adequate knowledge base to provide patient care. Level and years of experience are another factors that can greatly impact how pharmacists make clinical decisions, and this

can be related to how comfortable they are. A new graduate pharmacist may feel hesitant and unconfident in making a decision or feel that they are making the wrong decision. A pharmacist with over 20 years of experience can feel confident in their abilities and feel more comfortable in making a decision. However, it should be stated that no matter how experienced a pharmacist is, they should be confident about their knowledge base and skills before making any decision. Perception of professional responsibilities has been found to be a driving factor toward not making clinical decisions as well. In the United Kingdom, pharmacists and nurses can participate in a prescribing program to gain independent prescribing authority; however, there was still some hesitancy to prescribe, despite having the knowledge of certain disease states such as urinary tract infections (UTIs). There was an avoidance of making the final decision because the pharmacists and nurses believed that doctors should be the ones initiating medications [17].

Patient Factors A patient's socioeconomic status, race, gender, preferences, values, and other characteristics (e.g., comorbidities, pregnancy status, allergies) can all affect the treatment the patient will receive [18]. These patient factors can influence decision-making and some could potentially limit the number of options available. For example, a patient with low socioeconomic status may not be able to afford an expensive medication that could be very beneficial. Another example could be a patient who prefers to take medications without gluten, which can sometimes limit treatment options.

Clinical Decision-Making in Medicine

Physicians have a responsibility to ensure that their patients are receiving the appropriate treatment for the appropriate condition. When it comes to clinical decision-making, physicians are more worried about making the right diagnosis. Without the right diagnosis, the wrong treatment will be prescribed, and the patient will have poor outcomes.

Making a Diagnosis and Treatment Plan

It is important to remember that diagnosis is a process and involves the use of knowledge and critical reasoning skills. Some diagnoses can be made quickly, but others can take time. In instances, where symptoms are non-specific or there could be multiple diagnoses, a physician can communicate with the patient about what they know, order and review tests, consult with other colleagues, observe, and follow up with the patient over time to check for disease changes. Conducting a thorough assessment and having effective communication with the patient can help establish a good

relationship and set realistic expectations for management [19]. There are several approaches a physician may use in diagnosing, which are discussed below.

Since physicians focus more on the diagnosis of a medical condition, the process of diagnosing a patient can follow these four steps: *information gathering*, *hypothesis generation*, *hypothesis testing*, and *reflection*. Hypothetico-deductive is the term that has been used to define this process. *Data gathering* is done through asking open-ended questions to gather the appropriate data to make conclusions as to what the problem might be. The patient's medical history, medications, allergies, and results of the physical exam are gathered. During a physical exam, the physician may do a review of the systems, which involves a head-to-toe assessment, depending on the symptoms. Examining the patient carefully is paramount for physicians to prevent inaccuracies in diagnosis. In *hypothesis generation*, the physician may come up with several differential diagnoses. At this point of the process, the physician may be more susceptible to cognitive biases and may end up using shortcuts to arrive at a diagnosis (see section "Cognitive Biases"). One such cognitive bias that is commonly used by physicians is pattern recognition (see Table 5.2). A couple of problems exist with coming up with differential diagnoses. One is that there would not be enough time to explore each possible diagnosis. The other problem is that the diagnoses that seem the most likely may be given more emphasis and focus in the beginning which could lead to a very early diagnosis. One strategy can involve doing something about the diagnoses that can be treated today to at least come up with a treatment plan, and in the meantime, other diagnoses can be explored and ruled out [20]. Some other strategies used for hypothesis generation include ruling out the worst-case scenario and the Casablanca strategy. The Casablanca strategy involves "rounding up the usual suspects" by ordering lab tests or other tests to see how the presented problem may evolve, decline, or declare themselves. This strategy may be used when there is uncertainty about the diagnosis. In *hypothesis testing*, bloodwork, confirmatory testing, or diagnostic imaging may be done. Non-specific blood work may be done at first to see a complete blood count and differential, and then additional blood work may be done later depending on the results, the patient's history, and physical exam [21]. Lastly, *reflection* is examining the results, which can lead to a final diagnosis [22, 23].

Another approach that can be used in medicine for making clinical decisions is looking at probabilities. This is to help select an appropriate test for the patient and help determining how useful the results will be [20]. Pretest probability is the probability that the patient has the disease before knowing the results, and this is based on their current symptoms. When the pretest probability is low, the chance of having the disease is low, but there is a higher chance of false positivity. Therefore, a physician may think about pretest

probabilities in order to eliminate unnecessary lab tests that could potentially waste time waiting for the results. For example, a patient can present with symptoms of strep throat, and the physician can use the symptoms to calculate a Centor Score. The Centor Score is a quick and reliable tool used to predict if a patient could have streptococcal pharyngitis from culture results. The higher the score, the higher the likelihood of having a strep infection, and one may consider sending the patient for rapid strep testing and/or culture. A higher score can correlate with a higher pretest probability, so the physician can consider additional lab work. However, if the score is very low, it is unlikely to be strep throat, and that diagnosis could be eliminated [24].

How severe the illness is can be a driving factor in making decisions for the patients. After gathering the required information, an assessment is done to determine how severe the illness is. The physician may have found that the patient requires intervention that cannot be obtained from the community and thinks the patient requires close monitoring to prevent significant morbidity and death. Therefore, immediate medical intervention and hospitalization are required for the patient. If there is moderate illness, the physician may decide to prescribe a treatment that can be accessed in the community. For patients who have mild illness, treatment may not be required and instead, the patients may be advised to self-monitor. Something else physicians may consider in their decisions is how much the illness is impacting the patient's quality of life. If there is severe impairment in function, treatment may be warranted. For example, a patient is diagnosed with depression but has only mild impairment in daily function; therefore, pharmacotherapy may not be needed at this time, but the patient can be referred to a psychologist for psychotherapy, as it is first line in the treatment of depression. Some physicians may also establish thresholds for treatments, meaning that once the threshold is passed, treatment can be started. If a treatment has shown good benefit for a disease and has a low risk for adverse effects, the threshold is low. On the other hand, if the treatment has been shown to have very minimal benefit and has a higher risk for adverse effects, the threshold for starting treatment is higher. For example, second-generation antihistamines are effective for most symptoms of allergic rhinitis and are well tolerated. Another example would be the use of oral antifungals for onychomycosis, but they have minimal benefit and higher risk for adverse effects (such as liver damage) when used long term [20].

Evidence-based medicine combines the best available evidence with clinical expertise and patient factors to make clinical decisions. It has been found that physicians use references (e.g., MEDLINE) that give them fast answers instead of accessing databases that can give well-researched evidence. DynaMed® uses synthesized evidence to make appropriate recommendations for certain medical conditions

and could be a way to avoid getting answers from unreliable sources. In a randomized trial by Alper et al., primary care clinicians were able to modify their answers and their clinical decisions after using DynaMed® without increasing search time [25]. However, it is important to note that clinical practice guidelines are there to *guide* clinicians, and they must use their clinical judgment to ensure that the treatment is appropriate for their patients since clinical practice guidelines typically do not account for inter-patient variability. Guidelines provide broad outlines on how decisions should be made, but it is up to the physician to think to what extent the recommendations apply to their patients.

Factors That Affect Clinical Decision-Making in Medicine

Making clinical decisions is more than just using knowledge and skills. There are many clinical and non-clinical factors and things to consider when making clinical decisions. When making a decision regarding the treatment, physicians are essentially deciding the indication of the treatment based on the medical condition. When they are in a difficult situation where the diagnosis is unknown or uncertain, prescribing decisions can be changed depending on how the condition evolves [26]. They may decide to prescribe or not prescribe treatments even when there is no clear diagnosis. The various factors that can influence a physician's clinical decision will be highlighted below.

Patient Factors In practice, a physician should be participating in shared decision-making by taking into account the patient's values and preferences. A patient may prefer to take or refuse treatment, and the physician should take that into consideration; however, some patients may prefer treatment when it may not be required. For example, treatment with antimicrobials is not usually required for a cold that is viral, but the patient may want them prescribed. In these cases, the physician should weigh the benefits of the medication versus the risks (e.g., antibiotic resistance, adverse effects). Perhaps the physician may end up prescribing unnecessary treatment for the satisfaction of the patient. The patient's presentation of their symptoms should be taken into account as well. If the symptoms are severe, then the physician may be more inclined to prescribe treatment. Adherence to treatment is another factor that might influence treatment decisions [27]. If a patient is less likely to be adherent to a medication that needs to be taken four times a day, then a once-daily medication can be an alternative. The cost of the medication is another factor that affects treatment decisions. Patients with low socioeconomic status or patients with an insurance plan that does not cover the treatment are less likely to take the

treatment if they cannot afford it. Therefore, a low-cost alternative would be prescribed.

Intrinsic Factors There are many physicians that are specialized in clinical areas that often require a referral from a primary care provider. For example, a patient with complicated cardiac history may be referred to a cardiologist by their family physician. Physicians' education and training in medical school, residency, fellowship programs, along with experience, values, and willingness to continuously learn, all have an impact on how they make clinical decisions [27]. More experienced physicians are likely to be more confident in diagnosing patients and prescribing treatment, whereas a new physician may take more time in coming up with a diagnosis, affecting their treatment decisions. A physician's values are also a very important factor to take into account. For example, a physician who does not believe in abortion has the right to exert their conscientious objection; however, they must assist the patient in finding a physician who can prescribe abortion medication. Willingness to learn involves communicating with other colleagues or healthcare professionals, searching for clinical evidence, and ensuring they are up to date with current guidelines [27]. There may be instances where physicians will ask the pharmacist to recommend a medication, as pharmacists are the drug experts, and a pharmacist's recommendation can drive the physician to prescribe that medication.

Contextual Factors The practice site where the physician practices, whether it is a private practice or hospital, is a factor that influences the behaviors around clinical decision-making. Certain practice sites can get a huge diversity of patients or a subset of a patient population. The practice site can also have certain clinical infrastructure including information technologies, medical equipment, care processes, support staff, resources, and workload that can impact clinical decisions. For example, a huge workload and the pressure to limit appointments can lead to errors in decision-making as they may have less time to do a literature search or may not use clinical practice guidelines or use cognitive short-cuts. Pharmaceutical industries may have effects on prescribing decisions of physicians as well. A representative from a pharmaceutical company will come and do a presentation on a drug and promote it. They may also provide samples, evidence, conduct meetings, sponsor medical education, and provide gifts [26].

Components of Clinical Decision-Making

An effective approach to reaching a decision involves a systematic process of collecting the relevant information, assessing it appropriately, and coming up with reasonable options. Pharmacists can use any process they want to make

clinical decisions as long as they are organized and clear in their process. They must also take into account the various factors that go into making a decision.

Tools/Strategies for Clinical Decision-Making

A commonly used approach that pharmacists may find useful is to first collect the data. Depending on why the patient is seeking care, various assessments are then done. After doing an assessment, reasonable alternatives are thought out and discussed with the patient. The pros and cons of each alternative may be discussed as well. Shared decision-making is then facilitated and the best alternative is chosen for that patient. The monitoring and follow-up plan are created and implemented following the interaction with the patient.

Patient assessment tools for new prescriptions, refills, adaptations, renewals, comprehensive medication reviews, and prescribing for both community and hospital settings are available. These tools provide guidance on how a pharmacist should communicate with patients and other healthcare professionals and provide checklists to ensure pharmacists are thorough in their assessments [28, 29]. Chat, check, and chart is another simple tool from the Alberta College of Pharmacy (ACP) that was created to improve the efficiency in doing a patient assessment. This tool could be useful if the pharmacy is very busy and there is not a lot of time to interact with patients, but it is thorough enough to get all the required information. The "Chat" component involves the three prime questions to gather information and it can be used for new prescriptions and refills. The three prime questions are: "What are you taking the medication for," "How were you told to use it," and "What were you told to expect?" These questions were designed to assess what the patient knows about their therapy and to fill in some knowledge gaps. The "Check" component involves the assessment of drug therapy and helps to identify any drug therapy problems. It involves asking about indication, efficacy, safety, and adherence (which will be discussed later). Finally, "Chart" is documentation of the interaction and what occurred. This would involve writing the data, assessment, and plan (DAP note) and putting it on the patient's file [30, 31].

Other tools that can aid in clinical decision-making involve the use of drug interaction or dose checkers. When there is a drug interaction, the pharmacist can determine if the interaction is significant enough to cause harm or reduce the benefit of other medications. There are medications that are absolutely contraindicated when taken with other medications, which makes it easier to make a decision. However, when two drugs have an interaction that is not contraindicated, this is where clinical judgment comes in. It is impor-

tant that the pharmacist also looks at the patient in front of them because drug interactions may be significant for some patients but not others. Whether a patient experiences the effects of drug interactions can depend on many factors such as their comorbidities and other medications. UpToDate Lexidrug® (formerly Lexicomp®) and Micromedex® are examples of useful drug interaction checkers pharmacists can use in their practice. When checking for drug interactions, it is important to make sure all of the patient's medications are accounted for. Searching a patient's electronic medical record can help in finding out if they are getting their medications from another pharmacy to avoid any missed drug interactions that can cause harm. EPIC® is an electronic health record that is used by many hospitals and health systems across North America. In this system, one chart can go through multiple areas in which patients receive their care including several emergency departments, community physician locations, ambulatory surgery, and clinic areas [32]. This system contains templates and forms for clinical documentation, chronic disease registries, and clinical decision support tools. Some of these tools include references, reminders, alerts, and guides. This can help healthcare professionals ensure they are making appropriate decisions for their patients [33].

Clinical decision support tools use algorithms and evidence-based recommendations to aid clinical decisions. By integrating patient data with clinical knowledge, these systems enhance healthcare delivery. They match patient information to a computer database to generate recommendations, allowing professionals to combine their expertise with algorithmic suggestions. Such tools include order sets, alerts, reminders, computerized guidelines, and workflow aids. EPIC® is an example of this type of system. Another example of a tool that uses algorithms is apps such as Bugs & Drugs®. These apps are clinical decision-making tools in that they provide recommendations of antimicrobials for certain infections based on certain criteria. Other advantages of having computerized clinical decision support tools are a lower risk of medication errors, adherence to clinical guidelines, and diagnostic support. A study found that electronic clinical decision-making algorithms were useful in a healthcare setting with low resources and lack of support [34]. However, some disadvantages include disrupting workflow, a disconnect with face-to-face interactions, alerts that may not be important, placing too much trust on the computer system, and technical maintenance [35].

Shared Decision-Making Tools

One way to involve patients in clinical decision-making is by the use of patient decision aids. These types of aids can assist decision-making for screening, treatment, and other options.

With patient decision aids, the patient can be informed on outcomes, benefits, and risks, and the frequency and likelihood of these components. The patients will then have realistic expectations of their therapy. The Ottawa Hospital Research Initiative website provides patient decision aids that can be useful for patients having doubts about whether they should screen for certain diseases or take certain treatments. Another method is using the three-talk model, which includes team talk, option talk, and decision talk. Team talk involves listing alternatives, offering support, establishing goals of therapy, and reassuring the patient. Option talk is checking knowledge of the disease or options, describing each option, exploring the risks and benefits of each option, providing support, and summarizing. Decision talk focuses on patient preferences and making a decision [36]. A 1000-person tool is another decision aid that shows the risks of screening for certain diseases or taking a medication by showing 1000 shaded figures, and the ones that are colored in illustrate the risk or benefit. For example, if 1000 men were screened for a prostate-specific antigen (PSA), 720 men will have a negative result so that translates to 720 figures colored in grey. One-hundred and seventy-eight will have a positive result, where further follow-up indicates no prostate cancer, so this would translate into 178 figures colored in blue. The Canadian Task Force on Preventive Health Care has several of these 1000-person tools that can be discussed with patients who are concerned about the risks of a treatment [37].

Gathering Information

The data would include information about the patient's chief complaint, demographics, medical conditions, medications (including recently started or discontinued medications, adverse effects to medications), family history, functional history, social history (e.g., smoking, alcohol use, recreational drug use, occupation), allergies, lab test results, review of systems (e.g., head-to-toe assessment), physical exam, vital signs, immunization history, medication adherence, and medication management (e.g., physicians involved in care, administration, monitoring). Depending on what the complaint or drug therapy problem is, additional questions may be asked focusing on the issue. If the patient believes they are having an adverse drug reaction to a medication, the pharmacist would do an assessment of the reaction and determine if it is drug-related. If there is a complaint about the medication not working, the pharmacist may assess how the patient is taking the medication and determine if there needs to be a change in therapy or educate the patient. If a patient has a minor ailment, the use of the SCHOLAR/E (symptoms, characteristics, history, onset, location, aggravating factors, remitting factors, explanatory model) can

help to gather information on the details of the ailment. SOCRATES (site, onset, character, radiation, associated features, time course, exacerbating or alleviating factors, severity) and PQRST (palliation, quality, region, severity, timing) are other mnemonics that can be used to assess pain management. See Chaps. 1 and 2 of this book for further details about principles of patient assessment.

Determining Drug-Related Problems and Goals of Therapy

There are four main types of drug-therapy problems: indication, efficacy, safety, and adherence. Any time there is a new prescription, refill, or if the patient has a problem with their medication, indication, efficacy, safety, and adherence should always be assessed. For *indication*, the main concern is whether the drug the patient was prescribed is appropriate in the first place. Is the drug indicated for this specific condition? Is it a first-line therapy? If there is no approval for the drug for the condition, is there evidence for off-label use? Has the patient started the therapy? Can additional therapy be added to optimize management? For *efficacy*, lab tests, physical exam, or relief of symptoms are all assessed within a specific time frame depending on what the condition is. If there is only suboptimal improvement, a dose increase, adding on another drug, or alternative therapy may be indicated. If the maximum dose has already been attained, then either adding on another drug or switching to another drug may be appropriate. In *safety*, adverse effects and interactions with other medications are assessed or monitored. If the patient is at risk of an adverse effect because the dose is too high, consider factors such as weight, age, organ function (e.g., kidney or liver disease), duration of therapy, and drug pharmacokinetics. In these cases, a pharmacist can adapt the dose, contact the prescriber, provide education (e.g., non-pharmacological intervention), or prescribe a different/additional drug. The same thing goes for potential drug interactions, think about if the interaction is significant and if it can be managed. For *adherence*, consider dosage frequency, confusing directions, cost/access, adverse effects, patient preferences, beliefs, values, age, memory, and ability to self-administer drugs. If a patient has trouble remembering to take it, pill packaging, cell phone reminders, caregiver support, or changing to a drug with lower frequency of administration can be done. Motivational interviewing and addressing barriers to taking medications are other strategies that can be used as well. See Chaps. 1 and 2 of this book for further details about principles of patient assessment.

After determining the drug-related problems, the goals of therapy are then discussed. Goals of therapy are parameters that are used to assess efficacy and safety of treatment, and they must be realistic for the patient to achieve. Each drug-

related problem will have their own goals of therapy. Goals can include curing a condition, prevention, slowing progression, reducing/eliminating symptoms, or normalizing a lab value.

Scientific Evidence

Evidence-based medicine (EBM) is the process of combining the best available evidence, clinical expertise, and patient factors to make a clinical decision [38]. EBM is more than just applying the first available evidence blindly to patient-specific problems. It involves the following steps: creating a clinical question, gathering evidence, evaluating the quality and validity of the evidence, determining how applicable the evidence is to the patient. With these steps, clinicians can gather enough information that can be useful to answer the formulated question for the patient. When determining the applicability of the results, it is important to remember that the study population may have several characteristics that are different from the patient that the clinician is working with. Therefore, clinical judgment is still required, and this is one of the reasons why evidence cannot just be blindly applied to every patient. Some of the disadvantages of EBM are that it can be time-consuming, and sometimes the clinical question may not have literature that is available. Clinical practice guidelines also provide evidence-based recommendations that can be applied to certain clinical scenarios. With guidelines, they also provide the level of evidence to tell the reader about the quality of the evidence that was used to make the recommendation. As with EBM, guidelines should not be blindly followed because some recommendations can be based on low-quality evidence, and some guidelines do not account for patient characteristics that are not similar to the populations used to create the guidelines [39].

Patient Factors

A clinician should factor in the patient's medical history, such as existing conditions and medication use, when making clinical decisions. Cost and insurance coverage are relevant considerations for patients, particularly if affordability is a concern. These factors may influence the treatment options recommended by clinicians. Socioeconomic status is defined as a social standing that combines education, income, and occupation. An individual's socioeconomic status can be indicative of inequities when accessing resources. Many patients may not have opportunities or resources they can access, which can affect the quality of health care they receive and can lead to poorer outcomes in health [40]. Eliciting patient preferences and beliefs is also an important thing to consider in making decisions. Non-medical factors,

also known social determinants of health, are a group of factors that can have a huge impact on health outcomes as well as clinical decision-making. The World Health Organization (WHO) lists 10 social determinants of health [41]:

- Income and social protection
- Education
- Unemployment and job insecurity
- Working life conditions
- Food insecurity
- Housing, basic amenities and the environment
- Early childhood development
- Social inclusion and non-discrimination
- Structural conflict
- Access to affordable health services of decent quality

Depending on the practice site, there may be patients who present with these barriers; therefore, the clinician should be able to tailor their care to each individual to determine the appropriate treatment plan.

Care Plan and Documentation

After a clinical decision has been made, a monitoring and follow-up plan should be discussed with the patient. Patients should be counseled on what to look for in terms of efficacy (e.g., resolution of symptoms) and safety (e.g., most common side effects), what serious adverse effects to look out for, other non-pharmacological measures to take, and when to seek medical attention. Monitoring can also come in the form of reviewing lab tests after a certain period of time. Follow-up can include giving the patient a phone call in a few days, scheduling an in-person appointment, or making sure patients follow up with other healthcare professionals involved in their care. Notifying the patient's primary care provider and other health care professionals is the next step after the plan has been formulated, so that everyone involved in the patient's care knows what is going on. The last part of the process in clinical decision-making is to do documentation. There are several ways to document a patient interaction. The most commonly used format in pharmacy is the DAP note consisting of data, assessment, and the plan. All the data that were gathered from the interaction (e.g., medications, results from lab tests) are written in the data section. Assessment includes interpretation of the data, DRPs identified, if medical conditions are managed, identifying outcomes and goals, and identifying if the patient requires education. The plan consists of recommendations, regimens for drug therapy, patient education and referrals, monitoring

parameters, and follow-up. Another form of documentation healthcare providers can use is SOAP (subjective, objective, assessment, and plan). This is similar to the DAP note except for the data section, where the subjective and objective are in separate sections. The subjective data would include the patient's feelings, experiences, beliefs as well as chief complaint, history of present illness, medical history, medications, allergies, and review of systems. The objective data would include vital signs, physical findings, lab tests, imaging results, other diagnostic results, and review of documentation from other clinicians. For physicians, the assessment part can include the differential diagnosis and listing the problems in order of importance [42]. The plan would be very similar to the DAP note.

Decision-Making in the Era of Artificial Intelligence

Clinical decision-making is a cornerstone of pharmacy practice, requiring the integration of evidence-based guidelines, pharmacotherapeutic knowledge, and patient-specific factors to optimize medication use. In recent years, artificial intelligence (AI) has emerged as a powerful adjunct to this process, offering the ability to process large datasets and generate clinically relevant predictions in real time. AI tools, including machine learning algorithms and natural language processing, can support pharmacists by predicting adverse drug reactions, identifying complex drug–drug interactions, and guiding individualized dosing strategies. These technologies expand the pharmacist's capacity to detect patterns and risks that may be overlooked through traditional methods, thereby augmenting, but not replacing, clinical judgment [43].

Despite these opportunities, several challenges must be addressed before AI can be fully integrated into clinical decision-making in pharmacy. Algorithms are inherently dependent on the quality and representativeness of training data; thus, biases and inequities in underlying datasets may perpetuate health disparities if not carefully managed. Moreover, the “black box” nature of many AI models raises concerns regarding transparency, accountability, and ethical use in patient care. Importantly, reliance on AI should not diminish the critical thinking, contextual interpretation, and clinical judgment that define pharmacy practice. Instead, AI should be positioned as a supportive tool that enhances the pharmacist's decision-making capacity. Future research, regulatory frameworks, and education efforts must ensure that AI implementation in pharmacy practice is both evidence-based and ethically grounded, ultimately advancing patient-centered care [44].

Conclusion

With an expanded role in patient care, pharmacists are well equipped with the knowledge and skills to make clinical decisions for their patients. A pharmacist should know the basic principles of clinical decision-making, critical thinking, and shared decision-making and be able to apply these principles in practice. A pharmacist should be able to recognize the most common cognitive biases that can affect clinical judgment and decision-making. Having this awareness can lower the chances of making an inappropriate decision and ensure that a thorough assessment is being done. When taking on the decision-making process, there are many factors and barriers that can affect the ability of the pharmacist to make a decision, which include the environment they work in, intrinsic factors, and relationships with other prescribers and patients. Even though pharmacists can participate in the decision-making process, some pharmacists have a reluctance in doing so. With changes in the university curriculum and an increase in confidence over time, detailed assessments and clinical decision-making should come much easier to pharmacists. Differences exist in clinical decision-making with other health professionals as well. While pharmacists focus more on drug therapy, physicians focus their decision-making around diagnoses. Just like pharmacists, physicians are also prone to many cognitive biases that should be addressed and avoided. When making clinical decisions, the pharmacist should be aware of the tools and strategies that are available that can aid in decision-making. Being aware of all the components that go into the process of decision-making is essential so that the pharmacist can be organized and detailed in their assessments. With these skills, pharmacists will be able to make appropriate decisions that minimize harm and maximize benefit to the patient.

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Part II

Symptoms Assessment

Headache Assessment

6

Sherif Hanafy Mahmoud

Chapter Objectives

1. Describe the main types of headache disorders and their characteristics.
2. Assess patients presenting with headache.
3. Identify the red flags in patients presenting with headaches that prompt referral to health care practitioners or the emergency department.

Background

A patient comes to your pharmacy requesting something for his headache. What information do you need to conduct a proper assessment? To answer this question, pharmacists need to have a reasonable background about different types of headache disorders and how each type is being managed. In addition, gathering relevant patient-specific information spanning from symptom assessment to past medical history is essential to decide the proper course of action, such as recommendation of an abortive therapy vs. referral.

Headache is an ache in the head or, in other words, pain in the head. It is one of the most common symptoms encoun-

tered by healthcare practitioners. According to the international classification of headache disorders version III (ICHD-III), headache is divided into three distinct categories [1]. The first category is “primary headaches,” which includes headaches not attributed to another disorder. Primary headaches are the most common types of headaches and include tension-type headaches (TTH), migraines, and cluster headaches. The second category is “secondary headaches.” In this category, headache is essentially a symptom of an organic or psychiatric illness or induced by a substance (or drug) and/or its withdrawal. For proper assessment of patients’ headaches, awareness of the possible causes of secondary headaches is essential. Secondary headaches range from an adverse reaction to a drug the patient is taking to a more serious life-threatening condition. The management of secondary headaches entails the management of the disease rather than the headache itself. For example, if a patient experiences headaches due to his uncontrolled hypertension, controlling the patient’s blood pressure will potentially control the patient’s headaches. The third category is headache secondary to painful cranial neuropathies such as trigeminal neuralgia. Table 6.1 depicts a summary of the classification of headache disorders.

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Table 6.1 Classification of headache disorders [1]

Primary headaches	Headaches not attributed to another disorder such as: Migraine Tension-type headache (TTH) Cluster headaches Other primary headaches
Secondary headaches	Headache secondary to other disorders such as: Adverse reactions to drugs. Essentially, any drug could cause headache. It is more important to confirm the temporal relation between drug initiation and symptom onset Medication overuse headache: rebound headaches due to the overuse of headache abortive therapies for more than 10–15 days per month Trauma Cerebrovascular causes, e.g., intracerebral hemorrhage, ischemic stroke, subarachnoid hemorrhage Infection, e.g., common cold, flu Brain tumors Other organic illnesses, e.g., hypertension
Painful cranial neuropathies, other facial pains and other headaches	Trigeminal neuralgia Other headaches

Primary Headaches

Figure 6.1 depicts the diagnostic criteria for the three main types of primary headaches: TTH, migraine, and cluster headache. TTH is the most common primary headache disorder. Most people experience it, with at least one attack in a lifetime. TTH may occur at any age, but new onset is less common over the age of 50 years. It is usually mild to mod-

erate in intensity and feels like there is a squeezing band around the head, dull, and nonpulsating. It is always felt by both sides of the head (bilateral) and rarely associated with other symptoms. A headache attack typically lasts from 20 min to up to 1 week. It is generally self-limiting and treated by over-the-counter (OTC) analgesics such as acetaminophen and ibuprofen.

Migraine is the second most common primary headache disorder. The World Health Organization has considered migraine as one of the top 20 disabling illnesses worldwide. It is experienced by 15–20% of all females and 5–10% of all males, with onset always below the age of 50 years. It is usually moderate to severe in intensity, throbbing, and pulsating. It is usually felt by one side of the head (unilateral) and might be associated with nausea, vomiting, photo, and/or phonophobia. Migraine could be associated with or without aura. Auras are sensory perceptions that occur before the headache attack, such as hearing sounds or seeing lights. A headache attack typically lasts from few hours to up to 3 days. It is generally treated with NSAIDs and triptans. Due to the severe disabling nature of the attacks, patients might benefit from prophylactic therapies.

Cluster headaches are the most disabling type of primary headaches; fortunately, it is not as common as TTH or migraine. The overall incidence is 0.1% and is more common in males than females. It is characterized by very severe penetrating unilateral pain around the orbital area with associated autonomic symptoms such as lacrimation, sweating, and nasal congestion. A headache attack typically lasts from 15 min up to 3 h. The short-lived nature of the headache attack makes conventional oral abortive therapies not practical or ineffective. By the time the drug is absorbed by the gastrointestinal tract, the headache attack could have been subsided.

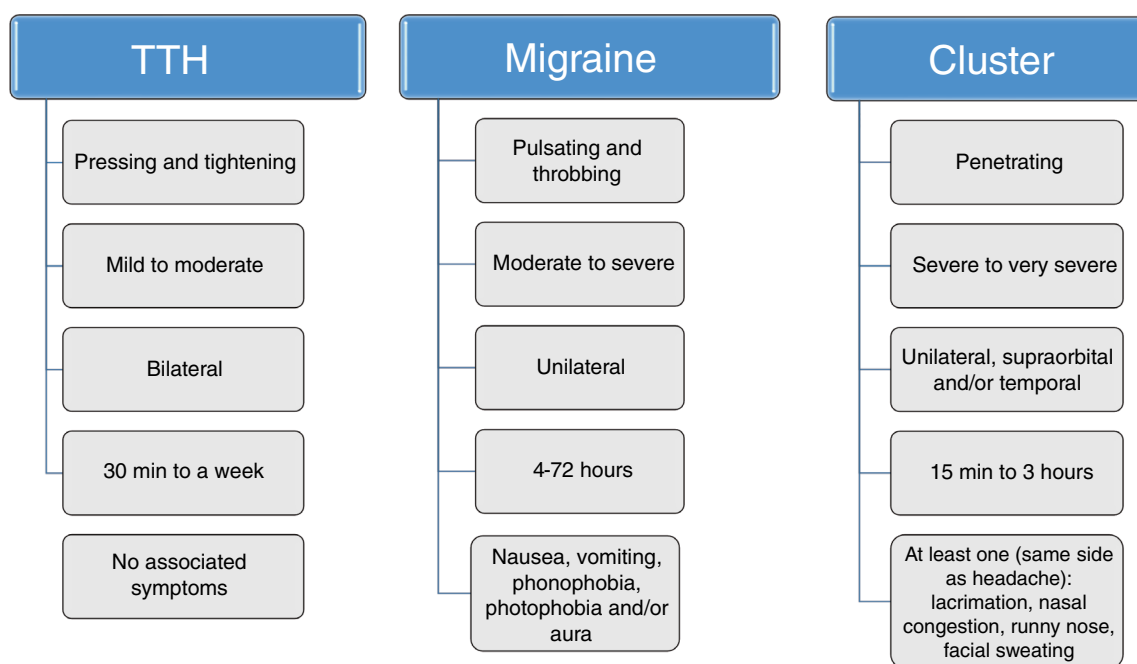


Fig. 6.1 Diagnostic criteria for the three main types of primary headaches: tension-type headache (TTH), migraine, and cluster headache

Symptom Assessment (SCHOLARE)

To allow proper assessment of patients' headaches, the following information will need to be collected. If the patient was previously diagnosed with a headache disorder, a headache diary detailing the onset and severity of the attacks, preceding symptoms, aggravating factors, and medications taken with and without relief will be very helpful to assess the patient. Figure 6.2 depicts a flow chart describing assessment of patients presenting with headache in the pharmacy.

Symptoms

- *In addition to your headache, did you experience any other symptoms?*
- *Did you experience any nausea or vomiting?*
- *Do you have any sensitivity to light or noise?*

Clarifying the symptoms associated with the patient's headache will help in the differential diagnosis of the headache type (see Table 6.1 and Fig. 6.1) and identify the presence of any red flags.

Characteristics

- *Please describe your headache. Is it throbbing, pulsating, or band-like?*
- *Was your recent headache different from the ones you experienced before?*
- *On a scale of 1–10, how severe is your headache?*
- *How often do you get these headaches?*
- *How long do your headaches last?*

Clarifying the characteristics, severity, duration, and frequency of the attacks will help in the differential diagnosis of the headache type and identify the presence of any red flags.

History

- *How long have you been having these headaches?*
- *Did this happen in the past? Was it different?*
- *Did you recently hit your head or have any injury?*
- *What were you doing when you got the headache attack?*

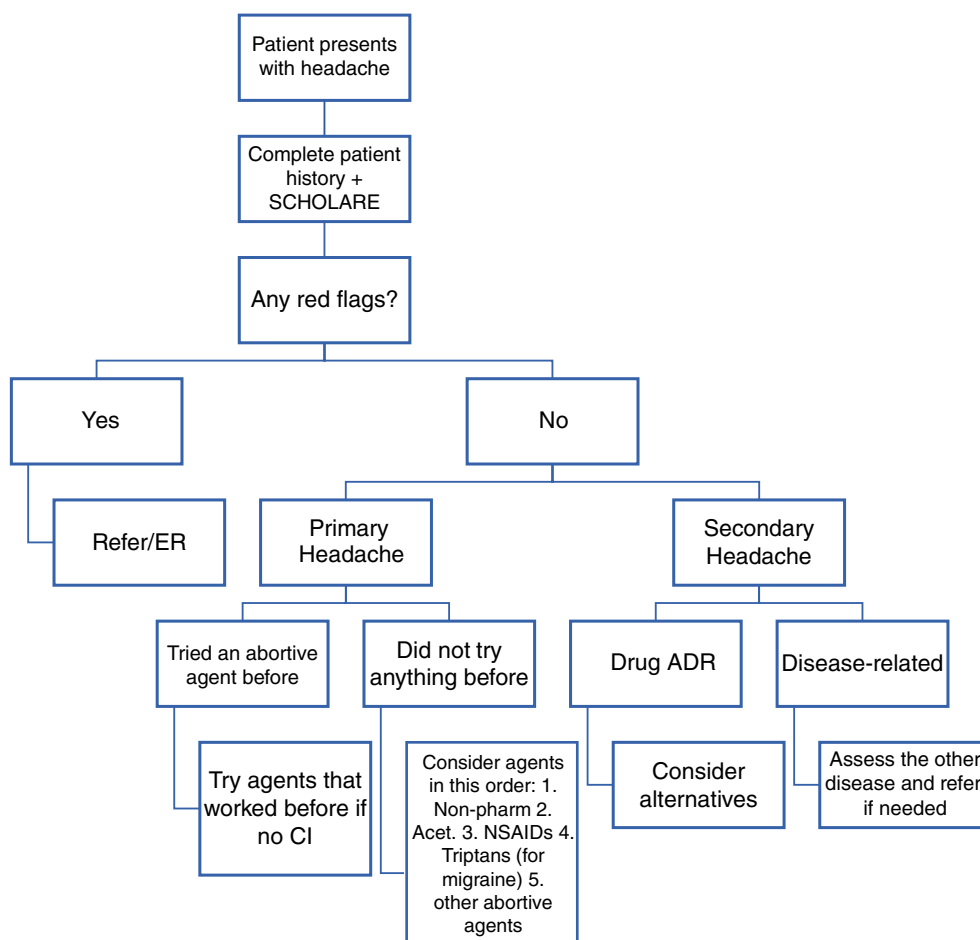


Fig. 6.2 Assessment of patients presenting with headache in the pharmacy. Acet. acetaminophen, ADR adverse drug reactions, CI contraindications, ER emergency department, Non-pharm nonpharmacological

management, NSAIDs nonsteroidal anti-inflammatory drugs, SCHOLARE Symptoms, Characteristics, History, Onset, Location, Aggravating factors, Remitting factors and Explanatory model

Knowing the history of the patient's headache will help determine whether the headache occurs in an atypical pattern. In addition, above questions will help identify any triggers or external precipitants.

Clarifying the location of the patient's headache will help in the differential diagnosis of the headache type (see Fig. 6.1).

Onset

- Was this your first headache attack?
- When did your headaches start?
- Was your headache gradual or abrupt?

Clarifying the age at onset and temporal evolution of the patient's headache will help in the identification of the presence of any red flags.

Location

- Describe the location of your headache. Is it unilateral or bilateral?

Aggravating Factors

- What makes your headache worse? A discussion about patient's triggers.

Although primary headaches are not attributed to a certain cause, there are factors that could precipitate or aggravate the attacks. To illustrate, TTH and migraine could be precipitated by mental stress, smoking, fatigue, lack of sleep, weather changes, and prolonged poor body posture associated with excessive use of computers. In addition, some kinds of food might precipitate a headache attack in certain individuals, such as tyramine-containing food and chocolate. Furthermore, headaches might be associated with menses and oral contraceptive use in some patients. Fasting and caf-

feine intake abstinence in coffee drinkers could also precipitate headaches in some individuals. With regard to cluster headache, vasodilators and alcohol intake could precipitate or aggravate cluster attacks. Identification of those triggers will aid in tailoring the nonpharmacological management. For example, maintaining adequate sleep could help with headaches aggravated by lack of sleep. In addition to avoiding triggers, other nonpharmacological measures include biofeedback and relaxation therapy.

Remitting Factors

- *What makes your headache better? Pharmacological and nonpharmacological.*
- *What did not work for you?*
- *Was the medication effective in aborting the headache attack completely?*
- *Were there any side effects from the medications you took to relieve your headache attack?*

Various treatment options are available for headache disorders [2, 3]. Table 6.2 summarizes the abortive agents used in controlling primary headache disorders. It is necessary to check the patient's previous experience with abortive agents. What worked for the patient before might work again, and vice versa. In addition, if the patient experienced an adverse

or allergic reaction from the drug before, alternative options might need to be considered. To illustrate, if acetaminophen did not work for a patient recently diagnosed with migraine, the patient might try any of the over-the-counter NSAIDs, given that they are not contraindicated. In addition to checking medication efficacy, the dose taken needs to be checked. Some patients might be under- or overdosing their medications and this might affect assessment. For example, sumatriptan recommended oral dosage range is 25–100 mg/dose with maximum 200 mg/day. A dose of 25 mg is less likely to be effective than 50 mg. On the other hand, a 100 mg dose is more likely to be effective, but at the same time, it is more likely for the patient to experience adverse reactions. If the patient tried a 25 mg dose and it did not work, a 50 mg dose might be tried. Conversely, if the patient tried a 100 mg dose and it did not work, sumatriptan is likely ineffective and alternative agents need to be tried.

- *How often do you take the headache medications?*

It is important to ask about the frequency of abortive agents. This will determine if the patient is at risk of or actually experiencing medication overuse headache (MOH). Generally, MOH is defined as the recurrence of headache attacks for more than 15 days per month, which gets worse with administration of abortive agents [1]. Patients at risk are those who use triptans, opioids, opioid combinations, and barbiturates for more than 10 days per month or those who use acetaminophen or NSAIDs for more than 15 days per month. Those patients could benefit from prophylactic therapies.

- *Are you on any medication to prevent further headache attacks? If yes, what is the dose and for how long have you been on it?*

Patients diagnosed with primary headaches that are severe enough to limit their daily activities, those who experience frequent headaches, and/or those for whom most of the abortive agents are not effective or contraindicated might benefit from headache prophylaxis. Due to the severity of migraine and cluster headaches, prophylactic therapies are common in these conditions. Patients on prophylactic therapies need to be assessed for the appropriateness of the agent used in light of the patient's comorbidities, efficacy, and safety. Table 6.3 depicts prophylactic agent selection suggestions based on the clinical setting and patient comorbidities. These suggestions could be helpful while assessing patients on or need prophylactic therapies. With regard to efficacy, it generally takes several weeks for the agent to provide a benefit. If the prophylactic agent has failed, with adequate trial at target dose for 2 months, a trial of another or additional agent is recommended.

Table 6.2 Abortive agents used in controlling primary headaches

Headache type	Abortive agents (suggested oral dose)
Tension-type headache	Acetaminophen (325–1000 mg, max 4 g per day) NSAIDs, e.g., ibuprofen (200–600 mg) and naproxen (250–500 mg)
Migraine	Acetaminophen (325–1000 mg, max 4 g per day) NSAIDs, e.g., ibuprofen (200–600 mg) Triptans: Sumatriptan (oral, 25–100 mg, max 200 mg/24 h; subcutaneous 6 mg, max 12 mg/24 h; nasal solution 5–20 mg, max 40 mg/24 h) Naratriptan (1–2.5 mg, max 5 mg/24 h) Almotriptan (6.5–12.5 mg, max 25 mg/24 h) Eletriptan (20–40 mg, max 40 mg/24 h) Frovatriptan (2.5 mg, max 5 mg/24 h) Zolmitriptan (oral and nasal spray 2.5–5 mg, max 10 mg/24 h) Rizatriptan (5–10 mg, max 20 mg/24 h) Triptans + NSAIDs Gepants Ubrogepant Rimegepant Ditans Lasmiditan Ergot derivatives
Cluster headache	Inhaled oxygen Subcutaneous sumatriptan Zolmitriptan nasal spray

NSAIDs nonsteroidal anti-inflammatory drugs

Table 6.3 Prophylactic drug treatment strategies for migraine based on the clinical setting and comorbidities

Clinical setting/ comorbidities	Suggested agents
Healthy	Propranolol
Comorbid depression	Tricyclic antidepressants, Venlafaxine
Comorbid hypertension	Propranolol, nadolol, metoprolol, candesartan, and lisinopril
Comorbid insomnia	Tricyclic antidepressants, gabapentin, and melatonin
Comorbid obesity	Topiramate
Comorbid epilepsy	Valproic acid, topiramate, gabapentin
Low adverse effects	Magnesium citrate, riboflavin, butterbur, Coenzyme Q10
Failed 3 or more trials of prophylactic therapies	May consider calcitonin gene-related peptide monoclonal antibodies (e.g., erenumab, fremanezumab, galcanezumab, eptinezumab) or calcitonin gene-related peptide receptor antagonists (e.g., atogepant, rimegepant)

Explanatory Model

- *What do you think might have caused your headaches?*
- *What do you worry about most regarding your headache?*
- *What kind of treatment do you think would be most helpful?*

Patient-Specific Characteristics

In addition to assessing the patient's symptoms, a knowledge of the patient's medical and medication history allows proper selection and/or assessment of the appropriate abortive agent. The following examples illustrate how patient-specific characteristics are essential in headache assessment:

- **Age:** The age of the patient at headache onset is important to determine if there is any red flag. New-onset headache at ages older than 50 or younger than 5 years is considered a red flag and prompts referral to a health care professional for further assessment.
- **Pregnancy status:** Nonpharmacological measures and acetaminophen are the abortive agents of choice in pregnant women.
- **Past medical history:** Identifying patient's comorbidities will help in recognizing the possible causes of secondary headaches and the presence of any comorbidities that contraindicate the use of specific abortive agents. For example, a history of peptic ulcer disease precludes the use of NSAIDs, and a history of ischemic heart disease precludes the use of triptans and ergot derivatives. Table 6.4 summarizes the precautions and contraindications of commonly used headache abortive agents. Another example, a patient with a history of hypertension

Table 6.4 Precautions and contraindications of commonly used abortive agents

Abortive agent	Precautions and contraindications that pharmacists need to be aware of when assessing patients with headaches
Acetaminophen	Precautions Maximum dose is 4 g per day from all products containing acetaminophen to avoid hepatotoxicity Heavy alcohol use Patients with liver disease Contraindications Allergy to acetaminophen Avoid in patients with severe active liver disease
NSAIDs	Precautions Risk for or presence of cardiovascular disease History of gastrointestinal bleeding and peptic ulcer disease Patients at risk of hyperkalemia Liver impairment Asthma (contraindicated in aspirin-sensitive asthma) Contraindication Allergy to NSAIDs or ASA Active bleeding Active gastrointestinal bleeding and peptic ulcer disease Renal impairment
Triptans	Precautions Poorly controlled hypertension Smoking Contraindications Ischemic heart diseases Cerebrovascular diseases such as previous stroke Peripheral vascular diseases

Above list does not cover all contraindications and precaution. Consultation of the drug monograph is advised.

NSAIDs nonsteroidal anti-inflammatory drugs; ASA acetylsalicylic acid

could prompt the pharmacist to assess the patient's blood pressure control as a potential cause to the patient's headache.

- **Medication history:** Identifying patient's current medications will help recognize the possibility of drug-induced headaches or any possible drug interaction with the abortive agents. Essentially, any drug could cause headache. It is more important to confirm the temporal relation between drug initiation and symptom onset. In addition, it is recommended that pharmacists to always check for the presence of any clinically significant drug interactions when assessing their patients.

Red Flags

It is very important to determine if the patient's headache could be caused by an underlying medical condition, which could be, on some occasions, life-threatening. Presence of any of the following red flags prompts referral to a health care practitioner or the emergency department:

- *New-onset headache at ages older than 50 years:* New-onset headache at age older than 50 years should prompt the pharmacist to refer the patient to the physician or a specialist for further assessment, as this could be potentially caused by an organic illness or a space-occupying lesion such as brain tumors.
- *New-onset headache in immunosuppressed individuals:* New-onset headache in immunocompromised patients could be caused by central nervous system infections such as meningitis or brain abscess and should be ruled out.
- *New-onset severe headache in pregnant women:* Severe headache during pregnancy needs referral to rule out eclampsia or cerebral venous sinus thrombosis.
- *New-onset severe and abrupt “thunderclap” headache:* Acute-onset severe headache could be caused by potentially life-threatening conditions such as subarachnoid hemorrhage, intracerebral hemorrhage, meningitis, or presence of mass-occupying lesion. It is strongly recommended that the patient should seek immediate medical attention and go to the emergency department to be examined. For example, many patients with subarachnoid hemorrhage, a life-threatening brain bleed, present with severe thunderclap headache as the only symptom.
- *Increased frequency or increased severity of headaches:* Progressive headache symptoms could indicate space-occupying lesions such as brain tumors, brain abscesses, and chronic subdural hematomas (seen in patients on anticoagulants and those with recent trauma to the head). Furthermore, progressive headaches could indicate medication overuse headache. This needs to be assessed through identifying the number of abortive therapies the patient is taking per month. Ruling out secondary causes of headache in patients with MOH is essential.
- *Headache in patients with recent head trauma.*
- *Significant change in the pattern of the headaches the patient is getting:* Changes in headache pattern could indicate a more serious condition and should be closely examined.
- *Presence of other symptoms* that indicate a more serious cause of the patient’s headache, such as stiff neck, altered level of consciousness, fever, motor weakness, and other focal neurological symptoms, prompts referral.

Additional Assessment Considerations

Since primary headaches are not attributable to any underlying etiology, normal physical examination is expected. However, if secondary causes are expected, further investigations are warranted. Examples of further investigations include imaging such as head computed tomography (CT scans), CT angiography, and magnetic resonance imaging (MRI scans). In addition, lumbar puncture, dental examination, and endocrine, biochemical, infection, and oncological workup are recommended.

Follow-up assessment is recommended for patients with headaches. Pharmacists should advise patients, especially the ones with frequent headaches to maintain a headache diary detailing their episodes, possible precipitating and ameliorating factors. In addition, headache severity and frequency, the dose of the abortive agent, response to therapy, presence of any adverse reactions need to be noted in the headache diary.

Clinical Pearls

- Pharmacists play an important role in identifying red flags in patients presenting with headaches.
- Assessment of patients presenting with headaches involves the following two key steps:
 - (a) Assessment of the characteristics and history of patients’ headaches and the presence of any associated symptoms.
 - (b) Assessment of the appropriateness of the abortive and prophylactic agents, when applicable.
- Pharmacists need to assess for medication overuse headache in patients frequently taking abortive agents for their headaches.

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Cough Assessment

7

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Chapter Objectives

1. Outline common etiologies of cough.
2. Assess patients with cough.
3. Identify patients who may benefit from symptomatic therapy for cough.
4. Identify red flag symptoms that indicate need for referral and urgent assessment.

Background

Cough is a very common symptom and is often the top reason patients will seek care from a healthcare provider [1–5]. It can range in chronicity (acute to chronic) and in severity (mild to severe) [6]. The cough reflex serves a purpose, and that is to clear the airway from bacteria, debris, and secretions. The physiologic pathway that results in cough is rather complex. Humans have cough receptors in the respiratory tract, pericardium, diaphragm, pleura, esophagus, and stomach. There are chemical and mechanical receptors that respond to a variety of stimuli, which results in an impulse that travels through the vagus nerve to the “cough center” in the medulla. An efferent signal then travels down to the spinal motor nerves to the expiratory musculature, producing cough [7]. Despite serving a purpose physiologically, cough can be an unpleasant symptom and, depending on severity and chronicity, can significantly impair health-related quality of life [8, 9]. There are many possible causes for a troublesome cough, and assessing for these becomes important to ensure optimal management and prevent complications. The American College of Chest Physicians (ACCP) suggests

that assessing the duration of symptoms is the most useful first step in assessing patients who present with cough. Based on duration, cough is classified as acute (<3 weeks), subacute (3–8 weeks), and chronic (>8 weeks). Duration of symptoms is key because acute, subacute, and chronic cough each tend to have distinct etiologies [6, 10].

Pharmacists can play an important role in the assessment and management of patients with cough. They can identify patients who need urgent medical attention, those who are appropriate for self-care, and anyone in between. Cough assessment involves gathering a focused history from the patient, concentrating on clinical features, exposure history, presence or absence of red flags, and key patient-specific factors. Once the pharmacist has gathered the necessary information, they can move on to next steps, such as recommending a nonprescription medication, prescribing or de-prescribing a medication, or referral to the family physician or the emergency department, as appropriate.

Etiology

Acute cough is most commonly caused by viral infections, followed by exacerbations of asthma or chronic obstructive pulmonary disease (COPD), or pneumonia. Other conditions to consider, depending on symptomatology, include pulmonary embolism (PE), or heart failure (HF). Tuberculosis (TB) should be considered in endemic areas or high-risk populations, regardless of symptom duration [6, 11]. Similarly, *subacute cough* is commonly postinfectious or secondary to asthma or COPD exacerbations. Upper airway cough syndrome (UACS) is another common cause of subacute cough (see below) [6, 11]. *Chronic cough* is most commonly caused by UACS, asthma, GERD, or nonasthmatic eosinophilic bronchitis (NAEB), or a combination of these [6, 11]. Furthermore, medications can be associated with causing acute, subacute, and chronic cough. Therefore, the main etiologies of cough can be grouped into the following gen-

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eral categories: infection-related, preexisting condition-related, and medication-related cough.

Infection-Related Cough

Upper Respiratory Tract Infection

Upper respiratory tract infections (URTI) are most commonly caused by viral pathogens and often present as common cold-like symptoms, which may include acute cough, rhinorrhea, sneezing, sore throat, mild fever, malaise, and headache. Cough often lingers after the other symptoms have resolved. This is called a postinfectious cough and is a common cause of subacute cough. Postinfectious cough is thought to be caused by a combination of postnasal drip and a direct sensitizing effect of the virus on airway receptors [7, 12, 13]. Both of these mechanisms fall under the umbrella of upper airway cough syndrome (UACS), which is discussed in the preexisting condition-related cough section. Table 7.1 shows the infection-related syndromes that cause cough. It should be noted that while the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) virus, which caused the COVID-19 pandemic, now mostly causes relatively mild URTI-like symptoms, it still has the potential to cause more severe, influenza-like illness or lower respiratory tract disease in some patients. Features relatively unique to SARS-

CoV-2 infection include loss of taste or smell (although these findings are not always present). Additionally, some patients may go on to experience the post-infectious syndrome “Long COVID,” which can present in a variety of ways, often involving a chronic cough [14]. Although very rare in North America until recently, due to the success of vaccination programs, measles has been making a resurgence in recent years. Measles is highly contagious and commonly presents with URTI symptoms, including cough, and often presents with fever and a characteristic generalized maculopapular rash [15]. Patients with suspected measles should be directed to call the local health authority or health information line for further directions to minimize the risk of spreading the virus.

Pneumonia

Pneumonia is an infection of the pulmonary parenchyma and has the potential to cause severe illness and death. Symptoms of pneumonia include productive cough, fever, malaise, fatigue, pleuritic chest pain, and dyspnea [16]. If pneumonia is suspected, referral is required for further work-up and treatment. “Atypical” organisms (see Table 7.1) can cause a different constellation of symptoms, which tend to be more nonspecific and less severe (e.g., general malaise, no dyspnea, or chest pain) with a more protracted onset and disease course. They may present with acute or persistent cough, either productive or nonproductive [17].

Influenza

Influenza is an important viral pathogen that causes outbreaks in a seasonal pattern, usually peaking in winter months. It is commonly spread through droplet transmission. Main symptoms include cough, fever, myalgia, rhinorrhea, headache, and fatigue. Influenza is usually self-limiting in previously healthy individuals. Despite this, certain strains can cause high rates of morbidity and mortality among young adults and children [18]. Usual risk factors for complications from influenza are:

- Age ≥ 65 years old
- Nursing home residence
- Pregnant or postpartum (within 2 weeks after delivery)
- Children < 5 years old (especially < 2 years old)
- Chronic medical conditions such as asthma, COPD, diabetes mellitus (DM), cardiovascular disease, immunosuppression, or obesity

Complications include pneumonia, hospitalization, and worsening of preexisting conditions. An estimated 300,000–640,000 influenza-related deaths occur each year worldwide. Diligent hand hygiene and annual vaccination are the best safeguards against influenza [18, 19].

Table 7.1 Infection-related causes of cough

Syndrome	Organisms
URTI (“common cold”)	Rhinovirus Coronavirus including COVID-19 Parainfluenza virus Respiratory syncytial virus Others
Influenza	Influenza A or B
Acute bronchitis	Viruses cause $\geq 90\%$ of cases, see above pathogens
Pneumonia	Viruses (URTI organisms and influenza) Bacteria (community-acquired pneumonia): <i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> <i>Moraxella catarrhalis</i> Atypical organisms: <i>Mycoplasma pneumoniae</i> <i>Chlamydia pneumoniae</i> <i>Legionella</i> spp
Mycobacterial pulmonary disease (e.g., tuberculosis)	<i>Mycobacterium tuberculosis</i> <i>Mycobacterium avium</i> complex Other nontuberculous mycobacteria
Pertussis (Whooping cough)	<i>Bordetella pertussis</i> <i>Bordetella parapertussis</i>

URTI upper respiratory tract infection. Note: the list is not all-inclusive

Mycobacterial Pulmonary Disease (e.g., Tuberculosis)

The most significant pulmonary mycobacterial disease, tuberculosis (TB), is caused by the bacteria *Mycobacterium tuberculosis*. Around 90% of people who get infected never develop the disease and instead have what is called latent TB infection (LTBI). Five percent of people with LTBI do develop active TB when it becomes reactivated, and 5% of people who get infected do develop active TB within 18–24 months of infection. Active TB most commonly affects the lungs but can also infect the brain, kidneys, and bones. Productive chronic cough with or without hemoptysis is one of the common symptoms of active TB. Persistent fever, weight loss, night sweats, chest pain, and fatigue are other common symptoms [20–22]. Risk factors for acquiring TB include:

- Homelessness, incarceration, or illicit drug use
- Immunosuppression, chronic kidney disease on hemodialysis (CKD-HD), or diabetes mellitus (DM)
- Residence or travel to an area with high rates of TB. In Canada, this includes First Nations reserve communities

Any patient with a cough lasting longer than 2–3 weeks should be referred and investigated for TB if they have any additional TB symptoms.

Pulmonary infections caused by other mycobacteria besides TB (referred to as nontuberculous mycobacteria (NTM)) can present very similarly to TB; however, these do not have the same level of communicability and, as such, do not carry the same public health risk. However, these infections can be similarly difficult to treat and are associated with health burdens of their own [23]. Patients will typically be evaluated for TB first, but in patients who have TB-like symptoms and have screened negative for TB, it could still be of value for them to be worked up for NTM infections.

Pertussis (“Whooping Cough”)

Pertussis, also known as “whooping cough,” is caused by the bacteria *Bordetella pertussis* and sometimes *Bordetella parapertussis* [24]. It is a highly contagious and underdiagnosed condition. Cough is one of the most common symptoms, and it can last up to 8 weeks or more. This cough generally involves severe hacking or bursts of rapid coughing, sometimes with vomiting after a coughing fit. Some, particularly younger patients, may experience the hallmark inspiratory “whooping” sound, following the characteristic rapid coughing fits. Young infants are more likely to have an atypical presentation (e.g., whoop and post-cough vomiting being absent), but these patients are at highest risk. Any patient with cough suspected to be pertussis should be referred for further evaluation and management [11, 25].

Preexisting Condition-Related Cough

Asthma

Asthma exacerbation can present with acute cough, as well as wheezing, breathlessness, and chest tightness. Inflammation, bronchospasm, and excess mucous trigger these symptoms [26, 27]. Severe asthma exacerbations can be life-threatening, and can occur in patients with any baseline level of severity. Patients at high risk for asthma-related death include those who have:

- Had a previous severe exacerbation
- Overuse of short-acting beta-agonist (SABA)
- Severe asthma
- Recurrent hospitalizations or ED visits in the past year
- Use of illicit drugs or having psychiatric illness or psychosocial problems
- Poor adherence to treatment plans
- Obesity
- Denial of illness or severity of illness

For patients who are not at high risk for fatal exacerbation, mild exacerbations can be managed at home, as long as they respond to initial therapy (e.g., 2–6 puffs of salbutamol every 20 min when necessary). After initial therapy, patients should contact their physician for further management, such as consideration for oral corticosteroids. For more severe exacerbations (marked breathlessness, impaired speech, accessory muscle use, or drowsiness), patients should use initial therapy with salbutamol as above and proceed to the ER urgently for assessment and management. Peak expiratory flow (PEF) is a valuable parameter for monitoring in asthmatics, both acutely in exacerbations and chronically for assessing long-term control. Of note, chronic cough is also common in asthma. Cough and other symptoms are often worse at night and early morning. For a patient with asthma with worsening symptoms such as chronic cough, breathlessness, declining PEF readings, or increased use of rescue medication, these findings would indicate that a lack of control exists. In this case, their preventive medication regimen should be reevaluated and likely stepped up. These patients should be referred to their family physician or specialist in a timely manner [26, 27].

Chronic Obstructive Pulmonary Disease

Acute exacerbation of COPD (AECOPD) is defined as a change in respiratory status that exceeds normal day-to-day variations and requires additional therapy. AECOPD is marked by three cardinal symptoms: worsened dyspnea, increased sputum production, and increased sputum purulence [28, 29]. More than 80% of exacerbations can be managed in the outpatient setting, with intensified bronchodilators,

prednisone, and with or without antibiotics [29]. Patients with AECOPD generally need to be referred. Patients who might need ventilatory support (e.g., hypoxia, resting dyspnea, RR > 20–30, cyanosis, increased work of breathing) or those who cannot manage their symptoms at home should be urgently referred to their physician or to the emergency department.

Heart Failure

Acute decompensated heart failure (ADHF) is another cause of acute cough. All patients with ADHF should be referred to either their physician or to the ED, depending on the severity of symptoms. Many patients will require hospitalization, including those with dyspnea at rest, hypotension, altered mentation, >2 kg weight gain, or worsened peripheral edema. In-hospital mortality for these patients can be high [30]. Severe pulmonary edema can cause a productive cough with frothy pink liquid sputum. Other symptoms of acute or chronic heart failure include fatigue, weakness, nausea, decreased appetite, shortness of breath, and reduced exercise tolerance. Cough can be a chronic symptom of heart failure and is often worse at night, as it is due to fluid overload.

Pulmonary Embolism

PE is an important consideration in patients with acute cough. PE can cause significant morbidity and mortality, especially when left untreated. Virchow's triad represents three broad categories of risk for thrombosis: endothelial injury, blood flow stasis, and hypercoagulability. Recent deep vein thrombosis (DVT), trauma, surgery, immobility, and current diagnosis of cancer are examples of risk factors for PE. Patients with PE often present with dyspnea, pleuritic chest pain, and cough, sometimes with hemoptysis. Any patients with acute cough who may be at risk for PE should be referred to the ED for prompt assessment [31].

Upper Airway Cough Syndrome

Upper airway cough syndrome (UACS) is the most common cause of chronic cough [11]. It was previously referred to as postnasal drip, but was renamed UACS as it is unclear if this cough mechanism is indeed postnasal drip or direct irritation or inflammation of upper airway cough receptors. UACS is caused by a heterogeneous group of rhinosinusitis illnesses, including allergic rhinitis, vasomotor rhinitis, and sinusitis. UACS is usually diagnosed after sequential empiric therapies are tried, such as antihistamines or decongestants [10, 11]. Referral to primary care provider is generally required.

GERD

Gastroesophageal reflux disease (GERD) is one of the most common causes of chronic cough, causing up to 30–40% of cases. Many of these patients experience dyspepsia, heart-

burn, or a sour taste. However, up to 40% of patients with cough due to reflux do not have any traditional GERD symptoms. It is thought that refluxed acid and stomach contents stimulate receptors in the upper and lower respiratory tract, leading to cough [32]. ACCP recommends empiric antireflux therapy for patients who have chronic cough and classic symptoms of GERD. If there is no other explanation for the cough after evaluation, a trial of antireflux therapy is also recommended for patients with chronic cough, even if they do not have GERD symptoms [10]. Antireflux therapy should include lifestyle modification (weight loss if overweight, limiting fat intake, smoking cessation, etc.) as well as proton pump inhibitor use. Further options include metoclopramide and antireflux surgery [33].

Other Conditions Associated with Chronic Cough

Other conditions associated with chronic cough include non-asthmatic eosinophilic bronchitis, also known as cough-variant asthma. Patients with this condition typically have normal spirometry and respond to corticosteroids (inhaled or systemic) [11]. Patients may find that this condition is self-limiting, episodic, or persistent without treatment, and sometimes long-term corticosteroid treatment may be required. Other possible conditions associated with chronic cough include malignancy, bronchiectasis, exposure to irritants (such as cigarette smoke), or psychogenic or habitual cough. Patients with chronic cough should typically be referred for workup for these or other potential causes.

Medication-Related Cough

Cough is listed as a possible adverse effect for many medications. In order to assess whether cough could be due to a medication, one must consider the temporal relationship between initiation of the medication and cough. Observed frequency of cough associated with the medication or class of medications provides another important clue. Table 7.2 lists drugs that have been associated with cough [34].

Angiotensin-Converting Enzyme (ACE) Inhibitor-Induced Cough

ACE inhibitors are one of the most commonly implicated medications for drug-induced cough. It has been described as a persistent dry cough, sometimes with a scratching or tingling sensation in the throat [36]. The incidence is anywhere from 5% to 35% of patients taking this class of medication. The exact mechanism of this adverse effect is unknown. Bradykinin and substance P likely play a role. These molecules are normally broken down by ACE, and thus build up in the presence of an ACE inhibitor. It has been proposed that bradykinin sensitizes airway sensory nerves, possibly through increasing prostaglandin levels [36, 37].

Table 7.2 Medications that have been associated with cough [34, 35]

Medication groups associated with cough	ACE inhibitors Antiretrovirals Antifungals Beta-blockers Chemotherapy agents Inhaled medications Immunosuppressants Liposomal drug formulations Monoclonal antibodies (“-mab”) Recombinant DNA drugs (e.g., eltrombopag, filgrastim, erythrocyte-stimulating agents, insulin glargine, dornase alfa) NSAIDs Tyrosine kinase inhibitors (“-inib”)
Individual drugs associated with cough >10%	Atovaquone Cinacalcet Desloratadine Glutamine Immune globulin Nicotine Octreotide Pamidronate Sacubitril and valsartan (9%)

ACE angiotensin-converting enzyme, DNA deoxyribonucleic acid, NSAIDs nonsteroidal anti-inflammatory drug

The onset of ACE inhibitor-induced cough ranges from hours to months after initiation of therapy. After stopping the ACE inhibitor, resolution of cough usually occurs within 1–4 weeks but may take up to 3 or more months. Cessation of ACE inhibitor is recommended for all patients with chronic cough who are taking an ACE inhibitor [36]. Upon rechallenge, cough will recur in approximately 67% of patients. Rechallenge may be reasonable in patients with compelling indication for ACE inhibitor therapy [38]. Most patients can be switched to an angiotensin II receptor blocker (ARB), as these agents have proven efficacy for many of the same indications, and are much less likely to cause cough [36].

Symptom Assessment (SCHOLAR)

Patient assessment using the SCHOLAR approach provides a systematic framework to elicit necessary information, clarify the differential diagnosis, and identify red flag features. Figure 7.1 depicts the initial assessment and general management approach to cough. The following questions are suggested in order to assess patients presenting with cough.

Symptoms (Main and Associated)

- Please describe your symptoms.
- Have you experienced urinary incontinence associated with your cough?

- Have you experienced vomiting due to your cough?
- Do you have any sharp chest pain associated with your cough?
- Do you have a fever or chills?
- Do you have sore muscles (myalgias)?
- Do you feel weak or tired?
- Do you feel short of breath? Are you limited in the amount or intensity of physical exertion you can perform?
- Have you noticed any wheezing?
- For asthmatics (if they monitor PEF): has your PEF declined from baseline?
- Do you have heartburn symptoms?
- Has your voice become hoarse?
- Have you noticed any swelling in your legs or feet? Have you gained weight?
- Do you have any difficulties or pain with swallowing?

It is essential to ascertain the main and associated symptoms. Further questions identify red flags (see “Red Flags” section). If the cough is severe enough to cause vomiting, this may be a case of pertussis and referral is required. Additionally, cough severe enough to cause urinary incontinence requires primary care referral. These questions will also help identify patients with asthma exacerbation.

Characteristics

- Are you producing any sputum? How much sputum comes up?
- Is it a dry hacking cough?
- If you do have sputum, what does it look like? What color is it? Is there blood in it? Is it pink and frothy?
- Describe the type of physical activity you can do without getting short of breath
- For patients with COPD: Have you noticed an increase in the amount of sputum production? Has your sputum changed color or consistency?

It is helpful to know whether the cough is productive or not, as this may guide therapy if the patient wants to take a medication for their cough. The characteristics questions further identify patients who need to be referred (see “Red Flags” section).

History

- Have you had a similar illness like this before?
- Do you smoke cigarettes, cigars, pipes, or any other products?
- What is your past medical history? The following conditions are particularly relevant to cough: COPD, asthma,

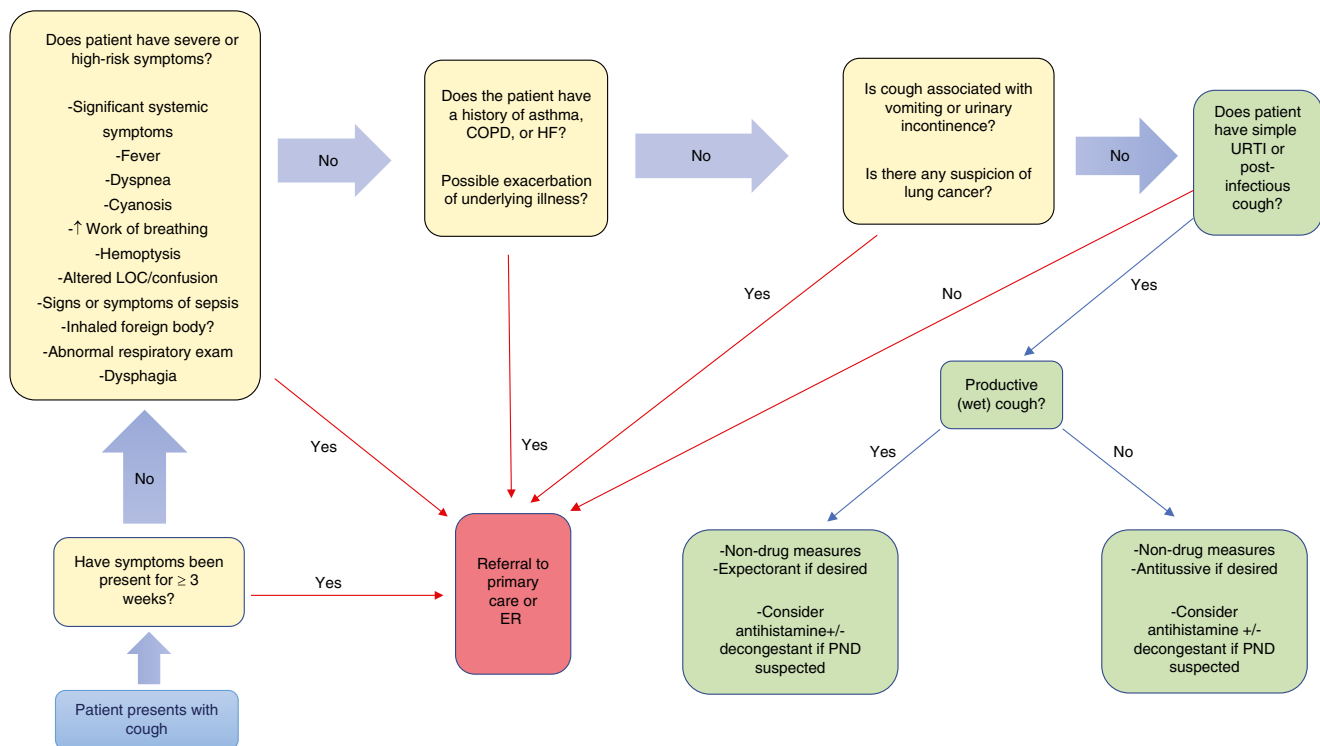


Fig. 7.1 Initial assessment and general management approach to cough. COPD chronic obstructive pulmonary disease, HF heart failure, LOC level of consciousness, PND postnasal drip, URTI upper respiratory tract infection

HF, history of allergies, history of DVT or PE, and history of cancer.

- What medications do you take?
- Have you recently started taking any new medications?
- What is your travel history? Have you recently moved from or traveled to an area where TB is endemic (see [20, 21])?
- Have you had any recent sick contacts at school, work, or at home?

History questions will elicit important information about medical history, and medications, including any potential for medication-induced cough. Each of these helps identify what the etiology of the cough is, as well as management options. If sick contacts are identified, this can help identify the cause of cough, as well. For example, if there is a “cold” going around at school, and the symptoms are consistent with this, the patient very likely has a viral URTI. Anyone with cough who has risk factors for, or additional symptoms of, TB should be referred for evaluation [6].

Onset

- How long have you had your cough?
- Is your cough worse at night?

- How has your cough changed over time (e.g., has it been getting better, worse, staying the same?)
- For other associated symptoms, how long have you been experiencing those? How have they changed over time?

The duration of symptoms is a key piece of information that helps narrow the differential diagnosis greatly. A cough that is worse at night could be due to certain causes (e.g., asthma or HF). For associated symptoms, the timeline and trajectory of these could assist with your assessment of their relevance to the cough-associated problem.

Location

- If the patient has chest pain, explore its location.

Aggravating Factors

- Can you think of anything that makes your cough worse?
- Does exercise or cold air make it worse?
- Does laughing or talking make it worse?
- Does lying down make it worse?
- Are you exposed to noxious chemicals at your job?
- Have you noticed any environmental triggers?
- Do you have any allergies?

Identifying triggers can help the patient to avoid them, if possible. This can also help identify the etiology of the cough, and whether the patient requires referral.

Remitting Factors (Treatment Options)

- Have you tried any pharmacologic or nonpharmacologic remedies? Has anything helped? Did anything not help?
- Did you experience any side effects?

Nonpharmacological Measures

Humidifying the air and staying hydrated may provide relief for some individuals. Honey, either on its own or mixed with hot water and lemon, is a strategy that may provide some soothing relief as well. Smoking cessation is a great option for anyone who is ready and willing to try a quit attempt. Many people actually experience an increase in sputum production in the first few days of a quit attempt, but they should be reassured that this is temporary and that cough will be greatly reduced in the long run by smoking cessation. For patients with allergies or sensitivities, avoiding triggers can make a big difference for cough and associated symptoms. This often involves a trial-and-error approach. Hand hygiene and covering one's mouth when coughing are extremely important strategies for preventing the spread of infection.

Pharmacological Measures

Many patients may wish to try medication to alleviate their symptoms. It is important to distinguish a productive ("wet") cough from a nonproductive ("dry") cough. It should be noted that the evidence of effectiveness for antitussives and expectorants is lacking. For patients with a productive cough, antitussive medications should be avoided, as they can prolong and worsen their illness [39]. For these patients, an expectorant such as guaifenesin can be tried, which has a mechanism of action of loosening phlegm and helping to expel it. For patients with a nonproductive cough, an antitussive such as dextromethorphan can be tried to help lessen their coughing. Table 7.3 depicts the pharmacologic options for cough in adults. Cough and cold medications are not appropriate for over-the-counter use in children <6 years old, as per Health Canada guidelines. Serious adverse effects have occurred in young children. This warning includes antihistamines, antitussives, expectorants, and decongestants. Use of these medications in children must be in conjunction with a primary care provider [40]. For bacterial infections,

Table 7.3 Pharmacologic options for cough in adults [39, 43]

Drug	Dose	Comments
Dextromethorphan	10–20 mg by mouth every 4 h as needed	Antitussive, not recommended for use in productive cough Structurally related to codeine CYP 2D6 metabolism (genetic polymorphisms common) Screen for drug interactions (may precipitate serotonin syndrome)
Codeine	10–20 mg by mouth every 4 h as needed	Antitussive, not recommended for use in productive cough Not safe in pediatric population due to CYP 2D6 polymorphisms and potential for supratherapeutic levels in fast metabolizers Deaths have occurred
Guaifenesin	200–400 mg by mouth every 4 h as needed	Expectorant. Loosens secretions to aid in their expulsion
Diphenhydramine	25 mg by mouth every 4 h	Any first-generation antihistamine is suitable for use in cough Anticholinergic properties are useful for reducing secretions and postnasal drip Anticholinergic adverse effects of first-generation agents (e.g., sedation, dry mouth) may outweigh the potential beneficial effects
Cetirizine	10 mg by mouth daily	Other second-generation antihistamine. May be beneficial if allergic component to symptoms (e.g., allergic rhinitis) and would avoid the sedating effect of first-generation agents
Pseudoephedrine	30–60 mg by mouth every 4–6 h as needed	Decongestants can be tried for UACS

CYP 2D6 cytochrome P450 isoenzyme 2D6, UACS upper airway cough syndrome

such as community-acquired pneumonia, antibiotics will likely be required. Choice of antibiotics depends on severity of illness, previous antibiotic exposure, etc. and is beyond the scope of this chapter. For patients with influenza, oseltamivir could be considered if symptom duration is <48 h or for those at risk of complications from influenza. Oseltamivir may decrease symptom duration by 12–72 h [41]. Similarly, nirmatrelvir/ritonavir could be considered in some high-risk patients with COVID infection, if started within 5 days of symptom onset [42].

Red Flags

Red flags signal a need for referral to another healthcare practitioner or the emergency department in some cases. Pharmacists can play a key role in identifying patients who have red flag features.

- Cough ≥ 3 -week duration

Persistent cough >3 -week duration without an obvious cause (e.g., recent URTI) needs to be worked up for identification of cause, particularly if not improving or getting worse.

- Significant systemic illness
- Change in mental status
- New or worsening dyspnea (breathlessness)
- Pleuritic chest pain
- Prolonged or high fever
- Abnormal respiratory exam (e.g., wheezing, crackles, stridor)
- Increased work of breathing (e.g., respiratory rate >20 breaths/minute, using accessory muscles to breathe, unable to speak normally)
- Cyanosis (e.g., bluish or purple discoloration of lips/mouth, or fingers/hands, which may feel cold to the touch)
- Hemoptysis
- Suspicion of inhaled foreign body
- New or worsening dysphagia

The preceding symptoms all indicate a degree of severity that warrants urgent/emergency department attention. Significant systemic illness, prolonged/high fever, change in mental status, increased work of breathing, and cyanosis may be signs and symptoms of sepsis and/or impending respiratory failure. Pleuritic chest pain and hemoptysis require evaluation to rule out PE and assess for pneumonia, both of which require urgent/timely therapy.

- Urinary incontinence or vomiting associated with coughing
- Suspicion of lung cancer

For patients with the preceding signs and symptoms, timely evaluation is required by primary care provider or more urgently if the situation requires.

- Worsened HF symptoms: shortness of breath, orthopnea (e.g., requiring more pillows than normal to sleep), paroxysmal nocturnal dyspnea, peripheral edema, or weight gain.
- AECOPD (see three cardinal symptoms in AECOPD section)

- Asthma exacerbation not responding to initial therapy (see asthma exacerbation section above). If risk factors for fatal asthma exacerbation (see asthma section above), emergency attention is required.

The above represent worsened chronic conditions. Severity of symptoms, as well as patient risk factors, will dictate whether timely primary care or emergency referral is required. Many of these patients will require hospitalization.

Monitoring and Follow-Up

For patients with acute cough (<3 -week duration), frequency of monitoring and follow-up will depend on initial assessment and triaging (i.e., referral to primary care vs. urgent care vs. self-care). For example, for many patients who are appropriate for self-management, and who have trialed a cough medication, it would be prudent to follow up with them in the next 1–7 days. Effectiveness-monitoring parameters would include frequency of cough, quality and quantity of sleep, and amount of sputum production. Safety monitoring parameters would include common side effects of whatever medication was chosen. For example, safety monitoring for codeine would include questioning for drowsiness or constipation. For dextromethorphan, one would need to ask about serotonin toxicities if the patient is taking any other serotonergic drugs.

For patients with mild asthma exacerbation that is being managed at home, follow-up should occur within 12–24 h to ensure the patient is improving and does not require further assessment and management.

For AECOPD or HF exacerbation, frequent follow-up would be needed for patients who have been sent with a home management plan from their primary care provider, to ensure that continued home management is appropriate and that they are improving. These patients would likely receive additional/intensified therapies, such as oral prednisone $\pm$ antibiotics for AECOPD and intensified diuretic therapy for HF. These patients are often complex and thus require diligent monitoring based on severity of illness and medications prescribed. Monitoring parameters would include target symptoms (cough, dyspnea, daily weights, and edema for HF), as well as side effects from additional or intensified therapies.

ACCP recommends using a validated tool to assess patients at follow-up. Figure 7.2 shows examples of validated tools for measuring severity of cough as well as cough-related quality of life. The scales may be used at baseline and at follow-up visits to measure treatment effects.

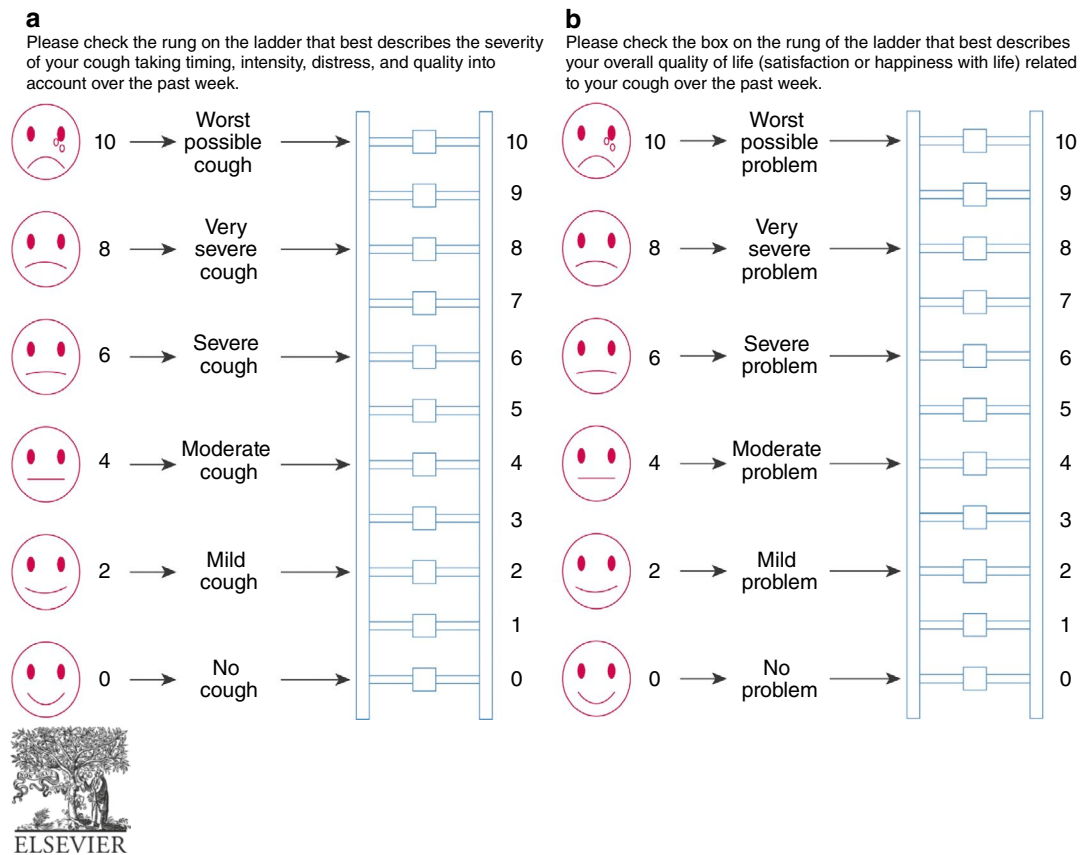


Fig. 7.2 Representative Punum ladders to assess (a) cough severity or (b) overall quality of life. (Reprinted from Irwin et al. [6], Copyright 2025, with permission from Elsevier)

Clinical Pearls

Pharmacists can play an important role in assessing and managing cough. A systematic approach to cough assessment is recommended and includes:

- Determine duration of cough.
- Look for red flags and refer, if present.
- Use SCHOLAR or other structured questions to elicit key information such as symptoms, characteristics, and past medical history.
- If patient is appropriate for treatment at home, elicit their goals of therapy, and recommend symptomatic therapies if appropriate and desired.
- Monitoring and follow-up will be based on treatment plan, severity of symptoms, and underlying comorbidities.

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Nausea and Vomiting Assessment

8

Tara Leslie

Chapter Objectives

1. Describe the etiology, key symptoms, and treatment principles for patients presenting with nausea and/or vomiting due to common causes seen in pharmacy practice.
2. Apply a systematic approach to assess patients presenting with nausea and/or vomiting in order to determine the probable cause and provide appropriate guidance for next steps.
3. Identify the red flags in patients presenting with nausea or vomiting that should trigger referral to other health-care practitioners or urgent medical care.

Background

Nausea and vomiting are common conditions with variable etiology. In some cases, they are self-limiting but, in others, can be associated with an urgent and/or serious medical disorder. Pharmacists need to have awareness of the different potential causes of nausea and/or vomiting and the associated symptoms for each. Using a stepped approach to gathering important information, the pharmacist can begin to delineate the cause of the nausea and/or vomiting and quickly refer if a critical underlying issue is plausible. If serious concerns have been ruled out or appear unlikely given the presentation and assessment, the pharmacist can take next steps to develop an appropriate treatment plan and/or provide reassurance.

Nausea can be defined as the inclination to vomit or a feeling in the throat or epigastric area that vomiting is imminent [1]. Vomiting is defined as the ejection or expulsion of

gastric contents through the mouth with involuntary muscle contractions [1, 2]. Vomiting differs from regurgitation, where forceful contractions are absent [2]. The vomiting center (VC), within the lateral reticular formation of the medulla, organizes the vomiting response upon activation by afferent impulses. Impulses may originate from gastrointestinal tract (GIT) vagal afferents, extramedullary central nervous system (CNS), visual, or vestibular stimuli, or the chemoreceptor trigger zone (CTZ). The CTZ can be triggered by endogenous or exogenous substances as it is located within the area postrema and hence lies partially outside of the blood–brain barrier [3, 4]. The pathogenesis of vomiting involves many different receptors including serotonergic, dopaminergic, histaminic, and muscarinic [3, 4].

Several serious conditions requiring urgent medical care such as appendicitis, pancreatitis, stroke, and myocardial infarction can present with nausea and/or vomiting as part of their constellation of symptoms. Other less concerning causes include pregnancy, onset of an acute self-limiting gastric infection, exposure to nauseating medications, recent surgery, or experiencing different types of motion or electronic stimuli (motion sickness). Table 8.1 lists medication-related, disease-related, and other causes of nausea and/or vomiting.

It is important to note that nausea and/or vomiting can be distressing and unpleasant for patients even if the symptoms are due to self-limiting conditions. Further, if unresolved or prolonged, symptoms can lead to dehydration, hypovolemia, and consequential electrolyte, metabolic, and hemodynamic concerns. Ultimately, the goal is to treat the underlying cause of the nausea and/or vomiting as appropriate, eliminate the symptoms, and correct any sequelae such as dehydration.

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Table 8.1 Causes of nausea and/or vomiting

Medication causes	Disease causes	Other causes
Antimicrobials	Cardiac	Motion sickness
Cephalosporins	Acute myocardial infarction	Nausea and/or vomiting of pregnancy
Ketoconazole	CNS causes	Radiation therapy
Macrolides	Head trauma/brain injury	Recent excessive ingestion of alcohol
Metronidazole	Migraine	Recent surgery—postoperative nausea/vomiting (PONV)
Penicillin	Meningitis	
Antineoplastics	Stroke	
Cisplatin	Gastrointestinal	
Cyclophosphamide/anthracycline combinations	Appendicitis	
Dacarbazine	Cholecystitis	
NSAIDs	Gastroenteritis (viral, bacterial, parasitic)	
Diclofenac	Intestinal obstruction	
Ibuprofen	Pancreatitis	
Naproxen	Peptic ulcer disease	
Opiates	Genitourinary	
Codeine	Urinary tract infection (UTI)	
Fentanyl	Pelvic inflammatory disease (PID)	
Hydromorphone		
Morphine		
Oxycodone		
Others		
Amiodarone		
Bromocriptine		
Cyclosporine		
Fluoxetine (and other SSRIs)		
Iron supplements		
Ketamine		
Metformin		
Mycophenolate		
Nitrous oxide		
Potassium supplements		
Ropivacaine (and other anesthetics)		

NSAIDs nonsteroidal anti-inflammatory drugs, SSRIs selective serotonin reuptake inhibitors

Common Causes of Nausea and/or Vomiting in Pharmacy Practice

Gastroenteritis

Nausea and/or vomiting are symptoms commonly present in gastroenteritis, defined as inflammation of the stomach, small intestine, and/or large intestine [5, 6]. In addition to nausea and/or vomiting, most patients with gastroenteritis also present with diarrhea, and some may report fever, malaise, and/or cramping abdominal pain [5–7].

Most cases of acute gastroenteritis are self-limiting and caused by an acute infectious process. The infective source is most commonly of viral origin (70%) and can include norovirus, rotavirus, and adenovirus [5, 6]. However, bacterial pathogens such as *Shigella*, *Salmonella*, *E coli*, or *Campylobacter* or parasites such as *Giardia*, *Amebiasis*, or

Cryptosporidium can also be implicated [5]. Patients with acute gastroenteritis may report exposure to a sick friend/family member, travel, or ingestion of potentially spoiled food (e.g., attendance at a potluck) within their recent history. Although the exact mechanism is unknown, vomiting due to gastroenteritis is believed to be related to stimulation of the vagus nerve or serotonin receptors in the GIT, which send impulses to the VC [6].

The primary supportive treatment of acute gastroenteritis is largely preventing dehydration or providing appropriate rehydration [5]. Patients with severe dehydration require urgent referral for additional assessment and IV rehydration. For oral rehydration, options include commercial electrolyte replacement solutions (preferred), diluted fruit drinks, or diluted commercial sport drinks [5]. Given the numerous receptors involved in the pathogenesis of vomiting, appropriate use of agents such as dimenhydrinate, ondansetron, or metoclopramide can be useful to halt nausea and/or vomiting symptoms.

Motion Sickness

Nausea, with or without vomiting, is the signature symptom of motion sickness, a syndrome coinciding with the experience of certain types of motion [8] or visual experiences [9]. Examples of activities related to motion sickness are travel by aircraft, boat, or motorized vehicle, going on an amusement park ride, or the use of technology with 3D displays. The pathogenesis of motion sickness is thought to be related to a mismatch between the sensory and actual patterns of vestibular and visual systems [8, 9]. Motion sickness often starts with minor symptoms such as a feeling of gastric fullness, malaise, or drowsiness, and then progressing to nausea [8]. Patients may also experience belching, headache, sweating, dizziness, hypersalivation, and facial pallor [8, 9].

If the motion sickness stimuli cannot be avoided, preventative strategies should be employed. Non-pharmacologic suggestions include habituation by slowly introducing the motion, maintaining a view of the true visual horizon, or choosing a location in the vehicle that minimizes the motion such as over the wing of an airplane [8]. If medication is required, efficacy is improved if combined with behavioral strategies and if administered prior to motion exposure or quickly after the onset of symptoms. Transdermal scopolamine was historically the preferred choice for prevention of motion sickness. However, marketing was recently discontinued in Canada, leading to limitations in access [10]. It may still be available in other countries. Other pharmacologic options include dimenhydrinate or promethazine [8].

Medication-Induced Nausea and Vomiting

Exposure to medications can be a common cause of nausea and/or vomiting and occurs through a variety of mechanisms. Some medications, such as antibiotics, can be locally irritating to the GIT, leading to abdominal discomfort followed by nausea and/or vomiting. Symptom onset typically occurs shortly after ingestion, and patients may feel relief after emesis. Preventative measures include administration of the offending medication with food and a full glass of water. If ineffective, avoiding the medication in sensitive individuals could be considered if an appropriate alternative agent exists.

Nausea and/or vomiting are common side effects experienced by patients initiated on opioids. This effect appears to be dose-related, and tolerance usually (but not always) occurs within days to weeks after introduction [11]. The exact etiology of opiate-induced nausea and/or vomiting (OINV) is uncertain. Postulated mechanisms include a combination of heightened vestibular sensitivity, direct exposure with the chemoreceptor trigger zone, and slowed gastric emptying [11].

Perhaps the most notorious group of medications known to cause nausea and/or vomiting is cytotoxic antineoplastic agents. The etiology of antineoplastic-induced nausea and/or vomiting (AINV) is complex and not fully delineated. Signals from the chemoreceptor trigger zone, cerebral cortex, GIT, and other areas are sent to the VC, causing subsequent signals to effector organs leading to nausea and/or vomiting. Numerous neurotransmitter receptors are involved such as serotonin (5-HT₃), dopamine, and neurokinin 1 (NK1) [12]. The emetic potential of the antineoplastic agent(s) to be administered is the most important factor to consider when determining the requirement and regimen for AINV prophylaxis. A classification system of emetogenicity of anticancer drugs was developed by Hesketh et al. in 1997 [13] and further refined most recently by Jordan et al. in 2017 [14]. Each agent is categorized as high, moderate, low, or minimal emetic risk based on the agent's potential to cause vomiting in patients who don't receive prophylaxis [13, 14].

AINV can be classified as acute, delayed, anticipatory, breakthrough, or refractory. Acute AINV occurs within the first 24 h after administration of the medication with the intensity peaking at 5–6 h. Delayed AINV occurs more than 24 h after drug administration and can last for 6–7 days [12]. Delayed nausea is more common than acute and can often be more severe and treatment resistant. Anticipatory AINV is considered a conditioned response and develops prior to a patient's next chemotherapy treatment [12]. Breakthrough AINV is a term for nausea or vomiting that occurs despite appropriate prophylaxis. Patients receiving high or moder-

ately emetogenic antineoplastic therapy require combination therapy in order to prevent acute and delayed AINV as well as an agent to treat breakthrough nausea [15, 16]. Hesketh et al. have published updated guidelines in 2017 and 2020 on the prescribing of antinauseant/antiemetic combination regimens for AINV and radiation-induced nausea and/or vomiting (RINV) [15, 16]. Combinations for prevention of AINV with highly emetogenic chemotherapy may include a serotonin (5-HT₃) antagonist such as ondansetron, a neurokinin 1 (NK1) receptor antagonist such as aprepitant, dexamethasone, and olanzapine [15, 16].

Postoperative Nausea and Vomiting

Nausea and/or vomiting is a common complication following surgery, affecting approximately 30% of postoperative patients and up to 80% of those at high risk [17, 18]. Poorly controlled postoperative nausea and vomiting (PONV) can lead to dehydration, improper wound healing, aspiration, and delays in hospital discharge [17, 18]. The etiology of PONV is complex, involving stimuli from the CTZ as well as afferents from the GIT, cerebral cortex, and vestibular system. The VC is activated by cholinergic, dopaminergic, histaminergic, and/or serotonergic receptors [18].

Patients who are female, nonsmokers, have a history of PONV or motion sickness, and are less than 50 years of age are at higher risk for PONV. Higher risk is also associated with the use of general anesthesia (versus regional), longer surgery length and durations of general anesthesia exposure, anesthesia involving volatile agents and/or nitrous oxide, the use of intra- or postoperative opioids, and certain types of surgeries such as intra-abdominal and gynecologic. PONV risk mitigation strategies include using regional anesthesia when possible, ensuring adequate hydration, minimizing use of nitrous oxide and/or volatile anesthetics, employing opioid sparing analgesia strategies, and using propofol for induction and maintenance of anesthesia [17, 18].

Due to the multifactorial etiology of PONV, combination pharmacotherapy, targeting different receptors, is employed for at-risk patients. Anti-emetic agents used for PONV include 5-HT₃ antagonists, dexamethasone, dimenhydrinate, metoclopramide, phenothiazines, and NK-1 receptor antagonists [17, 18]. Apfel et al. developed a PONV risk stratification tool to help guide prophylactic antiemetic prescribing [19]. Two anti-emetics from different classes should be considered for adults with one to two risk factors of female sex, history of PONV or motion sickness, nonsmoking status, or use of postoperative opioids. Whereas, three to four agents, from different classes, should be considered in patients with three or all four risk factors [17, 19].

Nausea and Vomiting of Pregnancy

Nausea and/or vomiting are common symptoms of pregnancy, affecting 50–80% of pregnant people [20]. In some cases, nausea and vomiting of pregnancy (NVP) is mild and temporary, but for others it can be more severe, debilitating, and long lasting [20]. Although commonly referred to as morning sickness, NVP can occur throughout the day. Symptoms typically start weeks after conception, peak around 10–16 weeks gestation, and subside around 20 weeks gestation [7, 21]. Hyperemesis gravidarum (HG) is a severe and persistent type of NVP that can lead to dehydration, weight loss, and electrolyte imbalance that can potentially harm both the patient and unborn child [20]. The pathogenesis of NVP has been postulated to be related to a rise in human chorionic gonadotropin. However, the evidence to support this theory is controversial, and hence the exact etiology remains unknown [22].

People of childbearing potential who present with nausea and/or vomiting should be asked about possible pregnancy including the date of last menses. If pregnancy is suspected, but not yet confirmed, a pregnancy test should be recommended. The outcome will help inform the differential diagnosis and assist in selecting safe treatment options as appropriate. To avoid misdiagnosis, other potential causes should be assessed and ruled out prior to concluding symptoms are due to pregnancy alone. If symptoms are negatively impacting the pregnant person's quality of life or if hydration or nutritional status is of concern, the patient should be referred. In addition, referral for prenatal care should be provided for all patients suspecting or recently discovering a pregnancy.

Dietary and lifestyle changes are often the first treatment strategy for NVP. Adequate rest, avoiding strong odors, separating solid food intake from liquid intake, eating smaller bland meals more frequently, and avoiding fatty foods are reasonable recommendations, although limited evidence exists to support them [20, 21]. Ginger, vitamin B6 (pyridoxine), and acupressure that stimulates the pericardium P6 acupressure point are also reasonable first-line options for mild to moderate symptoms [21]. Given the devastating historical impact of thalidomide, hesitancy exists with the use of pharmacotherapy for NVP in both patients and health care providers. However, if symptoms persist or are moderate to severe, pharmacotherapy with agents such as doxylamine + pyridoxine, dimenhydrinate, or ondansetron may be considered. The Society of Obstetricians and Gynecologists of Canada (SOGC) published the Clinical Practice Guidelines for NVP in 2016, authored by Campbell et al. [20]. Within these guidelines, a treatment algorithm is provided to assist in NVP pharmacotherapy decisions [20].

Symptom Assessment (SCHOLAR)

Nausea and/or vomiting is a common presentation that can be associated with a vast array of underlying causes. The etiology can vary from a mild self-limiting illness to a serious condition requiring urgent referral. A careful and thorough symptom assessment using a systematic process is essential. Information such as the characteristics of the nausea and/or vomiting, the symptom onset and behaviors, other associated symptoms, and patient history provide key clues to the underlying etiology and appropriate next steps. These features and the associated etiology and next steps are summarized in Table 8.2.

Table 8.2 Description of key factors for differential diagnosis of nausea and vomiting (NV)

Description of additional symptoms and/or characteristics	Possible cause	Management strategies
Often presents with symptoms of diarrhea and may also present with headache, cramping abdominal pain, fever, malaise Onset is often following recent contact with a sick friend/family member, recent travel, or recent ingestion of potentially spoiled food/drink	Gastroenteritis	Assess severity and duration Consider pharmacotherapy with antiemetic/antidiarrheal or referral if IV hydration or culture required
Motion-related (aboard plane, ship, car) or electronic stimuli-related Symptoms can include vertigo, belching, sweating, dizziness, hypersalivation, and facial pallor	Motion sickness	Preventative strategies; avoiding motion/stimuli (if reasonable), habituation, or pharmacotherapy
Recent ingestion of medications, administration of chemotherapy, radiation therapy, or surgery requiring general anesthetic	Medication-induced NV or antineoplastic-induced NV or radiation-induced NV or postoperative NV	Assess severity and duration Determine offending agent and explore strategies to mitigate symptoms or avoid the agent If these strategies are not reasonable/appropriate, consider pharmacotherapy to prevent and/or treat symptoms

(continued)

Table 8.2 (continued)

Description of additional symptoms and/or characteristics	Possible cause	Management strategies
Timing of nausea/vomiting may or may not be in the morning Amenorrhea or positive pregnancy test Onset is prior to 12 weeks gestation Lack of other causes of NV	NV of pregnancy	Consider lifestyle/dietary measures or pregnancy safe pharmacotherapy if ineffective Referral for prenatal care and/or refractory symptoms
Severe abdominal pain Onset may be abrupt	Appendicitis, acute cholecystitis, pancreatitis, intestinal obstruction	Prompt referral
Polydipsia and/or polyuria Possible altered mental status and/or fruity breath odor History of diabetes	Diabetic ketoacidosis (DKA)	Prompt referral
Dyspepsia, heartburn, or epigastric pain Symptoms of hematemesis, coffee-ground vomitus, bloody stools, melena (dark tarry) stools	Gastrointestinal bleed, peptic ulcer disease, cancer	Consider pharmacotherapy for mild symptoms of dyspepsia and referral in the case of more serious symptoms Urgent referral required for hematemesis
Symptoms include headache, neurologic symptoms, and/or disorientation Emesis can be projectile Possibly following trauma to the head	Intracranial disorders, stroke	Urgent referral for additional investigations/workup
Headache possibly with aura and/or photophobia May have nausea without emesis or headache may diminish after emesis Patient history of migraine	Migraine	Treat underlying cause of migraine Referral if concerns of intracranial disorder
Chest pain, may be described as crushing, may radiate to jaw or left arm	Myocardial infarction	Urgent referral for additional investigations/workup
Symptoms may include headache, stiff neck, fever	Meningitis	Urgent referral for additional investigations/workup
Symptoms may include fever, dysuria, vaginal discharge, and/or suprapubic or flank pain	UTI or PID	Referral for additional workup and antimicrobial treatment

PID pelvic inflammatory disease, *UTI* urinary tract infection

Symptoms

- In addition to the nausea and/or vomiting, did you experience any other symptoms?
 - Do you have changes in your stools such as diarrhea or constipation?
 - Do you have a fever?
 - Do you have any pain? If yes, can you describe the location, type, and intensity of the pain?
 - Do you have any neurologic symptoms such as headache, dizziness, or confusion?

Clarifying the symptoms associated with the nausea and/or vomiting will help to determine the cause and identify red flags. Intense pain can be impetus for urgent referral. Abdominal pain can be associated with appendicitis, acute cholecystitis, or pancreatitis. Lower abdominal, flank, or pelvic pain can be associated with a genitourinary type infection, while chest pain can be associated with an acute myocardial infarction. Diarrhea and fever often accompany nausea and/or vomiting in gastroenteritis. Constipation can be indicative of a bowel obstruction, whereas melena stools (black tarry appearing feces) can suggest bleeding in the GIT. Nausea and/or vomiting can accompany migraine headache. However, worrisome neurologic symptoms such as confusion with or without headache warrant urgent referral.

Characteristics

- Are you experiencing nausea, vomiting, or both?
 - Do the symptoms occur in a certain pattern? For example, at the same time each day?
- Can you describe the nausea?
 - Is it constant or periodic?
 - If periodic, how frequent?
 - On a scale of 1–10, how severe is the nausea?
- Can you describe the vomiting?
 - How frequently (or how many times) have you vomited?
 - What does your vomitus look like? (food present, blood present, coffee-ground appearance, or other)
 - Does it happen without warning and/or projectile?

The characteristics of the nausea and/or vomiting, such as appearance, severity, frequency, and timing, provide helpful details to assist in identifying a cause. Nausea can be present without vomiting in medication-related causes or motion sickness. Projectile vomiting without previous nausea can be a symptom of an intracranial disorder.

Vomitus with bright red blood (known as hematemesis) or a coffee-ground appearance is associated with a bleeding gastric or duodenal ulcer. When a large amount of blood is

being lost, hematemesis may be present and should be considered a medical emergency. The coffee-ground appearance results from a smaller volume of blood sitting in the GIT and becoming partially digested. In both cases referral is prudent.

Frequency of vomiting can provide important insight into the patient's current hydration status and possible sequelae of dehydration. In addition, it informs the clinician of the patient's ability to tolerate oral medications if pharmacotherapy is indicated after completion of the assessment. In NVP, symptoms are sometimes (but not always) more intense in the morning. Severity helps establish a numerical baseline for symptom comparison after intervention.

History

- Has the nausea and/or vomiting happened before?
- Have you eaten/drunk anything out of the ordinary (pot-lucks, restaurants, questionable water, excessive alcohol, etc.)?
- Have you taken any new medications or recreational drugs?
- Have you traveled recently? Where and when?
- Have you had any recent contact with someone having similar symptoms?
- Have you recently injured yourself or hit your head?
- If the patient has a uterus and is of childbearing age: Is there any chance you are pregnant? And (if appropriate) when was your last menses?
- Was the person recently discharged/transferred from surgery?

A positive travel history, contact with a similarly sick individual, or exposure to potentially contaminated food/water can suggest infectious gastroenteritis. Assessment of other features such as accompanying symptoms, time since onset, symptom severity, and location of travel (if applicable) is important in determining if a patient with suspected gastroenteritis requires referral for cultures, antibiotics, or IV rehydration.

Recent ingestion of nauseating medications implies a medication cause. Nausea and/or vomiting are common after acute alcohol intoxication and can also be related to recreational drug use. Patients suspected to be at risk of alcohol poisoning or drug overdose require immediate medical attention. Patients with nausea and/or vomiting related to trauma, specifically a head injury, should also be referred to an acute care medical facility. Potentially pregnant patients may be experiencing NVP but other causes should also be explored. Recent surgery can suggest PONV and additional questions

to explore timing and details of the surgery would be helpful for the assessment.

Onset

- When did the nausea and/or vomiting begin?
 - For patients receiving antineoplastic medications, when did the nausea/vomiting begin in relation to your chemotherapy treatment (within 24 h, after 24 h, before the treatment began)?
- Did the nausea and/or vomiting start suddenly or gradually?
- Are there certain times of the day when the nausea is worse or vomiting is more frequent?

Vomiting that has persisted for an extended period of time can lead to concerns of dehydration and electrolyte disturbances. Sudden onset can trigger suspicion that the cause is recent exposure to an infectious source or nauseating substance. Although not always, NVP symptoms may be more frequently experienced in the morning. Onset is important for determining the type of AINV (acute, delayed, breakthrough, or anticipatory) for patients receiving chemotherapy treatment.

Location

- Not applicable

Aggravating Factors

- What makes the nausea and/or vomiting start or worsen?
 - Are the symptoms triggered or worsened by certain odors?
 - Are the symptoms associated with specific types of motion or activities (electronic games/simulations)?
 - Do the symptoms occur after ingestion or administration of certain foods or medications?
 - In the case of AINV, are there specific triggers that worsen your nausea and/or vomiting (such as the anticipation of the next chemotherapy administration)?

Responses to these questions can further assist the clinician to identify the cause of the symptoms (e.g., motion sickness, NVP, or medication-induced nausea and/or vomiting). It can also provide insight into preventative measures that can be employed in the future.

Remitting Factors

- Have you tried anything for nausea and/or vomiting in the past? (Explore pharmacologic and non-pharmacologic interventions.)
 - Did any of these interventions diminish or eliminate the nausea and/or vomiting?
 - Did any interventions make the nausea and/or vomiting worse?
 - Did you experience any side effects from the intervention(s) that helped?
 - (If a medication(s) has helped in the past) How often do you take the antiemetic medication?

Various treatment options exist for nausea and/or vomiting with varying efficacy depending on the underlying cause of the nausea and/or vomiting (Table 8.3). Hence, treatment recommendations including the choice of antiemetic(s) must first be guided by the cause (e.g., motion sickness, NVP, or AINV). Changes in the treatment plan, such as the pharmacotherapy agent or dose, are influenced by the successes and failures of past or current interventions. For example, an assessment of antiemetics used previously is extremely important for patients with AINV to determine how to treat breakthrough symptoms and escalate prophylactic treatment for future chemotherapy cycles. In many cases, women with NVP should have failed non-pharmacologic options prior to starting a pregnancy-safe pharmacotherapy agent.

Table 8.3 Summary of select antiemetics/antiemetics [10, 15, 17, 20, 21, 23, 24]

Antiemetic agent(s)	Clinical use in adults	Select warnings
Dexamethasone	Prevention of AINV in combination with other antiemetics Prevention of PONV	Side effects include hyperglycemia, hypertension, fluid retention, insomnia, mood changes
Dimenhydrinate	Management of nausea/vomiting related to GE, RINV, PONV, DINV Management of vertigo or nausea/vomiting related to motion sickness Management of NVP ^a	Contraindications include narrow-angle glaucoma, chronic lung disease, difficulty in urination due to prostatic hypertrophy Due to CNS depressant effects, avoid with alcohol May cause drowsiness
Doxylamine succinate (with pyridoxine)	Management of NVP ^a	Contraindications include uncontrolled asthma, narrow-angle glaucoma, stenosing peptic ulcer, and coadministration with monoamine oxidase inhibitors (MAOIs) Common side effects include somnolence
Ginger ^b	Management of NVP	Can potentiate bleeding risk associated with antiplatelet or anticoagulant drugs Side effects include dyspepsia, diarrhea, mouth irritation
Metoclopramide	Prevention of PONV or AINV Treatment of AINV ^a , RINV ^a Management of NVP ^a	Can elevate prolactin levels and/or cause extrapyramidal symptoms Side effects include diarrhea and drowsiness
Neurokinin 1 (NK1) receptor antagonist Aprepitant Fosaprepitant Netupitant	Prevention of AINV in combination with other agents Prevention of PONV (aprepitant only) ^a	Several drug interactions: aprepitant is a substrate, moderate inhibitor, and inducer of CYP 3A4 and an inducer of CYP 2C9
Phenothiazines Prochlorperazine Chlorpromazine Promethazine	Management of nausea and/or vomiting Treatment of AINV (prochlorperazine) ^a Management of motion sickness (promethazine) ^a	Can cause extrapyramidal reactions, tardive dyskinesia, QTc prolongation Use with extreme caution in the elderly Contraindicated in severe depression and Parkinson's disease
Pyridoxine (Vitamin B6) ^b	Management of NVP	Side effects are minimal but may include drowsiness, headache, neuropathy
Serotonin (5HT ₃) antagonists	Ondansetron	Prevention of AINV and RINV Prevention and treatment of PONV Undifferentiated N/V managed in the ER setting ^a Management of NVP ^a
	Granisetron	Prevention of AINV and RINV
	Palonosetron	Prevention of AINV
Scopolamine base (transdermal patch) ^{b,c}	Prevention of symptoms of motion sickness	Contraindications include angle-closure glaucoma; prostatic hyperplasia; paralytic ileus; myasthenia gravis May cause drowsiness, orthostatic hypotension, and dry mouth Elderly patients are at increased risk of CNS adverse effects Patients should wash hands well immediately after applying patch

AINV antineoplastic-induced nausea and vomiting, DINV drug-induced nausea and/or vomiting, ER emergency room, GE gastroenteritis, NVP nausea and vomiting of pregnancy, PONV postoperative nausea and vomiting, RINV radiation-induced nausea and vomiting

^aNot an approved Health Canada indication (off-label use)

^bHealth Canada considers it to be a natural health product

^cCurrently unavailable in Canada

Patient-Specific Characteristics

Patient factors are important considerations for assessing the need for referral or proper selection of antiemetics.

Pregnancy

As mentioned earlier, patients of childbearing potential should be questioned regarding possible pregnancy and if uncertain, a pregnancy test should be done. For pregnancy at less than 20 weeks gestation, NVP is a reasonable diagnosis when all other causes of nausea and/or vomiting have been ruled out. If an antiemetic is required, selection should be done based on both the cause of the nausea and consideration to safety for both patient and unborn child.

Age

Very young or elderly patients may be at higher risk of dehydration and resulting complications.

Past Medical History

Identifying the patient's comorbidities can assist in determining potential underlying causes of nausea and/or vomiting and aid in appropriate antiemetic selection. For example, consider a patient with diabetes. Nausea and/or vomiting could be associated with a serious cause such as diabetic ketoacidosis. In addition, diabetic patients are much more likely to have quick complications from dehydration or reduced oral intake. Another example is a patient with a notable history of cardiac arrhythmia. Serotonin (5-HT₃) antagonists, such as ondansetron, have a known risk of QT prolongation and may be an inappropriate choice.

Medication History

Obtaining a best possible medication history is important to assess medication causes of nausea and/or vomiting and also to avoid potential drug interactions with antiemetic agents. For example, identifying the specific antineoplastic drugs a cancer patient is receiving is important in order to assess their emetogenic potential and appropriate prophylaxis. In less extreme cases, a medication history may elucidate the offending agent, and an alternative administration strategy or therapeutic agent may be easily initiated. For drug interaction considerations, an example is a patient with motion sickness who is already taking anticholinergic medications. This patient may be unable to tolerate the additional anticholinergic effects of scopolamine.

Red Flags

It is essential that assessment of the patient with nausea and/or vomiting include considerations for underlying causes that can be serious and, in some cases, life-threatening. Presence of any of the following red flags indicates that prompt referral to an urgent care center is warranted.

- *Severe pain:* Severe pain in the chest, abdomen, or pelvis can indicate a serious underlying cause such as myocardial infarction, appendicitis, pancreatitis, cholecystitis, peptic ulcer disease, intestinal obstruction, or pelvic inflammatory disease.
- *Head trauma or neurologic symptoms or disorientation:* A history of head trauma or the presence of neurologic deficits requires urgent referral to assess for stroke or a brain injury. Disorientation with neurologic symptoms, fever, and stiff neck is associated with meningitis.
- *Symptoms suggesting infection requiring antibiotics:* Presence of additional symptoms such as fever with dysuria and/or urinary frequency suggests a urinary tract infection. Presence of fever with vaginal discharge and suprapubic pain suggests pelvic inflammatory disease. If gastroenteritis due to a serious bacterial or parasitic cause is suspected, patients should be referred for additional workup.
- *Vomit containing blood, resembling ground coffee, and/or melena stools:* Nausea and/or vomiting with melena stools and/or vomitus that contains blood or has the appearance of coffee grounds are suggestive of gastrointestinal bleeding.
- *Severe, refractory, and/or prolonged nausea and/or vomiting:* Persistent vomiting can lead to dehydration, electrolyte disturbances, and metabolic issues. Referral for possible parenteral hydration and treatment is required.
- *Dehydration:* Patients with signs of substantial dehydration such as dry mucus membranes, reduced urine output, and/or mental status changes should be referred.

Additional Assessment Considerations

Additional workup, including physical exam, may be indicated depending on the working differential diagnosis. When the suspected cause of nausea and/or vomiting is parasitic or bacterial gastroenteritis, stool and/or blood cultures may be appropriate. In the event of significant dehydration, laboratory tests are warranted to explore potential electrolyte or metabolic disturbances. If a serious underlying cause is suspected, additional tests are often required to explore and/or confirm the diagnosis. These may include diagnostic imaging, endoscopic procedures, laboratory tests, cultures, or an electrocardiogram.

Follow-up assessment is recommended for patients with nausea and/or vomiting. Patients with AINV and an ongoing chemotherapy treatment schedule should be encouraged to keep a diary of nausea and vomiting episodes. Patients with other medication-induced nausea and/or vomiting should be monitored for the effectiveness of the intervention (change in administration instructions or change in therapeutic option) at minimizing or eliminating nausea/vomiting symptoms while still ensuring the original treatment goals are achieved. Patients with suspected viral gastroenteritis should be monitored for resolution of symptoms and adequate rehydration. As part of managing NVP, patients should be monitored for symptom improvement and maintenance of adequate nutrition, hydration, and prenatal care.

Clinical Pearls

- Pharmacists are very accessible to the public and often a patient's first point of contact when seeking health care for nausea and/or vomiting.
 - Pharmacists need to be knowledgeable in the assessment, treatment, and monitoring for common and self-limiting causes of nausea and/or vomiting.
 - Identification of red flags is important to ensure patients are referred to urgent or prompt medical attention when required.
- Assessment of nausea and/or vomiting involves the following key steps:
 - Symptoms, characteristics, history, and onset to ascertain the probable cause of nausea and/or vomiting and recognize possible red flags
 - Aggravating and/or remitting factors to further clarify the possible cause and inform the decision of next steps
 - Development of an individualized treatment and monitoring plan informed by the cause of the nausea and/or vomiting, along with important patient-specific factors

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Chapter Objectives

1. Outline common etiologies of acute diarrhea.
2. Assess patients with diarrhea.
3. Identify patients who are candidates for pharmacological therapy (antibiotics, antidiarrheal), and nonpharmacological therapy (oral rehydration solution or other means of fluid repletion).
4. Identify red flag symptoms that prompt referral and urgent assessment.

Background

Diarrhea is a common ailment that patients often seek treatment for. The US data suggest an incidence of 0.6 bouts of diarrhea per person per year [1]. Canadian sales for over-the-counter (OTC) remedies for diarrhea exceeded \$50 million CAD in 2008 [2]. Pharmacists play an important role in the assessment and management of patients with diarrhea. They can identify patients who need urgent medical attention, those who are appropriate for self-care, and anyone in between. Diarrhea assessment involves gathering a focused history from the patient, concentrating on clinical features, exposure history, presence or absence of red flags, and key patient-specific factors. Once the pharmacist has gathered the necessary information, they can move on to the next steps such as prescribing an OTC medication, or referral to the family physician or the emergency department.

Diarrhea is a syndrome characterized by a clinically significant increase in bowel movement frequency or volume. It is often defined as ≥ 3 loose bowel movements in a 24 h

period, or >250 g stool/day. Acute diarrhea is defined by symptoms lasting up to 14 days, whereas persistent diarrhea lasts 14–30 days, and chronic diarrhea has >30 days duration [1]. Diarrhea can also be classified according to the underlying mechanism into osmotic, secretory, and inflammatory. Osmotic diarrhea results when nonabsorbable, osmotically active substances are present and draw excess water into the intestinal lumen (e.g., magnesium hydroxide). Secretory diarrhea occurs when intestinal secretions into the lumen exceed absorption. This can be in response to toxins from enteric pathogens, or due to malabsorption of luminal molecules such as bile salts in the setting of decreased absorptive area post bowel resection. Inflammatory diarrhea occurs when the intestinal epithelium is damaged by inflammation, resulting in exudation of blood and pus into the lumen. Infection and inflammatory bowel diseases (IBD) such as Crohn's disease and ulcerative colitis are common causes of inflammatory diarrhea [3].

Etiology

Acute diarrhea can be grouped into the following categories: infectious, drug-related, and disease-related.

Infectious Diarrhea

Infection is the most common cause of acute diarrhea. In the United States, the incidence of acute infectious diarrhea has been reported by the Centers for Disease Control and Prevention (CDC) to exceed 40 million cases annually, costing over \$150 million in healthcare expenditure [4]. The infectious agents can be viral, bacterial, or protozoan. Table 9.1 depicts examples of the most common pathogens. All infectious agents of acute diarrhea can be acquired from either improperly handled food (e.g., poultry, shellfish, imported fruits and vegetables) or fecally contaminated water sources. In most cases, stool culture and microscopy

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Table 9.1 Common infectious diarrhea pathogens and exposure sources [1, 4–6]

Organism group	Organism	Possible exposure source
Viruses	Norovirus	Outbreaks common in restaurants, healthcare facilities, schools, daycares, etc.
	Rotavirus	
	Enteric adenovirus	
Bacteria	<i>Eschericia coli</i> (enterotoxigenic)	Travel to resource-limited areas
	<i>Shigella</i> spp.	
	<i>Campylobacter</i> spp.	
	<i>Salmonella</i> (non-typhoidal)	
	<i>Clostridioides difficile</i>	Antibiotic exposure, hospitalization, gastric acid suppression, and immunosuppression
Protozoa	<i>Giardia lamblia</i>	Daycare, drinking untreated stream water (known as “beaver fever”)
	<i>Cryptosporidium</i> spp.	Public swimming pools, daycare
	<i>Entamoeba histolytica</i>	Travel to resource-limited areas

spp. species

fail to identify the causative agent and are not routinely recommended. In cases associated with bloody diarrhea, or other severe symptoms (e.g., ≥ 6 stools per day, fever, signs of dehydration, severe abdominal pain), or a duration of symptoms > 7 days, referral is warranted. These patients will likely benefit from microbiological testing and/or antimicrobial therapy [4].

***Clostridioides Difficile* Infection**

Clostridioides difficile is an important pathogen that causes significant morbidity and mortality. An estimated 37,900 cases of *C. difficile* infection (CDI) occurred in Canada in 2012, costing over \$280 million CAD [7]. Diarrhea is the most common symptom, with colitis, toxic megacolon, need for surgery, and death as possible complications. *C. difficile* is an anaerobic, gram-positive, sporulating, toxin-producing organism that is commonly present in the human gastrointestinal (GI) tract. Many people are asymptomatic carriers, due to the protective effects of our normal flora. Antibiotic therapy is a major risk factor for CDI, due to disruption of our microbiome. Virtually any antibiotic can predispose one to CDI. The following are antibiotics that have been strongly associated with CDI [8, 9]:

- Fluoroquinolones (moxifloxacin > ciprofloxacin > levofloxacin)
- Cephalosporins
- Clindamycin

Antibiotics with relatively lower risk for causing CDI [9]:

- Aminoglycosides
- Trimethoprim-sulfamethoxazole
- Tetracyclines
- Metronidazole

In addition to antimicrobials, observational data suggest an association between proton pump inhibitors (PPIs) and CDI, highlighting the importance of PPI deprescribing when appropriate [8]. Decreased killing of spores due to reduced gastric acid is thought to be the mechanism underlying this association.

Ingestion of *C. difficile* spores is the primary mode of transmission. The spores are easily spread because they remain viable on inanimate objects for long periods of time. *C. difficile* spores are not reliably removed by hand sanitizer, so hand washing with soap and water is preferred in the setting of confirmed or suspected CDI. *C. difficile* has become an important pathogen in the community as well as the hospital [8]. Pharmacists play a critical role in identifying patients who are at risk for CDI. For any patient with recent antibiotic exposure, and new-onset diarrhea, CDI must be considered and differentiated from non-CDI antibiotic-associated diarrhea (AAD). This can be difficult, as up to 35% of patients who take antibiotics will get diarrhea, most of whom do not have CDI [10, 11]. In general, AAD is milder and of shorter duration. Patients with CDI must not receive loperamide, diphenoxylate, or other antimotility agents, as they can precipitate toxic megacolon. Since the infection can progress to severe illness, curative antibiotics should be started promptly once CDI is suspected and they should not be withheld for stool-testing results [8]. Reassessment of the culprit antimicrobials is also important, and they should be discontinued if their risks outweigh their benefits. There is insufficient evidence at this time to support routine use of probiotics for the treatment or prevention of CDI [8]; therefore, they are not recommended. Probiotics could risk the development of bacteremia in immunocompromised patients [9].

Action items for suspected CDI:

- Refer to family physician or emergency room for stool sample, further assessment, and prompt initiation of antibiotic treatment when appropriate (metronidazole or oral vancomycin).
- Assess current antibiotics for appropriateness (i.e., assess for opportunity to hold causative antibiotics).
- Hold PPIs if they are not needed.
- Avoid loperamide and other antimotility agents.

After an initial episode of CDI, up to 25% of individuals will experience a recurrence. Risk factors for recurrence include age > 65 years, need for ongoing antibiotics during therapy for CDI, and immunosuppression [12]. Recurrent CDI is associated with 33% increased risk of mortality at 180 days compared to those who do not suffer a recurrence [8].

Traveler's Diarrhea

Up to 40–60% of travelers going from resource-rich to resource-limited countries will develop diarrhea [13]. The symptoms are usually mild and self-limited. Most cases should not be treated with antibiotics. Antimicrobial therapy should be reserved for those with more severe symptoms such as fever, blood and pus in the stool, or diarrhea that substantially interferes with travel activities. Symptomatic therapies such as loperamide or bismuth can be used to treat mild-to-moderate symptoms. Antimotility agents should not be used for those with severe symptoms,

unless antibiotics are also prescribed [13]. Of note, bismuth subsalicylate has been studied as a prophylactic agent against traveler's diarrhea (TD), with a success rate of 61% when taken at the recommended dose of two tablets four times a day, which equals 2.1 g per day [4]. An oral vaccine for prevention of TD is available; however, overall efficacy and costliness limit its use.

Medication-Related Diarrhea

Any newly started medication should be considered as a possible culprit for new onset diarrhea. Many medications can cause this adverse effect, especially at the beginning of therapy. Table 9.2 lists drugs that are known to be common or important causes of diarrhea. In addition to the drugs in Table 9.2, opioid withdrawal is a medication-related cause of diarrhea that is frequently overlooked.

Table 9.2 Drugs implicated to cause diarrhea [3, 14, 15]

Type of diarrhea	Implicated drugs
Osmotic diarrhea	Acarbose Antibiotics ACE inhibitors Enteral feeds Oral magnesium supplements or laxatives Sugar alcohols (mannitol, sorbitol, xylitol) Osmotic laxatives (e.g., lactulose, PEG 3350)
Secretory diarrhea	5-ASA derivatives Antibiotics Anticholinergics (constipation > diarrhea) [16] Antineoplastics Gold salts Metformin Digoxin Calcitonin Carbamazepine Cholinesterase inhibitors Colchicine Cimetidine Caffeine Prostaglandins (e.g., misoprostol) Simvastatin NSAIDs Stimulant laxatives Theophylline
Diarrhea due to altered motility	Cholinesterase inhibitors Cholinergic drugs (e.g., bethanechol) Irinotecan Macrolide antibiotics Metoclopramide Thyroid hormones

(continued)

Table 9.2 (continued)

Type of diarrhea	Implicated drugs
Inflammatory diarrhea	Antibiotics Albendazole Cimetidine Carbamazepine Chemotherapy agents Cocaine Etidronate Flutamide Gold salts Statins Immunosuppressants Itraconazole Methyldopa NSAIDs Isotretinoin Olmesartan (sprue-like enteropathy) PPIs Ranitidine SSRIs Stimulant laxatives Tyrosine kinase inhibitors TMP/SMX
Fatty diarrhea	Aminoglycosides Gold salts Cholestyramine Colchicine HAART Laxatives Methyldopa Octreotide Orlistat Tetracyclines

ACE angiotensin-converting enzyme, *HAART* highly active antiretroviral therapy, *NSAIDs* nonsteroidal anti-inflammatory drugs, *PEG* polyethylene glycol, *PPIs* proton pump inhibitors, *SSRIs* selective serotonin reuptake inhibitors, *TMP/SMX* trimethoprim-sulfamethoxazole, *5-ASA* 5-aminosalicylic acid

Disease-Related Diarrhea

All patients with persistent and chronic diarrhea should be referred to a physician for further assessment. This is because chronic illnesses are more likely to be the cause. Inflammatory bowel diseases like Crohn's disease and ulcerative colitis have severe sequelae if left untreated. Chronic secretory diarrhea can be caused by pancreatic insufficiency (malabsorption), as well as neuroendocrine tumors (e.g., carcinoid syndrome). Fecal impaction seen in severe constipation can lead to overflow diarrhea, as unformed stool bypasses the impaction. Celiac disease, food intolerance, diverticulitis, and certain malignancies are all further examples of chronic conditions associated with diarrhea [3]. Detailed discussion of these disease states is beyond the scope of this chapter.

Symptom Assessment (SCHOLARE)

Patient assessment using the SCHOLARE approach provides a systematic framework to elicit necessary informa-

tion, clarify the differential diagnosis, and identify red flag features. Figure 9.1 depicts the initial assessment and general management approach to diarrhea. The following questions are suggested in order to assess patients presenting with diarrhea:

Symptoms (Main and Associated Symptoms)

- *Please describe your symptoms.*
- *Do you have a fever?*
- *Have you been vomiting as well?*
- *Do you have any viral symptoms like sore throat, cough, or myalgias?*
- *Do you have severe abdominal cramping?*
- *Do you have signs or symptoms of dehydration (dry mouth, dark urine, decreased urine output, thirst, dizziness, weight loss)?*

Exploring the patient's main and associated symptoms will help clarify the differential diagnosis as well as identify

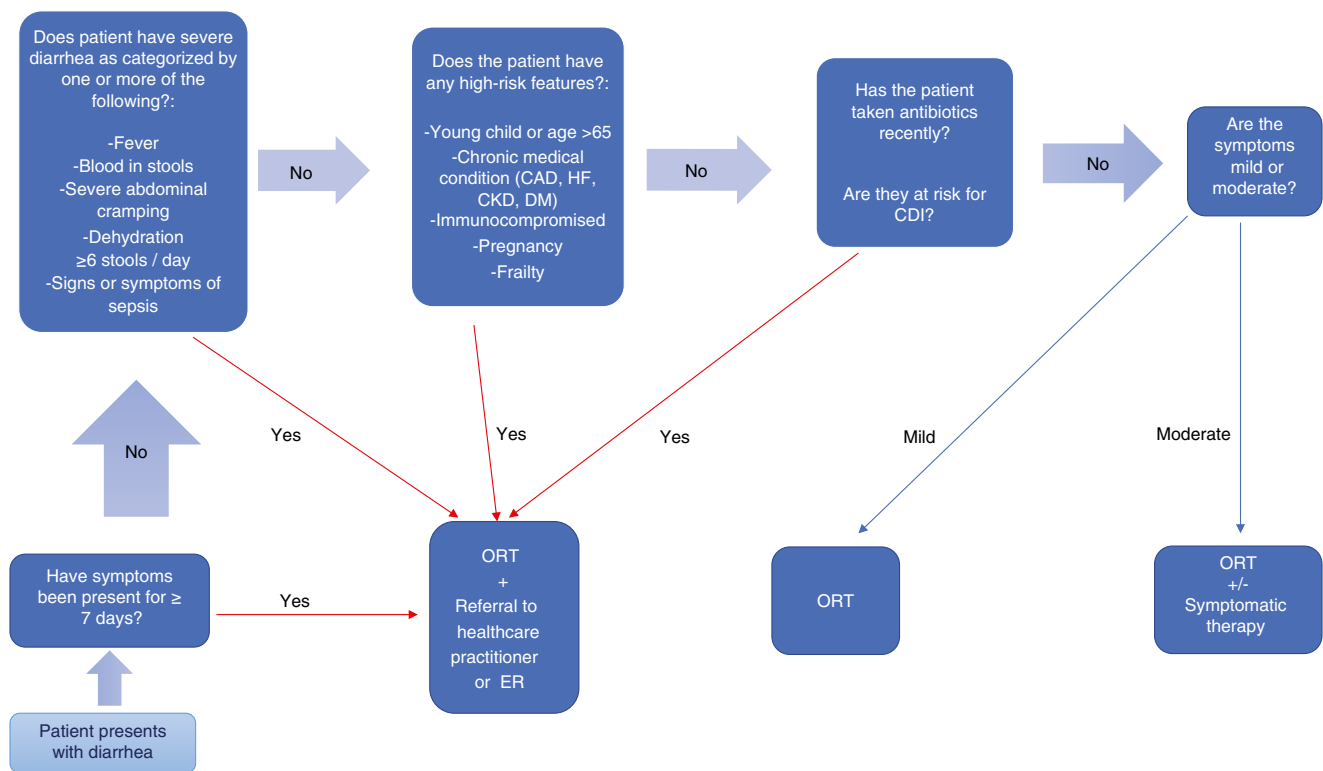


Fig. 9.1 Initial assessment and general management approach to diarrhea. CAD coronary artery disease, CDI *Clostridioides difficile* infection, CKD chronic kidney disease, DM diabetes mellitus, ER emergency room, HF heart failure, ORT oral rehydration therapy

red flag symptoms such as dehydration, or fever. Questions about symptoms related to viral infection may yield clues regarding infectious etiologies.

Characteristics

- Are your stools watery? Or are they more formed?
- Is there any blood in your stools?
- Are your stools mucousy, fatty, or containing pus?
- How many loose stools per day? ≥ 6 ?
- Are they large or small volume?

Exploring symptom characteristics further helps elucidate infectious etiology as well (viral vs. bacterial vs. noninfectious). These questions will also identify red flag features and severe symptoms. Bloody or purulent stools indicate a higher likelihood of bacterial infection [17].

History

- Have you had similar illnesses like this before?
- Did you take any antibiotics in the last 3 months?
- Have you done any traveling recently?

- Have you been hospitalized recently?
- Do any of your friends and family have a similar illness? Are you aware of any diarrhea outbreaks at work or school?
- Did you eat any food that could have made you ill?
- Have you recently started taking any new medications?
- What medications do you take?
- Do you have any drug or food allergies?
- What is your past medical history?

The history questions will elicit important information, such as antibiotic, travel, hospitalization, or other exposure histories. These will identify patients at risk for CDI or AAD, traveler's diarrhea, or food-borne illness. Certain chronic diseases are associated with diarrhea (e.g., IBD, irritable bowel syndrome, and HIV), as are certain medications. Additionally, certain medications may require reassessment in terms of dosing if the diarrhea is severe enough to cause acute kidney injury (AKI). For example, significant diarrhea may affect the pharmacological effects of warfarin, and the international normalized ratio (INR) should be rechecked in these cases.

Onset

- *When did the diarrhea start?*
- *Was the onset gradual or abrupt?*

It is important to know the duration of symptoms to categorize as acute, persistent, or chronic. Symptoms >7 days require referral to physician. Onset of illness is an important distinguishing feature.

Location

If abdominal cramping is present, explore the location of the cramps.

Aggravating Factors

- *What makes your diarrhea worse?*
- *Have you noticed any foods that cause or worsen your diarrhea?*
- *Does the patient have baseline characteristics that predispose to diarrhea?*
- *Does the patient have any risk factors for diarrhea complications?*

Make note if the patient has any chronic disease or food intolerances that could be contributing to their presentation. These patients will require further assessment by their family physician or specialist. See Red Flags section for high-risk patient-specific factors that warrant referral to the physician or the emergency department.

Remitting Factors (Treatment Options)

- *Have you tried any pharmacologic or nonpharmacologic strategies?*
- *Did anything help?*
- *Did you experience any side effects? If yes, what are these?*

It is important to know what the patient has tried, since this can help guide therapy.

Explanatory Model

- *What do you think might have caused your diarrhea?*
- *What kind of treatment do you think would be most helpful?*

Nonpharmacologic Measures

Maintaining fluid status/rehydration is the cornerstone of therapy for patients with diarrhea. For most adults with mild-to-moderate illness, oral rehydration therapy (ORT) can be delivered in the form of drinking adequate fluids, and by eating salty foods like saltine crackers and soups. Oral rehydration solution (ORS) should be considered for infants, the elderly, and anyone with substantial watery diarrhea [18]. If baseline weight is known, serial weight measurements can be an excellent way to gauge fluid losses [5]. In general, ORS is comprised of water, sugar, and electrolytes. ORS uses the sodium–glucose cotransporter in the small intestine to facilitate the absorption of water along with sodium and glucose. The World Health Organization (WHO) endorses the use of ORS, and they have their own formulation. WHO reformulated their ORS in 2002 to have reduced osmolality (245 vs. 311 mOsm/L). They have found that the reduced osmolality ORS reduced stool output by 20%, reduced vomiting by 30%, and reduced the need for intravenous fluids (IVF) by 33%, when compared to the higher osmolality solution [19]. There are several commercially available ORS, and there are also recipes for homemade solutions. Commercially available preparations are generally preferred to avoid mixing errors. They include Gatorlyte®, Hydralyte®, and Pedialyte®. The following is a common recipe for a homemade ORS [16]:

- 1/2 teaspoon of salt
- 2 tablespoons of sugar
- 1 L safe drinking water

Fluid replacement should account for initial losses, maintenance, and ongoing losses [17]. Online calculators are available [20]. For children, specific age-based recommendations are available from the Canadian Pediatric Society [21]. For patients with severe dehydration, IVF must be used initially to correct fluid deficits.

Pharmacologic Measures

If self-care measures are deemed appropriate, there are a number of pharmacological options that may be useful, as outlined in Table 9.3. It is important to rule out contraindications for antidiarrheal medications. Antimotility agents such as loperamide can precipitate toxic megacolon in patients with CDI, and are thus contraindicated for patients with suspected or known CDI. Loperamide can also prolong the duration and lead to more severe illness in patients with dysentery, so it should be avoided if any symptoms such as bloody or mucousy stools, fever, or severe abdominal cramping are present [17]. Antimotility agents can also mask fluid losses due to pooling of fluids in the intestines. It should be noted that diarrhea that persists for more than 48 h after ini-

Table 9.3 Medication options for diarrhea [9, 13, 17]

Drug	Dose (adults)	Comments
Loperamide	4 mg po × 1, then 2 mg as needed after each loose stool, maximum 16 mg/day	Antimotility agent Contraindicated in CDI
Diphenoxylate/atropine	5 mg (diphenoxylate) po 4×/day as needed	Antimotility agent Contraindicated in CDI Atropine is added to discourage misuse May be less effective than loperamide
Bismuth subsalicylate	524 mg (2 tablets) po every 30–60 min as needed (up to 4.2 g/day)	Antisecretory and antimicrobial action Space from other medications

tiating loperamide merits medical attention [5]. Severe diarrhea should not be managed with loperamide or other symptomatic therapies on their own. See the “Red Flags” section for characteristics of severe diarrhea that warrant referral to the physician or the emergency department. Antimicrobial therapy is not indicated for most cases of acute diarrhea, and must be used judiciously to prevent antimicrobial resistance and CDI. In cases of severe diarrheal illness, or in patients at high risk for complications, antibiotic therapy is often utilized [6].

Red Flags

Red flags signal a need for referral to another healthcare practitioner, or the emergency room in some cases. Pharmacists play a key role in identifying patients who have red flag features.

Features of Severe Diarrhea [6]:

- Fever (≥ 38.5 °C).
- Bloody stools.
- Severe abdominal cramping.
- ≥ 6 loose stools in 24 h period.
- Symptoms lasting ≥ 7 days.
- Associated continuous vomiting
- Symptoms of dehydration (dark urine, reduced urine output, marked thirst, dizziness, dry mouth, decreased skin turgor, weight loss).

Any of the preceding symptoms indicate that the patient is experiencing severe diarrhea. Bacterial causes are more common in severe acute diarrhea compared to non-severe, especially when there is blood or pus in the stool, or if the patient is febrile. Further evaluation and management is required, and may include stool culture, stool for ova and parasites, *C. difficile* testing, abdominal X-ray, or endoscopy.

The microbiology tests will help guide antibiotic therapy if required. Intravenous fluids are essential if the patient is severely dehydrated. Worldwide, diarrhea causes 2.2 million deaths each year, largely due to dehydration.

High-Risk Patient-Specific Factors [1, 6]:

- Young children or age > 65
- Immunocompromised status
- Chronic illness such as cardiovascular disease, chronic kidney disease (CKD), or diabetes mellitus (DM1 or DM2)
- Frailty
- Pregnancy

These patients will be more sensitive to the physiologic changes that come with acute diarrhea, including dehydration, fluid shifts, and invasive infection. Closer monitoring is required, in addition to appropriate goal-directed therapies.

Monitoring and Follow-Up

The frequency and duration of follow-up should be guided by the severity of illness, presence or absence of dehydration symptoms, and presence of red flag features. These factors will determine what parameters to monitor as well. In all cases, patients should monitor how many bouts of diarrhea they are having per day, as well as stool characteristics such as consistency. The following cases will help illustrate possible follow-up plans:

- Ahmed is a 60-year-old male with a history of hypertension, on ramipril and hydrochlorothiazide, presents to your pharmacy with diarrhea that started 36 h ago. He has had three loose stools in the past day, but feels well other than the diarrhea (afebrile, no dehydration symptoms). His current blood pressure is 120/85 mmHg (baseline $\sim 130/90$ mmHg). After you have decided on a treatment plan together, when would you want to follow-up?
 - Since Ahmed has no red flag features, he is appropriate for self-care. However, he is almost at the cutoff age, and he is also on two medications that could complicate his picture (ramipril increases his risk for acute kidney injury (AKI); hydrochlorothiazide increases the risk for hyponatremia, and both increase his risk for hypotension). It would be prudent to follow-up in the next 24–48 h to make sure his diarrhea is improving (e.g., number of loose stools per day), his blood pressure is not dropping, and he continues to be afebrile with no signs of dehydration, cramping, and blood in the stools. Additionally, inquire about any possible medications side effects (if pharmaco-

logical therapy was started), how much fluid he has been able to keep down, and whether he is able to eat normally.

- Olivia is a 30-year-old female with no past medical history and takes no regular medications. She has been having two loose stools per day for the past 2 days. She is afebrile, has no signs or symptoms of dehydration, but she does have mild abdominal cramping. Once you have decided on a treatment plan together, when would you want to follow-up with Olivia?
 - Since Olivia is young, with mild symptoms and no red flag features, it is appropriate to wait for a longer period of time to follow-up. It is reasonable to check in after 4 or 5 days to make sure that her diarrhea is resolving. In addition, it is essential to check the number of loose stools/day, presence of any red flag features, and any side effects from medication if one was started.

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Constipation Assessment

10

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Chapter Objectives

1. Define constipation and identify its main causes.
2. Assess patients presenting with constipation.
3. Identify the red flags in patients presenting with constipation that prompt referral to healthcare practitioners.

Background

A patient comes to your pharmacy complaining of infrequent defecation or constipation. What information do you need to conduct a proper assessment? To answer this question, pharmacists need to have a reasonable background about how constipation is defined and caused. In addition, gathering relevant patient-specific information spanning from symptom assessment to past medical history is essential to decide the proper course of action, such as recommendation of a pharmacological or non-pharmacological therapy versus referral for further assessment.

Constipation is one of the common symptoms encountered by pharmacists in their day-to-day practice. It has been defined by the American College of Gastroenterology as an “unsatisfactory defecation and is characterized by infrequent stools, difficult stool passage, or both” [1]. According to the Canadian Digestive Health Foundation, one-fourth of Canadians experience symptoms of constipation to some degree [2]. Constipation is more prevalent among women, adults over 65, nonwhite individuals, those with low-fiber diets, people of lower economic status, and those with sedentary lifestyles [3].

Etiology and Diagnosis

Determining the underlying cause of constipation is essential for proper assessment and subsequent management. Based on etiology, constipation is classified into primary and secondary. Primary constipation is not attributed to an identified cause, and this includes functional (normal transit), slow transit, and obstructive constipation [4]. On the other hand, secondary constipation is the one attributed to a precipitating cause. Table 10.1 depicts secondary causes of constipation.

While assessing a patient presenting with constipation, clinicians need to determine if the patient *actually has constipation*. Acute or short-term constipation generally involves symptoms of reduced defecation frequency (fewer than three defecations per week or more than 4 days with no bowel movement), unsatisfactory evacuation, straining, and/or sitting in the toilet for a long time with no bowel movement. On the other hand, functional chronic constipation is diagnosed if the patient has two or more of the following symptoms in more than a quarter of the defecations for at least 3 months (Rome IV diagnostic criteria) [4, 5]:

- Low bowel movement frequency (less than three times per week)
- Straining
- Hard or lumpy stools
- Feeling of incomplete defecation
- Feeling of obstruction in the anorectal area
- Use of manual measures to facilitate defecation such as the use of digital evacuation and pelvic floor support

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Table 10.1 Example causes of secondary constipation

Drugs	Opioid analgesics Calcium channel blockers, e.g., verapamil Drugs with anticholinergic properties such as diphenhydramine, dimenhydrinate, tricyclic antidepressants Iron products, e.g., ferrous gluconate, sulfate and fumarate Calcium carbonate Aluminum-based antacids Levodopa/carbidopa Bismuth subsalicylate Cholestyramine Sucralfate Resins
Disease	Dehydration Diabetes mellitus Hypothyroidism Cancer, e.g., colon cancer Gastrointestinal diseases, e.g., irritable bowel syndrome, intestinal obstruction Depression Anxiety Autonomic neuropathy Parkinson disease Spinal cord injuries
Other causes	Advanced age Lack of time and suppression of the urge to defecate Low-fiber, high-fat or high-sugar diet and reduced fluid intake

Symptom Assessment (SCHOLARE)

Proper assessment requires taking a history regarding the characteristics of patients' constipation and the presence of any associated symptoms. This is in addition to identifying red flags that prompt referral to healthcare practitioners. A knowledge of the patient's medical and medication history allows proper selection and/or assessment of the appropriate pharmacological agents. In addition, it helps in identifying the possible precipitating factors for secondary constipation, which could be attributed to drugs or diseases. Figure 10.1 is a flow chart describing the assessment of patients presenting with constipation in the pharmacy. The following questions are suggested in order to assess patients with constipation:

Symptoms and Characteristics

- *Please describe your constipation. What do you mean by constipation?*
- *In addition to your constipation, did you experience any other symptoms? This might include nausea, vomiting, reduced appetite, rectal bleeding, abdominal pain ... etc.*
- *If you pass any stools, can you describe its consistency and color?*
- *Do you pass gas?*

Clarifying the characteristics and frequency of bowel movements and associated symptoms will help in determining if the patient actually has constipation (see etiology and diagnosis section) and identifying the presences of any red flags (see red flags section).

History and Onset

- *How long have you been having constipation?*
- *Did this happen in the past? Was it different?*
- *Was there any recent change in your diet? Or fluid intake?*
- *Please describe your normal bowel habit. How many times per day or week do you normally have bowel movement?*
- *Was your constipation onset abrupt or gradual?*

Location

If abdominal pain is present, explore the location.

Aggravating Factors

- *What makes your constipation worse? A discussion about patient's triggers.*
- *Did you recently start on any medications, over-the-counter drugs, or herbals?*
- *Describe your diet and fluid intake.*

Remitting Factors

- *Have you tried anything for your constipation? Pharmacological and non-pharmacological.*
- *What worked and what did not work for you?*

Several *non-pharmacological* measures that could be helpful in ameliorating constipation include the following:

- Increased fluid, fiber, prune, fruits, and vegetable intake.
- Maintain a regular bowel routine, e.g., going to the wash-room at the same time every day.
- Avoid suppressing the urge to defecate.
- Exercise.
- Increase calories for patients with low intake.

Multiple *pharmacological* measures are available for the management of constipation. Table 10.2 summarizes the currently available pharmacological choices for constipation. It is recommended that pharmacological options to be tried if non-pharmacological management was ineffective. Options

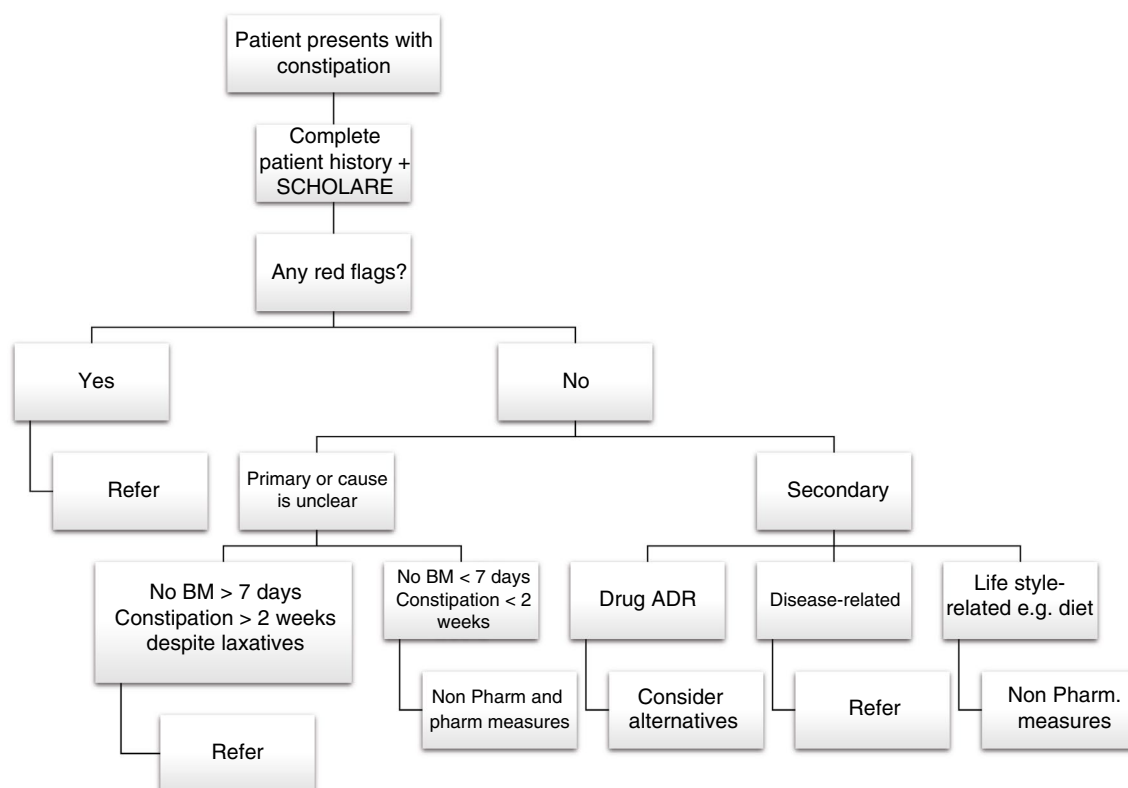


Fig. 10.1 Assessment algorithm for patients presenting with constipation in the pharmacy. ADR adverse drug reaction, BM bowel movement, SCHOLAR Symptoms, Characteristics, History, Onset, Location, Aggravating and Remitting factors, Explanatory model

Table 10.2 Pharmacological options for constipation

Class	Agent (Adult Dose)	Comments
Bulk laxatives	Psyllium (2.5–7.5 g po once to 3 times daily) Bran (1–7 g once to twice daily)	Suitable for long-term use
Stool softeners (emollients)	Docusate sodium (100 mg po twice daily) Docusate calcium (240 mg po twice daily)	They are no more effective than placebo Not recommended
Osmotic laxatives	Polyethylene glycol 3350 (17 g po daily) Lactulose (15–30 ml po once or twice daily) Glycerin (1 adult suppository rectal when needed)	Suitable for long-term use An option for constipation secondary to opioids use
Stimulant laxatives	Senna 2 tablets po at bedtime Bisacodyl (10 mg suppository rectal when needed; 5–10 po daily)	An option for constipation secondary to opioids use
Enemas	Phosphate enema (120 mL rectally) Mineral oil enema (120 mL rectally)	
Others (second-line agents) to be tried if above agents are ineffective	Methylnaltrexone (subcutaneous injection; dose depends on patient's weight)	For constipation secondary to opioids use
	Linaclotide (145 µg po daily)	For chronic constipation Expensive
	Prucalopride (1–2 mg po daily)	For chronic constipation Second line if other therapies fail
	Naloxegol (12.5–25 mg po daily)	For constipation secondary to opioids use

include bulk, osmotic, and stimulant laxatives. Stool softeners such as docusate sodium have been found to be no more effective than a placebo and should not be recommended.

Explanatory Model

- *What do you think might have caused your constipation?*
- *What do you worry about most regarding your constipation?*
- *What kind of treatment do you think would be most helpful?*

Patient-Specific Characteristics

In addition to assessing constipation and its associated symptoms, a knowledge of the patient's medical and medication history allows proper selection and/or assessment of the appropriate pharmacological agents. In addition, it helps in identifying the possible precipitating factors for secondary constipation (see Table 10.1). The following examples illustrate how patient-specific characteristics are essential in constipation assessment:

- **Age:** Individuals older than 65 years are at higher risk of developing constipation than younger ones. This is because they are more likely to have multiple comorbidities and multiple drugs.
- **Pregnancy status:** Increased pressure on the abdomen, combined with hormonal changes and possible calcium intake, increases the propensity to constipation in pregnant women. Non-pharmacological measures should be tried first. If constipation persists, bulk laxatives could be tried followed by osmotic laxatives if needed.
- **Past medical history:** Identifying a patient's comorbidities will help in recognizing the possible causes of secondary constipation. Ideally, controlling the underlying conditions, if possible, could resolve constipation. The use of laxatives in the interim is recommended.
- **Medication history:** Identifying a patient's current medications will help recognize the possibility of drug-induced constipation (see Table 10.1). It is more important to confirm the temporal relation between drug initiation and symptom onset. If one of the patient's medications is the culprit, alternative therapy could be considered, if possible.
- **Diet:** Low-fiber, high-fat, or high-sugar diets and reduced fluid intake increase the propensity for constipation. Patients should be advised to increase their fiber and fluid intake.

Red Flags

It is very important to determine if the patient's constipation could be caused by an underlying medical condition. Presence of any of the following red flags prompts referral to a healthcare practitioner:

- Blood in stool or rectal bleeding
- Recent surgery (especially abdominal surgery)
- Family history of colon cancer
- Continuous abdominal pain
- Constipation for more than 2 weeks or no defecation for more than 7 days with the use of laxatives
- Presence of any associated symptoms indicating a more serious illness, e.g., fever, altered mental status, signs of severe dehydration, persistent vomiting
- Anemia
- Unexplained weight loss
- Age less than 2 years

Additional Assessment Considerations

Further assessment for patients with red flags is recommended. Assessment considerations include, but are not limited to, the following:

- Physical examination of the abdomen and rectal area
- Laboratory investigations to determine the secondary causes of constipation such as thyroid function tests, electrolytes, complete blood count, and test for occult blood
- Colonoscopy or sigmoidoscopy

Follow-Up and Monitoring

The goals of therapy are to ameliorate constipation through achieving satisfactory optimal defecation frequency, avoid complications such as hemorrhoids secondary to straining, and avoid adverse reactions of drug therapy. Achieving the "optimal" defecation frequency goal should be individualized according to the patient's normal bowel habits. For example, for patients having a bowel movement daily, it is reasonable to target a daily bowel movement. On the other hand, it is reasonable to target three defecations per week for patients having three to four defecations per week.

At follow-up assessment, patients deemed suitable for non-pharmacological and/or pharmacological measures should be asked if their constipation persists. Presence of constipation for more than 2 weeks or no bowel movements for more than 7 days despite the use of laxatives should

prompt referral to healthcare practitioners. A laxative is considered ineffective if there is no response following a trial period of 2–4 weeks at the recommended dose. In addition to checking for symptom control, pharmacists need to monitor if the patient experiences adverse reactions to pharmacological management. Adverse reactions include, but are not limited to, abdominal cramping, bloating, flatulence, diarrhea, and nausea.

Clinical Pearls

- Pharmacists play an important role in identifying red flags in patients presenting with constipation.
- Assessment of patients presenting with constipation involves assessment of the characteristics and history of patients' constipation and the presence of any associated symptoms.
- Pharmacists need to assess for medication-induced constipation and determine the need for alternative therapies.

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Chapter Objectives

1. Describe the epidemiology, etiology, risk factors, and pathophysiology of heartburn.
2. Assess adult patients presenting with heartburn in the community pharmacy setting.
3. Identify alarm features in adults presenting with heartburn that prompt referral to a physician or emergency department.
4. Conduct follow-up assessment of a patient on long-term proton pump inhibitor (PPI) therapy with a view towards reducing the dose or stopping the PPI altogether.

Background

A patient comes into the pharmacy requesting your advice for treatment of heartburn symptoms. What is the process for assessment and determining if self-treatment or pharmacist-prescribed therapy is appropriate? To answer this question, pharmacists must have a fundamental knowledge of the common underlying causes of heartburn, potential differential diagnoses, and be able to gather relevant patient-specific information to determine if the patient can safely pursue self-care, be managed by a pharmacist, or requires referral to a physician. The presumptive diagnosis of gastroesophageal reflux disease (GERD) may be established in patients with typical heartburn and regurgitation without alarm symptoms. In physician-centric clinical practice guidelines, it is suggested that empiric proton pump inhibitor (PPI) therapy can be initiated in those with GERD symptoms without further

workup [1–3]. Therefore, pharmacists are uniquely placed to play a role in the evaluation of patients with heartburn symptoms and direct individuals to effective over-the-counter therapies such as antacids, alginates, histamine 2 receptor blockers (H2RA), and PPIs for self-treatment [4–7]. Pharmacists in some jurisdictions may prescribe prescription medications (e.g., H2RA, PPI) for pharmacist-supervised therapy according to minor ailment treatment protocols/colaborative practice agreements, or refer patients with alarm symptoms for physician care [8, 9].

Classification of Upper GI Symptoms: GERD vs. Dyspepsia

Heartburn, defined as a burning sensation in the retrosternal area (behind the breastbone), and *regurgitation*, defined as the perception of flow of refluxed gastric content into the mouth or hypopharynx, are the primary symptoms of *gastroesophageal reflux disease* (GERD) [10]. GERD is defined by consensus as “A condition that develops when the reflux of gastric contents into the esophagus causes troublesome symptoms and/or complications.” [10] Notably, some degree of gastroesophageal reflux, or movement of gastric contents into the esophagus is physiologic [11]. Heartburn is considered troublesome if mild symptoms occur 2 or more days per week, or moderate/severe symptoms occur more than one day per week [10]. In situations where endoscopic evaluation is appropriate, GERD can be further classified as nonerosive reflux disease (i.e., the presence of symptoms without visible erosions) or erosive reflux disease (i.e., symptoms and erosions are present on endoscopy; aka erosive esophagitis) [1].

In addition to heartburn and regurgitation, GERD may manifest with other “atypical” or “alarm” symptoms, which should prompt referral to a physician or an emergency room for further investigation depending on the severity of the condition (Table 11.1). For example, chest pain may be a symptom of GERD and the pain may cardiac or noncardiac in nature. It is essential to distinguish cardiac from

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Table 11.1 Gastroesophageal reflux disease: typical symptoms, atypical symptoms, and alarm features and that require referral to a physician

Typical symptoms	Atypical and/or nonspecific symptoms	Alarm symptoms
Heartburn (A burning sensation in the retrosternal area (i.e., behind the breastbone) Regurgitation (The perception of flow of refluxed gastric content into the mouth or hypopharynx)	Chest pain Nausea Belching Epigastric pain Respiratory symptoms (recurrent cough, hoarseness, wheeze, rhinosinusitis) Choking attacks, especially at night Nocturnal awakening	Persistent vomiting (not associated with cannabis use) Gastrointestinal bleeding (hematemesis, melena) Iron deficiency anemia Involuntary weight loss (>5% over 6–12 months) Difficult/painful swallowing (dysphagia, odynophagia) Epigastric mass Family history of esophageal or gastric cancer New onset of symptoms $\geq$ 50 years of age

Adapted from Boardman 2015 [6], Armstrong 2016 [7], and Hunt 2017 [28], ACG/CAG Dyspepsia guidelines [14]

noncardiac chest pain before considering GERD as a cause of chest pain [2]. Dysphagia, chronic cough, asthma, and laryngitis are possible so called “extra-esophageal” symptoms [2]. GERD may also present with atypical symptoms including dyspepsia, epigastric pain, nausea, bloating, and belching but these symptoms also overlap with other conditions [2]. The main complications of GERD include erosive esophagitis, Barrett’s esophagus, esophageal strictures, esophageal carcinoma, and extra-esophageal symptoms as described above [1].

The presence of alarm features may suggest the presence of complications or malignancy. As shown in Table 11.1 the main alarm symptoms are:

- Vomiting (persistent, not associated with cannabis use)
- Evidence of gastrointestinal bleeding, (e.g., hematemesis, melena, hematochezia, occult blood in stool)
- New onset dyspepsia in patients $\geq$ 50 years of age
- Anemia
- Anorexia (loss of appetite)
- Weight loss (unexplained)
- Dysphagia (Difficulty swallowing) or odynophagia (pain when swallowing)
- Gastrointestinal cancer in a first degree relative

Patients presenting to family physicians with upper GI symptoms often present with multiple upper GI symptoms, making symptoms-based diagnosis in these patients difficult in practice [12, 13]. While heartburn is typically indicative of GERD, there is considerable overlap with dyspepsia symptoms. *Dyspepsia* is defined as predominant epigastric pain for at least 1 month [14]. Dyspepsia may be associated with other GI symptoms such as epigastric fullness, nausea, vomiting, or heartburn, provided epigastric pain is the patient’s primary concern [14]. Upon presentation and in the absence of investigation, patients may be labelled as having “uninvestigated” dyspepsia. Approximately 40% of patients with uninvestigated dyspepsia have an underlying organic cause for their symptoms (e.g., peptic ulcer disease, reflux esophagitis, gastroesophageal malignancy, drug induced dyspepsia, biliary pain,) while up to 60% have “functional” dyspepsia, that is, dyspeptic symptoms with no underlying organic cause on diagnostic evaluation [15]. It has been generally accepted that endoscopy is not routinely needed in those with a clinical diagnosis of functional dyspepsia [16]. Current dyspepsia clinical practice guidelines conditionally suggest endoscopy to exclude upper GI malignancy in dyspepsia patients age 60 or over based on very low quality evidence. [14] In dyspepsia patients under the age of 60, they suggest endoscopy is not necessary to investigate alarm features (conditional recommendation, moderate quality of evidence) and rather suggest noninvasive *H. pylori* testing in these patients (Strong recommendation, high quality evidence). Further, in those who are *H. pylori* negative or remain symptomatic after *H. pylori* eradication they strongly recommend empirical physician supervised PPI therapy based on high quality evidence. Others have suggested that in areas with known local prevalence of *H. pylori* <20%, proceeding straight to empiric PPI therapy in place of *H. pylori* testing as the preferred approach [17].

Epidemiology

GERD symptoms are prevalent worldwide. For example, a recent systematic review of 28 studies of GERD prevalence suggested that GERD defined as at least weekly heartburn and/or regurgitation is present in 18–28% of Americans and between 9% and 26% of Europeans [18]. The Canadian Consensus Conference Guideline states that “GERD is the most prevalent acid related disorder in Canada” (Level II-1, A evidence) [19]. This is primarily based on a population survey of 1000 Canadians where 17% reported heartburn in the previous 3 months; and 13% experiences moderate/severe upper GI symptoms weekly [20]. Clinically troublesome heartburn is seen in about 6% of the American population [21].

Etiology and Risk Factors

The development of GERD reflects an imbalance between aggressive and physiological defense mechanisms in the gastrointestinal tract [22]. The main causes and risk factors are listed in Table 11.2. Esophagitis, a complication of GERD, occurs when refluxed gastric acid and pepsin cause inflammation of the esophageal mucosa, causing erosions and ulcers [23].

The two main pathophysiologic mechanisms of GERD are gastroesophageal junction incompetence and impaired esophageal acid clearance [22]. Gastroesophageal junction incompetence may be caused by several mechanisms including spontaneous and transient relaxation not associated with swallowing, low resting lower esophageal sphincter (LES)

pressure, anatomic disruption of the gastroesophageal junction (e.g., hiatus hernia) and transient increase in intra-abdominal pressure (e.g., straining, pregnancy) [23]. Improper esophageal clearance of gastric fluid may be due to ineffective esophageal motility, or retrograde flow associated with hiatus hernia; and diminished salivation or salivary neutralizing capacity. While the extent of symptoms and mucosal injury is proportional to the frequency of reflux events, the duration mucosal acidification and the caustic potency of refluxed fluid, gastric acid hypersecretion is not usually a dominant factor in GERD [23].

There are several factors that may exacerbate or trigger GERD symptoms [22, 23].

- Lifestyle factors: Abdominal obesity, eating large meals, alcohol consumption, smoking, caffeine, stress.
- Pregnancy, gastric hypersecretory states, delayed gastric emptying, disruption of esophageal peristalsis.
- Certain foods (fatty or fried foods, coffee/tea or other caffeinated beverages, spicy foods, acidic foods: e.g., citrus, tomatoes, onions, or others: chocolate, mint.)
- Bending over or lying down after eating; wearing tight fitting clothing.

There are several risk factors for GERD. They include being overweight or obese, diet, pregnancy, smoking, and other medical conditions such as Crohn's disease, hypothyroidism, and use of medications [24, 25].

Table 11.2 Possible causes of GERD [24, 25, 28]

Drug causes	Decreased lower esophageal sphincter pressure
	Anticholinergic Alpha agonists Beta agonists Calcium channel blockers Estrogen Opioids (e.g., morphine) Theophylline Nitrates (e.g., isosorbide mononitrate) Benzodiazepines (e.g., diazepam) Clomipramine Barbiturates Direct irritation of mucosa Bisphosphonates Iron (e.g., ferrous sulfate) Potassium supplements Ascorbic acid Erythromycin, tetracycline, doxycycline, clindamycin Quinidine Chemotherapy (e.g., paclitaxel) Direct irritation of mucosa and COX inhibition Aspirin NSAIDs Corticosteroids
Disease causes	Mechanical or functional impairment of the esophagogastric junction, e.g., hiatus hernia Obesity Gastric reductive surgery Pregnancy
Other causes	Smoking Alcohol Foods High fat content, chocolate, peppermint (lower LES pressure) Spices, onions, citrus juice, coffee (direct mucosal irritation) Cola, beer, milk (stimulate acid secretion) Body position Bending over, lying down

NSAID nonsteroidal anti-inflammatory drug, COX cyclooxygenase, LES lower esophageal sphincter, GERD gastroesophageal reflux disease

Categorization and Management of Presumptive GERD

Numerous clinical practice guidelines exist regarding the management of GERD [1–3, 19, 26–28] and several have been specifically tailored to address the community pharmacist management of heartburn/GERD [4–7, 16, 29–32]. Care pathways or collaborative practice agreement algorithms to direct pharmacist assessment and prescribing of prescription medication are also available [33]. The various lifestyle and pharmacologic options for the management of heartburn and GERD are shown in Table 11.3 [34, 35].

Classically, symptom severity, frequency, and duration have been important in determining whether self-treatment is appropriate or referral for prescription PPI therapy should be pursued. GERD symptoms have been typically characterized as mild or moderate/severe [19].

- *Mild GERD* is characterized by infrequent reflux symptoms (<3 times per week), of low intensity (e.g., 1–3 of 10) and short duration (e.g., <3 months), and with minimal long-term effect on activities of daily living or quality of life.

Table 11.3 Non-pharmacologic and pharmacologic management strategies for gastroesophageal reflux disease

Non-pharmacologic strategies [34]	Examples
Lifestyle and dietary modification	Weight loss for those who are overweight Elevate the head of the bed by 10–20 cm particularly if nocturnal symptoms are present Elimination of dietary triggers in those who note correlation with GERD symptoms and experience symptomatic improvement Avoid eating up to 3 h before bedtime for those with nocturnal GERD Avoid lying down after meals Stop smoking Avoid alcohol Avoid tight clothing across the waist
Pharmacologic choices [35]	Generic name and adult dose
Antacids	Magnesium-aluminum hydroxide (e.g., Maalox, Mylanta) See label instructions for dosing Calcium carbonate 200–400 mg PRN (max 2 g elemental calcium in 24 h)
Alginates	Sodium alginate (e.g., Gaviscon 2–4 tsp QID) Sucralfate 1 g TID or QID
Histamine 2 receptor blockers	Ranitidine (OTC: 75 mg daily) 150 mg BID ^a Famotidine (OTC: 10 mg daily) 20 mg BID Nizatidine 150 mg BID Cimetidine 600 mg BID
Proton pump inhibitors ^b	Esomeprazole (OTC: 20 mg daily x 2 weeks) 40 mg daily Omeprazole 20 mg daily (OTC: 20 mg daily x 2 weeks) 20 mg daily Pantoprazole 40 mg daily Rabeprazole 20 mg daily Lansoprazole 30 mg daily Dexlansoprazole 30 mg daily
Potassium-competitive acid blockers ^c	Vonoprazan 20 mg daily
Other	Pepto Bismol 30 mL (2 tablets) q30–60 min PRN (Max 8 doses/day)

GERD gastroesophageal reflux disease, PRN as required, OTC over the counter

Note: There is limited evidence to support the effectiveness of lifestyle changes aside from weight loss and elevating the head of the bed

^aHealth Canada stopped the distribution of ranitidine products in September 2019, due to potentially high concentrations of the potential carcinogen N-nitrosodimethylamine (NMDA). As of January 2020, some companies have resumed ranitidine sales subject to additional safety testing [64]

^bDuration of prescription PPI therapy for GERD is 4–8 weeks

^cVonoprazan is not yet approved or available for use in Canada

- *Moderate or severe GERD* is characterized by more frequent, intense or prolonged symptoms. (e.g., daily attacks of reflux pain, symptoms present for >3 months, pain intensity 7–10 of 10, or symptoms that interfere with daily activities and occur at night.)

Based on these definitions, self-treatment with lifestyle modification and OTC medications is recommended for those with mild symptoms, while the recommended treatment for those with severe heartburn is once daily prescription PPI therapy [19, 36]. Patients with severe symptoms may have esophageal erosions characteristic of Barrett's esophagus. However, the relationship between presenting symptomatic frequency or severity and severity of esophageal injury is weak (i.e., those with less severe symptoms may have these conditions as well) [7].

In approaches that take into consideration the availability of OTC PPI therapy, heartburn is classified as either occasional (i.e., episodic) or frequent [6, 7, 16, 27, 29–31]. For example, the World Gastroenterology Organization (WGO) guidelines for community-based management of gastrointestinal conditions classify heartburn as being episodic when heartburn is mild or moderate, but infrequent [27]. Frequent heartburn is 2 or more days per week [7, 27]. In this approach, referral is suggested if symptoms have lasted for 3 months, or when severe or nocturnal heartburn is present [27]. The WGO guidelines suggest use of antacids, alginates, or OTC H2RA for those with mild or moderate episodic heartburn (i.e., <1x/week) and OTC PPI therapy for those with frequent heartburn [27]. In Canada, OTC PPI therapy with omeprazole or esomeprazole are indicated for the treatment of frequent heartburn only [37, 38]. While some are conservative and suggest referral and no OTC PPI therapy when symptoms have lasted >3 months, are severe or nocturnal [27], others suggest that an OTC PPI is appropriate but also to concurrently refer patients with severe GERD symptoms [4]. Most recent pharmacist specific algorithms recommend self-treatment with an OTC PPI trial for 2–4 weeks and no referral for those with frequent or bothersome symptoms [6].

Contemporary clinical practice guidelines from the American College of Gastroenterology and American Gastroenterological Association continue to recommend that treatment for GERD can be started without further investigations if no alarm features are present and that an 8-week course of PPI therapy is preferred [2, 3]. Those who fail to respond are sent for further investigation once being PPI-free for 2–4 weeks to determine the underlying cause of symptoms. They also suggest trying to discontinue therapy after 8 weeks if response to reduce long-term use of PPI's [2].

Care pathways/collaborative practice agreement protocols designed to facilitate pharmacist assessment and prescribing, are designed to consider self-management, OTC PPI therapy and provision of prescribed PPI therapy and are consistent with OTC PPI and physician created guidelines. Typically, they recommend OTC treatments and a step-up approach for those with infrequent symptoms and starting 4–8 weeks of prescription strength rather than OTC PPI therapy for those with frequent or moderate symptoms.

Diagnosis

The most useful tool in the diagnosis of GERD is the history [19, 39]. The presumptive diagnosis of GERD is often based on clinical symptoms alone in patients with classic symptoms of heartburn and/or regurgitation [10]. It is commonly reported that the presence of heartburn or acid regurgitation has a high specificity (89% and 95% respectively), but a low sensitivity (38% and 6%) to diagnose GERD [39]. A more recent systematic review of seven studies found that heartburn and regurgitation had a sensitivity of 30% to 76% and a specificity of 62% to 96% for the presence of erosive esophagitis [40]. Upper gastroesophageal endoscopy is not required to make a diagnosis of GERD. However, it can detect esophageal manifestations of GERD and identify upper GI tract malignancy. There are usually no physical signs of GERD and the role of physical assessment in the evaluation of heartburn, regurgitation or GERD is limited to evaluation and inspection to exclude other medical problems such as cardiac disease, asthma, or cancer [28].

The American College of Gastroenterology and American Gastroenterological Association GERD guidelines both suggest empiric PPI therapy is a reasonable approach to confirm GERD when it is suspected in patients with typical symptoms [2, 3]. A meta-analysis suggested that an empiric trial of PPI therapy had a 78% sensitivity and 54% specificity for predicting a diagnosis of GERD using pH monitoring as the reference standard [41]. However, the associated likelihood ratios (LR+ 1.7, LR- 0.41) indicate that both positive and negative responses have small effects on changing the probability of a GERD diagnosis.

Further investigations that may be requested by a physician to evaluate heartburn or GERD include endoscopy and esophageal pH monitoring. Barium radiographs should not be performed to diagnose GERD [2]. Endoscopy is recommended in patients with alarm symptoms or nonresponders. Ambulatory 24-hr pH monitoring and impedance monitoring are indicated in patients whose atypical reflux symptoms call a GERD diagnosis into question, in those who fail standard medical therapy but have normal endoscopy, and as preoperative evaluation before anti-reflux surgery [42]. *H. pylori* screening is not recommended in GERD. Eradication of *H. pylori* infection is not routinely required as part of anti-reflux therapy [1]. However, *H. pylori* testing is carried out when peptic ulcer disease is suspected.

The differential diagnosis for GERD includes cardiac disease, peptic ulcer disease, infectious esophagitis (e.g., *Candida*, herpes simplex), pill esophagitis, eosinophilic esophagitis, Schatzki ring, dysphagia caused by esophageal motility disorder, esophageal obstruction, esophageal cancer, reflux hypersensitivity, dyspepsia (e.g., functional dyspepsia), and biliary colic [23, 28].

Initial Assessment of New Self-Diagnosed Heartburn Symptoms in Adults

GERD management has been recognized as an area where pharmacists may have a large impact on meeting patient and health system needs [43, 44]. With the availability of over-the-counter PPI therapy for the short-term management of frequent heartburn and facilitating legislation allowing pharmacists in most Canadian jurisdictions to prescribe prescription PPI therapy, consensus treatment algorithms [4, 6, 16, 31, 33, 45] have been created to guide pharmacist assessment of heartburn symptoms in adults (Fig. 11.1). While each is different, all share commonalities in their approach to the patient presenting with new heartburn symptoms.

- **Assess the symptoms:** Confirming the indication for therapy by enquiring about the nature of the patient's symptoms [4]. Enquire if the patient has been previously diagnosed with GERD by a physician. Remember, while many patients may have other upper GI complaints, in GERD heartburn and regurgitation will be the predominant symptom.
- **Determine whether referral to a physician is required:** Referral is required in patients with alarm features, those with a familial or personal history of gastrointestinal cancer, age of onset of symptoms at age ≥ 50 , or in situations where patients have been nonresponsive to prior attempts at self-treatment with OTC H2RA or PPIs. If the patient's dominant symptoms are consistent with heartburn and regurgitation and the patient does not have any alarm or atypical symptoms, the community pharmacist can make a presumptive diagnosis of GERD.
- **Assess medical and medication history:** To assess drug-induced causes of reflux and assess the appropriateness of treatment alternatives.
- **Establish symptom frequency and severity:** Enquire about the characteristics and onset of symptoms to determine if patients have episodic or frequent heartburn. While some self-care treatment algorithms suggest OTC PPI therapy for patients with episodic heartburn [4, 6], it is recommended to treat episodic heartburn initially with antacids, alginates or H2RA and reserve OTC PPI therapy for patients with frequent heartburn [6, 27, 37, 38]. In select jurisdictions, treatment of frequent heartburn may be managed by 4–8 weeks of pharmacist prescribed PPI therapy similar to clinical practice guidelines for physician directed therapy.

Table 11.4 depicts a list of potential questions that will aid in proper assessment of adult patients with heartburn symptoms.

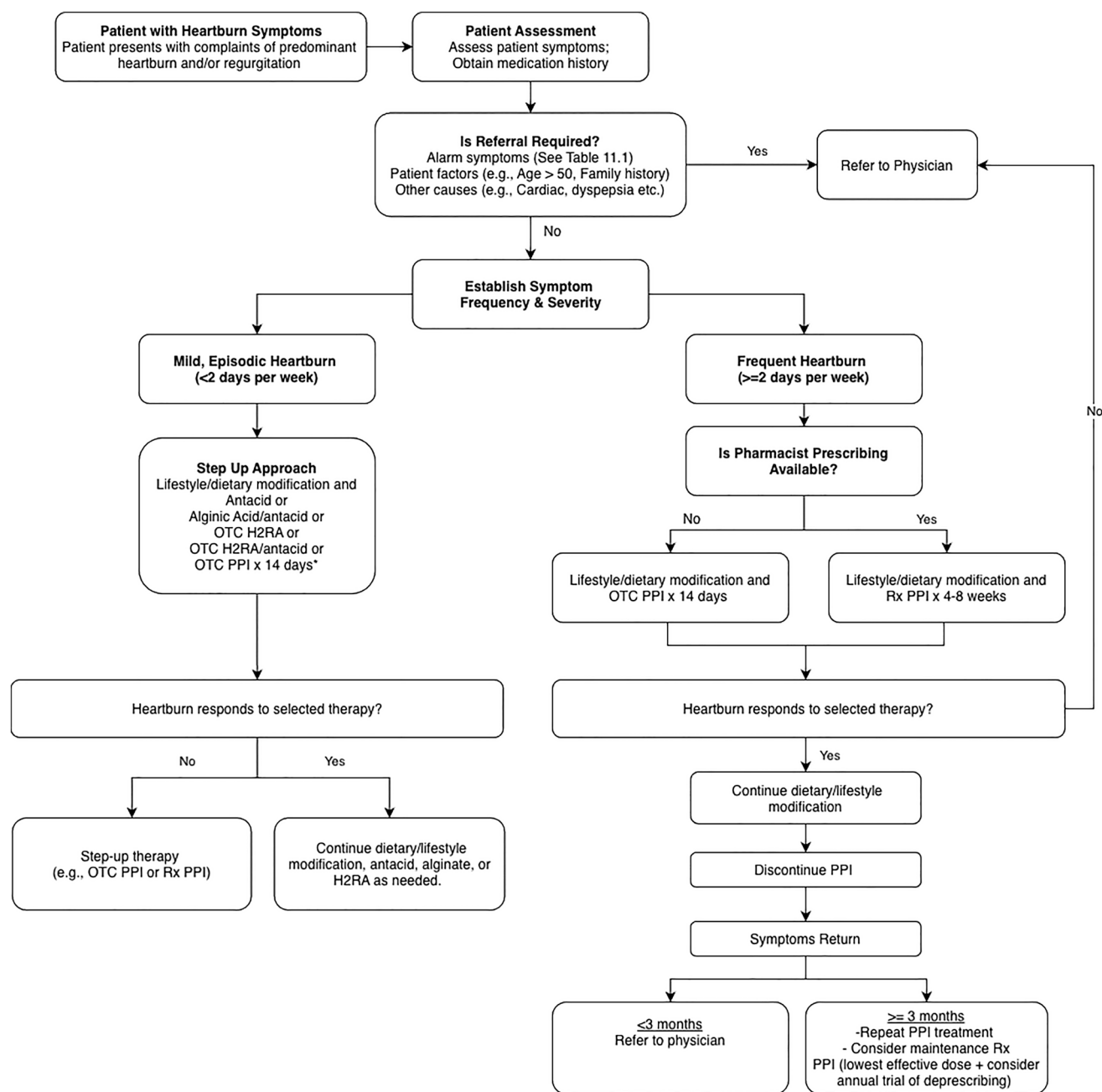


Fig. 11.1 Algorithm for self-care and pharmacist-based management of heartburn. OTC over the counter, H2RA histamine 2 receptor antagonist, PPI proton pump inhibitor (Adapted from: Hunt 2014 [27], Boardmann 2014 [5], and Katz 2022 [2]). *The indication for OTC PPI in Canada and the USA is for frequent heartburn occurring 2 or more

days per week. In other parts of the world, they are indicated for short term treatment of reflux symptoms regardless of frequency and are an option for patients with occasional symptoms that are very bothersome or who do not wish to take antacids/alginate or H2RAs [5]

Table 11.4 Potential questions to assess patients presenting with heartburn symptoms [6, 7]

Step	Question	Notes
Assess the symptoms	What is the nature of your symptoms?	Typical symptoms of heartburn and regurgitation suggest a presumptive diagnosis of GERD
	Have you been previously diagnosed with GERD by a physician?	Heartburn symptoms are substernal in location and may travel up the esophagus to the pharynx Atypical symptoms prompt referral Predominant epigastric pain symptoms just below the ribs in the upper central area of the abdomen (i.e., dyspepsia) require referral
Rule out referral for alarm symptoms	Are you experiencing difficulty swallowing or having pain when you swallow?	Presence of alarm features prompts referral
	Do you have any signs of internal GI bleeding? (e.g., black tarry stools, vomiting blood)	
	Are you over the age of 50 and are these new symptoms?	
	Have you lost a significant amount of weight (i.e., >5%) without trying?	
	Are you vomiting? If so, is it persistent?	
	Do you have a history of anemia, or signs of anemia?	
	Do you have a first-degree relative with a history of gastric and or esophageal cancer?	
Medical and medication history	For females: Are you pregnant?	Management starting with antacids according to a step-up approach is appropriate in pregnancy
	What medications are you currently taking?	
	Are you on chronic NSAID or ASA therapy?	
	Medical History? What other medical conditions has your doctor diagnosed?	
Characteristics and onset	How often do you get (heartburn) symptoms?	Episodic heartburn is mild or moderate, but infrequent and may be treated with antacids, alginates, H2RA, and OTC PPI
	On a scale of 1–10, how severe is your heartburn?	
	How long do your symptoms last?	Frequent heartburn is 2 or more days per week and may be treated with OTC or prescribed PPI
	Do your symptoms happen at night?	
	Do your symptoms effect your daily activities, work productivity?	
	How long have you been having (heartburn) symptoms? (i.e., when did your symptoms start)	
	Is this your first episode of heartburn, or is this a relapse of symptoms?	
	Have you ever had an endoscopy before?	
Aggravating/remitting factors	Have you tried any lifestyle changes or medications that have made your symptoms better or worse?	Retreatment with OTC PPI is appropriate for patients whose symptoms recur more than 3 months after stopping previously effective over the counter proton pump inhibitor therapy Patients with refractory symptoms require referral

GERD gastroesophageal reflux disease, PPI proton pump inhibitor, OTC over the counter, GI gastrointestinal

Symptoms and Alarm Features

The pharmacist should start by asking the patient to describe their symptoms and listen for cues regarding typical, atypical, and alarm symptoms. When evaluating heartburn symptoms, it is essential to differentiate scenarios where patients are experiencing new undiagnosed symptoms, or whether they have physician-diagnosed GERD. Among the first questions, the pharmacist should enquire about atypical and alarm symptoms (as shown in Table 11.1) to identify any Red Flags and if any are present, the patient should be referred to their physician. Patients with atypical or extraesophageal symptoms such as chest pain, chronic cough,

hoarseness, sore throat, shortness of breath, and wheezing should be referred to a physician for further workup [7].

Reflux-like symptoms can occur in other GI conditions (e.g., functional dyspepsia) and the treatment options are different [7]. If the description of symptoms is more consistent with dyspepsia, then peptic ulcer disease, GI cancer, and functional dyspepsia are part of the differential diagnosis. In this case, referral to a physician for *H. pylori* testing, potential endoscopy or other testing is recommended [14]. As part of the process, the pharmacist should be aware of other possible causes of dyspepsia, including cardiac or hepatobiliary causes of epigastric pain.

History

Gathering a medical and medication history is necessary in patients with heartburn symptoms to identify potential drug induced dyspepsia and assess the appropriateness of the different therapeutic options. The use of chronic NSAID or ASA therapy is a well-known risk factor for peptic ulcer disease and may suggest that dyspeptic symptoms are related to peptic ulcer disease rather than GERD. Several other classes of drugs are known to cause dyspepsia and esophageal complications, and these should be ruled out (See Table 11.2). Additionally, there are several clinically important drug interactions that pharmacists should aim to avoid when suggesting therapy for GERD. For example, an acidic stomach environment is required for absorption of specific antiretroviral drugs and in these situations proton pump inhibitor therapy may be absolutely contraindicated [46]. Another notable drug interaction of PPIs is the interaction of omeprazole and clopidogrel post cardiac stent [47]. Several classes of medications (e.g., fluoroquinolones, tetracyclines) are known to have clinically significant interactions with antacids [48].

Assessment of Patients with Recurrent Symptoms

Clarifying if the patient is experiencing a first episode or recurrence of heartburn symptoms helps the pharmacist orient themselves during the patient assessment and in the determination of whether or not repeated self-treatment is appropriate. The assessment of patients with recurrent symptoms is consistent with the initial assessment of new patients described above. However, the pharmacist must gather more history regarding the resolution of symptoms and previous treatment to make appropriate recommendations. A history of a previous endoscopy likely indicates an established history of GI disease, and in this case, self-treatment may not be appropriate.

Patients who have refractory symptoms (i.e., persistent symptoms despite appropriate treatment) should be referred to a physician [6, 37]. This may relate to incorrect diagnosis, nonadherence, or inadequate acid suppression. Patients who have initiated OTC PPI therapy and had a resolution of GERD symptoms but experience a recurrence >7 days but less than 3 months despite a trial of over the counter PPI therapy, are recommended to be referred to their family physician [6]. If symptoms recur more than 3 months after stopping previously effective over the counter PPI, retreatment is appropriate [6]. For pharmacist prescribed therapy, inadequate response after 4–8 weeks of PPI treatment should also prompt referral to a physician [33]. Recurrence after 3 months may be eligible for retreatment with previously

used medication. If recurrence is unexplained, referral to physician is suggested.

Assessment and Management of Heartburn in Pregnancy

Heartburn is common in pregnancy, with a reported incidence between 17% and 45% [49]. It typically is more common in the latter stages of pregnancy. For example, studies have found the prevalence to increase from 22% in the first trimester, to 39% in the second trimester, and 72% in the third trimester [50]. The cause of heartburn is multifactorial. Pregnancy likely aggravates GERD due to a decrease in lower esophageal sphincter pressure caused by changes in hormonal status, displacement of the lower esophageal sphincter into the thoracic cavity, and increased intraabdominal pressure [49, 51]. Heartburn usually resolves after delivery [52].

Pharmacist tailored OTC PPI treatment algorithms generally do not specifically address management of heartburn in pregnancy [4, 6, 16], but the 2013 WGO guideline recommends that self-treatment is not appropriate [27]. Product monographs for the OTC PPI available in Canada suggest pregnant or breastfeeding females talk to their doctor or pharmacist before a trial of therapy [37, 38]. However, some minor ailment pharmacist prescribing algorithms for GERD management permit pharmacist directed therapy, including PPI therapy to be initiated [33].

The evidence for lifestyle and pharmacologic management of heartburn in pregnancy has been summarized elsewhere [49, 53]. However, establishing the efficacy of interventions in this context has been limited by a lack of trials in this population. A step-up approach beginning with lifestyle and dietary modification, followed by antacids, alginates, or sucralfate, then H2RA, and finally PPI is recommended [28, 54]. Antacids and alginates are safe in pregnancy and breastfeeding. Calcium based antacids are preferred because adverse effects are rare and they have been shown to be beneficial in the prevention of hypertension and pre-eclampsia and reduce the composite outcome of maternal death or serious morbidity [55]. Sodium bicarbonate and long-term or high doses of magnesium trisilicate should be avoided [28]. There is a lack of clinical trial evidence regarding the efficacy of H2RA in heartburn in pregnancy [49]. Despite this, ranitidine or famotidine can be combined with antacids if antacids alone are insufficient. Famotidine and ranitidine are both considered compatible with pregnancy and are considered by experts as safe in pregnancy [52, 56].

There is limited evidence for proton pump inhibitors in pregnancy. However, proton pump inhibitors appear to be safe in pregnancy [57, 58]. A group of experts made a strong recommendation based on moderate quality evidence that

there is no contraindication for the use of Category B OTC PPIs for heartburn during pregnancy [29]. Historically, omeprazole was the only PPI considered FDA category C based on potential embryotoxic and fetotoxic effects in animal studies and similar concerns in human case reports. However, the standard FDA pregnancy category letters for prescription medications (i.e., A, B, C, D, and X) has been eliminated [61]. Available data suggests that omeprazole is not teratogenic in humans and recognize that omeprazole has the largest reported safety experience as it has been used in the largest number of treated patients [59]. Authoritative resources currently classify omeprazole, pantoprazole, and lansoprazole as “human data suggests low risk.” [60]

Follow-Up Assessment

Newly Initiated Therapy for Heartburn

Follow-up assessment of patients initiated on OTC treatments for heartburn is not mandatory in all situations as the goal of self-treatment is to have the patient become symptom free. If patients are started on OTC PPI, inform them to call back to the pharmacy/consult a physician if symptoms have not resolved in 2–4 weeks’ time as this suggests the presence of peptic ulcer disease or erosive esophagitis. A potential risk of continuing self-treatment with OTC PPI beyond 2 weeks is delaying presentation for early esophageal or gastric cancer, but this has not been borne out in clinical practice [29]. Return of symptoms after a period of months after successful therapy may be an indication for another course of therapy [6]. Referral to a physician is suggested if:

- Patients do not respond to initial OTC PPI therapy.
- Individuals have persistent (>1 month) or recurrent symptoms after use of a PPI [29].
- More than one course of OTC PPI treatment every 4 months is necessary [37, 38].

For those prescribed PPI therapy by a pharmacist, follow up should be completed at 4 weeks to assess symptom resolution. If therapy is appropriate and continued, follow-up should also occur at 8 weeks to discuss deprescribing of therapy.

Patients on Long-Term PPI: Deprescribing

While therapy for GERD is typically limited to 4–8 weeks, chronic use of PPI therapy is problematic [61]. As the adverse consequences of long-term PPI therapy have become better recognized, deprescribing (i.e., stopping, stepping down, or reducing doses) of PPI therapy in appropriate situations has

become an area of focus [62]. An evidence-based guideline targeted at adults over 18 years of age taking a continuous PPI for longer than 28 days for GERD or esophagitis suggests assessing the indication for PPI therapy as the starting point in determining if it is appropriate to deprescribe PPI therapy [61]. Long-term PPI therapy is appropriate for patients with Barrett esophagus, severe esophagitis (grade C or D on endoscopy), documented history of bleeding GI ulcers, and chronic NSAID users with bleeding risk.

An attempt to stop or reduce PPI therapy should be attempted at least once per year in most patients [63]. Questions will assist in determining if long-term PPI therapy continues to be appropriate or if de-prescribing is acceptable are as follows:

- Why are you taking a PPI?
- Have you ever had an upper GI endoscopy before?
- Do you have a history of upper GI bleeding? Ever been hospitalized for GI bleeding?
- Are you currently taking an NSAID (on chronic NSAID therapy)?

Recognize that caution is needed when deprescribing PPI as a systematic review of stopping or lowering the dose of PPI in adults compared to long term daily PPI, suggests that on demand prescribing may lead to an increase in GI symptoms [62].

Clinical Pearls

- Assessment of heartburn is aimed at determining if self-treatment (including OTC PPI) is appropriate, whether pharmacist-directed management with prescription PPI therapy, or physician referral is required. Consider age, comorbid conditions, concomitant medications, presence of alarm symptoms, and other risk factors [6]
- Alarm features and chest pain should be evaluated promptly by a physician to rule out cardiovascular disease, upper GI carcinoma, and peptic ulcer disease [16, 32].
- A presumptive diagnosis of GERD can be based on clinical symptoms of heartburn and/or regurgitation without further investigation or referral.
- In jurisdictions where permitted, pharmacists may prescribe acid suppressing therapy including once daily PPI therapy for 4–8 weeks. Refer those who are not responsive for physician care.
- Alternatively, if pharmacist prescribing is not available, pharmacists can recommend 2 weeks of OTC PPI therapy to those with frequent heartburn symptoms.
- Point-of-care tools are available to assist in shared decision making regarding stopping or reducing doses of PPI

therapy at [Deprescribing.org](https://deprescribing.org) <https://deprescribing.org> and Choosing Wisely Canada <https://choosingwiselycanada.org/recommendation/gastroenterology/>

Links to Major GERD Clinical Practice Guidelines

- American Gastroenterological Association Clinical Practice Update on the Personalized Approach to the Evaluation and Management of GERD: Expert Review (2022) [3]
- American College of Gastroenterology: Guidelines for Diagnosis and Management of Gastroesophageal Reflux Disease (2022) [2]
- World Gastroenterology Organization Global Guidelines. GERD Global Perspective on Gastroesophageal Reflux Disease (2017) [28].
- Canadian Association of Gastroenterology: Canadian Consensus Conference on the management of Gastroesophageal Reflux Disease in Adults, Update. (2005) [19]
- Canadian Association of Gastroenterology: Pharmacist-specific Guidelines for the Medical Management of GERD in Adults (2008) [32]

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Fever Assessment

12

Mark Diachinsky

Chapter Objectives

1. Describe the etiology, pathophysiology, and categories of fever in different age groups.
2. Conduct initial assessment of patients presenting with fever.
3. Identify red flags in patients presenting with fever that prompt referral to other health-care practitioners or the emergency department.
4. Apply the general principles of antipyretic drug therapy and follow-up assessment or referral in patients with persistent fevers.

Background

A parent comes into your pharmacy requesting something to treat their child's fever. What information do you need to gather to conduct a proper assessment of the child? It is important for pharmacists to have an understanding of the common causes of fevers. In addition, it is important to gather relevant, patient-specific information on symptom assessment and a thorough past medical history in order to make an appropriate recommendation for therapy or referral.

Definition and Epidemiology

In its simplest definition, fever is any abnormal elevation of core body temperature above the usual range for an individual as a result of the body increasing the temperature set point. Fever, also known as pyrexia, is a complex physiological response that results from the activation of numerous physiological, endocrinologic, and immunologic systems [1]. It is

most often associated with infectious sources but can be attributed to noninfectious diseases as well. It is a common reason for patients to seek medical advice, especially in children. In the United States, it accounts for up to 25% of emergency department (ED) visits in children, 15% in the elderly, and 5% in adults [2, 3]. In Canada, it is among the top reasons for ED visits for patients under 5 years of age [4].

Pathophysiology of Fever

Fever is the result of a biological response mediated and controlled by the central nervous system (CNS), largely in response to circulating cytokines and prostaglandins [5]. The hypothalamus is responsible for regulating body temperature through input from nerve receptors in the skin that measure the surrounding environment relative to the temperature of the blood surrounding the hypothalamus. In response to a trigger, the inflammatory cascade releases cytokines, which leads to prostaglandin-2 being released from peripheral tissues, which subsequently raises the hypothalamic temperature set point via cAMP release [3].

Other symptoms often accompany fever due to the physiological response of the body working to maintain the elevated temperature set by the hypothalamus. For example, shivering increases heat production through increased activity of the muscles along with vasoconstriction of peripheral blood vessels to conserve heat [6]. An increase in heart rate may also accompany fevers, anywhere from approximately three to ten beats per minute [7, 8]. Febrile seizures are seizures associated with fevers in childhood without a known cause (i.e., CNS infection or epilepsy). It occurs in 3–5% of children aged 6 months–6 years, and the likelihood of a recurrence of a febrile seizure is higher after an initial episode [9]. Febrile seizures can be distressing for children and caregivers, and although antipyretic therapy is commonly initiated to prevent the recurrence of febrile seizures, its efficacy has not been demonstrated, and its use is generally not

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recommended for these patients [2, 9]. Other symptoms may include somnolence and malaise, cold extremities, a hot forehead, and a subjective feeling of cold [6].

Fever is a normal physiological response that aids the immune response. It is usually self-limiting and generally harmless. Presence or absence of fever can be used to monitor the progression of infections. Pharmacological management of fever is mainly for the relief of discomfort associated with fever. Therapy includes the use of antipyretics, such as acetaminophen and nonsteroidal anti-inflammatory drugs (NSAIDs). Alternating antipyretics is not routinely recommended due to the risk of adverse effects or error in dosing, especially in children. If significant discomfort is present prior to the next dose of an antipyretic, alternating antipyretics should only be considered if proper monitoring of compliance can be ensured. Table 12.1 provides a list of commonly used antipyretics. Various formulations of acet-

aminophen and ibuprofen are available, and patients should be counseled to read the label to ensure the appropriate dosage. Various cough and cold medicines also contain acetaminophen or ibuprofen, so patients should be advised of the risk of multiple sources of acetaminophen or ibuprofen. Dosages for children are determined based on body weight, and caregivers should be counseled to ensure the correct dosage and formulation are given. In particular, careful attention to liquid concentrations is important, as some medications can come in multiple different concentrations, and this is a common cause of dosing errors. Non-pharmacological management may include maintaining ambient temperatures, avoidance of physical exertion, removal of excess clothing or bedding, and increasing fluid intake [10].

Hyperthermia

Fever differs from hyperthermia as it results from the body maintaining a higher body temperature through thermoregulatory effectors [5]. In contrast to fever, hyperthermia is an increase in body temperature due to dysfunctional thermoregulation [2]. Hyperthermia is the result of increased heat production or decreased heat dissipation, and therefore a loss of the body's responsiveness to environmental thermal conditions [6]. These heat-related illnesses are often divided into heat exhaustion and heat stroke, or malignant hyperthermia. Heat exhaustion is less severe and results in a core temperature between 37 and 40 °C, whereas in a heat stroke, the core temperature exceeds 40 °C [11]. Symptoms of heat exhaustion include anxiety and confusion, dizziness, fatigue, weakness, nausea, headache, hypotension, and cutaneous flushing. Symptoms of heat stroke include anhidrosis, cardiac arrhythmias, hyperventilation, and more severe mental status changes such as ataxia, coma, irritability, and possible seizures [11]. Management of heat exhaustion includes removal of the person from the hot environment and hydration with continued monitoring. If the condition progresses to heat stroke, the patient requires more immediate cooling, and medical attention must be sought immediately. Hyperthermia can be a life-threatening event and requires immediate attention and first aid.

Diagnostic Criteria

Surprisingly, considering fever is a cardinal symptom reported to clinicians, there has been long-standing controversy of the criteria that defines fever [1]. *Steadman's Medical Dictionary* defines fever as "a bodily temperature above the normal of 98.6 °F (37 °C)"; however, textbooks in medicine and physiology vary in their definition of the upper limit of normal body temperature from 37.1 to 38 °C

Table 12.1 Commonly used antipyretics

Drug class	Dose	Adverse reactions
Acetaminophen	<i>Children:</i> 10–15 mg/kg/dose Q4-6H PO/PR PRN Maximum 5 doses per day or 75 mg/kg/day (not exceeding adult doses) <i>Adults:</i> 325–650 mg Q4-6H PO/PR PRN Maximum 4 grams per day	Hepatotoxicity (*note different product concentrations available: 32 mg/mL, 80 mg/mL)
NSAIDs	Ibuprofen <i>Children:</i> 5–10 mg/kg/dose Q6-8H PO PRN Maximum 4 doses per day or 40 mg/kg/day (not exceeding adult doses) <i>Adults:</i> 200–400 mg Q6-8H PO PRN Maximum 3.2 grams per day Acetylsalicylic acid (ASA) <i>Children:</i> NOT RECOMMENDED <i>Adults:</i> 325–650 mg Q4-6H PO PRN Maximum 4 grams per day Naproxen <i>Children greater than 12 years and adults:</i> 220 mg Q8-12H PO PRN Maximum 440 mg per day	GI discomfort, GI bleeding, dizziness, headache, diarrhea, skin rash, allergic reactions, reduced renal function, water retention, platelet dysfunction, Reye's syndrome (ASA in children less than 18 years old) (*note different product concentrations available: 20 mg/mL, 40 mg/mL)

GI gastrointestinal, NSAIDs nonsteroidal anti-inflammatory drugs, PR rectal route, PRN when necessary, Q every, H hours, PO oral route

(98.8–100.4 °F) [1, 12]. The most utilized definition of a fever is a core body temperature of 38 °C (100.4 °F) or higher [2]. A body temperature greater than or equal to 41.5 °C (106.7 °F) is considered hyperpyrexia [3]. However, consideration of the patient is important, as fever is an elevation of core temperature that is normal for an individual and defining a general upper limit of normal for a population may be misrepresentative [1]. It is also important to note that the above definitions refer to core body temperatures, whereas measured temperatures at a particular site (e.g., rectal or axilla) are an estimation of core temperature, and there are substantial differences between sites.

While early investigations by Carl Wunderlich in 1868 demonstrated a normal core body temperature of 37 °C, a more recent study showed that core temperatures vary based on age, sex, and time of day [13]. Mackowiak et al. investigated 148 healthy men and women aged 18–40 years and found an overall mean oral temperature of 36.8 °C, with a range of 35.6–38.2 °C. This same study looked at variations during the day and reported a maximum oral temperature (within the 99th percentile) of 37.2 °C in the morning and 37.8 °C in the afternoon. The authors concluded that, for young healthy adults, a fever may be more accurately defined as an early morning temperature of greater than or equal to 37.2 °C or a temperature of greater than or equal to 37.8 °C at any time during the day [13]. Additionally, studies have shown that women have higher normal body temperatures than men [13–15]. For women, body temperature varies due to the menstrual cycle, with about 0.4 °C higher body temperature during the luteal phase compared to the follicular phase [16]. Elderly populations are generally thought to have lower body temperatures than younger adults; however, comparisons between groups of young and elderly subjects showed that the group of elderly subjects had lower axillary and oral temperatures but had similar rectal temperatures to the young subjects [1, 17]. The normal body temperature in children is not as well defined, although generally tend to be higher than adults [18]. Overall, infants have higher mean body temperatures that begin to decrease toward adult ranges beginning at approximately 1 year of age, until stabilizing around 13–14 years of age in females and 17–18 years of age in males [1]. Healthy infants can have normal body temperatures as low as 36 °C during sleep and temperatures as high as 37.8 °C during active parts of the day, including after feeding [2]. Due to the considerable variation between individuals and within the same individual, it is not surprising that it is difficult to define an upper limit of normal. While the most common definition of a fever is 38 °C or greater, it is important to consider individual differences, and therefore clinical decisions should be based on the patient's normal temperature variations when available.

Methods of Measurement

Body temperature is most commonly measured in units of Celsius or Fahrenheit, and often the two are used interchangeably. A conversion table between Celsius and Fahrenheit is provided in Table 12.2. The most common methods of measuring body temperature are via the rectum, mouth, axillary (armpit), tympanic (ear), or skin (often forehead, or temporal artery). As previously noted, these measurements provide an estimation of core temperature and vary among sites. While rectal measurements are often considered the gold standard [2, 19–21] because they provide the most accurate estimation of core temperature, this method is also discouraged by other guidelines because of safety concerns, practical issues, and the discomfort it may cause (both physical and psychological) [22, 23]. Additionally, rectal measurements are contraindicated in immunocompromised or neutropenic patients due to the risk of bowel perforation and infection [24]. Oral thermometers are often recommended due to their accuracy; however, they are on average 0.5 °C lower than rectal measurements [2]. This method is not considered suitable for children less than 5 years of age and may be uncomfortable for patients [2]. Axillary temperature measurements are often preferred for their practicality and reasonable accuracy (however, it is also not as sensitive as rectal measurements) [2]. Axillary temperatures may be a more suitable method for infants as they appear to be more reliable and are within 0.25–0.5 °C of rectal temperatures. Axillary temperatures have a greater difference from rectal temperatures in older children (0.5 °C or greater) [2]. Infrared tympanic thermometers are a popular choice among patients, especially parents, due to their ease of use with young children. However, they are not accurate in infants less than 3 months and also have a low sensitivity [2]. Body temperature can also be measured directly from

Table 12.2 Conversion table from Celsius to Fahrenheit for temperature (T) measurements

Celsius (°C)	Fahrenheit (°F)
35.5	95.9
36.0	96.8
36.5	97.7
36.8	98.2
37.0	98.6
37.5	99.5
38.0	100.4
38.5	101.3
39.0	102.2
39.5	103.1
40.0	104
$T_{(°F)} = T_{(°C)} \times 1.8 + 32$	
$T_{(°C)} = (T_{(°F)} - 32)/1.8$	

Table 12.3 Recommended sites of temperature measurement

Age	Recommended technique
0–2 years	1. Rectal (most accurate) 2. Axillary (armpit) Tympanic thermometers not recommended
2–5 years	1. Rectal (most accurate) 2. Axillary (armpit) or tympanic (ear) 3. Hospital setting: temporal artery
> 5 years	1. Oral (most accurate) 2. Axillary (armpit) or tympanic (ear) 3. Hospital setting: temporal artery

Adapted from the *Canadian Pediatric Society* [21]

Table 12.4 Normal temperature ranges

Method of measurement	Normal range of temperature
Rectal	36.6–37.9 °C (97.9–100.2°F)
Tympanic (Ear)	35.8–37.9 °C (96.4–100.2°F)
Oral (Mouth)	35.5–37.5 °C (95.9–99.5°F)
Axillary (Armpit)	36.5–37.5 °C 97.8–99.5°F)

Adapted from the *Canadian Pediatric Society* [21]

central sites. Central sites include intravascular, urinary bladder catheter, esophageal, and also include the rectum [23]. These are invasive methods, and other than via the rectum, would have to be done under trained health professionals within institutional settings, or an intensive care unit in a hospital. For the ambulatory patient or patients seen in the community, the only feasible central site of measurement is via the rectum. A summary of the recommended sites of measurement for children are shown in Table 12.3. Normal temperature ranges for each site of measurement are depicted in Table 12.4.

Etiology

Fever can be caused by infectious and noninfectious causes. Infectious causes are usually due to bacterial or viral infections, but there can be other causes of infections. Noninfectious causes can include immune-mediated and inflammatory causes, as well as certain drugs, diseases, and malignancies. When the cause of a fever cannot be determined, it is called “fever without source” (FWS) [2]. Height of fever may not define the severity of illnesses; however, there is an association with a greater risk of serious bacterial infections when temperatures are greater than 39 °C, especially in children less than 6 months [2]. Serious bacterial infections can include urinary tract infections, pneumonia, sepsis, or meningitis. Temperatures above 41 °C have been associated with a higher risk of meningitis [2]. Temperature alone does not define serious bacterial infections, and infections can occur in patients with normal temperatures. Physical signs such as pallor, mottled appearance, blue skin color, reduced activity, poor feeding, no smile, decreased response to stimuli, weak high-pitched cry in infants, tachy-

Table 12.5 Possible causes of fevers

Infectious causes	Bacterial (e.g., UTIs, otitis media, pneumonia, bacteremia, bacterial meningitis) Viral (e.g., influenza, HIV, viral meningitis, COVID-19) Parasitic (e.g., malaria, toxoplasmosis, giardiasis) Arthropod (e.g., Lyme, Rocky Mountain spotted fever) Fungal (e.g., candidiasis, blastomycosis, histoplasmosis)
Noninfectious causes	Malignancy (e.g., leukemia, lymphoma) Autoimmune (e.g., rheumatoid arthritis, systemic lupus erythematosus) Drug reaction (e.g., allergic reaction, metabolic consequences of a drug, side effect) Vaccinations (more common in children) Environmental fever (e.g., heat stroke, excess exercise) Hyperthyroid or thyroid storm Neurologic (e.g., intracranial hemorrhage) Pulmonary embolus Myocardial infarct Renal infarct Blood transfusion reaction Factitious fever (e.g., Munchausen’s vs. Munchausen’s by proxy) Neuroleptic malignant syndrome Malignant hyperthermia Serotonin syndrome

Adapted from [3]

HIV human immunodeficiency virus, *UTIs* urinary tract infections

pnea, tachycardia, capillary refill time greater than 3 s, and a reduced urine output may be indicative of a serious bacterial infection and should prompt a referral [2]. Symptoms such as vomiting, drowsiness, confusion/irritability, stiff neck/joints and pale blotchy skin may also be signs of sepsis or meningitis. Elderly patients may have less intense fevers in response to infection and may become hypothermic when infected [10, 25]. They have greater morbidity and mortality from infections, and individuals older than 60 years require more careful monitoring [10, 25]. Older individuals are also more susceptible to heatstroke [11]. Table 12.5 outlines possible infectious and noninfectious causes of fever. Noninfectious causes including pulmonary embolism (PE), intracranial hemorrhage (ICH), neuroleptic malignant syndrome (NMS), thyroid storms, heatstroke, and some drug fevers can be more life-threatening [3]. Hyperpyrexia (body temperature greater than 41.5 °C) may occur in sepsis but are more common with those with ICH, neuroleptic malignant syndrome (NMS), and heatstroke [3]. It can also occur with malignant hyperthermia, serotonin syndrome, thyroid storms, anticholinergic toxidrome (e.g., tricyclic antidepressants), or sympathomimetic toxidrome (e.g., amphetamines, cocaine) [3]. Drug fevers are febrile responses related to the administration of a drug [26] and can be caused by various mechanisms. More serious examples of drug fevers include neuroleptic malignant syndrome, serotonin syndrome, or

toxic ingestion. The prevalence of drug fever is unknown in the general population but occurs in approximately 10% of hospital admitted patients [26]. These fevers may occur at any time during drug therapy but frequently occur after 1–2 weeks of therapy; however, long-term drug therapy should not exclude the possibility of drug fevers [26]. These fevers can sometimes be accompanied by a maculopapular

rash. Examples of drugs that can commonly cause fevers and possible known mechanisms can be found in Table 12.6. Furthermore, certain patterns of fevers can also be observed with various diseases (Table 12.7). Patterns may include different rates of temperature changes, fevers during specific times of the day, every 3 or 4 days, in sporadic episodes, or for long periods of time [1].

Table 12.6 List of drugs implicated to cause fever

Altered thermoregulatory mechanisms	<i>Increase body metabolism:</i> Exogenous thyroid hormones (e.g., levothyroxine), monoamine oxidase inhibitors <i>Vasoconstriction, limiting heat dissipation:</i> Epinephrine, cocaine, amphetamines <i>Impair sweating:</i> Anticholinergics (e.g., atropine, antihistamines, tricyclic antidepressants, phenothiazines, butyrophenones) <i>Blocking histamine 2 (H2) receptors in the hypothalamus:</i> H2 receptor antagonists (e.g., cimetidine) <i>Raising catecholamine synaptic concentrations (sympathomimetic poisoning syndrome):</i> Cocaine, methamphetamines, amphetamines <i>Altered function of eccrine sweat glands causing oligohydrosis:</i> carbonic anhydrase inhibitors (e.g., acetazolamide) Weak carbonic anhydrase inhibitors (e.g., topiramate and zonisamide) <i>Altered cytokine levels:</i> Clozapine
Drug administration-related fever	Vancomycin (due to pyrogenic impurities), cephalothin, pentazocine (vaccinations, IM injections), amphotericin B, bleomycin, cytarabine
Fevers relating to the pharmacologic action	Jarisch-Herxheimer reaction due to release of substances from dying organisms (e.g., treatment of syphilis, borreliosis, leptospirosis, brucellosis, trypanosomiasis) Damage of malignant cells by chemotherapeutic agents
Idiosyncratic reactions	<i>Malignant hyperthermia caused by a genetic disorder of calcium regulation:</i> In genetically susceptible patients exposed to inhaled anesthetic agents (halothane, enflurane, isoflurane, methoxyflurane, sevoflurane, cyclopropane, diethyl ether, ethylene), depolarizing muscle relaxants (succinylcholine, decamethonium, gallium), possibly anticholinesterases, ketamine, digoxin, potassium, theophylline, atropine, and glycopyrrolate <i>Neuroleptic malignant syndrome caused by a reduction of dopamine at the synapse of D2 receptors in the CNS:</i> Phenothiazines (fluphenazine, perphenazine, trifluoperazine); butyrophenones (haloperidol, droperidol); thioxanthenes (thiothixene); dibenzazine derivatives (olanzapine, clozapine); tricyclic antidepressants (amoxapine, desipramine, nortriptyline, amitriptyline, clomipramine, imipramine) <i>Serotonin syndrome caused by excessive serotonergic activity in the CNS:</i> Selective serotonin reuptake inhibitors (citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline); serotonin and norepinephrine reuptake inhibitors (desvenlafaxine, duloxetine, venlafaxine); serotonergic drugs (tryptophan, amphetamines, tricyclic antidepressants, trazodone, dextromethorphan, meperidine, fentanyl, tramadol, monoamine oxidase inhibitors, buspirone, lithium, triptans, serotonin receptor antagonists, dihydroergotamine). <i>Red cell hemolysis in G6PD deficiency causing release of endogenous pyrogens:</i> Sulfonamides (sulfamethoxazole-trimethoprim, dapsone), antimalarials, nitrofurantoin, quinidine, chloramphenicol, rasburicase
Hypersensitivity reactions/immune-mediated	<i>Reported drugs (bold indicates most commonly reported drugs):</i> allopurinol, aminoglycosides, aminosalicic acid, amphotericin B, antihistamines, asparaginase, atropine, azathioprine, barbiturates, bleomycin, blood transfusions, carbamazepine, cephalosporins, chlorambucil, chloramphenicol, chlorpromazine, cimetidine, clindamycin, colistin, corticosteroids, cytarabine, daunorubicin, diltiazem, folate, haloperidol, heparin infusion, hydralazine, hydroxyurea, imipenem, interferon, iodides, isoniazid, intravenous immune globulin, labetalol, levamisole, macrolides, mebendazole, methyldopa, metoclopramide, nifedipine, nitrofurantoin, NSAIDs, novobiocin, pamidronate, para-aminosalicylic acid, penicillins, phenytoin, procainamide, procarbazine, propylthiouracil, prostaglandin E2, quinidine, ritodrine, rifampin, salicylates, 6-mercaptopurine, streptokinase, streptomycin, streptozocin, sulfonamides, tetracyclines, thioridazine, tolmetin, triamterene vancomycin, vitamin preparations

Adapted from [26]

Note: **BOLD** type indicates drug reported more commonly

CNS central nervous system, G6PD glucose-6-phosphate dehydrogenase, NSAIDs nonsteroidal anti-inflammatory drugs

Table 12.7 Patterns of fevers associated with specific diseases

Pattern	Possible cause
Continuous (sustained) fever, with slight remissions not exceeding 2 °F (16.7 °C)	Lobar and gram-negative pneumonia, rickettsioses, typhoid fever, CNS disorders, tularemia, and falciparum malaria
Intermittent (septic, quotidian, “picket fence”) fever with wide fluctuations, usually normal or low in the morning and peaking between 4:00 P.M. and 8:00 P.M.	Localized pyogenic infections and bacterial endocarditis. Malaria may present as quotidian (daily spike), tertian (spike every third day), or quartan (spike every fourth day) Acute brucellosis often as intermittent Salmonellosis, military tuberculosis, double malaria infections, and gonococcal and meningococcal endocarditis often present as double quotidian (two daily spikes)
Saddle-back (biphasic) fever: several days of fever, a distinct reduction in febrile levels for approximately 1 day, and then several additional days of higher fever	Dengue and yellow fever, Colorado tick fever, relapsing fever, Rift Valley fever, influenza, poliomyelitis, and lymphocytic choriomeningitis
Intermittent hectic (Charcot’s) fever: sporadic episodes of fever; periods of normal temperature and recurrence of fever	Cholangitis (usually associated with cholelithiasis, jaundice, and leukocytosis)
Pel-Ebstein fever, characterized by weekly or longer periods of fever and equally long afebrile periods, with repetition of the cycle	Hodgkin’s disease, brucellosis due to <i>Brucella melitensis</i> , and relapsing fever Occasionally in tuberculosis
Reversal of diurnal pattern of fever (typhus inversus), with the highest temperature elevation in the early morning hours rather than during the late afternoon or evening	Occasionally in military tuberculosis, salmonellosis, hepatic abscess, and bacterial endocarditis
Jarisch-Herxheimer reaction, with sharply increased elevation of temperature and exacerbation of other clinical abnormalities	Occurs several hours after beginning penicillin treatment for syphilis, in leptospirosis and tick-borne relapsing fever Following tetracycline or chloramphenicol therapy for acute brucellosis

Adapted from [1]
CNS central nervous system

Symptom Assessment (SCHOLAR)

For proper assessment of a patient’s fever, the following information should be collected in order to ensure more urgent follow-up is not required. If the patient is 3 months of age or younger, they should be referred to an urgent care center immediately.

Symptoms

- In addition to fever, did you experience any other symptoms?
- Are you experiencing any nausea or vomiting, diarrhea, runny nose, or cough?
- Are you experiencing any pain?
- Do you have any stiff joints, particularly of the neck?
- Are you feeling more tired than usual, confused, unable to eat or drink?
- For infants, do they have a bulging fontanelle?

Clarifying the symptoms associated with the patient’s fever will help identify the presence of any red flags that prompts referral.

Characteristics

- How many days have you had a fever?
- What is the duration of the fever(s), is there a pattern (evening only, morning and evening)?
- How high was the highest temperature measured?

Clarifying the characteristics, height of fever, and patterns of fever will help determine severity of fevers, differential diagnosis of fever vs. hyperthermia, and identifying the presence of any red flags.

History

- Were you doing any exercise or were you out in hot weather prior to/during your fever?
- Have you recently been around any sick contacts?
- Have you recently traveled outside of the country?
- Have you recently received any treatment for cancer or any other immunosuppressant?

Knowing the history of the patient’s fever, as well as their relevant medical background, will help determine the severity of the fever and the need to refer the individual to a physician, or seek immediate medical attention.

Onset

- When was your first recorded fever?
- How quickly did your temperature elevate?
- Has your fever abated at all?

Clarifying the immediacy of fever onset and how quickly the fever elevated will help identify the presence of any red flags.

Location

- What was the site of temperature measurement (oral, tympanic, axillary, rectal, etc.)?

Clarifying the location of the temperature measurement will help determine the accurate core temperature and the presence of any red flags.

Aggravating Factors

- What makes the fever worse?
- Does the fever occur after administration of a particular medication?

Fever is caused by infectious and noninfectious causes. Fevers that are attributable to medications may occur at any time. In hyperthermia, the patient may not be able to tolerate hot environments and will show a progression of hyperthermia symptoms.

Remitting Factors

- What makes it better? Pharmacological and non-pharmacological.
- Were any antipyretics (NSAIDs or acetaminophen) trialed?
- Was the medication effective in alleviating the fever, or symptoms associated?
- How often are you taking medication to alleviate the fever?

NSAIDs and acetaminophen are common antipyretics used to alleviate fever. They are not effective at alleviating hyperthermia, such as heat exhaustion or heat stroke, as these are conditions associated with an inability of the body to regulate cooling, not due to an elevated core temperature set point by the hypothalamus. Antipyretics may not abate the fever completely but may provide some relief of fever. Information about the duration and dose of antipyretics is important to determine if side effects may occur, such as renal or GI toxicity from NSAID administration, or hepatic toxicity by acetaminophen.

Patient-Specific Characteristics

In addition to assessing the patient's symptoms, a knowledge of the patient's medical and medication history determines the most appropriate plan of management. The following examples illustrate how patient-specific characteristics are essential in fever assessment.

- *Age:* Fever in infants less than 6 months is considered a red flag and prompts referral to a health-care professional for further assessment due to the risk of a serious bacterial infection. Elderly with fevers are also at a greater risk of morbidity and mortality due to infection, and individuals older than 60 years require more careful monitoring and are more likely to require a referral.
- *Pregnancy status:* Pregnant women may require more careful assessment and antipyretic therapy. Studies suggest that fever during the first trimester may be associated with increased risk of neural tube defects and congenital abnormalities [10]. Acetaminophen is the preferred pharmacological therapy, as use of ASA and NSAIDs are associated with side effects, including interference with labor, premature closure of the ductus arteriosus causing pulmonary hypertension in the infant, displacement of bilirubin from protein binding sites in vitro, and inhibition of platelet aggregation in the mother and child if taken close to the time of delivery [10].
- *Past medical history:* Identifying comorbidities may help determine whether an infection is causing fever or if there is a noninfectious cause. Patients who recently had surgery are more at risk of an infection and are more susceptible to fever in general and would require referral [27]. Patients with malignancies or other immunocompromising diseases, such as HIV, are at risk of serious bacterial infection and require prompt referral. Recent travel may be associated with more serious infections and is also a red flag. Some comorbidities may contraindicate the use of specific antipyretics, such as a history of peptic ulcer disease, and therefore, the patient should avoid the use of NSAIDs.
- *Medication history:* Identifying the patient's current medications will help recognize the possibility of drug-induced fever or any possible drug interactions with antipyretic therapy. Many drugs are known to cause drug fever that may occur at any time during drug therapy. It is also important to obtain an accurate medication history to determine the presence of any red flags. For example, patients undergoing treatment with chemotherapy or other immunosuppressive therapies are at risk of serious infections and require prompt referral.

Red Flags

The presence of any of the following red flags prompts referral to a health-care practitioner or the emergency department. These red flags may indicate that the patient has a life-threatening condition.

- *Fever in children less than 6 months old:* Children less than 6 months are at greater risk of serious bacterial infections and should be referred to the emergency department for a septic workup. This often includes urine cultures, blood cultures, and a possible lumbar puncture to rule out meningitis.
- *Fever in adults older than 60 years:* Adults greater than 60 years are less likely to have benign causes of fevers and are at greater risk of morbidity and mortality. They require careful monitoring and may require prompt referral to a physician.
- *Fever greater than 40.5 °C:* Patients with fever greater than 40.5 °C are at risk of more serious infections, including meningitis, and should be referred to a physician for further assessment.
- *Fever for greater than 72 h:* Fevers lasting longer than 72 h can indicate a more serious infection and the patient should be referred to a physician for further assessment. Children with fevers lasting more than 5 days should be assessed for Kawasaki's [22]. Additionally, fevers that persist for an additional 24 h after recommending antipyretic therapy or without any obvious cause should also be referred.
- *Fever associated with a stiff neck:* These symptoms may accompany meningitis. Prompt referral to the emergency department for further assessment, cultures, imaging, and possible lumbar puncture to rule out meningitis is warranted.
- *Fever with seizures:* Febrile seizures occur more commonly in children aged 3 months to 5 years [27] and are rarely associated with permanent seizure disorders or neurologic damage. Patients with febrile seizures require assessment by a physician and may require further care/follow-up assessments by a physician.
- *Fever with localized pain, swelling, or heat:* This may be indicative of a thromboembolism, including deep vein thrombosis (DVT) or more serious pulmonary embolism (PE), which should be urgently assessed by a physician, and may require further diagnostic imaging and bloodwork.
- *Fever in patients who have recently traveled:* Recent travelers are at risk of more serious infection, in particular, people who have traveled to developing countries. Incubation times vary and infections can appear shortly after exposure to months following travel. Early evaluation should be completed by a physician, especially in

patients that have visited areas with malaria in recent months. Often, the physician will need to alert public health officials if the traveler was contagious while traveling or infected with a pathogen that is a public health concern such as yellow fever or Ebola.

- *Fever in immunosuppressed individuals:* Patients who are immunocompromised are at greater risk of serious infections and require prompt referral to the emergency department for initiation of antibiotics and septic workup. Assessment may include urine and blood cultures, as well as possible diagnostic imaging such as chest X-rays to rule out respiratory infections.
- *Fever in patients who had recent surgery or dental procedure:* Patients are at risk of serious infections or abscesses (which may be difficult to manage with antibiotics alone) after surgical procedures. For example, patients who have undergone recent dental surgery may have infections as severe as endocarditis. These patients should be promptly referred to the emergency department.
- *Fever with lethargy, poor oral intake, difficult to arouse:* These symptoms are often more associated with serious bacterial infections, such as bacteremia or meningitis. Additionally, children that appear very ill, are inconsolable/excessively fussy, or exhibit any symptoms that are worrisome to parents have a higher likelihood of serious infections. Prompt referral to the emergency department is recommended for additional workup and possible imaging.
- *Fever in patients who have recently eaten poorly cooked meat or fish:* Raw foods can put patients at risk of serious infections such as *Salmonella* or *E. coli* O157:H7 (also called the Hamburger Disease). Prompt referral to the emergency department is recommended as these illnesses can cause severe organ damage or death if not treated promptly.

Physical Assessment Skills

Temperature can be measured by a pharmacist when necessary, but a patient can complete this task independently at home with a thermometer. Methods of measurements are shown in Table 12.3.

Additional Assessment Considerations

Since fever can be caused by infections, further investigation to determine the source of the infection is warranted. Rapid antigen testing is available for community pharmacists to rule out group A streptococci (GAS) throat infection. This is in addition to COVID-19 testing. Urine cultures and blood cultures may be required for moderate to severe infections,

or for septic workup. Imaging such as chest X-rays or computed tomography (CT scans) may be completed to diagnose lung infections. If concerned about meningitis, a lumbar puncture may be completed. Laboratory investigations may also be required to determine the risk or presence of infection, as well as response to antimicrobials, particularly by monitoring a complete blood count (CBC) often with a differential to examine the white blood cell (WBC) breakdown.

Clinical Pearls

- Fever is among the most common complaints encountered by health-care professionals and pharmacists are often an early point of contact in the primary care setting.
- Fever is a complex physiological response. There is no definite “set temperature” that defines fever in all individuals. Fever is an elevation of core temperature, which varies among individuals and at different times of the day.
- It is important to know variations of temperatures at different sites of measurement.
- Often fevers are not serious and it is not always necessary to treat a fever with antipyretic therapy. Discomfort from fever can be managed with medications; however, patients should be referred to a physician if requiring medications for more than 3 days.
- Acetaminophen and ibuprofen should not be alternated for the treatment of fever.
- Due to the various formulations available, patients should be counseled on reading the label to ensure the correct dosage, especially for children, as different product concentrations may be available. Dosage should be determined based on the child’s weight and instructions on the label, with appropriate measuring devices.
- Pharmacists need to complete accurate information gathering in order to identify the need for physician assessment based on red flags and other patient information. The pharmacist’s role is to support the relief of discomfort caused by a fever and to rule out more serious infections or hyperthermia, which can lead to significant morbidity and mortality if not promptly managed.

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Chapter Objectives

1. Develop a systematic approach to a dermatological assessment.
2. Identify the key questions that make up a dermatologic history.
3. Utilize appropriate terminology in the description of skin lesions.
4. Describe the morphology of common drug eruptions.
5. Describe initial steps in management of drug eruptions.

Background

The purpose of this chapter is to enable pharmacists to perform a thorough dermatological history and examination, to use a common set of reference terms to describe skin findings, and to recognize some of the most common dermatological conditions pharmacists may encounter. It is estimated that one in six (15%) visits to physicians involves dermatological problems and pharmacists are often the first point of care for many patients seeking a diagnosis and treatment of many common skin conditions [1, 2].

Symptom Assessment

To allow proper assessment of patients presenting with a dermatological presentation, both subjective and objective data must be collected. As a pharmacist, a thorough medical history and physical examination will constitute the dermatological evaluation.

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Complete Medical History: History of Presenting Dermatological Issue

Explore the patient's story in relation to their dermatological issue by obtaining relevant information about the presenting illness and any associated or underlying features. Pathology can be divided into two categories, limited exclusively to the skin or a manifestation of a systemic process. Abnormal findings may represent a disease process limited to the skin such as a dermatitis associated with poison ivy exposure or it may refer to a systemic illness, such as varicella rash where other organs or body processes may be involved. The line of questioning should glean data that helps identify the manifestation as local or systemic, which is paramount for successful diagnosis and treatment.

Symptoms

Common symptoms to inquire about during the history would include pain, burning, stinging, and pruritus. The symptoms should be categorized by when they first manifested, where they are located, if they are intermittent and how severe they are. These symptoms can be described subjectively through questioning the patient (e.g., SCHOLAR) or through patient questionnaires (such as the Dermatology Quality Life Index (DLQI) [3] or through objective assessment instruments if possible (e.g., using the Psoriasis Area Severity Index score (PASI) to grade psoriasis) [4]. Pruritis is a common symptom in most dermatologic conditions such as urticaria and dermatitis. Pain is a less frequently reported symptom and is often secondary to localization of purulent material or nerve inflammation, such as in herpes zoster infection [5]. Additional symptoms may include gastrointestinal disturbances, rheumatological inflammation in bones and joints, neurological changes in sensorimotor function, and/or psychological manifestations like anxiety and depression. To gain a holistic understanding of the impact of the conditions, assessment of other body systems may be appropriate as applicable through a thorough review of systems.

Characteristics

It is important to determine if there have been any appreciable changes over time. Any lesion on the body that has recently grown rapidly, changed color, become asymmetric, or bled or crusted should make you suspect malignant transformation. Cancer would be a less likely concern in a lesion that has remained unchanged for a few or more years; although some patients, especially the elderly, may not be aware of such changes. With generalized rashes, one should ask about where it first arose and how it has progressed over time. A rash such as measles classically begins on the face and then spreads to the trunk and limbs whereas a fixed drug eruption often affects one single area of the skin and does not change location.

Initial changes or lesions should also be elicited as sometimes when patients present to their health-care practitioner their rash has changed and the morphology might differ. For instance, bullae related to impetigo may have been present initially but might have ruptured by the time of examination.

Onset

A key part of history taking is to determine when a lesion or rash first appeared. It may have appeared acutely within recent days or, conversely, may have been present for many years. If the latter, ask why the patient is seeking medical evaluation at this time. A correlation between the onset of the lesions with any particular event or exposure should be explored. In some cases, the patient will have already noted a clear association (e.g., a rash, which appeared after the ingestion of a medication). Often, there will be a need to ask several questions to elicit such associations, as patients may be unaware of the importance of various factors. For example, if an allergic contact dermatitis is suspected, the list of possible exposures to consider may be extensive, including cosmetics, detergents, topical medications, and occupational and recreational exposures. Additionally, exposure to sunlight may be an inciting or exacerbating factor; or a recent flu-like illness may lead you to suspect a viral exanthem.

Aggravating/Remitting Factors

It is important from a diagnosis perspective to ask if the patient has received remittance or exacerbation of the condition spontaneously or with any particular therapeutic agents. For example, rosacea may become exacerbated through a host of environmental factors including sun exposure, topical steroid use, alcohol use, caffeine consumption, and other. Improvement with avoidance of these trigger factors and topical or systemic rosacea therapy would help to confirm this diagnosis versus another condition, such as lupus erythematosus. Response, or lack thereof, to therapy may also suggest one diagnosis over another. Mistakenly treating a cutaneous tinea infection with topical steroids might lead to

mild temporary improvement, however, continued therapy with a steroid will result in subsequent aggravation and worsening of the condition as the tinea infection will continue to expand.

Complete Medical History: Medical, Medication, Family and Social History

Medical History

When gathering a medical history, obtain a list of chronic medical conditions, recent illnesses, and medical procedures, which may suggest a systemic cause of the skin findings. Often skin diseases can be precipitated by a manifestation of an underlying medical condition or a fungal, bacterial, or viral infection. One should also obtain a history of all relevant allergies to both medications and environmental factors.

Medication History

It is essential to make note of current prescription and non-prescription medications, including systemic, injectable, and topical therapy. Gather information on the duration of use, adherence to medications, adverse drug reactions the patient is experiencing, and recently discontinued medications. Some medications are more likely than others to cause dermatologic manifestations, but almost any agent could be implicated. Although newly prescribed medications (taken for days or weeks) are the most likely to cause adverse drug reactions, even those taken continuously for months or years may sometimes cause reactions. Remember that a complete drug history includes the seven “I”s [6]:

- Instilled (eye drops, ear drops)
- Inhaled (nasal and oral)
- Ingested (capsules, tablets, syrup)
- Inserted (suppositories and ovules)
- Injected (IM, SC, IV)
- Incognito (herbs, nontraditional medicine, homeopathic, vitamins, over the counter)
- Intermittent (patients taking medications on an intermittent basis)

Family and Social History

Gather a brief synopsis of the presence or absence of illnesses in the patient’s first-degree relatives (i.e., parents, siblings, and children). This may reflect a contagious etiology or common exposure among household members or may suggest the possibility of a hereditary condition such as psoriasis or atopic dermatitis.

Inquiring about a patient’s social history, which contains their use of alcohol, tobacco, and illicit drug use as well as nutrition and exercise may be helpful in identifying contrib-

uting factors of their dermatological condition and help in the selection of an appropriate therapeutic agent for treatment. In addition, information regarding employment and living conditions may be included and will further contribute to the holistic understanding of the etiology of the patient's dermatologic concern.

Physical Assessment

The extent of the physical exam will depend on the pharmacist's comfort level, nature of practice, and clinical setting. A full dermatological examination would include the skin, hair, nails, and mucous membranes (mouth, eyes, nose, and genitalia). Steps 1 through 4 offer a systematic approach to conducting a dermatologic assessment. The degree of physical examination must be deemed reasonable for the practice setting, purposeful for the treatment of the patient, and supported by the skill set of the pharmacist. An entire body examination is usually unnecessary for a readily identified localized process. However, some findings may be missed if the patient is not examined beyond the most apparent pathology, or that which the patient points out. A medical gown and private examination room should be made available for the patient if the examination goes beyond readily accessible and visible areas. Physical assessment should be performed in a room with sufficient lighting and a penlight can be helpful in evaluating lesions within the mouth or nose. A magnifying glass and ruler may be of assistance in further identifying and quantifying characteristics of the dermatological presentation [6].

Step 1: Physical Examination

Initially, one should assess the overall appearance and disposition of the patient. Essentially, one should see if a patient appears well or if there are signs of a more systemic illness with labored breathing, diaphoresis, sallow skin color, or shortness of breath. If there are any red flags or severe systemic illness, a patient should be instructed to see their family physician or proceed to the emergency room. If the patient is generally well, a full dermatological exam is comprised of an inspection of the skin, hair, and nails. A systematic approach starting from the head and following through to the toes is recommended (Table 13.1). Lesions may be described as outlined in *STEP 2* and if a lesion such as a nevus (mole) is present, make note of any abnormal characteristics, using elements of the acronym *ABCDEs of Melanoma* as warning signs for nevi that warrant further examination (Table 13.2) [7]. Inspect the hair and make note of color, texture, and distribution of the hair. In addition, it may be beneficial to look at the hair shafts which may elude to disorders of increased

Table 13.1 Head-to-toe approach of a dermatological physical examination

Body area	Notes
(a) Scalp and hair	Remove any hair bands or accessories. Make note of alopecia (hair loss) or thinning. Hair morphology of thickness, length, breakage can also be assessed. Assess the scalp for any erythema, scaling, scarring, erosions, or other growths
(b) Head and neck	Includes eyes, nose, and mouth. Flashlight or magnifying glass may be used to assist in visualization. Assess for any lesions or inflammatory rashes. Outer ear and external ear canal as well as behind the ears. In some cases, even inner ear canals can be involved with rashes like psoriasis and discoid lupus. Neck should also be visualized
(c) Torso (back, chest and abdomen)	Remember to inspect the axillae and under the breasts, and in skin folds of obese patients. Once again, note any rashes or lesions
(d) Arms, hands, and fingernails	Inspect sides of fingers and web spaces. Look at dorsum and palms. When examining nails also look at the surrounding area and cuticles (periungual)
(e) Genitalia/buttocks	Follow site policy. Advised to have another HCP chaperone present. Pubic area and labia in women. Pubic area, scrotum, and penis in men
(f) Legs, feet, and toe nails	Evaluate the feet in the same manner as the hands, including interdigital areas and the soles

HCP health-care provider

fragility of hair or parasitic infestations. Lastly, inspect the nail for hypertrophy, subungual hyperkeratosis, abnormal shape or curvature, pitting, and color change. Evaluate the nail bed for separation from the nail plate and hemorrhage. Lastly, nail folds should be inspected for erythema, inflammation, swelling, and tenderness.

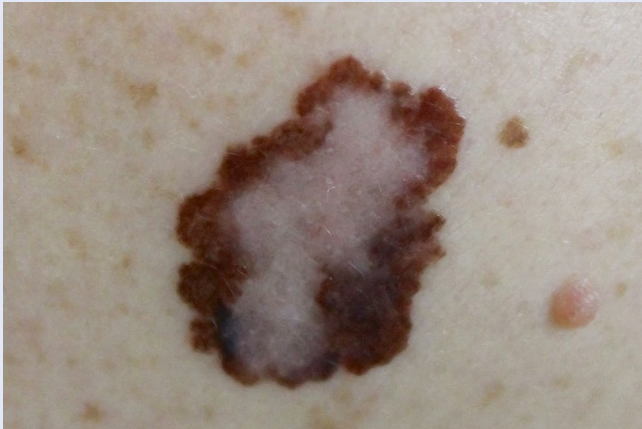
Step 2: Describe and Document Morphology [8]

The word morphology is used by dermatologists to describe the use of descriptors to accurately characterize and document skin lesions. The morphologic characteristics of skin lesions are key elements in establishing the diagnosis and communicating skin findings. Follow the approach outlined in steps (a) through (f) as depicted in Fig. 13.1.

Number and Distribution

There may be one or more lesions in a localized area, or numerous in several areas. The distribution often helps to suggest a plausible etiology. For example, a systemic manifestation of a viral illness like pityriasis rosea will often cause a papulosquamous eruption (both papules and scales present) limited mostly to the trunk, often in a Christmas tree pattern. An eruption confined to one single dermatome (the cutaneous distribution of a single spinal nerve root) is classic for a herpes zoster infection. Lesions on sun-exposed areas,

Table 13.2 ABCDEs of melanoma^a

Characteristic	Notes	Image
Normal	Symmetric, even border, uniform coloring, smaller than 6 mm, no evolution over months/years	
A. Asymmetry shape	Melanoma lesions are often irregular, or not symmetrical, in shape. Benign moles are usually symmetrical	
B. Border	Typically, noncancerous moles have smooth, even borders. Melanoma lesions usually have irregular borders that are difficult to define	

(continued)

Table 13.2 (continued)

Characteristic	Notes	Image
C. Color	The presence of more than one color (blue, black, brown, tan, etc.) or the uneven distribution of color can sometimes be a warning sign of melanoma. Benign moles are usually a single shade of brown or tan	
D. Diameter	Melanoma lesions are often greater than 6 mm in diameter (approximately the size of a pencil eraser)	
E. Evolution	The evolution of your mole(s) has become the most important factor to consider when it comes to diagnosing a melanoma. If a mole has gone through recent changes in color and/or size, and/or elevation or if patient experiences any new symptom such as bleeding, itching, or crusting, bring it to the attention of a dermatologist immediately	

^aImages, used under license from [Shutterstock.com](https://www.shutterstock.com)

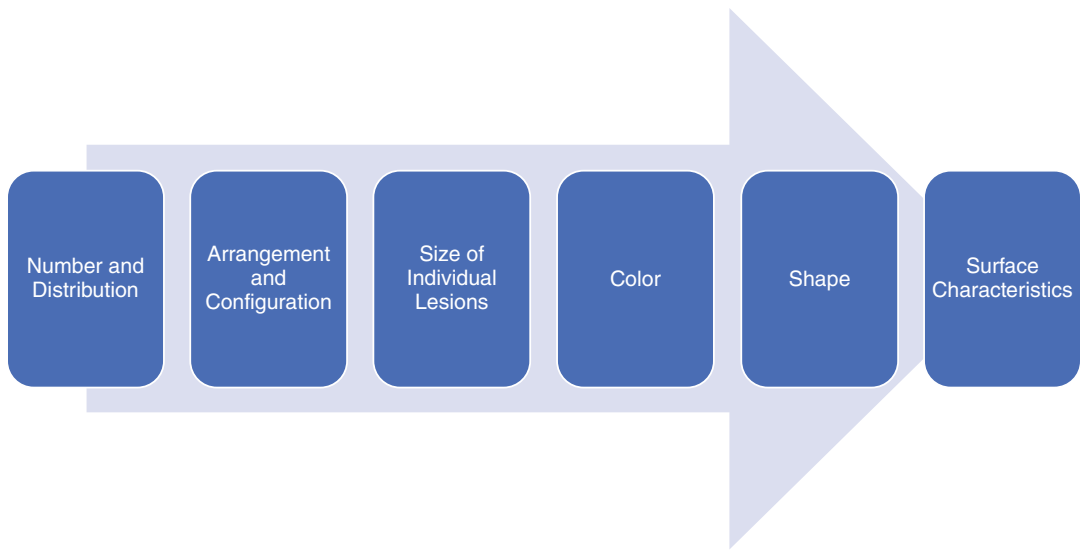


Fig. 13.1 Approach to describing the dermatological lesions

such as the face, dorsal hands, V of the neck and upper chest may suggest a photosensitivity (sun-induced) reaction like polymorphous light eruption.

Arrangement and Configuration

Some eruptions can make interesting and characteristic patterns on the skin that can also provide a clue as to their etiology. For instance, an acute dermatitis with well-demarcated linear borders often suggest a contact allergen exposure. Other configurations may be labeled annular (circular) or serpiginous (wavy). Granuloma annulare (an inflammatory dermatosis) and cutaneous tinea infections can both be annular helping us to narrow down our differential diagnosis when these lesions are present.

Size of Individual Lesions

If several lesions are present, they may be similar in size, or there may be a range of sizes. If the latter, you should indicate the range (i.e., 5–15 mm). Nevi on the body are a common example where you will have a tremendous range of sizes of lesions.

Color of Lesion

Skin lesions may present in a multitude of colors and uniformity. Hypopigmented skin is an area which is lighter in color than the surrounding skin and hyperpigmented areas are darker in color than the surrounding skin. Noting uniformity (or nonuniformity) of color is important. It is important to establish a baseline skin color to allow for accurate assessment of changes (use of an objective tool rather than a subjective description is recommended) and not to equate race with skin color [9].

Erythema, defined as the reddening of the skin, is one of the most frequently observed clinical signs associated with dermatological injury, irritation, inflammation, or infection resulting from vasodilation and increased blood flow to the affected area.

The appearance of erythema can vary significantly based on the individual's underlying skin pigmentation. In light skin tones, erythema commonly presents as shades of pink, red, or purple. However, in dark skin tones, these color changes may be less visually apparent or may manifest as a subtle violaceous (purplish) hue surrounding inflamed or injured areas. This variation in presentation can lead to under-recognition or misinterpretation of erythema in individuals with richly pigmented skin [10–12].

Given the potential for erythema to be missed in richly pigmented skin, it is essential to inquire about and evaluate associated clinical signs and symptoms that may indicate inflammation or infection. These may include [10–12]:

- Localized pain, burning, or itching
- Increased warmth of the skin

- Swelling, which may appear as smooth, shiny, or tight skin
- Scaling, crusting, or erosions
- Systemic symptoms such as general malaise or fatigue

Colour presentation can vary widely depending on the lesion and the individual's skin tone. For example, a lesion may present with an erythematous (red) ring at the periphery and a flesh-colored center, suggesting a possible annular or inflammatory process. Another lesion might appear as uniformly light brown, or it may display irregular shades of brown within the same lesion; both could be rightly labelled hyperpigmented, but a more precise description is necessary to distinguish the two. In contrast, areas of complete depigmentation, which appear starkly white compared to the surrounding skin, often point to conditions such as vitiligo.

Shape and Lesion Morphology

Lesions on the skin are categorized in terms of the primary morphology and secondary features. Primary lesions are initial lesions that have not been altered by trauma or manipulation and have not regressed. The most common primary morphological terms include: Macules, patches, papules, and plaques. Secondary lesions develop during the evolutionary process of skin disease or are created by manipulation or complication of primary lesion (e.g., rubbing, scratching, scaling, and infection). Table 13.3 describes the various types of primary and secondary lesions and common differential diagnoses based on morphology.

Surface Characteristics

On occasion, primary and secondary morphological terms might not give enough information to render an accurate diagnosis. Other terms can be used to describe the surface morphology of a lesion or rash. Terms such as smooth, rough, shiny, dull, waxy, verrucous (warty), keratotic (thickened skin), and filiform (finger- or thread-like) can all be applied. For example, the surface of a wart can be described as being rough, verrucous, keratotic, and/or filiform. A mole, on the other hand, would normally be described as smooth or domed.

Step 3: Palpate the Lesion [1]

When applicable, palpate the lesion. Palpation should always be conducted while wearing gloves. Never palpate an inflamed area, because this may cause a possible infection to spread deeper and produce intense pain. When palpating, carefully run your index finger over the lesion and note the texture of the surface (i.e., rough or smooth). At times you may be able to distinguish a slightly raised lesion from a flat one by careful palpation with the eyes closed. Next, deter-

Table 13.3 Description of primary and secondary lesions and common differential diagnoses^a

Term	Description	Image	Example
Primary lesions			
Macule	Flat lesion, less than 1 cm		Freckle (small, pigmented lesions on sun-exposed areas) Viral exanthem (erythematous rashes arising focally and spreading) Idiopathic guttate hypomelanosis (small depigmented white spots on sun-exposed areas)
Patch	Flat lesion, greater than 1 cm		Vitiligo (depigmented areas of skin and hair) Viral exanthems (erythematous rashes, most often have fever or systemic symptoms) Alopecia areata (smooth, bald patches in hair-bearing areas)
Papule	Raised lesion, less than 1 cm		Wart (domed or filiform papules with thrombosed capillaries and surrounding callus) Insect bite/sting (discrete, painful, or itchy lesion) Basal cell carcinoma (telangiectatic, pearly, domed nodule)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Plaque	Raised lesion, greater than 1 cm		Psoriasis (scaling, erythematous rash usually seen on extensor surfaces) Atopic dermatitis (itchy, thickened skin, and accentuated skin markings usually in flexural areas) Granuloma annulare (raised, annular rash on dorsal hands or feet) Tinea cruris (raised, annular rash with scaling borders)
Vesicle	Fluid-filled lesion, less than 1 cm		Herpes simplex virus (discrete, clustered, painful blisters) Dyshidrotic eczema (small blisters on the sides of fingers and toes, very itchy) Varicella zoster virus (widespread blisters in chicken pox and dermatomal blisters in shingles, usually painful)
Bullae	Fluid-filled lesion, greater than 1 cm		Bullous pemphigoid (autoimmune blistering disease) Bullous impetigo (large, flaccid blisters often seen in children) Edema blisters (seen in the legs of patients with severe stasis dermatitis and lower limb edema)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Nodule	Elevated, solid lesion, greater than 1 cm, usually in the dermis or subcutaneous lesion		Dermatofibroma (firm, skin colored to red to brown nodule in the dermis, usually after trauma) Acne (firm, painful lumps under the surface of the skin, unable to express discharge or fluid) Lipoma (soft, subcutaneous, rubbery swellings)
Cyst	A papule or nodule that contains fluid or semi-solid contents		Epidermoid inclusion cyst (subcutaneous, mobile lesion filled with keratin) Acne (subcutaneous nodules, often not as mobile, filled with pus)
Pustule	A discrete lesion that contains purulent material or neutrophils		Acne (small lesions on the surface where pus can be expressed) Pustular psoriasis (can develop pustules, mainly on palms and soles) Folliculitis (pustules located discretely at sites of hair follicle openings)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Wheal	Raised erythematous lesions with surrounding hypopigmentation		Chronic spontaneous urticaria (idiopathic eruption of wheals and possibly angioedema, with itch) Bites and stings (usually discrete, itchy areas following exposures) Drug eruption (widespread, itchy, inciting drug usually started within several days or weeks of outbreak)
Secondary lesions			
Scale	Desquamated skin cells that peel and flake off		Seborrheic dermatitis (greasy yellow flaking in glabella and nasolabial folds) Tinea infections (annular plaques with peripheral scale and central clearing)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Lichenification	Thickened areas of skin with enhanced skin markings		Lichen simplex chronicus (chronically excoriated papules and plaques of skin)
Crust	Plasma that has exuded and solidified into scale		Impetigo (honey-colored crusting on an erythematous base)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Erosion	Superficial loss of surface epithelium		Herpes simplex virus (eroded and/or crusted solitary or clustered lesions on mucosal surfaces like the lip)
Ulceration	Deeper loss of both surface epithelium and dermis		Venous leg ulcer (open wounds on a background of edematous skin with varicosities and possibly hemosiderin deposition)
Atrophy	Loss of epidermis, dermis, or subcutaneous tissue		Lichen sclerosus (pink to erythematous thinning of skin with white scarred and/or atrophic areas)

(continued)

Table 13.3 (continued)

Term	Description	Image	Example
Hypertrophy	Thickening of epidermis, dermis, or subcutaneous tissue		Keloid scar (pink to skin colored to hyperpigmented papules and plaques, usually on upper torso and ears)

^aImages, used under license from [Shutterstock.com](https://www.shutterstock.com)

mine the consistency by pressing on the lesion and then palpating it between your fingers. Terms used to describe consistency include rock-hard, firm, rubbery, fluctuant, and soft. Also, note how far the lesion extends below the skin surface by feel and if it is fixed in place or freely mobile.

Laboratory and Diagnostic Tests

In most cases, a good history and physical examination can often help the practitioner to arrive at a diagnosis and management plan. However, in some cases further diagnostic testing may be required. If a diagnosis is unclear, the patient might have to see their physician or dermatologist for a skin biopsy. The skin biopsy is one of the most valuable diagnostic tools available and can greatly aid in arriving at a timely and correct diagnosis. Other diagnostic tests include swabbing skin lesions or taking skin scrapings to rule out bacterial, viral, and fungal infections. Bloodwork can also be of great utility in diagnosing skin conditions associated with systemic features or underlying internal medical conditions. In some cases, patients are also sent for skin prick allergy testing and/or patch testing when allergies are believed to be involved in their presentation.

Pharmacist's Role in the Monitoring and Therapy of Dermatological Diseases

Pharmacists and other health-care providers may encounter numerous challenges in the treatment of dermatological diseases. Patient surveys have found treatment nonadherence rates as high as 73% in certain dermatological diseases such

as in the psoriasis population, which can create a vicious cycle between disease exacerbation and treatment failure leading to poor patient outcomes [13]. The most commonly cited reasons for patient nonadherence include frustration with medication efficacy, inconvenience of administration, and fear of adverse effects [13]. Poor adherence is both a cause and consequence of inadequate treatment efficacy as the patient perceives it. When developing a treatment plan, it is paramount to engage the patient in treatment decisions and understanding any reasons why the patient might or might not adhere to any specific therapy.

Pharmacists are in an ideal position to enhance adherence to therapy. Factors that are associated with better adherence include choosing the simplest therapy and dosing regimen that suits the patient's lifestyle, providing easy-to-understand instructions for medication use, and appropriately conveying information about therapy risk. It has also been shown that a collaborative model of management that consists of regular follow-up visits and good health-care practitioner-patient relationships further improves adherence [14].

Managing patient expectations and conveying the chronic nature of particular dermatological diseases (e.g., acne, rosacea, and psoriasis) is instrumental in enhancing the patient's commitment to therapy. Key goals of dermatological disease therapy are generally to clear skin lesions, mitigate systemic and local symptoms, and improve quality of life. However, treatment satisfaction is an important factor that may be overlooked. Treatment satisfaction encompasses efficacy and safety as well as convenience of the therapeutic regimen, ease of administration, and cost or coverage of medication/therapy. In some instances, treatments may take several days or up to several weeks to see maximal benefit. Inadequate efficacy or loss of efficacy should be evaluated thoroughly

by addressing the patient's adherence to the regimen before making or suggesting a change in treatment approach. Importantly, patients should be encouraged not to abruptly discontinue their medication unless directed by a health-care provider as this can also trigger a relapse in disease control. Adherence to medications should be assessed at routine refills as well as a periodic review of the medication profile to identify any medications that have the potential to exacerbate the dermatological condition. Regular follow-ups are not only beneficial in enhancing adherence but also allows for early detection and intervention to manage the comorbidities associated with various dermatological conditions (e.g., psoriatic arthritis with psoriasis, depression associated with chronic urticaria) [15]. Addressing all these facets of therapy are vital to designing a treatment plan that will complement the patient's routine and treatment the patient would be willing to use.

When pertinent, reinforcement of healthy lifestyle behaviors and providing solutions to avoid established triggers as applicable should all be part of a comprehensive treatment plan. Adjustments in therapy should be considered depending on patient response (efficacy and adverse effects). Referrals to a specialist should be considered when disease is extensive, distressing, and unresponsive to current therapy or if the patient is experiencing unmanageable adverse reactions to topical or systemic therapy. Table 13.4 provides a

summary of common adverse effects and parameters for monitoring of commonly prescribed medications in dermatology.

Drug Reactions

The purpose of this section is to help pharmacists develop a clinical approach to the evaluation and initial management of patients presenting with drug reactions. It is always astute to consider drugs as the cause of a skin eruption. Most cutaneous drug reactions are inflammatory, generalized, and symmetrical. Diagnosis is established by clinical features, including morphology and timing.

The most common types of drug reactions include exanthematous or morbilliform eruptions, urticaria, fixed drug eruptions, drug-induced hypersensitivity syndrome (DIHS), also called drug reaction with eosinophilia and systemic symptoms (DRESS) and epidermal necrolysis, which includes Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). Some individuals may be at a higher risk for the development of drug reactions than others (Table 13.5). Risk factors include female sex, prior history of drug eruptions, recurrent drug exposure, and different genetic HLA types. Drug-induced skin reactions can be classified according to timing into immediate reactions and delayed

Table 13.4 Adverse effects and monitoring parameters for commonly prescribed dermatological agents [19–21]

Drug	Examples ^a	Possible adverse effects and suggested monitoring parameters ^{a,b}	Additional notes (clinical pearls) ^{a,b}
<i>Topical</i>			
Corticosteroids	<i>Low potency:</i> Hydrocortisone <i>Medium to high potency:</i> Beclomethasone Betamethasone Mometasone <i>Ultra-high potency:</i> Clobetasol	Application site: Irritation (due to vehicle most often) Pruritus/burning Folliculitis Contact dermatitis Hypertrichosis Acneiform eruptions Pigmentation change Miliaria Atrophy Striae Telangiectasia; purpura; impaired wound healing; steroid rosacea	Caution in active viral, fungal, parasitic infections, infections that may be exacerbated by immunosuppression (e.g., HSV, VZV). Caution with periocular use with history of cataracts and/or glaucoma Systemic corticosteroid side effects (Cushing's syndrome, adrenal crisis) with use of vast topical quantities
Retinoids	Tretinoin Tazarotene Adapalene	Application site: Erythema/peeling Burning/stinging Pruritus/dryness Tenderness/pain Contact dermatitis Increased photosensitivity (less with adapalene); skin fissure/cheilitis	Tazarotene contraindicated in pregnancy Topical retinoids not used in pregnancy

(continued)

Table 13.4 (continued)

Drug	Examples ^a	Possible adverse effects and suggested monitoring parameters ^{a,b}	Additional notes (clinical pearls) ^{a,b}
Topical antibiotics	<i>Macrolides:</i> Erythromycin Clindamycin	Application site: Dryness Contact dermatitis	Enhanced efficacy when used in conjunction with benzoyl peroxide
Other antibacterials	Benzoyl peroxide	Application site: Contact dermatitis Stinging/erythema Burning/pruritus Dryness/peeling Increased risk of photosensitivity reactions; bleaching of hair and clothing; lingering odor	Caution if sensitivity to balsam of Peru or cinnamon
Calcineurin inhibitors	Tacrolimus Pimecrolimus	Application site: Burning/stinging Soreness Pruritus Tingling Erythema Flu-like symptoms; acne/folliculitis development due to occlusive formulation	Warning: rare cases of skin malignancy and lymphoma have been reported in patients treated with topical calcineurin inhibitors. Avoid continuous long-term use Controversial as human studies have shown no further risk Do not use in immunocompromised patients and active infection
Exfoliants/comedolytic agents	Salicylic acid	Application site: Irritation Burning Erythema Peeling Increased risk of photosensitivity reactions	
Janus kinase (JAK) inhibitor	Ruxolitinib, delgocitinib	Application site: Redness, pruritus, or tingling	Ruxolitinib: serious warnings and precautions box: Serious infections, malignancies, blood clots, and cardiovascular events Delgocitinib does not have a serious warnings and precautions box
Phosphodiesterase-4 (PDE-4) inhibitor	Roflumilast	Application site: Pain at the cream application site Headache Vomiting Weight loss Insomnia	Possible drug interactions: Erythromycin, enoxacin, oral ketoconazole, fluvoxamine, cimetidine Contraindicated in patients with moderate to severe liver impairment
Other	Dapsone	Application site: Peeling Dryness Erythema Burning Irritation	Can be combined with adapalene and benzoyl peroxide
<i>Systemic</i>			

(continued)

Table 13.4 (continued)

Drug	Examples ^a	Possible adverse effects and suggested monitoring parameters ^{a,b}	Additional notes (clinical pearls) ^{a,b}
Retinoids	Isotretinoin Acitretin	Hyperlipidemia; depression; increased photosensitivity; dermatitis; alopecia; dry eye/blurred vision; reduced visual acuity; dry mucosa/lips; epistaxis/cheilitis; oropharyngeal pain; URTI; arthralgia/myalgia; auditory impairment; inflammatory bowel disease Lab tests to monitor: Liver tests (AST, ALT, ALP) Lipids (TG, LDL, cholesterol) Blood glucose Thyroid (TSH) CBC (complete blood count) with diff Pregnancy testing (hCG)	Pregnancy risk factor X (do not use in pregnancy or breast feeding). Must abstain from pregnancy for 3 years after use of acitretin Do not use if sensitivity to other retinoids, vitamin A, or its metabolites Monitor consumption of other sources of vitamin A (risk of hypervitaminosis A) Avoid use with alcohol. For acitretin, female patients should abstain from ethanol or ethanol-containing products during therapy and for 2 months after discontinuation Caution in patients with predisposition or history of bone loss Do not use concurrently with tetracyclines due to risk of pseudotumor cerebri May cause depression, mood disturbances, and suicidal ideation. Discontinue use and refer for further assessment
Antibiotics	<i>Tetracyclines:</i> Tetracycline Doxycycline Minocycline	Fatigue/drowsiness Headache/dizziness Vertigo/ataxia Increased photosensitivity Acne N/V/D/anorexia Hyperpigmentation (minocycline) Esophagitis/esophageal ulcer	Not to be used in pregnancy/lactation, in children <8 yo (unless these are the only effective or safest option) Autoimmune syndromes: Lupus-like, hepatitis, and vasculitis autoimmune syndromes (including serum sickness, e.g., fever, arthralgia, and malaise) have been reported; discontinue if symptoms occur and assess liver function tests May be associated with increases in BUN (blood urea nitrogen). Use caution in patients with renal impairment as this may lead to azotemia, hyperphosphatemia, acidosis, and possibly to drug accumulation and potential hepatotoxicity Risk of myasthenia gravis Risk of pseudotumor cerebri (concurrent use with systemic retinoids)
	<i>Macrolides:</i> Erythromycin Clarithromycin Azithromycin	Rash Tinnitus, hearing loss GI effects include abdominal pain and cramping, N/V/D, jaundice, hepatitis Arrhythmia Seizure Weakness Super infection (<i>C. difficile</i> , candida)	Caution in patients with liver dysfunction, history of cholestatic jaundice, or hepatic dysfunction with prior azithromycin use Concurrent use with CYP3A4 inhibitors

(continued)

Table 13.4 (continued)

Drug	Examples ^a	Possible adverse effects and suggested monitoring parameters ^{a,b}	Additional notes (clinical pearls) ^{a,b}
Immunosuppressants	Methotrexate (MTX)	Bone marrow suppression; malaise/fatigue; chills/fever Neurotoxicity; drowsiness/dizziness GI (nausea, vomiting, diarrhea); shortness of breath; headaches; rashes, oral ulcerations; hepatotoxicity; pneumonitis; nephritis Labs: Liver tests (AST, ALT, ALP) Albumin SCr/urea CBC diff Pregnancy (hCG)	Do not use in pregnancy/lactation Folic acid should be given on days before/after MTX. Supplementation of MTX with folic acid improves tolerability
	Cyclosporine	Hyperlipidemia; neurotoxic; tremor/headache; gum hyperplasia URTI; hypertension (monitor blood pressure); GI effects N/V/D/abdominal pain Nephrotoxic; hyperkalemia; hyperuricemia; myalgia; malignancy (skin cancer especially); bone marrow suppression Labs: Potassium/magnesium LFT/bilirubin Lipids SCr (fasting)/urea Urinalysis CBC diff	Do not use in pregnancy/lactation (females of childbearing potential), immunodeficiency syndromes, alcoholism (not absolute), liver disease, renal impairment Monitor for blood dyscrasias Drug interactions (see product monograph)
Janus kinase (JAK) inhibitor	Abrocitinib Upadacitinib Baricitinib Ritlecitinib	The exact risk and frequency of side effects can vary between different JAKIs.	Serious warnings and precautions box: Serious infections, malignancies, blood clots, and cardiovascular events Patients on JAKIs require regular monitoring, including blood tests for blood counts, liver function, and lipids, as well as vigilance for symptoms of infection or other serious events Oral Janus kinase inhibitors (JAKIs) are generally not recommended or are contraindicated during pregnancy and breastfeeding Different JAKIs are approved for different indications
Other	<i>Apremilast</i>	Headache URTI Weight loss GI effects N/V/D Depressed mood Tachyarrhythmia Labs: SCr (not routine use, only with those with kidney failure)	Do not use in uncontrolled infection, immunodeficiency (except autoimmune disease), malignancy (except nonmelanoma skin cancer), uncontrolled HTN, reduced renal function, pregnancy, and lactation

(continued)

Table 13.4 (continued)

Drug	Examples ^a	Possible adverse effects and suggested monitoring parameters ^{a,b}	Additional notes (clinical pearls) ^{a,b}
Biologics	<i>TNFi (tumor necrosis factor inhibitor) inhibitors</i> Adalimumab Etanercept Infliximab	Injection site reactions Infusion reaction (infliximab only) Risk of infections (URTI, LRTI, skin, others) TB reactivation Oral herpes Oropharyngeal pain Arthralgia Headache/fatigue Skin rashes Lupus-like syndrome Bone marrow suppression <i>Labs (prescreening):</i> CBC with diff Chest X-ray TB test HBV, HCV, HIV Antinuclear antibody (ANA)	Do not use or suspend treatment in severe active infection (sepsis, active TB, opportunistic infections, hepatitis) Contraindicated in demyelination disease. Monitor for neurological disease Contraindicated in heart failure (NYHA (New York Heart Association) class III/IV) Caution and monitor in patients with history of previous lymphoma or malignancy Caution in pregnancy and lactation Review patient's vaccination history prior to initiating therapy
	<i>IL-17A inhibitors:</i> Ixekizumab Secukinumab Brodalumab	Injection site reactions Development of infection (increased rates of staph and candida infections) Neutropenia; fatigue; nausea/diarrhea Arthralgia/myalgia Exacerbation of inflammatory bowel disease <i>Labs (prescreening):</i> CBC with diff Chest X-ray TB test HBV, HCV, HIV ANA	Do not use or suspend in active infections (see TNFi inhibitors) Active inflammatory bowel disease (brodalumab is contraindicated) Brodalumab and risk of suicidal ideation (black box warning; see product monograph) Caution and monitor in patients with history of previous lymphoma or malignancy Caution in pregnancy and lactation Review patient's vaccination history prior to initiating therapy
	<i>IL12/23 inhibitors</i> Ustekinumab <i>IL 23 inhibitors</i> Guselkumab	Injection site reactions Increased incidence of infections; headache; nausea/diarrhea; arthralgia Neurotoxicity (rare); pneumonitis (rare); malignancy (unclear risk) <i>Labs (prescreening):</i> CBC with diff Chest X-ray TB test HBV, HCV, HIV ANA	Do not use or suspend in active infections (see TNFi inhibitors) Caution and monitor in patients with history of previous lymphoma or malignancy Caution in pregnancy and lactation Review patient's vaccination history prior to initiating therapy

ANA antinuclear antibody, ALP alkaline phosphatase, ALT alanine aminotransferase, AST aspartate aminotransferase, D diarrhea, GI gastrointestinal, hCG human chorionic gonadotropin, HSV Herpes simplex virus, HTN hypertension, LRTI lower respiratory tract infection, N nausea, SCr serum creatinine, TSH thyroid stimulating hormone, URTI upper respiratory tract infection, VZV varicella zoster virus

^aList not all-inclusive

^bIf known hypersensitivity to the active ingredient or any other components of the formulation, use is contraindicated

Table 13.5 Patient risk factors for drug reactions [16]

Risk factors for drug reactions	Notes
Female	
Prior history of drug reaction	
Recurrent drug exposure	Repeated courses of therapy with the same drugs or related drugs are associated with higher rates of adverse drug reactions
HLA type	HLA-B*1502: carbamazepine and SJS/TEN in Han Chinese, SE Asians HLA-B*5701: abacavir and DRESS in whites, Hispanics HLA-B*5801: allopurinol and SJS/TEN in Han Chinese, Taiwanese, Thai
Certain disease states	Reactions to aminopenicillins occur more commonly in patients with Epstein Barr virus (EBV) infection HIV-positive patients have high rates of dermatologic reactions to sulfonamides and other drugs

reactions. Immediate reactions occur less than 1 h after the last administered dose, such as flushing from niacinamide. Delayed reactions generally occur after 1 h, but usually more than 6 h and occasionally weeks to months after the start of administration [6].

The role of allergy testing is of limited value in evaluating adverse cutaneous reactions to medications as most drug-related reactions cannot be reproduced through percutaneous testing. Penicillin is the exception to this rule. Penicillin skin testing is the preferred method of evaluation of possible type I, IgE-mediated penicillin allergy (urticaria due to penicillin) [6, 16].

Drug Timelines [6]

The most important information in determining if a rash is medication-related is its timing. When starting a timeline, start with the onset of the rash as day 0, and work backward and forward creating a drug timeline that encompasses all the medications, herbals, supplements, and over-the-counter products the patient has been taking (Fig. 13.2). The inciting culprit drug is usually one that has been initiated prior to day 0 and overlaps day 0 and possibly beyond.

Exanthematous Drug Eruptions

Exanthematous eruptions are the most common of all cutaneous drug eruptions (~90%) and the manifestation is generally limited to the skin. The lesions present as widespread erythematous macules, which initially appear on the trunk and spread centrifugally to the extremities in a symmetric fashion.

The patient may indicate pruritus and a fever may be present. Skin lesions usually appear more than 2 days after the drug has been started, mainly around days 8–11, and occasionally persist several days after having stopped the drug [6].

The clinical course of the eruption usually resolves in a few days to a week after the medication has been discontinued. In some cases, the medication may be continued by the care provider if the eruption is not too severe and the medication cannot be substituted. The eruption generally resolves without sequelae though extensive scaling and desquamation may occur. Management of exanthematous eruptions usually consists of a course of topical steroids, with the addition of oral antihistamines if warranted.

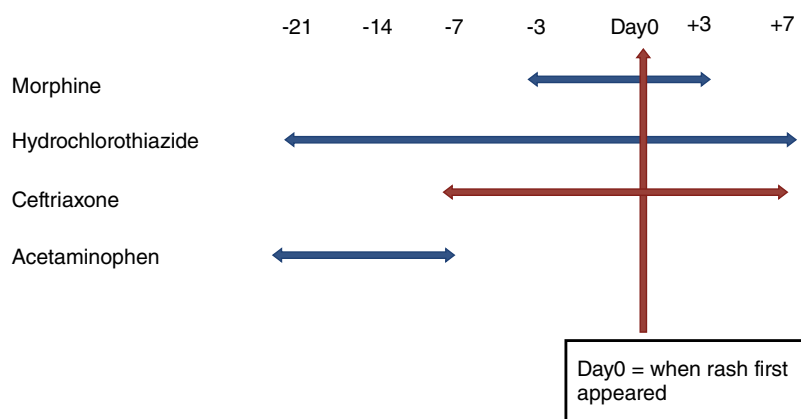
Urticarial Eruptions [17]

Urticaria refers to the development of wheals on the skin and angioedema refers to the development of deeper subcutaneous swelling in the skin. Acute urticaria is defined as lasting less than 6 weeks and chronic lasting greater than 6 weeks. Clinically, this manifests as itchy, erythematous, edematous papules, and plaques, often surrounded by a vasoconstricted halo. Individual lesions often last less than 24 h and are characterized by spontaneous appearance and resolution in completely random areas on the skin and/or subcutaneous tissues. Anaphylaxis is a severe form of these reactions often characterized by swelling in the throat, difficulty breathing, hypotension, and possibly even death if medical treatment is not provided. These reactions can occur immediately after the administration of a food or medication, or can sometimes be delayed as well. Table 13.6 depicts drugs commonly implicated in Urticaria, Angioedema, and Anaphylaxis.

Management includes identifying and trying to eliminate any potential offending agent, and avoiding re-exposure or re-challenge in the future. First-line medical management includes the use of second-generation, nonsedating H1 antihistamines at regular or supra-therapeutic doses. In cases of chronic urticaria, where first-line treatments fail after maximal titration, patients can be initiated on the biologic omalizumab. In cases of anaphylaxis, patients should be given an epinephrine injection (e.g., EpiPen®) to inject and proceed immediately to the hospital.

Fixed Drug Eruption [6, 18]

A fixed drug eruption is an adverse drug reaction characterized by the formation of an erythematous patch or plaque that will recur at the same site with re-exposure to the drug. This distinguishing feature is why this eruption is classified as “fixed.” Early lesions are sharply demarcated erythematous macules and become edematous, forming a plaque,

Fig. 13.2 Example of drug timeline**Table 13.6** Drugs commonly implicated in urticaria, angioedema, and anaphylaxis^a

Drug	
Antibiotics Penicillins, cephalosporins, sulfonamides	Anti-hypertensives Angiotensin-converting enzyme inhibitors, calcium channel blockers
Histamine-releasing drugs Opiates, amphetamines, aspirin	Oral contraceptives

^aNot an inclusive list

which may evolve to become a bulla (blister) and then an erosion. The lesions are commonly solitary, however, they may be multiple with random distribution. In previously sensitized individuals, lesions may occur from 30 min to 8 h after ingesting the drug. A list of commonly implicated drugs are listed in Table 13.7.

The clinical course of the eruption usually resolves days to weeks after the drug is discontinued. Non-eroded lesions can be treated with a potent topical corticosteroid ointment. Eroded cutaneous lesions can be treated with a protective or antimicrobial ointment and dressing until the site has re-epithelialized. Patients may experience pain with the eruptions, especially for lesions on mucosal regions. In these cases, pain management should be addressed.

Drug-Induced Hypersensitivity Syndrome [6, 16]

Drug-induced hypersensitivity syndrome (DIHS) is also known as drug reaction with eosinophilia and systemic symptom (DRESS). Drug-induced hypersensitivity syndrome (DIHS). A skin eruption is accompanied by systemic symptoms and internal organ involvement (e.g., liver, kidney, heart, and bone marrow). The eruption typically presents as an exanthem, erythematous centrofacial swelling with fever, malaise, and lymphadenopathy. Over 70% of patients will present with eosinophilia as well. In addition, liver function

Table 13.7 Drugs commonly implicated in fixed drug eruptions^a

Drug	
Phenolphthalein (laxatives)	Barbiturates
Tetracyclines	NSAIDs
Metronidazole	Salicylates
Sulfonamides	Food dyes (yellow)

NSAIDs nonsteroidal anti-inflammatory drugs

^aNot an inclusive list

test abnormalities and/or hepatosplenomegaly are helpful diagnostic tools. More specifically, the clinician should order CBC, liver function tests (LFTs), BUN, and creatinine as the liver, kidney, and bone marrow are common targets. Signs and symptoms of (DIHS/DRESS) typically begin in the third week after the start of the medication or after increasing a medication dose, however, the range may be as little as 1 week and up to 12 weeks. The fatality rate of DIHS/DRESS may be up to 10%. See Table 13.8 for commonly implicated drugs.

Signs and symptoms of DIHS/DRESS may persist and recur for many weeks after the cessation of drug treatment. It is paramount that all suspect medications are stopped or substituted and all nonessential medications are discontinued. Therapy may consist of the use of topical steroids, systemic antihistamines, and, in severe cases, systemic steroids may be initiated to avoid impending organ failure. There is a need to continue monitor organs for functional decline.

Epidermal Necrolysis Spectrum [6, 16]

The epidermal necrolysis spectrum consists of both Steven-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) which are both acute life-threatening mucocutaneous reactions and constitute a dermatologic emergency. Patients are classified into one of three groups according to body surface area (BSA) involvement (SJS <10%, SJS/TEN 10–30%, TEN >30%). Mortality rates vary from 5% to 12% for SJS and greater than 20% for TEN [6]. Increasing age, significant

Table 13.8 Drugs commonly implicated in DIHS/DRESS^a

Drugs	
Allopurinol	Anticonvulsants Phenytoin Carbamazepine Lamotrigine
Antibiotics Sulfonamides Penicillin Minocycline Metronidazole	NSAIDS Sulindac Diclofenac Meloxicam
Anti-TB drugs Isoniazid	Anti-HIV drugs Abacavir

DIHS drug-induced hypersensitivity syndrome, DRESS drug reaction with eosinophilia and systemic symptoms, NSAIDS nonsteroidal anti-inflammatory drugs

^aNot an inclusive list

Table 13.9 Drugs commonly implicated in toxic epidermal necrolysis^a

Drug	
Sulfa antibiotics, sulfasalazine	NSAIDs
Allopurinol	Nevirapine
Tetracycline	Thiacetazone
Anticonvulsants (carbamazepine, lamotrigine, phenobarbital, phenytoin)	

NSAIDS nonsteroidal anti-inflammatory drugs

^aNot an inclusive list

comorbid conditions, and greater extent of skin involvement correlate with poorer prognosis. Over 100 different drugs have been associated with SJS/TEN, and the highest risk agents are listed in Table 13.9.

These two conditions represent similar processes but differ in severity based on the body surface area that is involved. They are characterized by extensive necrosis and detachment of the epidermis and mucosal surfaces. Clinical presentation typically begins within 8 weeks after the onset of drug exposure. Mucous membrane involvement (buccal, ocular, genital), fever, headache, rhinitis, and myalgias may precede the lesions by 1–3 days. Eruption is initially symmetric and distributed on the face, upper trunk, and proximal extremities. Early skin lesions are characterized by erythematous, irregularly shaped, dusky red to purpuric macules, which coalesce as they progress. Lesions evolve to flaccid blisters and the necrotic epidermis is easily detached at pressure points or by frictional trauma, revealing large areas of exposed, red, sometimes oozing dermis.

Early recognition and the withdrawal of the offending drug(s) is paramount to prognosis. In case of doubt, all nonlife-sustaining drugs should be discontinued. Care should proceed in a burn unit for patients with >25–30% BSA involvement and extensive supportive care is needed.

Clinical Pearls

- Always take a comprehensive history and perform a physical examination if possible when a patient presents with a rash or lesion in order to determine if this is an issue that can be dealt with at the pharmacy level or should be referred to a physician
- Use correct, precise, and common terminology when describing rashes and lesion to ensure accurate records and to be able to communicate with other health-care practitioners using a common language
- If patients present with a rash, along with systemic symptoms such as fever, facial edema, or malaise, refer to a physician for prompt assessment
- Never forget the 7 “I”s when looking for causative agents of drug reactions
- When creating a drug timeline, always use day 0 as your starting point and work forward and backward to determine the culprit medication
- When uncertain of a definitive diagnosis, or if patients are unresponsive to therapy, pharmacists should refer patients for further assessment and workup through their general practitioner or dermatologist

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Suggested Resources

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Part III

Chronic Illnesses Assessment

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Chapter Objectives

1. Describe the diagnostic criteria and tests for diabetes.
2. Describe glycemic control targets for different populations with diabetes.
3. Describe treatment options for patients with diabetes.
4. Apply various tests to assess glycemic control.
5. Describe hypoglycemia, its symptoms and its treatment, and how to avoid it.

Background

Diabetes mellitus is a group of metabolic diseases characterized by elevated blood glucose levels (hyperglycemia) which could be caused by flaws in the secretion of insulin, its action, or both [1, 2]. An estimated one in ten people in the world were living with diabetes in 2021 [3]. This proportion is expected to reach 15% by 2045 because of the rise in the obesity and physical inactivity levels [3–5]. Due to its chronic nature and the severe complications associated with it, diabetes carries a health and a financial burden on the affected individuals, their care givers, and society as a whole [5]. Blood sugar levels that are not within the target range put patients at higher risk for microvascular and macrovascular complications [5]. Pharmacists are frontline primary healthcare pro-

viders who see patients with diabetes frequently. Their interventions in patients with diabetes are well supported by high-level evidence in the literature. This evidence combined with their interest in caring for patients with diabetes puts them in a key position to join the fight against diabetes.

Diabetes can be divided into:

1. *Type I diabetes*

Accounts for 5–10% of the total diabetes cases and occurs when β -cells in the pancreas are destructed by a cellular-mediated autoimmune attack or other unknown etiology [2, 5].

2. *Type II diabetes*

Accounts for around 90% of the total diabetes cases [2]. It encompasses insulin resistance and insulin deficiency at varying degrees [2, 5].

3. *Gestational diabetes*

“Any degree of glucose intolerance with the onset or first recognition during pregnancy” [2, 5].

4. *Other specific types of diabetes*

This includes a variety of conditions that are caused by genetics, other diseases, or substance use [2, 5].

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Case Finding and Diagnosis

Case Finding

Early identification and management of diabetes can help achieve the treatment targets and delay/prevent long-term complications [5]. The patient identification process can be time consuming and frustrating because of the poor yield when healthcare professionals use traditional screening meth-

ods [6]. Screening methods include applying tests to entire populations to determine prevalence or probability that the individual will have a disease regardless of the presence or absence of risk factors [6]. In order to improve the yield and the patient identification process as a whole, case finding (a focused approach) is suggested. This is a targeted approach using demographics, risk factors, and/or symptoms to decide whether to apply a test or proceed with further testing [6].

Type I Diabetes

Due to the lack of evidence for type I diabetes prevention and the fact that various serological biomarkers are not widely available, general screening recommendations cannot be made [5]. However, family history of type I diabetes (with special attention to the sex of the family member and their age at onset) can help in estimating the risk of developing type I diabetes [7].

Type II Diabetes

All individuals who are 40 years or older should be assessed using fasting plasma glucose (FPG) or glycated hemoglobin (A1C) every 3 years [8]. The same recommendation applies for those who are at high risk of developing diabetes using a risk assessment calculator. The Canadian Diabetes Risk Assessment Questionnaire (CANRISK) has been validated for assessing diabetes risk in Canada [9]. More frequent (every 6–12 months) and/or earlier testing, using FPG, A1C, or two-hour plasma glucose (2hPG) in a 75 g oral glucose tolerance test (OGTT), should be considered in individuals who are at a very high risk of developing diabetes using a risk assessment calculator or those who have at least one of the following risk factors [5]:

- History of
 - Prediabetes (impaired glucose tolerance, impaired fasting glucose, or A1C 6.0–6.4%)
 - Gestational diabetes
 - Delivering macrosomic infants
- Elements of metabolic syndrome (see below)
- Medications associated with hyperglycemia (e.g., statins, glucocorticoids)
- Medical conditions associated with diabetes (e.g., polycystic ovary syndrome, psychiatric disorders)
- End organ damage associated with diabetes (micro- and macrovascular complications)
- Members of high-risk population
 - Indigenous
 - Hispanic
 - South Asian
 - African
 - Asian
 - Arab

Diagnosis

The following tests are used as the diagnostic criteria for diabetes [2, 5]:

1. FPG (no caloric intake for at least 8 h)
2. 2hPG in a 75 g OGTT
 - Plasma glucose concentration is measured 2 h after taking a glucose solution (75 g anhydrous glucose dissolved in water)
3. A1C (a measurement of the average glucose control over the previous 3 months)

According to the result of the aforementioned tests, individuals can be classified as outlined in Table 14.1 [2, 5].

Prediabetes refers to individuals with impaired fasting glucose (IFG) (6.1–6.9 mmol/l), impaired glucose tolerance (IGT) (7.8–11.1 mmol/l), and/or A1C between 6.0% and 6.4% [5]. Individuals with prediabetes are at high risk of developing diabetes and its complications. As such, they could benefit from cardiovascular (CV) risk factor modification [5]. Prediabetes usually occurs in the context of the metabolic syndrome, a multifaceted condition that is characterized by a group of abnormalities which include elevated lipid levels, elevated blood pressure, and abdominal obesity [5]. Metabolic syndrome is defined as having at least three of the following disorders [10]:

1. Elevated waist circumference, depending on the patient's ethnicity:
 - Canada, United States: ≥ 88 cm for females and ≥ 102 cm in males
 - Europe, Middle East, Sub Sahara Africa, and Mediterranean ≥ 80 cm for females and ≥ 94 cm in males
 - Asia and South and Central America ≥ 80 cm for females and ≥ 90 cm in males
2. Blood pressure ≥ 130 mm Hg systolic and/or ≥ 85 mm Hg diastolic
3. Fasting plasma glucose ≥ 5.6 mmol/l
4. Serum triglycerides ≥ 1.7 mmol/l
5. HDL < 1.3 mmol/l in females and < 1 mmol/l in males

Table 14.1 Diagnostic tests results classification [2, 5]

Test result	FPG (mmol/l)	2hPG (mmol/l)	A1C (%)
Normal	≤ 6.0	< 7.8	< 6.0
Prediabetes	6.1–6.9	7.8–11.1	6.0–6.4
Diabetes	≥ 7.0	≥ 11.1	≥ 6.5

2hPG two hours plasma glucose, A1C glycated hemoglobin, FPG fasting plasma glucose

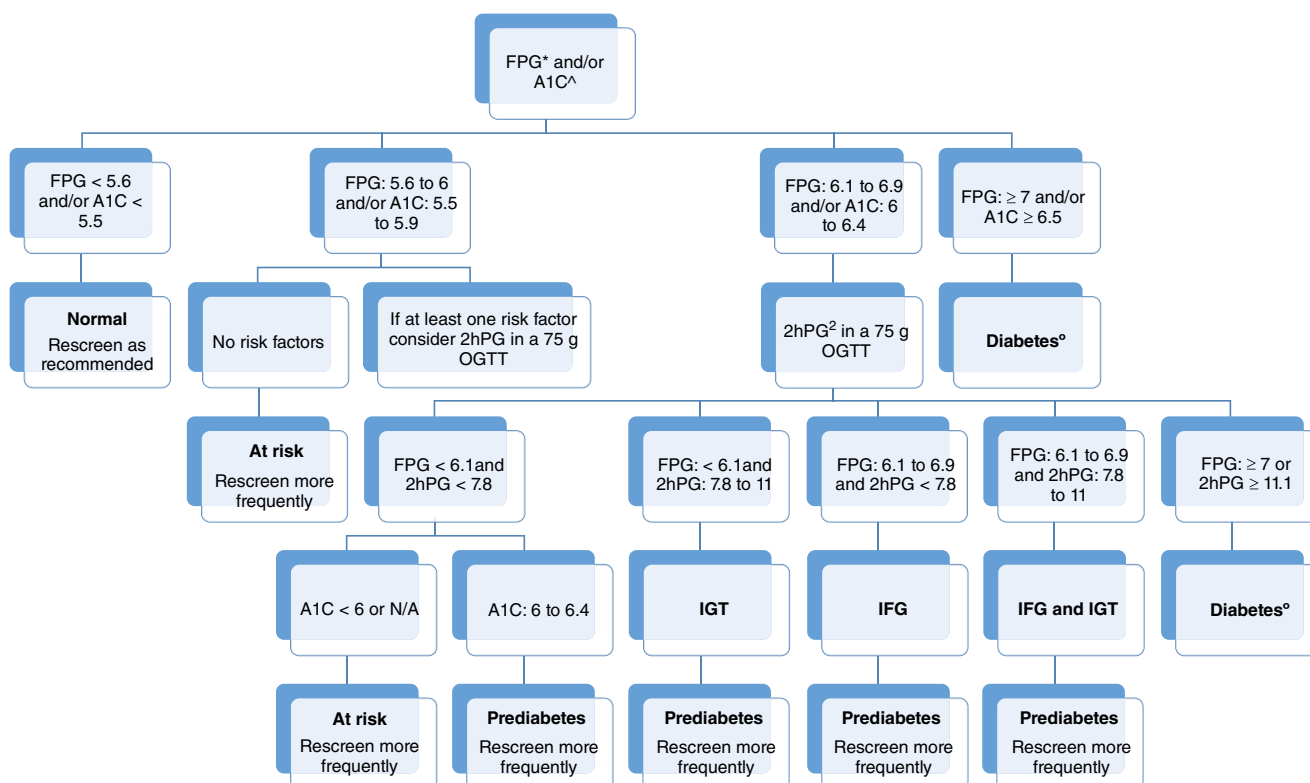


Fig. 14.1 Case finding and diagnosis algorithm for type II diabetes [5]. *FPG's unit is mmol/l, ^A1C's unit is %, ^2hPG's unit is mmol/l. °If the individual had no symptoms of hyperglycemia but one test result was indicating diabetes, the test should be repeated on another day to con-

firm the diagnosis. 2hPG two hours plasma glucose, A1C glycated hemoglobin, FPG fasting plasma glucose, IFG impaired fasting glucose, IGT impaired glucose tolerance, N/A not available

The decision of which diabetes diagnostic test to use is up to the clinician's judgement [5]. If the results of two of the tests are available and are indicating diabetes then the diagnosis is confirmed [5]. While, if the individual had no symptoms of hyperglycemia but only one test result was indicating diabetes, then that test should be repeated on another day to confirm the diagnosis [5]. If the random plasma glucose (any time of the day without considering the time since the last meal) test result was ≥ 11.1 mmol/l but the individual had no hyperglycemia symptoms, an alternate test (FPG, 2hPG, or A1C) is required to confirm the diagnosis [5]. Figure 14.1 illustrates the case finding and diagnosis algorithm for type II diabetes [5].

Management

Type I Diabetes

Insulin should be started at the day of diagnosis for patients with type I diabetes [5]. Table 14.2 lists the types of insulin based on their onset of action [5]. Insulin can be divided into **basal** (refers to intermediate or long-acting insulins which

Table 14.2 Insulin types based on their onset of action

Onset of action	Insulin
Rapid-acting insulin analogues	Aspart
	Aspart (faster-acting)
	Lispro U-100
	Lispro U-200
	Glulisine
Short-acting insulins	Regular (Novolin® ge Toronto, Humulin®-R.)
	Regular (Entuzity® (U-500))
Intermediate-acting insulins	NPH (Humulin®-N, Novolin® ge NPH)
Long-acting insulins	Glargine U-100
	Glargine U-300
	Glargine biosimilar
	Detemir
	Degludec U-100
	Degludec U-200
	Insulin regular-NPH (Humulin® 30/70)
Premixed insulins	Insulin regular-isophane (Novolin® ge 30/70, 40/60, 50/50)
	Biphasic insulin aspart
	Insulin lispro/lispro protamine

NPH neutral protamine Hagedorn insulin

provides glucose control in the fasting state and between meals) and **bolus** (refers to rapid or short-acting insulins which is administered to control the glucose rise after meals or correct hyperglycemia) [5].

Insulin regimens should be individualized based on age, general health, lifestyle, diet, hypoglycemia awareness, ability for self-management, adherence, and social and financial aspects [5]. The most successful insulin regimens for managing type I diabetes are those that combine basal and bolus insulin. Such regimens attempt to mimic the pancreas' normal activity in a person without diabetes [5].

Bolus insulin dose may vary between meals depending on the carbohydrate intake and glycemic index (a scale that ranks carbohydrate-rich foods by how much they raise blood glucose levels compared to a standard food), exercise, time since the last insulin dose, and blood glucose levels [5]. Injections of rapid-acting insulin analogues before meals resulted in lower glucose levels after meals and improved overall glycemic control [11–14]. Faster-acting insulin aspart can be administered at the start of the meal or up to 20 min after the start if necessary [15]. The rest of the rapid-acting insulin analogues should be administered 0–15 min before starting the meal, while short-acting insulins should be administered 30–45 min before the meal [5]. When compared to short-acting insulin, insulin aspart and lispro have been associated with improved glucose control after meals, A1C [14, 16] and quality of life [17], and reduced nocturnal hypoglycemia [14, 16]. It has been reported that insulin glulisine is most effective when given before meals and is equivalent to insulin lispro in glycemic control [11, 18]. When compared to insulin aspart, faster-acting insulin aspart had an earlier onset of action and showed superior glucose control after meals and non-inferior A1C reduction in patients with type I diabetes [15].

When combined with bolus insulin, insulin detemir and insulin glargine (U-100) were associated with better glycemic control and less hypoglycemia (including nocturnal) when compared to neutral protamine Hagedorn (NPH) insulin (once or twice daily) [19–26]. It has been reported that bio-similar glargine (same amino acid sequence but produced in a different process than glargine) had similar efficacy and safety outcomes in adults with type I diabetes who were switched from insulin glargine (U-100) [27]. When compared to insulin glargine (U-100), insulin glargine (U-300) had consistent, gradual, and extended flat release from subcutaneous tissues, longer duration of action (>30 h), similar effect on A1C and hypoglycemia (one study reported less nocturnal hypoglycemia), and less weight gain [28–32]. When compared to insulin glargine and insulin detemir, insulin degludec had longer duration of action (42 h), similar glycemic control, lower nocturnal hypoglycemia, and less basal and total insulin dose [29–40]. Degludec (U-100) and (U-200) have similar glucose control effects and half-lives (29).

The choice of insulin regimen should be accompanied with ongoing comprehensive education about caring for and using insulin; self-monitoring blood glucose (SMBG); preventing, recognizing, and treating hypoglycemia; adjusting food intake and exercise; and managing diabetes on sick days [5].

Type II Diabetes

Type II diabetes treatment regimens and glycemic targets should be individualized [5]. Such treatment regimens should aim to avoid and treat hyperglycemia and reduce the risk of microvascular and macrovascular complications [41].

A1C Target

While a target of A1C $\leq 7\%$ has been recommended for most patients with diabetes [41–44], a stricter target of $\leq 6.5\%$ has been recommended for patients with type II diabetes, who are at low risk for hypoglycemia, in order to reduce their risk of retinopathy and chronic kidney disease (CKD) [44, 45]. A less stringent target of 7.1–8% has been recommended for patients who are functionally dependent to reduce their risk of hypoglycemia and prevent overtreatment. A similar reasoning has been used to justify a target of 7.1–8.5% in patients who have limited life expectancy, a history of severe hypoglycemia, especially if combined with hypoglycemia unawareness and/or frailty with/without dementia [5]. Table 14.3 lists the A1C targets for different populations.

It is important to keep in mind that the treatment targets in older adults depend on their clinical frailty index [5]. Table 14.4 lists the A1C targets in older adults based on the clinical frailty index.

Agent Choice

Table 14.5 lists all the medication classes and agents that can be used in the management of type II diabetes. The choice of the treatment regimen at the diagnosis of type II should depend on the difference between the patient's A1C and their individual target and the presence of symptomatic hyperglycemia and/or metabolic decompensation [5]:

Table 14.3 A1C targets for different populations

Population	A1C target
Most adults with type I or type II	$\leq 7.0\%$
Adults with type II to reduce the risk of CKD and retinopathy if at low risk of hypoglycemia	$\leq 6.5\%$
Functionally dependent	7.1–8.0
History of recurrent severe hypoglycemia	7.1–8.5
Limited life expectancy	7.1–8.5
Frail older adults and or dementia	7.1–8.5
End of life	Measurements are not recommended

A1C glycated hemoglobin, CKD chronic kidney disease

Table 14.4 A1C targets in older adults based on the clinical frailty index

Clinical frailty index	A1C target
Functionally independent (1–3)	≤7%
Functionally dependent (4–5)	7.1–8%
Frail with/without dementia (6–8)	7.1–8.5%
End of life	Measurement not recommended Avoid symptomatic hyperglycemia or any hypoglycemia

A1C glycated hemoglobin

- If current A1C is <1.5% of the individualized target:
 - Healthy lifestyle interventions (exercise, weight management, and healthy eating) with/without metformin should be started.
 - If glycemic target is not achieved within 3 months, metformin should be started (if not initiated already) or its dose should be increased.
 - If glycemic target is not achieved within 3 months, a second agent should be added.
- If current A1C is ≥1.5 of the individualized target:
 - Healthy lifestyle interventions with metformin should be started.
A second agent could be considered when starting the treatment regimen.
 - If glycemic target is not achieved within 3 months, a second agent should be added.
- If the patient has symptomatic hyperglycemia and/or metabolic decompensation (dehydration, diabetic ketoacidosis, hyperglycemic hyperosmolar state):
 - Healthy lifestyle interventions with insulin and with/without metformin should be started.
 - If glycemic target is not achieved within 3 months, a second agent should be added.

If not contraindicated, metformin is considered the first line of treatment in patients with type II diabetes [5]. The presence of established cardiovascular or renal disease or cardiovascular risk factors should govern the choice of the second line agent [5]. If the patient has established cardiovascular disease, renal disease or two cardiovascular risk factors in those who are 60 years or older, then we should choose an agent with demonstrated cardiorenal vascular benefits such as GLP-1 receptor agonists and SGLT2 inhibitors [48–50]. If the patient does not have any of the aforementioned conditions, The choice of the second line agent should consider the patient's medical history, social and work factors, their preferences and values, and the agent's characteristics [5]. Table 14.6 lists the second line classes and the considerations that could impact their choice in the treatment. Table 14.7 summarizes the renal dosage adjustment of antihyperglycemic agents.

Table 14.5 Medication classes used in the management of type II diabetes [5, 46, 47]

Medication class	Medications (total adult daily dose)
Biguanide	Metformin (500–2550 mg)
DPP-4 inhibitors	Alogliptin (25 mg) Linagliptin (5 mg) Saxagliptin (5 mg) Sitagliptin (100 mg)
GLP-1 receptor agonists	<i>Short-acting</i> Exenatide (10–20 mcg) Lixisenatide (10–20 mcg) <i>Longer-acting</i> Dulaglutide (0.75–1.5 mg once a week) Exenatide extended release (2 mg once a week) Liraglutide (0.6–1.8 mg once a day) Semaglutide (0.25–1 mg once a week) GLP-1 receptor agonist+GIP Tirzepatide (2.5–15 mg once a week)
SGLT2 inhibitors	Canagliflozin (100–300 mg) Dapagliflozin (5–10 mg) Empagliflozin (10–25 mg) Ertugliflozin (5–15 mg)
Alpha-glucosidase inhibitor	Acarbose (50–300 mg)
Insulin secretagogue	Gliclazide (80–320 mg)
Sulfonylureas	Gliclazide modified-release (30–120 mg) Glimepiride (1–8 mg) Glyburide (2.5–20 mg)
Insulin secretagogue	Repaglinide (1.5–16 mg)
Meglitinides	
Thiazolidinedione	Pioglitazone (15–45 mg) Rosiglitazone (4–8 mg)
Rapid-acting insulin analogues	Aspart Aspart (faster-acting) Lispro U-100 Lispro U-200 Glulisine
Short-acting insulins	Regular (Novolin® ge Toronto, Humulin®-R,) Regular (Entuzity® (U-500))
Intermediate-acting insulins	NPH (Humulin®-N, Novolin® ge NPH)
Long-acting insulins	Glargine U-100 Glargine U-300 Glargine biosimilar Detemir Degludec U-100 Degludec U-200
Premixed insulins	Insulin regular-NPH (Humulin® 30/70) Insulin regular-isophane (Novolin® ge 30/70, 40/60, 50/50) Biphasic insulin aspart Insulin lispro/lispro protamine

DPP-4 inhibitors dipeptidyl peptidase 4 inhibitors, *GIP* glucose-dependent insulinotropic polypeptide, *GLP-1* glucagon-like peptide-1, *SGLT2* sodium-glucose cotransporter 2

Table 14.6 Second-line classes and the effects that could impact their choice in the treatment in type II diabetes

Effect	DPP-4 inhibitors	GLP-1 receptor agonists	SGLT2 inhibitors	Alpha-glucosidase inhibitor	Sulfonylureas	Meglitinides	Thiazolidinedione	Insulin
Cardiovascular outcomes	Neutral	CV benefit: liraglutide Semaglutide Neutral: exenatide extended-releases, lixisenatide	CV benefit: canagliflozin and empagliflozin Ertugliflozin					Neutral: glargine Degludec (non-inferior to glargine)
Hypoglycemia	Rare	Rare	Rare	Rare	Yes	Yes	Rare	Yes
Weight	Neutral	Weight loss	Weight loss	Neutral	Weight gain	Weight gain	Weight gain	Weight gain
Glycemic efficacy ^a	Moderate	Moderate to high	Moderate to high	Mild	Moderate	Moderate	Moderate	Moderate to very high
Other considerations ^b	- Rare joint pain - Caution with saxagliptin and heart failure	- Subcutaneous injection - GI adverse events - Cases of gallstone disease - Contraindicated with personal/family history of medullary thyroid cancer or MEN 2	- Reduced progression of nephropathy and heart failure hospitalization with empagliflozin, dapagliflozin, and canagliflozin in those with clinical CVD - Genital and urinary tract infections - Hypotension - Dose-related changes in LDL-C - Rare cases of diabetic ketoacidosis (which may occur without hyperglycemia) - Caution with renal dysfunction, loop diuretics, and older adults - Increased risk of fractures and lower extremity amputation (avoid if prior amputation) with canagliflozin - Dapagliflozin not to be used with bladder cancer	- Common GI adverse events - Requires 3 times daily dosing - Requires using glucose if hypoglycemia occurs	- Relatively rapid blood glucose lowering - Glyburide associated with more hypoglycemia than gliclazide and glimepiride	- Reduced postprandial glycemia but usually requires 3–4 times daily dosing - Repaglinide contraindicated when co-administered with clopidogrel or with gemfibrozil	- Require 6–12 weeks for maximal effect - Mild increase in HDL-C - May induce edema and/or congestive heart failure - Rare occurrence of macular edema - Higher occurrence of fractures - Pioglitazone not to be used with bladder cancer - Controversy regarding MI risk for rosiglitazone	- Subcutaneous injection - No dose ceiling - Flexible regimens

DPP-4 inhibitors dipeptidyl peptidase 4 inhibitors, *GLP-1* glucagon-like peptide-1, *SGLT2* sodium-glucose cotransporter 2

^aWhen added to metformin

^bSee Table 14.7 for renal dose adjustment

Table 14.7 Renal dosage adjustment of antihyperglycemic agents

eGFR (ml/min)	Medication use		
	No dose adjustment	Reduce dose	Use alternative agent
≥60	Dose adjustment not necessary		
45–59 CKD 3A	Acarbose, metformin, linagliptin, dulaglutide, exenatide (caution when eGFR <50), liraglutide, lixisenatide, gliclazide (caution), glimepiride (caution), repaglinide, TZDs (caution), insulins Ertugliflozin	Alogliptin (12.5 mg daily), [eGFR <50 saxagliptin (2.5 mg daily), sitagliptin (50 mg daily)], canagliflozin (100 mg daily, do not initiate), empagliflozin (do not initiate)	Glyburide, dapagliflozin
30–44 CKD 3B	Acarbose, linagliptin, dulaglutide, exenatide (caution), liraglutide, lixisenatide, gliclazide (caution), glimepiride (caution), repaglinide, TZDs (caution), insulins	Metformin (500–100 mg daily), alogliptin (12.5 mg daily), saxagliptin (2.5 mg daily), sitagliptin (50 mg daily)	Glyburide, canagliflozin, dapagliflozin, empagliflozin, ertugliflozin
15–29 CKD 4	Linagliptin, dulaglutide, liraglutide, repaglinide (caution), TZDs (caution), insulins (caution)	Alogliptin (6.25 mg daily), saxagliptin (2.5 mg daily), sitagliptin (25 mg daily)	Acarbose, metformin, exenatide, lixisenatide, gliclazide, glimepiride, glyburide, canagliflozin, dapagliflozin, empagliflozin Ertugliflozin
<15 or dialysis CKD 5	Linagliptin (caution), dulaglutide (caution), repaglinide (caution), TZDs (caution), insulins (caution)	Alogliptin (6.25 mg daily), sitagliptin (25 mg daily)	Acarbose, metformin, saxagliptin, exenatide, liraglutide, lixisenatide, gliclazide, glimepiride, glyburide, canagliflozin, dapagliflozin, empagliflozin Ertugliflozin

CKD chronic kidney disease, eGFR estimated glomerular filtration rate, TZD thiazolidinedione

The choice of the second agent should be based on the following considerations [5]:

- *The presence/absence of clinical CVD* [5]
- Avoiding hypoglycemia
- Avoiding weight gain (in patients who are overweight)
- Adequate glycemic efficacy
- Patient values and preferences
- Level of kidney function (based on eGFR)
- Co-morbidities
- Cost, availability, and coverage
- Planning pregnancy

Assessment of Patients with Diabetes

It is important to remember not to be solely glucocentric (focused only on blood glucose management) when managing patients with diabetes [51]. The “*ABCDEs*” approach [52] can help to address all aspects of CV risk. This includes assessing A1C, blood pressure, cholesterol, drugs for CVD reduction, exercise/eating, screening for complications, smoking cessation, and self-management [52]. Table 14.8 lists the “*ABCDEs*” of diabetes care.

Adherence

Patient’s adherence to treatment regimen should be monitored and assessed at each encounter with the healthcare professional [53]. Such close monitoring can help answer questions/concerns that patients may have about the disease, the treatment regimen, and the complications [54]. Patients reported that they appreciated receiving such care and compassion [55].

Certain medications should be withheld in periods of acute illness [56]. The abbreviation SADMANs has been used to refer to those medication classes [56]. This includes sulfonylureas, ACE-inhibitors, diuretics, direct renin inhibitors, metformin, angiotensin receptor blockers, nonsteroidal anti-inflammatory drugs, and SGLT2 inhibitors [56].

Control and Monitoring

A1C

A1C is a measurement of the average glucose control over the previous 3 months [57]. As such, it is used as an indicator

Table 14.8 The ABCDES of diabetes care [52]

		Assessment
A	A1C	Assess A1C level (see individual targets above) Assess hypoglycemia and driving safety if patient is on insulin or insulin secretagogue
B	Blood pressure	Assess BP (target <130/80 mmHg) Assess the risk of falls if patient is on treatment
C	Cholesterol	Assess LDL (target <2.0 mmol/L)
D	Drugs for CVD reduction	If patient has CVD, use ACE-inhibitor/ARB, statin, and ASA (if A1C is not at target use antihyperglycemic agents with demonstrated cardiovascular benefits) If patient has diabetes complications, use ACE-inhibitor/ARB and statin If patient is ≥ 40 years and has type II diabetes, use statin If patient is ≥ 55 years and has risk factors, use ACE-inhibitor/ARB
E	Exercise and eating	Assess diet (target: healthy diets) Assess exercise (target: 150 min of moderate-vigorous exercise per week and 2–3 times per week of resistance exercise)
S	Screening for complications	Assess heart using ECG every 3–5 years if patient is >40 years or has diabetes complications Assess feet using microfilament/vibration every year or more frequently if there are any abnormalities Assess kidney function and status (eGFR and ACR) every year or more frequently if there are any abnormalities Assess eyes using dilated retinal exam every year or more frequently if there are any abnormalities
S	Smoking cessation	Ask permission to support, give advice, and provide therapy to help the patient quit
S	Self-management	Assess for all aspects that can prevent patient from achieving their goals, e.g., cost/coverage, mental health, stress

A1C glycated hemoglobin, ACE angiotensin converting enzyme, ACR random urine albumin/creatinine ratio, ARB angiotensin II receptor blocker, BP blood pressure, CVD cardiovascular disease, eGFR estimated glomerular filtration rate, mmHg millimetres of mercury

for the treatment effectiveness [5]. A1C should be tested regularly (at least every 3 months) when treatment targets are not achieved or when the treatment regimen is being changed [5]. Target A1C should be attained within 3–6 months of initiating or adjusting the treatment regimen [5].

Blood Glucose Monitoring

Capillary blood glucose (CBG) monitoring, intermittently scanned continuous glucose monitoring (icCGM), and real-time continuous glucose monitoring (rtCGM) are measures that provide information about glycemic control [5]. When dispensing a glucometer or testing strips, it is vital to make sure that the patient is capable of testing their blood glucose

successfully [54]. CBG frequency should be individualized based on the diabetes type, treatment regimen (including diabetes treatment or medications that could affect glycemic control), glycemic control, proneness and awareness to hypoglycemia, and acute illness [5]. Information about prandial glucose control can be obtained with testing before and after the same meal, while bedtime to morning testing will provide information on basal control [58].

Kidney Function and Status Assessment

Monitoring kidney function and status plays an essential role in diabetes treatment, as it could affect the treatment regimen and the medication doses [59]. Indeed, it has been reported that pharmacists ordering kidney function and status tests [eGFR and random urine albumin/creatinine ratio (ACR)] helped uncover a high percentage of unrecognized chronic kidney disease [60].

Treatment decisions should not be made solely based on one kidney function test result, as it could be affected by certain medications (e.g., fenofibrates), conditions (kidney injury, amputation), or diet (high-protein diet) [61].

Hypoglycemia

Hypoglycemia is defined as a combination of low plasma glucose level (<3.9 mmol/L), presence of neuroglycopenic (difficulty concentrating, speaking, confusion, headache, dizziness, weakness, drowsiness, and vision changes) or autonomic (sweating, hunger, trembling, palpitations, nausea, tingling, and anxiety) symptoms, and symptoms that respond to treatment (administration of carbohydrate) [5, 62].

According to the International Hypoglycemia Study Group [62] the severity of hypoglycaemia should be characterized into three levels based on the presence and severity of associated symptoms rather than by glucose levels alone. Therefore, hypoglycemia can be divided into the following [62]:

- Level 1: The patient has autonomic symptoms but is able to self-treat, does not have neuroglycopenic symptoms or changes to mental status, and plasma glucose levels are below normal (often 3.0–3.9 mmol/L).
- Level 2: The patient has neuroglycopenic symptoms with or without autonomic symptoms, but is able to self-treat their symptoms, and plasma glucose levels are below normal (often <3.0 mmol/L)
- Level 3: Patient usually needs assistance from another person as they may be unconscious and experiencing neuroglycopenic symptoms resulting in significantly altered mental/physical status, and plasma glucose levels are below normal (independent of glucose level)

It is essential to try to avoid hypoglycemia, especially in those who are more prone to hypoglycemia [5]. Testing plasma glucose is recommended, when possible, to confirm hypoglycemia and prevent overtreatment [5]. Treatment should be based on the severity of hypoglycemia [5]:

- Level 1 or 2: 15 g of carbohydrates (15 g glucose tablets, 150 ml of juice or soft (non-diet) drink, 15 ml (1 tablespoon) of honey, 5 cubes of sugar, 6 Lifesavers™)
 - Plasma glucose should be tested within 15 min, if plasma glucose is still < 3.94 mmol/l another 15 g of carbohydrate should be administered.
- Level 3 (conscious patient):
 - Able to swallow: Option 1: 20 g of carbohydrate preferably as glucose tablets or equivalent; Option 2: glucagon 1 mg subcutaneously/intramuscularly or glucagon 3 mg intranasally. Plasma glucose should be tested within 15 min, if plasma glucose is still < 3.9 mmol/l another 15 g of carbohydrate should be administered.
 - Unable to swallow: Administer glucagon 1 mg subcutaneously/intramuscularly or glucagon 3 mg intranasally.
- Level 3 (unconscious): If there is intravenous access, 10–25 g of glucose should be given intravenously over 1–3 min. Glucagon 1 mg subcutaneously/intramuscularly or glucagon 3 mg intranasally should be given if there is no intravenous access. Emergency services should be called. Patient should discuss this episode with the healthcare team.

The patient should have a meal or a snack once hypoglycemia is reversed. If the next meal is more than an hour away, the patient should have a snack that includes a protein source and 15 g of glucose [5]. Individuals who are on agents that may cause hypoglycemia (e.g., insulin, insulin secretagogues) should be educated about hypoglycemia prevention, recognition, and treatment [5]. Such education should be revisited regularly [5]. Patients should also be educated about driving instructions to make sure that they are not posing risks on the others or themselves [54].

Complications

Blood sugar levels that are not within the target range can lead to injury to vasculature [63]. Damage to the large blood vessels may lead to macrovascular complications, while microvascular complications may occur if the damage occurs to the small blood vessels [63]. Macrovascular complications include cardiovascular diseases such as heart attack and stroke, while retinopathy, nephropathy, and neuropathy are considered microvascular complications [64]. See Table 14.8 for the frequency of assessment for complications.

Advise for Patients with Diabetes

Foot Care

Patients with diabetes should do the following foot care activities every day [65]:

- Check their feet for cuts, cracks, bruises, blisters, sores, infections, or any unusual markings.
- Apply skin lotion on heels and toes and wipe out excess lotion.
- Change socks.

They should also trim their toe nails straight across and clean any cuts or scratches with mild soap and water and cover with dry dressing. Healthcare professionals should be contacted immediately if there is pain, swelling, warmth, or redness [65]. Patients are advised wear comfortable and supportive shoes, avoid high heels, buy their shoes in the late afternoon, and avoid extreme temperatures [65].

Vaccination

Patients with diabetes should receive their influenza vaccine every year [5]. They should also get their pneumococcal vaccination if they are 18 years or older and again when they are over 65 years [5]. Herpes zoster vaccine is recommended for individuals who are 60 years or older [5]. All children and those in high-risk groups (does not specify patients with diabetes) for hepatitis B should be vaccinated [5].

Driving

If patients with diabetes are planning on driving they should [5]:

- Measure their blood glucose immediately before driving and every 4 h on long drives.
 - Keep emergency supply of fast-acting carbohydrates.
 - Keep supplies of meals and snacks for longer drives and take regular rests.
 - Measure more frequently if there are factors that could increase the likelihood of hypoglycemia (e.g., exercise).
- Stop the car in a safe location, if hypoglycemia develops.
 - Switch off the car engine.
 - Treat hypoglycemia.
 - Consider waiting 40 min before driving again.

Mental Health

- Diagnosis and management of diabetes are associated with significant stress for the person and caregivers, lower perceived quality of life, stigma, financial burden, worsen psychosocial symptoms (such as depressive and anxious symptoms), and may increase the person's risk for psychiatric disorders [5].
- Screening and assessment for symptoms of distress or common psychiatric disorders (e.g., mood and anxiety disorders) of all individuals with diabetes, as well as the carers of those with diabetes, are recommended [5].
- Person-centered strategies including cognitive behavior therapy, motivational interviewing, and building self-management skills should be integrated into primary care where appropriate [5].

Remission of Type 2 Diabetes

- Remission of type 2 diabetes is defined as reaching A1C targets without any antihyperglycemic medications for 3 months or more [5]. The A1C threshold for remission to:
 - Prediabetes is A1C between 6.0% and 6.4%.
 - Normal glucose concentrations is A1C less than 6.0%.
- If remission is achieved, A1C should be measured at least every 6 months to assess persistence of diabetes remission and monitor for potential relapse [5].

Clinical Pearls

- Pharmacists are frontline primary healthcare providers who see patients with diabetes frequently. Their interventions in patients with diabetes are well supported by high-level evidence in the literature. This evidence combined with their interest in caring for patients with diabetes and the current family physician crisis put them in a key position to join the fight against diabetes.
- Before making any decisions, pharmacists should listen to the patient to evaluate their knowledge and perspective about the situation.
- Initial assessment should include a review of the ABCDEs of the patient with the creation of SMART goals and an ACTION plan to help resolve the identified issues.
- Pharmacists are encouraged to:
 - Assess adherence, glycemic control, hypoglycemia, and any other adverse events or complications during follow-up assessments.
 Follow-up should always include communication within other members of the healthcare team to ensure continuity of care.

- Provide written sick day management instructions so that patients will know which medications to withhold in cases of acute illness.
- Teach about hypoglycemia when appropriate and reinforce regularly.

Teaching should include:

- Hypoglycemia prevention, symptom recognition, and treatment
- Driving instructions to make sure that they are not posing risks on others or themselves

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Hypertension Assessment

15

Ann Thompson and Peter Hamilton

Chapter Objectives

1. To provide an approach to assessing a patient with hypertension including key patient history information to gather at both initial assessment and follow-up visits.
2. To outline the various methods for assessing blood pressure (BP) and their role in diagnosis and monitoring for those with hypertension.
3. To define thresholds for diagnosing hypertension and target BP once a treatment plan is initiated.
4. To outline appropriate follow-up and monitoring parameters for patients with hypertension.

Background

Hypertension is one of the most common illnesses encountered by Canadians, affecting approximately 25% of adults. The negative impacts of hypertension are increased rates of cardiovascular disease, notably stroke, heart failure, atrial fibrillation, chronic kidney disease and death. Pharmacists, partnering with patients, can assess, educate, and help patients better manage this condition through a variety of strategies. This chapter will outline patient assessment considerations to enable pharmacists to provide care to patients with hypertension.

Epidemiology and Etiology

Hypertension is the elevation of systolic or diastolic blood pressure, or both, above normal levels. Hypertension, defined as drug treatment for high BP or BP $\geq 140/90$ mmHg, has a high prevalence in Canada and is present in 22.6% of adults [1]. This is a condition most commonly associated with older adults, and those aged 65+ have a prevalence of 50% and this rises with aging. The prevalence of hypertension in diabetics is also high, with 67.1% having hypertension (defined as drug treatment for high BP or BP $\geq 130/80$ mmHg). In Canada, despite advances in hypertension management, almost two-third of the patients have their blood pressure within target. Notably, self-reported hypertension prevalence has increased approximately twofold in the last two decades. Despite this, age-standardized mortality rates are falling in hypertensive Canadians (from 9.4 to 7.9 deaths per 1000 individuals) due to improved management and better BP control rates [2].

Most cases of hypertension are primary, with unknown cause, with recognition that the condition seems to involve increased sympathetic neural activity and increased angiotensin II and aldosterone activity. Secondary hypertension likely accounts for 10–15% of cases. Examples of secondary hypertension include:

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- Primary hyperaldosteronism
- Renal artery stenosis
- Chronic obstructive sleep apnoea

Risk Factors

The biggest risk factor for developing hypertension is increasing age, a non-modifiable risk. Once over the age of 65, gender does not seem to make a difference in the prevalence of hypertension. A number of modifiable factors can contribute to hypertension, such as excess salt intake, weight gain and obesity, sedentary lifestyle, obstructive sleep apnoea, select medications (e.g., NSAIDs, corticosteroids, stimulants, select antidepressants, such as monoamine oxidase inhibitors and serotonin and norepinephrine reuptake inhibitors, cyclosporine, oral contraceptives and sex hormones), and other substances (excess alcohol intake, licorice root, stimulants such as cocaine).

Presentation

Typically, hypertension has an asymptomatic presentation (30–40%) and is identified by routine measurement of BP. While hypertension is commonly believed to be one of the causes of headaches, the correlation is not established. One study evaluated the prevalence of headache in those with hypertension using office and 24-h ambulatory BP measurement. There was no difference in the prevalence of headaches, migraines or analgesics use between subjects who were hypertensive and normotensive. Given the prevalence of both headache and hypertension, it is difficult to establish causality [3]. In another cohort study assessing symptoms of those taking and not-taking antihypertensive therapy, symptoms reported most commonly by patients not taking therapy include dizziness (53%), headache (51%), tiredness (51%), palpitations (35%), and nervousness/restlessness (31%). Intensity of these symptoms was generally mild-moderate (e.g., pain due to headache rated as 4/10). If present and linked to hypertension, symptoms may resolve with reduction in BP [4].

Measuring Blood Pressure

The measurement of BP can occur in two places: in-office and out-of-office. Within office or clinician settings, the preferred BP measurement method is with an automated machine, unattended, that takes multiple readings, typically three to five in total. This method is called automated office

BP measurement, or AOBP. The less preferred in-office method (for accuracy) is manual or automated measurement done with the clinician present. If manual measurement is performed, the correct technique is outlined by Hypertension Canada: (<https://hypertension.ca/guidelines/supplementary-tables/#suptbl1b>) [5].

Out-of-office measurements are now preferred for three reasons. First, it eliminates white coat hypertension (which is a phenomenon associated with elevated BP in a clinician setting, with home or out-of-office BP readings that are lower). Second, more readings can be generated over a greater time frame to demonstrate the pattern and temporal trends associated with BP. Last, it reduces clinician measurement error due to inappropriate technique. There are two methods for obtaining out-of-office readings: ambulatory blood pressure measurement (ABPM), typically done over 24 h, and home blood pressure measurement (HBPM). Ambulatory BP measurement, if available, is the gold standard test for diagnosis of hypertension; however, it is not always readily available and/or may cost money (depending on coverage in local jurisdiction). This test typically measures BP every 20 min during the day and every 30 min during sleep hours. Hypertension Canada outlines the recommended technique on their website: <https://hypertension.ca/guidelines/supplementary-tables/#suptbl1c>

Home BP monitoring, if done properly, is an accurate predictor of the level of BP control (compared to ABPM) and is a very useful to guide decision-making between patients and clinicians.

BP readings are most accurate and have the potential to reduce overtreatment due to measurement error, when the following conditions are met:

- Using a validated electronic device. Hypertension Canada has a list of validated devices on their website at: <https://hypertension.ca/healthcare-professionals/recommended-devices>. Accuracy of home monitors can be determined by comparing to a machine of known accuracy. Electronic devices that are not accurate cannot be re-calibrated for accuracy.
- Using the appropriate cuff size as per the manufacturer guidelines. A cuff that is too small can overestimate BP, whereas too large a cuff can underestimate BP.
- Being comfortable prior to measurement (i.e., no acute pain, bladder empty, comfortable temperature).
- Sitting at rest, with back supported, for 5 min prior to taking reading. If the back cannot be supported, as is the case with most pharmacy BP measurement kiosks, one study has demonstrated that this does not lead to big differences in BP accuracy (diastolic BP was shown to only increase by approximately 2 mmHg when the back was unsup-

ported, and there was no statistical difference in systolic BP) [6].

- Supporting arm at heart level prior to taking BP. A new study evaluated the impact of having the arm at midheart level, supported on a desk, compared to the hand resting on the person's lap, or unsupported at the person's side. Both the latter 2 positions showed increases in both systolic BP and diastolic BP. The systolic increase was by 3.9 mmHg for the lap, and 6.5 mmHg for the unsupported arm, compared to the arm supported on the desk. The diastolic increases were 4.0 and 4.4 mmHg for each position respectively [7].
- Not talking while taking the BP measurement.

The accuracy of home monitors can be checked by comparing readings to those of a machine of known calibration. For BP machines that are inaccurate, there is no mechanism to “fix” them and a new monitor (with the appropriate cuff size) will need to be purchased. Pharmacists can support patients in home BP monitoring by ensuring they have adequate training and know the appropriate technique. They can also provide information about how to interpret BP readings [8].

Treatment Goals: BP Thresholds and Targets

Pharmacologic treatment thresholds in Canada are recommended based on baseline cardiovascular (CV) risk. Table 15.1 outlines BP thresholds to initiate pharmacologic therapy, and BP targets to achieve, if possible [8]. Even if targets are not achieved, there is a significant reduction in the risk of CV events with a 10–15% lowering of BP from baseline.

Table 15.1 Blood pressure (BP) thresholds to initiate pharmacologic therapy and recommended office BP treatment targets

Population	BP thresholds	BP treatment targets
High risk (defined as presence of CVD or subclinical CVD, CKD, estimated cardiovascular risk $\geq 15\%$, or age ≥ 75)	SBP ≥ 130 mmHg	SBP ≤ 120 mmHg (based on AOBP)
Diabetes	$\geq 130/80$	$\leq 130/80$
Low risk (defined as no target organ damage and low risk according to a CV risk estimation calculator)	$\geq 160/100$	$\leq 140/90$ ^a $\leq 135/85$ if AOBP
All others	$\geq 140/90$	$\leq 140/90$ ^a $\leq 135/85$ if AOBP

BP blood pressure, CVD cardiovascular disease, CKD chronic kidney disease, SBP systolic blood pressure, AOBP automated office blood pressure

^aBP target if the method of AOBP is used

Diagnosis

For persons presenting with features of hypertensive urgency or emergency, or if BP exceeds 180/110, then the diagnosis of hypertension is confirmed, and immediate treatment and ongoing monitoring are required. For all others, the diagnosis of hypertension is best made using out-of-office BP measurements. Figure 15.1 depicts the current diagnostic algorithm as outlined in the Canadian hypertension guidelines [8].

Management

Behavior modification is a cornerstone to the treatment and prevention of hypertension. Recommendations include:

1. Sodium reduction, to a level not exceeding 2 g sodium per day. Even if this goal cannot be achieved, reductions in dietary sodium have been unequivocally shown to decrease BP. Sodium restriction also enhances the hypotensive effect of diuretics and reduces urinary loss of potassium.
2. Moderate to intense exercise for at least 30 min most days of the week (and at a minimum, 4 days weekly).
3. Weight loss to achieve a healthy body weight, if required. Reducing central adiposity is a goal.
4. Abstaining from alcohol consumption, or reducing to two drinks per day or less.

Pharmacologic treatment is instituted when the benefits of treatment are anticipated to exceed the harms associated with therapy. Some representative classes and agents to choose from are given in Table 15.2. Hypertension guidelines can be consulted for treatment recommendations for both uncomplicated hypertension and hypertension associated with various comorbidities. In 2024, the European Society of Cardiology has released their latest guidelines [9], and Hypertension Canada, at the time of writing, is in progress with updated guidelines. Commonly, hypertension can be controlled with behavior modification and 1–3 antihypertensive agents, with robust evidence for the use of thiazide diuretics, dihydropyridine calcium channel blockers (CCBs), and/or angiotensin converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) either alone or in combination [9].

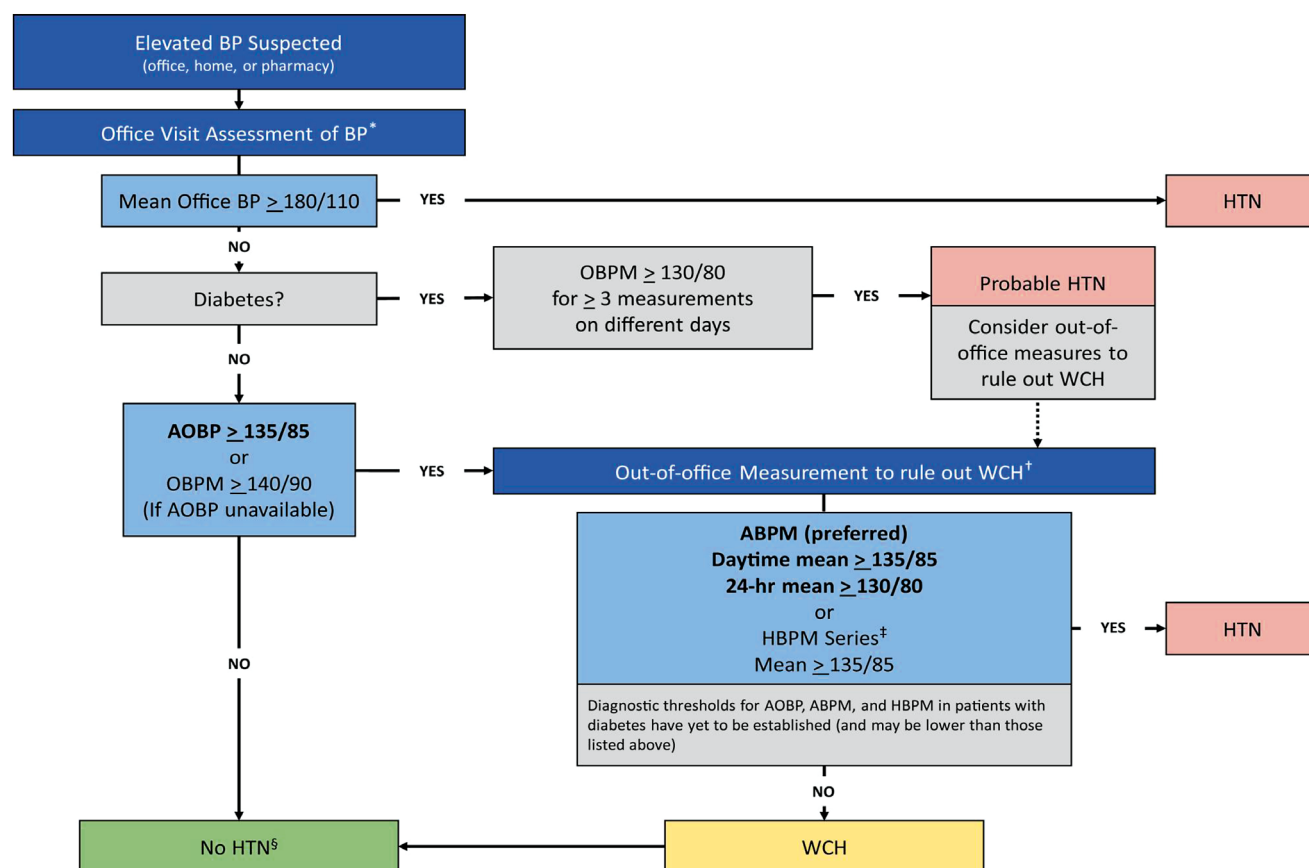


Fig. 15.1 Hypertension diagnostic algorithm for adults. ABPM ambulatory blood pressure monitoring, AOBP automated office blood pressure, BP blood pressure, OBPM office blood pressure measurement,

WCH white coat hypertension. (Reprinted from Rabi et al. [8], Copyright 2025, with permission from Elsevier)

Table 15.2 Classes of drugs typically used in the management of primary hypertension

Drug class	Drug names (select)
Thiazide diuretics (*agents with longer duration)	Indapamide ^a 1.25–2.5 mg daily Chlorthalidone ^a 12.5–25 mg daily Hydrochlorothiazide 12.5–25 mg daily
Angiotensin converting enzyme inhibitors (ACEi)	Perindopril 2–8 mg daily Lisinopril 5–40 mg daily Ramipril 2.5–10 mg daily Enalapril 2.5–20 mg daily
Angiotensin receptor blockers (ARBs)	Telmisartan 40–80 mg daily Candesartan 8–32 mg daily Irbesartan 150–300 mg daily Valsartan 80–320 mg daily
Dihydropyridine calcium channel blockers (DHP-CCBs)	Amlodipine 2.5–10 mg daily Nifedipine XL 20–120 mg daily
Non-dihydropyridine calcium channel blockers (NDHP-CCBs)	Diltiazem-extended release 180–360 mg daily Verapamil-extended release 120–360 mg daily

Initial Assessment of a Patient Newly Diagnosed with Hypertension

Given the prevalence of hypertension, pharmacists are accessible to screen for and educate patients about hypertension. If a patient does not have a diagnosis of hypertension, and high BP readings are detected, pharmacists should work with primary care clinicians to develop an appropriate care plan. An initial assessment by the pharmacist should include the following:

Patient History

- History of present illness: Note the course of hypertension and if the patient has ever experienced a hypertensive emergency or urgency, which indicates greater risk for complications of hypertension. When blood pressure is severely elevated (per the diagnostic algorithm, $\geq 180/110$), the diagnosis can be made in one visit, but for most, it is a pattern of sustained elevation (multiple

readings above the patient's target, either gathered through 24-h ABPM or many readings by HBPM). Hypertension during pregnancy increases the risk of developing sustained hypertension. Additionally, the longer hypertension has been present (untreated), the likelihood of target organ damage is greater. It is therefore important to try and discern when hypertension presented; one question could be "do you recall the last time your BP was normal?"

- (b) Past medical history: History of cardiovascular disease (CVD) and/or target-organ damage (such as microalbuminuria or left ventricular hypertrophy) are prognostic indicators of higher future risk of CVD/complications from hypertension and warrant treatment with pharmacologic therapy in addition to behavior modification. Consider the possibility of secondary causes if BP not easily controlled with behavior modification and ≥ 3 medications. Most common secondary causes include obesity, renal failure, primary hyperaldosteronism, sleep apnea, renal artery stenosis, and excessive alcohol use.
- (c) Age of onset: Younger onset leads to greater cumulative risk. Patients with a family history of premature hypertension in one parent (defined as onset < age 55) have a two- to threefold increase in risk of developing hypertension, and if both parents had premature hypertension, the risk can be up to 20-fold for developing hypertension.
- (d) Cardiovascular (CV) risk assessment if CVD or target organ damage are not already present to determine global CV risk and ascertain BP threshold for treatment and BP target. CV risk assessment can be performed using a variety of risk calculators. One on-line CV risk/benefit calculator, CardioRisk, is available at <https://www.circl.ubc.ca/cardiorisk-calculator.html> and includes a simple Framingham Risk Score as well as a dyslipidemia risk calculator. One benefit of the latter calculator is it provides the clinician and patient with cardiovascular disease risk estimation over 10-years, vascular age (based on patient-specific data), and pharmacologic and non-pharmacological treatment options.
- (e) Social history: alcohol and tobacco use, diet (with focus on sodium intake), drug coverage, and stress levels.
- (f) Medication use: Determine current prescription and non-prescription medication use. Note if any medications are known to increase BP and assess if discontinuation (or dose decrease) is an option. Medications known to increase BP are listed in Table 15.3.
- (g) Medication allergy or intolerance history: Determine if any antihypertensive agents have been trialed and not tolerated and document the nature of allergy/intolerance. Consider possibility that intolerance reactions may be related to use of a high dose and assess if there is willingness to trial the agent at a lower dose if felt to be beneficial for BP lowering.

Table 15.3 Examples of medications that may cause an increase in BP

Antidepressants (monoamine oxidase inhibitors (MAOIs), serotonin-norepinephrine reuptake inhibitors (SNRIs), selective serotonin reuptake inhibitors (SSRIs)	Calcineurin inhibitors (cyclosporine and tacrolimus)
Corticosteroids	Erythropoietin
Oral contraceptives and sex hormones	Midodrine
Non-steroidal anti-inflammatory drugs (NSAIDs, oral); topical NSAIDs have much lower impact on BP and are generally considered a safer alternative for OA pain	Stimulants including cocaine and decongestants (phenylephrine, pseudoephedrine)
Licorice root	Second-generation antipsychotics (e.g., clozapine, olanzapine)

BP Assessment

AOBP is recommended for in-office assessment and, subsequently, obtain out-of-office measurements (if patient is willing and able). For patients presenting with hypertensive emergency (e.g., symptoms of acute coronary syndrome, acute left ventricular failure, aortic dissection, stroke), refer for immediate medical attention. Home BP measurement should include two readings taken twice daily (approximately 12 h apart to inform diurnal pattern of BP) for 7 days. Days 2–7 (24 readings in total) are averaged to provide overall level of control, and if discrepancies exist between morning and evening readings, the average for each time point can be calculated. Hypertension Canada has a downloadable BP log (https://hypertension.ca/wp-content/uploads/2017/11/HTC_BloodPressureLog_ENG_PREVIEW-1.pdf) that can be printed and given to patients. It is important that BP is measured using a proper technique. Table 15.4 outlines the impact that improper technique can have on BP measurements [10].

Laboratory Values and Diagnostic Assessment

If lab values or diagnostic tests are not available, advise the patient that these are needed as part of the diagnostic workup for hypertension. In some jurisdictions, pharmacists can order laboratory tests as part of completing a patient assessment.

- (a) Review parameters, which may impact current and/or the selection of drug therapy, such as serum creatinine and electrolytes. Abnormalities in parameters necessitate physician referral for management.
- (b) Check urinalysis for presence of proteinuria.
- (c) Review parameters that impact CV risk such as lipids and fasting glucose/A1C.
- (d) Baseline ECG to evaluate rate, rhythm, presence of ischemia, and left ventricular hypertrophy (LVH).

Table 15.4 Factors that can increase blood pressure measurements [10]

Factor	Increase on SBP (mm Hg)	Increase on DBP (mm Hg)
“White coat” reaction: to physician	11–28	3–15
Cuff too small	10	2–8
Legs crossed	8–10	4–5
Talking or active listening	7	8
Arm unsupported	1–7	5–11
“White coat” reaction: to nonphysician	1–22	2–7
Smoking within 30 min	6	5

DBP diastolic blood pressure, SBP systolic blood pressure

Additional laboratory work-up is needed for secondary causes if resistant hypertension is present.

Referral

Patients with uncontrolled resistant hypertension should be referred to either a hypertension clinic, or internal medicine specialist to work-up secondary causes. Patients could be referred to healthcare practitioners to have relevant diagnostic tests (such as ECHO, renal ultrasound, as appropriate) ordered and/or assessed (if already done). A physician or nurse practitioner should complete an eye, neck, cardiac, and abdominal exam.

Physical Assessment Skills by Pharmacist

Pharmacists should initially assess blood pressure and heart rate. Blood pressure assessment can be done using patient-reported home BP measurements or within the practice setting with preference given to AOBP compared to manual measurement. It could be argued that BP should only be done outside the office, but that may not be possible for some patients, making the use of in-office measurement necessary. Using correct BP technique is important to avoid over- or underestimation of BP. Check for hypotension and orthostatic changes in patients who report a history of dizziness. Orthostatic hypotension is defined as either a drop in systolic BP of ≥ 20 mmHg or diastolic BP of ≥ 10 mmHg after 1 min when a patient goes from supine or seated position to standing position. Make sure the patient has adequate support when standing to minimize risk of fall. Prior to completing the two BP measurements (one seated and one standing), inform the patient that you will be asking for their symptoms during the position transition and, if experienced, the time for resolution. This should be documented in your patient

assessment. Secondly, pharmacists can assess extremities looking for peripheral edema. If present, it should be noted if this is pitting or non-pitting. Pitting edema is most commonly caused by heart failure, renal disease or venous insufficiency, with non-pitting edema commonly caused by dihydropyridine calcium channel blockers and possibly NSAIDs (if fluid retention).

Follow-Up Assessments

Adherence

Adherence to antihypertensive therapy is known to be poor (estimated to be 50–70%). Reasons are multifactorial and may include (1) the asymptomatic nature of the condition, (2) drug adverse effects and/or cost and (3) lack of perceived benefits of treatment. Fortunately, there are strategies to help patients adhere to their treatment plan. These include:

- Using medication regimens that fit the patient’s daily routine. Once daily medications improve adherence by simplifying medication taking. Once daily medications also tend to be longer acting, which facilitates BP control.
- Using single pill combination (SPC) tablets to minimize pill burden. There are many combination tablets available and their use may be cost saving.
- Using adherence packaging (dosettes, blister packs) and electronic adherence aids (e.g., reminders on phone via apps that track BP).
- Promoting home BP monitoring so that patients are more engaged in their BP control and can see the effect of their treatment regimen.
- Monitoring effects of treatment regimen more frequently, especially during first 3 months. Pharmacist call-backs have demonstrated benefits to adherence.
- Pharmacist participation in the management of blood pressure, mainly through prescribing of antihypertensives, up-titration of antihypertensives as required to achieve BP goals and reinforcing the treatment plan through more frequent monitoring.

Adherence should be assessed at each clinician visit so that achieved BP is interpreted in the context of medication-taking behavior. Not surprisingly, non-adherence is a contributor to poor BP control, and if present, reasons for non-adherence should be explored so that possible solutions can be implemented. Pharmacists should work together with patients to determine the reasons for non-adherence and use open-ended questions to explore how adherence could be improved [11–13].

Blood Pressure Control

At each follow-up visit, including pick-up of refills (if practicing in a community pharmacy), the efficacy of the treatment regimen can be assessed. Key elements of patient assessment include:

- *Efficacy assessment:* The degree of BP control achieved based on patient-specific target can be assessed by reviewing home BP logs. In general, if BP is at or below target 80% of the time, BP control is good. For patients not achieving this, intensification and/or additional therapy is needed. In some cases, this can be achieved by using longer-acting agents that have a more durable BP-lowering effect. Also, if an added treatment does not seem to be contributing to any BP lowering, rather than adding on therapy, it can be substituted for an alternate BP-lowering agent.
- *Safety assessment:* Determine if BP is too low for the individual patient (either persistently or intermittently), as evidenced by BP logs and reports of dizziness/light-headedness and general malaise and/or fatigue. If symptoms present, consider other contributors, such as acute illness or other medical conditions, that may also lead to these symptoms. If hypotension is suspected to be the main cause, encourage more frequent home BP monitoring to correlate symptoms with BP and de-prescribe anti-hypertensive therapy as required until symptom improvement. Ongoing home BP monitoring will guide future adjustments to the treatment plan.

Box 15.1 depicts an example of a patient assessment.

Box 15.1 Patient Assessment Using a Home BP Log

The following case illustrates an approach to a patient with hypertension:

A 62-year-old Japanese female patient you have been following brings in her home BP log from the last week. She was recently diagnosed with hypertension (BPs often 170–180 mmHg systolic), started on drug therapy and has been recording home BPs. She is asymptomatic, although initially noticed she was urinating more frequently, which has now normalized.

Pertinent Background Details

- *Chief Complaint:* Soliciting your advice on the effectiveness of her drug therapy for hypertension—she thought this drug would normalize her BP.

Past Medical History: Hypertension, treated for 1 month

Social: Drinks about 14 glasses of wine per week (approximately 2 per evening), non-smoker (quit 30 years ago, prior to having her children), obese for ++ years (BMI 32)

No family history of premature CVD

Medications: Indapamide 1.25 mg qam (started 4 weeks ago)

Does not use any OTC or herbal products

Behavior Modifications: Nil

Select Labs/Diagnostics (from 2 Weeks Ago):

Scr 68 $\mu\text{mol/L}$ (eGFR 87 mL/min/1.73m²), urinalysis reveals 1+ proteinuria

K⁺ 3.6 mmol/L (3.5–5.0) [was 4.0 mmol/L 3 months ago]

Na⁺ 138 mmol/L (133–146 mmol/L)

ECG: HR 72, normal rhythm, no LVH

BPLog

Date	Day number	Time	Heart rate	BP reading #1		BP reading #2	
				SBP	DBP	SBP	DBP
May 10	Day 1 morning	0655	80	155	95	152	93
	Day 1 evening	1800	85	162	90	160	89
May 11	Day 2 morning	0700	82	157	89	159	92
	Day 2 evening	2200	78	172	88	167	91
May 12	Day 3 morning	0630	91	165	88	165	90
	Day 3 evening	1900	96	168	87	166	89
May 13	Day 4 morning	0605	88	180	92	171	92
	Day 4 evening	1730	86	152	97	158	99
May 14	Day 5 morning	0645	85	170	99	165	96
	Day 5 evening	2230	91	166	95	166	94
May 15	Day 6 morning	0700	88	148	94	152	89
	Day 6 evening	2300	92	146	94	145	92
May 16	Day 7 morning	0640	81	157	96	160	94
	Day 7 evening	2040	83	155	97	154	97

(continued)

Box 15.1 (continued)**Patient Assessment Approach***Review BP log:*

Calculation of average BP, averaging days 2–7 (as per Hypertension Canada guidelines).

Average *morning* BP (12 readings over 6 days) is 162/93.

Average *evening* BP (12 readings over 6 days) is 160/93.

Overall BP control (24 readings over 6 days) is 161/93.

Pulse within normal limits.

Is it required to calculate a morning and evening BP average?

Not necessarily, but patients can have BP that is variable across the day, with one time period seemingly revealing a higher BP. This can influence recommendations to dose antihypertensive medication(s) at a certain time of day. Upon visual inspection, this BP log reveals similar BPs, regardless of time. When that is the case, an overall average is sufficient (notation can be made indicating that BPs are consistently elevated across the day).

BP measurement assessment (to assess accuracy of BP log)

Is the patient using proper technique to measure BP? Assess the following:

1. Using a monitor that is accurate when measured against one of known calibration?
2. Using an upper arm-based BP cuff, if appropriate size is available? If not, a wrist cuff would be appropriate.
3. Using appropriate set-up prior to measurement? (i.e., putting cuff on arm in appropriate location and tension, sitting with back supported, arm supported at heart level, feet flat on ground, with no crossed legs, not talking, no coffee/tobacco within previous 30 min)

Effectiveness/safety of current treatment

Taking indapamide for 4 weeks, which is enough time to elicit maximum response. Patient is tolerating and adherent. Urination has subsided.

Serum potassium has lowered, as expected with thiazide diuretics. It is still within normal range. She has 1+ proteinuria, likely due to uncontrolled hypertension.

Assessment/recommendations

- Home BP monitor is accurate and patient using correct technique.
- Important to calculate CVD risk; need to order lipid panel and A1C (or request that a person who can order labs complete this)
- BP above target of <135/85 (and if high risk, may consider lower SBP target of <120); requires additional therapy (many options exist, and using a long-acting, single-pill combination can improve adherence and efficacy). Rx for perindopril/indapamide 8/2.5 1 tab daily
- Behavior modification should be discussed, specifically, sodium and weight reduction, exercise, limiting alcohol intake (to nine or less drinks/week or lower, which may be increasing her BP), DASH diet.
- Re-check Scr and lytes (specifically K⁺ and Na⁺) in 10–14 days.
- Continue to monitor home BP, especially 3–4 weeks after new medication/behavior modifications introduced. Measure BP twice daily with average of 2 readings each time.

Adverse Reactions of Hypertension Drug Therapy

While antihypertensive therapies are generally well-tolerated, side effects and/or laboratory abnormalities can emerge, necessitating a change in therapy. For pharmacists working in a community pharmacy setting, adherence can be checked based on refill records, which if poor, may be a clue about poor treatment tolerability. Monitoring is important to check for anticipated possible effects, as well as effects that are not anticipated. Table 15.5 outlines the four first-line drug classes and common or anticipated adverse effects as well as management strategies. Fortunately, there are many antihypertensive drug agents available to prescribing clinicians allowing for flexibility to address adverse effects. It is important to work with patients to routinely assess for drug-therapy complications and assure them that other options exist, with the hopes of improving adherence to the prescribed treatment regimen.

Table 15.5 List of most common drug impacts in patients with hypertension

Drug class	Common adverse reactions/precautions	Management strategy
Thiazide diuretics	Hypokalemia (if severe, may present as muscle cramping) Hyponatremia Hyperuricemia and hypercalcemia Hyperglycemia (in non-DM patients)	Know baseline values. Monitor electrolytes 1–2 weeks after initiation of therapy or dose changes. If baseline $K^+ < 4$ mmol/L, recommend to prescribe in conjunction with an ACEi/ARB (as a SPC) or a potassium-sparing diuretic Lower dose (and do not exceed maximum recommended dose, which can exacerbate adverse reactions) Switch to a different class
Angiotensin converting enzyme inhibitors (ACEi)	Hyperkalemia (dose-related) Dry cough May cause transient increase in serum creatinine (Scr), which should resolve Do not use in pregnancy	Reduce dose if possible Prescribe with a thiazide or loop diuretic, which can normalize hyperkalemia If cough develops, switch to an ARB If Scr increases >30%, refer for investigation of renovascular hypertension
Angiotensin receptor blockers (ARBs)	Hyperkalemia (dose-related) May cause transient increase in serum creatinine (Scr), which should resolve Do not use in pregnancy	Reduce dose if possible Prescribe with a thiazide or loop diuretic, which can normalize hyperkalemia If Scr increases >30%, refer for investigation of renovascular hypertension
Dihydropyridine calcium channel blockers (DHP-CCBs)	Pedal edema bilaterally, non-pitting (dose-dependent)	Reduce dose Adding a venodilator to regimen; most commonly studied is ACEi
Non-dihydropyridine calcium channel blockers (NDHP-CCBs)	Bradycardia (dose-dependent) Pedal edema bilaterally, non-pitting (dose-dependent)	Monitor heart rate (HR) and reduce dose if <50 or if patient feels unwell/fatigued at higher HR Try dosing at bedtime to minimize pedal edema

Complications of Hypertension

Hypertension is a leading risk factor for disability and death from cardiovascular causes. The potential complications of uncontrolled hypertension are listed below:

Cardiac/vascular

- Left ventricular hypertrophy, atrial dilatation and fibrillation
- Diastolic and systolic heart failure
- Accelerated coronary artery disease, myocardial infarction and systemic atherosclerosis
- Aortic dissection and aneurysm

Renal

- Proteinuria
- Hematuria
- Renal failure

Central Nervous System

- Hemorrhagic, lacunar and ischemic stroke
- Posterior reversible encephalopathy syndrome (PRES)
- Seizure
- Vascular dementia
- Retinopathy
- Postural hypotension and syncope

Other

- Erectile dysfunction

Postural hypotension is a risk of treatment, especially if lower BP targets are being attempted. With the publication of the SPRINT and ACCORD trials, SBP < 120 mmHg reduces CVD and is therefore recommended for high-risk patients (on the assumption they tolerate the lower SBP). In the SPRINT trial, the risk of hypotension was higher (about double) in those achieving a SBP target of less than 120 mmHg compared to those achieving a SBP between 130 and 140 mmHg (1.8% vs. 0.8%). Syncope was also increased in patients achieving a SBP < 120 mmHg (1.6% vs. 0.6%). Injurious falls (requiring ER visit or hospitalization) were the same between groups [14]. Pharmacists can play a key role in monitoring patients for hypotension during clinic appointments or when patients pick up refills for antihypertensive medications. Assessment can consist of BP measurements and/or patient history that describes the nature,

frequency and duration of hypotensive episodes. Deprescribing antihypertensive drug therapy is appropriate if patients are experiencing symptomatic hypotension.

Clinical Pearls

- Pharmacists can play a key role in assessing patients (both initially and during follow-up) with hypertension to ensure treatment regimens are effective, safe and adhered to. As patient status can change, routine monitoring is important.
- Out-of-office BP measurement is important to monitor BP control. Use this method routinely to inform and justify changes to treatment regimens.
- Drug therapy must be tailored to ensure maximum effectiveness and minimal side effects. Short acting antihypertensive drugs should be avoided. The use of long-acting drugs, in combination, is preferred.
- The average antihypertensive drug will lower BP by about 10%. If the BP at the onset is well above target, combination therapy can be started.
- Changes to pharmacologic regimens can take up to 3–4 weeks to reach maximum treatment effects, so resist the urge to adjust regimens at intervals that are shorter.
- Use caution when lowering BP in patients with a vascular obstruction such as aortic stenosis or carotid/renal artery stenosis.
- ARB and ACEi are teratogenic and must be used with caution in women of childbearing age.

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Suggested Reading

- www.hypertension.ca. Hypertension Canada website, which provides education and recommendations for healthcare professionals, patients and partners about hypertension.

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Chapter Objectives

1. To define heart failure (HF)
2. To review the diagnosis of HF, including common signs and symptoms, risk factors, and common diagnostic tests
3. To highlight goals of therapy in patients with HF.
4. To outline a general approach to a patient with HF, including initial assessment and ongoing evaluation and monitoring.
5. To provide an approach to pharmacotherapy in HF, including parameters for initiation and titration of therapy.

Background

Heart failure (HF) is common with over 600,000 people living with the disease in Canada. Every year, approximately 50,000 people are newly diagnosed with HF [1]. Both the prevalence and incidence of heart failure is growing secondary to higher survival post myocardial infarction, better diagnosis and treatments for HF and an aging population. HF is burdensome to the healthcare system, costing an estimated \$2.8 billion dollars yearly which is largely driven by hospitalizations.

HF is a complex clinical syndrome in which there is a decrease in cardiac output that results in a reduced ability to meet the metabolic demands of the body. The proposed universal definition describes HF as a clinical syndrome with symptoms and/or signs caused by a structural and/or functional cardiac abnormality and corroborated by elevated natriuretic peptide levels and/or objective evidence of pulmonary or systemic congestion [2]. It is classified as a syndrome, because regardless of the cause, it presents with a similar group of signs and symptoms. Heart failure can pres-

ent acutely or chronically and can involve either or both ventricles. Ejection fraction (EF), or the amount of blood pumped out of a filled ventricle with each heart beat, is often used to further define HF. Heart failure with preserved ejection fraction (HFpEF) is a patient with an EF > 50%, and HF with reduced ejection fraction (HFrEF) is a patient with an EF < 40%. There is also a small subgroup of patients with EF 41–49% called HF with midrange ejection fraction (HFmrEF) and those with HF with improved EF (HFimpEF) that have improved from HFrEF (≥ 10 point increase from baseline LVEF, and a second measurement of LVEF > 40%). Specific management of heart failure is dependent on symptoms, etiology, and ejection fraction. This chapter will focus on the approach to the patient with chronic HFrEF, as this subset of HF is most applicable to the majority of pharmacists and the available robust evidence-based pharmacotherapy. However, the general assessment approach can be applied to the overall syndrome of HF.

Initial Assessment

The general approach to a HFrEF patient includes the following:

- Assess signs and symptoms of HF to determine baseline and functional capacity (an individual's ability to perform tasks and activities in daily life)
- Assess fluid status (signs and symptoms of congestion)
- Determine if there are any risk factors for HF that can be managed
- Determine the need for medical management:
 - Assess any contraindications/allergies
 - Consider any co-morbidities/other factors (other medications, drug interactions, cost, adherence) that might change therapy and doses selected
 - Assess baseline parameters for drug therapy prior to drug initiation/dose selection (e.g., serum creatinine

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(SCr), potassium, blood pressure (BP), heart rate, ECG abnormalities, etc.)

- Follow-up to monitor drug therapy and ensure optimization of medical therapy

The case described in Box 16.1 illustrates these points and some other considerations when approaching a patient with HF.

Box 16.1 Patient Care

You are the pharmacist in a heart failure clinic. A patient with a recent emergency room visit for heart failure has been referred to the clinic for assessment and treatment. You find the following admission history in the patient chart.

CC: Paul Smythe is a 70-year-old male presenting with worsening SOB and fatigue over the past few months.

HPI:

Mr. Smythe has been experiencing increased SOB on exertion for the last 2–3 months. He recalls first noticing his symptoms on his regular morning walk which is about 8 blocks. He began to get “short on air.” At first it just made him slow down, but soon it was making him stop after a couple of blocks. He tried taking his puffer with no improvement. About 3 weeks ago, he had to call a taxi to take him home because he was too SOB to continue. It was this time that he decided to go to the ED. He also notices SOB climbing one flight of stairs at home or doing any type of exertion, including things like taking out the garbage or mowing the lawn. He has seen his family doctor a couple of times and received two courses of antibiotics, for a presumed upper respiratory tract infection related to his COPD, without any improvement. Prior to this, he was rarely limited by SOB. He would only notice “mild” SOB if he climbed >2 flights of stairs or was doing heavy lifting. He walked daily and did all his own house and yard work. His main limitation was the pain in his left knee for his arthritis, which he was awaiting replacement surgery.

He also describes being “drained of energy.” He reports sleeping poorly and would get SOB any time he tried to lay flat in bed. At one point, he recalls sleeping with four pillows and eventually needing to sleep in his recliner to avoid being SOB. He has noticed swelling in both of his legs, which at one time precluded him from wearing any of his shoes. He describes a poor appetite and notes that even when he is hungry, he gets full quickly and is unable to eat much. Despite

this, he reports his pants getting tighter and his weight going up. He has gained ~7 kg over 2 months.

Two weeks ago, he went to the ED because the breathing got so bad. He was given furosemide 40 mg IV $\times$ 2 and then oral (40 mg BID) and referred urgently to the Heart Failure Clinic for assessment and treatment. At that time, his ECG reported NSR and a left bundle branch block (LBBB) and his CXR showed pulmonary edema. He was also noted to have 2+ pitting edema to the shin bilaterally. His symptoms have improved since this visit, but he is “nowhere near normal.” He can only walk one block without stopping and still requires two pillows to sleep, and the swelling in both of his feet is still present. His weight is down ~3 kg.

He denies chest pain, palpitations, syncope, presyncope or light-headedness. He has a “morning cough” that is productive but has been present and stable since he was diagnosed with COPD. However, he did notice more coughing when he needed to sleep in his recliner.

Past Medical Hx:

HTN $\times$ 20 years; has a home BP monitor—150–160/70–80 mmHg $\times$ 1 year; previously 160–180/70–80 mmHg

Dyslipidemia $\times$ 6 years

DM2 $\times$ 2 years; does not have home monitor

OA of left knee $\times$ 12 years

COPD $\times$ 4 years; “mild,” no hospital admissions

Family Hx: Mother died of MI at 82 years old

Social Hx:

Retired. Widowed. Lives alone in a house; prior to symptoms, able to maintain on own

Smoking: Quit smoking 20 years ago; 40 pack-years

EtOH: 2 drinks per week, never a heavy drinker; illicit drugs: none

Exercise: walked 30 min daily prior to symptoms starting

Diet: 3 meals per day; a lot of prepared frozen dinners; eats out weekly; no added salt

Allergies: NKDA

Medications:

Amlodipine 5 mg PO daily ($\times$ 15 years)

Hydrochlorothiazide 25 mg PO daily ($\times$ 10 years)

Atorvastatin 40 mg PO daily ($\times$ 6 years)

Furosemide 40 mg PO BID ($\times$ 2 weeks)

Metformin 1000 mg PO BID ($\times$ 2 years)

Tiotropium 18 μ g inhaled daily ($\times$ 4 years)

(continued)

Box 16.1 (continued)

Salbutamol 1–2 puffs inhaled QID prn (never uses, other than recently with increasing SOB)

Ibuprofen 200–400 mg PO TID prn (generally uses 400 mg qhs and then prn q2days)

Vital Signs:

- BP sitting = 130/80 BP standing = 118/78 HR sitting = 94 HR standing = 100

Ht = 160 cm Wt = 86 kg (Baseline: 78 kg)

Physical Examinations:

- *RESP*: course crackles to the bases bilaterally
- CVS*: S1, S2, S3, no murmurs, JVP 7 cm above the sternal angle (ASA), + HJR

GI/GU: ascites

MSK: ++ pitting edema to shins bilaterally

Labs:

	1 day ago	1 month ago	6 months ago
Na	132	138	134 mmol/L
K	3.5	3.9	4.4 mmol/L
Cl	104	101	106 mmol/L
SCr	130	98	70 μ mol/L
Hgb	145	138	140 g/L
WBC	7.0	7.8	7.9×10^9 /L
PLT	317	325	330×10^9 /L
BNP	2200	–	– pg/ml
TSH	3.00	–	– mU/L

Investigations:

- *CXR*: cardiomegaly, interstitial pulmonary edema

ECG: normal Sinus rhythm, left ventricular hypertrophy, left bundle branch block, QRS 130 msec, HR = 80 bpm

Cardiac PET Scan: normal perfusion, EF = 35%

Echocardiogram:

Ventricle: Mildly dilated left ventricle, concentric LVH, global LV hypokinesis, EF = 30%, some diastolic impairment

Atria: dilated left atrium

Valves: Aortic valve sclerosis, mild mitral regurgitation

Case abbreviations: BID, twice daily; BNP, B-type natriuretic peptide; BP, blood pressure; CC, chief complaints; Cl, chloride; COPD, chronic obstructive pulmonary disease; CVS, cardiovascular system; CXR, chest X-ray; DM2, diabetes mellitus type 2; ECG, electrocardiography; ED, emergency department; EF, ejection fraction; ETOH, alcohol; GI, gastrointestinal; GU, genitourinary; Hgb, hemoglobin;

HJR, hepatjugular reflex; HPI, history of present illness; HTN, hypertension; JVP, jugular venous pressure; K, potassium; LVH, left ventricular hypertrophy; MI, myocardial infarction; MSK, musculoskeletal; Na, sodium; NSR, normal sinus rhythm; OA, osteoarthritis; PET, positron emission tomography; PLT, platelets; QID, four times daily; RESP, respiratory; SCr, serum creatinine; SOB, shortness of breath; TID, three times daily; TSH, thyroid-stimulating hormone; WBC, white blood cells

Diagnosis**What Information from Mr. Smythe's Case Do We Have That Would Be Used in the Diagnosis for HF?**

The diagnosis of HF is clinical. It involves taking a careful history, reviewing the signs and symptoms of the patient, performing a physical examination, reviewing risk factors and using diagnostic tests such as blood tests and echocardiography. It is a constellation of these signs and symptoms that leads to a diagnosis of HF.

What Signs and Symptoms Does Mr. Smythe Have That Are Consistent with HF?

The most common symptoms (experienced by the patient) of HF including those experienced by Mr. Smythe are depicted in Table 16.1. The most common signs (detected by someone other than the patient) of HF, including those present in Mr. Smythe are depicted in Table 16.2. The signs and symptoms may vary among patients and will depend on the acuity of the situation and whether it is right-sided or left-sided HF. It is especially important to note that not all HF involves congestion all the time. Signs and symptoms related to congestion of peripheral tissues (leg swelling, ascites, HJR, elevated JVP) are related to right-sided heart failure, which may occur with left-sided HF or in isolation and can be improved/reversed with fluid management (e.g., diuresis). The most common cause of right sided heart failure is left sided heart failure. On the other hand, left-sided HF is characterized by a decrease in cardiac output manifesting as activity intolerance and decreased tissue perfusion and/or pulmonary congestion causing impaired gas exchange (cyanosis/hypoxia) and/or pulmonary edema (cough, orthopnea, paroxysmal nocturnal dyspnea [PND]). If the patient does not have any right-sided HF, they will not have the classic signs of periph-

Table 16.1 Common symptoms of heart failure

Symptoms	Mr. Smythe	Case-specific examples
Shortness of breath (SOB)	Yes	Change from baseline activity: Walking 8 blocks to 1 blocks Climbing >2 flights of stairs to <1 flight New limitation doing yard work
Cough	Yes	Preexisting cough but worse with orthopnea
Swelling (leg, abdomen)	Yes	Legs: Swelling in both of his legs, at times precluded wearing shoes Abdomen: Poor appetite, even with hunger, feels full quickly and is unable to eat much (early satiety) Reports pants getting tighter and weight going up (~7 kg in 2 months) despite decrease intake and less activity Weight loss post diuretic treatment (~3 kg)
Orthopnea	Yes	Reports sleeping poorly, SOB any time laid flat in bed Sleeps with 2 pillows, down from 4 pillows/sleeping in his recliner
Paroxysmal nocturnal dyspnea (PND)	No	—
Fatigue	Yes	Describes being “drained of energy”
Confusion	No	—

Table 16.2 Common signs of heart failure

Sign	Mr. Smythe	Case-specific examples
Lung crackles/rale	Yes	Crackles
Elevated jugular venous pressure (JVP) > 4 cm above the sternal angle (ASA)	Yes	7 cm ASA
Positive hepato-jugular reflex (HJR)	Yes	
Peripheral edema	Yes	2+ pitting edema bilaterally to mid-shin Ascites
Displace cardiac apex	Not noted	
S3, S4 heart sounds	Yes	S3
Tachycardia (HR > 120)	No	
Heart murmur	No	
Low blood pressure	No	
Chest X-ray findings:		
Cardiomegaly	Yes	
Pleural effusion	No	
Pulmonary edema	Yes	
Pulmonary venous redistribution	No	
ECG		
Q waves	No	
Left ventricular hypertrophy (LVH)	Yes	
Left bunch branch block (LBBB)	Yes	
Tachycardia	No	

HR heart rate

eral edema. Mr. Smythe has many signs and symptoms that are consistent with a clinical diagnosis of HF, many of which are consistent with acute congestion.

Clinical Pearl

Shortness of breath can be subjective. Listen carefully to the patient and include questions about daily activities in your assessment. When assessing SOB, establish baseline activity and timelines for the onset/change in symptoms. Always relate activities specifically to ones that the patient regularly performs.

What Diagnostic Tests Did Mr. Smythe Have to Contribute to the Diagnosis of His HF?

Diagnostic tests that add to the clinical diagnosis of HF are shown in Table 16.3. A specific laboratory test that can be used to evaluate HF is the B-type natriuretic peptide (BNP) or the N-terminal-pro-BNP (NT-proBNP) test. BNP is a hormone released by ventricular cardiomyocytes in response to ventricular stretching caused by increased blood volume. This test has a number of applications (diagnosis, prognosis, therapy decision making etc.), but its most practical everyday clinical application to date is its use as a tool to rule in or out HF if the diagnosis is in doubt (Table 16.4). Based on Mr. Smythe's signs and symptoms, a BNP was not necessary to make a diagnosis of HF (BNP = 2200 pg/ml). However, if his symptoms were inconsistent and if there was a question that his dyspnea was from a COPD exacerbation (note that he was treated with antibiotics twice for similar symptoms), then a BNP might be helpful in determining the cause of dyspnea.

Table 16.3 Diagnostic tests that add to the clinical diagnosis of heart failure

Diagnostic tests	Indication	Mr. Smythe
Chest X-ray findings:		
Cardiomegaly	Possible dilated ventricle	Yes
Pleural effusion	Pulmonary congestion	No
Pulmonary edema	Pulmonary congestion	Yes
Pulmonary venous redistribution	Pulmonary congestion	No
ECG findings		
Q waves	Possible ischemic	No
Left ventricular hypertrophy (LVH)	Possible pressure overload	Yes
Left bunch branch block (LBBB)	Possible dilated ventricle	Yes
Tachycardia	Sympathetic overdrive	No
BNP > 500 pg/ml	Fluid overload causing ventricle to stretch	Yes

BNP B-type natriuretic peptide

Table 16.4 Levels for BNP and NT-pro-BNP indicating heart failure [2]

BNP	NT-pro-BNP
<100 pg/ml—acute HF decompensation unlikely	<300 pg/mL—acute HF decompensation unlikely
	>450 pg/mL—HF likely (age < 50)
>500 pg/ml—HF likely	>900 pg/mL—HF likely (age 50–75)
	>1800 pg/mL—HF likely (age > 75)

BNP B-type natriuretic peptide, NT-pro-BNP N-terminal-pro-BNP

What Tests Has Mr. Smythe Had That Are Consistent with a Clinical Diagnosis of HF?

Because HF is a clinical diagnosis (e.g., based on history, risk factors, signs, and symptoms), diagnostic tests, including laboratory parameters, are mainly used to help determine the etiology or confirm clinical suspicion if the diagnosis is unclear (e.g., inconsistent signs and symptoms). Based on Mr. Smythe's history and signs and symptoms, the diagnosis of HF can be made. However, the treatment approach may be different depending on the type of HF present, which would require further evaluation.

The most common diagnostic test used to evaluate the function of the heart is echocardiography, or ultrasound of the heart. This modality can assess the structure of the heart (e.g., valves, muscle thickness, chamber sizes) as well as the function of the heart (e.g., ejection fraction (EF), filling pressure, etc.). There are a number of parameters used in the assessment of a patient with HF, but the most commonly used parameter is the EF, or the ratio of blood ejected from the heart relative to the amount presented to the left ventricular. Other tests that may be used to evaluate a patient with HF and determine etiology might include coronary catheterization, magnetic resonance imaging (MRI), stress testing (exercise stress test, MIBI scan, PET scan, etc.) and other laboratory parameters.

Mr. Smythe's EF is 30%, which is consistent with HFrEF (normal EF $\geq$ 60%). Of note, his results also indicated some diastolic dysfunction, which is common and consistent with his history of hypertension. Systolic and diastolic dysfunction can co-exist.

Risk Factors and Etiology

Does Mr. Smythe Have Any Risk Factors for Developing HF?

There are many risk factors for developing heart failure (Table 16.5). Mr. Smythe's risk factors include hypertension, diabetes, and use of nonsteroidal anti-inflammatory drugs (NSAIDs).

Table 16.5 Risk factors for developing heart failure (HF)

Hypertension	Family history of HF
Ischemic heart disease	Smoking
Valvular heart disease	Dyslipidemia
Diabetes	Drugs: NSAID, COX2 inhibitors, thiazolidinediones
Heavy alcohol use	Chemotherapy
Sleep apnea	Obesity

COX cyclooxygenase, NSAIDs nonsteroidal anti-inflammatory drugs

Table 16.6 Common causes of heart failure

Ischemia cardiomyopathy (36–59%)
Nonischemic cardiomyopathy
Hypertensive heart disease (4–78%)
Valvular cardiomyopathy (7–31%)
Idiopathic dilated cardiomyopathy
Familial cardiomyopathy
Inflammatory cardiomyopathy: viral, peripartum
Restrictive cardiomyopathies: sarcoidosis, hemochromatosis
Toxic cardiomyopathy: alcohol, radiation, chemotherapy
High-output cardiomyopathy: tachycardia-induced cardiomyopathy (e.g., atrial fibrillation), thyrotoxicosis

What Is the Cause of Mr. Smythe's HF?

The most common cause of HF is ischemic heart disease, followed by hypertension and valvular heart disease. Other causes are listed in Table 16.6. Given Mr. Smythe's high risk for coronary artery disease, this would need to be ruled out first as the most common cause of HF. Mr. Smythe did have a cardiac PET scan performed, indicating a low probability of ischemia, and therefore the most likely culprit in this case would be hypertension that appears to be poorly controlled. He does, however, have a number of other risk factors that could have contributed.

Classification

What Class of HF Is Mr. Smythe?

Patients can be classified by both their symptoms and stage of disease (Table 16.7) [2]. Because Mr. Smythe has symptoms of HF and a reduced EF, he would be classified as Stage C. His functional class has been somewhat variable through the course of the last couple of months, starting at II then progressing to III–IV and now back to II–III. Functional class is typically variable and fluctuates as the course of disease fluctuates. It also tends to be somewhat subjective depending on the provider. However, regardless of its limitations, it is used clinically to give an overall description of functional capacity of patient with HF at baseline and from visit to visit (e.g., valuable for monitoring). Notably, this

Table 16.7 Classification systems used in HF [3]

ACC/AHA stages	New York Heart Association Functional Class (NYHA FC)
A: High risk for HF; no structural heart disease; no symptoms of HF	None
B: Structural heart disease; no symptoms of HF	I: No limitation in physical activity. Ordinary activity does not cause symptoms
C: Structural heart disease; symptoms of HF (prior or current)	I: See above II: Slight limitation in physical activity. Comfortable at rest. Ordinary physical activity results in symptoms III: Marked limitation in physical activity. Comfortable at rest. Less than ordinary physical activity results in symptoms
D: Refractory HF requiring specialized care	IV: Unable to carry in any physical activity without symptoms or symptoms at rest

ACC/AHA American College of Cardiology/American Heart Association

classification system has also been commonly used as entry criteria into clinical trials evaluating pharmacotherapy in HF.

Clinical Pearls

When assessing functional capacity, first ascertain the types of activities a patient performs in their daily lives.

Do they do any regular exercise? Walking? Do they have stairs?

Do they do common housework? Vacuuming? Making beds? Carrying laundry?

Do they do any yardwork? Shoveling? Mowing the lawn? Gardening?

Often patients with HF have many co-morbidities. Determine what their main limitation is when accounting for their functional capacity. For example, if a patient with HF and knee arthritis notes that they can only walk one block, ask them why they stop. Is it knee pain, or SOB, or both?

Prognosis

What Is Mr. Smythe's Prognosis?

Chronic HF typically has a poor prognosis, which is often worse than many cancers. On average, 50% of patients with HF will die within 5 years. If hospitalized, the in-hospital mortality rate is 4–7% [1]. The mode of death is typically pump failure or arrhythmia. The course of disease is usually progressive with increasing frequency of acute events resulting in increased hospitalizations and mortality.

There are a number of risk stratification tools that can be used to identify patients at high risk; however, they are beyond the scope of this chapter.

Management

What Are the Goals of Therapy for Mr. Smythe?

The goals of therapy when treating a patient with HF vary depending on the clinical situation. Priorities may also change depending on whether the focus is on an acute exacerbation, chronic management, or end-stage disease and palliative care. Only a very small percentage of patients will receive ventricular assist devices or cardiac transplantation. The overall goals include the following:

- Decrease mortality.
- Decrease morbidity.
 - Decrease hospitalizations.
 - Decrease HF exacerbations.
 - Relieve/improve symptoms of HF.
 - Improve functional capacity.
 - Improve quality of life.
- Manage risk factors that may cause or worsen HF.

Given Mr. Smythe's current situation, all of these goals would be applicable.

How Should Mr. Smythe Be Managed?

The approach to treatment in HF involves both drug and non-drug measures as well as the consideration of whether the patient is in the acute or chronic phase of heart failure. Nondrug therapy includes sodium and fluid restriction, exercise, daily weights and device therapy (e.g., biventricular pacing, internal cardio-defibrillator).

Pharmacotherapy can be split into two categories, acute and chronic treatment. Acute treatment can include intravenous modalities such as inotropes and nitroglycerin; however, these treatments are outside the scope of this chapter. We will focus on the treatment of the ambulatory HF patient.

Acute Exacerbation of Heart Failure

An acute exacerbation of HF is typically characterized by worsening symptoms and congestion. Congestion is evaluated by doing an assessment of the patient's volume status, which includes the following parameters: JVP, HJR, pulmonary percussion and auscultation, and examining the lower limbs for edema. Pharmacists can be taught to perform these physical examination skills. However, also being familiar

Table 16.8 Signs and symptoms consistent with congestion

Signs consistent with congestion	Mr. Smythe	Case-specific examples
Lung crackles/rales	Yes	Crackles
Elevated jugular venous pressure (JVP); >4 cm above the sternal angle (ASA)	Yes	7 cm ASA
Positive hepato-jugular reflex (HJR)	Yes	
Peripheral edema	Yes	2+ pitting edema bilaterally to mid-shin ascites
S3 heart sounds	Yes	S3
Chest X-ray findings:		
Pleural effusion	No	
Pulmonary edema	Yes	
Pulmonary venous redistribution	No	
Symptoms consistent with congestion	Mr. Smythe	Case-specific examples
Shortness of breath	Yes	Change from baseline activity: Walking 8 block to 1 blocks Climbing >2 flights of stairs to <1 flight New limitation doing yard work
Cough	Yes	Preexisting cough but worse with orthopnea
Swelling (leg, abdomen)	Yes	Legs: Swelling in both of his legs, at times precluded wearing shoes Abdomen: Poor appetite, even with hunger, feels full quickly and is unable to eat much (early satiety) Reports pants getting tighter and weight going up (~7 kg in 2 months) despite decrease intake and less activity Weight loss post diuretic treatment (~3 kg)
Orthopnea	Yes	Reports sleeping poorly, SOB any time laid flat in bed Sleeps with 2 pillows, down from 4/recliner
Paroxysmal nocturnal dyspnea (PND)	No	NA
Rapid weight gain	Yes	Reports pants getting tighter and weight going up (~7 kg in 2 months) despite decrease intake and less activity Weight loss post diuretic treatment (~3 kg)

with these findings when they are performed by others can help the pharmacist make decisions regarding drug therapy. Mr. Smythe has many signs and symptoms consistent with fluid overload (Table 16.8).

Table 16.9 Common diuretics used in the treatment of HF

Drug	Starting dose	Maximum dose
Loop diuretics		
Furosemide	20–40 mg daily-BID	200 mg/day (caution with >120 mg)
Bumetanide	0.5–1 mg daily-BID	10 mg/day
Ethacrynic acid	25–50 mg daily-BID	400 mg/day (200 mg BID)
Thiazide diuretics		
Metolazone	2.5 mg daily	20 mg/day

BID twice daily

Typically, loop diuretics, most commonly furosemide, are the mainstay of treatment in patients that are fluid overloaded. Occasionally, loop diuretics may be combined with potent thiazides (e.g., metolazone) for additive effects and diuresis, especially in the case of diuretic resistance. This strategy is used sparingly due to frequent electrolyte abnormalities. The goal of therapy when using diuretics is to relieve congestion and improve symptoms. Starting dose (Table 16.9) will depend on patient symptoms, degree of volume overload, renal function, prior response to diuretics. Doses may need to be higher in cases of poor renal function. Diuretic dose should be reassessed often with a goal of the lowest effective dose or elimination of the agent completely if tolerated (and only using it as needed). Minimizing diuretic therapy will also support adding and up-titrating lifesaving medications. Many of these medications also have additive diuretic effects (e.g., ARNI, MRA, SGLT2i) and can augment their effects.

Clinical Pearls

- Lower limb edema can be caused by a number of disease processes, including drugs (e.g., calcium channel blockers, NSAIDs, pregabalin, etc.). Remember to comprehensively assess the patient and put peripheral edema in context with other signs and symptoms of HF.
- Peripheral edema from an acute exacerbation of HF may take days or weeks to resolve and should never be used in isolation as the only marker of fluid overload. Consider all sign and symptoms when assessing fluid status and response to diuretics.
- Diuretics can often precipitate or worsen preexisting gout. Be sure to assess if a patient has a history of gout and if they do, that they are on prophylactic medication (e.g., allopurinol)
- Sometimes other medications that lower blood pressure may require a dose reduction in order to adequately remove fluid using higher doses of diuretics. Just remember to minimize diuretics as tolerated post-acute exacerbation and restart/up-titrate required medications to doses prior to the event if possible.

Chronic Heart Failure

Chronic therapy can be started in the acute phase provided the patient is hemodynamically stable. The general approach to pharmacotherapy is outlined in Fig. 16.1 [3, 4]. Pharmacotherapy is the cornerstone of HF management. Combination therapy is used in most patients to target either the negative or augment the positive neurohormonal effects (e.g., renin–angiotensin–aldosterone system, sympathetic nervous system, natriuretic peptide system) of a failing heart. Table 16.10 details the specific drug classes and parameters to consider when initiating and monitoring as part of the HF patient assessment. Figures 16.2 and 16.3 provides a concise summary of the same. Both starting and target doses of these agents are listed in Table 16.11. The doses listed reflect tho-

ses used in clinical trials that showed improvements in patient outcomes.

Therapy can be grouped into medications for all patients with HFrEF (core medications) that have been proven to decrease CV mortality and hospitalizations for heart failure (HHF) and medications for select patients (individualized medications) that have either only been shown to improve HHF alone or improve symptoms. Core medications should be focused on for all HFrEF patients with the goal to initiate all four and uptitrated to target or maximum tolerated doses within 3–6 months of heart failure diagnosis. The order of initiation and titration can be tailored based on patient characteristics and co-morbidities. The general principle is to concurrently start and titrate them, rather than individually initiating and uptitrating one at a time.

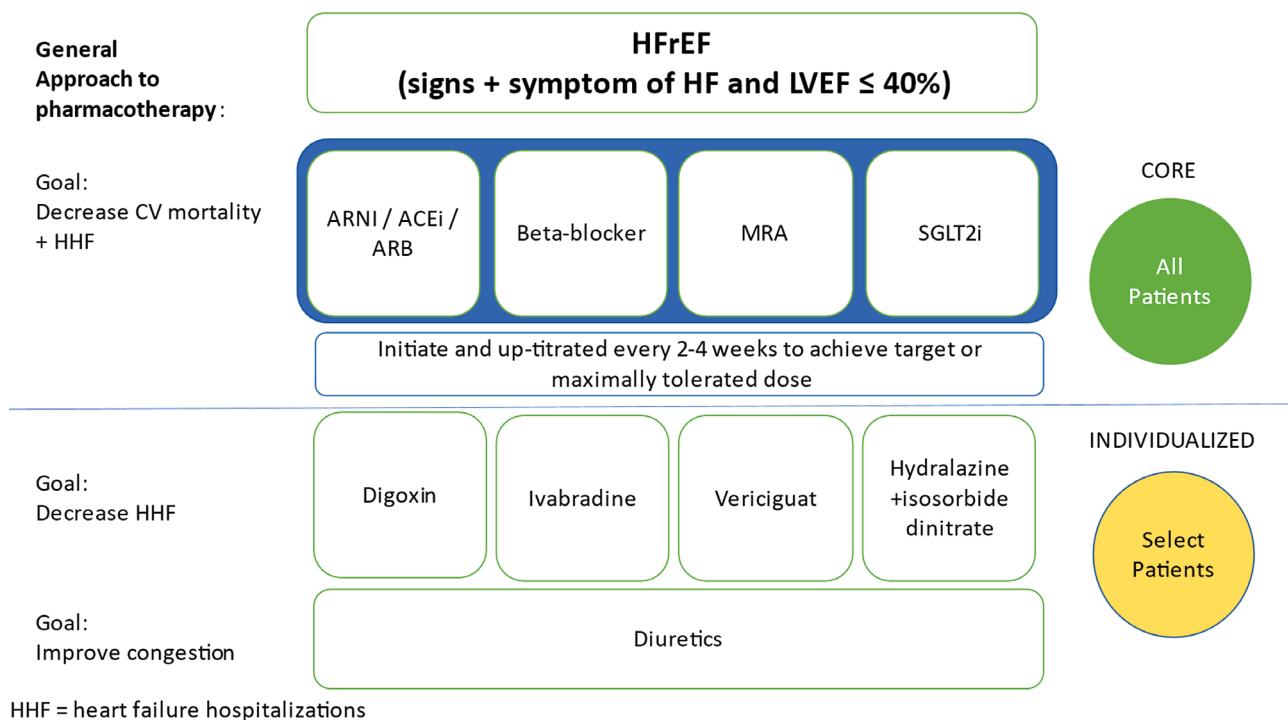


Fig. 16.1 Pharmacological approach to patients with HFrEF

Table 16.10 Heart failure medications considerations for initiation and monitoring

Drug class	Initiation	Monitoring
ACEI/ARB	Labs: Establish baseline SCr and K⁺ Avoid in those with SCr > 220umol/L and K > 5.0 mmol/L Vitals: Establish baseline blood pressure Avoid in those with SBP < 100 mmHg Other: When starting an ACEi, document the presence or absence of cough <i>prior to initiation</i> Review other medications for drug interactions, including those that increase potassium and affect renal function	Cough and angioedema: Cough, characterized as dry and persistent, occurs in 10–20% of patients receiving ACE is and does not require discontinuation unless it is bothersome to the patient In patients with angioedema from an ACEi (uncommon <1%), an ARB can be used cautiously as there have also been reports of angioedema with ARBs Renal function and electrolytes (K⁺): Monitor SCr and K⁺ 1–2 weeks after initiation and dose increases Watch for trends in the K⁺: if continuously increasing or there is a significant change, pre-emptive adjustments may be required to prevent hyperkalemic (e.g., diet, other medications) Concurrent MRA therapy requires closer monitoring of K⁺ and Scr A change in Scr of up to 30% acceptable with introduction of ACEi/ARB. There is no immediate need to decrease the drug dose if the increase stabilized, but closer long-term monitoring should be considered Blood pressure: Monitor with every visit and within 1 week of initiation or dose change Exaggerated BP lowering may occur if starting at high doses or combined with diuretic therapy Instructions on sick day management should be provided
ARNI (Sacubitril/ Valsartan)	Labs: Establish baseline SCr and K⁺ Avoid in those with SCr > 220umol/L and K > 5.0 mmol/L Vitals: Establish baseline blood pressure Avoid in those with SBP < 100 mmHg ARNI can have more of a hypotensive effect compared to ACEi/ARBs, dose selection is dependant on baseline ACEi/ARB dose Fluid status should be assessed prior to initiation as ARNI may promote natriuresis and require diuretic reassessment Other: Review other medications for drug interactions, including those that increase potassium and affect renal function When switching from ACEi to ARNI (or vice versa), a 36 h wash period is required to avoid and increase risk of angioedema. This is not necessary when switching from ARB to ARNI (as ARNI contain an ARB)	Side effects and tolerability similar to ACEi/ARB: If patient has angioedema to ACEi or ARB, ARNI is contraindicated Cough may occur but less likely than with ACEi (ACEi>ARNI>ARB) Renal function and electrolytes: Monitor SCr and K⁺ 1 week after initiation or dose increase, after dose stabilization, monitor every 3 months Blood pressure: ARNI can have more of a hypotensive effect compared to ACEi/ARBs Instructions on sick day management should be provided
B- blockers	Vitals: Establish baseline heart rate and blood pressure Avoid if HR < 60 bpm or SBP < 100 mmHg Avoid if heart block (and no pacemaker in-situ) Other: Should be started at very low doses and titrated slowly to prevent decompensation or worsening of HF initially In reactive airway disease, use a selective β-blocker cautiously (e.g., bisoprolol or metoprolol) Reassess the need for other AV nodal blocking drugs used in combination (e.g., digoxin, amiodarone)	Heart rate and blood pressure: Consider dose reduction if HR < 50 bpm If patient has low HR (HR 50–60) but is asymptomatic, there is no need to decrease dose Fluid retention: Transient fluid retention can occur when increasing β-blocker, and patient may require a diuretic dose change

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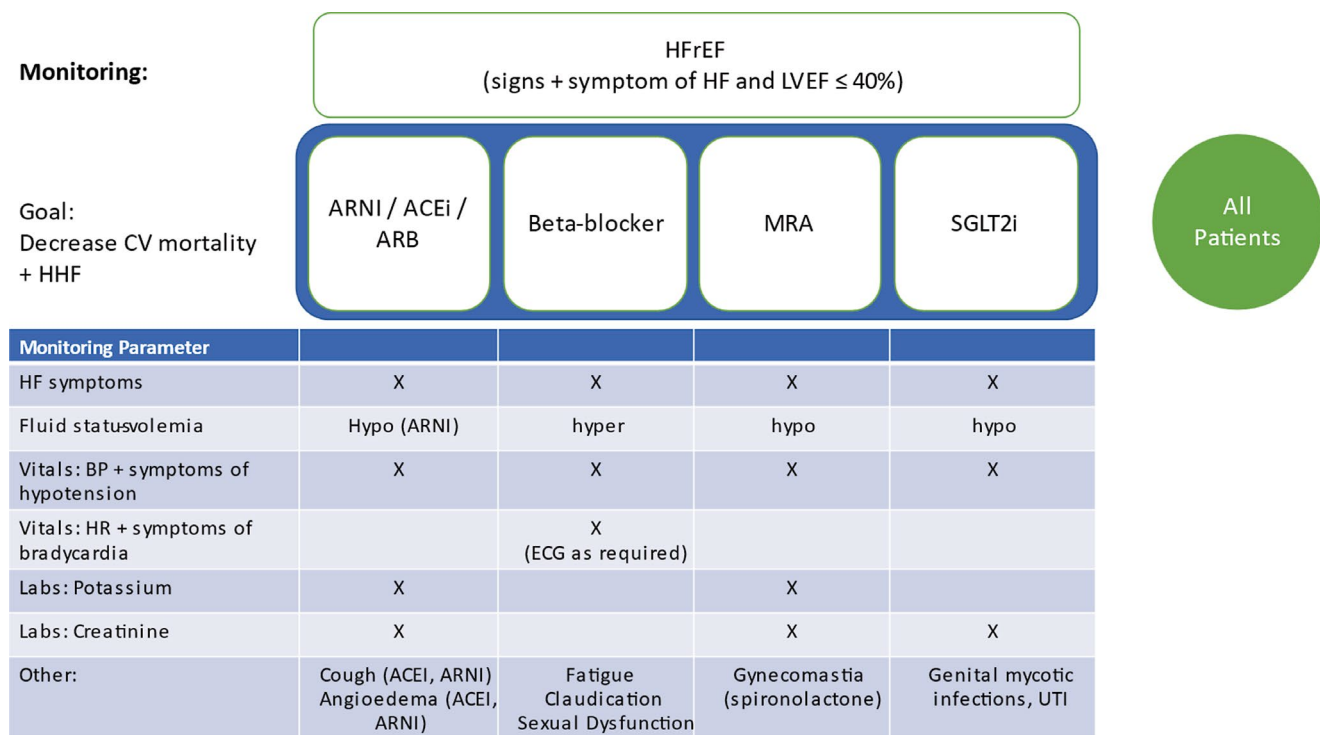
Table 16.10 (continued)

Drug class	Initiation	Monitoring
MRA	Labs: Establish baseline SCr and K⁺ Avoid in those with SCr > 220 μmol/L or GFR < 30 mL/min) and K > 5.0 mmol/L Vitals: Establish baseline blood pressure Other: Review other medications for drug interactions, including those that increase potassium and affect renal function	Renal function and electrolytes: Monitor K⁺ and SCr 2–3 days after initiation then 1 week after initiation. Check monthly × 3 months and then every 3 months thereafter. More frequent monitoring may be required in acute illness and those at high risk (DM, CKD, older) Hormonal side effects: Spironolactone may cause hormone-related side effects (e.g., gynecomastia (9%), impotence, post-menopausal bleeding) Eplerenone is a more selective mineralocorticoid receptor antagonist Instructions on sick day management should be provided
SGLT2i	Labs: Establish baseline SCr Avoid initiation with an eGFR < 20 mL/min/m² Vitals: Establish baseline blood pressure Other: Fluid status should be assessed prior to initiation as SGLT2i may augment diuresis and require diuretic reassessment In those with diabetes, adjustment of other hypoglycemics may be required with initiation	Renal function: Monitor SCr in 2–4 weeks after initiation and then every 3–6 months A 15% decrease in eGFR is acceptable Blood pressure: SGLT2i have limited blood pressure effects, unless there is underlying hypovolemia. If exaggerated BP effect, reassess fluid status/diuretic dose Instructions on sick day management should be provided
Solubale guanylate cyclase stimulator (vericiguat)	Labs: Establish a baseline hemoglobin Vitals: Establish baseline blood pressure	Blood pressure: Monitor with every visit and within 1 week of initiation or dose change. Monitor for symptoms of hypotension
I _f inhibitor (ivabradine)	Vitals: Establish baseline heart rate Indicated in those with a HR > 70 bpm on maximally tolerated β-blocker therapy Other: Ivabradine works on the SA node; therefore, the patient must be in normal sinus rhythm	Heart rate: Target HR = 50–60 bpm Monitoring HR and adjusting the dose ivabradine every 2 weeks may be required
Vasodilators (hydralazine / nitrates)	Used if the patient is unable to tolerate ACEI/ARB/ARNI (significant worsening renal function or persistent hyperkalemia despite dose reductions of RAAS agent, elimination and modification of other contributing factors) Vitals: Establish a baseline heart rate	Blood pressure Tolerance/administration: Avoid continuous (24 h) use of nitrates because patients can develop tolerance
Digoxin	Labs: Establish baseline SCr and K Dose may need to be adjusted based on renal function Vitals: Establish baseline heart rate	Heart rate Electrolytes and kidney function: Monitor K⁺ and SCr when changing digoxin dose, diuretic dose or dehydrating illness Increased monitoring may be required if the patient has decreasing or fluctuating renal function, elderly, or low body weight Levels: Digoxin levels are not done routinely and are not used to guide treatment. Levels are only done if one is suspecting toxicity

(continued)

Table 16.10 (continued)

Drug class	Initiation	Monitoring
Diuretics	Diuretics are used to manage symptoms of fluid overload and congestion Labs: Establish baseline SCr, sodium, and K Vitals: Establish baseline: Blood pressure Weight Other: Document if gout is present, consider prophylaxis	Electrolytes and kidney function Monitor SCr and electrolytes 2–3 days; then 1 week after initiation and dose changes Note renal function may also deteriorate with acute HF, so a fluid assessment is essential to assessing the cause of renal dysfunction Blood pressure Postural blood pressure (and heart rate) may help assess fluid status Weight Daily morning weight should be monitored in patients with HF with fluid retention or congestion that is not easily controlled with diuretics or in patients with significant renal dysfunction Monitor weight more closely in unstable or frail patients—rapid weight gain (1.5–2 kg) should prompt a rapid medical visit Symptoms Diuretics are titrated to symptom relief. When euvoemia is achieved, dosing is titrated to the minimum effective dose to maintain euvoemia, which may include no diuretic at all Some patients may be managed with diuretic as needed and given parameters as to when the diuretic should be taken. (e.g., take 20 mg of furosemide when >1 kg gained in 3 days)

**Fig. 16.2** Summary of monitoring parameters for core HFrEF medications

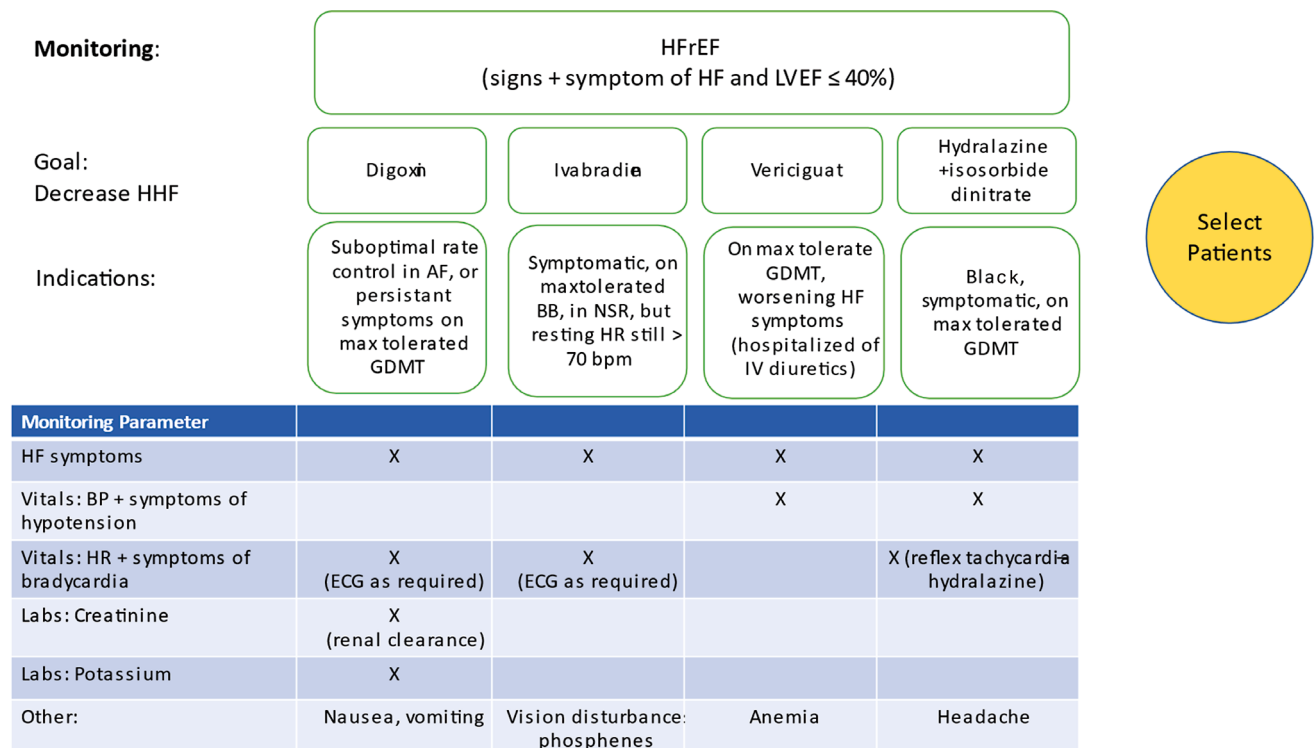


Fig. 16.3 Summary of monitoring parameters for individualized HFrEF medications

Table 16.11 Evidence-based drugs: start and target doses as shown in large clinical trials

Drug	Starting dose	Titration	Target dose
Angiotensin converting enzyme inhibitors (ACEi)			
Captopril	12.5 mg TID	Titrate every 2–4 weeks	25–50 mg TID
Enalapril	1.25–2.5 mg BID		10 mg BID
Lisinopril	2.5–5 mg daily		20–35 mg daily
Perindopril	2–4 mg daily		4–8 mg daily
Ramipril	1.25–2.5 mg BID		5 mg BID
Trandolapril	1–2 mg daily		4 mg daily
Angiotensin receptor blockers (ARB)			
Candesartan	4–8 mg daily	Titrate every 2–4 weeks	32 mg daily
Valsartan	40 mg BID		80 mg BID
Angiotensin receptor blocker/neprilysin inhibitor (ARNI)			
Sacubitril/ valsartan	50–100 mg BID	Titrate every 3–6 weeks	200 mg BID
Beta-blockers			
Carvedilol	3.125 mg BID	Titrate more slowly (may temporarily worsen HF symptoms), every 4 weeks	25 mg BID
Bisoprolol	1.25 mg daily		50 mg BID if >85 kg
Metoprolol*	6.25–12.5 mg BID		10 mg daily
Mineralocorticoid receptor antagonists (MRA)			
Spironolactone	12.5 mg daily	Titrate every 2–4 weeks	50 mg daily
Eplerenone	12.5–25 mg daily		50 mg daily
Sodium glucose co-transporter-2 inhibitors (SGLT2i)			
Dapagliflozin	10 mg daily	Not necessary	10 mg daily
Empagliflozin	10 mg daily	Not necessary	10 mg daily
I_f inhibitor			
Ivabradine	2.5–5 mg BID	Titrate every 2 weeks	Target to heart rate 50–60 bpm Maximum dose: 7.5 mg BID

(continued)

Table 16.11 (continued)

Drug	Starting dose	Titration	Target dose
Soluble guanylate cyclase stimulator			
Vericiguat	2.5 mg daily	Every 2 weeks	10 mg daily
Vasodilators			
Isosorbide dinitrate	20 mg TID	Titrate every 2–4 weeks	40 mg TID Equivalent dose: NTG patch $\approx$ 0.8–1.0 mg/h Isosorbide-5-mononitrate $\approx$ 60 mg daily
Hydralazine	37.5 mg TID		75–100 mg TID-QID

Physical Assessment Skills

Physical examination of the patient with HF will depend on the training of the individual pharmacist. All pharmacists should be able to perform blood pressure (supine, sitting, and standing) and heart rate to assess pharmacotherapy in HF. Complete fluid assessment may be limited to advanced trained pharmacists. Most pharmacists should be able to assess for pitting edema in the lower extremities. Advanced trained pharmacists may also be able to do a focused pulmonary examination (assessing for pulmonary edema, pleural effusions, crackles, and rales) and cardiovascular examination (extra heart sounds, including S3, S4, and JVP). Additionally, a focused gastrointestinal exam, including an assessment for ascites, hepatomegaly, and the HJR may also be performed for complete fluid assessment as necessary.

Follow-Up Assessment

Follow-up is dictated by the patient's signs and symptoms and by any changes in medication. Follow-up should include an assessment of the patient's fluid status followed by review of their HF medications including assessment of adherence, target dose achievement and adverse effects (see Table 16.10). Adherence in the HF patient is complicated by multiple co-morbidities and polypharmacy. Pharmacists are well-positioned to address and manage adherence issues in these patients. It has been shown that pharmacists can identify patients at risk and improve adherence in this population [5, 6].

Clinical Pearls

- Minimize pill burden and streamline regimens where possible.

- If a patient experiences symptomatic hypotension, consider staggering doses of anti-hypertensives (e.g., am and pm dosing) and reassessing nonessential anti-hypertensives.
- Consider gout prophylaxis in patients on diuretics and prone to gout.
- Many patients may have lower urinary tract symptoms (e.g., incontinence, frequency, etc.). Discuss management of diuretic therapy to avoid nonadherence.

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Chapter Objectives

1. Describe the risk factors, clinical presentation, and diagnosis of asthma.
2. Complete an initial assessment of patients newly diagnosed with asthma.
3. Conduct a follow-up assessment of patients with asthma.

Background

Asthma is a chronic respiratory condition characterized by chronic inflammation, variable airflow limitation, and airway hyperresponsiveness with increased bronchial secretions [1]. The clinical symptoms of asthma are shortness of breath, wheeze, chest tightness, and/or cough. The symptoms of asthma and the airflow limitation vary over time and in intensity. As of 2020, asthma affects 8.7% of Canadians aged 12 and over [2]. There are different phenotypes of asthma however, the clinical utility of phenotype classification to guide therapy is not well understood, and therefore, will not be explored in this chapter. This chapter will review how to confirm the diagnosis of asthma, assessing asthma control and appropriate management, and determining follow-up.

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Risk Factors

Risk factors that have been identified to contribute to asthma development include heredity, personal and family history of allergies or asthma, high level of exposure to airborne allergens (pets, house dust mites, cockroaches and mould), frequent respiratory infections early in life, tobacco smoke exposure in utero or in the home environment, and obesity [1, 3].

Diagnosis

The initial diagnosis of asthma should be based on the clinical presentation and confirmation of variable airflow limitation prior to therapy initiation in those 6 years of age and older [1]. The onset is more common in children and young adults but can develop at any age [3]. In children less than 6 years of age, clinical history, risk factors, physical exam, response to treatment and exclusion of alternate diagnosis are used as spirometry, lung function tests, cannot be performed in this age group [4].

Clinical Symptoms

The symptoms of asthma are shortness of breath, wheeze, a feeling of chest tightness, and cough [1, 4]. A pattern of asthma symptoms that would support the diagnosis include:

- varying over time and intensity (not always the same)
- worse at night or when waking up
- triggered by physical activity, laughter, allergens, irritants or cold air
- present or worsen with viral infections

Patients that have respiratory infections that last longer or are recurrent should be asked about asthma symptoms. For those with adult-onset asthma exposure history is key at

Table 17.1 Symptoms and possible exacerbating factors of asthma [5, 6]

Symptoms	Potential aggravating factors
Cough	Allergens, e.g., animal dander, pollens, rats, cockroaches, mold, certain foods
Shortness of breath (SOB)	Smoke, e.g., cigarette smoke, vaping, environmental smoke
Chest tightness	Respiratory tract infections (viral and bacterial)
Wheezing (a high-pitched sound secondary to airway narrowing)	Sinusitis
	Reflux
	Physical exertion
	Emotional distress
	Cold weather
	Occupational exposure to irritant chemicals
	Some drugs such as nonselective beta blockers, nonsteroidal anti-inflammatory drugs (NSAID)

work and at home. Pharmacists can play a key role to help identify patients with symptoms suggestive of asthma and refer them for further assessment and diagnosis. Table 17.1 depicts a summary of asthma symptoms and possible exacerbating factors.

Confirmed Variable (Previously Referred to as Reversible) Airflow Limitation

For the diagnosis of asthma, in addition to symptoms, confirmation of variable expiratory airflow limitation is needed [1, 4]. Lung function testing (spirometry) is the preferred method and is readily available in Canada. On spirometry testing the patient's baseline lung function is measured. The patient is then given a bronchodilator, usually salbutamol 200–400 mcg, and the lung function is then remeasured. In adults, an increase in the patient's forced expiratory volume in 1 second (FEV1) or forced vital capacity (FVC) by $\geq 12\%$ and an increase of at least 200 mL supports the diagnosis of asthma. In children 6 years and older an increase in FEV1 predicted by $\geq 12\%$ supports the diagnosis of asthma.

Less commonly for diagnosis, spirometry may be done at baseline and then repeated after 4 weeks of therapeutic treatment. In patients where the initial tests are negative, tests can be repeated during symptoms or early morning. Other diagnosis methods such as positive bronchial challenge test or peak expiratory flow measurements may be used in cases where spirometry is unavailable or difficult to capture a patient's symptoms for example if symptoms predominantly around an allergen at home. As a pharmacist, when assessing patients with asthma, it is important to ensure a proper diagnosis by confirming spirometry has been done or refer the patient for appropriate diagnosis.

Differential Diagnoses

There are a number of comorbidities that mimic asthma symptoms and/or worsen actual asthma. These conditions should be identified to avoid misdiagnosis and unnecessary treatment. Examples of these comorbidities include rhinosinusitis, gastroesophageal reflux disease (GERD), vocal cord dysfunction, heart failure, pulmonary hypertension, chronic obstructive pulmonary disease (COPD), congenital heart disease, bronchiectasis, and cystic fibrosis [1].

Diagnosis Summary

Pharmacists can play a key role in identifying patients that would benefit from referral for asthma diagnosis. When caring for patients with asthma the initial assessment should include confirming that the asthma diagnosis has been made appropriately based on both clinical symptoms and documentation of variable expiratory airflow limitation. This is important to ensure patients receive appropriate treatment for their condition and prevent airway remodeling and avoid misdiagnosing.

Assessment of Symptom Control

Asthma control and risk for exacerbation should be assessed to ensure an appropriate therapeutic approach at diagnosis and during every subsequent patient encounter [4]. The goals of asthma management are to achieve symptom control and to prevent or minimize future risk of short- and long-term complications, morbidity and mortality. GINA 2024 outlines criteria for asthma control (Fig. 17.1). Deviation from these control criteria (e.g., daytime symptoms more than 3 days per week or absence from work due to asthma) suggests loss of control and increased risk of future exacerbations. Examples of questions that can be asked:

- Daytime symptoms: How many times in a week do you/patient have a cough, wheeze, shortness of breath, heaving breathing, or chest tightness?
- Nighttime symptoms: Do you/ the patient wake up at night because of asthma?
- Reliever medication: How often do you/patient use reliever doses?
- Activity Limitation: How does asthma impact your/patient's activity and exercise?

Asthma control could also be assessed using other validated patient questionnaires such as Asthma Control Test

A. Recent asthma symptom control (but also ask the patient/caregiver about the whole period since last review#)						
In the past 4 weeks, has the patient had:				Well controlled	Partly controlled	Uncontrolled
• Daytime asthma symptoms more than twice/week?	Yes ☐ No ☐	}	None of these	1–2 of these	3–4 of these	
• Any night waking due to asthma?	Yes ☐ No ☐					
• SABA* reliever for symptoms more than twice/week?	Yes ☐ No ☐					
• Any activity limitation due to asthma?	Yes ☐ No ☐					
B. Risk factors for poor asthma outcomes						
Assess risk factors at diagnosis and periodically, particularly for patients experiencing exacerbations.						
Measure FEV ₁ at start of treatment, after 3–6 months of ICS-containing treatment to record the patient's personal best lung function, then periodically for ongoing risk assessment.						
a. Risk factors for exacerbations						
Uncontrolled asthma symptoms: Having uncontrolled symptoms is an important risk factor for exacerbations. ²⁵						
Factors that increase the risk of exacerbations even if the patient has few asthma symptoms†						
SABA over-use: High SABA use (≥3 x 200-dose canisters/year associated with increased risk of exacerbations, increased mortality particularly if ≥1 canister per month) ^{86–89}						
Inadequate ICS: not prescribed ICS, poor adherence, ⁹⁰ or incorrect inhaler technique ⁹¹						
Other medical conditions: Obesity, ^{92,93} chronic rhinosinusitis, ⁹³ GERD, ⁹³ confirmed food allergy, ⁹⁴ pregnancy ⁹⁵						
Exposures: Smoking, ⁹⁶ e-cigarettes, ⁹⁷ allergen exposure if sensitized, ^{98,99} air pollution ^{99–102}						
Psychosocial: Major psychological or socioeconomic problems ^{103,104}						
Lung function: Low FEV ₁ (especially <60% predicted), ^{96,105} high bronchodilator responsiveness ^{93,106,107}						
Type 2 inflammatory markers: Higher blood eosinophils, ^{93,108,109} high FeNO (adults with allergic asthma on ICS) ¹¹⁰						
Exacerbation history: Ever intubated or in intensive care unit for asthma; ¹¹¹ ≥1 severe exacerbation in last year ^{112–113}						
b. Risk factors for developing persistent airflow limitation						
History: Preterm birth, low birth weight and greater infant weight gain, ¹¹⁴ chronic mucus hypersecretion ^{115,116}						
Medications: Lack of ICS treatment in patient with history of severe exacerbation ¹¹⁷						
Exposures: Tobacco smoke, ¹¹⁵ noxious chemicals; occupational or domestic exposures ⁵²						
Investigation findings: Low initial FEV ₁ , ¹¹⁵ sputum or blood eosinophilia ¹¹⁶						
c. Risk factors for medication side-effects						
Systemic Frequent OCS, long-term, high-dose and/or potent ICS, P450 inhibitors ¹¹⁸						
Local: High-dose or potent ICS, ^{118,119} poor inhaler technique ¹²⁰						

See list of abbreviations (p.11). *Based on SABA (as-needed ICS-formoterol reliever not included); excludes reliever taken before exercise (see Assessing asthma symptom control, p.38).

#In addition to assessing recent asthma symptom control, also ask the patient about symptom control over the whole period since their last clinical review. There are no validated tools for assessing long-term symptom control, i.e., over periods longer than 4 weeks.

†'Independent' risk factors are those that are significant after adjustment for the level of symptom control. Cytochrome P450 inhibitors such as ritonavir, ketoconazole, itraconazole may increase systemic exposure to some types of ICS and some LABAs; see drug interaction websites and p.122 for details. For children 6–11 years, also refer to Box 2-3, p.40. See Box 3-5, p.55 for specific risk reduction strategies.

Fig. 17.1 GINA assessment of asthma control at clinical visits in adults, adolescents, and children aged 6–11 years. GINA ©2024 Global Initiative for Asthma, reprinted with permission. Available from www.ginasthma.org

(ACT), Asthma Control Questionnaire (ACQ). Pharmacists are uniquely positioned to provide care and ongoing assessment and monitoring to patients with asthma. Pharmacists are accessible healthcare practitioners. The simple act of assessing asthma control and inhaler technique at every opportunity can provide a chance for conversation on control and identification of deterioration. As a result, pharmacists can provide meaningful support.

Studies have shown that many patients believe that their asthma is controlled; however, their symptoms indicate otherwise [7]. Deaths due to asthma have occurred even in those thought to have mild asthma. The majority of these deaths occur outside the healthcare setting and involve patient's underestimating or delaying treatment for flare-ups, and this confirms the need for regular assessment.

Lack of symptom control may suggest the need for intensification of asthma pharmacotherapy [1]. Before intensifying asthma management in patients presenting with uncontrolled or partly controlled disease, an assessment of potential factors of therapy failure should be conducted. These factors include confirming asthma diagnosis, correct inhaler techniques, assessing and managing comorbidities, inability to afford medication, lack of understanding of the role of controller medications and relievers, adherence to pharmacotherapy and trigger avoidance. Exceeding doses of short-acting bronchodilators can be a sign of uncontrolled asthma or exacerbation requiring management in a facility with advanced treatment options (e.g., oxygen, noninvasive ventilation). Reliever overuse could be a red flag that prompts referral to the patient's prescriber. Respiratory specialists are generally very happy to receive communication from community pharmacists to ensure that patients are able to follow the plan they agreed to. If no Asthma Action Plan has been provided, the pharmacist can suggest one or fill one out based on the treatment that controlled the patient's asthma previously. If the patient has never achieved control, an escalation in therapy or referral may be warranted.

Assessment of Exacerbation Risk

An exacerbation is an increase in asthma symptoms [4]. A mild exacerbation is an increase in symptoms that can be managed without systemic steroids or an emergency department/hospital visit. A severe exacerbation is one that requires systemic steroids or an emergency department/hospital visit. Having visited a hospital for asthma in the past year increases the risk of asthma-related death. Patients that have exacerbations requiring systemic steroids over their life are at risk of long-term side effects from those steroids, including osteoporosis, fragility fractures, cataracts, glaucoma, and adrenal suppression [1]. Poor asthma control and several other risk factors can increase the risk of asthma exacerbations [1, 4].

A pharmacist, in addition to assessing symptom control, should be assessing for risk of exacerbations on an ongoing basis. Factors that increase the risk of exacerbations include a history of exacerbations in the last year, poorly controlled asthma, excessive short-acting beta 2 agonist use (> 2 per year), poor medication adherence, absence of an ICS prescription, incorrect inhaler technique, exposure to symptom triggers, low FEV1 (especially <60% predicted), low socioeconomic status, comorbidities such as obesity and psychological concerns, current smoking, and concurrent food allergies. Patient's using ≥ 1 SABA canister per month are at risk of mortality as it indicates uncontrolled asthma.

Addressing exacerbation risk factors is important, and treatment strategies need to be tailored to the risk factors. Examples of strategies include encouraging quitting smoking, supporting weight reduction, and having an asthma action plan. Of note, smoking cessation and/or avoiding second-hand smoke can be very useful in improving asthma control. Pharmacists not only can provide behavioral support and suggestions/prescriptions for medications, but also they can refer patients and families to other existing resources such as provincial or national Quit programs or Help Lines, national associations such as the Lung Association and Employee and Family Assistance programs.

Management

The goals of asthma management are to achieve symptom control and to prevent or minimize future risk of short- and long-term complications, morbidity and mortality. Non-pharmacological and pharmacological strategies are needed to achieve these goals. Non-pharmacological strategies address the treatment of exacerbation risk factors as discussed above. Effective self-management plans include asthma action plans, medication review, and self-management strategies. Asthma actions plans have shown to reduce exacerbations [8]. An asthma action plan allows the patient to monitor for symptoms and provides directions for self-management and referral. Asthma management guidelines for baseline asthma and exacerbations are available and detailed from the Canadian Thoracic Society (CTS) and the Global Initiative for asthma (GINA). Figure 17.2 depicts an example of an asthma action plan.

Pharmacological Options for Asthma

Asthma pharmacotherapy is generally divided into two main classes: symptomatic therapy (*relievers*) and maintenance therapy (*controllers*). There are at least 60 different inhalers when considering the different inhaler devices, medications, and combinations of products available. Up-to-date inhaler

Date: November 27, 2025

Name: Hady Halawa

Usual Symptoms: Cough, SOB, wheeze

Triggers: dust, smoke, molds

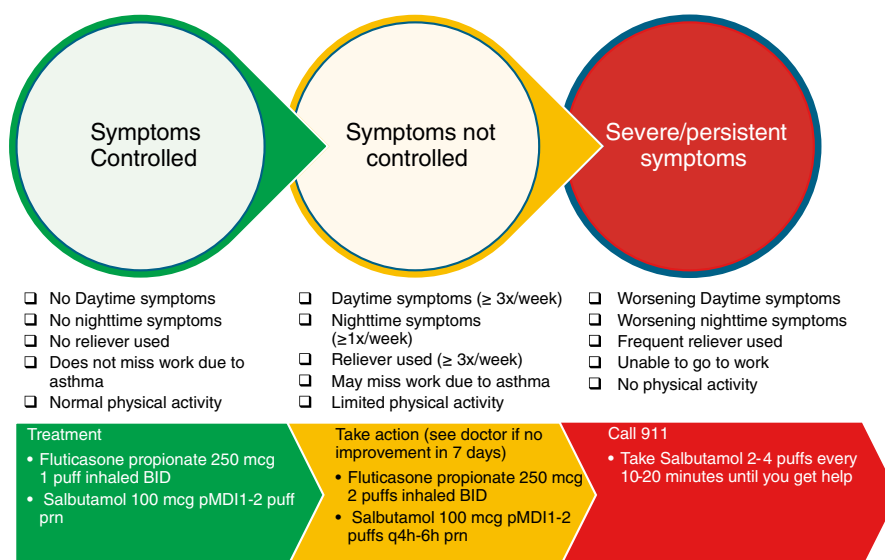


Fig. 17.2 An example of an asthma action plan. BID, twice daily; pMDI, pressurized metered dose inhaler; SOB, shortness of breath

options and appropriate inhaler techniques are available through different organizations including Lung Saskatchewan, RESPTREC, and RQESR (primarily French). Table 17.2 summarizes asthma treatment options available in Canada, which has been growing and will continue to grow. Table 17.2 depicts the comparative dosing categories of inhaled corticosteroids [4].

Relievers are used, when needed, for quick relief of asthma symptoms. Relievers are available as inhaled formulations. *Anti-Inflammatory Reliever (AIR)* refers to an inhaler that contains both a rapid acting bronchodilator and ICS. All patients should have a reliever prescribed.

Maintenance Therapy with ICS is the cornerstone of asthma management given the characteristic of chronic inflammation. Controllers are regularly used medications to address the chronic inflammation and prevent the progression to airway remodelling. Controllers are generally inhaled, but there are other options that could be oral (e.g., leukotriene receptor antagonists) or injectable (e.g., biologics). *Maintenance and reliever therapy (MART)* usually refers to ICS-formoterol combinations that are used both as the every-day controller inhaler and as needed therefore serving a dual purpose.

For mild asthma, in those 12 years of age and older, scheduled ICS plus as needed SABA/Fast acting beta agonist (FABA) (ex. Fluticasone scheduled daily and as needed salbutamol) and FABA/ICS as required (ex. Budesonide/formoterol 1 puff as needed) had similar rates of asthma exacerbations requiring oral corticosteroids, both approaches having lower exacerbation rates than SABA treatment alone [9]. FABA/ICS as required was slightly favored, though low certainty evidence, for asthma-related hospital admission or emergency department or urgent care visit. Asthma control,

though small difference, favored the ICS scheduled with as needed SABA/FABA. Different guidelines prioritize differently; thus, this may lead to different recommendations from various organizations.

Patients having an exacerbation should increase reliever use, increase ICS use if able, and review response [1]. For patients with PEF or FEV₁ less than 60% of personal best or not improving after 48 h should be considered for systemic steroids (e.g., prednisone 50 mg oral daily for 5–7 days) and referral to a prescriber for an assessment.

It is important to draw attention to the difference between asthma control and asthma severity. Asthma control refers to the current symptom burden. Asthma severity is defined by the treatment step needed to achieve asthma control. Initial therapy depends on the patient's asthma control. ICS containing treatment should be initiated for best treatment outcomes [1]. The Canadian Thoracic Society and GINA guidelines outline initial management options. Ongoing asthma management decisions depend on the patient's asthma control.

Severe asthma represents approximately 5–10% of asthmatic patients. Severe asthma is defined by CTS as “Asthma which requires treatment with high-dose ICS, as outlined in Table 17.3, and a second controller for the previous year, or systemic corticosteroids for 50% of the previous year to prevent it from becoming ‘uncontrolled,’ or which remains ‘uncontrolled’ despite this therapy” [4]. It is important to determine if asthma is indeed severe or rather uncontrolled asthma. Lack of asthma control is more commonly the cause of persistent symptoms and exacerbations. Prior to labelling asthma as severe asthma practitioners should ensure that comorbidities are ruled out and managed, medication adherence is confirmed, inhaler technique is appropriate, the diag-

Table 17.2 Asthma treatment options available in Canada as of 2024

Category	Class	Drug name	Comments
Relievers	Short-acting beta 2 agonists (SABA)	Salbutamol	
		Terbutaline	
	Anti-inflammatory reliever/MART	Budesonide/formoterol	Budesonide/formoterol can be used as both controller and reliever
	Short-acting muscarinic antagonists (SAMA)	Ipratropium	Alternative reliever in patients experiencing adverse effects from SABA (tremors, tachycardia)
Controllers (inhaled)	Inhaled corticosteroids (ICS)	Beclomethasone	Cornerstone of asthma controller therapy
		Budesonide	Available in multiple formulations: pMDI, DPI and oral inhalation solution (budesonide)
		Ciclesonide	
		Fluticasone furoate	
		Fluticasone propionate	
		Mometasone	
	Long-acting beta 2 agonists (LABA)	Formoterol	These should not be used without concurrent ICS due to increased mortality risk [10]
		Salmeterol	
	ICS/LABA combination	Budesonide/formoterol**	** Can be used as MART as formoterol is fast acting Mometasone/formoterol can be used as both in regions where low dose formulation is available (not currently available in Canada)
		Mometasone/formoterol	
		Fluticasone furoate/vilanterol	
		Fluticasone propionate/salmeterol	
		Mometasone/indacaterol	
	Long-acting muscarinic antagonists (LAMA)	Tiotropium	
Controllers (oral)	Leukotriene receptor antagonists (LTRA)	Montelukast	
	Methyl xanthines	Theophylline	Limited use due to adverse effects
Others	Oral corticosteroids (OC)	Prednisone	Taken for short period of time in case of acute exacerbations
Biologics (injectables)	IgE antibody	Omalizumab—SC	Biologic therapy reserved for patients with severe asthma and prescribed by a specialist.
	IL-5 inhibitor	Mepolizumab/benralizumab—SC	
		Reslizumab—IV	
		Dupilumab—SC	
	anti-thymic stromal lymphopoeitin	Tezepelumab—SC	

DPI dry powder inhaler, pMDI pressurized metered dose inhaler, *prn* when necessary, SMI soft mist inhaler

Table 17.3 Comparative dosing categories of inhaled corticosteroids [4]

ICS name	ICS daily dose in mcg					
	Pediatric (6–11 years)			Adults and adolescents (≥ 12 years)		
	Low	Medium	High	Low	Medium	High
Beclomethasone	≤200	201–400 ^a	>400 ^a	≤ 250	251–500	>500
Budesonide	≤400	401–800	>800	≤400	401–800	>800
Ciclesonide	≤200	201–400 ^a	>400 ^a	≤200	201–400	>400
Fluticasone furoate	N/A	N/A	N/A	100		200
Fluticasone propionate	≤200	201–400	>400 ^a	≤ 250	251–500	>500
Mometasone	100	≥200–<400	≥400	100–200	>200–400	>400

ICS inhaled corticosteroids

^aThese doses are not approved for use in children in Canada

nosis is correct, and exposure to allergens or triggers (e.g., smoke or occupational irritants) is addressed [1]. Once patients are on appropriate treatment, the goal for patients with severe asthma is still to achieve well-controlled asthma.

Inhaler Devices

Multiple inhaled medication delivery devices are on the market, including metered dose inhalers, dry powder inhalers, soft mist inhalers, and nebulizers. Nebulizers are primarily used in acute care settings for asthma patients.

- (a) Metered dose inhalers: are a pressurized system that delivers aerosolized medication using propellants. Often used with a holding chamber [11].
- (b) Dry powder inhalers: have a dry powder that is breathed in using the patient's own inspiratory force. Many devices fall in this category. Handihaler, Breezehaler, and Aerolizer all have capsules that are punctured prior to inhalation of the dry powder. Turbuhaler, RespiClick, Twisthalers all contain a reservoir of medication. Diskus, Genair, and Ellipta devices have single dose blisters in the inhaler.
- (c) Soft Mist Inhaler have a spring to aerosolize the medication [12]

Asthma Self-Management Education and Correct Inhaler Technique

The need for self-management education is a place where pharmacists can have a great impact by ensuring asthma education, providing and explaining an asthma action plan, providing inhaler teaching, and optimizing inhaler technique. One of the most important factors contributing to asthma treatment success is inhaler administration using the correct technique. Wrong inhaler technique might result in treatment failure secondary to suboptimal delivery of the drug or adverse reactions secondary to incorrect administration instructions. The pharmacist toolkit for assessing and teaching inhaler technique should contain:

- Placebo devices that are used to demonstrate the correct technique with or without a valved holding chamber
- Device instruction sheets in languages that are used in the pharmacist's practice
- A Peak Flow Meter sample for demonstration (used for monitoring, not diagnosis)
- Sizing chart for mask sizes for chambers, if needed
- Copies of patient education booklets, sheets, airway pictures, list of applications for recording medications and/or symptoms

It is important to be familiar with variations in steps for all devices, and these are best learned through the manufacturer's patient package inserts, single page device sheets or videos from reliable sources such as the Lung Association or Asthma Society. Use caution if using resources that are not validated by these organizations or your regional health authority. Some videos on YouTube use alternate incorrect steps even though they are supposedly from "healthcare providers"

The following are examples of resources and tools used for asthma teaching:

- "I Can Control Asthma" website: It contains helpful resources such as an asthma education pamphlet that is available in multiple languages, pediatric asthma action plan and instructional videos. Website address: <https://www.ucalgary.ca/icancontrolasthma/>.
- The Lung Association: If people have difficulty learning to use devices, there are videos on proper inhaler technique available on this website. Web address: <https://www.lungsask.ca/>
- Manufacturers often have good patient education videos or printed device instruction sheets, sometimes in multiple languages.
- Asthma Canada—Breath Easy

It is important to counsel patients that there are often videos available on the internet that might show incorrect inhaler techniques. Only use resources from a reliable organization website.

Follow-Up Assessment

Follow-up assessments are essential to check control given its variable nature and propensity to be aggravated by a myriad of factors (see Table 17.1). Patient progress regarding asthma management can be more easily implemented with regular assessments. Travel, new employment, renovations at home or work and addition of new respiratory irritants such as inhaling cannabis or electronic vaporizer fumes can present risk for poor asthma control. Each interaction a patient has with healthcare providers should trigger a brief follow-up. Patients with persistent poorly controlled asthma should be referred for assessment by a specialist.

Of note, many children under the age of 6 outgrow their symptoms. For children with well-controlled asthma with no exacerbations for 3–6 months but exposed to their usual triggers, a monitored trial off of medication can be attempted [4].

Peak Flow Meter

In asthma, many patients use a peak flow meter, which assesses how quickly they can exhale. Patients should know what their peak expiratory flow (PEF) is when they are not experiencing any symptoms, also known as their personal best. A decrease in a patient's PEF can show a worsening in asthma control. From a pharmacy perspective, patients who have asthma and use a peak flow meter could have their normal value recorded while completing a care plan in order to monitor disease progression. PEF more than or equal to 90% of personal best is one of the indicators of asthma control. Readings from 90–60% (in conjunction with symptoms assessment) may suggest the need to escalate therapy. Not everyone requires a PEF meter, as they cost and require monitoring and recording.

Adverse Reactions

Pharmacists play an important role in monitoring how patients tolerate asthma drug therapy, as this can be a common barrier for nonadherence. Oral thrush is a potential adverse reaction of ICS. Oral thrush can occur in asthma patients especially if they do not rinse, gargle and spit out water after administering the medication. ICS can also cause hoarseness of voice. If the rinses do not help, switching to another device, such as an MDI with a chamber could greatly reduce this adverse effect. In addition, some people experience coughing or throat irritation when taking dry powder inhalers. Adjusting the force of their inhalation can reduce this problem. This will require assistance from the pharmacist to demonstrate and observe the changes made by the individual. This is where having placebo devices is crucial. Table 17.4 summarizes some adverse reactions of asthma pharmacotherapy [5]. Patients requiring systemic corticosteroids (oral or IV) because of frequent exacerbations are at risk of long term adverse effects and thus imperative to reduce risk of exacerbations.

Vaccinations

Routine vaccinations, including influenza and COVID-19, are recommended in patients with asthma. In patients with severe asthma, recent or current wheezing or on high inhaled or oral corticosteroids important to check vaccine monographs to ensure safety, for example, live attenuated influenza virus is contraindicated in this population. Patients with asthma are at higher risk of complications from respiratory illnesses [13]. RSV vaccine is also recommended for those 75 years of age and older with asthma. The National Advisory

Table 17.4 Adverse reactions of asthma pharmacotherapy^a [5, 6]

Drug class	Possible adverse effects
Short-acting beta 2 agonists (SABA)	Tremors, tachycardia, palpitation, nervousness, headache
Short-acting muscarinic antagonists (SAMA)	Dry mouth, metallic taste, upper respiratory tract infection (URTI)
Inhaled corticosteroids (ICS)	Oral thrush, hoarse voice, dysphonia, sore mouth, URTI
Leukotriene receptor antagonists (LTRA)	Headache, abdominal pain
Long-acting beta 2 agonists (LABA)	Headache, pharyngitis
Long-acting muscarinic antagonists (LAMA)	Dry mouth, metallic taste, URTI, headache
Systemic corticosteroids	Taking frequent courses of oral corticosteroid can lead to many adverse effects such as osteoporosis, cataracts, gastric ulcers, sleep difficulties, weight gain, hyperglycemia, and increased complications from respiratory infections

^aThis list is not comprehensive; consult individual drug monographs for a complete list of possible adverse effects

Committee on Immunization (NACI) recommends that asthma control should be optimized prior to any vaccination. In addition, some patients may be on chronic oral corticosteroids and this might affect eligibility for vaccination. Generally, a person who is on less than 20 mg of prednisone for less than 14 days may receive any inactive immunization [14]. Pharmacists need to consult the policies for their health authority and manufacturer's recommendations with regard to vaccination eligibility.

Environmental Impact

Metered dose inhalers use propellants such as hydrofluorocarbon propellants that act as a greenhouse gas [15]. Emissions from 100 doses delivered via MDI are equivalent to a 290 km car ride. MDIs are a common type of inhaler used due to the ability to use with a valve holding chamber which eases administration. Over 95% of the propellant emission occurs when the patient is using the inhaler. It is imperative that patients know how to use their inhaler appropriately. Inhalers should also be disposed of appropriately and brought to designated pharmacies for appropriate recycling. Dry powder inhalers (DPI) have lower carbon footprint than a metered dose inhaler and may be a consideration when prescribing medications.

Clinical Pearls

- Pharmacists play an important role in the management of patients with asthma.
- Self-management strategies that include inhaler teaching, asthma action plan, and self-monitoring plan are essential for patients
- Assessment at follow-up visits includes review of inhalers and patient technique, symptom control, exacerbation risk and events, and adverse events
- Ensure each asthma patient has an Action Plan and knows what to do in case of lack of symptom control.

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Chronic Obstructive Pulmonary Disease Assessment

18

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Chapter Objectives

1. Describe the epidemiology, risk factors and pathophysiology of chronic obstructive pulmonary disease (COPD).
2. Screen patients with symptoms of COPD and refer undiagnosed individuals for diagnostic testing and promptly refer patients with red flag symptoms to the appropriate level of care.
3. Assess patients with COPD for appropriateness of current drug therapy.
4. Describe ongoing management strategies of COPD.

Background

Chronic obstructive pulmonary disease (COPD) is a preventable and treatable condition characterized by progressive air-flow limitation that is not fully reversible and is associated with an abnormal inflammatory response of the lungs to harmful particles or gases [1]. It is a disease with significant burden to patients as well as healthcare systems. Pharmacists play an important role in identifying patients who are at risk for COPD, referring for diagnosis and managing drug therapy. Pharmacists assess drug effectiveness and safety but also assess for adherence and proper use of inhalers, which are the

mainstay of COPD pharmacotherapy. Pharmacists can enhance these patients' quality of life by serving as one of the first points of contact for initial assessment and as accessible, reliable healthcare providers for ongoing follow-up.

Epidemiology

COPD is the third leading cause of death worldwide [2]. It is also the second leading cause of hospitalization in Canada, only behind childbirth [3]. The number of Canadians aged 35 and older with a COPD diagnosis rose by 82% (from 1.1 million to just over 2 million) between 2000 and 2013, driven by the aging population [4]. It will continue to be a significant public health concern due to the historically high smoking rates. Despite the increasing number of patients diagnosed with COPD, it is generally underdiagnosed [2]. Given the potential for inappropriate diagnosis, pharmacists play a key frontline role to identify and refer patients for timely and appropriate diagnosis.

Risk Factors

COPD is a treatable and preventable disease in a vast majority of cases. Early identification of at-risk patients is important because COPD is a persistent and progressive disease. Pharmacists play a significant role in addressing risk factors to limit disease progression.

Risk factors for COPD can be environmental or intrinsic. Environmental factors are primarily cigarette smoking (responsible for at least 75% of cases), occupational exposure (e.g., dust, chemicals), and air pollution. Intrinsic host factors include a rare genetic predisposition called alpha-1-antitrypsin (AAT) deficiency, airway hyper-responsiveness, and impaired lung growth [1]. Inhalation of noxious particles, such as cigarette smoke and environmental pollutants, triggers inflammation in the lungs. This leads to oxidative stress, excessive mucus production, impaired tissue repair,

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and airway remodeling. The cumulative exposure to inhaled particles—the onset of smoking at an earlier age, a higher cumulative exposure measured in pack-years, and continuing to smoke—negatively impacts mortality. E-cigarette aerosols, though newer, are increasingly recognized as harmful to pulmonary cells, potentially contributing to similar inflammatory pathways [5].

Pathophysiology

COPD involves airflow obstruction as a result of airway inflammation. There is narrowing of airways due to mucus production, impaired ciliary function, and smooth muscle thickening. At the alveolar level, gas exchange is impaired because of irreversible enlargement of airspaces, destruction of alveolar walls with fibrosis, and loss of elastic tissue meaning a loss of recoil tension necessary to support expiration. This leads to mucus plugging, air trapping, lung hyperinflation, and progressive ventilation/perfusion inequalities. Over time, destruction of vascular beds, hypoxemia, and carbon dioxide retention exacerbate the disease [1].

Screening

As frontline healthcare providers, pharmacists can be an accessible initial point of contact for patients. Screening for COPD is an important step in assessing patients, especially those who have avoided or have limited experience within the healthcare system.

The following are examples of possible screening questions [6–8]:

1. *Do you currently smoke or have you smoked in the past?*
2. *Have you ever been exposed to secondhand smoke or to other air pollutants such as dust or chemicals?*
3. *How often are you short of breath? Has this seemed to change over time?*
4. *What makes your shortness of breath worse?*
5. *Do you cough or wheeze regularly? How long have you had these symptoms?*
6. *Do you cough up sputum (phlegm, mucus)?*
7. *Do you or anyone in your family have a lung disease?*
8. *Do you get frequent colds that persist longer than those of other people you know?*

Individuals with dyspnea, chronic cough or sputum production, recurrent lower respiratory tract infections, or exposure to previously discussed risk factors—especially those over 40 years of age—should be considered for referral to primary care providers for spirometry testing [2].

Patient History

Additional targeted information that is valuable for COPD assessment includes the following:

- *Social history:* exposure to environmental or occupational risk factors such as smoking.
- *Family history:* note history of COPD or respiratory diseases, including AAT deficiency.
- *Past medical history:* history of exacerbations or hospitalizations for respiratory disorders in the last 12 months. History or active diagnosis of asthma should also be elicited. COPD is often associated with comorbidities like ischemic heart disease, anemia, cachexia, osteoporosis, depression, anxiety, and cancer [1, 2].
- *Symptoms:* Evaluate difficulty sleeping (due to coughing, breathlessness, or sleep apnea), fatigue (from breathing effort, poor oxygen exchange, or polycythemia), and the impact on daily functioning, especially in vulnerable populations. Assess severity of symptoms (see below). Assess anxiety or depression that may be related to other symptoms of COPD.
- *Concerning symptoms:* Watch for weight loss, anorexia (suggesting severe disease or another diagnosis), and ankle swelling (possible right heart failure or cor pulmonale) [2].
- *Red flag signs and symptoms* [1, 2]: Immediate referral is needed for excessive shortness of breath, respiratory rate >24 breaths per minute, chest pain, blood in sputum, and change in mental status (e.g., confused and/or drowsy), peripheral arterial oxygen saturation (SpO₂) <92%, or other signs of hypoxia (i.e., cyanosis).

Diagnosis and Initial Assessment

Spirometry

Spirometry is a common pulmonary function test that measures how much and how quickly air can be expelled from the lungs, distinguishing obstructive (air cannot get out) from restrictive (air cannot get in) lung diseases by assessing lung volumes. Results are compared to predicted values from healthy individuals. The test takes 15–20 min and carries no risk for patients [9].

FEV₁ is the forced expiratory volume in 1 s [1]. FVC is the forced vital capacity, the total volume of air exhaled after maximum exhalation [1]. A postbronchodilator FEV₁/FVC ratio of less than 0.70 confirms obstructive disease and is required for a COPD diagnosis [2]. Limitation of using FEV₁/FVC is that older patients may be overdiagnosed and those of younger age may be underdiagnosed [2], as such

Table 18.1 Classification of airflow obstruction in COPD (FEV₁/FVC <0.7) [1, 2]

Grade	Spirometry
GOLD 1 (mild)	FEV ₁ is ≥80% predicted
GOLD 2 (moderate)	FEV ₁ is 50–79% predicted
GOLD 3 (severe)	FEV ₁ is 30–49% predicted
GOLD 4 (very severe)	FEV ₁ < 30% predicted

FEV₁ post-bronchodilator forced expiratory volume over 1 s, FVC forced vital capacity

there may be diagnosticians that use less than the fifth percentile (LLN). Before spirometry confirmation, clinicians may refer to the condition as “presumed COPD” in situations where the diagnostic tests cannot be performed immediately (i.e., patient having severe symptoms or during admission to hospital). FEV₁ is used at follow up around week 12–16 post exacerbation as it is predictive of all-cause mortality [2]. Degree of airflow obstruction or limitation is categorized as shown in Table 18.1 [1, 2].

COPD Classification

The most commonly agreed upon parameters used for COPD classification are *symptoms* and *frequency of exacerbations*, which are used by both GOLD and CTS [2, 6]. The differences between the GOLD Report and CTS guidelines for COPD classification and treatment recommendations are in the utilization of FEV₁ and blood eosinophil levels [2, 6]. The CTS guideline uses exacerbations, symptoms, and FEV₁ to categorize patients into mild, moderate, and severe categories and does not incorporate blood eosinophils into the classification [6]. The GOLD Strategy uses symptom and exacerbation frequency information to divide patients into ABE classification and also uses *blood eosinophil levels* to determine therapeutic management [2]. Refer to the most updated categorization algorithm parameters set by these organizations. Classifying COPD patients into appropriate categories is useful for pharmacists to help direct initial therapy selection and to continually evaluate the effectiveness of a patient’s current treatments.

Exacerbations

A patient’s exacerbation history is the most heavily considered parameter for determining therapy. The most effective predictor of future exacerbation events is the patient’s history of exacerbations [10]. Patients with a history of moderate to severe exacerbations are at higher risk for future

Table 18.2 COPD exacerbation risk classification [2, 6]

Risk of future COPD exacerbations	Definition
<i>Low risk</i>	0 or 1 moderate exacerbations ^a (not leading to hospitalization)
<i>High risk</i>	≥ 1 leading to hospitalization ^a or ≥ 2 moderate exacerbations ^a (i.e., exacerbation requiring oral steroid and/or oral antibiotics but not leading to hospital admission [2])

^aExacerbation events in 1 year

exacerbations, and subsequent morbidity, disease progression, and mortality [2]. The treatment for patients with high risk of future exacerbations is now escalated sooner and are now treated aggressively regardless of symptom burden [2, 6]. See Table 18.2 below for exacerbation risk classification and associated definitions.

Symptoms and Symptom Severity

Key symptoms in COPD include chronic cough, sputum production, and progressive shortness of breath. When evaluating a patient with a chronic cough, other conditions must be considered, such as asthma, lung cancer, tuberculosis, heart failure bronchiectasis, interstitial lung disease, chronic allergies, postnasal drip syndrome, gastroesophageal reflux, upper airway cough syndrome, medication-induced cough [e.g., from angiotensin converting enzyme (ACE) inhibitors] and idiopathic cough [2].

The tools used to measure symptom severity are the modified Medical Research Council (mMRC) dyspnea scale and the COPD Assessment Test (CATTM).

- The mMRC dyspnea scale classifies patients according to the severity of symptoms which ranges from 0 to 4 (Table 18.3) [2]. The mMRC grade is predictive of mortality risk [2].
- CATTM is a questionnaire addressing the impact of COPD over and above dyspnea. It includes eight statements about symptoms and activities. The patient scores each statement on a scale of 0–5, with a cumulative score of 0–40 [11]. The patient can perform the questionnaire independently [11] which may be useful in a clinic or pharmacist assessment setting. A score change of 2 or more units in 2–3 months suggests a significant change in health status [11]. CATTM can be found at <http://www.catestonline.org> [11] and used if abiding by the Terms of Use [12].

Table 18.3 GOLD mMRC Dyspnea Scoring Scale [2]

GOLD mMRC scoring	Definition
Grade 0	Breathless with strenuous exercise
Grade 1	Shortness of breath when hurrying on the level or walking up a slight hill
Grade 2	Slower than people of the same age on the level or stops for breath when walking at own pace on the level because of breathlessness
Grade 3	Stops for breath after walking about 100 meters or after a few minutes on the level
Grade 4	Too breathless to leave the house or breathless with dressing

GOLD Global Initiative for Chronic Obstructive Lung Disease, *mMRC* modified Medical Research Council dyspnea scale

Table 18.4 Symptom burden classification [2, 6]

Symptom burden	mMRC or CAT TM scores
Low	mMRC 0–1 or CAT TM < 10
Moderate to high	mMRC ≥ 2 or CAT TM ≥ 10

They enable pharmacists to effectively assess symptom burden and subsequently classify patients between low or moderate-severe categories as depicted in Table 18.4 [2, 6].

Other Considerations in Assessment

Physical Examination

Via inspection, pharmacists can observe any of the following findings consistent with COPD: barrel chest, accessory respiratory muscle use, pursed lip breathing, prolonged expiration, cyanosis (bluish discoloration of the skin and lips), tachypnea (respiratory rate > 20 breaths per minute), or signs of being a smoker: smell of cigarette smoke, yellow stained fingers and nails, patients may also appear older than their actual age.

Auscultation might identify breath sounds more commonly associated with COPD like rhonchi, wheezing, or crackles [13]. Auscultation may also reveal diminished breath sounds [13].

Laboratory

Arterial blood gases (ABGs) assess pulmonary function and gas exchange during acute COPD exacerbations but are not typically used for chronic management [1]. Hypoxemia refers to low blood oxygen levels (partial pressure of oxygen, PaO₂), while hypercapnia—high blood carbon dioxide levels (PaCO₂)—indicates hypoventilation and is common in advanced COPD [1, 14]. The normal range of PaO₂ is 80–100 mmHg and PaCO₂ is 35–45 mmHg [14]. Although no specific test measures low tissue oxygenation (hypoxia) directly, chronic hypoxia can trigger increased red blood cell production, leading to polycythemia, evidenced by elevated hematocrit or hemoglobin levels.

Imaging

If available, chest x-ray might show hyperinflation and depressed diaphragm. A computed tomography scan may also show emphysematous changes.

Management

Management involves pharmacologic and nonpharmacologic measures for both prevention and treatment. The goals of COPD therapy are to manage symptoms and reduce risks as described in Table 18.5. Figure 18.1 illustrates the CTS

Table 18.5 COPD goals of therapy [1, 2, 15]

Symptoms	Decrease or alleviate breathlessness and other respiratory symptoms Improve exercise tolerance and daily activity Improve overall health status Support measures for palliative and end-of-life care
Risk	Slow/prevent disease progression with an optimized, tailored inhaler regimen Reduce the frequency and severity of exacerbations Treat exacerbations and complications of the disease Recognize and treat COPD comorbidities (cardiovascular disease, depression, osteoporosis, and smoking-related malignancies) Reduce mortality

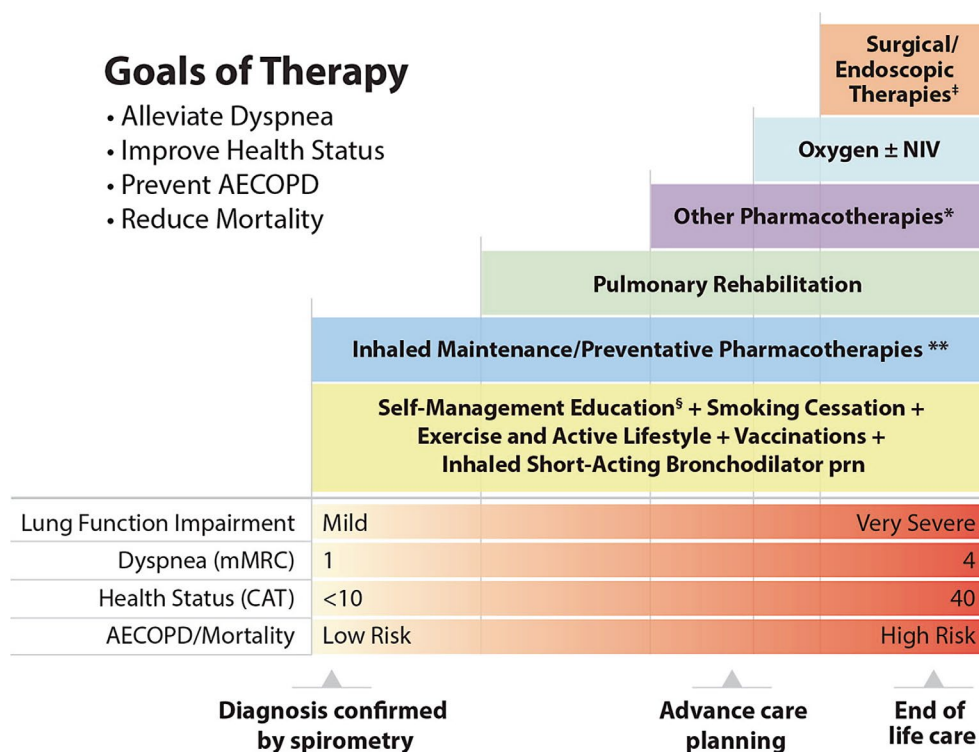


Fig. 18.1 Integrated Comprehensive Management of COPD. Integrated comprehensive management of COPD includes confirming COPD diagnosis with postbronchodilator spirometry, evaluation and on-going monitoring of dyspnea/symptom burden and risk of exacerbations and use of both pharmacologic and nonpharmacologic interventions to alleviate dyspnea/symptoms, improve health status, prevent AECOPD and reduce mortality. The approach should not be viewed as “stepwise” and may not necessarily occur in the order they appear for all patients. Self-Management Education includes optimizing inhaler device technique and [re-]review, assessment and review of medication adherence, breathing and cough techniques, early recognition of AECOPD, written AECOPD action plan and implementation (when appropriate), promoting physical activity and/or exercise, and other healthy habits including diet and smoking cessation. **Inhaled Maintenance/Preventative Pharmacotherapies are long-acting muscarinic antagonists (LAMA)

and/or long-acting beta₂-agonists (LABA) with or without inhaled corticosteroids (ICS). ICS monotherapy should NOT be used in COPD management. *Other pharmacotherapies include oral therapies (prophylactic macrolide, and PDE-4 inhibitor, mucolytic agents for patients with chronic bronchitis), alpha-1-antitrypsin augmentation therapy for documented severe A1AT deficiency, and opioids for severe refractory dyspnea (see prior CTS Guideline). †Surgical therapies may include lung transplantation and lung volume reduction (including with endoscopic valves). A1AT alpha-1 antitrypsin, AECOPD acute exacerbation of COPD, CAT COPD assessment test, COPD chronic obstructive pulmonary disease, CTS Canadian Thoracic Society, mMRC modified Medical Research Council, prn as-needed, NIV noninvasive ventilation. (Reprinted from Bourbeau et al. 2023 [6]. With permission from Elsevier)

COPD Integrated Comprehensive Management method which is not to be viewed as a stepwise approach as it is aimed at addressing impairment, mitigating the risk of adverse clinical outcomes, and providing meaningful clinical benefits [6].

Non-pharmacologic Strategies [1, 2]

- **Smoking cessation:** It is beneficial at any stage of COPD to promote smoking cessation, with or without pharmacologic intervention. Smoking cessation has the greatest ability to improve symptoms, decrease exacerbations, affect the course of progression of disease, and reduce mortality [2]. Pharmacists are in a unique position to play a pivotal role in smoking cessation by utilizing the 5A model (Ask, Advise, Assess, Assist, Arrange) [1, 2]. Counseling performed by health professionals increases quit rates, even 3 min has been shown to improve cessation rates. E-cigarettes or “vaping” are not recommended as smoking cessation strategies [2]. Patients should also be encouraged to avoid smoking other substances, including marijuana, as this exposes the lung to harmful particles [2].
- **Shielding measures** [2]: Encourage masking, minimizing contact with sick individuals, and hand hygiene.
- **Vaccinations** [16]: Follow the latest guidelines from the National Advisory Committee on Immunization (NACI).

All COPD patients should receive pneumococcal, annual influenza, and COVID-19 vaccines if able. The respiratory syncytial virus (RSV) vaccine is advised for older adults, especially if covered by insurance or they are able to afford it.

- *Avoiding certain drugs:* Some drugs can worsen COPD. Antitussives can impair the natural mechanisms to clear secretions, sedating medications (opioids, hypnotics) can decrease the respiratory drive, and first generation antihistamines can thicken mucus, so use these with caution [1]. Recent guidelines state that COPD patients with indications for beta-blocker therapy should not be managed differently solely because of their COPD diagnosis [4, 6].
- *Self-management and education* [2]: Patients should be educated on COPD and proper use of inhaler devices. Education on self-management strategies, like action plans, by qualified healthcare professionals improves quality of life, improves skills, reduces respiratory-related hospital admissions [2].
- *Physical activity* [2]: Exercise should be encouraged as poor endurance is a strong predictor of mortality. Training can be enhanced by optimizing bronchodilators.
- *Other treatments:* *Pulmonary rehabilitation* programs improve dyspnea, health status, and exercise tolerance in stable patients and reduce hospitalization in the first 4 weeks after an exacerbation of COPD [2]. It also improves depression and anxiety [2]. *Supplementary oxygen* improves survival in patients with chronic respiratory failure [2]. Note that pulse oximetry does not provide information on PaCO₂ or pH [2]; patients with PaO₂ < 92% should be referred for additional workup.

Inhaled Medications

Pharmacological management of COPD includes inhaled beta-agonists, muscarinic agonists, and can include inhaled corticosteroids in some patients. There are short acting and long acting of beta-agonists (SABA, LABA) and of muscarinic antagonists (SAMA, LAMA). The term bronchodilator refers to beta-agonists or muscarinic antagonists. Table 18.6 outlines the different classes of inhaled drugs utilized in the management of COPD. There are reliable resources describing inhaled devices and corresponding medication(s). These can be accessed through the following links: [RESPTREC COPD Inhaled Medications](#) [17], [RxFiles Asthma & COPD: Inhalation Devices Chart](#) [18], and [Inhalopédia](#) [19]. GOLD guidelines and the CTS guidelines update recommendations frequently so it is highly encouraged to refer to most recent publications for directing therapy. See Fig. 18.2 for CTS COPD Classification and corresponding treatment recommendations.

Previously, controversy surrounded the use of inhaled corticosteroids (ICS) as add-on treatment as they carry the potential for side effects such as pneumonia and cataracts. Evidence from the IMPACT Trial and ETHOS Trial supports ICS as part of triple therapy (ICS+LABA+LAMA) regimens at a fixed dose due the benefit of preventing moderate to severe exacerbations and hospitalizations and reducing all cause mortality in patients at high risk of exacerbations compared to dual therapy (LABA/LAMA) [20, 21]. In these patients it was seen that the benefits outweigh the risks: in the IMPACT trial the number of patients on triple therapy to cause one episode of pneumonia in 1 year was 33 (NNH = 33), whereas it took 4 patients on triple therapy to prevent one moderate to severe exacerbation in 1 year (NNT = 4) [20], the ETHOS trial needed 7 patients on triple therapy to prevent one moderate to severe exacerbation per year (NNT = 7) [21]. Single-entity triple therapy (SITT) is

Table 18.6 Classes of inhaled drugs used in the management of COPD

Drug class	Drug names
<i>Inhaled beta-agonists</i>	
Short-acting (SABA)	Salbutamol, terbutaline
Long-acting (LABA)	Formoterol, salmeterol
<i>Inhaled muscarinic antagonists</i>	
Short-acting (SAMA)	Ipratropium
Long-acting (LAMA)	Tiotropium, glycopyrronium, aclidinium, umeclidinium
<i>Inhaled combination beta-agonists/muscarinic antagonists</i>	
Short-acting (SABA/SAMA)	Salbutamol/ipratropium
Long-acting (LABA/LAMA)	Formoterol/aclidinium, olodaterol/tiotropium, indacaterol/glycopyrronium, vilanterol/umeclidinium
<i>Inhaled combination beta-agonists/corticosteroids^a</i>	
Long-acting (LABA/ICS)	Salmeterol/fluticasone propionate, formoterol/budesonide, vilanterol/fluticasone furoate
<i>Inhaled triple combination inhalers</i>	
Long-acting (LABA/LAMA/ICS)	Formoterol/glycopyrronium/budesonide, indacaterol/glycopyrronium/mometasone Vilanterol/umeclidinium/fluticasone

^aLABA/ICS dual therapy may be still seen in practice, though LAMA/LABA is recommended instead [2, 6]

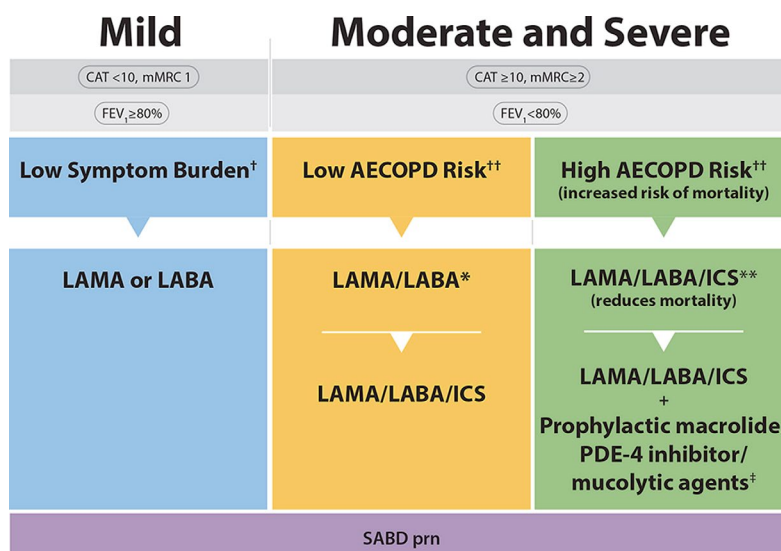


Fig. 18.2 CTS 2023 COPD Pharmacotherapy. This figure promotes an evidence-informed approach that aligns proven effective treatments with spirometry, symptom burden, risk of future exacerbations and mortality risk. Because of the clinical heterogeneity in COPD, spirometry should not be used in isolation to assess disease severity and this is why it is also important to perform a thorough clinical evaluation of the patient, including symptom burden and risk of exacerbations that permits the implementation of treatments that are specific for subpopulations. SABD prn (as needed) should accompany all recommended therapies across the spectrum of COPD. [†]Symptom burden encompasses shortness of breath, activity limitation, and impaired health status. ^{††}Individuals are considered at “Low Risk of AECOPD” if ≤ 1 moderate AECOPD in the last year (moderate AECOPD is an event with prescribed antibiotic and/or oral corticosteroids) and did not require hospital admission/ED visit. Individuals are considered at “High Risk of AECOPD” if ≥ 2 moderate AECOPD or ≥ 1 severe exacerbation in the last year (severe AECOPD is an event requiring hospitalization or ED visit). *LAMA/LABA single inhaled dual therapy is

preferred over ICS/LABA inhaled combination therapy considering the additional improvements in lung function and the lower rates of adverse events such as pneumonia. ICS/LABA combination therapy should be used in individuals with concomitant asthma. There is no universally accepted definition of concomitant asthma. The 2017 CTS Position Statement on COPD Pharmacotherapy provides guidance on the assessment of patients who may have concomitant asthma. **Triple inhaled ICS/LAMA/LABA combination therapy should preferably be administered in a single inhaler triple therapy (SITT), and not in multiple inhalers (see text), although we acknowledge that some patients continue to prefer separate inhalers. +Oral pharmacotherapies in this group include prophylactic macrolide, and PDE-4 inhibitor and mucolytic agents for patients with chronic bronchitis. CAT COPD assessment test, mMRC Modified Medical Research Council, SABD prn short-acting bronchodilator as needed, AECOPD acute exacerbation of COPD, ED emergency department, LAMA long-acting muscarinic antagonist, LABA long-acting β_2 -agonist, ICS inhaled corticosteroid. (Reprinted from Bourbeau et al. 2023 [6]. With permission from Elsevier)

recommended over regimens with multiple devices for COPD as this provides a simplified regimen [6]. Guidelines still recommend against ICS monotherapy for COPD [2, 6]. LABA+ICS dual therapy may have been initiated previously for some patients, though guidelines now suggest that if patients require ICS, triple therapy (ICS + LABA+LAMA) is more appropriate [2, 6]. GOLD guidelines outline an algorithm to follow for patients currently on LABA+ICS [2]. LABA+ICS may be appropriate if the patient has concurrent asthma (or formerly referred to as asthma COPD overlap syndrome); treatment of patients with both COPD and asthma should primarily follow asthma guidelines [2]. When patients are treated with triple therapy, ICS should not be withdrawn unless they experience significant side effects [2, 6]. Patients with serum eosinophils ≥ 300 cells/ μ L have the highest risk of experiencing an increase in exacerbations if ICS is withdrawn as part of de-escalation attempts [2, 6].

Other Pharmacotherapies

Treatment with oral and other therapies for COPD have evolved in recent years, many of them having significant side effects or costs and are usually reserved for patients unresponsive to inhaled therapy. The following are typically prescribed by specialists:

Systemic steroids: Used for moderate to severe exacerbations as they improve lung function (FEV₁) and oxygenation as well as decrease recovery time [2]. Use is limited to the short term due to the broad range of side effects.

Prophylactic azithromycin: The most commonly prescribed oral medication for chronic use. A dose of 250 mg daily or 500 mg given three times per week for 1 year (often prolonged duration) in patients with frequent exacerbations reduced the risk of exacerbations [2]. It works by suppressing pro-inflammatory cytokines, but smoking reduces its effectiveness [1]. Drawbacks include antibiotic resistance and risks such as QTc prolongation and hearing loss.

Phosphodiesterase inhibitors (PDE4i): Roflumilast reduces moderate and severe exacerbations in patients with chronic bronchitis and severe COPD by reducing inflammation, with no bronchodilator effect [2].

N-acetylcysteine (NAC): A mucolytic that can be added for patients to reduce exacerbations. CTS 2023 suggests adding it for patients who exacerbate despite triple therapy with LAM/LABA/ICS [6] and GOLD 2025 discusses its use in patients not receiving ICS to reduce exacerbations and improve health status [2].

Vitamin D: Some evidence suggests that supplementation can help decrease exacerbation rates in patients with deficiency [2]. It is estimated that 32% of Canadians have levels that are considered insufficient (<50 nmol/L) and 10–12% have severe deficiency (<30 nmol/L) [22]. These patients may have comorbidities where vitamin D supplementation is also beneficial.

Dupilumab: A biweekly injection blocking interleukin-4 and -13 receptors. It can be added for patients with blood eosinophils (≥ 300) and experiencing exacerbations despite LABA+LAMA+ICS. For these patients it reduces exacerbations, improves lung function, and overall health [2].

Theophylline: Not commonly prescribed due to a narrow therapeutic window and multiple significant drug interactions.

Ensfentrine: A nebulized dual PDE3/PDE4 inhibitor with anti-inflammatory activity and bronchodilator effects. GOLD guidelines recommend it as an add-on for dyspnea [2] but is not yet available in Canada.

Follow-Up Assessment

To optimize therapy, regular follow-up is essential to monitor efficacy, exacerbations, adverse effects, and, most importantly, adherence. GOLD 2025 has a thorough checklist for practitioners to use during follow up assessment [2].

Adherence

One of the most common problems with inhalers is adherence and appropriate inhaler technique. Incorrect inhaler use is a potential obstacle to achieving COPD control [6]. Patients should be assessed at each refill for *how* and *when* they are using their inhalers to ensure optimal drug therapy [2]. Pharmacists can utilize the teach-back method to evaluate a patient's technique. Common mistakes are incorrect type of inhalation for the device (i.e., slowly breathing in a dried powder inhaler etc.) and not rinsing after using ICS combinations. A good resource with consolidated technique information is the [RxFiles Inhaler Technique](#) document [23]. There are many alternative inhaler devices available

that allow for modification of therapy based on the patient's ability to administer medications [17, 18]. When assessing adherence, consider the patient's manual dexterity to load capsules into inhaler device chambers or ability to twist Respimat devices. Inhalation strength, particularly the inspiratory flow rate, is crucial for the effective delivery of medications via dry powder inhalers (DPIs). Consider coordination of hand movements and breaths when using a metered dose inhaler (MDI), most patients would benefit from the recommendation of a spacer/valved holding chamber (VHC) to ensure optimal drug delivery. Other questions to consider are whether once daily versus twice daily dosing is important (i.e., dry powder inhaler that is once daily or MDI that is twice daily). Some patients may benefit from using only one inhaler type (i.e., MDI for long-acting medications and for short acting medications). There is substantial evidence that favors single-inhaler maintenance therapy over multiple-inhaler regimens as they are possibly more effective, are more convenient, and have better adherence [6].

COPD Control

Signs and Symptoms

Assess if there have been any changes to symptoms from baseline (cough, sputum production, dyspnea, etc.) and severity using the same scale used previously (mMRC, CATTM). COPD is a progressive disease therefore, overtime, it is expected that the patient will require more therapy to control symptoms, especially if underlying modifiable risk factors are not addressed.

Follow Up Pharmacological Treatment

The GOLD initiative and CTS provide recommendations for treatment strategies to address ongoing management of therapy based on the patient's existing regimen as well as the parameters of symptom burden exacerbations as discussed in the COPD Classification section. Due to emerging evidence and evolving algorithms, this chapter encourages readers to become familiarized with the updated recommendations frequently, especially if making therapy suggestions or performing prescribing activities.

COPD Action Plan

Patients with COPD should be provided with a written, individualized action plan [2, 6]. An example can be found at the following link: [My Action Plan](#) [24]. These plans empower patients by teaching self-management strategies and recognizing early signs of acute exacerbations. Action plans

Table 18.7 List of adverse reactions of select drug therapies in patients with COPD [15]

Drug	Common adverse reactions/precautions
Beta-agonists	Tremors, hypokalemia, tachycardia, and palpitations
Muscarinic antagonists	Dry mouth, metallic taste, urinary retention, if released into the eye—glaucoma and mydriasis. Potential to increase risk of cardiovascular events
Inhaled corticosteroids	Oropharyngeal candidiasis, dysphonia, cough ^a Systemic side effects (adrenal insufficiency, osteoporosis) with long-term high-dose usage
Azithromycin	Gastrointestinal intolerance, QTc prolongation
PDE4i (roflumilast)	Nausea, diarrhea, abdominal pain, weight loss, decreased appetite, headache, dizziness, insomnia, anxiety, depression (avoid in patients with history of suicidal ideation)
Oral/systemic corticosteroids	<i>Short-term use:</i> GI upset, hypertension, sodium/water retention, hyperglycemia, nausea, mood changes, insomnia, appetite stimulation, leukocytosis, flushing <i>Long-term use</i> [15, 25]: Cushing's syndrome, adrenal suppression, gastrointestinal ulcerations, osteoporosis, cataracts, weight gain, immune suppression and decreased response to vaccines (>14 days of prednisone >20 mg or equivalent for adults)

^acan be mitigated by using MDI+spacer or rinsing after use. PDE4i phosphodiesterase-4 inhibitors

describe usual daily symptoms (e.g., sputum color, activity level), steps to take if symptoms change for 2 days, and when to seek urgent care. Short counseling on their use, along with ongoing support, reduces in-hospital healthcare utilization but increases antibiotic and corticosteroid use [2]. The main actions or strategies include increasing short-acting bronchodilator(s), providing a prescription for an antibiotic or a short course of oral prednisone (or providing contact information for a prescriber), and possibly adjusting supplemental oxygen (by a qualified professional). Pharmacists can complete, reinforce, and update the drug therapy components of action plans during follow-ups.

Adverse Drug Reactions

Table 18.7 lists the adverse reactions of certain drug therapies in patients with COPD.

Disease Complications

Acute Exacerbation of COPD

The definition of an acute exacerbation of COPD (ECOPD or AECOPD) is an event characterized by increased dyspnea

and/or cough and sputum that worsens in <14 day [2]. A significant proportion of ECOPD are caused by viruses [2, 26].

An ECOPD has been likened to a “lung attack,” where an exacerbation is to COPD what a myocardial infarction is to coronary artery disease [27]. It is an acute event with worsening of symptoms that are beyond the day-to-day normal variation [1]. COPD exacerbations also increase the risk of myocardial infarction, arrhythmias, and stroke, with the highest risk occurring during the first few weeks, but persisting up to 1–2 years [2]. Decompensated heart failure, pneumonia, and pulmonary embolism may mimic an ECOPD and can occur concomitantly. Goals of therapy for ECOPD are to prevent hospitalization, acute respiratory failure, and death and to return symptoms to baseline [1].

Classification

One approach for classifying an exacerbation of COPD is to evaluate using the cardinal symptoms: increased dyspnea, increased sputum volume, and increased sputum purulence [1, 26]. Mild is one cardinal symptom (with one of: upper respiratory infection within 5 days, fever, increased wheezing or cough, or increased RR >20% from baseline), moderate is two cardinal symptoms, and severe is three cardinal symptoms [1]. This information can be readily obtained in an outpatient setting.

The GOLD 2025 guidelines utilize the proposed ROME criteria to classify an exacerbation into three distinct categories according to the following parameters: dyspnea, oxygen saturation, respiratory rate, heart rate, serum C-reactive protein (CRP), and arterial blood gases [2]. Some of these parameters can only be obtained in an inpatient setting; in depth threshold descriptions can be found in the GOLD 2025 Report [2].

Management of Acute Exacerbation of COPD [1, 2]

Treatment of a mild ECOPD involves an increase in short acting bronchodilator therapy (SABD). A moderate exacerbation is treated with SABD, antibiotics [1, 26], and possibly systemic corticosteroids (i.e., oral prednisone 40 mg or equivalent daily for 5 days [2]). Severe ECOPD requires hospitalization and may involve oxygen support or mechanical ventilation for acute respiratory failure. Pharmacologic management includes scheduled short-acting bronchodilators, antibiotics, and systemic corticosteroids [2]. Though there are no clinical trials evaluating long acting inhaled medications during an exacerbation, it is recommended to continue

these treatments during an exacerbation [2]. Refer to local guidelines for antibiotic selection.

Long-Term Complications of COPD

As the disease progresses, and if not managed, the patient may develop secondary pulmonary hypertension, which can result in cor pulmonale (right-sided heart failure) [2], and patients can move on to respiratory failure and death [1].

Palliative and End of Life Care

COPD patients suffer from many elements of disease that impact their life. They are less likely than lung cancer patients to receive help with symptomatic components like fatigue, dyspnea, depression, anxiety and/or insomnia [2]. It is important to support COPD patients with symptom relief, emotional support and quality of life even before end-of-life. Pharmacists can help recognize these elements and help provide care or connect patients with appropriate practitioners. Opioids can be used for refractory dyspnea in end stage disease [6]. Low dose short acting morphine or hydromorphone have been utilized.

COPD and the Environment

Extreme weather events have become more frequent, intense, and widespread due to climate change. COPD patients are at an increased risk of death from exposure to extreme temperatures; extreme cold has more risk than heat for deaths while heat waves are associated with more hospitalizations. These risks are higher in regions with historically temperate climates. Lower temperatures are associated with increased sputum production. Higher temperatures with increased SABA use and decrease in FEV₁ [2]. To prevent morbidity and mortality associated with temperature, patients should maintain bedroom temperatures above 18°C during cold weather and keep living spaces below 32°C and sleeping spaces below 24°C during high temperatures [2].

There are significant differences in the carbon footprint between the inhaled devices. The use of a propellant, device materials, and whether the device may be recycled impact this. Information can be found via this link: [BC Guidelines COPD Medication Table](#) [28]. Dry powder inhalers have lower carbon footprint than a metered dose inhaler but considerations must be made regarding patient's inspiration strength. It is recommended that those without the inspiration strength for DPI, to switch to MDI with a spacer device or soft mist inhaler. Improper inhaler technique and poor

peak inspiratory flow are associated with higher COPD-related healthcare utilization, and subsequent increase in CO₂ footprint from hospitalization [2].

Clinical Pearls

- Pharmacists play an important role in the identification of COPD. Refer patients for diagnosis who have symptoms and risk factors associated with COPD (exposure history, dyspnea, sputum production, cough, and/or recurrent lower respiratory infections).
- Assess therapy regimens, advocate for appropriate changes based on current guidelines, and streamline where possible (i.e., consolidate into a single device vs. multiple inhalers).
- Optimize inhalers based on patient factors such as ability to manipulate inhaler devices, breath coordination, frequency of dosing, and cost.
- Follow-up visits and prescription refills should include assessing inhaler technique and adherence, monitoring symptom control, evaluating recent exacerbations, screening for adverse effects, and updating COPD action plans.

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Chapter Objectives

1. Describe the epidemiology, etiology, risk factors, and pathophysiology of epilepsy.
2. Describe the classification of different seizure types.
3. Conduct initial assessment of patients newly diagnosed with epilepsy.
4. Apply the general principles of antiseizure medications therapy in the follow-up assessment of patients with epilepsy.

Background

Epilepsy is a neurological condition characterized by the occurrence of epileptic seizures. It is important to differentiate seizures from epilepsy. A seizure is “a symptom” of an illness, while epilepsy is a disease. Seizures or spells have been defined by the International League Against Epilepsy (ILAE) as “A transient occurrence of signs and/or symptoms due to abnormal excessive and synchronous neuronal activity in the brain” [1]. Seizures can be provoked secondary to a mechanical or metabolic insult to the brain or can be unprovoked. If a patient experienced more than one unprovoked seizure more than 1 day apart and/or is at high risk of experiencing further unprovoked seizures, he or she is considered to have epilepsy. In addition, patients who have a constellation of symptoms where epileptic seizures are a common occurrence (epilepsy syndromes such as Lennox-Gastaut syndrome) are considered to have epilepsy [2]. Epilepsy affects approximately 50 million individuals worldwide, ranking among the most prevalent neurological disorders globally. While 10% of the population will experience a seizure at some time point, the prevalence of epilepsy ranges from 400 to 1000 per 100,000 population [3]. Around 1% of

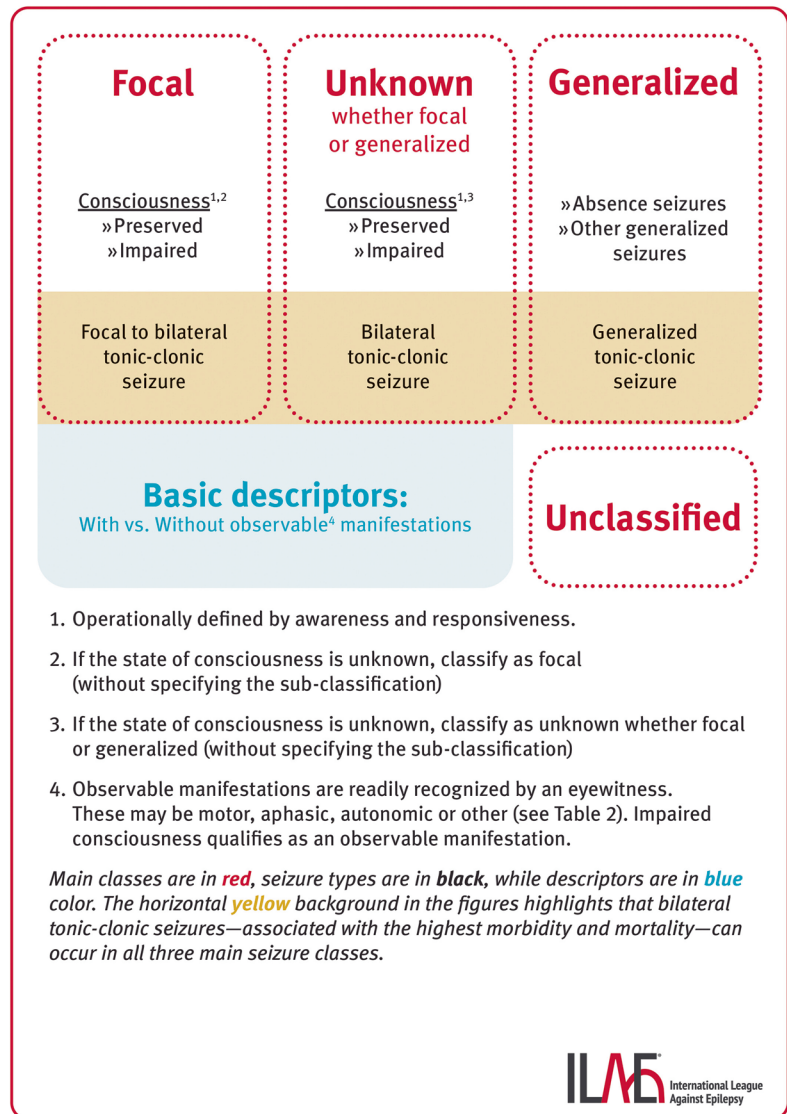
Canadians are diagnosed with epilepsy with an annual incidence of 50–60 per 100,000 population. The highest incidence of epilepsy occurs at ages younger than 10 and older than 65 years [4]. Epilepsy has a great impact on the patients’ quality of life. Patients with epilepsy might have their independence compromised, high unemployment rate, and tend to be socially isolated. For example, patients with uncontrolled seizures lose their driving privileges, unable to swim alone or care for their children by themselves. Current estimates indicate that as many as 70% of individuals with epilepsy may achieve seizure freedom with accurate diagnosis and appropriate treatment. Therefore, pharmacists can play an important role in the management of patients with epilepsy. They are able to assess the patients for the presence of any clinically significant drug interactions, adverse reactions, and red flag symptoms secondary to the antiseizure medications (ASMs). In addition, they aid in the interpretation of ASMs concentrations (therapeutic drug monitoring).

Classification of Seizure Types

According to the 2025, ILAE updated seizure classification, seizures are classified into three main classes: focal, unknown and generalized (Fig. 19.1) [5]. Generalized seizures account for approximately 30% of seizure types, while focal seizures account for 70% of seizures. Compared to generalized seizures, focal seizures (formerly called partial seizures) involve only a certain part of the brain. They are further classified according to the patient’s consciousness and the presence or absence of motor and other symptoms. Based on patient’s consciousness, focal seizures are categorized into focal seizures with preserved consciousness (simple partial seizures) and focal seizures with impaired consciousness (complex partial seizures). In simple partial seizures, patients are usually aware of their surroundings and the seizures typically last for less than 60 s. Those seizures could be motor such as muscle twitching, sensory, and/or an autonomic. On the other hand, complex partial seizures are associated with

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Fig. 19.1 Seizure classifications. (Reprinted from Beniczky et al. [5], under the terms of the Creative Commons CC BY license (John Wiley and Sons) (Copyright 2025))



altered level of consciousness and behavioral arrest. They typically last for 1–2 min followed by postictal confusion. Motor automatism such as chewing movement or lip smacking is a relatively common feature in patients exhibiting complex partial seizures.

In generalized seizures, there is a widespread involvement of the brain, and seizures could be convulsive (motor involvement) or nonconvulsive (non-motor). The most common motor generalized seizure is the generalized tonic-clonic convulsion (GTC, formerly called grand-mal seizure). GTC is typically associated with loss of consciousness with hyperextension of the body followed by rhythmic full body contractions (tonic and clonic phases, respectively). GTC

generally lasts for 1–2 min followed by postictal confusion, stupor, and headache. On the other hand, the most common non-motor generalized seizure is the typical absence seizure (previously known as petit-mal seizure). It is more common in children than adults, and it is characterized by sudden and brief impairment of awareness lasting for several seconds. Myoclonic seizures are another form of generalized seizures that are characterized by sudden brief muscle contractions (milliseconds) without loss of consciousness. Awareness of the patient's seizure type is essential because some drugs are not effective or even may aggravate certain seizure types. For example, phenytoin and carbamazepine are not effective in controlling myoclonic and absence seizures.

Etiology and Risk Factors

A wide range of etiologies have been implicated to cause seizures and epilepsy:

- Genetic causes.
- Traumatic brain injury.
- Central nervous system (CNS) infections, e.g., meningitis and encephalitis.
- Progressive CNS diseases such as brain tumors and Alzheimer disease.
- Metabolic derangements, e.g., hypoglycemia, hyponatremia, and uremia. Because laboratory abnormalities could precipitate seizures, it is important to rule them out while assessing patients presenting with seizures.
- *Drugs and their withdrawal*: Numerous drugs could precipitate or aggravate seizures in patients with epilepsy. These include illicit drugs and drugs prescribed for various illnesses. Pharmacists need to be aware of these drugs when assessing patients presenting with seizures or have a history of epilepsy. Table 19.1 summarizes drugs potentially implicated to lower seizure threshold in susceptible individuals. To illustrate, Fig. 19.2 depicts drugs implicated to have caused seizures in a retrospective study of 386 cases of drug-induced seizures in the state of California, USA [6]. It is worth to mention that bupropion, an antidepressant, was the most commonly impli-

cated drug to precipitate seizures. Therefore, bupropion should be avoided, if possible, in patients with a history of seizures and epilepsy. In addition to drug-induced seizures, other agents could precipitate seizures if withdrawn abruptly such as in the case of benzodiazepines and alcohol withdrawal.

- Additional risk factors could precipitate or aggravate seizures in susceptible individuals such as fatigue, sleep deprivation, flickering lights, and stress. These factors need to be taken into consideration when assessing patients with epilepsy.

Table 19.1 Drugs potentially implicated to lower seizure threshold in susceptible individuals

Drug class	Examples
Antidepressants	Generally, all antidepressants might lower the seizure threshold and should be used with caution in patients with history of seizures and epilepsy Bupropion is highly implicated to aggravate seizures
Antipsychotics	Quetiapine, haloperidol, clozapine
Immunosuppressives	Tacrolimus, cyclosporine
Illicit drugs	Amphetamines, cocaine
Antimicrobial agents	Beta-lactam antibiotics, fluoroquinolones, acyclovir (generally if given at high doses in patients with impaired renal function)
Analgesics	Tramadol, meperidine

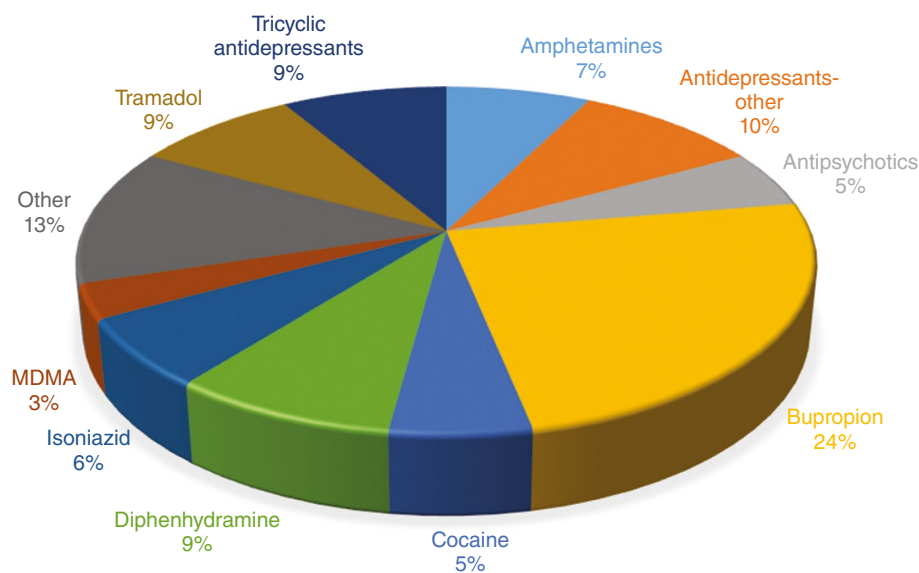


Fig. 19.2 Breakdown of drug-induced seizure by drug type. MDMA, methylenedioxymethamphetamine. (Reprinted by permission from Springer Nature: Thundiyil et al. [6], Copyright 2007)

Diagnosis

History is the most important component in diagnosis. Witnesses are usually asked to describe what happened exactly with the timing, if possible, and if there is any associated postictal confusion. It is suggested not to ask the witnesses to name the seizure as they might be mistaken. In addition to describing the spell, the following could be helpful in assessing patients presenting with seizures:

- Presence or absence of auras.
- Circumstances surrounding the seizure that could have precipitated the event.
- History of drug or alcohol abuse.
- History of childhood epilepsy and childhood illnesses such as meningitis, encephalitis, and febrile seizures.
- History of head injury, stroke, brain tumor, or any systemic conditions that might affect CNS and may precipitate seizures.

Physical examination is often not helpful as it happens after the seizures have subsided. The clinician can look for evidence of tongue biting, incontinence, postictal confusion, and any evidence of neurological symptoms. In addition, a comprehensive lab panel looking for any precipitating factors should be done to rule out seizures secondary to electrolytes and metabolic derangements such as hypoglycemia and hyponatremia. Additional diagnostic modalities, which include electroencephalography (EEG) (however, it is normal in 20% of patients) and imaging such as head computed tomography (CT) and magnetic resonance imaging (MRI) are utilized for differential diagnosis.

Pathophysiology

Normal neuronal activity is usually non-synchronized. In other words, some neurons are inhibited and other are excited during the relay of information throughout the brain. Seizures occur when neurons are activated synchronously. To simplify this, seizures are the result of an imbalance between excitatory and inhibitory mechanisms. Examples of these excitatory mechanisms include activation of the excitatory glutamate receptors and enhanced sodium and calcium influx into the neurons. On the other hand, inhibitory mechanisms involve inhibition or malfunction of the gamma aminobutyric acid (GABA) neurotransmission and potassium channels. This imbalance could be caused by genetic mutations leading to ion channels defect (genetic etiology of epilepsy)

or secondary to an insult to the brain as in the case of traumatic brain injury or stroke. The use of ASMs aims to re-establish the balance between the inhibitory and excitatory transmission by either augmenting inhibitory mechanisms or inhibiting excitatory mechanisms.

Management

Patients with epilepsy should have individualized goals of therapy tailored to fit their seizure type, etiology, comorbidities, personal preference and lifestyle. Ideally, complete seizure freedom is the ultimate goal; however, despite advances in epilepsy management, almost one-third of patients fail to achieve complete seizure freedom. Therefore, in some patients, the goal is reducing the frequency of seizures, by a certain percentage, rather than achieving complete seizure control. A balance between seizure control and adverse effects of ASMs needs to be maintained with an ultimate goal to improve the patient's quality of life. Treatment goals need to be reviewed and revised at every follow-up visit.

Non-pharmacological measures include counseling patients to avoid seizure triggers such as sleep deprivation and heavy alcohol intake. In addition, patients need to be advised to consult their pharmacist or healthcare provider when starting or stopping any over the counter medications and complementary and alternative medicine (CAM) products such as herbal supplements as those might have the propensity to aggravate seizures or interact with the patient's ASMs. Other non-pharmacological management includes the use of ketogenic diet, vagus nerve stimulation and epilepsy surgery.

Pharmacological management typically involves the use of antiseizure medications (ASMs), formerly referred to as antiepileptic drugs (AEDs). These agents aim to re-establish the balance among inhibitory and excitatory mechanisms. They augment the inhibitory mechanisms by enhancing the inhibitory GABA neurotransmission or blocking the excitatory sodium channels, calcium channels, or glutamate receptors. Table 19.2 summarizes the different ASM classes and their recommended adult dosage range. In addition to their classifications based on their mechanism of action, ASMs can be classified into old and new. Old ASMs are generally metabolized by the liver and prone to multiple drug interactions. On the other hand, new ASMs are usually eliminated through the kidneys and tend to have fewer drug interactions; however, there are exceptions, so it is important to consult the specific drug monograph for detailed information about ASM disposition.

Table 19.2 Classes of antiseizure medications (ASMs) used in the management of seizures

Drug class	Individual ASMs and adult dosage range in patients with normal liver and kidney function
Sodium channel blockers	Phenytoin 4–7 mg/kg/day Carbamazepine 400–1200 mg/day Oxcarbazepine 600–2400 mg/day Eslicarbazepine 400–1600 mg/day Lamotrigine 50–500 mg/day Lacosamide 100–600 mg/day Zonisamide 100–600 mg/day
SV2A modulators	Levetiracetam 1000–3000 mg/day Brivaracetam 50–200 mg/day
GABA-A receptor agonists	Clobazam 5–80 mg/day Clonazepam 1.5–20 mg/day Phenobarbital 2–3 mg/kg/day Primidone 250–2000 mg/day
Other GABA modulators	Vigabatrin 1000–3000 mg/day
T-type calcium channel blockers	Ethosuximide 500–1500 mg/day
AMPA receptor blockers	Perampanel 2–12 mg/day
Others/multiple mechanisms of actions	Valproic acid/divalproex 15–60 mg/kg/day Topiramate 200–400 mg/day Gabapentin 900–3600 mg/day Pregabalin 150–600 mg/day Rufinamide 400–3200 mg/day Stiripentol 50 mg/kg/day Cenobamate 200–400 mg/day

AMPA receptor is a glutamate receptor, GABA gamma amino butyric acid, SV2A synaptic vesicle glycoprotein 2A

Initial Assessment of a Patient Newly Diagnosed with Epilepsy

Pharmacists play an important role in the management of patients with epilepsy. They are able to assess the patients for the presence of any clinically significant drug interactions, adverse reactions, and red flag symptoms secondary to the ASMs therapy. In addition, they aid in the interpretation of ASMs concentrations (therapeutic drug monitoring). ASMs are generally recommended in patients who experience two or more unprovoked seizures or experience seizures secondary to brain insults and are at high risk of experiencing further seizures. Similar to other medical conditions, proper initial assessment of patients newly diagnosed with epilepsy requires collection of a complete relevant history (demographics, history of present illness, medical history, social history, medications, adherence concerns, and physical examination findings). Initial assessment includes the following:

- *Assessment of the appropriateness of the selected ASM given the patient's seizure type(s):* Assessment of the appropriateness of the ASM involves making sure that the

Table 19.3 Effectiveness of antiseizure medications (ASMs) in various seizure types

Seizure type	Active ASMs
Focal seizures	Generally, most of the available ASMs are active against focal seizures Phenytoin, phenobarbital, primidone, carbamazepine, eslicarbazepine, oxcarbazepine, lamotrigine, lacosamide, levetiracetam, brivaracetam, clobazam, perampanel, valproic acid/divalproex, vigabatrin, topiramate, gabapentin, cenobamate
Generalized tonic-clonic seizures	Phenytoin, carbamazepine, valproic acid, levetiracetam, lamotrigine, clobazam, topiramate, perampanel
Myoclonic seizures	Valproic acid/divalproex, levetiracetam, lamotrigine, topiramate, clobazam, rufinamide
Absence seizures	Ethosuximide, valproic acid/divalproex, lamotrigine

ASM is effective for the patient's particular seizure type. ASMs are generally effective in many seizure types. However, many exceptions exist. To illustrate, due its unique mechanism of action, ethosuximide is only effective in controlling absence seizures. On the other hand, phenytoin and carbamazepine are ineffective and may aggravate absence and myoclonic seizures. Table 19.3 depicts a general outline of the effectiveness of ASMs in various seizure types. It is important to note that first-line recommendations vary among different guidelines [7].

- *Assessment of the appropriateness of the selected ASM given the patient's characteristics:* Patients' age, sex, comorbidities, concomitant medications, plans for pregnancy, drug coverage, and adherence issues are very important factors that need to be considered in the initial assessment of patients with epilepsy. Table 19.4 summarizes the patients' specific factors that need to be assessed with individual ASMs.
- *Assessment of the appropriateness of the dosage regimen of the selected ASM:* Generally, patients start with an ASM monotherapy at fraction of the target dose to minimize adverse reactions. Then, the dose gets titrated up based on the patient's tolerance and seizure control. The rate of the dose titration depends on the urgency for seizure control. If rapid seizure control is required as in the case of emergencies, some ASM can be started directly at target dose. These include phenytoin, phenobarbital, valproic acid, and levetiracetam. However, if there is no urgency, it is preferable that all ASMs to be titrated slowly. On the other hand, other agents should be titrated slowly in order to avoid adverse reactions and enhance tolerance. For example, carbamazepine needs to be titrated slowly to avoid gastrointestinal adverse reactions and drowsiness. Other examples include topiramate and lamotrigine. While topiramate slow titration potentially helps alleviat-

Table 19.4 Summary of patient-specific factors that need to be taken into consideration with individual antiseizure medications

ASM	Patient factors to be considered
All ASMs	<i>PMH</i> : Caution in patients with hepatic and renal impairment <i>MH</i> : Consult a drug interaction reference with any changes in patient's medications regimen <i>Pregnancy</i> : The lowest effective ASM dose in pregnant women is recommended; monotherapy preferred; valproic acid should be avoided <i>SH</i> : Caution with alcohol intake and to be avoided, if possible. Excessive intake should be avoided
Brivaracetam	<i>PMH</i> : Caution in patients with psychiatric history, hepatic and renal impairment
Carbamazepine	<i>PMH</i> : Caution in patients with hepatic impairment, hyponatremia, and cardiac diseases <i>Race</i> : Asians have increased propensity for SJS. HLA-B*1502 allele screening is recommended to determine the risk <i>MH</i> : It is a liver microsomal enzyme inducer. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM
Clobazam	<i>PMH</i> : Caution in patients with history of drug misuse (dependence risk) and hepatic impairment
Clonazepam	<i>PMH</i> : Caution in patients with history of drug misuse (dependence risk), respiratory problems, hepatic and renal impairment. Avoid in patients with acute narrow angle glaucoma
Eslicarbazepine	<i>PMH</i> : Caution in patients with hepatic and renal impairment. <i>MH</i> : It is a liver microsomal enzyme inducer. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM <i>Coverage</i> : Expensive
Gabapentin	<i>PMH</i> : Caution in patients with renal impairment
Lacosamide	<i>PMH</i> : Caution in patients with cardiac conduction problems (it causes PR interval prolongation), hepatic and renal impairment
Lamotrigine	<i>PMH</i> : Caution in patients with renal and hepatic impairment. <i>MH</i> : Slower titration schedule and lower target dose if the patient is on valproic acid. It has multiple drug interactions; consult drug interaction reference when lamotrigine gets started
Levetiracetam	<i>PMH</i> : Caution in patients with psychiatric illnesses, behavioral problems, and renal impairment
Oxcarbazepine	<i>PMH</i> : Caution in patients with hepatic and renal impairment <i>MH</i> : It is a liver microsomal enzyme inducer. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM <i>Coverage</i> : Expensive
Perampanel	<i>PMH</i> : Caution in patients with hepatic and renal impairment <i>Coverage</i> : Expensive
Phenobarbital	<i>PMH</i> : Caution in patients with hepatic and renal impairment, cardiac disease, diabetes, hyperthyroidism, and severe anemia <i>MH</i> : It is a liver microsomal enzyme inducer. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM
Phenytoin	<i>PMH</i> : Caution in patients with hepatic and renal impairment, cardiac disease, low albumin, and hyperthyroidism <i>Sex</i> : Due to its long-term side effects, it is not a preferred agent for chronic use in females. It causes coarsening of facial features, hirsutism, and gingival hyperplasia <i>Dose titration note</i> : It exhibits nonlinear pharmacokinetics: Dose increases are not associated with proportional increase in drug level. Dose changes should not exceed 50–100 mg/day at a time <i>MH</i> : It is a liver microsomal enzyme inducer. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM
Topiramate	<i>PMH</i> : Caution in patients with history of renal stones, metabolic acidosis, hepatic and renal impairment
Valproic acid/ divalproex sodium	<i>PMH</i> : Caution in patients with hepatic and renal impairment <i>MH</i> : It is a liver microsomal enzyme inhibitor. Always consult a drug interaction reference when it is initiated or a new drug added to existing regimens containing this ASM <i>Pregnancy</i> : Avoid during pregnancy. The least favorable ASM option in women in childbearing period

ASM antiseizure medication, *MH* medication history, *PMH* past medical history, *SJS* Stevens-Johnson syndrome

ing the cognitive impairment associated with its use, lamotrigine slow titration potentially protects against possible dermatological reactions that could be life threatening. If those agents are initiated at full dose, pharmacists might need to flag it to the prescriber for re-assessment.

Follow-Up Assessment

Adherence

Adherence in taking ASMs as prescribed is very essential for proper seizure control. As with other conditions, pharmacists play an important role in promoting patient compliance. All the general factors that affect patient adherence such as knowledge of the disease, regimen simplification, and dose reminder tools apply to patients with epilepsy. One of the factors that could specifically affect adherence in epilepsy patients is the temptation to self-stop the ASM if patient's seizures are well controlled for a while. Patients need to be aware that they should not stop their medications without consulting their neurologist. Abrupt stop of ASM medications might lead to withdrawal seizures and even might lead to status epilepticus, a life-threatening medical emergency. Neurologists might assess titrating down ASMs if the patients are seizure-free for 2 or more years for epilepsy with non-acquired causes or for 6–12 months for epilepsy secondary to acquired causes.

Control

Symptoms

It is important to assess seizure control and determine if the patient's specific goals are achieved, e.g., complete seizure freedom or and 50% reduction in the frequency of seizures. If seizure control is not adequately achieved, other factors need to be assessed before judging therapy failure such as patient adherence, ASM dose adequacy, presence of drug interactions, and presence of possible seizure triggers. In addition, worsening seizure control should prompt the pharmacist to refer the patient to seek medical attention especially if there is not apparent cause for the patient's worsening seizure control. If ASM monotherapy fails to control patient's seizures, another monotherapy could be tried before adding a second agent (ASM polytherapy). This approach will have the advantage of minimizing adverse reactions, avoiding drug interactions, better adherence, and less cost. In some patients, polytherapy might be needed especially if two or three monotherapy alternatives failed to control the patient's seizures.

Laboratory

In specialized practice settings, pharmacists might aid in the therapeutic drug monitoring of ASMs. Many of the ASMs have suggested reference ranges. A reference range is the range of drug concentrations below which the ASM is most probably ineffective, and above which it is most probably toxic. It is important to mention that it is not carved in stone. To illustrate, the drug can be effective at concentrations below the reference range, or toxic at concentrations within the reference range. Reference ranges can be used as a tool, rather than an ultimate target, as it is important to *treat the patient, not the level*. If we can measure a drug concentration, it does not mean that we need to measure it and routine levels are not recommended. Drug levels are usually measured at steady state. However, pre-steady state levels can be beneficial in some situations to determine the adequacy of the dosage. Drug levels need to be requested only if indicated. For example, the suggested reference range for total phenytoin concentration is 40–80 $\mu\text{mol/L}$. If a patient has no adequate seizure control while on phenytoin, measuring phenytoin concentration will be helpful to determine if there is a room to increase the dose. The following are the indications for ASM level:

- Change in seizure frequency.
- Suspected dose-related adverse reactions or toxicity.
- Administration of multiple interacting drugs.
- Checking adherence.
- Formulation and/or route change.
- Conditions of altered pharmacokinetics, e.g., elderly, pregnancy.
- Determination of the individual's therapeutic range.

Adverse Reactions

Pharmacists play an important role in monitoring how patients tolerate drug therapy. Monitoring ASM therapy involves asking questions about adverse drug reactions and checking some laboratory test. ASMs' adverse reactions are divided into acute and chronic. Acute adverse reactions are the ones generally experienced by patients within the first few weeks of therapy. They are further divided into concentration-dependent where dose reduction results in amelioration of the symptoms and idiosyncratic. The latter are not dose-dependent and range from mild to life threatening. On the other hand, chronic adverse reactions are usually apparent with the long-term administration of ASMs. Table 19.5 summarizes the common adverse reactions, lab monitoring, and red flag complication of currently available ASMs.

Table 19.5 Common adverse reactions, lab monitoring, and red flag complications of individual antiseizure medications (ASMs)

ASM	Symptoms and laboratory monitoring	Red flag complications (prompt referral)
All ASMs	<i>ADR</i> : Hypersensitivity reactions, drowsiness, dizziness, fatigue, ataxia, and symptoms of suicidal ideations <i>Lab</i> : LFT, CBC, SCr	DRESS Hypersensitivity reactions Suicidal ideations Hepatotoxicity Signs of ASM toxicity, e.g., excessive sedation, ataxia
Carbamazepine, eslicarbazepine, oxcarbazepine	GI upset, blurred vision, diplopia, behavioral changes Electrolytes; thyroid function Reference range 20–50 µmol/L Periodic eye examination	Bleeding Hematological abnormalities
Clobazam, clonazepam, phenobarbital	Behavioral changes and signs of dependence; changes in respiratory status Phenobarbital reference range 40–170 µmol/L	Signs of dependence Respiratory depression
Lacosamide	Symptoms of heart block (e.g., slow or irregular pulse, headache) ECG in patients at risk of cardiac disorders or on concomitant medications that prolong the PR interval	Symptoms of atrioventricular (AV) block
Lamotrigine	Close monitoring for dermatological reactions	Skin reactions Symptoms of meningitis Hematological abnormalities
Levetiracetam	Behavioral and psychiatric changes; SCr	Behavioral and psychiatric changes
Perampanel	Behavioral and psychiatric changes	Behavioral and psychiatric changes
Phenytoin	Nystagmus, slurred speech, blurred vision, confusion Gingival hyperplasia, hirsutism, acne, coarsening of facial feature Reference range: Total 40–80 µmol/L; free 4–8 µmol/L	Hematological abnormalities
Topiramate	Weight loss; hydration status, sweating changes, or increased body temperature Serum bicarbonate Periodic eye examination	Renal stones Metabolic acidosis
Valproic acid/divalproex sodium	GI upset, tremors, weight gain Motor and cognitive function Serum ammonia Reference range 350–700 µmol/L	Hematological abnormalities Pancreatitis

ADR adverse drug reactions, *ASM* antiseizure medication, *CBC* complete blood count, *DRESS* drug reaction with eosinophilia and systemic symptoms, *LFT* liver function tests, *SCr* serum creatinine

Complications

The main complication of epilepsy is worsening seizure control due to disease progression. In addition, patients might have injuries inflicted while they are seizing. In general, worsening seizure control secondary to no apparent causes should prompt referral.

- Assessment at follow-up involves exploring patient's adherence, seizure control, and the presence of any drug-related adverse reactions or toxicity.
- Pharmacists need to be aware of the rare but serious red flag complications of ASMs such as DRESS syndrome, blood abnormalities, liver failure, and suicidal ideations.

Clinical Pearls

- Pharmacists play an important role in the management of patients with epilepsy.
- Initial assessment of patients with epilepsy involves assessment of the appropriateness of the selected ASM given the patient's seizure type and patient's characteristics and assessment of the appropriateness of the dosage regimen of the selected ASMs.

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Osteoporosis Assessment

20

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Chapter Objectives

1. Assess fracture risk in patients using bone mineral density and risk factors.
2. Identify potential complications from osteoporosis drug therapy.
3. Assess appropriate monitoring parameters in osteoporosis including labwork and imaging.

Background

Osteoporosis is characterized by low bone mass and bone tissue deterioration leading to compromised bone strength and increased risk of fracture. Over 200 million people are affected by osteoporosis worldwide [1]. Osteoporosis affects both women and men, with approximately one in three women and one in five men at risk for an osteoporotic fracture in their lifetime [1]. Osteoporosis can occur from loss of estrogen in menopause and age-related changes in bone or from secondary causes such as certain disease or medications (Tables 20.1 and 20.2). The most common sites for osteoporosis fractures are in the wrist, spine, hip, and ribs. It is estimated that over 80% of all fragility fractures occurring in postmenopausal women are from osteoporosis [2]. A fragility fracture is defined as a low trauma fracture occurring from standing height or less. Fractures are associated with pain, disability, disfigurement, and loss of independence, as well as significant cost to the health care system. Once an individual sustains a fragility fracture, they are at increased

Table 20.1 Risk factors for osteoporosis

Individual characteristics
Age (>50 years old)
Family history (especially parent hip fracture history)
Personal history of fracture after age 40 years
Vertebral fractures
Race/ethnicity (Caucasian, East Asian)
Low body weight or major weight loss
Lifestyle factors
Current smoker
Alcohol intake >3 or more drinks per day
Low calcium intake
Low vitamin D intake or deficiency
Low physical activity
High caffeine intake
Hypogonadal states
Women: Early menopause (<45 years), premature ovarian insufficiency, previous amenorrhea (anorexia, athletic)
Men: Hypogonadism
Endocrine disorders
Hyperparathyroidism
Hyperthyroidism
Cushing's syndrome
Diabetes type 1 and 2
Rheumatologic disorders
Rheumatoid arthritis
Systemic lupus
GI disorders
Celiac
Inflammatory bowel disease
Other malabsorption diseases
Other diseases/disorders
Chronic kidney diseases
COPD
HIV
Malignancies (i.e., multiple myelomas, leukemia, lymphoma)

COPD chronic obstructive pulmonary disease, *GI* gastrointestinal

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Table 20.2 Medications associated with bone loss or fractures

Glucocorticoids (>3 months cumulative dose in past year at prednisone dose of >5 mg daily)
Anticonvulsants
Aromatase inhibitors
Antiandrogen therapy
Excess thyroid replacement
Anticoagulants (long-term heparin therapy)
Chemotherapy
Depo-medroxyprogesterone
Gonadotropin-releasing hormone (GnRH) agonists
Selective serotonin reuptake inhibitors
Proton pump inhibitors
Preformed retinol/vitamin A supplements >10,000 IU
Thiazolidinediones
Antiretrovirals (tenofovir, certain protease inhibitors)

risk for another fracture, with 2.5-fold increased risk of fracturing after a hip fracture and nearly 5-fold risk after vertebral fracture [3, 4]. Fractures are also associated with increased mortality. The risk of mortality in 1 year after a hip fracture in women is 28%, and this risk is even higher in men at 37% [5]. Despite the societal impact of fractures, a care gap with osteoporosis management exists with less than 20% of people who have had a fragility fracture receiving treatment [6].

Pathophysiology

Bone remodeling is a continuous process, which is tightly regulated resulting in a normal balance of bone resorption and bone formation for optimum bone strength. Any imbalance of the normal bone remodeling process can lead to excessive resorption or decreased bone formation leading to inadequately filled bone remodeling pits. If this process continues, this can result in lower bone mineral density and microarchitecture deterioration, leading to increased risk of fracture. The bone remodeling process is complex and is controlled by various hormones (i.e., estrogen, testosterone, parathyroid hormone, calcitonin, etc.), cytokines (i.e., RANKL), and other compounds (i.e., growth factor, vitamin D, prostaglandin, etc.). Osteoblasts produce RANKL (receptor activator of nuclear factor kappaB ligand), a cytokine involved in differentiation, maturation, and activation of osteoclasts. RANKL binds to the receptor RANK found on the osteoclast surface to start the osteoclast differentiation, activation, and bone resorption process. Factors that affect this regulation can lead to an imbalance in normal bone physiology, such as loss of estrogen during menopause, aging, certain conditions, or medications.

Diagnosis

Identifying individuals at risk for further bone loss and fractures and targeting osteoporosis treatment to the individual patient is the key in preventing future fractures. Diagnosis of osteoporosis is established by fracture risk assessment, quantifying risk using validated 10-year fracture risk assessment tools such as FRAX, and bone mineral density measurement with a dual-energy X-ray absorptiometry (DXA). Also, spinal X-rays may be warranted to check for vertebral fractures. Assessment for fracture risk should begin in postmenopausal females and males aged 50 years and older, or consider assessment in younger patients if presence of secondary causes [6]. Osteoporosis is defined as a T-score of 2.5 or more standard deviations below the peak bone mass for young adults, and a T-score of -1 to -2.5 is considered osteopenia (low bone mass). However, treatment decisions should be guided based on 10-year fracture risk assessment and individual patient characteristics (i.e., risk profile, other medical conditions, and preferences) [6].

Management

The goal of therapy is to prevent further bone loss and prevent future fractures. All patients should receive education on their fracture risk, treatment options including preventative measures and osteoporosis medications, and goals of therapy. The treatment plan should be individualized to the patient to ensure adherence given the complexity of many of the medication regimens.

Non-pharmacological Therapy

General preventative measures include ensuring adequate calcium and vitamin D intake, exercise (includes weight bearing, resistance, and balance), smoking cessation, limiting caffeine intake to <400 mg daily (approximately four cups of coffee), and preventing falls. Osteoporosis Canada currently recommends following Health Canada recommendations for dietary allowance for calcium and vitamin D. For calcium Health Canada recommends 1000 mg per day for males over 50 years and 1200 mg of calcium per day for females >50 years and males >70 years. Diet sources of calcium are preferred to supplements, however supplements can be added to meet the dietary amounts. Vitamin D recommendations include 600 IU per day for individuals over 50 years and 800 IU for anyone over 70 years. To meet this require-

ment it is recommended that all adults over the age of 50 take a minimum 400 IU vitamin D along with dietary amounts of vitamin D [6]. Hip protectors have been shown to reduce hip fractures in patients in long-term care facilities [6]. Any secondary causes of osteoporosis should also be identified and treated before starting on osteoporosis treatment.

Pharmacological Options

There are a number of pharmacologic options to treat osteoporosis (Table 20.3). Most options available are antiresorptive agents including bisphosphonates (alendronate, risedronate, zoledronic acid [ZA]), RANK-Ligand inhibitor (denosumab), selective estrogen modulator (raloxifene), and menopausal hormone therapy (MHT). There are two anabolic agents available in Canada: recombinant parathyroid hormone (teriparatide) and a sclerostin inhibitor (romosozumab). All of these agents have evidence of vertebral fracture reduction, while most, except for raloxifene, also have non-vertebral and hip fracture reduction in randomized control trials [6]. Some studies have also shown a reduced mortality rates in patients at high risk of fractures. There are a range of formulations, routes of administration, and dosing regimens to choose from, allowing for individualization to enhance adherence. Osteoporosis Canada recommends the following pharmacologic interventions to initiate therapy for preventing fractures, for most patients (6):

- Bisphosphonates (alendronate, risedronate, ZA) for people meeting criteria for initiation of pharmacotherapy. Bisphosphonates initial treatment should be 3–6 years and then therapy stopped for a drug holiday (see under drug holiday).
- Menopausal hormone therapy for postmenopausal females aged less than 60 years or less than 10 years since menopause who need menopause symptom relief.
- Denosumab in people who have contraindications to bisphosphonates, do not tolerate bisphosphonate or have issues with access. Denosumab therapy should be long term and it is important to not delay dosing for more than 1 month due to increase in bone loss and vertebral fractures with discontinuation. Denosumab should be switched to an antiresorptive when discontinuing.
- Raloxifene for postmenopausal women who have contraindications to other osteoporosis medications or do not tolerate or choose not to use another agent.

The Osteoporosis Canada guidelines also recommend to seek advice from a menopause expert to consider anabolic therapy in people who have a had a severe recent vertebral fracture (within the last 2 years) or more than one vertebral fracture and with T-scores ≤ -2.5 or less. Anabolics should be transitioned to an antiresorptive therapy when discontinuing (2 years for teriparatide, 1 year for Romosozumab).

Pharmacologic therapy is recommended in all individuals who have a 10 year fracture risk of $\geq 20\%$ or if they have a T-score of ≤ -2.5 and are aged 70 years and over as these individuals have the largest benefit with fracture reduction. Individuals with a previous hip or spine fracture or 2 or more fractures should also be started on pharmacotherapy. FRAX with BMD is the preferred method to calculate 10 year fracture risk for determining pharmacotherapy initiation. For individuals with 10 year fracture risk of 15–19.9% or with T-scores ≤ -2.5 and less than 70 years, these individuals are considered to have an intermediate benefit and pharmacotherapy can be suggested.

Table 20.3 Classes of drugs used in the management of osteoporosis

Drug class	Drug	Dose
<i>Antiresorptive agents</i>		
Bisphosphonates	Alendronate	70 mg po once weekly ^a 10 mg po daily
	Risedronate	35 mg po once weekly ^a 5 mg po daily 150 mg po monthly
	Zoledronic acid	5 mg intravenous infusion once yearly
RANK ligand inhibitor	Denosumab	60 mg subcutaneous injection every 6 months
Selective estrogen receptor modulator	Raloxifene	60 mg po daily
Hormone therapy	Various estrogen products	Conjugated estrogen 0.3–0.625 mg daily, oral 17 beta-estradiol 0.5–1 mg, 25–50 µg patch Transdermal gel 1–2 pumps
<i>Anabolic agent</i>		
Recombinant parathyroid hormone	Teriparatide	20 µg subcutaneous daily
Sclerostin inhibitor	Romosozumab	210 mg subcutaneous injection once monthly

^aMost common dose

Initial Assessment of a Patient

The initial assessment of patients diagnosed with osteoporosis and those people at risk of osteoporosis is similar. Pharmacists play an important role in the screening and management of these patients. Pharmacists are in an ideal position to screen patients who are at high risk for fractures and provide tailored patient education on osteoporosis risk factors, preventative measures, and medication options, as well as recommend follow-up with their primary care provider for bone mineral density testing [7]. Pharmacists can assess medication appropriateness for patients on osteoporosis

medications by assessing drug interactions, medical contraindications, adverse reactions, and long-term risks with medications. Furthermore, they can monitor and support patients with adherence to osteoporosis medication or if experiencing adverse effects. As with any other chronic medical condition, initial assessment of patients begins with a collection of complete relevant history (demographics, history of present illness, medical history, social history, medications (prescription and nonprescription), social history, laboratory, and physical exam findings). Additionally, fracture history, historical height loss, and history of falls should be collected.

- *Assessment of patient for fracture risk (not yet diagnosed with osteoporosis):* All postmenopausal females and males over the age of 50 years should be screened for risk factors for bone loss or fractures. The purpose of risk factor assessment is to identify patients who may be at risk for fractures, as well as who should be assessed for bone mineral density with DXA. Risk factor assessment includes capturing all of the following information:
 - Personal fracture history—assess if fragility fracture (low vs. high trauma fractures); type of fracture to determine if related to osteoporosis.
 - Medication history (focus on medications that are linked with bone loss or fractures such as glucocorticoids, aromatase inhibitors, anticonvulsants, diuretics, and others—see Table 20.2).
 - Family history of osteoporosis (parental history of hip fracture).
 - Smoking status, alcohol intake.
 - Menstrual status in women (assess age of menopause to determine if early menopause or postmenopausal).
 - Medical conditions that may lead to bone loss or fractures (i.e., RA, hypogonadism, chronic renal failure, GI malabsorption [e.g., celiac, IBD], hyperparathyroidism, and others—see Table 20.1).
 - Assess for other potential causes of low bone mass (e.g., history of anorexia).

Osteoporosis Canada recommendations for who should get a BMD include anyone 70 years of age and over. For anyone younger than 70 years they should get a BMD depending on the presence of risk factors (age 65–69 years with 1 risk factor and age 50–64 years with 2 or more risk factors)0.70.

The two Fracture Risk Assessment tools used in Canada are the Fracture Risk Assessment (FRAX) created by the

World Health Organization and the Canadian Association of Radiologists and Osteoporosis Canada (CAROC) . The CAROC uses BMD (femoral neck T-score), age, sex, fragility fracture, and glucocorticoid use to stratify patients into low (<10%), moderate (10–20%), and high (>20%) risk of fracturing within the next 10 years. FRAX also includes these risk factors but also captures BMI, family history (parental hip fracture), current smoker, alcohol intake, and secondary osteoporosis. The inclusion of BMD with the FRAX is optional. The FRAX is the preferred tool to use in Canada

- *Assessment of the appropriateness of the selected osteoporosis medication given the patient's future fracture risk.* Assessment of the appropriateness of an osteoporosis medication involves an understanding of the patient's risk of fracturing within the next 10 years using a Fracture Risk Assessment Tools (FRAX with BMD preferred), BMD results, history of fragility fractures, historical or prospective height loss, fall risk, and osteoporosis risk factors (medical conditions, medications that can lead to further bone loss). In some patients, especially if reported height loss or back pain, vertebral imaging may be required to diagnose vertebral fractures. Additionally, it requires an understanding of the evidence for fracture risk reduction (vertebral, hip, and non-vertebral fractures) for each of the pharmacologic options (see Table 20.3). Decisions on who to treat are as follows:
 - Largest benefit of treatment:* defined as individuals with 10 year fracture risk $\geq 20\%$ or if 70 years and greater and the T-score ≤ -2.5 or if previous hip or spine fracture or if multiple fractures (2 or more fractures). Recommend pharmacotherapy.
 - Intermediate benefit of treatment:* defined as individuals with 10 year fracture risk of 15–19.9% or if less than 70 years and the Tscore ≤ -2.5 . Suggest pharmacotherapy. .
 - Do not recommend pharmacotherapy:* in individuals with 10 year fracture risk of <15% or Tscores greater than -2.5 . <15%
No pharmacologic therapy is required. General preventative measures should be considered.
- *Assessment of the appropriateness of the selected osteoporosis medication given the patient's characteristics.* A detailed patient history, focusing on the clinical risk factors for bone loss or fractures, is important to identify future fracture risk as well as to help choose among the therapy options. In addition, risk factor for falls is impor-

Table 20.4 Patient considerations when initiating therapy

Medication	Factors to consider
Oral bisphosphonates	No issues swallowing or having medications get “stuck” in their throat Adequate renal function (calculated CrCl >30 mL/min) Need to space from all food and drinks (other than plain water) for 30 min after taking dose Previous major jaw surgery and/or jaw radiation (if yes, then should be avoided)
IV zoledronic acid	Adequate hydration prior to administration Adequate renal function (calculated CrCl >30 mL/min)
Denosumab	Willingness to have injections
Teriparatide	Willingness to inject on a daily basis Cost 2-year maximum lifetime usage
Romosozumab	Willingness to have injections No history of CV disease or stroke Cost 1-year maximum lifetime usage
SERMs	No history of VTE
HT	No history of VTE

HT hormone therapy, SERMs selective estrogen receptor modulators, VTE venous thromboembolism

tant to capture, especially modifiable risk factors which can be addressed. Patient's preference for therapy, including routes of administration and dosing regimens (i.e., once weekly vs. every 6 months vs. once a year), should be considered. Comorbidities, concomitant medications, compliance issues, and drug coverage should be reviewed to rule out contraindications, potential drug interactions, and future adherence problems. Table 20.4 outlines patient-specific factors to consider when starting on specific osteoporosis medications.

- *Assessment of the appropriateness of the selected osteoporosis medication based on potential issues that may occur with each medication.* When initiating an osteoporosis medication, also consider the potential for adverse effects with specific agents in an individual patient, including long-term risks. For example, capturing swallowing difficulties or esophageal dysmotility in patients can help decide among the options (e.g., alendronate is associated with esophagitis). Bisphosphonates and denosumab have been associated with a risk of osteonecrosis of the jaw and atypical fractures with long-term use, though these events are extremely rare, it may be helpful to identify patients at increased risk such as patients with a history of major or planned dental surgery. Table 20.5 includes a list of adverse effects with osteoporosis medications.

Table 20.5 List of most common drug complications in patients with osteoporosis

Drug	Common adverse reactions and laboratory monitoring/precautions	Red flag complications
Bisphosphonates (alendronate, risedronate, zoledronic acid)	Upper GI symptoms (monitor weight loss if severe GI symptoms) Muscle aches/pains Hypocalcemia—Monitor calcium and albumin Infusion-related complications (only ZA) Flu-like symptoms, bone/muscle pain Renal toxicity (only ZA IV)	Atypical femoral fracture—Sudden onset of thigh pain Osteonecrosis of the jaw—Inform dentist prior to major dental work
Denosumab	Dermatologic reactions (eczema, rash) Muscle pain Theoretical risk of increase in infections	Atypical femoral fracture—Sudden onset of thigh pain Osteonecrosis of the jaw—Inform dentist prior to major dental work
Teriparatide	Orthostasis after injection—Sit or lay down on first administration Leg cramps Headache, dizziness Hypercalcemia—Monitor calcium and albumin	Theoretical risk of osteosarcoma
Romosozumab	Joint pain Headache Hypocalcemia	CV disease Stroke
SERMs	Hot flushes, sweating Leg cramps Peripheral edema	VTE (increased risk in people with previous history of VTE)
HT	Breast tenderness Bloating, water retention Headache	VTE (increased risk in people with previous history of VTE)

HT hormone therapy, SERMs selective estrogen receptor modulators, VTE venous thromboembolism, ZA zoledronic acid

Physical Assessment Skills

Physical assessment can be used to determine the extent of osteoporosis deformities (i.e., kyphosis), as well as assessing for vertebral fractures. The assessment of height loss and postural changes is important in the initial examination of a patient. Proper technique for measuring height is key; a wall-

mounted stadiometer is preferred over other measurement devices. Patients should stand in their bare feet with heels touching the wall, as well as the back or buttocks, while looking straight ahead with their chin horizontal to the ground. This measurement is sometimes difficult to do if a patient is significantly kyphotic, as they may not be able to get their back or buttocks to touch the wall. Measurement of height and height loss is a red flag for a potential vertebral fracture. Current height can be compared to the person's height at the age of 20; if there has been a 4 cm or greater loss of height, further investigation is warranted, such as a lumbar and thoracic spine X-ray to determine if there are vertebral fractures. If the patient is uncertain of their height at age 20, any recent documentation of height can be used; a loss of 2 cm or more would indicate further investigation is warranted.

For assessment of vertebral fractures, other physical exam techniques are included below:

- Occiput to wall distance can be measured as well. This is an indicator of potential thoracic vertebral fractures. Have the patient stand with their ankles and buttocks touching the wall, and keeping their chin horizontal, then have them try to touch the back of their head to the wall. Ideally, this distance should be 0 cm, but a distance of 5 cm or greater may indicate thoracic spine fractures.
- Rib–pelvis distance is another measure of kyphosis. Have the patient stand facing away from you with their arms outstretched. The space between the bottom of the ribs and the top of the pelvis should be at least two finger-breadths apart.
- The spine can also be percussed by thumping a closed fist along the midline. This should be done relatively gently. Pain that is localized to the midline of the spine is also indicative of potential vertebral fractures.

To coincide with assessment of osteoporosis, assessment of falls can be useful in determining which patients are at increased risk of fracture. The most predictive factor for someone having a future fall is if they have had a fall in the past. Other risk factors include polypharmacy, certain medications (benzodiazepines, diuretics, psychotropics), use of a gait aid, and cognitive impairment. The Timed Up and Go (TUG) test is one quick and easy to administer assessment that can be used as a fall screen. To administer the TUG test: (1) have the person stand up from the chair, (2) walk 3 meters at normal pace, (3) turn, (4) walk back to the chair at normal pace, and (5) sit down again. If it takes the person more than 12 s to complete this test, they are considered at higher risk of a fall.

Follow-Up Assessments

Adherence

Bisphosphonates are generally associated with a high rate of noncompliance due to many factors such as: osteoporosis is a symptomless condition, administration and spacing of doses from other medications/food can be difficult to manage, and patients may not remember to take extended form dosing (i.e., once weekly). Compliance and appropriate administration (spacing of medications from food, etc.) of bisphosphonates should be reinforced at refills. Bisphosphonate compliance is improved with once weekly or once monthly dosing; however, rates still remain quite poor. For the medications administered less frequently, such as denosumab and zoledronic acid, patients may need reminders that a dose is due.

Control–Efficacy

Signs and Symptoms

In follow-up monitoring for osteoporosis, patients should be asked about any new fractures (i.e., while on therapy) or if they have new onset of back pain. Height should again be measured to determine if any height loss has occurred since the previous visit. Any new fractures since the last visit should be assessed, as well as a determination of whether the fracture was a fragility fracture. A fracture while on therapy, if the patient has been adherent, may be considered a failure to therapy. This may mean the patient requires a change in therapy, depending on a thorough assessment of adherence and risk factors. In addition, a reassessment of risk factors should occur, and a subsequent re-calculation of FRAX and/or CAROC to determine if the patient's risk status has changed. An increase in risk may mean reassessment of therapy, for example, a decrease in fracture risk (moving from moderate to low) could mean therapy can be discontinued. Patients may lower their risk by quitting smoking, lowering alcohol intake, and being off corticosteroid therapy.

Laboratory

Renal function should be reassessed at follow-up as well, especially if the patient is at risk of poor renal clearance of bisphosphonates. Renal function should be calculated prior to a dose of zoledronic acid being administered as well. Other lab work such as calcium and albumin can be reviewed, but not necessarily needed if patient is doing well and has no other comorbidities. Markers of bone turnover (NTX, CTX) are not generally recommended for routine monitoring of efficacy of therapy.

Imaging

In general, after the start of therapy, you can repeat a DXA in 1–3 years. For higher-risk patients, especially those on high-dose corticosteroids, a DXA may be done after 1 year to ensure response is adequate and to determine if the BMD is dropping over time. A change in BMD can be reported with the diagnostic report, and this is most accurate if the patient continues to have their BMD measured with the same device; therefore, it is recommended when possible that patients have their follow-up DXA scans done at the same clinic as the previous scan. Thoracic and lumbar spine X-rays can be done at the same time as a follow-up DXA or if the patient develops new back pain as vertebral fractures can be silent and the patient may not have any symptoms.

Adverse Drug Reactions

As a key part of assessment, pharmacists should determine how patients are able to tolerate the medications prescribed for their osteoporosis. Adverse drug reactions or side effects can lead to noncompliance with medications and put the patient at risk of fractures. Table 20.5 shows some of the commonly occurring adverse reactions with osteoporotic medications as well as red-flag complications that occur rarely, but patients should be advised of when starting these medications.

Drug Holidays

There is controversy surrounding when to stop bisphosphonates in patients at mild to moderate risk of fracture. Long-term use of bisphosphonates is associated with a higher risk of atypical femoral fracture and osteonecrosis of the jaw, and as such, long-term therapy should be re-evaluated approximately every 3–6 years after initiation of bisphosphonate therapy. In patients with a lower fracture risk, they can be reassessed after 3 years, however, those with a higher risk should likely stay on for longer. Discontinuation of denosumab can lead to an abrupt decline in a patient's BMD and a potential increase in fracture risk. As such, drug holidays are generally not recommended with denosumab therapy without a plan for sequential therapy with another osteoporosis medication. If a patient sustains an atypical femoral fracture while on either bisphosphonate or denosumab therapy,

continued treatment with anabolic agents should be considered, dependent on their calculated fracture risk.

Disease Complications

The major complications related to osteoporosis are fractures. These fractures occur in bone that is primarily trabecular bone, such as the vertebrae and the hips. Once people suffer from one fragility fracture, they are at a much higher risk of future fractures—this is termed the fragility cascade; having one fracture predicts the chance of a person having a future fracture. If a patient suffers a fracture while on appropriate therapy, the first consideration is if the patient has been adherent. If they have not been adherent to therapy, a discussion of how to improve adherence would be beneficial or a consideration of changing therapy to something that is easier for the patient to manage should be discussed. For example, patients who find it difficult to separate doses of oral bisphosphonates from other medications and/or food may find that IV zoledronic acid once a year is easier to manage. If the patient was adherent to oral bisphosphonate therapy, and spacing it appropriately from food and calcium, a change in therapy is likely warranted after a fragility fracture. Moving from a bisphosphonate to denosumab is definitely a consideration at this time. The use of anabolics is limited mainly due to cost but can be considered in very fragile patients who are considered to suffer fragility fractures while on bisphosphonate or denosumab therapy.

Clinical Pearls

- Assessment of people at risk of osteoporosis and those diagnosed with osteoporosis should include an assessment of risk factors, including medications that may contribute to bone loss and/or increase fracture risk.
- BMD is important both for initial diagnosis and assessment of risk.
- Vertebral fractures should be screened for at follow-up visits because patients may have one without knowing.
- Monitoring patients at follow-up should include an assessment of adverse effects from osteoporosis medications, as well as an adherence check.

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Chapter Objectives

1. Describe the epidemiology, etiology, clinical presentation, and diagnosis of rheumatoid arthritis.
2. Describe the goals of therapy and the management strategy for rheumatoid arthritis.
3. Conduct an initial assessment of a patient newly diagnosed with rheumatoid arthritis.
4. Conduct a follow-up assessment of a patient on disease-modifying antirheumatic drug therapy, considering the regimen's effectiveness and safety and the patient's ability to adhere.

Background

Epidemiology and Etiology

Rheumatoid arthritis (RA) is a chronic autoimmune disease characterized by inflammation, pain, stiffness, and progressive joint destruction. RA is the most common inflammatory arthritis, affecting approximately 1 out of every 100 Canadians. It can occur at any age, with increasing prevalence between 40 and 60 years of age, and affects women two to three times more than men. Indigenous peoples are at a greater risk of developing RA and often have a more serious disease course, due to various factors, including systemic racism, intergenerational trauma, and geographical challenges that impact the care they receive [1, 2]. Individuals with RA are more likely to miss work and have difficulties maintaining employment, which often results in substantial productivity losses that impact quality of life.

The exact cause of autoimmune inflammation associated with RA is unknown; however, it is thought to result from a complex interaction between genetic and environmental risk factors. Cigarette smoking increases disease susceptibility and may increase the risk of greater disease severity. Certain infections (viral and bacterial) have also been thought to trigger RA, but no single agent has proven to be responsible.

Pathophysiology

In RA, the immune system can no longer differentiate self from nonself (foreign) tissues and attacks the synovial membrane, the layer of tissue that lines joints and secretes synovial fluid. The synovium becomes thickened and inflamed forming pannus (a layer of granulation tissue), which invades and destroys the cartilage and eventually the surface of the bone leading to joint destruction. While the factors that initiate the inflammatory response are unknown, the activation of T cells and B cells, along with the production of pro-inflammatory cytokines, leads to progressive bone breakdown and joint destruction. Some patients with RA form antibodies called rheumatoid factor (RF) and anti-citrullinated protein antibody (ACPA). RF is positive in 60–70% of patients with RA, and seropositive patients tend to have a more aggressive disease. High levels of ACPA are also indicative of aggressive disease and poorer outcomes; measurement is useful in the differential diagnosis of early polyarthritis, because of the relatively high specificity for RA, but is not useful for ongoing monitoring of disease progression.

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Clinical Presentation

The symptoms and course of RA varies from patient to patient and are characterized by periods of flares, varying in length and severity, which interrupt stable periods of low disease activity or remission. The majority of patients develop symptoms gradually, with the predominant symptoms being joint pain, stiffness, and swelling. Morning stiffness greater than 1 h is characteristic of RA. Typically, the metacarpophalangeal (MCP), proximal interphalangeal (PIP), metatarsophalangeal (MTP), and wrist joints are affected early in the disease; however, RA may affect almost any joint in the body with the exception of the thoracic and lumbar spine, the distal interphalangeal (DIP) joints, and the first carpometacarpal (CMC) joint.

Patients may present with or develop a number of extra-articular manifestations. Nonspecific symptoms might include fatigue, weakness, and anorexia. A number of patients develop rheumatoid nodules (non-painful, pea- to mothball-sized subcutaneous nodules on pressure points or in the lungs), pleuritis, pleural effusions, and vasculitis. Sjögren's syndrome associated with RA leads to a combination of dry eye (keratoconjunctivitis sicca) and dry mouth. Rheumatoid arthritis in association with splenomegaly and neutropenia is known as Felty's syndrome.

Patients may also have laboratory abnormalities. Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP), nonspecific markers of inflammation, are commonly elevated and patients may experience anemia of chronic disease. When aspirated, synovial fluid is often thickened and cloudy. Erosions will be seen on radiographs at the first visit in 20% of patients and will be present in up to 70% of patients at 1 year if left untreated.

Osteoarthritis and RA are the most common types of arthritic disorders, but they differ significantly in presentation (Table 21.1). Since management of the two conditions also differs significantly, early evaluation and diagnosis are essential to optimize patient care.

Table 21.1 Comparison of rheumatoid arthritis and osteoarthritis

Characteristic	Rheumatoid arthritis	Osteoarthritis
Joint pain	With activity and at rest	With activity
Joint symptoms	Pain, swelling, warmth, stiffness	Pain, bony enlargement
Joint pattern	Symmetric	Asymmetric
Morning stiffness	≥60 min	<30 min
Systemic symptoms	Common, especially fatigue	Absent
Acute phase reactants (CRP, ESR)	Elevated	Normal

CRP C-reactive protein, ESR erythrocyte sedimentation rate

Diagnosis

The diagnosis of RA should be considered in patients with persistent bilateral swelling or inflammatory pain in multiple joints. Criteria revised and developed in 2010 note a diagnosis of “definite RA” based on the confirmed presence of synovitis in at least one joint, absence of an alternative diagnosis better explaining the synovitis, and achievement of a total score of 6 or greater (of a possible 10) from individual scores in four domains: number and site of involved joints (range 0–5), serological abnormality (range 0–3), elevated acute-phase response (range 0–1), and symptom duration (two levels; range 0–1) [3]. The new criteria are intended to help identify patients earlier in the course of disease so that treatment may be initiated as soon as possible. Where there is diagnostic doubt, x-ray, ultrasound, or magnetic resonance imaging (MRI) can be used to improve the certainty above clinical criteria alone [4].

Prognosis

Clinical features associated with poor prognosis include early functional limitations, extra-articular disease, positive RF or ACPA, and early erosions on x-ray. However, clinical outcomes in RA have improved significantly with changes in drug therapy and treatment approach. Outcomes are dependent on a number of factors: degree of disease activity, pre-existing joint damage, psychological health, and comorbidities. Left untreated, inflammation and joint damage can readily lead to functional disability that can impact mobility and ability to work, resulting in significant financial and productivity losses, as well as emotional and social burdens affecting quality of life.

Patients with RA are at increased risk of serious comorbidities. Half of all deaths are cardiovascular-related and patients with poor disease control are at increased risk of lymphoproliferative disorders (e.g., lymphoma, leukemia, and multiple myeloma) and lung cancer. Infections (especially upper and lower respiratory), osteoporosis, depression, and fibromyalgia also occur more frequently. To date, patients with RA have experienced reduced life expectancy and although recent Canadian evidence suggests that the risk of death in patients with RA is improving over time, it still remains elevated compared to the general population (40–50% more deaths) [5].

Management

Optimal management of RA requires rapid and sustained suppression of inflammation with immunomodulatory disease-modifying antirheumatic drugs (DMARDs). Evidence suggests that there is a “therapeutic window of opportunity” where the disease is more amenable to treatment, and aggressive therapy within this window can slow disease progression and long-term structural damage. RA patients treated with aggressive therapy within 3 months of symptom onset have better outcomes than patients treated later after symptom onset [6]. Tight control using a “treat to target” approach tailored to the disease activity of the individual patient and aimed at achieving remission is critical for optimizing long-term outcomes [7–9]. There are a number of treatment approaches aimed at achieving tight control (e.g., step-up, combination, and step-down therapy), and the best approach is not clearly defined. What is important is frequent assessment of disease activity, usually every 3–6 months, with modification of DMARD therapy as necessary to achieve treatment goals. Determination of “response” and attainment of remission are measured using tools such as the Disease Activity Score (DAS-28), Simplified Disease Activity Index (SDAI), or Clinical Disease Activity Index (CDAI), all disease activity scores using a 28-joint physical exam, as well as the ACR-20 (American College of Rheumatology) that measures 20% improvement in various components. The DAS score is calculated using an ESR or CRP, while the SDAI and ACR-20 scores include a CRP (the CDAI does not include a laboratory marker). These scores are also used to determine eligibility for and ongoing reimbursement of the DMARD therapies used to treat RA by provincial and private drug benefit plans.

Non-pharmacological Therapy

A number of non-pharmacological measures play a role in the comprehensive management of RA in addition to drug therapy. All patients should receive education about their disease, treatment options, and goals of therapy. Helping patients and their families understand RA will empower them to take an active role in their care. Patients may also benefit from education programs, such as those offered by the Arthritis Society of Canada. Other non-pharmacological options include:

- Physical therapy.
- Occupational therapy.
- Weight reduction/nutritional therapy.
- Emotional and psychological supports.

Pharmacological Therapy

There are a number of medications available for RA treatment (Table 21.2). Methotrexate is the anchor drug in the treatment of RA and is the drug of first choice, given alone or as combination therapy. Ideally, methotrexate should be started within the first 3 months of symptom onset and rapidly titrated to 20–25 mg/week as tolerated, ideally by subcutaneous route [10]. Patients who have markers of a poorer prog-

Table 21.2 Classes of disease-modifying antirheumatic drugs (DMARD) used in the management of rheumatoid arthritis

Drug class	Drug	Dose
<i>Conventional synthetic DMARD</i>		
	Hydroxychloroquine	200–400 mg daily po Maximum: 5 mg/kg ABW
	Leflunomide	10–20 mg daily po
	Methotrexate (MTX)	Initial: 10–25 mg weekly po, subcut or IM (may increase by 5 mg q1–4 weeks to achieve maintenance dose) Maintenance: 15–25 mg weekly po, subcut or IM (for po doses >15 mg, divided q12h doses are better absorbed and tolerated) Add folic acid 1–5 mg daily (may skip day of MTX administration) to reduce adverse effects (GI, liver toxicity, mouth sores) Consider trial of parenteral methotrexate for patients who do not exhibit a clinical response or experience GI intolerance to po therapy
	Sulfasalazine	<i>Initial:</i> 500 mg BID po and then increased to maintenance dose of 1 g BID po (<i>always provide the enteric-coated tablet for better tolerability</i>)
<i>Biologic DMARD</i>		
B cell depleters	Rituximab	1 g × 2 doses 2 weeks apart IV; infusions are given with 100 mg of methylprednisolone Doses should be repeated every ~6 months
IL-6 inhibitors	Tocilizumab	Dosing for IV administration: 4 mg/kg q4 weeks IV, infused over 1 h; may increase to 8 mg/kg IV q4 weeks if response is inadequate Dosing for subcut administration: 162 mg weekly For patients <100 kg start with 162 mg subcut every other week and increase to weekly based on response
	Sarilumab	200 mg q2 weeks subcut

(continued)

Table 21.2 (continued)

Drug class	Drug	Dose
T cell inhibitors	Abatacept	Dosing for initial IV infusion: <60 kg: 500 mg; 60–100 kg: 750 mg; >100 kg: 1 g After initial dosing, administer at 2 and 4 weeks, and monthly thereafter Dosing for subcut administration: 125 mg weekly starting within 24 h of IV loading dose; the same weekly subcut dose is recommended even if IV loading dose is not provided
	TNF-inhibitors	
	Adalimumab	40 mg q2 weeks subcut
	Certolizumab	400 mg at weeks 0, 2, and 4 and then 200 mg q2 weeks subcut. May give 400 mg q4 weeks subcut as maintenance dose
	Etanercept	25 mg twice weekly or 50 mg once weekly subcut
	Golimumab	50 mg once monthly subcut; may increase to q3weeks based on response 2 mg/kg IV at 0 and 4 weeks, then q8 weeks thereafter
	Infliximab	3 mg/kg IV at 0, 2, and 6 weeks, and q8 weeks thereafter For incomplete response, dose may be increased to 10 mg/kg and/or the frequency may be increased up to q4 weeks
<i>Targeted synthetic DMARD</i>		
JAK-inhibitor	Baricitinib	2 mg daily po
	Tofacitinib	5 mg BID po or XR 10 mg daily po
	Upadacitinib	ER 15 mg daily po

ABW actual body weight, BID twice daily, ER extended release, IL interleukin, IM intramuscular, IV intravenous, JAK janus kinase, po oral, subcut subcutaneous, TNF tumor necrosis factor, XR extended release

nosis, moderate-high disease activity, and recent onset of disease may be considered for initial combination DMARD therapy to increase the likelihood of attaining rapid, tight control of symptoms [7].

NSAIDs (e.g., naproxen 500 mg po BID) or corticosteroids (e.g., prednisone 20 mg po daily ×2 weeks, then tapered off over 6 weeks, or methylprednisolone 60 mg IM × 1) may be used as “bridging therapy” for symptomatic relief if needed. They provide quicker relief of symptoms compared with DMARDs, which take weeks to months to achieve maximal benefit. Short-term use of NSAIDs or high-dose corticosteroids can also be used to manage acute flares. However, it is important to note that NSAIDs have no impact on disease progression and corticosteroids are associated with a number of unwanted effects with long-term use.

A number of natural health products are thought to have anti-inflammatory effects and thus may be of benefit in the treatment of RA. Evidence suggests that omega-3 fatty acids (fish oils) (but not omega-6 or omega-9 fatty acids) may improve the symptoms of early RA and reduce the rate of DMARD failure [11]. Other agents, such as glucosamine and chondroitin, turmeric, and ginger, while generally well tolerated, lack evidence of benefit in RA. Patients who wish to try these agents should be encouraged to objectively monitor their symptoms and set a date to reassess effectiveness (e.g., 3 months), with discontinuation if there is no improvement. Products thought to boost the immune system (e.g., ginseng, echinacea, alfalfa sprouts) are not recommended as they may interfere with DMARD therapy. *Tripterygium wilfordii* products should not be recommended as use is associated with a number of severe side effects.

Initial Assessment of a Patient Newly Diagnosed with Rheumatoid Arthritis

Pharmacists can play an important role in the management of patients with RA, through assessment of medication appropriateness and patient education. They should assess the appropriateness of the initial DMARD regimen, along with any prescribed symptom management (or bridging) therapy, and are ideally positioned to support patient’s adherence to therapy, including helping them navigate adverse effects and drug interactions. As with patients with other chronic medical conditions, initial assessment of patients newly diagnosed with RA should begin with the collection of relevant medical history data (history of present illness, medical history, allergies, medications (prescription and nonprescription), social history, laboratory and physical exam findings). Initial assessment should include the following considerations:

- *Assessment of the appropriateness of the prescribed DMARD regimen given the patient’s presentation:* Understanding the patient’s history (disease onset and pattern of clinical findings (signs and symptoms)) will help the pharmacist determine the appropriateness of the initial DMARD regimen, as well as response (effectiveness) at follow-up. There is no known “best” treatment regimen for RA; rather, any of the evidence-based treat-to-target strategies may be used, with patient reassessment every 3–6 months, with or without bridging therapy:
 - *Step Up:* initial monotherapy (methotrexate with dose optimization unless contraindicated), with further DMARD therapies added if necessary.
 - *Combination:* initial combination of DMARD therapies (including methotrexate with dose optimization

- unless contraindicated, along with hydroxychloroquine and/or sulfasalazine), with addition or replacement of a DMARD as necessary.
- *Step Down*: initial biologic DMARD (bDMARD) (with methotrexate with dose optimization unless contraindicated) and/or prednisone, with the aim of discontinuing the bDMARD/prednisone once remission is achieved.
 - Note: Most provincial and private drug benefit plans require a trial of methotrexate in combination with two other conventional synthetic DMARDs (csDMARD) therapies (i.e. “Triple Therapy”) for 12 weeks prior to approval of bDMARD use.
 - *Assessment of the appropriateness of the selected DMARD regimen given the patient’s characteristics*: In addition to consideration of severity of illness, a patient’s preferences with respect to route of administration and common adverse effects can often be taken into consideration along with comorbidities, allergies, concomitant medications, plans for pregnancy, and drug coverage. Table 21.3 outlines patient-specific factors that should be taken into consideration with individual DMARD therapies.
 - *Assessment of the appropriateness of the dosage regimen of the selected DMARD regimen*: Generally, patients start at the target dose; however, some DMARDs are titrated to minimize adverse effects. One such example is sulfasalazine, which should be titrated over at least 1 month to avoid GI adverse effects. Although many rheumatologists initiate methotrexate at full dose (i.e., 25 mg weekly), others will titrate from as low as 7.5 mg weekly.
 - *Assessment of the appropriateness of the bridging regimen*: Given the delayed onset of effectiveness of csDMARD therapies, a full-dose NSAID or tapering dose of prednisone is commonly prescribed to improve patient’s pain, inflammation, and function in the short term. The lowest possible dose and duration should be utilized.

Table 21.3 Summary of patient-specific factors that should be taken into consideration with individual DMARD therapies

DMARD	Patient factors to be considered
<i>Conventional synthetic DMARDs</i>	
Hydroxychloroquine	PMH: G6PD deficiency (contraindicated); risk of retinal toxicity increased with dose >5 mg/kg ABW, renal impairment, preexisting ocular disease, concomitant tamoxifen use (along with age >60, hepatic impairment) DI: Additive hypoglycemia with antidiabetic agents SH: Tobacco use reduces effectiveness Pregnancy/breastfeeding: Considered safe
Leflunomide	PMH: Test for HBV, HCV (and HIV in high-risk individuals) prior to initiation DI: Lower dose (e.g., 20 mg every other day) can be considered when used in combination with methotrexate; cholestyramine utilized to eliminate leflunomide from biliary circulation SH: Excessive alcohol intake should be avoided; one to two drinks per week generally considered safe Pregnancy: Contraindicated, as it causes birth deformities in animals; no reported birth deformities in humans
Methotrexate	PMH: Caution in patients with renal impairment (contraindicated GFR <10 mL/min), chronic liver disease, preexisting immunodeficiency or blood disorders (leukopenia, thrombocytopenia); test for HBV, HCV (and HIV in high-risk individuals) prior to initiation DI: Lower dose (e.g., 15 mg) is often prescribed when used in combination with leflunomide; concomitant use of NSAIDs, PPIs, and penicillin is generally considered SAFE; TMP/SMX should be avoided SH: Excessive alcohol intake should be avoided; one to two drinks per week generally considered safe Pregnancy: Can cause miscarriage or birth deformities (people who have the potential to become pregnant must use effective birth control and discontinue ≥3 months prior to conception)
Sulfasalazine	PMH: G6PD deficiency (contraindicated) Allergies: Sulfonamide allergy (salicylate allergy can be desensitized to enable use) DI: Antibiotics may reduce the gut bacteria needed to cleave sulfapyridine (active drug) from 5-ASA, reducing effectiveness Pregnancy/breastfeeding: Considered safe
<i>Biologic DMARDs</i>	
All	PMH: Hepatitis B or C (unless receiving/received effective antiviral treatment), other active infection; csDMARDs and non-TNF α are preferred if previous treated or untreated skin cancer (melanoma or nonmelanoma) DI: Use with any other bDMARD or tsDMARD Coverage: Expensive
B cell depletor Rituximab	PMH: Preferred if history of treated lymphoproliferative disorder. Typically reserved for difficult to treat RA that has not responded to other bDMARDs. Allergies: Murine protein Pregnancy: Contraindicated

(continued)

Table 21.3 (continued)

DMARD	Patient factors to be considered
IL-6 inhibitors (sarilumab, tocilizumab)	PMH: Exposure to tuberculosis DI: Active inflammation (IL-6) downregulates CYP enzyme activity, thus IL-6 inhibition may lead to improved drug metabolism Pregnancy: Risk may outweigh benefit; no pattern of congenital anomalies or increased risk of spontaneous abortion seen in limited data to date
T cell inhibitor Abatacept	PMH: COPD, exposure to tuberculosis Pregnancy: Risk may outweigh benefit; no pattern of congenital anomalies or increased risk of spontaneous abortion seen in limited data to date
TNF inhibitors (adalimumab, certolizumab, etanercept, golimumab, infliximab)	PMH: Heart failure (NYHA class III-IV), multiple sclerosis, untreated hepatitis C, exposure to tuberculosis, previously treated lymphoproliferative disorder (csDMARD, abatacept, tocilizumab preferred), serious infections (csDMARD, abatacept preferred) Allergies: Murine protein (infliximab) Pregnancy: Risk may outweigh benefit; no pattern of congenital anomalies or increased risk of spontaneous abortion seen in limited data to date; some providers consider certolizumab pegol and etanercept to be compatible with pregnancy
<i>Targeted synthetic DMARD</i>	
JAK inhibitors (baricitinib, tofacitinib, upadacitinib)	PMH: Exposure to tuberculosis, untreated hepatitis B or C DI: Affected by strong CYP3A4 inhibitors and CYP 3A4/2C19 inhibitors Pregnancy: Contraindicated

ABW actual body weight, DMARD disease-modifying antirheumatic drug, DI drug interaction, IL-6 interleukin-6, JAK janus kinase, NSAID non-steroidal anti-inflammatory drug, PMH past medical history, PPI proton pump inhibitor, SH social history, TMP/SMX trimethoprim/sulfamethoxazole, TNF tumor necrosis factor

Physical Assessment Skills

In specialized practice settings, pharmacists might perform the physical assessment skills necessary to utilize various validated disease activity scoring tools, such as the DAS28 using either CRP or ESR, CDAI, and SDAI. Each of these assessment tools incorporates a 28-joint count (including MCPs, PIPs, wrists, elbows, shoulders and knees) and a patient and provider global assessment of disease activity (e.g., “Considering all of the ways your arthritis has affected you, how would you rate yourself on a scale of 0 to 10”). It is important to note that this does not mean that the numerous additional joints that may be affected by RA are unimportant, and they should also be considered as part of a global functional assessment, with or without further physical exam.

Clinical examination is the foundation of RA management. Each joint is inspected for deformity, alignment, and swelling. Joint lines and presence of synovitis are assessed by palpation, along with an assessment of tenderness. Passive and active range of motion should also be assessed and can readily be done in the primary care setting. For example, the pharmacist could ask a patient to attempt to make a fist or perform a ‘finger tuck’ test; wrist, elbow, and knee flexion and extension; and shoulder elevation via flexion or abduction and internal and external rotation.

Follow-Up Assessments

Control

Signs and Symptoms

Although the general practice pharmacist may not have the training and experience necessary to complete a tender and swollen joint count assessment, they can and should assess signs and symptoms of patients with rheumatoid arthritis as part of their routine follow up, for example at medication refill encounters. The ultimate goal is remission, defined by ACR/EULAR as a tender and swollen joint count, CRP (in mg/dL), and patient global assessment of ≤ 1 each [7]. Even without physical exam skills, pharmacists can ask key questions to determine the patient’s response to therapy, targeting a lack of signs and symptoms of the disease. For example, “Considering all of the ways your arthritis affects you, how do you feel your arthritis has been over the past week on a scale of 0 to 10” could be used for someone still experiencing signs and symptoms. For someone who has low disease activity or who perhaps is more stoic, a question like “Is there anything you’re not doing or need help with right now because of your joints” gives the clinician the opportunity to discuss patient goals and/or get a better sense of ongoing functional limitations.

Pharmacists practicing in primary care can also readily assess presence and duration of morning stiffness (“*When you wake up in the morning, are you more stiff than later in the day?*”) And if so, “*How long does it last?*”), pain level (“*How would you rate your pain on a scale of 0 to 10, where 10 would mean that you could not get out of bed?*”), and range of motion (hands, wrists, shoulders) as previously outlined.

The Health Assessment Questionnaire Disability Index (HAQ) is a patient-reported measure of disease activity that provides information about a patient’s functional status (basic activities of daily living). It includes 20 questions related to self-care (e.g., dressing, hygiene, eating, gripping, and typical household activities) that are rated with respect to difficulty in completing the specified task, along with notation of any aids/devices utilized or help from another person required. This tool is commonly utilized by rheumatologists but can be readily adopted by pharmacists in primary care to track improvement from DMARD therapy [12, 13].

If the patient’s DMARD regimen (mono- or combination therapy) fails to achieve remission or low disease activity, then a change should be made, often with the addition of a disease-modifying therapy to a well-tolerated, partially effective regimen. Pharmacists may identify the ongoing use of an NSAID or cycling use of prednisone as a signal of a patient with uncontrolled disease; this is another opportunity to conduct a patient assessment and potentially intervene.

Laboratory

Markers of systemic inflammation, CRP and ESR, are also used to measure effectiveness of DMARD therapy and are part of the DAS28 and CDAI assessment tools. A CRP <10 mg/L is a requirement in the ACR/EULAR Boolean-based definition for remission [8, 14].

Diagnostic Imaging

Radiographs of the hands and feet are recommended as often as every 6–12 months in patients with recent onset, active disease and in whom bony erosion is suspected despite minimal physical findings or functional limitations [4, 7]. A change in therapy should be considered in patients with radiographic progression despite a reasonable clinical response. However, use of radiographs should be minimized in those with established disease and those in remission.

Other imaging modalities, such as ultrasound and MRI, can also be used to guide changes in DMARD therapy [4, 7]. Ultrasonography is particularly helpful to confirm clinical examination findings (whether synovitis is present) and to guide intraarticular corticosteroid injections, as it can be conducted in the clinic setting.

A lack of progression on radiographs of the hands and feet is also required for ongoing reimbursement of bDMARD and targeted synthetic (tsDMARD) DMARD therapies by some provincial and private drug benefit plans.

Adverse Drug Reactions

Pharmacists play an important role in monitoring how patients are responding to and tolerating drug therapy. Monitoring DMARD therapy involves inquiring about tolerance and adverse events, reviewing associated laboratory results, and encouraging patients to contact a healthcare provider if there are concerns. Like all medications, treatments for RA carry risks of adverse events; however, the risk of joint damage and permanent disability from RA is much greater than the risk of severe side effects. The majority of side effects are uncommon, improve over time, and/or are reversible upon discontinuation. Table 21.4 summarizes the common adverse reactions and recommended lab monitoring, as well as rare but serious adverse events of currently available medications used to treat RA.

Patients with RA are at an increased risk of infections, particularly upper respiratory infections, and complications of those infections, due to the disease itself as well as the therapies used to manage it (bDMARDs and tsDMARDs more so than csDMARDs). Infectious symptoms may be diminished or masked in patients on advanced therapies (b/tsDMARDs). Patients should be advised to hold or delay the dosing of an advanced therapy if they have a fever or require antibiotics for any reason (including topical therapies) and may resume once they are feeling well and their antibiotic treatment has been completed.

In addition to recommendations for the general adult population, which include annual influenza and COVID-19 vaccination, patients with RA should receive pneumococcal (conjugate-13 followed at least 8 weeks later by polysaccharide OR conjugate-20 (alone)) and herpes zoster (once ≥50 years or prior to JAK inhibitor initiation) vaccinations [15]. Patients who are at high risk should also receive the hepatitis B vaccine. These and other non-live vaccines may be given prior to initiation of or during DMARD therapy, recognizing that the immune response may not be as pronounced or last as long on DMARD therapy. The exception is rituximab, which, due to its mechanism of action, substantially reduces immunogenicity and vaccines should be withheld for 6 months following its administration. Once DMARD therapy has been started, live vaccines are not recommended due to the risk of causing infections. Live vaccines should be administered at least 4 weeks prior to initiation of bDMARD or tsDMARD therapy or at least 1 month after stopping csDMARD, at least 1 week after stopping tsDMARD, and at least 1 dosing interval for bDMARD therapies [16]. For more information on vaccine recommendations, see the Canadian Immunization Guide [15].

bDMARDs and tsDMARDs can also increase the risk of reactivation of latent tuberculosis infection (LTBI). Prior to starting therapy, patients will be screened for LTBI via skin testing and chest x-ray. Patients who screen positive will be

Table 21.4 Common and rare but serious adverse reactions and monitoring of DMARD therapy

	Adverse reactions		
Drug	Common	Rare but serious	Monitoring
Conventional synthetic DMARDs			
Hydroxychloroquine	GI (cramping, diarrhea), rash, headache, hyperpigmentation, photosensitivity	Corneal and retinal deposition (higher risk with prolonged use of higher doses ^b); hypoglycemia	Baseline ophthalmologic exam then annually after 5 years ^b of treatment
Leflunomide	GI (diarrhea, nausea, decreased appetite/weight loss); less frequently rash, increased blood pressure, hair thinning/loss, increased liver enzymes	Bone marrow suppression, hepatic toxicity, pulmonary infection/fibrosis <i>Alcohol restriction may minimize hepatic toxicity</i>	Blood pressure, CBC, ALT, SCr, Plt, Alb q2–4 weeks for ≥3 months, then q2–3 months
Methotrexate	GI* (nausea, diarrhea), feeling unwell/tired for 24–48 h post dose; less frequently headache, hair thinning/loss*, mouth sores/ulcers*, increased liver enzymes*. <i>*folic acid administration helps reduce adverse effects related to the disruption of the cell cycle</i>	Bone marrow suppression, hepatic toxicity, renal toxicity, pneumonitis <i>Alcohol restriction may minimize hepatic toxicity</i>	CBC, ALT, SCr, Plt, Alb q2–4 weeks for ≥3 months, then q2–3 months
Sulfasalazine	GI (nausea, cramping, diarrhea, dyspepsia), rash, headache, photosensitivity	Bone marrow suppression	CBC, Plt, ALT q2–4 weeks for ≥3 months, then q3 months
Biologic DMARDs			
B cell depletor (rituximab)	Infusion reactions ^c (flushing, itching, decreased blood pressure, etc.), CNS (fatigue, chills, headache), GI (nausea, cramping, diarrhea), rash	Bone marrow suppression, severe infusion reaction, PML, severe skin rash, arrhythmias	Baseline CBC, ALT, hepatitis B and C serology Repeat CBC and LFTs q2–4 months
IL-6 inhibitors (sarilumab, tocilizumab)	Injection site reactions (subcut administration), increased blood pressure, increased cholesterol	Infusion reactions (IV administration), bone marrow suppression, liver toxicity, renal toxicity, GI perforation, increased risk of malignancies,	Baseline CBC, ALT, SCr, lipid profile, hepatitis B and C serology, screen for latent TB Repeat CBC, ALT, SCr in 1–2 months then q3–6 months. Lipid profile in 1–2 months, then q6 months
T cell inhibitor (abatacept)	Injection site reactions (subcut administration), headache, nausea, nasopharyngitis	Infusion reactions (IV administration), exacerbation of COPD, increased risk of lymphoma, leukemia, and other malignancies	Baseline CBC, ALT, SCr, hepatitis B and C serology, screen for latent TB Repeat CBC, ALT, SCr in 1–2 months then q3–6 months
TNF inhibitors (adalimumab, certolizumab, etanercept, golimumab, infliximab)	Injection site reactions (SC administration), headache, nausea, rhinitis, cough	Infusion reactions (IV administration ^c), increased risk of lymphoma, leukemia and other malignancies	Baseline CBC, ALT, SCr, hepatitis B and C serology, screen for latent TB Repeat CBC, ALT, SCr in 1–2 months then q3–6 months
Targeted synthetic DMARDs			
JAK inhibitors (baricitinib, tofacitinib, upadacitinib)	GI (nausea, dyspepsia, diarrhea), headaches, increased blood pressure/heart rate, increased cholesterol	Bone marrow suppression, hepatic toxicity, renal toxicity, GI perforation, increased risk of lymphomas and other malignancies May increase risk of VTE and/or MI especially in those with risk factors	Baseline CBC, ALT, SCr, lipid profile, hepatitis B and C serology, screen for latent TB. Repeat CBC, ALT, SCr in 1–2 months then q3–6 months. Lipid profile in 1–2 months, then q6 months

ABW actual body weight, Alb albumin, ALT alanine aminotransferase, CBC complete blood cell count, CNS central nervous system, DMARDs disease-modifying antirheumatic drugs, GI gastrointestinal, IBW ideal body weight, IV intravenous, JAK janus kinase, Plt platelets, PML progressive multifocal leukoencephalopathy, SCr serum creatinine, subcut subcutaneous, TB tuberculosis, TNF tumor necrosis factor

^aCumulative dose >1000 g, doses >5 mg/kg ABW—6.5 mg/kg IBW, or 400 mg/day for >5–7 years

^bYearly if high risk (liver/kidney disease, obesity, >60 years old, preexisting eye disease)

^cPre-treatment: acetaminophen, diphenhydramine and IV methylprednisone

required to take a 9-month course of isoniazid (or alternate therapy), initiated 1–3 months prior to starting the bDMARD or tsDMARD therapy.

Determining whether there is a risk of specific malignancies related to DMARD therapies remains an area of research; however, any risk must be balanced with the recognition that tight control of systemic inflammation also reduces the risk of RA-associated malignancies. Patients who have an active malignancy often have their DMARD therapy regimen adjusted, just as certain DMARDs are used preferentially in patients who have a history of various types of malignancy (see Table 19.3).

Complications

In addition to risk of infection and malignancy as outlined above, patients with RA are also at risk of:

Joint Deformity

Likely, the most recognized complication of RA is joint deformity, which can lead to functional limitations and long-term disability.

CVD

RA and atherosclerosis both involve activation of T cells, production of proinflammatory cytokines, and elevated CRP, thus it makes sense that the high systemic inflammatory burden of uncontrolled RA contributes to an increased risk of cardiovascular events and independently exacerbates modifiable risk factors like insulin resistance, dyslipidemia, and hypertension [17]. The best way to reduce cardiovascular risk in patients with RA is the achievement of tight control of both their arthritis (remission) and known modifiable risk factors (targets as outlined in guidelines). While both MTX and TNFi have demonstrated benefit in reducing cardiovascular events in observational studies, [18, 19] higher dosed JAK inhibitors have been associated with an increased risk of VTE and myocardial infarction in patients with risk factors (age >65 years, known atherosclerotic disease, smoking history) [20].

Adherence

Patient adherence with their prescribed drug regimen is essential for tight control of the disease and symptom management, and pharmacists play an important role in helping patients achieve optimal adherence. There are a number of factors that may affect medication adherence in RA. All patients should be counseled on the proper dosing schedule for each of their medications as a number of commonly used agents are not dosed daily (e.g., methotrexate, bDMARDs).

When monitored properly by clinical and laboratory assessment, the majority of side effects are rare and the most common ones improve over time and/or are reversible. Patients should be encouraged to complete their routine blood work and attend follow-up appointments with their rheumatologist even when they are feeling well. If a patient feels they are experiencing a side effect related to their medication, they should be encouraged to discuss the issue with one of their healthcare providers, as there may be strategies to reduce their burden. Medications used to treat RA can impair the body's ability to fight infections. Patients who have a fever or believe they have an infection should contact their primary care provider. Those who are having surgery, or have been prescribed an antibiotic and have questions about how long to hold their therapy, should be counseled to contact their rheumatologist for instructions on how to manage their RA treatment. Finally, the majority of the bDMARDs and tsDMARDs are associated with significant costs; medication coverage issues should be assessed and addressed to ensure optimal compliance.

Clinical Pearls

- Pharmacists can play an important role in the management of patients with rheumatoid arthritis.
- Initial assessment involves assessment of both the prescribed DMARD regimen, which should be targeting remission, and any necessary bridging therapy given the patient's clinical findings.
- Assessment at follow-up includes the patient's ability to adhere to therapy, objective measures of disease activity, and discussion of any adverse effects, along with monitoring laboratory measures of DMARD effectiveness and safety.

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Chapter Objectives

1. Describe the disease burden, epidemiology, diagnosis, prognosis, and management of major depressive disorder and other depressive disorders.
2. Conduct initial assessment of patients who present with new diagnosis of depression or who are suspected of having depression but have not been formally diagnosed.
3. Outline monitoring parameters and follow-up plans for patients using pharmacological therapy to manage their depression.
4. Provide guidance to patients on how to manage partial/non-response or adverse effects of medications.

Background

Depression is one of the most common and debilitating mental health conditions, affecting individuals of all ages, from children to the elderly. According to recent estimates, over 7.6% of Canadians aged 15 and older experienced a major depressive episode in 2022, reflecting an increase from 4.7% a decade earlier [1]. Globally, the prevalence of depressive disorders is approximately 4%, with rates spiking by 28% during the COVID-19 pandemic due to the associated social, economic, and health-related stressors [2, 3]. Despite growing awareness of mental health, significant treatment gaps persist, with nearly half of those affected by depression remaining untreated [4]. Barriers such as limited access to mental health professionals, long wait times, and systemic inequities contribute to this shortfall.

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Pharmacists are well equipped to both screen patients for the presence of depressive symptoms and monitor therapy to ensure that remission is being achieved while ensuring patients do not experience adverse effects [5]. In several Canadian provinces, the expanded scope of pharmacy practice allows pharmacists to initiate or adjust therapy under certain circumstances, further integrating them into the mental health care team [6].

The term “depression” encompasses a range of disorders with distinct characteristics but similar management approaches. These disorders differ in their duration, timing, and precipitating factors. This chapter focuses primarily on major depressive disorder (MDD) but also includes pertinent details about other types. Common subtypes and specifiers include [7]:

- Major depressive disorder.
- Persistent depressive disorder (PDD)/dysthymia.
- Premenstrual dysphoric disorder (PMDD).
- Peripartum depression.
- Depression due to medication/other substance or medical condition.
- Major depressive disorder with seasonal pattern.

Epidemiology and Burden of Illness

Recent data indicates that approximately 14% of Canadians aged 15 and older have experienced a major depressive episode at some point in their lives [8]. On an annual basis, about 7.6% of this population reports having a major depressive episode, with young women disproportionately affected. In fact, the prevalence among young women has doubled, rising from 9.0% in 2012 to 18.4% in 2022 [1].

The prevalence of depression varies among different age groups. Young adults aged 15 to 24 exhibit higher rates, with 11.1% reporting a major depressive episode in the past year [1]. This elevated rate may be attributed to financial, academic, and social stressors common in this demographic. In

contrast, older adults aged 65 and above have a lower prevalence, with 4.7% experiencing a major depressive episode in the past year [1]. The rates of depression may be higher in individuals older than 75 years, with a 2010 meta-analysis indicating that the prevalence of depression is 7.2% in this age group. This meta-analysis contained data from Canada, the United States, and multiple European countries. Depression remains a significant cause of disability in the elderly, often leading to impairments in physical, emotional, social, and functional status [9].

There is also a significant burden of illness in children, with 2.8% of those under 13 being affected and 5.7% of adolescents aged 13–18 [10]. In children, depression may contribute to behavioral challenges and impair normal psychosocial development. Depression in younger patients may become more difficult to treat the longer it continues and can persist into adulthood. Children who have depression are 4 times more likely to develop depression as an adult compared with those who do not [7].

The prevalence of MDD is higher among racialized, Indigenous, gender minority, and certain religious populations, who also face significant barriers to mental health care. Common obstacles include cost, stigma, cultural beliefs, language differences, fear of judgment, unequal service distribution, and fragmented health systems [11].

The burden of a depressive disorder falls on the patient, their friends and family, and society as a whole. Depression has a major impact on physical, emotional, social, and functional status. The effects on mood can be obvious but can also lead to difficulty in interpersonal relationships, isolation from friends and family, and conflict with those around them. Patients also frequently experience bothersome physical symptoms. Many patients have impairment in their ability to work or attend school. This manifests as absences from work, missed classes, and decreased productivity when they are present. It is often made worse by the specific symptoms such as lack of motivation, cognitive dysfunction, fatigue, and insomnia. The average individual with MDD will have 34.4 days per year where they are unable to fulfill major roles [12] and experience a “healthy life expectancy” that is shortened by up to 6 years [13]. More recent metrics from 2019 demonstrate that depression ranked second among medical causes of years lived with disability (YLDs) worldwide [2]. In 2001, Health Canada predicted that the financial burden of depression each year was \$14.4 billion [14], which includes costs associated with treatment, loss of productivity, premature death, and medications. Given that the prevalence of depression in the overall population has increased since then, the current economic burden is likely greater.

Importantly, individuals with MDD have a fivefold higher risk of suicide attempts than the general population, and in Canada, approximately 15 men per 100,000 and 5 women per 100,000 die by suicide, totaling over 4000 deaths

annually. About half of these individuals suffer from MDD, highlighting the significant impact of this disorder on Canadians [15].

Etiology

The etiology of depressive disorders is unclear, and although there are some prevailing theories, they all have both supporting and contradicting evidence. More research is needed to determine the true cause of depression.

The monoamine theory was one of the early theories of depression and was proposed by a psychiatrist named Joseph Schildkraut in 1965 [16]. He concluded that depression came from a lack of monoamine neurotransmitters, specifically serotonin and norepinephrine. In animal and human studies, he observed that drugs such as monoamine oxidase inhibitors, which increased norepinephrine and serotonin, led to improved mood. Also, drugs such as reserpine, which inhibit these neurotransmitters, led to depressed mood [17]. However, there are some inconsistencies with this theory. In more targeted studies meant to support this hypothesis, results were inconsistent and the changes that were seen were not specific to depression over other mood disorders. Also, if it was a simple matter of replacing neurotransmitters, patients could expect immediate effect on taking their first antidepressant dose, when we know that it takes weeks to see benefits. In more recent studies on amine neurotransmitters, the role of dopamine has become a focus since its actions are so closely entangled with those of serotonin and norepinephrine. Many antidepressants have some effect on dopamine pathways [18]. While the monoamine theory remains a cornerstone in depression etiology, a recent systematic umbrella review has questioned the strength of the “serotonin deficiency” model, suggesting that depression may be more multifactorial than a simple chemical imbalance [19].

Another theory concerns neuroendocrine mechanisms related to cortisol and impairment in monoamine pathways. Depressive episodes are often preceded by stressful events, which the body responds to by producing cortisol. Persistent states of stress lead to desensitization of norepinephrine and serotonin receptors and dysregulation of the stress response [20]. Cortisol levels increase when serotonin and norepinephrine act on cells in the hypothalamus to release corticotrophin-releasing hormone (CRH) which starts the cascade that releases cortisol. When CRH was injected into the brains of experimental animals, they exhibited some of the common depressive symptoms seen in humans such as decreased energy, increased anxiety, and lack of appetite [17].

Recently, research has been focused on theories of trophic hormones and degeneration in the hippocampus as mecha-

nisms that produce depressive conditions. Two substances being investigated are brain-derived neurotrophic factor (BDNF) and glutamate. BDNF levels are associated with mood: depressed patients have lowered levels, and those who are treated with antidepressants see recovery in their levels. Glutamate's proposed role is related to excitotoxicity and long-term potentiation of N-methyl D-aspartate (NMDA) receptors, but the exact mechanism is unclear. Notably, chronic hypercortisolemia—reflected by the failure of cortisol suppression on the dexamethasone suppression test—may exacerbate both glutamate excitotoxicity and reductions in BDNF. This aligns with neuroplasticity and neuronal regeneration theories supported by imaging studies showing atrophy and decreased functional activity in the hippocampus and prefrontal cortex in depression [18]. It is proposed that through the actions of serotonin, norepinephrine, and BDNF, antidepressants contribute to the regrowth of neurons. This theory also aligns with the time course usually needed to see remission. It may also explain why some patients find exercise to be effective, because exercise is also thought to promote neurogenesis [17].

Additionally, emerging evidence points to a significant role for inflammation and immune dysregulation in the pathophysiology of depression, highlighting how pro-inflammatory cytokines and microglial activation can influence neurotransmission and mood regulation [21]. Furthermore, the gut–brain axis has gained attention as studies indicate alterations in gut microbiota and related immunoneuroendocrine pathways may contribute to the onset and

progression of depressive disorders [22]. Further research in these key areas are ongoing to fully understand their potential role in depression.

Risk Factors

The development of depression is complicated and related to genetic, environmental, and social factors. Although we cannot say for certain that a factor has caused depression in a patient, there are some risk factors that have been identified:

- Young adults and elderly.
- Female sex.
- Gender dysphoria.
- Previous depressive episode or substance use disorder.
- Family history of depression, suicide, or substance abuse.
- Low socioeconomic status.
- History of adverse childhood events/maltreatment.
- Use of certain medications and other substances (Table 22.1).
- Chronic medical conditions or disability (see Table 22.1).
- Social factors/Stressful life events (see Table 22.1).
- Presence of other psychiatric conditions, for example, anxiety, substance use disorder.
- Insomnia and night-shift employment.
- Hormonal changes, for example, puberty, peripartum, menopause, andropause.

Table 22.1 Factors implicated in developing major depressive disorder

Drugs [18, 23, 24]	Medical conditions [18, 25]	Social factors [15]
Alcohol	Alzheimer's/dementia	Bereavement
Anticonvulsants	Cancer	Divorce
Antirejection drugs	Chronic pain	Domestic violence
Beta-adrenergic blockers	Diabetes	Job strain/income inequality
Reserpine	Heart disease/stroke	Environmental disasters
Vesicular monoamine transporter 2 (VMAT2) inhibitors	HIV	Isolation/social rejection
Clonidine	Insomnia	Job loss
Corticosteroids	Multiple sclerosis	Social rejection
Isotretinoin	Parkinson's disease	Peer victimization/bullying/cyberbullying
Nicotine/tobacco	Spinal cord injury	Trauma
Oral contraceptives	Traumatic brain injury	Sedentary lifestyle/screen time
Gonadotropin-releasing hormone (GnRH) agonists	Seizures	
Substances/drugs of abuse	Hypothyroidism	
Tamoxifen	Withdrawal from substances	
Interferons		
Varenicline		

This table does not include every drug or disease state that is associated with depression but covers the most significant contributors

Presentation

As the most accessible health care providers, pharmacists are in an ideal position to recognize patients who may be experiencing symptoms of depression. Pharmacists might be the first persons to notice something as small as a previously very adherent patient no longer picking up medications, for example. Patient presentation can be highly varied and initial symptoms may be non-specific, so having a high index of suspicion is valuable. A depressed patient will not always complain of the typical “depressed mood and loss of interest in activities.” Patients may mention fatigue, memory problems, stress, “mood swings,” trouble with interpersonal relationships, or have a history of multiple medical visits with unclear symptoms and no real resolution [26]. Insomnia can be what causes many patients to first seek help, so it is important to probe those who are seeking over the counter sleep remedies for the presence of other symptoms. Box 22.1 presents some patients who would require further assessment for depression.

It is important to remember that not all patients will present with mood symptoms and may instead have physical complaints, including vague aches and pains, gastrointestinal disturbances, weight changes, or headaches. When assessing patients with physical symptoms that do not have a clear cause, it is important to consider that they may indicate depression.

Children and adolescents tend to present with more behavioral problems, anxiety, irritability, somatic symptoms, phobias, and social withdrawal. As children and adolescents age, there is an increase in melancholy, psychosis, and suicidality [10]. Pharmacists may be able to intervene by guiding parents to access needed supports for their children and for themselves.

Box 22.1: Example Scenarios of Patients Who Would Require Further Assessment for Depression

1. AC is a 30-year-old man who was brought to the hospital after mentioning a suicide plan to his partner, who was very worried. His partner says that 3 months ago, he lost his job, and so he spends his days “just sitting on the couch staring at the TV.” He is often short-tempered and does not spend time with friends because it is “too much work.” AC has also lost some weight but is not sure exactly how much; however, he says his clothes feel very loose.
2. ZT is a 16-year-old girl who comes to the pharmacy 1 day and asks what you suggest to help her sleep. She says she hasn’t really slept well in

about 3 weeks because she keeps waking up throughout the night. She reveals that she recently (~1 month ago) came out to her family as bisexual, and although her father is supportive, her mother has made many derogatory comments. When you ask further, she says she has been feeling “pretty down lately.”

3. TS is a 22-year-old transgender woman who arrives at the pharmacy to pick up her hormone replacement therapy (HRT) prescription. She mentions feeling increasingly anxious and isolated because her family is not supportive of her transition and often misgenders her. TS describes difficulty sleeping for the past month, frequent episodes of crying, and a loss of interest in activities she once enjoyed, such as gaming with friends and volunteering. She confides that she’s started missing her medical appointments because she feels “nothing is going to help anyway” and worries she “doesn’t belong anywhere.”
4. When you are filling a prescription for your patient LN, he mentions that he hopes he is “not like his mother” when he gets older because she is “cranky, never leaves the house, and stopped doing the things she used to love.” He tells you that although she has been widowed for 20 years, she was not like this until 2 years ago when she moved to a seniors’ apartment. Since then she has been difficult to be around and complains of constant aches and pains that won’t go away no matter what she tries.

Diagnosis

Pharmacists are not usually responsible for making the official diagnosis of depression; however, they play a vital part in recognizing potential symptoms and ensuring patients receive appropriate care. As frontline healthcare providers, pharmacists in Canada are increasingly involved in collaborative primary care services, which may include screening, initial assessment, and referral for patients who present with symptoms of depression. Staying informed about diagnostic criteria and other disorders that may mimic or co-occur with depression is therefore crucial for pharmacists to effectively guide patient care and support multidisciplinary management.

Depression is characterized by a persistent depressed mood, and/or loss of interest in previously enjoyed activities, as well as other symptoms that are listed in Table 22.2. The

Table 22.2 Common symptoms in patients with depressive disorders

Psychological symptoms	Physical symptoms
Depressed mood	Insomnia
Loss of interest	Restlessness, agitation
Feeling overwhelmed	Weight gain or loss without trying
Feeling excessively guilty	Headaches, backaches
Anxiety	Vague aches and pains
Withdrawal from social life	Slowed movements or constant motion
Suicidal thoughts or actions	

diagnostic criteria for all depressive disorders are included in the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (DSM-V). It describes the symptoms, time frame, and other qualifiers that are needed for proper diagnosis.

Major Depressive Disorder

To be considered a major depressive disorder, five or more criteria must be present over a period of 2 weeks, with at least one of depressed mood or loss of interest. These symptoms are not related to a normal response to an event such as bereavement or a natural disaster, cause clinically significant distress or impairment in daily life, and are not due to another medical condition or to use of a substance. The patient also should not have had a manic or hypomanic episode before, and their symptoms are not better explained by another disorder [7]. The mnemonic SADIFACES [27] can help one remember the diagnostic criteria for a major depressive episode (Table 22.3). For further information, refer to the DSM-V.

Major depressive episodes (MDE) are classified based on severity. A *mild* MDE includes 5–6 symptoms, and the patient is usually still able to function relatively normally, although it may take more effort than usual. The symptoms themselves are often felt by the patient to be mild. A *moderate* episode usually involves 6–8 of the diagnostic criteria, and the patient may feel that the symptoms are of a medium level of severity and are having some impact on function. A *severe* episode includes all or almost all of the diagnostic criteria, and severe enough to majorly affect the person's ability to function, or they are unable to function at all [27]. Some may also consider an episode to be severe any time there is suicidal ideation present.

Other Depressive Disorders

Persistent depressive disorder (PDD), replacing what has commonly been called *dysthymia*, involves a more chronic course of depression, presenting with 2 or more depressive symptoms for an extended period of time (1 year for chil-

Table 22.3 SADIFACES mnemonic for major depressive episode criteria [28]

S	Sleep—increased or decreased
A	Appetite—Increased or decreased
D	Depressed mood
I	Interest lost (anhedonia)
F	Fatigue or decreased energy
A	Anxiety/agitation
C	Concentration difficulties
E	Esteem/excessive guilt or worthlessness
S	Suicidal thoughts or actions

“D” and “I” are bolded because at least one of the two is required, in addition to other symptoms, to diagnose depression

dren, 2 years for adults) [7, 15]. Individuals presenting with PDD are less likely to achieve remission and show lower response rates overall, demanding diligent follow up and a focus on improved quality of life. It is currently estimated that 20–30% of individuals presenting with depression will follow a more chronic course of PDD [15].

Premenstrual dysphoric disorder (PMDD) is diagnosed when there is a pattern in multiple cycles where at least five of the symptoms are present in the week before menstruation starts, and they start to remit once it has started, and symptoms are absent in the week post-menses [7].

Peripartum depression is a major depressive episode that starts within 4 weeks of delivery, but otherwise follows similar criteria to typical MDE [7].

Depression due to substance/medication includes the same features as MDD but is associated with taking or withdrawing from a substance and lasts beyond what is normally expected based on the pharmacology of the substance. The onset is within 1 month of taking a substance that could reasonably cause depression and the pattern does not better reflect another depressive disorder [7].

In *depression due to another medical condition*, the important qualifier is that the pathophysiology of the depressive symptoms is directly related to the medical condition, for example, traumatic brain injury, stroke, or hypothyroidism. It can be very difficult to determine if the depressive episode is caused by the medical condition or if the medical condition is simply the stressor, which precipitated a depressive episode [7].

MDD with Seasonal Pattern describes major depression, which recurs and remits in a pattern with seasons. It is often present in the fall and winter, and then remits on its own in the spring and summer [7].

Depression Rating Scales

Several clinician and patient-rated scales have been validated in depression. They are useful in both diagnosis and follow-up of patients. Different scales have been developed for

adults, elderly patients, and children [28]. The gold standard scales for adults have historically included the HAM-D, BDI, and IDS/QIDS, which remain highly regarded for research or specialized clinical assessment. However, the PHQ-9 has become the preferred option in recent guidelines due to its ease of use and strong psychometric properties [15, 29]. The shorter PHQ-2 can serve as a quick initial screening tool, helping identify patients who need further evaluation with the full PHQ-9. In fact, the CANMAT guidelines recommend beginning with the PHQ-2—then proceeding to the PHQ-9 if the PHQ-2 score is ≥ 2 —because this approach offers similar sensitivity and higher specificity compared to using the PHQ-9 alone [15, 30]. For geriatric patients, the GDS is the preferred scale, and the CDRS-R is the most commonly used for children. Some of the more commonly used scales are listed here:

- 9-Item Patient Health Questionnaire (PHQ-9)
- 2-item Patient Health Questionnaire (PHQ-2)
- Hamilton Depression Rating Scale (HAM-D)
- Beck Depression Inventory (BDI)
- Inventory of Depressive Symptomatology (IDS/QIDS)
- Zung Self-Rating Depression Scale
- Geriatric Depression Scale (GDS)
- Children's Depression Rating Scale Revised (CDRS-R)
- Kutcher Adolescent Depression Scale (KADS)

These scales will be discussed further in the section on initial assessment.

Differential Diagnosis

Since there is a high rate of concurrent psychiatric disorders with depression, it is important to test for other disorders while assessing for depression. Other psychiatric disorders can also present very similarly to depression. Anxiety symptoms can be present in patients who have depression, including symptoms such as sleep disturbances, difficulty concentrating, fatigue, irritability, and restlessness, but may also be due to comorbid generalized anxiety disorder, obsessive-compulsive disorder, or another anxiety disorder. It would make sense to consider an anxiety disorder in a patient being treated for depression if all other symptoms responded to treatment, but symptoms of anxiety persisted. Likewise, symptoms associated with attention-deficit/hyperactivity disorder (ADHD), such as inattention, decreased motivation, and impaired executive function can sometimes be mistaken for (or coexist with) depressive symptoms and should be considered and ruled out if appropriate treatment strategies for depression do not adequately resolve such symptoms.

Bipolar disorder should also be considered since these patients are often first treated when they present with a

depressive episode. When given antidepressants, patients who have bipolar disorder often have a rapid switch to mania or hypomania, which can be very dangerous. Past episodes that resemble mania or hypomania should be enquired about to minimize the risk of this switch occurring. In addition to psychiatric causes, physical causes of symptoms should be considered. Anemia, dementia, and various endocrine conditions can produce depressive symptoms in some patients and should be ruled out with appropriate testing before diagnosing depression. Ruling out hypothyroidism is especially important because it often presents as depression but will remit once thyroid replacement therapy is started.

Investigations

Labs to be drawn to rule out physical causes:

- Thyroid-stimulating hormone (TSH) for hypothyroidism.
- Complete blood count (CBC) for anemia.

Labs to be drawn to rule out physical causes, only when medical history and examination suggest they may be indicated:

- Vitamin B12 level to assess for deficiency.
- Cortisol levels.
- Vitamin D levels.
- Testosterone levels.

Other investigations that may be done to confirm diagnosis of depression

- Mini-Mental State Evaluation to assess for dementia.
- Psychologic rating scales for anxiety, other mood disorders.
- Full medical history assessment to identify Parkinson's, stroke, or other disorders that directly cause depression.
- In individuals assigned female at birth, a full clinical assessment to identify the potential role of perimenopause or menopause in causing depressive symptoms.
- Medication history, as well as use of any other non-prescription or recreational substances.

Prognosis

Untreated episodes often last 6 months or longer, while on treatment the average time to remission is 20 weeks (4.6 months). Prognosis will vary based on the characteristics of the episode as well as patient adherence and response to treatment. Those who have a better response to treatment generally have a lower risk of relapse and a better prognosis

than those who have more treatment-resistant depression [25]. In half of patients, their first episode will be their only episode. The other half of patients will have persistent (15%) or recurrent (35%) episodes. With pharmacotherapy, up to two-thirds of patients will eventually have remission of the episode [31], but there is still a 30–40% chance of recurrence [32]. In children and adolescents who have one episode, the rate of relapse is as high as 70% [10]. Even once patients are in remission, 90% will experience some residual symptoms [32].

Management

Treating depression is important because although some episodes will remit on their own, the longer an episode continues untreated, the harder it will be to treat in the future (Fig. 22.1). Given the impact on quality of life and the higher risk of self-harm or suicide, it is preferable for patients with depressive disorders to be treated. Depression can be managed through both pharmacological and non-pharmacological means. In addition to nonpharmacological and pharmacological interventions, psychoeducation and self-management are central to managing MDD, as they equip patients with practical knowledge about depression and empower them to actively engage in problem-solving, lifestyle modifications, and behavioral activation. Numerous resources—including the CHOICE-D patient guide—provide accessible tools to support patients in effectively implementing self-management strategies [33].

Short-term goals of therapy (acute phase of treatment) are focused on the current episode and usually include a time frame of 8–16 weeks or until remission is achieved. Main goals include:

- Addressing patient safety.
- Alleviating symptoms.
- Avoiding medication adverse effects.
- Reaching remission.
- Restoring function and quality of life.

Long-term goals become the priority once remission is reached and they include:

- Maintaining remission, preventing relapse (continuation phase of treatment).
- Returning to full social, occupational, and interpersonal function.
- Preventing recurrence (maintenance phase of treatment).

Non-pharmacological Therapy

Non-pharmacological treatment can be very effective in managing acute depressive episodes as well as providing patients with coping mechanisms to use in future episodes. Cognitive behavioral therapy and other forms of psychotherapy are at least as effective as pharmacotherapy for mild-moderate episodes [34]; however, the cost and time commitment may be prohibitive to patients. Light therapy (10,000 lux of white light for 30 min daily in the morning) has been found effective in mild-moderate depression, especially seasonal affective disorder [35]. The evidence behind the benefits of exercise (30–40 min of low- to moderate-intensity exercise 3–4 times a week for a minimum of 9 weeks) in depression management is strong and is therefore recommended as a first-line treatment option in mild depression, and an effective adjunctive therapy in moderate-severe cases [15, 36].

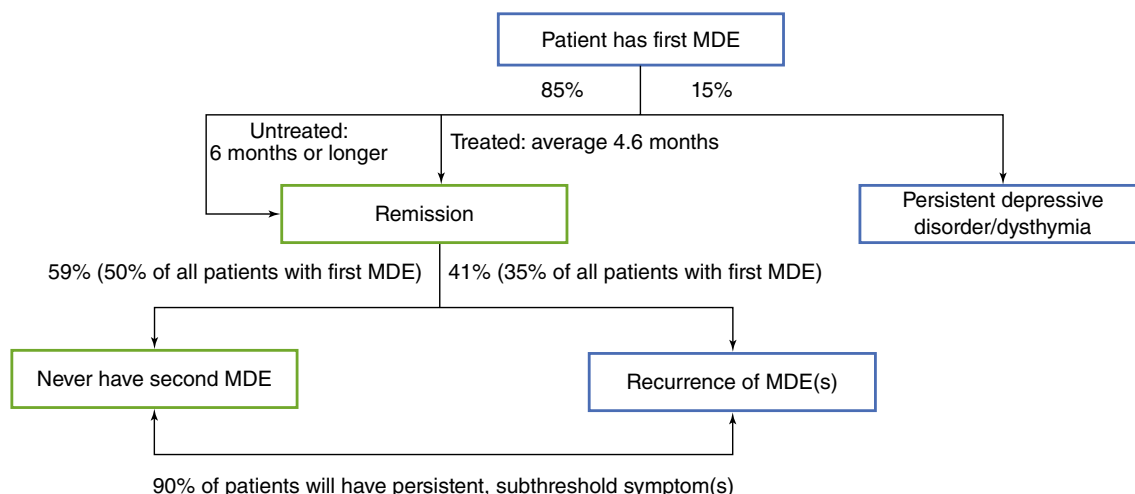


Fig. 22.1 Course of patients with a first major depressive episode. MDE major depressive episode as defined in the DSM-V. (Adapted from [25])

Additional adjunctive strategies such as sleep hygiene and cognitive behavioral therapy targeting insomnia (CBT-I), meditation and mindfulness-based practices, smoking cessation, and healthy eating (such as a Mediterranean-type diet) have all demonstrated benefit in depression management [15, 37].

Complementary and Alternative Medicine

As in most other medical conditions, patients are increasingly interested in using a variety of complementary and alternative medicines (CAM) and nutraceuticals to manage depression. To date, no treatment in this category currently matches the evidence level of first-line psychotherapy or pharmacotherapy for moderate to severe depression, and therapeutic dose ranges for CAM remain inconsistent. Consequently, CANMAT recommends CAM alone only for mild MDE (St. John's wort—first line; acupuncture—second line) and moderate MDE (St. John's wort—second line), while reserving its adjunctive use for moderate severity illness. L-methyl folate may be considered as a second-line adjunctive treatment [15].

Although saffron, lavender, and roseroot may reduce depressive symptoms in mild MDE according to meta-analyses, they show only modest benefit and have significant limitations in randomized controlled trials, thus remaining third-line options. Similarly, due to mixed or insufficient evidence, S-adenosyl-L-methionine (SAM-e), omega-3 fatty acids, and dehydroepiandrosterone (DHEA) are also recommended as third-line treatments for mild–moderate MDE. Patients should always be made aware of the comparative lack of regulation of these products and possibility of interactions with other medications [15].

Digital Health Interventions

Digital Health Interventions (DHIs), such as internet or mobile apps delivering evidence-based therapies (e.g., CBT, BA), offer convenient, real-time support and can be culturally adapted; however, many remain poorly validated, pose privacy concerns, and face high dropout rates. Daily mood monitoring may exacerbate self-stigma, and unmoderated forums risk misinformation, so mitigating strategies like therapist moderation, crisis guidance, and anonymity options are important [38]. Guided DHIs (e.g., iCBT programs such as Good Days Ahead, BounceBack, and Deprexis) tend to show stronger evidence for mild to moderate MDD, whereas unguided options (e.g., MoodGym, Headspace, and MoodKit) offer limited efficacy and are better suited to mild cases or use as adjuncts [39]. Overall, clinicians should weigh factors like the DHI's evidence base, usability, and safety features before recommending it to patients [15].

Neuromodulation Treatments

Neuromodulation (or neurostimulation) encompasses a range of treatments that alter brain activity through electrical or magnetic stimulation, generally reserved for cases where first-line psychotherapy and pharmacotherapy have not yielded sufficient benefit. Among noninvasive neuromodulation options, electroconvulsive therapy (ECT) remains the most effective for those who have not responded to other treatments and who present with severe symptoms (e.g., extreme suicidality, psychosis, rapid physical deterioration), achieving response rates of 65–75%—especially in older or more severely depressed patients. However, it requires general anesthesia and may lead to cognitive side effects. Repetitive transcranial magnetic stimulation (rTMS), another noninvasive modality, has fewer side effects than ECT but somewhat lower efficacy; newer variants like theta burst stimulation (TBS) allow shorter, more intensive protocols. Additional noninvasive approaches include transcranial direct current stimulation (tDCS) and magnetic seizure therapy (MST), both of which have potential benefits but remain less established. Surgical methods, such as vagus nerve stimulation (VNS) and deep brain stimulation (DBS), are generally considered after noninvasive approaches due to limited evidence, higher risk, and the need for an implanted device [15].

Pharmacotherapy

Every patient will respond differently to different agents, so there is no designated drug to be used as the first choice every time like there may be in other conditions. It is not currently recommended to routinely use pharmacogenetic tests for initial antidepressant selection due to modest effect sizes, potential delays, and inconsistent clinical benefits; the use of these tests should be reserved for specific cases of unusual adverse effects or poor treatment response [15]. It makes sense, then, to use the side effect and safety profile to guide the decision of which agent to use in a specific patient. This strategy may improve adherence, especially since patients tend to experience the side effects of a drug before the benefits [40]. The pharmacological treatments available in Canada are presented in Table 22.4. First-line treatments according to the Canadian Network of Mood and Anxiety Treatment (CANMAT) 2023 guideline update [15] include selective serotonin reuptake inhibitors (SSRIs), serotonin/norepinephrine reuptake inhibitors (SNRIs), bupropion, mirtazapine, vilazodone, and vortioxetine based on their efficacy and safety profiles. Other first-line agents that are not available in Canada include milnacipran, mianserin, and agomelatine. Selection of a specific agent depends on indi-

Table 22.4 Pharmacotherapy choices for depressive disorders [15, 41, 42]

Class	Specific agents (usual adult dose)
Selective serotonin reuptake inhibitors (SSRI)	Citalopram (20–40 mg daily) Escitalopram (10–20 mg daily) Sertraline (50–200 mg daily) Paroxetine (20–50 mg daily) Fluoxetine (20–60 mg daily) Fluvoxamine (100–300 mg daily)
Serotonin/norepinephrine reuptake inhibitors (SNRI)	Venlafaxine (75–225 mg XR daily) Desvenlafaxine (50–100 mg daily) Duloxetine (60 mg daily) Levomilnacipran (40–120 mg daily)
Serotonin-2 antagonist/reuptake inhibitor (SARI)	Trazodone (150–300 mg daily in 1–3 divided doses with majority of dose at bedtime)
Serotonin-1A agonist/reuptake inhibitor	Vilazodone (20–40 mg daily)
Serotonin modulator and stimulator (SMS)	Vortioxetine (10–20 mg daily)
Noradrenergic/specific serotonergic agent (NaSSA)	Mirtazapine (15–45 mg qhs)
Norepinephrine/dopamine reuptake inhibitor (NDRI)	Bupropion (150–300 mg SR bid, 150–300 mg XL daily)
Tricyclic antidepressants (TCA)	Amitriptyline (50–300 mg daily divided bid or qhs) Nortriptyline (50–150 mg daily divided bid or qhs) Clomipramine (50–300 mg qhs) Imipramine (50–200 mg qhs) Desipramine (50–200 mg qhs) Doxepin (50–200 mg qhs) Trimipramine (75–200 mg daily)
Monoamine oxidase inhibitors (MAOI)	Phenelzine (15 mg bid-tid) Tranylcypromine (10 mg bid-tid)
Reversible inhibitor of monoamine oxidase-A (RIMA)	Moclobemide (150–300 mg bid after meals)
Atypical antipsychotics	Aripiprazole (2–10 mg daily) Quetiapine (150–300 mg XR daily) Risperidone (1–3 mg daily) Brexipiprazole (0.5–2 mg daily) Olanzapine (2.5–10 mg daily) Ziprasidone (20–80 mg bid)
Other adjuncts	Buspirone (20–30 mg daily divided bid-tid) Lithium (600–1200 mg daily, target serum level 0.5–0.8 mmol/L) Triiodothyronine (T3) (25–50 mcg daily) Esketamine (56–84 mg intranasally 2×/week for weeks 1–4, then 1–2×/week; 28 mg initial dose if Japanese ancestry) Racemic ketamine (0.5 mg/kg IV over 40 mins, repeat at 0.5–1 mg/kg 1–3×/week) Lamotrigine (100–300 mg daily) Modafinil (100–400 mg daily) Pramipexole (1–2 mg bid)

vidual patient factors, which will be discussed later. Older classes of medication such as tricyclic antidepressants (TCAs) and monoamine oxidase inhibitors (MAOIs) are reserved for second- and third-line options because they tend to have a greater side effect burden for patients even though they have good efficacy. Various adjunctive medications, such as atypical antipsychotics, are available to use further along the treatment pathway in more difficult to treat cases.

Duration of Treatment

For patients using pharmacotherapy, treatment should be continued for at least 6 months after remission and can be continued for up to 2 years or lifelong in those with a high risk of recurrence [15]. An initial treatment phase of 6–12 months is now recommended, demonstrating improved treatment outcomes compared to stopping before 6 months. Patients who are at high risk of recurrence and will likely benefit from an extended duration of therapy are those who have one or more of the risk factors found in Table 22.5. The duration for non-pharmacological treatments is less clear, but often they can be decreased in frequency once remission is reached [43]. Interventions such as exercise and healthy diet do not have a defined duration but should be continued for their benefits to overall health.

Patients should be aware of the likely duration when therapy is started. Since the duration of maintenance therapy only starts once patients actually reach remission, the total length of treatment is usually not initially known. For example, Patient A may reach remission from their episode in 6 weeks, and Patient B may take 6 months. If both of these patients were unlikely to experience a relapse, they would both have at least 6 months of maintenance therapy, but Patient A would be treated for at least 7.5 months total and

Table 22.5 Factors to guide consideration of extended treatment [15]

Factors
History of childhood mistreatment ^a
Multiple, repetitive episodes
Unable to reach full remission on treatment, or have residual symptoms ^a
Severe episodes, especially if suicidality, psychosis, major impact on function
Persistent depressive disorder or dysthymia
Other comorbid psychiatric condition or physical condition that contributes to symptoms
Continued emotional/psychological stressors, lack of social support

^aThere is substantial evidence that these factors increase the likelihood of recurrence

Patient B for at least 12 months. Also, since stopping therapy prematurely is associated with poorer outcomes, [15] patients should know that they should complete the full duration of maintenance therapy after remission is reached.

Initial Assessment of a Patient Newly Diagnosed with Depression

Pharmacists have an important role to play in ensuring that pharmacotherapy is optimized once a patient has been diagnosed with depression and prescribed treatment. After assessing the history of the present illness, a careful assessment of their overall history must be done. This includes their full medical, medication, social, and family history, as well as assessing their allergies, functional status, medication adherence, and medication-taking beliefs. Reviewing physical exam findings and any lab values or diagnostic imaging will ensure there are no other drug-related problems that need to be addressed. When gathering this information from patients, it should be done in a conversational manner that focuses on building rapport with the patient. Patients may be uncomfortable discussing their mental health and medical history, so pharmacists should offer to move the conversation to a more private location, especially if working in a busy community pharmacy.

Patient assessment should include the following areas, and information pertinent to depression should be clearly documented to facilitate future follow-up. By gathering this information, it allows you to determine the patient's baseline and any current risks, as well as ensuring that therapy is indicated, effective, safe, and that they can adhere to it.

Gather Information About Depression

Ask about specific symptoms, and what led the patient to seek treatment. This may reveal which symptoms are the most bothersome to the patient and help to guide therapy selection. Determine if this is their first episode or if they have had previous episodes. If there have been past episodes, ask about duration and management. Ask about functional impairment and if the patient has had to miss days of work or school, or has trouble taking care of themselves or their children. Exploring functional impairment may reveal specific patient goals for recovery.

Depression rating scales can help to collect this information, as well as provide a method of monitoring improvement over time. CANMAT strongly recommends the use of Measurement-Based Care (MBC), which involves regularly administering validated rating scales, reviewing the results with patients, and using this information alongside clinical judgment to guide treatment decisions [15]. Using the same

scale each time, pharmacists can systematically track patient progress, detect nonresponse early, and adapt therapy accordingly. Try to use the same scale for the same patient every time so you can track their progress. In a community pharmacy setting, for example, a brief self-rated scale such as PHQ-9 can be completed while the patient waits for a new or refill prescription, yielding real-time feedback. Clinician-rated scales may be more suited to primary care or appointment-based practice settings because of the time and training needed for proper administration. However, either type of scale can be used regardless of practice setting, and each has benefits and limitations (Table 22.6). Regardless of the scale used, the routine use of validated scales to collect and monitor symptoms has been shown to improve medication adherence and overall outcomes in depression management [15].

Majority of depression scales are also available in multiple languages, allowing culturally competent care to be provided to patients. It is important to engage your patient in deciding which language of preference they would feel most comfortable with when selecting a scale.

Additional scales directed at quality of life and global functioning, such as the WHO Disability Assessment Scale and the EuroQoL-5D (EQ-5D), may also be employed to assess depression impact and progress across multiple domains [44, 45].

Suicide Risk

Given that half of individuals in Canada that die by suicide suffer from MDD, suicide risk assessment and tailored safety planning is a critical component of depression assessment [15]. All patients with depression should be assessed for risk of suicide, homicide, and self-harm at all stages of care. Some factors that should be considered when assessing the risk in a specific patient are included in Table 22.7. When approaching the subject of suicide with patients, the strength of the therapeutic relationship is essential: if a patient trusts you to have their best interests in mind, they may be more willing to tell you about suicidal thoughts. When patients do report suicidal ideation, the priority is keeping the patient safe. Questions that help to quickly identify patients who need immediate intervention include:

Do you have a specific plan?

Do you have the materials to carry out that plan?

If you leave here/go home, will you be safe?

Resources for patients who report suicidal ideation are available in the form of crisis phone lines, online, mental health professionals, and emergency rooms when needed. These resources will differ based on your location, so phar-

Table 22.6 Characteristics of common depression assessment scales

Scale	Scoring	Benefits	Limitations
HAM-D [28]	Clinician-rated Each item is scored from 0 to 4, then the points are totaled. Score of <7 indicates no depression, or remission if patient is on treatment. Mild depression is 7–17, moderate is 18–24, severe depression is >24 For the 21-item version, do not include items 18–21 in the score	Interview guide available, increases reliability High level of agreement between different reviewers and high test-retest reliability Widely used, most clinicians familiar	Some items include multiple concepts Does not assess atypical symptoms well Different symptom domains carry different weight (can overemphasize insomnia)
BDI [28]	Self-rated BDI-II is current version Each item scored from 0 to 3, points are totaled. Scores <11 indicate normal mood, 11–16 is mild mood disturbance, 17–20 is borderline depression, 21–30 is moderate depression, 31–40 is severe depression, and > 40 is extreme depression	Measure current symptoms Does not require training to administer Patients can complete privately so may be more honest Aligns with DSM-V criteria	Patients must be able to understand the language it is written in Physical illnesses may artificially elevate score Length may be overwhelming to some patients
IDS/QIDS [28]	Clinician-rated IDS-30 (30 items) and QIDS (16 items) Self-rated IDS-SR (30 items) and QIDS-SR (16 items) The scores are added up for each item according to the scoring guide. The guide and interpretation of scores for the different versions are found at: http://www.ids-qids.org	Better captures milder depression Includes atypical symptoms Each symptom domain has equal weight Items measure only one concept each	Scoring is complicated and requires scoring guide to do properly 30-item versions can take a long time to administer
Zung SDS [28]	Self-rated Each item scored 1–4, based on how often the item applies. For items 1, 3, 4, 7–10, 13, 15, and 19, a score of 1 indicates “a little of the time,” and 4 indicates “most of the time.” for items 2, 5, 6, 11, 12, 14, 16, 17, 18, and 20, the scoring is reversed with 1 applying to “most of the time” and 4 applying to “a little of the time.” Scores of 50–69 indicate depression, and >70 is severe	Captures a broader variety of symptoms compared to BDI Moderate correlation with the HAM-D	Has not been revised since initial publication in 1965 so it is not often used clinically Does not indicate mild depression categories Minimal assessment of atypical symptoms
PHQ-9 [46]	Self-rated Questions 1, 2: One or both is rated as 2 or 3 Questions 1–9: At least 5 questions rated as 2 or 3; however, question 9 can be rated as 1, 2, or 3 Question 10: Rates as “somewhat,” “very,” or “extremely difficult” To calculate score, total up each column. Scores of 10–14 indicate mild depression, 15–20 moderately severe, and > 20 as very severe	High sensitivity and specificity for diagnosing major depression Quickly rated and scored Includes patient’s view on level of impairment Provides treatment recommendations based on score	To score properly, need to have the appropriate forms with the correct format Does not include physical symptoms, anxiety
GDS [47]	Clinician-rated or self-rated Answers associated with depression are “yes” to items 2–4, 6, 8, 10–14, 16–18, 20, 22–26, and 28 and “no” to 1, 5, 7, 9, 15, 19, 21, 27, 29, and 30 For each answer that matches the previous answers, 1 point Scores 0–9 indicate no depression, 10–19 mild, 20–30 severe	Yes/no format is easily understood Uses items that are phrased to be acceptable in elderly patients Distinguishes non-depressed, mildly, severely depressed patients	Questions about somatic symptoms correlated poorly with total score so not included Not yet commonly used in trials so may be hard to compare trials with GDS scores
CDRS-R [48]	Clinician-rated Items scored from 1 to 6 or 1 to 7 and scores from each question added up Scores 28 or less indicate remission or no depression, while scores >40 indicate depression	Combination of reports from child, observer, parents, school, and others Good at detecting response to treatment	May take a long time to administer Requires training to properly administer
KADS [49]	Self-rated 11-item version has better sensitivity than other versions Each item rated 0–3 and score totaled; however, total score does not correspond to diagnostic categories and is meant to be a measure of tracking symptoms in a specific patient	Uses language that is familiar to adolescents Correlates well with clinician assessment of severity Good sensitivity to change	Has not been directly correlated with other scales No validated diagnostic categories associated with score, so cannot be used to diagnose

HAM-D Hamilton Depression Rating Scale, *BDI* Beck Depression Inventory, *IDS/QIDS* Inventory of Depressive Symptomatology/Quick Inventory of Depressive Symptomatology, *Zung SDS* Zung Self-report Depression Scale, *PHQ-9* 9-item Patient Health Questionnaire, *GDS* Geriatric Depression Scale, *CDRS-R* Children’s Depression Rating Scale Revised, *KADS* Kutcher Adolescent Depression Scale

Table 22.7 Factors that increase suicide risk [15, 50]

Factors	
<i>Age</i>	<i>Comorbidities</i>
Older males	Concurrent substance use disorders (especially alcohol use disorder)
Adolescents	Other mental health condition, especially cluster B personality disorders
	Sleep disorders
<i>Social or environmental factors</i>	Chronic pain or painful medical conditions
Higher degree of isolation, hopelessness, loneliness	Cancer
Lack of support network	Psychotic symptoms, anxiety
Family history of suicide	Traumatic events and/or post-traumatic stress disorder
High degree of impulsiveness	Past suicide attempts or self-harm
Stressful life events (e.g., financial stress and victimization)	Suicidal ideation with a well-developed plan and/or intent to act

macists should be aware of local resources and have contact information on hand to provide to patients.

A pharmacy-specific consideration for patients who report suicidal ideation is the supply of medication to provide. In a patient that you are concerned about, it might be appropriate to only provide a 1-week supply of medication at a time, or even dispense daily if you believe they are especially high risk. It may also guide the choice of therapy based on how great of a risk intentional overdose would pose. For example, an overdose of amitriptyline is much more likely to be fatal than one with citalopram [51].

Consider Patient Medical and Medication History

The goal of collecting this information is to ensure that there are no medical conditions or medications the patient is taking that may interfere with, or be affected by, their antidepressant. Also, if a patient has responded well to an antidepressant in the past, they may respond well to that same agent again. Antidepressants have the potential for many drug interactions, so knowing a patient's full medication list is important. Some of the common interactions are listed in Table 22.8. Look for medical conditions and medications that may contribute to depression (refer to Table 22.1), as well as those that are contraindications to certain medications (Table 22.9). One consideration that is often not assessed initially is the patient's risk of QTc interval prolongation. In patients who are at high risk (elderly, females, electrolyte abnormalities, recent myocardial infarction, use of other QTc prolonging agents, congenital long QTc) and are starting on antidepressants known to prolong QTc inter-

val (i.e., citalopram and escitalopram), an ECG should be done at baseline, and with dose adjustments if warranted.

Pregnancy and Breastfeeding

There is little evidence for most antidepressants in pregnancy. The most clinical experience and evidence exist for SSRIs, so they are often chosen if a drug must be started. In particular, citalopram and sertraline are supported with the most available evidence [54]. The evidence that exists does indicate some level of risk, but it needs to be balanced against that of untreated depression in the mother [55]. In patients already using antidepressants who become pregnant, the decision is made based on a discussion between the mother and clinician. For women already on an antidepressant that get pregnant, the risk of relapse is approximately sixfold in those with a history of severe, recurrent depression who discontinue treatment compared to those who continue treatment [56].

Consider Appropriate Dose

Many antidepressants, such as SSRIs, are started at half of the therapeutic dose to make sure the patient will tolerate it, and then increased after a week. For other agents like bupropion, the starting dose may be therapeutic. Since many are metabolized by the liver, it is important to check for dosing adjustments that need to be made in cases of liver damage. This applies to most SNRIs, vilazodone, bupropion, and more. Kidney function can also guide dose and is important to consider for all agents but particularly paroxetine, SNRIs, TCAs, and mirtazapine. Some antidepressants will also have maximum doses based on age or comorbidities. For example, the maximum dose of citalopram in hepatic impairment is 20 mg/day, even though the usual maximum is 40 mg/day [51]. Recommended dosing in common comorbidities can be found in individual drug monographs.

Consider Patient Preference

Pharmacists engaged in treatment recommendations should use shared decision-making to engage patients in decisions about their care by outlining the various evidence-based treatments for depression (e.g., lifestyle modifications, psychotherapy, pharmacotherapy, neuromodulation, CAM) and addressing any misconceptions they may hold [15]. For example, the difference between physical dependence and addiction should be explained in a way the patient can understand if this is a concern to them. Concerns about treatment side effects should also be discussed so that patients can decide what side effects they are willing to tolerate. Present an appropriate treatment option and explain

Table 22.8 Summary of drug interactions for antidepressant classes [52, 53]

Class	Interactions
All antidepressants	<i>Serotonin syndrome</i> MAOIs, linezolid, methylene blue (contrast medium), triptans, and many others Lithium and tryptophan may enhance serotonergic effects <i>St. John's wort</i> Herbal product that acts on serotonin, monoamine oxidase, and other targets to produce antidepressant effects Can cause serotonin syndrome, also has many interactions via liver microsomal enzymes Should not be used in combination with antidepressant drugs <i>Antiplatelet/anticoagulants/nonsteroidal anti-inflammatory drugs</i> Serotonin has an effect on platelet aggregation so increased bleeding risk with these combinations
Selective serotonin reuptake inhibitors (SSRI) Citalopram Escitalopram Sertraline Paroxetine Fluoxetine Fluvoxamine	<i>Cytochrome P450 liver microsomal enzymes</i> Many interactions since SSRIs are both substrates and inhibitors; check interaction reference every time Fluvoxamine and fluoxetine higher risk Paroxetine inhibition of 2D6 can have major consequences on drugs activated by it (e.g., tamoxifen, codeine) <i>Anticholinergic drugs</i> Paroxetine is highest risk <i>Additive sedation</i> Paroxetine and fluvoxamine highest risk <i>QT interval prolonging agents</i> Citalopram highest risk
Serotonin/ norepinephrine reuptake inhibitors (SNRI) Venlafaxine Desvenlafaxine Duloxetine Levomilnacipran	<i>Cytochrome P450 liver microsomal enzymes</i> Substrates and inhibitors of CYP isoforms, including 3A4, 2D6, and 1A2 Check drug interactions reference when combining with inducers, inhibitors, other substrates <i>Drugs with sympathomimetic effects</i> Increase heart rate and blood pressure
Serotonin-2 antagonist/reuptake inhibitor (SARI) Trazodone	<i>QT prolonging agents</i> <i>Inducers of CYP450 2D6 and 3A4</i> <i>Induces metabolism of p-glycoprotein substrates</i> <i>Additive sedation</i> CNS depressants <i>Anticholinergics</i> <i>Taking with food increases extent of absorption but decreases peak concentration and slows onset</i>
Serotonin-1A agonist/reuptake inhibitor Vilazodone	<i>Cytochrome P450 liver microsomal enzymes</i> Substrate of 3A4 (major), 2C19 and 2D6 (minor) Moderate inhibitor of 2C19 and 2D6 Maximum dose 20 mg if strong 3A4 inhibitor present <i>Slight inhibition of p-glycoprotein, monitor drugs with narrow therapeutic range (e.g., digoxin)</i> <i>Food increases absorption by about 50% so vilazodone should be taken with food</i>
Serotonin modulator and stimulator (SMS) Vortioxetine	<i>Cytochrome P450 liver microsomal enzymes</i> Metabolized by 2D6 and many other isoenzymes but does not induce or inhibit them Reduce dose by half if strong 2D6 inhibitor started
Noradrenergic/specific serotonergic agent (NaSSA) Mirtazapine	<i>Cytochrome P450 liver microsomal enzymes</i> Substrate of 2D6 and 3A4 <i>Enhances anticoagulant effect of warfarin, could be clinically significant in some patients</i> <i>Additive impairment with agents that cause sedation or motor impairment (ex: Alcohol, benzodiazepines)</i>
Norepinephrine/dopamine reuptake inhibitor (NDRI) Bupropion	<i>Cytochrome P450 liver microsomal enzymes</i> Substrate of 2B6 Inhibitor of 2D6 Metabolism induced by HIV protease inhibitors <i>Avoid combination with other drugs that predispose patients to seizures (ex: Steroids, antipsychotics, tramadol, alcohol)</i>

(continued)

Table 22.8 (continued)

Class	Interactions
Tricyclic antidepressants (TCA) Amitriptyline Nortriptyline Imipramine Clomipramine Desipramine Doxepin Trimipramine	<i>Cytochrome P450 liver microsomal enzymes</i> Substrates of 2D6, 1A2, 2C19, and others Inhibitors of 2D6 Clomipramine interaction with grapefruit juice <i>Can exaggerate vasopressor response (ex: Epinephrine)</i> <i>Agents that prolong QT interval</i> <i>Anticholinergic agents</i> <i>Additive sedation and CNS depression (e.g., alcohol, opioids, benzodiazepines)</i> <i>Dopaminergic effects exaggerated with dopamine increasing agents</i>
Monoamine oxidase inhibitors (MAOI) Phenelzine Tranylcypromine	Numerous severe interactions with other medications and food, check product monograph and drug interaction resource every time and provide patient with a list of medications and foods to avoid
Reversible inhibitor of monoamine oxidase A (RIMA) Moclobemide	<i>Cytochrome P450 liver microsomal enzymes</i> Substrate and inhibitor of 2C19 and 2D6 <i>Anticholinergic agents</i> <i>Hypertensive effects with alpha/beta-agonists, amphetamines, buspirone, methylphenidate</i>

Table 22.9 Patient-specific factors to guide selection of antidepressant drugs [41, 52]

Class	Absolute contraindications	Relative contraindications/cautions	Other patient factors to consider
All classes of antidepressants	Combination with MAOI Patient with bipolar depression and no additional mood-stabilizing therapy Prior allergic reaction to that medication	Pregnancy and breastfeeding Renal or hepatic impairment Seizure disorder or history	Comorbid conditions Risk in overdose Financial burden and drug coverage
Selective serotonin reuptake inhibitors (SSRI) Citalopram Escitalopram Sertraline Paroxetine Fluoxetine Fluvoxamine	Class: Use within 1 week of MAOI (5 weeks if fluoxetine) and RIMA Citalopram and escitalopram: Known or congenital QT prolongation Paroxetine: Pregnancy	Class: Bleeding disorders Severe liver dysfunction Diabetes Citalopram and Escitalopram Use of other QT-prolonging agents; max dose 20 mg (cit) and 10 mg (escit.) Paroxetine: Elderly	Class: Indicated for anxiety disorders Safer in overdose compared to TCAs Citalopram, escitalopram, fluoxetine, fluvoxamine: Health Canada approved for use in patients <18
Serotonin/ norepinephrine reuptake inhibitors (SNRI) Venlafaxine Desvenlafaxine Duloxetine Levomilnacipran	Class: Use within 1 weeks of MAOI and RIMA Desvenlafaxine Gastrointestinal narrowing/strictures, conditions that affect GI transit Duloxetine: Severe/end-stage renal impairment Hepatic failure Uncontrolled glaucoma Levomilnacipran: History of stroke, cardiovascular disease Past cardiac intervention NYHA class III or IV heart failure	Class: Bleeding disorders Hypertension Diabetes Venlafaxine: Poor adherence Desvenlafaxine: Cardiovascular, cerebrovascular conditions Lipid disorders	Desvenlafaxine, Levomilnacipran: Cost is important to consider, as they are more expensive than other agents Duloxetine: Indicated for chronic and neuropathic pain
Serotonin-2 antagonist/ reuptake inhibitor (SARI) Trazodone	Use within 1 week of MAOI and RIMA Men: Other risk factors for priapism (e.g., sickle cell disease)	Elderly patients Orthostatic hypotension Recent myocardial infarction	Insomnia in combination with other antidepressants or alone Manage “sundowning” in dementia
Serotonin-1A agonist/ reuptake inhibitor Vilazodone	Use within 2 weeks of MAOI and RIMA	Cardiovascular disease Hypertension Patient at risk of arrhythmia	Expensive, consider cost

(continued)

Table 22.9 (continued)

Class	Absolute contraindications	Relative contraindications/cautions	Other patient factors to consider
Serotonin modulator and stimulator (SMS) Vortioxetine	Use within 3 weeks of MAOI and RIMA	Severe hepatic impairment	Expensive, consider cost
Noradrenergic/ specific serotonergic agent (NaSSA) Mirtazapine	Use within 2 weeks of MAOI and 1 week of RIMA	Past myocardial infarction or angina History of stroke Prostatic hypertrophy	Sedation can benefit insomnia Helpful in patients who will not eat because stimulates appetite Weight gain can be significant
Norepinephrine/ dopamine reuptake inhibitor (NDRI) Bupropion	Use within 1 week of MAOI Current or past seizure disorder Patients undergoing withdrawal from alcohol or benzodiazepines Current or past bulimia or anorexia	Insomnia Anxiety Alcohol consumption Patients who have weight loss as a symptom Doses >150 mg daily in severe renal or hepatic impairment	Indicated for smoking cessation (brand: Zyban) More activating so beneficial in patients who are fatigued
Tricyclic antidepressants (TCA) Amitriptyline Nortriptyline Imipramine Clomipramine Desipramine Doxepin Trimipramine	Use within 2 weeks of MAOI (3 weeks for clomipramine and imipramine) and 1 week of RIMA Recent myocardial infarction Cross-reactivity within class for hypersensitivity reactions Patients at high risk of overdose	Any personal or family cardiovascular history Elderly Prostatic hyperplasia Glaucoma Thyroid dysfunction	Class: Benefit in neuropathic pain Doxepin Insomnia Amitriptyline, nortriptyline: Migraine prophylaxis
Monoamine oxidase inhibitors (MAOI) Phenelzine Tranylcypromine	Use of other serotonergic drugs within 1–2 weeks* (3 weeks for vortioxetine, clomipramine, and imipramine, 5 weeks for fluoxetine) Use of bupropion within 2 weeks Cerebrovascular or cardiovascular disorders Recurrent or frequent headaches Liver damage Blood dyscrasias Pheochromocytoma Tyramine-containing foods: Check reference	Any foods that are not clearly fresh and free from preservatives can present a risk, even if stored properly in a refrigerator Hypertension Overdose can be fatal	May have better response in catatonic or treatment resistant patients
Reversible inhibitor of monoamine oxidase A (RIMA) Moclobemide	Use of meperidine Use of other serotonergic drugs within 24 h (2 weeks for vilazodone) Use of bupropion within 24 h	Renal impairment Overdose can be fatal	Avoids the dietary restrictions of irreversible MAOIs

*Consult drug resources for specific washout periods

why you are recommending that specific approach; present at least one alternative treatment option to engage the patient in selecting a care plan that aligns with their individual values and needs. Emphasize the importance of achieving symptom remission and maintaining long-term mental wellness. Consider practical factors such as local resource limitations, and direct patients to accessible psychoeducational materials (e.g., CANMAT's free CHOICE-D guide at www.CANMAT.org) [33].

Consider Social and Family History

Assess the use of alcohol, caffeine, cannabis, and other recreational drugs. These substances may produce symptoms of depression or be used by patients to alleviate symptoms. Withdrawing from substances may also play a role. Ask about the patient's occupation, family life, financial situation, overall lifestyle, and any major stressors that exist for them. These factors can help in assessing appropriateness of

therapy and their ability to adhere to medication and engage in treatment. Patients who have a first-degree relative with depression or other mood disorders are more likely to have depression themselves [57]. Patients may also respond to the same medications that family members have responded to. A family history of other mood disorders can also provide alternate diagnoses to consider if symptoms that are not congruent with depression are present.

Consider Barriers to Adherence

Antidepressant adherence is notoriously poor. One in five patients will not be adherent to their antidepressant prescription [58]. By addressing factors of intentional and non-intentional non-adherence early on, many of them can be mitigated. Practically, consider if patients can actually take the medication. Certain medications, like fluoxetine and escitalopram, come in liquid or orally dissolvable forms to assist patients who cannot swallow pills. Also, many of the newer agents are very expensive compared to equally effective older agents. For example, citalopram and vortioxetine are both first-line agents, but based on Canadian pricing, citalopram would cost a patient around \$14/month, while vortioxetine would be closer to \$120/month [41]. Depressive symptoms may present their own barriers in terms of low motivation or cognitive impairment. Pharmacists should also explore any psychological or attitudinal factors that may affect a patient's willingness to use antidepressants, such as doubts about whether they truly need treatment, embarrassment or stigma associated with having depression, or general hesitancy toward medications.

Physical Exam and Labs/Diagnostic Imaging

These investigations may be done by the patient's physician in the process of diagnosis and are available on request or via a provincial electronic health record. They may also be ordered by other health professionals, including pharmacists working in a collaborative team environment or in alignment with their province's scope of practice. In addition to confirming the diagnosis of depression, baseline information is needed when patients are started on medication to monitor for adverse effects. Specific values that should be noted at initial assessment are thyroid-stimulating hormone (TSH), a complete blood count (CBC) and vitamin B12, liver enzymes (LFTs), and serum creatinine (SCr). Weight should also be documented, especially if starting on antidepressants associated with weight gain or loss. Glucose and lipid profiles should also be noted particularly if starting on antidepressants or other medications associated with weight gain or changes in these parameters (eg., mirtazapine, antipsychot-

ics). A baseline ECG should be assessed in individuals with cardiovascular disease, or other compelling risk factor(s), who may be starting on medications that can prolong the QTc interval. If baseline lab values and diagnostic tests are not available, pharmacists should ensure they are ordered in a timely manner.

Follow-Up Assessment

The style of follow-up assessments may vary based on the practice setting and can be done over the phone or via in-person visits. When a patient is newly started on treatment, they should be followed up with at least every 2 weeks, if not weekly, to check for tolerance and early response, as well as any adverse effects and to assess safety risk. Once they are stabilized on therapy, follow-up intervals can be extended to every month or even every 3 months if they are in the maintenance phase of treatment.

Suicide Risk

Monitor for suicidal behavior throughout antidepressant therapy, paying particular attention to the first month before and after initiating an antidepressant, as risk is elevated during these periods and declines with continued use. Discontinuation also poses heightened risk, so the month following the end of treatment should be approached with vigilance [15]. While most patients experience a decrease in suicidal thoughts on medication, a small subset—especially adolescents and young adults—may see a temporary increase in ideation when starting a new antidepressant [15]. CANMAT recommends routine suicide risk assessment for all patients, with enhanced monitoring during these high-risk windows. For younger patients, in particular, explain that if they notice any spike in suicidal thoughts, it may be a side effect that calls for urgent intervention, including implementing a safety plan and contacting crisis lines, or visiting an emergency department. Family and close friends should be engaged in the safety plan, if appropriate for a specific patient and with consent.

Adherence

Medication adherence is very important with antidepressants. Patients should be advised not to discontinue their treatment without consulting their psychiatrist or other original prescribers. Even if patients have achieved remission, they should continue their treatment for the entire duration of therapy because patients who discontinue treatment early have an overall worse prognosis and a higher likelihood of

relapse [59]. When asking about adherence, it is important to do so in a non-judgmental way that invites the patient to tell you about their concerns, instead of making them feel like they are in trouble for not taking their medication perfectly. Pharmacists should work with patients at each encounter to maximize their ability to take their medication through identifying barriers and helping the patient to find solutions. Adherence aids such as pill boxes and reminders can be helpful for patients who have trouble remembering to take their medication. Pharmacists should also ask if any medication side effects are contributing to non-adherence.

Efficacy

The goal of treating a depressive episode is to have a patient reach full remission. This is defined as no longer meeting diagnostic criteria for the depressive disorder, and a return to their normal level of functioning. This definition does include the possibility of some lingering symptoms, which many patients will have. Patients may not reach remission with one antidepressant, and in fact only about 1/3 will achieve remission with the first agent tried [31]. Most patients will take 6–8 weeks to reach remission once they are taking a therapeutic dose [15].

Response is defined as at least a 50% improvement on the score of a rating scale. This is why a baseline score is important and why the same scale should be used for the same patient each time. Partial response is defined as between 25% and 50% decrease in the score, while non-response is less than 25% change [31].

In the first 2–4 weeks of therapy, patients may have an early response (indicated by a 20% reduction in their symptom score [15]), and this is a positive predictor of reaching full remission with the agent they are on. The first symptoms that patients will notice changes in are often appetite, sleep, and motivation. Not having this early response does not mean they cannot respond, but if by 4 weeks they have seen no improvement, there is less chance of remission occurring with that particular agent [31]. Increasing the dose can be tried if early response is not seen and the drug is being tolerated well.

Repetition of depression scales, following MBC principles, is the most common way to assess control of depressive symptoms [15]. Since most scales consider symptoms over the past 1–2 weeks, they should ideally be assessed every 2 weeks. Pharmacists may not be seeing the patient that often, but scales should be administered as often as is practical based on your setting, and the scores recorded in a care plan. Pharmacists taking a more active role in prescribing for depression should ensure follow-up visits are scheduled based on these timelines to adequately monitor their treatment recommendations and implementation. Although blood

drug levels can be drawn, it is not recommended because it does not change management in most cases. It could be useful in a minority of patients if there are questions of adherence or suspicion that they are ultra-rapid or slow metabolizers [15]. There are no lab parameters that correlate well with efficacy of antidepressant therapy.

Even though remission/responses are defined numerically, also consider the patient and the specific goals that were set at the start of therapy. Ask about their progress toward these goals, especially if there were functional or symptom-related goals that they wanted to achieve. Even if they have reached the definition of remission on paper, if there are bothersome residual symptom(s), they should be addressed, both for patient function and because residual symptoms are considered a major risk factor for episode recurrence [15]. For some symptoms like insomnia, consider that they may indicate comorbid disorders. Patients with PDD may never reach full remission; when this is the case, the focus should shift to optimizing function and quality of life [15].

For Patients Who Do Not Reach Remission

Since many patients will not reach remission with the first medication, having a plan on how to address lack of response is a good idea. Before making any changes to antidepressant therapy, always ensure you have assessed adherence and that the trial was an adequate dose and duration. If psychotherapy has not yet been trialed, this may be a good time to explore it as it is better to initiate early in the treatment journey. It is also important to reassess diagnosis and any concurrent psychiatric and medical conditions, as well as life stressors, that could be impacting response to treatment.

If the medication dose has been subtherapeutic or it provided a partial response and was well-tolerated, increasing the dose within the therapeutic range is a reasonable next step. If increasing the dose is not necessary or not tolerated, the two main options are switching to a different antidepressant or adding on another therapy. Patient preference should always be considered when making further treatment recommendations. It is generally recommended to switch to another antidepressant if there are tolerability issues with the first one and to add an adjunctive medication if there is a partial response to the first antidepressant and it is well tolerated. Table 22.10 outlines additional considerations for switching versus adding on therapy. Clinical studies demonstrate superior efficacy with similar tolerability in favor of adjunctive treatment, but this should be considered alongside patient preference and individual benefits and risks.

If switching antidepressants is selected, consider choosing one with a different mechanism of action and stronger evidence of efficacy. If adding an adjunctive medication is

Table 22.10 Factors to consider when deciding whether to switch or add therapy [15, 58]

Favors switching	Favors adding
First antidepressant trial	2+ failed trials
Poorly tolerating current antidepressant	Low potential for interactions
High potential for drug interactions	Patient can manage added pill burden
High pill burden	Targeting specific symptoms (ex: Insomnia)
No response (<25%)	At least partial response (25–50%)
Less urgent need for control (no suicidality or severe functional impairment)	Need for control is more urgent
	Previous poor tolerance of discontinuation syndrome

selected, first-line recommendations include aripiprazole and brexpiprazole with second line options including bupropion, intranasal esketamine or intravenous ketamine, mirtazapine, quetiapine-XR, risperidone, olanzapine, lithium, modafinil, triiodothyronine [15].

Some patients may have depression that is more difficult to treat. The term “treatment resistant depression (TRD),” described as failure of 2 or more antidepressants, is less commonly used now due to its reliance on medication agents alone and the lack of hope associated with the term failure. “Difficult-to-treat depression (DTD)” has now been introduced, which describes a longer course of persistent depression that does not respond adequately to multiple standard treatments and is further along in the treatment trajectory [15]. TRD may still be used in certain clinical contexts (e.g., when referring to responses to medications alone) and research studies. This means that the patient has not reached full remission, but they may still have had some degree of improvement. While neuromodulation treatments for depression may be appropriate as first-line treatment for some patients (e.g., ECT for severe depressive episodes), CANMAT generally recommends these treatments after at least one switch or adjunctive strategy described above has been trialed due to feasibility issues with accessing treatment [15]. r-TMS and ECT are most commonly recommended as both first and second line options.

Strategies for Switching Agents

When patients are switching antidepressants for any reason, pharmacists are responsible for helping them through the process safely. It is important to consider the nature of the agents that are being used. If switching between different agents in the same class, often you can immediately replace one with a full therapeutic dose of the other because their similar mechanisms of action will minimize the risk of dis-

continuation syndrome, for example, citalopram to sertraline and venlafaxine to desvenlafaxine. The patient should still be aware that some withdrawal symptoms are possible. Another strategy is to cross taper the medications, which means that the new medication is started at the lowest available dose while still on the old medication and titrated up to the lowest therapeutic dose. Then, the old drug is tapered down slowly. This can be beneficial for patients who have more severe depression because it does not leave them with a “drug-free interval.” However, it can potentially lead to increased side effects while both medications are being used. This should never be done with MAOIs, as the risk of adverse effects is too high. When you are concerned about combining the agents, as with MAOIs, a washout period is necessary. When switching to or from MAOIs or moclobemide, the usual washout period is 2 weeks. There are exceptions (e.g., 5 weeks for fluoxetine to MAOI), so the monographs of both drugs should be checked every time.

An excellent resource that can guide switching antidepressants is www.SwitchRx.ca, and it can also be used to create tapering schedules if discontinuing an antidepressant altogether.

Adverse Reactions

Patients should be asked about any side effects they have noticed since starting their antidepressant medications. It is important to remember that many common side effects of antidepressants (e.g., nausea, headache, drowsiness) are more common in the first 2 weeks of starting an antidepressant but remit with continued use, while others may occur on an ongoing basis. When assessing an adverse effect that is thought to be related to a drug, ensure that the time course makes sense. Good examples are sexual dysfunction or insomnia, as both can be symptoms of depression, but could also be adverse effects of medication. Having a list of other medications they are taking will also help to identify if reactions are related to a drug interaction. It is important for pharmacists to ask about specific side effects because patients may not attribute a side effect to the medication or may not report it unless asked directly.

Some adverse reactions are common across all classes of antidepressants, while others may be class-specific. Since all current antidepressants have some role in increasing serotonin, associated adverse effects are possible for all classes. Bupropion tends to have less serotonergic effect than other antidepressants, so it may have lower rates of serotonergic side effects.

Some common adverse reactions are presented in Table 22.11, as well as those that are rare but severe enough that patients should be aware. A complete list of all adverse effects associated with each drug may be found in the individual monographs.

Table 22.11 Selected adverse effects of different antidepressants [18, 41, 52]

Drug or class	Adverse effect	Level of concern	Notes
All antidepressants	Serotonin syndrome	Rare but serious, requires referral	Usually occurs when multiple serotonergic medications are combined, or an overdose; rapid onset within 24 h Symptoms: Changes in mental status, diarrhea, hyperreflexia, agitation, tremor, seizures, arrhythmias, rhabdomyolysis, disseminated intravascular coagulation, respiratory arrest
	Suicidal ideation, behavioral disturbances	Rare but serious, requires referral	More common in adolescents/young adults More common in first few weeks of therapy Associated with agitation, akathisia, emotional lability, aggression, depersonalization Monitoring: Pharmacist to ask every 2 weeks until stable, then monthly
	Manic switch	Quite common, can be serious and requires referral	Occurs in patients with bipolar disorder only treated with antidepressants, or patients misdiagnosed with depression instead of bipolar disorder.
	Increased bleeding risk	Common, clinical significance varies	Concern when patient is also using anticoagulants, antiplatelets, or nonsteroidal anti-inflammatory drugs Ranges from mild bruising to severe bleeds Monitoring: Bleeding/bruising drug interactions
	Sexual dysfunction	Very common, may be mild or serious impairment	Patients rarely report without being asked Lower rate with mirtazapine, bupropion, moclobemide Common reason for non-adherence Monitoring: Pharmacist to ask, as patients rarely self-report
	Acute angle-closure glaucoma	Rare but serious, requires referral	Medical emergency, risk of permanent vision loss Symptoms: Rapid onset of severe pain in the eye(s), blurred vision, redness Patients who already have elevated intraocular pressure or narrow ocular angles are at higher risk
	Increased risk of fractures	Uncommon but can be serious	Especially in patients who are using antidepressants for long-term treatment Dizziness/drowsiness can contribute to fall risk and fractures
All SSRIs	Nausea, headache, anxiety, drowsiness, dizziness, sweating, tremor, insomnia	Common, usually mild	Tend to remit with continued therapy, approximately 2 weeks Usually can be alleviated with non-pharmacological methods or changing administration time
	Weight gain/loss	Uncommon, usually minor	Most SSRIs are weight-neutral; however, paroxetine associated with weight gain, fluoxetine with anorexia and weight loss Weight gain can be related to recovery of appetite when treatment is started
	Hyponatremia	Uncommon, can be serious and requires referral	Related to SIADH: Syndrome of inappropriate antidiuretic hormone More common in elderly Symptoms: Nausea, vomiting, confusion, muscle cramps/weakness, seizure Monitoring: Na ⁺ level at baseline and if symptomatic

(continued)

Table 22.11 (continued)

Drug or class	Adverse effect	Level of concern	Notes
Citalopram, escitalopram (less common)	QT prolongation	Common, clinical significance varies (escitalopram not considered clinically relevant [15])	QT prolongation is common and dose related, unless elderly/other risk factors, arrhythmia and sudden cardiac death very rare Symptoms: Palpitations, feeling faint/dizzy Monitoring: K+ and Mg2+, high-risk patients ECG done at baseline and if symptomatic
Fluoxetine	Stimulating	Common, mild	Most stimulating SSRI
Fluvoxamine	Nausea, sedation, constipation	Common, can be more significant	Worse initially, and should decrease within approximately 2 weeks
Sertraline	Sexual dysfunction, diarrhea	Common, can be more significant	Highest rate among SSRIs Diarrhea often remits in about 2 weeks but sexual dysfunction does not
Paroxetine	Anticholinergic: Dry mouth, dizziness, urinary retention, constipation, blurred vision	Common, can be more significant	Highest rate of all SSRIs Controlled release form might have less noticeable or less severe adverse effects than immediate release for some patients [60]
All SNRI	Same as SSRI class effects	Common, usually mild	Usually resolve in first 2 weeks of therapy
	Elevate blood pressure and heart rate	Common, usually mild	Tends to be sustained for duration of therapy Palpitations can also occur Monitoring: Baseline blood pressure and heart rate, patient to check regularly
	Dry mouth	Common, usually mild	Dose-related, may be worse with venlafaxine and duloxetine
	Nausea and GI upset	Common, can be significant	Starting at low dose and titrating up slowly will decrease, and it usually remits within 2 weeks of reaching therapeutic dose Less frequent with desvenlafaxine
Venlafaxine, Desvenlafaxine	Elevated LDL, triglycerides, liver function tests	Relatively common, usually not significant	Monitoring: Baseline lipid panel and liver function tests, repeat yearly unless symptomatic or other risk factors present
Duloxetine	Elevated liver function tests	Uncommon and usually minor	Monitoring: Baseline liver function tests, repeat if symptomatic
Levomilnacipran	Hyperhidrosis	Common, mild-moderate	Patient to report if bothersome, may be managed with non-pharmacological methods
Trazodone	Erectile dysfunction and urinary hesitancy	Uncommon but may be significant	Dose-related
	Priapism	Rare but serious, requires referral	Medical emergency and may lead to permanent impotence if not treated
	Orthostatic hypotension	Quite common, can be significant	Higher risk in elderly patients, can lead to falls
	Anticholinergic effects	Quite common, can be significant	Dose-related, elderly patients more sensitive
	Sedation	Common, significant impairment	Should be taken at bedtime based on degree of sedation May affect ability to drive or operate machinery safely
Vilazodone	QT prolongation	Uncommon, clinical significance varies	Arrhythmia and sudden cardiac death very rare, but should avoid using soon after cardiac event Symptoms: Palpitations, feeling faint/dizzy Monitoring: K+ and Mg2+, high-risk patients ECG done at baseline and if symptomatic
	Nausea, vomiting, diarrhea	Common, usually mild	Usually occurs during initial treatment period and titration and remits within 1–2 weeks
	Weight gain	Common, usually mild	Approximately 1–2 kg increase in body weight over 1 year of treatment
	Muscle spasm	Common, usually mild	Few patients in trials discontinued the medication due to muscle spasm

(continued)

Table 22.11 (continued)

Drug or class	Adverse effect	Level of concern	Notes
Vortioxetine	Hypertension	Uncommon, little clinical significance	More significant increases in blood pressure (2–5 mmHg systolic) in the first 14 days of treatment but returned to normal with longer term use Monitoring: Baseline blood pressure and patient to check regularly
	SSRI-like side effects	Uncommon, usually mild	Nausea comparable to SSRIs, but otherwise much lower incidence of adverse effects including sexual dysfunction, sedation, insomnia
Mirtazapine	Weight gain	Common, significant	Many patients gain greater than 7% of their current body weight Monitoring: Body weight at baseline and regularly
	Sedation	Common, moderately significant	More common at lower doses because at higher doses, noradrenergic effect counteracts sedation May impact ability to drive or operate machinery
Bupropion	Insomnia, agitation, tachycardia, diaphoresis	Common, moderately significant	Administering in the morning minimizes effect of insomnia Monitoring: Heart rate, sleep
	Seizures	Uncommon but serious	Dose related, see monograph for more information Monitoring: Electrolyte abnormalities increase seizure risk
	Weight loss	Common, may be significant	May also be associated with decreased appetite Risk in patients with eating disorders like anorexia/bulimia
All TCAs	Anticholinergic effects	Common, can be significant	High risk in elderly patients—Question indication Interactions with other medications that have same effect will be additive
	Antihistamine: Weight gain, dizziness, drowsiness	Common, can be significant	Weight gain may affect adherence significantly Drowsiness/dizziness can affect ability to drive or operate machinery Monitoring: Baseline weight
	Cardiotoxicity: Arrhythmias, orthostatic hypotension	Common, may be clinically significant	At normal doses as well as overdose Contributes to lethality of overdoses
All MAOIs, RIMA	Hypertensive crisis	Common, usually severe, requires referral	Higher risk with irreversible inhibitors but still possible with RIMA Be very careful with drug interactions

SSRI selective serotonin reuptake inhibitor, SNRI serotonin norepinephrine reuptake inhibitor, GI gastrointestinal, K⁺ potassium, Mg²⁺ magnesium, TCA tricyclic antidepressant, MAOI monoamine oxidase inhibitor, RIMA reversible inhibitor of monoamine oxidase-A

Discontinuation Syndrome

When discontinuing an antidepressant, symptoms associated with discontinuation syndrome are common. The symptoms include flu-like symptoms, insomnia, nausea, imbalance, sensory disturbances including shock-like sensations, and hyperactivity and can be remembered using the acronym “FINISH.” Symptoms are usually not severe but, in some cases, may be disabling and can last 1–3 weeks. This can occur with all antidepressants but SSRIs and SNRIs are especially implicated. It is generally recommended to reduce

the dose by 25–50% every 2–4 weeks to reduce the risk of discontinuation syndrome, however, agents with shorter half-lives, such as paroxetine, venlafaxine, desvenlafaxine, and duloxetine, are usually associated with worse symptoms, and need to be tapered more slowly [61]. A slower taper involves reducing the dose by 25–50% over several months. A slower taper should also be considered if a high dose of antidepressant has been used, if used for a long duration, if the patient is at high risk of relapse, or if withdrawal symptoms have previously been an issue [61]. If patients stop their antidepressant abruptly, they are at risk of experiencing dis-

continuation syndrome, and patients who miss doses may also experience a milder form. Even in patients who are tapered slowly, some symptoms can occur. One strategy to ease the process of discontinuation is to provide the patient with a single dose of fluoxetine, as its long half-life causes it to essentially taper itself off over the course of 3–5 weeks [62]. If symptoms are severe, the original antidepressant can be restarted to alleviate symptoms and discontinuation can be retried later if the patient is willing.

Complications of Disease

There are several complications associated with depression. In addition to the increased risk of suicide discussed earlier, depression is also associated with cardiovascular disease, stroke, pregnancy complications, falls in elderly patients, and poor recovery after surgery.

Children of parents with poorly managed depression may have their cognitive and interpersonal development affected, and can experience problems with emotional regulation, internalizing, and attachment disorders. They are also at a higher risk of developing depression themselves.

Many patients who experience one depressive episode will also experience relapse or recurrence at some point in their life. This may be symptoms, a full major depressive episode or even persistent depressive disorder. Relapse is often defined by symptoms presenting again while treatment is ongoing but the patient has reached remission. Recurrence is usually after the patient has been asymptomatic for a longer period and then experiences another episode [59]. These terms are often used interchangeably.

Clinical Pearls

- Pharmacists are in a perfect position to screen patients for depression. Having a strong therapeutic relationship with patients will encourage them to report distress to you.
- Stigma plays a big role in the number of patients who are unwilling to seek help for depression and other mental illnesses. Be a non-judgmental resource for your patients who may be experiencing depression.
- Talk to every patient with depression (and their family if possible) about suicide risk.
- Involving the patient in decisions about management will increase their satisfaction with therapy and their adherence.
- Symptoms of depression are not a normal part of aging and should not be ignored in the elderly.
- Many patient and clinician resources for depression exist. Some of them are listed here.

- Patients:
Canadian Mental Health Association.
Mood Disorders Society of Canada.
Centre for Addiction and Mental Health (CAMH).
Canadian Network for Mood and Anxiety Treatment
CHOICE.D Guide.
- Clinicians:
Canadian Network for Mood and Anxiety Treatment.
Switch Rx.
American Psychiatric Association.

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Chapter Objectives

1. To identify key features of chronic non-cancer pain as a pathological form of pain.
2. To review assessment strategies that allow the clinical pharmacist to assess pain and evidence of reduced coping in chronic pain patients.
3. To link pain and coping assessments to pharmacotherapy optimization and management.
4. To assist the patient in discovering new, meaningful life strategies for living with chronic pain.

Background

In 1931, physician, theologian, philosopher, and missionary Albert Schweitzer opined that “Pain is a more terrible lord of mankind than even death itself.” Worldwide, the prevalence of pain varies depending on the specific definition. A global 2021 study estimated the prevalence of pain varied substantially across countries, ranging from 9.9 to 50.3% with a mean of 27.5%. Similar to other studies, this study demonstrated that the odds of experiencing pain are higher for women, older persons, and those living in rural settings [1]. Chronic pain epidemiology varies depending on the definition and pain type. A recent US study estimated that 20.9% of US adults (51.6 million persons) experienced chronic pain, and 6.9% (17.1 million persons) experienced high-impact chronic pain, which results in a substantial restriction to daily activities [2]. Analysis from the global burden of disease in 2015 reported that low back and neck pain were ranked the fourth leading cause of disability-adjusted life years (DALYs) globally just after ischemic heart disease,

cerebrovascular disease, and lower respiratory infection, in thousands at 94,941.5 (95% uncertainty interval (UI) 67,745.5–128,118.6). This suggested that in 2015, over half a billion people worldwide had low back pain, and more than a third of a billion had neck pain of more than 3 months duration [3].

Based on the Global Burden of Disease Study 2016, pain and its related conditions have been attributed as the leading cause of disability. The most common chronic pain condition is recurrent tension-type headaches, affecting 1.9 billion people worldwide. In the United Kingdom, 13–50% of adults were estimated to be living with chronic pain, of which 10.4–14.3% have been moderately to severely disabled by the condition [4]. A 2019 analysis of Canadian data using Statistics Canada’s Health Survey estimated that “...7.6 million, or one in five people (across their lifespan), live with chronic pain” [5].

Chronic pain also demonstrates an age-related trend with the prevalence increasing through adult life, reaching a peak around the seventh decade [6]. Despite this age relationship, the global prevalence of chronic pain in young adults has been estimated to be 11.6% (95% CI 9.4–14.3), suggesting that chronic pain should not be overlooked in younger patients [7].

The classical definition of pain, as recommended by the International Association of Pain is “An unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage” [8]. However, this definition does not accurately describe chronic pain, as chronic pain is not simply the sustained activation of acute pain pathways. In addition, there is enormous heterogeneity observed in chronic painful syndromes, so the term chronic pain has little diagnostic value and even less value for the determination of pharmacotherapy. A detailed taxonomy of non-cancer-related pain syndromes with specific treatment recommendations is beyond the scope of this chapter. However, within the confines of chronic non-cancer pain (CNCP), fundamental pain assessment and pharmacotherapy recommendations can be provided for clinical pharmacists

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as part of their ongoing assessment of pain pharmacotherapy. As patients have greater access to pharmacists, good pain assessment information can be vital for pharmacotherapy monitoring and for a critical assessment alerting the patient and the patient's primary care or pain physician of developments that may require implementing changes to both drug and non-drug therapy. The current opioid crisis, along with increased prevalence of chronic pain after severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic [9], further emphasizes the need for pharmacists to play a key role in optimizing the therapeutic benefit from drug therapy while minimizing the risk of addiction, toxicity, and adverse reactions. Pharmacists have a vital role to play in this setting by optimizing the use of analgesic and adjunct medications and appropriately de-prescribing when harm outweighs benefits. It must be emphasized that the majority of people with chronic painful conditions experience significant pain that impairs their quality of life, resulting in significant physical disability and emotional distress. Chronic pain also affects the patient's significant others such as partners, relatives, employers, co-workers, and friends. An essential part of assessing the appropriateness of pharmacotherapy must include an examination by the clinical pharmacist of BOTH biological and psychosocial factors. Psychosocial factors, such as behavioral presentation, including their emotional state (e.g., anxiety, depression, and anger), perception and understanding of symptoms, and reactions to those symptoms by significant others, provide valuable indicators of a patient's ability to cope with the chronic pain. This multidimensional assessment provides valuable information for assessing the efficacy or the need for ongoing pharmacotherapy or signals the need for a professional psychosocial intervention, such as cognitive-behavioral. The clinical pharmacist must be aware of the need for an integrated interdisciplinary team approach for treating CNCP that must include non-pharmacological as well as pharmacological techniques.

Acute Versus Chronic Pain

Why is chronic pain so difficult to treat? It is difficult because chronic pain is not simply the sustained activation of acute pain pathways and processes. Unchecked pain barrage leads to changes in the central and peripheral processing mechanisms of pain. Sustained acute pain leads to chronic pain conversion, and pain transforms from a physiological adaptive mechanism designed to protect the body from harm to an evolving pathophysiological process. We refer the reader to the excellent review by Woolf [10]. An unchecked pain barrage may also lead to loss of inhibitory pain mechanisms and neurons, which may render analgesics ineffective. These features of acute and chronic pain can be clinically observed and are identified in Boxes 23.1 and 23.2.

Box 23.1: Acute Pain

- Features: Sharp and well-defined onset; short duration.
- Etiology: Nociceptive/inflammatory/neuropathic features are usually clearly identifiable.
- Function: Physiologically adaptive protective function.
- Associated with tissue damage: Disappears with tissue healing.
- Pharmacotherapy response: Responds well to analgesics and anti-inflammatory agents in usual doses.
- Key neuronal fibers: nociception, C; A- δ , pain. A- β touch fiber stimulation decreases pain (gated response).
- Examples: Acute postoperative pain, labor pain, trauma, fractures, soft tissue injury, flail chest, stab injuries.

Box 23.2: Chronic Pain

- Features: Duration greater than months. No clear onset or onset associated with acute pain. Associated with cancer, acute zoster, neurological diseases, and hematological disorders.
- Etiology: Inflammatory, neuropathic, dysfunctional, nociceptive/neuropathic features become mixed and complex.
- Function: Maladaptive, no longer serves a physiological function.
- Pharmacotherapy response: Poor or requires higher doses.
- Persists months beyond the usual course of acute disease, or a reasonable time for an injury to heal, usually taken to be 3 months.
- Key neuronal fibers, C; A- δ , pain. Stimulation of A- β touch fibers increases pain (loss of gated response).
- Pathological features: *Hyperalgesia*, increased pain from a stimulus that normally provokes pain; there may also be an expanded area of pain. *Allodynia*, pain due to a stimulus that does not normally provoke pain.
- Examples: Post-herpetic neuralgia, fibromyalgia, osteoarthritis,

Pain Type

As previously indicated, pain itself is both a sensory and an emotional response. There is enormous heterogeneity encompassed by the terms pain and chronic pain. For the purposes of this chapter, chronic pain will be taken as chronic non-cancer pain (CNC), as there are differences in the etiology and in the risk-benefit of pharmacotherapy compared to cancer pain. The pharmacist monitoring analgesia must look for the general features of nociceptive and neuropathic pain. The International Association for the Study of Pain (IASP) defines *nociceptive pain* as “pain that arises from actual or threatened damage to non-neural tissue and is due to the activation of nociceptors” [11]. Nociceptive pain denotes a normal functioning of the somatosensory nervous system in direct contrast to neuropathic pain. The IASP defines *neuropathic pain* as “pain caused by a lesion or disease of the somatosensory nervous system” and is further divided into central and peripheral neuropathic pain [12].

The IASP has recently added the term *nociplastic pain*, which arises from altered nociception despite no clear evidence of actual or threatened tissue damage, causing the activation of peripheral nociceptors or evidence of a disease or lesion of the somatosensory system causing the pain. By this very definition, nociplastic pain denotes a pathological condition for which drug responsiveness will vary. Critically, pharmacotherapy responsiveness is significantly different for nociceptive vs. neuropathic pain. Neuropathic pain responds poorly to standard opioid therapy, but it may respond to higher opioid doses of opioids with unique effects on other receptors such as methadone. Because CNC patients will usually present with a complex mixture of nociceptive, nociplastic, and neuropathic pain, pharmacotherapy will be multi-modal in approach. The need for non-pharmacological interventions such as cognitive-behavior therapy cannot be overemphasized due to the sensory-emotional effect of pain. An ideal situation would be the ability to prevent the development of chronic pain. Sadly, chronic pain is often realized very late in the biological cascade that results in chronicity. However, risk factors for acute to chronic conversion have been identified and are summarized in Table 23.1. Chronic pain also presents with additional risk factors that may aid the clinician in early identification, summarized in Table 23.2.

Table 23.1 Risk factors for acute–chronic conversion

An extensive list of risk factors has been identified, especially in the postsurgical literature, looking at the risk of acute to chronic conversion. The most common pooled risk factors include the following [14–16]:
Psychological vulnerability (catastrophizing).
Anxiety or depression.
Female gender.
Older age, younger age (adults) may be at higher risk for chronic post-surgical pain.
Genetic predisposition.
Inefficient diffuse noxious inhibitory control (DNIC).
Descending pathway of pain inhibition.
Nerve damage due to injury or surgery.
History of poor acute pain management.
History of poor response to common analgesics.

Table 23.2 Risk factors for chronic pain

Additional risk factors associated with chronic pain include the following [15]:
Ethnicity and cultural background.
Smokers (genetics +/- enzyme induction).
Alcoholism.
Lack of sunshine and vitamin D.
Lack of physical activity.
Socio-economics, deprivation.
Obesity.

Currently, pain researchers have proposed a biopsychosocial model. This model integrates the biology of chronic pain with the resulting disability as a multivariate, multidimensional interaction amongst biological, psychological, and sociological factors that interact and influence each other as shown in Fig. 23.1 [13].

Researchers have attempted to construct pain models that integrate biological with psychosocial elements. However, many clinicians are aware that patients may present with chronic disease elements that would be anticipated to result in a significant reduction in quality of life and may manifest as suffering. A model that demonstrates the complexity of this interaction is the biopsychosocial model of pain, which illustrates the scope of the condition for the patient. It further emphasizes the need for an interdisciplinary team to facilitate treatment of the patient. Pharmacotherapy itself is only a small part of the full treatment paradigm for chronic pain (See Fig. 23.1).

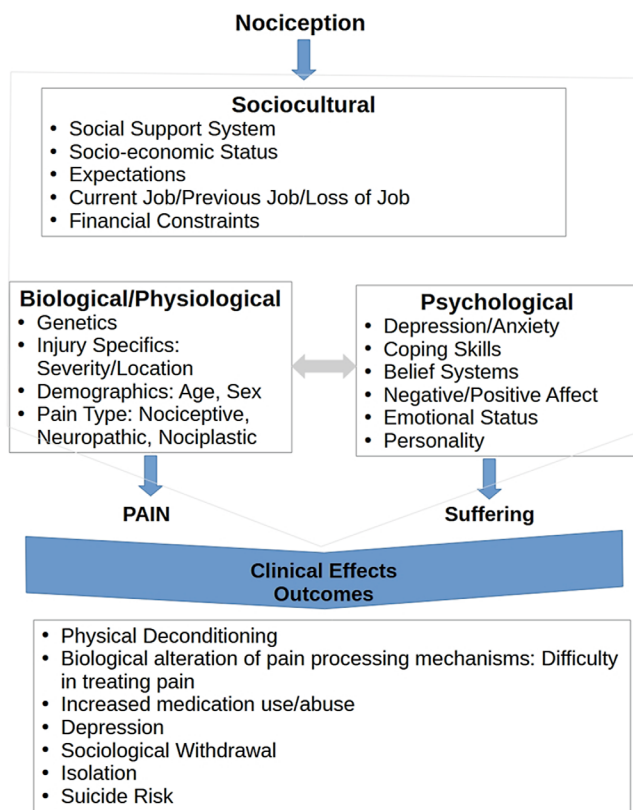


Fig. 23.1 Biopsychosocial model of pain. (Adapted from reference [13])

Goals of Therapy

Where possible, the goals of therapy should always attempt to treat the underlying cause of the pain, reduce pain, improve quality of life, and improve functioning. Pharmacists need to be aware that pain reduction alone is an insufficient goal. Improved quality of life needs to be a major focus for treating chronic pain because analgesics will likely only achieve modest pain reductions. Pharmacists need to work closely with patients to determine the patient's ultimate pain goals. Oftentimes, the patient may not be seeking direct pain relief but relief from the unremitting effects pain has on other physiological functions such as sleep and mood. Restoring previous function is often not possible, but it may be possible to improve physical, emotional, and social functioning to yield an improved quality of life through an improved ability to perform the activities to which the patient ascribes significant meaning. Pharmacists need to work with the patient to minimize inappropriate medication use while advocating for appropriate medication use.

Role of the Pharmacist in Management

A meta-analysis by Hadi et al. in 2014 concluded that pharmacist-led medication review reduces pain intensity and improves physical functioning and patient satisfaction [17]. Additionally, they concluded that there was weak evidence that a pharmacist medication review may also prevent or stop adverse events from occurring. Given the current opioid crisis and the ongoing clinical problem of poorly treated chronic pain, the pharmacist has a significant role to play. In a recent meta-analysis [18], the pooled analysis demonstrated that pharmacist-led interventions reduced the pain intensity by -0.22 with a 95% confidence interval of -0.35 to -0.09 . They further noted that the pharmacists were effective in achieving good opioid stewardship; however, the impact of the intervention on physical functioning, anxiety, depression, and quality of life revealed mixed results. Suggesting more research may be required. Medication review or reconciliation with the patient's pain goals and psychosocial functioning is a vital role for the pharmacist. Critically, the clinical pharmacist needs to be aware that chronic pain affects more than just the individual patient. Chronic pain has a pervasive and invasive effect on the patient's significant others, including spouses, partners, relatives, employers, coworkers, and friends. Thus, treatment must include not only a comprehensive assessment of the pathophysiology and biology of the pain but also an assessment, or at least an awareness, of the psychosocial and behavioral effects. Due to the multifaceted nature of pain as an emotional and sensory event, anxiety, depression, and anger are part of the assessment and treatment. Turk and Meichenbaum suggest that for chronic pain assessments, it is important to ascertain the following key factors [19, 20]:

1. What is the extent of the patient's physical impairment?
2. To what extent is the patient suffering, disabled, or unable to enjoy the usual activities?
3. Is there any evidence of symptom amplification?

These factors should be considered during the pharmacist's pain and medication review. Escalation in these domains, despite an appropriate trial of pharmacotherapy, could be used to alert the medical team that pharmacotherapy is no longer providing adequate benefit and emphasize the need for non-pharmacological interventions by other health professionals.

Table 23.3 Neuropathic Pain Symptom Inventory (NPSI) [27]

<i>Severity of spontaneous pain</i>		
Q1. Does your pain feel like burning?	0 no burning, 10 worst burning imaginable	
Q2. Does your pain feel like squeezing?	0 no squeezing, 10 worst imaginable squeezing	
Q3. Does your pain feel like pressure?	0 no pressure, 10 worst imaginable pressure	
Q4. During the past 24 h, has your spontaneous pain been present?	Permanently	()
	8–12 h	()
	4–7 h	()
	1–3 h	()
	<1 h	()
<i>Severity of painful attacks</i>		
Q5. Does your pain feel like electric shocks?	0 no electric shocks, 10 worst imaginable electric shocks	
Q6. Does your pain feel like stabbing?	0 no stabbing, 10 worst imaginable stabbing	
Q7. In the past 24 h, how many of these pain attacks have you had?	>20	()
	11–20	()
	6–10	()
	1–5	()
	None	()
<i>Severity of provoked pains</i>		
Q8. Is your pain provoked or increased by brushing the painful area?	0 no pain, 10 worst imaginable pain	
Q9. Is your pain provoked or increased by pressure on the painful area?	0 no pain, 10 worst imaginable pain	
Q10. Is your pain provoked or increased by contact with something cold on the painful area?	0 no pain, 10 worst imaginable pain	
<i>Severity of abnormal sensations</i>		
Q11. Do you feel pins and needles?	0 no pins and needles, 10 worst imaginable pins and needles	
Q12. Do you feel tingling?	0 no pins and needles, 10 worst imaginable pins and needles	
Total intensity scores	Subscores	
1. Q1 =	Burning (superficial) spontaneous pain: Q1 = /10	
2. (Q2 + Q3) =	Pressing (deep) spontaneous pain (Q2 + Q3)/2 = /10	
3. (Q5 + Q6) =	Paroxysmal pain: (Q5 + Q6)/2 = /10	
4. (Q8 + Q9 + Q10) =	Evoked pain: (Q8 + Q9 + Q10)/3 = /10	
5. (Q11 + Q12) =	Paresthesia/dysesthesia: (Q11 + Q12)/2 = /10	
(1 + 2 + 3 + 4 + 5) =	/100	

Note: Q4 and Q7 are for assessing the persistence of the pain, but do not get scored

Initial Pain Assessment

An initial pain assessment should seek to comprehensively understand the patient's pain syndromes with an integrated review of the pharmacological and non-pharmacological interventions that have been both successful and failed. Chronic Pain Assessment Tools and Scales are summarized in Box 23.3. As part of the patient's best-possible medication history (BPMH), pharmacists should assess the response the patient received from each medication. CNCP patients will often have good memory recall of drug therapy that failed and drug side effects that were particularly troublesome. Statements such as "would never try that drug again" or "that drug was useless" may indicate these issues. Pharmacists should look for patterns of responsiveness or refractoriness

in order to help assess pain type or detect patterns of abuse. As marijuana and marijuana products are now legal in Canada, medical marijuana prescriptions will no longer be required. Pharmacists need to add a discussion on marijuana use to their routine pain assessment. This assessment must be completely nonjudgmental, focusing first on whether marijuana has been used in the past or the patient is currently using marijuana. The next step is to determine if the marijuana product has been effective in reducing pain or symptoms related to pain such as anxiety and mood. The pharmacist should then probe for evidence of harm such as sedation, dizziness, or a deterioration in mood and increased isolation. Currently, evidence exists for the use of cannabinoids in neuropathic pain; however, a popular view exists espousing the use of marijuana as a panacea for all chronic

pain conditions [21]. Patients need to be advised on the current level of evidence and helped to evaluate their own marijuana consumption for efficacy and toxicity. Cannabinoid hyperemesis syndrome has been observed with increasing frequency. Patients present with increasing nausea and emesis that is relieved by taking hot showers. The syndrome resolves with supportive care and the cessation of marijuana products [22].

Ongoing Assessment of a Patient Diagnosed with Chronic Non-cancer Pain

Chronic non-cancer pain should never be viewed as static, nor should chronic pain be viewed as simply sustained activation of acute pain pathways. Chronic non-cancer pain represents a heterogeneous pathophysiological process of its own. Due to changes in central and peripheral pain processing, acute pain pharmacotherapy is often ineffective, especially in usual drug doses. Patients must be re-assessed in order to determine changes in pain processing. In addition, the occurrence of new diseases and complex changes may render previous treatments similarly ineffective. Clinical pharmacists must be able to provide sufficient assessment so that the efficacy and toxicity of pharmacotherapy can be correctly monitored and patients properly advised to seek additional treatment and/or support from their pain specialist, primary care provider, or psychologist. Pharmacists need to be aware that pharmacotherapy alone will be insufficient for the optimal treatment of chronic non-cancer pain. An interdisciplinary approach that includes all possible treatment modalities, including nerve blocks, surgical interventions, cognitive behavioral therapy, occupational therapy, physician therapy, and even spiritual support, must be considered using shared decision-making with the patient.

Comorbidities

Chronic pain patients often present with a significant number of physical and mental comorbidities. Some may be directly related to the chronic pain syndrome or may be associated with patient-specific genetics and/or the aging process. It is reported that 88% of chronic pain patients have additional chronic diseases [23, 24]. It is essential that the patient be treated holistically, considering all chronic conditions that can evolve and change rapidly. Too often, clinicians view chronic conditions as static and fail to recognize the significance of a new symptom or exacerbation of a chronic one. It is important to discuss changes in the status quo directly with the patient and be alert to the development of new conditions.

The above assessment tools have been well studied and applied in clinical practice. The clinical pharmacist will have to choose the best assessment strategy for the individual patient. A pain assessment with SCHOLAR or SOCRATES is recommended as the fundamental assessment tool for the initial visit and for follow-up as a quick pain review. The graded chronic pain scale is for establishing baseline and for follow-up with a goal of keeping the patient out of the Grade 3 High Impact Chronic Pain category. The neuropathic pain symptom inventory is especially helpful for monitoring patients where neuropathic pain is their primary symptom. All chronic pain patients should be monitored for the development of neuropathic elements to their pain.

Box 23.3: Chronic Pain Assessment Tools and Scales

A. Pain Overall Assessment.

1. *SCHOLAR*: Symptoms, Characteristics, History, Onset, Location, Aggravating, and Remitting Factors.
2. *The SOCRATES* acronym lends itself well to pain assessments: Site, Onset, Character, Radiation, Associations, Time Course, Exacerbating and Relieving Factors, Severity.

Comments on Use: It is best to use a consistent approach for assessment from one visit to the next to allow for serial assessments to be compared with less bias.

B. Pain Intensity Assessment: Numerical Scale

0: No Pain 10: Worst Pain

Comments on Use: Pain intensity should always be combined with assessments of the quality of life and coping with the pain. Be vigilant to detect patients that do not fully understand the scale.

C. Pain Type Determination

1. Descriptors of neuropathic pain: sharp, shooting, stabbing, burning, electric shocks.
2. Descriptors of nociceptive pain: aching, throbbing, steady, dull (if visceral: aching, gnawing, cramping, squeezing).

Comments on Use: Classifying the type of pain correctly can be used to help determine the appropriateness of pharmacotherapy

D. Pain Duration and Evolvement.

When did it start? How long? Has it changed over time?

E. Patient's pain goal

(continued)

Box 23.3 (continued)

1. Includes pain score.
2. Goals such as “sleeping through the night”.
3. Increased ability to perform activities of daily living (ADLs).

F. Pharmacotherapy review

1. Appropriate for pain type: nociceptive and neuropathic features.
2. Appropriate for pain intensity.
3. Opioid appropriateness review: Includes risk for addiction and abuse.
4. Cannabinoid review: neuropathic pain vs. personal trials of cannabidiol (CBD), tetrahydrocannabinol (THC).
5. Consider non-pharmacological interventions, including counseling, cognitive behavioral therapy (CBT), etc.

Comments on Use: Because pain can be both a sensory and emotional event, changes in pharmacotherapy for anxiety and depression may be indicative of inadequate pain control. Pharmacists should also optimize pharmacotherapy for anxiety and depression as part of a multidisciplinary team where appropriate.

G. Opioid Abuse Risk: Screener and Opioid Assessment for Patients with Pain (SOAPP)[®] Version 1.0-SF [25]

Please answer the questions below using the following scale:

0 = Never, 1 = Seldom, 2 = Sometimes, 3 = Often, 4 = Very Often

1. How often do you have mood swings?
2. How often do you smoke a cigarette within an hour after you wake up?
3. How often have you taken medication other than the way that it was prescribed?
4. How often have you used illegal drugs (for example, marijuana, cocaine, etc.) in the past 5 years?
5. How often, in your lifetime, have you had legal problems or been arrested?

Comments on Use: Assess all items answered with >3. It is best to avoid opioid trials in patients with high risk, but with appropriate clinical oversight, even patients with high risk may be managed on opioids. This requires significant patient-team cooperation and should only be attempted with an experienced pain physician.

H. Psychosocial Screening: ACT-UP [26]

1. Activities: How is your pain affecting your life (i.e., sleep, appetite, physical activities, and relationships)?
2. Coping: How do you deal/cope with your pain (what makes it better/worse)?
3. Think: Do you think your pain will ever get better?
4. Upset: Have you been feeling worried (anxious)/depressed (down, blue)?
5. People: How do people respond when you have pain?

Comments on Use: Increasing coping difficulties are a sign that the current treatment is inadequate. Drug therapy may require modification, but this should be combined with referral or recommendation for psychosocial treatment modalities from other healthcare professionals. Pharmacists are in a good position to advocate for these modalities.

I. Neuropathic Pain Symptom Inventory (NPSI)

The NPSI is summarized in Table 23.3. The NPSI should be performed only after a discussion on pain quality that identifies pain descriptors indicative of an increase in or the appearance of pain that is potentially neuropathic. The NPSI can then be performed and the results shared with the patient and the patient's pain physician for a more extensive and precise assessment in accordance with neuropathic pain diagnostic guidelines.

- J. *Graded chronic pain scale revised:* mild, bothersome, and high impact chronic pain [28] is summarized in Figs. 23.2 and 23.3.

1. In the past 3 months, **how often** did you have pain?

Never Some days Most days Every day

2. Over the past 3 months, **how often** did pain limit your life or work activities?

Never Some days Most days Every day

Now think about pain you have had during the past 7 days...

3. What number best describes **your pain, on average?**

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

No pain Pain as bad as you can imagine

4. During the past 7 days, what number best describes how **pain has interfered with your enjoyment of life?**

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Does not interfere Completely interferes

5. During the past 7 days, what number best describes how **pain has interfered with your general activity?**

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Does not interfere Completely interferes

6. Are you **not working or unable to work** due to pain or a pain condition?

Yes No

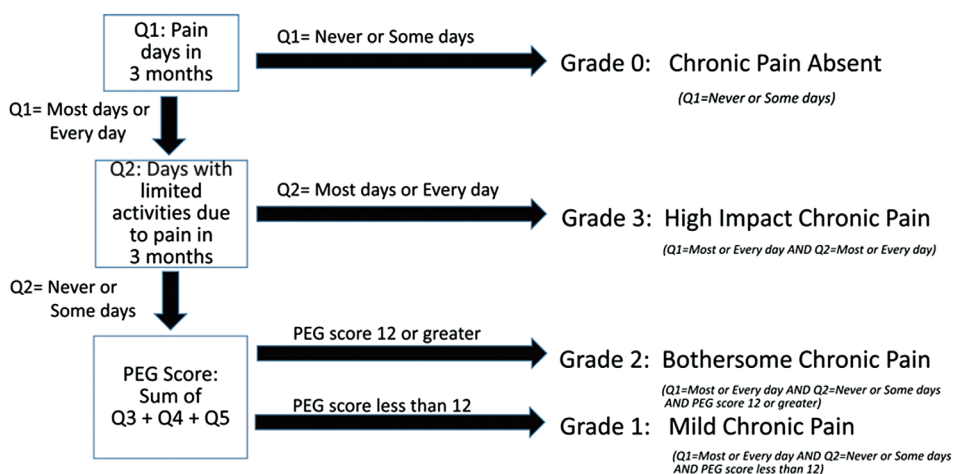
If you never had pain in the past 3 months, skip questions

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Fig. 23.2 Graded Chronic Pain Scale. (Reprinted with permission from Wolters Kluwer Health, Inc. Von Korff M, DeBar LL, Krebs EE, Kerns RD, Deyo RA, Keefe FJ. Graded chronic pain scale revised: mild, bothersome, and high-impact chronic pain. *Pain*. 2020 Mar 1;161(3):651–61)

Fig. 23.3 Chronic pain grade scoring rules for Graded Chronic Pain Scale.

PEG = Pain Average, Interference with Enjoyment of Life, interference with General activity. (Reprinted with permission from Wolters Kluwer Health, Inc. Von Korff M, DeBar LL, Krebs EE, Kerns RD, Deyo RA, Keefe FJ. Graded chronic pain scale revised: mild, bothersome, and high-impact chronic pain. *Pain*. 2020 Mar 1;161(3):651–61)



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Notes: GCPS-R items 1 and 2 were developed by United States government employees (National Center for Health Statistics) and are therefore in the public domain. The PEG (GCPS-R items 3-5) is a work product of United States government employees (Veteran's Health Administration) and is therefore in the public domain.

Treatment Options

General pharmacotherapy approaches to pain mechanisms are summarized in Tables 23.4 and 23.5.

Pharmacotherapy is only a part of the therapy that is used to treat CNCP. The perception and meaning of pain is a complex interaction of neuronal mechanisms and the emotional response to those stimuli. CNCP cannot be treated with drugs alone. Pharmacists need to be aware of non-pharmacological modalities, especially when a medication review reveals inadequate analgesia and the patient demonstrates signs and symptoms of increasingly maladaptive coping.

Cognitive and behavioral techniques, which may include distraction, goal setting, and exercises, have been demonstrated to be both safe and effective [20]. Exercise programs, tai chi, and yoga have also been shown to help reverse deconditioning and improve mood and the sense of well-being. A multidisciplinary team including psychologists, social workers, physiotherapists, and occupational therapists is essential for teaching appropriate exercises and coping strategies and providing emotional support [17, 20, 29].

In fact, most CNCP patients ultimately develop a strategy of coping that involves all these techniques and disciplines. Our goal as clinical pharmacists is to optimize the beneficial use of drug therapy in this process while emphasizing de-prescribing for maladaptive drug use to help the patient find the best balance of pharmacological and non-pharmacological techniques to improve quality of life.

Table 23.4 Pharmacotherapeutic recommendations by pain type [13, 21, 29, 30]

Nociceptive		Neuropathic	
Non-Inflammatory	Inflammatory	Peripheral	Central
Opioids: Morphine, Hydromorphone, Oxycodone, Fentanyl, Methadone			
Buprenorphine			
NSAIDs/Acetaminophen			
	Immunosuppressant		
	Anti-Inflammatory		
		$\alpha 2-\delta$ ligand: gabapentin, pregabalin	
TCAs			
SNRI			
(Especially if centralized)		TCA, SNRI, Cannabinoids,	

SNRI serotonin norepinephrine reuptake inhibitors, TCAs tricyclic antidepressants

Table 23.5 Drug class recommendations for chronic non-cancer pain

Drug class	Pain type and dosing recommendations
Acetaminophen	Mild to moderate nociceptive pain; optimize prior to adding other analgesics; monitor for hepatic damage
Oral NSAID: Ibuprofen, naproxen, diclofenac	Inflammatory pain, musculoskeletal pain. Use the lowest effective dose for the shortest time period Useful with inflammatory pain states. Note that the risk of GI and cardiovascular complications increases with increasing age
Topical NSAID: Diclofenac	Inflammatory, musculoskeletal pain. Decreased systemic exposure to the drug, with a decrease in GI and cardiovascular toxicity
Opioids	Nociceptive pain: Lowest effective dose, shortest duration. Adequate trial must include an assessment of quality of life and coping. Possible role for methadone and buprenorphine in neuropathic pain. Note in the case of opioid-induced hyperalgesia (OIH), pain may be increased with increasing opioid dose
Tricyclic antidepressants (TCA): Amitriptyline, nortriptyline	Neuropathic pain: Central and peripheral. TCAs are very effective if patients can tolerate significant anticholinergic effects. Caution in frail elderly: Assess anticholinergic burden
SNRI: Duloxetine	Neuropathic pain. May also be useful for osteoporotic pain due to changes in centralized pain processing.
Alpha-2 calcium blockers: Gabapentin, Pregabalin	Neuropathic pain: Peripheral and central. Use lowest effective dose
Cannabinoids/ standardized medical marijuana products	Neuropathic pain: Peripheral and central

NSAID nonsteroidal anti-inflammatory agent, SNRI serotonin norepinephrine reuptake inhibitors

Opioid Appropriateness

Opioid prescribing rates in Canada are among the highest in the world [31]. Substantial risks such as addiction and overdose accompany opioid therapy, and these risks have a significant individual and societal impact. In light of the opioid crisis, the Canadian Guideline for Opioid Use in Chronic Non-cancer Pain recommends the optimization of non-opioid strategies before considering opioid therapy [31]. Despite best efforts to optimize non-opioid pharmacotherapy and behavioral strategies, a proportion of patients will suffer from persistent problematic pain. It is important to acknowledge that there will be a subset of patients that will derive a functional benefit from opioid therapy and a subset that will suffer a serious adverse outcome, such as addiction and a fatal or nonfatal overdose. A recent meta-analysis in 2022 based on very low to low certainty evidence from placebo-controlled RCTs of 4 to 15 weeks duration suggests that strong opioid analgesics may provide a reduction in pain score that is clinically significant, but disability was not improved. Longer treatment durations were similarly not associated with clinical benefit. Use of strong opioids was associated with increased adverse events and cases of abuse and dependence with a possible increase in all-cause mortality [32]. These analyses emphasize the need for careful patient assessment with close follow-up that includes both pain reduction and functional improvement while monitoring for serious adverse outcomes.

Recent literature has identified patient factors that are associated with an increased risk of addiction and a fatal or nonfatal overdose, including current or past substance abuse disorder and an active psychiatric disorder [31]. The Opioid Risk Tool (ORT) has been used to screen for the risk of opioid misuse; however, its predictive value has not been validated. When considering a trial of opioid therapy, the medical history should include questions to screen for a current or past history of substance abuse and psychiatric disorders. Opioids are not recommended for patients with current or past substance abuse disorders [31]. For patients with an active psychiatric disorder, it is recommended that the condition be stabilized before a trial of opioids is initiated because of the impact on pain perception and expression [31].

When an opioid trial is initiated, there are a number of considerations for determining appropriateness (Table 23.6) [31–34]. Therapy should be prescribed at the lowest dose and for the shortest duration, usually no more than 4 weeks. Although the evidence is not strong, Naloxone kits should be considered for all patients, especially those who are identified as at risk of toxicity due to high dose, medical history, or comorbidities. Naloxone, an opioid antagonist, is available

Table 23.6 Opioid trial initial patient assessment

Indication	Effectiveness
Characteristics of pain Nociceptive vs. neuropathic (neuropathic is opioid resistant) Persistent vs. intermittent (continuous dosing for persistent pain) Opioid use in previous 6 months Non-pharmacological strategies	MED <50 mg if opioid naïve or <90 mg if on long-term opioid therapy Note: Some patients may benefit from higher doses; suggest a second opinion If the dose exceeds accepted guidelines Consider taper, rotation, or referral

Note: MED morphine equivalent doses

Table 23.7 Clinical features of opioid misuse

1. Altering the route of delivery
2. Accessing opioids from other sources
3. Unsanctioned use
4. Drug seeking
5. Repeated withdrawal symptoms
6. Accompanying conditions (addiction to another drug)
7. Deteriorating social function
8. Views on opioid medication

as kits in all Canadian provinces. The kit typically includes three doses of naloxone and the supplies to administer the injection in a patient's thigh. For those unfamiliar, Canada provides the Overdose Intervention App (OD+) that provides a guide for using naloxone. With patients' consent, education of the patient's family and support group is advisable. The opioid trial must seek to determine an improvement in both pain and functional goals against the risk of adverse outcomes, including drug-seeking behavior and a decrease in patient functioning. Communication and collaboration with the patient and other healthcare providers are essential. The patient interview should include a patient-centered functional assessment. Patients should be screened for the clinical features of opioid misuse (Table 23.7) [25, 34]. Tools such as the Screener and Opioid Assessment for Patients with Pain (SOAPP)[®] can be useful in recognizing aberrant drug-related behaviors [31]. The morphine equivalent doses (MED) should be calculated at each follow-up as the risk of unintentional overdose and death increases as the prescribed dose of opioids increases, with the most significant change in rates at >50 mg, if opioid naïve, and >90 mg MED, if on long-term opioid therapy [31]. It is essential to remember that higher MEDs are associated with increased risk of overdose and death, and that opioids themselves can cause hyperalgesia. The patient with escalating opioid doses presenting suddenly with increasing pain may be a clinical sign of overdose and not escalating pain [35].

Follow-Up Assessments

Patient-Specific Goals of Therapy

The clinician should spend time with the CNCP patient to identify the patient's goals of therapy. Unlike acute pain, cure and complete resolution of the pain may not be possible. A reduction in pain score may not translate directly to functional or quality of life goals. Therefore, the patient may seek moderation of the pain to enable them to perform specific activities. The clinical pharmacist should be aware of non-pharmacological treatments in order to advocate for alternatives to drug therapy that facilitate the achievement of patient goals. Certainly, a reduction in baseline chronic pain monitoring scores is a desirable endpoint. For example, a PEG score reduction from Grade 3 is a desirable outcome and indicates an improvement from high-impact chronic pain to more tolerable levels. The clinical pharmacist needs to be aware that any drug additions, deletions, or dosage adjustments influence monitoring scoring systems and seek to confirm if the patient associates these reductions with achievement of their specific goals. The clinician should also be aware that goals may change overtime as critically important life events unfold.

Adherence

Patient adherence in CNCP is almost always linked to the toxicity or efficacy of the drug therapy. Problematic or ineffective therapies are quickly identified by the patient and discarded. Patients may become quite manipulative to access the therapies that have historically worked the best with the least amount of side effects. When assessing drug adherence, pharmacists should evaluate dispensing time intervals, dispensing events at other pharmacies, double doctoring, and the use of other medications and alternative treatments that may alter the current drug therapy. This becomes especially important for the monitoring of both the safety and efficacy of long-term opioid therapy.

Pain Control

The clinical pharmacist needs to be alert for changes in the patient's pain syndrome. Due to the chronicity of the disease, patients and clinicians may conclude the disease is static with little need for close monitoring or therapy adjustment. CNCP should be viewed as a dynamic, ongoing disease state that will demonstrate acute exacerbations in response to concomitant disease or infections and may require aggressive management. A CNCP patient with an acute exacerbation in pain intensity must be immediately worked up to determine

if the pain represents an aggravation of the existing condition or the development of a new one. New pain symptoms require a thorough and complete new pain assessment with the intent to look for new underlying conditions. This includes appropriate laboratory and imaging studies. Patients with chronic pain may be at risk for a delay or underdiagnosis of new disease states. A good example occurs when the pain from CNCP masks the emergence of cancer.

Adverse Drug Reactions

The most common reasons for poor drug adherence are inadequate response (ineffective treatment) and adverse drug reactions. Common adverse reactions vary in severity and may be managed with careful dose titration. However, severe adverse drug reactions may warrant the discontinuation of the drug entirely. Table 23.8 provides a list of common adverse reactions and precautions combined with "red flag" complications that necessitate the discontinuation of that drug therapy.

Complications

CNCP often presents as a result of other disease states, but CNCP may also result in significant complications. Complications of CNCP may be related to deconditioning, hormonal effects of chronic pain and stress, and neuropsychiatric complications. CNCP results in a significant loss in activity, leading to an underappreciated increase in deconditioning. This results in an increased loss of mobility and obesity, and may exacerbate the development of concomitant diseases such as type 2 diabetes mellitus. Disruption of normal biomechanics can lead to muscle underuse and atrophy, with compensatory muscle overuse with tissue degeneration and damage. The situation is exacerbated by obesity, and the patient needs to be encouraged to undertake some form of exercise to prevent atrophy and damage, combined with efforts to attain a healthy weight.

CNCP is also associated with excess catecholamine production, especially adrenaline release combined with cortisol release. Since acute pain is designed for a protective purpose, chronic pain can result in an overactivation of the sympathetic nervous system and a perpetual state of arousal. This may feed directly into feelings of fear, anxiety, or dread. It may also contribute to the maintenance of the pain state.

Neuropsychiatric symptoms, especially in patients with unremitting symptom control and increased difficulty in coping, can result in insomnia, memory loss, cognitive decline, depression, and suicidal ideation. Escalating opioid doses with increasing generalized pain may indicate opioid-

Table 23.8 List of the most common and important drug complications in patients with CNCP^a

Drug	Common adverse reactions/precautions	Red flag complications
Opioids (morphine, hydromorphone, fentanyl, methadone, buprenorphine) Codeine (NOT recommended)	Sedation, dizziness, nausea, vomiting, constipation, sweating	Addiction/diversion Hallucinations Neurotoxicity: Hallucinations, hyperalgesia, myoclonus, seizures Hypogonadism QTc prolongation: Avoid in long QT syndrome Overdose: Respiratory depression Drug diversion CYP2D6 ultra-metabolizers: Overproduction of morphine: Overdose Poor metabolizers: Little or no morphine production: Underdose
Acetaminophen	Usually no adverse reactions but may cause nausea, vomiting, loss of appetite	Hepatic failure: Overdose or chronic dose
NSAIDs	GI: Vomiting, bloating, diarrhea, constipation, mucosal erosions Renal: ↓GFR, Na, and water retention, edema CVS thrombotic events, increased blood pressure, congestive heart failure, palpitations CNS: Headache, fatigue, insomnia, vertigo, seizures Others: Asthma attacks, urticaria, neutropenia	GI bleeding Renal failure Thrombotic events Frail elderly are at elevated risk
α-δ Ca ²⁺ blockers Gabapentin Pregabalin	Dizziness, drowsiness, weakness, tired feeling Nausea, diarrhea, constipation, blurred vision, headache, breast swelling, dry mouth, or Loss of balance or coordination	Ataxia with increased fall risk, especially in the frail elderly Delirium, dementia risk Suicidal ideation: Has been discounted by current research; however, suicidal risk should be assessed and monitored in CNCP
SNRI Venlafaxine Duloxetine	Dizziness, nausea, dry mouth, sweating, tiredness, insomnia, anxiety or agitation, constipation	Serotonergic syndrome: Diarrhea may be an early symptom Symptoms include high body temperature, agitation, increased reflexes, tremor, sweating, and dilated pupils
SNRI/opioid Tramadol (NOT recommended)		Serotonergic syndrome CYP2D6 ultra-metabolizers: CNS sedation, respiratory depression, death CYP2D6 poor metabolizers decreased opioid effect due to reduction in M1 metabolite
Tricyclic antidepressants Amitriptyline Nortriptyline	Anticholinergic side effects Dry mouth, constipation	Delirium Cognitive impairment
Cannabinoids Nabiximols (Sativex®)	Sedation Dizziness	Cannabinoid hyperemesis syndrome: Paradoxical increase in nausea and emesis often relieved by hot showers. Stop cannabinoid therapy

NSAID nonsteroidal anti-inflammatory agent

^aAdapted from Refs. [13, 21, 36–47]

induced hyperalgesia and attempts to use opioids as anxiolytics and should be carefully evaluated [22, 35]. The risk of an accidental or intentional opioid overdose increases with an increasing morphine equivalent daily dose. Pharmacists should pay critical attention to increasing depression and anxiety and be alert for comments related to a sense of hope-

lessness or futility and suicidal or end-of-life comments. In Canada, legislation that allows for medical assistance in dying (MAID) has brought the discussion of euthanasia more into the open. CNCP patients may ask their pharmacist about the drugs used for MAID, and the reason for this interest should be carefully explored with the patient.

Clinical Pearls

- Assess the efficacy of pharmacotherapy, including pain intensity, pain quality, AND psychosocial indicators of coping.
- Assess drug adherence by evaluating dispensing time intervals, dispensing events at other pharmacies, double doctoring, and the use of other medications that may affect opioid therapy.
- Opioid-naïve patient: Assess efficacy based on pain intensity score, and assess the effect on psychosocial aspects using ACT-UP. Look for improvement in quality of life as indicated by the ability to perform ADLs and meaningful activities. A decreased ability to participate in meaningful activities should be discussed with the patient and physician. The initial opioid dose is recommended to be <50 mg morphine equivalent dose (MED) per day. Duration of therapy should not exceed 1 week without reassessment. If doses are increased, they should be titrated upward, and the individual should be monitored for pain control and/or function improvement.
- For individuals using long-term opioid therapy, total doses should not typically exceed 90 mg MED/day, and the individual should not receive more than a 30-day supply. Assess as indicated for the opioid-naïve patient. Monitor for hyperalgesia, increased pain that may be generalized and not typical of usual chronic pain. Opioid neurotoxicity can induce hyperalgesia. Monitor for sedation, visual disturbances, hallucinations, and myoclonus, which could indicate opioid neurotoxicity.
- Assess patient use of marijuana products. What product are they taking? What is the THC, CBD content? Has the drug been effective in relieving pain? Relieving anxiety? What are the side effects? Dizziness? Sedation? Has the marijuana product allowed for a reduction in other prescription drug use? Monitor for ataxia, increased sedation, and confusion or delirium. Monitor for increased nausea with emesis relieved by hot showers, which could indicate cannabinoid hyperemesis syndrome.
- Assess the patient for the need for non-pharmacological interventions such as cognitive-behavioral techniques.
- *Constantly Assess—Reassess* for changes in the current disease and the occurrence of new diseases.

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Part IV

Specialized Assessments

Chapter Objectives

1. Discuss the clinical relevance of pharmacokinetic concepts.
2. Apply the basic pharmacokinetic principles in therapy assessments.
3. Identify the concepts and rationale for therapeutic drug monitoring (TDM).
4. Apply pharmacokinetic principles in TDM.

Background

Pharmacokinetics (PK) is the study of what happens to the drug in the body following its administration. It is the study of the drug journey from the time of its administration to its elimination. It involves four distinct processes: absorption, distribution, metabolism, and excretion (ADME). Understanding basic pharmacokinetic principles is essential in day-to-day clinical pharmacy practice (Table 24.1). It allows answering many clinical questions such as:

- *What is the equivalent intravenous dose of morphine, if my patient is currently on 10 mg oral morphine?*
- *A patient has been initiated on vancomycin, when should the drug level be measured?*
- *My patient had a serious adverse reaction to rivaroxaban. When will it be cleared from his system?*
- *Is administration of a loading dose for this medication required? If so, what is the recommended loading dose?*
- *Carbamazepine level is 20 $\mu\text{mol/L}$. What action should I take?*
- *I inadvertently missed my morning dose of medication. What is the recommended course of action in this situation?*

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Table 24.1 Applications of pharmacokinetics in therapy assessments

Pharmacokinetic applications	Pharmacokinetic (PK) parameter(s) utilized
Therapeutic drug monitoring	All PK parameters
Prediction of the extracorporeal drug removal by renal replacement therapies such as hemodialysis	Protein binding, volume of distribution, clearance
Estimation of intravenous (IV) to oral (PO) dose conversion	Bioavailability, salt factor
Estimation of the drug loading dose	Volume of distribution, target steady state concentration, bioavailability
Determination of the time it takes for the drug to reach steady state	Half-life
Determination of how long it takes for the drug to be out from the system	Half-life
Assessment of pharmacokinetic drug interactions	All PK parameters
Assessment of missed drug doses	Half-life, pharmacokinetic–pharmacodynamic relationship

This chapter will go over the basic pharmacokinetic principles by discussing the ADME system and its applications in patient's therapy assessment. Then, the rationale and approach to therapeutic drug monitoring will be discussed.

Basic Pharmacokinetic Concepts Related to ADME System

Absorption

Bioavailability

Following administration by any route, with the exception of the intravenous (IV) route, drugs need to be absorbed to reach the systemic circulation. The fraction of the administered drug that is absorbed is called bioavailability or "F," and it reflects the extent of drug absorption. This depends on the physicochemical characteristics of the drug and the characteristics of the absorption site, such as blood supply and pH. The bioavailability for the IV route is 100% because the

drug is directly delivered to the bloodstream. On the other hand, other routes such as oral, intramuscular, and subcutaneous routes generally have bioavailability values of less than 100%. Knowledge of the bioavailability is very helpful in day-to-day clinical practice. Bioavailability provides insight into the systemic exposure of a drug by indicating the extent to which it is absorbed into the bloodstream. This information is essential for determining equivalent dosages when transitioning between different routes of administration. To illustrate, Table 24.2 depicts the oral bioavailability values of select drugs. For drugs with almost complete oral bioavailability, such as phenytoin and valproic acid, as shown in the table, the IV to oral dose conversion ratio is one-to-one. For example, 500 mg per day of IV valproic acid is equivalent to 500 mg per day of the oral formulation. On the other hand, other drugs such as morphine and propranolol have significantly lower oral bioavailability secondary to the extensive first pass metabolism. While assessing patients' medications, pharmacists should expect lower intravenous dosages when compared to oral regimens for drugs with low oral bioavailability. For example, 30 mg oral morphine has been suggested to be equivalent to 10–15 mg of parenteral morphine. Furthermore, drugs that have no oral bioavailability (i.e., not absorbed) are not expected to be used orally to treat systemic illnesses. For example, vancomycin is not absorbed orally and should be given intravenously when used to treat systemic infections.

Salt Factor

The second concept related to absorption is the salt factor, or "S." Some drugs are available in different salt forms and/or pro-drugs. The salt factor is the fraction of the pro-drug or salt form that contains the pharmacologically active drug. With regards to clinical relevance, the salt factor is important in determining the equivalent dosages of different salt forms of the drug. For example, while the parenteral and oral extended release formulations of phenytoin contain phenytoin sodium salt, immediate release tablets and suspension contain the free phenytoin acid form. Phenytoin sodium contains 92% of the free phenytoin acid form. In other words, it has a salt factor of 0.92. However, this difference is not clinically significant in most patients owing to the long half-life of phenytoin. On the other hand, the salt factor for other drugs is clinically significant, and it should be included in equivalent dosage calculations. For example, aminophylline,

the ethylene diamine salt of theophylline, contains 80% theophylline (salt factor = 0.8). In other words, when switching aminophylline to theophylline, aminophylline dose needs to be multiplied by 0.8.

Rate of Absorption

The third concept is the rate of absorption of the drug. This is best illustrated when sustained release (SR) formulations of the drug are compared with immediate release ones. As shown in Fig. 24.1, immediate release formulations release the drug rapidly with the potential for higher maximum concentrations (C_{max}) than the SR formulation; however, their effect goes away quicker than the SR formulations. The rate of drug absorption determines the onset of effect and time to peak concentration and is beneficial in formulation conversions. For example, the extended release formulations of diltiazem slowly release the drug in the gastrointestinal tract, resulting in a prolonged duration of action and hence are administered once daily. Conversely, the immediate-release formulation of diltiazem delivers the drug rapidly; however, due to its short half-life, it requires administration four times per day. If a patient receiving diltiazem SR 240 mg orally once daily is unable to swallow tablets, transitioning to the liquid formulation may be considered an appropriate alternative. However, the 240 mg dose will need to be given in four divided doses; otherwise, giving the full liquid dose at once will put the patient at risk of bradycardia and hypotension. Pharmacists play an important role in managing drug formulation conversions. When evaluating patients' medication regimens, it is imperative to determine whether controlled-release medications are being administered correctly.

Pharmacokinetic Drug Interaction: Absorption

Pharmacokinetic drug interactions can occur at any stage during the journey of the drug in the body. Drug–drug and food–drug interactions might lead to alterations in the rate and extent of drug absorption. For example, co-administration of levothyroxine and iron supplements leads to reduced bio-

Table 24.2 Oral bioavailability of select drugs

Drug name	Bioavailability
Phenytoin	90–100%
Valproic acid	90–100%
Vancomycin	0%
Morphine	17–33%
Propranolol	25%

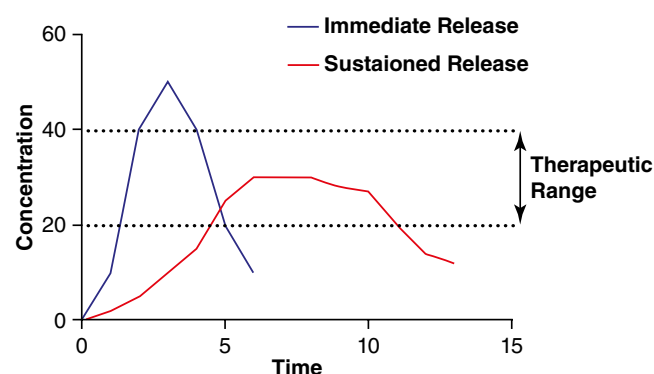


Fig. 24.1 Concentration vs. time profile of immediate and sustained release formulations

availability of levothyroxine and hence a reduction in its efficacy [1]. Administering levothyroxine and iron supplements should be spaced at least 4 h to avoid that interaction. Another example is phenytoin. Administering phenytoin along with feeds in tube-fed patients could drastically reduce phenytoin bioavailability by up to 80%, putting the patient at risk for breakthrough seizures. Holding feeds 2 h before and 2 h after phenytoin administration is recommended. Pharmacists need to be vigilant in identifying the potential for these interactions. Iron, calcium, aluminum, and magnesium salts, polystyrene, and cholestyramine are examples of agents commonly implicated in absorption interactions. It is important to check drug interaction reference and administration with food guides while assessing patients' medication regimens.

Distribution

Volume of Distribution

Once the drug is absorbed or given intravenously, it distributes via systemic circulation to different tissues in the body. The extent of drug distribution depends on many factors such as the drug's lipophilicity, tissue affinity and protein binding. Volume of distribution (V_d) is the apparent volume in which the drug distributes. It reflects the extent of drug distribution in the body. Drugs with a high V_d are more extensively distributed throughout body compartments compared to drugs with a low V_d . With regard to clinical relevance, V_d helps determine the loading dose. Administering loading doses of drugs could be of value to rapidly attain therapeutic concentrations. The intravenous loading dose can be calculated by multiplying the drug's desired steady state concentration (C_{ss}) by its volume of distribution (Eq. 24.1). If the same amount of the drug is added to two containers with different volumes, different concentrations will result. Therefore, drugs with high V_d require higher doses than drugs with low V_d to obtain the same concentration. To illustrate, as seen in Table 24.3, amiodarone has V_d of 66 litres per kilogram, suggesting extensive distribution in the body compartments. As a result, amiodarone needs to be loaded with a much higher dose compared to drugs with low V_d to rapidly target concentrations.

$$\text{Loading dose} = \text{desired } C_{ss} \times V_d \quad (24.1)$$

It is important to note that patient-specific factors that might alter V_d should be taken into consideration when pharmacists are assessing the appropriateness or calculating a drug loading dose. These factors include obesity, conditions of altered protein binding (see next section), patient's weight and conditions associated with altered body volume status such as pregnancy, cystic fibrosis and burns. Drug-specific information pertaining to V_d values in various disease states

Table 24.3 Protein binding and volume of distribution of select drugs

Drug	Volume of distribution	Percent protein binding
Phenytoin	0.7 L/kg	90%
Vancomycin	0.7 L/kg	55%
Levetiracetam	0.5–0.7 L/kg	<10%
Amiodarone	66 L/kg	>96%

could be found in the PK section of drug monographs and PK studies [2].

Protein Binding

Another important concept that has a major influence on the pharmacokinetics of drugs is protein binding. Drugs can bind to plasma albumin, alpha acid glycoprotein, or lipoproteins. Generally, basic drugs bind to alpha acid glycoprotein, while acidic ones bind to albumin. The higher the extent of protein binding is, the smaller the volume of distribution of the drug. Most drugs exist in two forms: free and bound. The bound fraction of the drug is restricted to plasma, while the free fraction is available for distribution to tissues. Generally, the free fraction is the pharmacologically active form and is the one that is available for elimination. Therefore, any factor that can alter protein binding has the potential to affect the drug's efficacy, clearance, and volume distribution. Table 24.4 depicts a summary of the factors that might affect protein binding of drugs that pharmacists need to be aware of.

Drug Concentrations

Drugs for which therapeutic drug monitoring is valuable typically have their recommended reference ranges reported. The reference range is the range of drug concentrations below which the drug is most probably ineffective and above which the drug is most probably toxic. It should be noted that reference ranges are primarily derived from limited evidence, such as retrospective studies, expert opinions, and population averages, and therefore should not be regarded as absolute values. They are *not carved in stone*. Drugs can be effective at concentrations below the recommended range or toxic at concentrations within the range. Therefore, reference ranges can be used as a tool to assess a patient's drug therapy rather than an ultimate goal to target (*treat the patient, not the level*). Measuring routine drug concentrations is not recommended. Drug concentrations are only measured if they are indicated (discussed later in the chapter) and are generally measured at steady state. However, pre-steady state levels can be beneficial in some situations to determine the adequacy of the dosage.

Pharmacokinetic Drug Interaction: Distribution

Generally, highly protein-bound drugs (protein binding $\geq 90\%$) are prone to drug interactions secondary to protein binding displacement. As seen in Table 24.3, phenytoin and

Table 24.4 Factors affecting protein binding of drugs

Altered plasma protein concentration	Conditions associated with low albumin concentration: Hepatic failure Nephrotic syndrome Burns Malnutrition Aging Critical illness This results in increase in the free fraction of acidic drugs Conditions associated with increase in alpha acid glycoprotein: Inflammatory condition such as rheumatoid arthritis This results in decrease in the free fraction of basic drugs
Displacement from plasma protein binding	Diseases associated with protein binding displacement Renal failure (uremic substances) Hyperbilirubinemia Drug–drug interactions

amiodarone are extensively protein-bound and are subject to protein binding displacement. On the other hand, levetiracetam and vancomycin are not highly protein bound, making them not prone to interactions via protein binding displacement. Drug-induced alteration in protein binding might be clinically significant with initiation of therapies or dose changes. With chronic administration, the increase in the free fraction will lead to an increase in the drug's volume of distribution and plasma clearance, often resulting in a clinically insignificant change in the drug's free fraction at steady state. Monitoring the free fraction of highly plasma protein-bound drugs could be of value in patients with suspected drug interaction at distribution and with conditions of altered protein binding, such as uremia and hypoalbuminemia. Unfortunately, free concentrations are not widely available, more expensive, and labor-intensive. Consulting a drug interaction resource when highly protein bound drugs are prescribed is recommended.

Metabolism and Excretion

Clearance

The disposition of drugs within the body involves metabolic conversion to inactive metabolites, excretion of the drug in its unchanged form, or a combination of these processes. The main organ for metabolism is the liver; however, drug metabolism could happen in other sites such as the gastrointestinal (GI) tract, blood, and kidneys. The main organ for drug excretion is the kidney; however, drug excretion could happen through other sites such as the GI tract and the lungs. Drug clearance is the volume of the blood cleared from the drug per unit time. Drug clearance depends on multiple fac-

Table 24.5 Extent of metabolism and elimination of select drugs

Drug	Percent metabolism	Percent elimination	Half-life
Valproic acid	>93%	<7%	5–20 h
Digoxin	20%	80%	1–2 d
Vancomycin	Negligible	90%	8 h
Gabapentin	0%	100%	5–7 h

tors such as drug metabolizing enzymes, protein binding, blood flow to the elimination organs and kidney and liver functions. Total clearance is the sum of the drug's renal and non-renal clearance. Appreciation of the renal and non-renal components of drug clearance (seen in drug monographs) are helpful in assessing the need for dosage adjustments in renal and/or hepatic impairment. To illustrate, as seen in Table 24.5, valproic acid is mainly metabolized with a negligible portion of the dose being excreted unchanged by the kidney. Therefore, it is expected that dosage adjustment is not required in patients with renal failure but might be needed in patients with liver disease. Conversely, vancomycin, digoxin, and gabapentin are primarily eliminated via renal pathways. It is advisable to adjust the dosage of these medications in accordance with renal function. In addition to the route of drug elimination, clearance is an independent determinant of the drug half-life (see next section).

Half-Life

Drug half-life ($t_{1/2}$) is the time taken for the drug concentration to be reduced into half. The value of the drug's half-life is dependent on the drug's clearance and volume of distribution. An increase in drug clearance, with no corresponding change in V_d , results in a shortened half-life due to more rapid elimination of the drug. Conversely, a decrease in clearance will prolong the drug's half-life. When the volume distribution increases with no clearance changes, the drug's half-life will increase. To simplify, drugs with large V_d are highly distributed in the body and are “*hiding*” in the tissues away from the elimination organs and thus have longer half-life. If both clearance and V_d change in the same direction, half-life is expected not to or minimally change due to the counteracting effects of clearance and volume distribution.

Drug half-life corresponding to the patient's renal and liver function could be obtained from drug monographs and utilized in clinical decision-making. However, it is important to note that reported half-life values are based on population parameters and do not necessarily reflect the individual's half-life. In some situations, the drug's half-life for an individual can be calculated if two drug level values are available. Equation 24.2 depicts the simplest equation used to determine the elimination rate constant (k) for drugs with first-order linear elimination kinetics, given that there is no dose administered between the two levels. C_1 is the first concentration, C_2 is the second concentration, and t is the time

interval in between. Then, half-life can then be calculated using Eq. 24.3.

$$\ln C_1 - \ln C_2 = kt \quad (24.2)$$

$$t_{1/2} = \frac{0.693}{k} \quad (24.3)$$

The concept of half-life is highly relevant in routine clinical practice. The half-life of a medication dictates the duration required for the drug to achieve steady state concentrations. Typically, a drug reaches steady state after three to five half-lives. For instance, the half-life of phenobarbital ranges from 2 to 6 days. Consequently, attainment of steady-state concentrations may require up to 3 weeks of continuous phenobarbital administration. Drugs similar to phenobarbital will require a loading dose if rapid attainment of target concentrations is needed. Furthermore, following discontinuation, it typically requires three to five half-lives for the drug to be eliminated from the body. This is useful in situations where knowledge of the duration of the drug in the body is needed, such as switching between agents with a similar adverse effect profile and in cases of drug toxicity.

Pharmacokinetic Drug Interaction: Metabolism and Excretion

The most clinically important pharmacokinetic drug interactions occur at the levels of metabolism and excretion. Such interactions often result from the concurrent use of agents that induce or inhibit liver microsomal enzymes and/or P-glycoprotein. Pharmacists should consult reputable drug interaction references whenever relevant medications are initiated, discontinued, or have their dosages adjusted, given their capacity to inhibit, induce, or serve as substrates for hepatic microsomal enzymes. The following are examples of drugs commonly implicated in drug–drug interaction at the metabolism level:

- *Liver microsomal enzyme inducers:* rifampin, phenytoin, fosphenytoin, phenobarbital, primidone, carbamazepine, St. John's wort.
- *Liver microsomal enzyme inhibitors:* clarithromycin, erythromycin, cimetidine, valproic acid, ketoconazole, ritonavir, voriconazole.
- *P-glycoprotein inducers:* carbamazepine, rifampin, St. John's wort.
- *P-glycoprotein inhibitors:* clarithromycin, erythromycin, amiodarone, ketoconazole.
- *Drug classes prone to these kinds of interactions:* immunosuppressives, antiepileptic medications, warfarin, direct oral anticoagulants, antiretrovirals, antifungals.

Drug–drug interactions at the excretion stage occur when one drug alters the elimination of another, often by affecting renal or biliary clearance. In the kidneys, many drugs are

actively secreted in the proximal tubules via transporters such as organic anion transporters and organic cation transporters. Competition for these transporters can reduce the excretion of one drug, leading to increased plasma concentrations and potential toxicity. For example, probenecid inhibits the renal secretion of penicillin. Changes in urinary pH can also influence drug ionization and reabsorption; alkalinizing the urine may enhance the excretion of weak acids, whereas acidifying the urine promotes elimination of weak bases. In the biliary system, drugs may compete for transporters, altering enterohepatic recycling. Such interactions are particularly important for drugs with narrow therapeutic ranges, where small changes in elimination can have significant clinical consequences.

Therapeutic Drug Monitoring: Rationale

Therapeutic drug monitoring (TDM) is *the process of applying pharmacokinetic principles and knowledge of drug concentrations along with clinical assessment to facilitate appropriate dosing of drugs in order to maximize efficacy and minimize adverse drug reactions*. The definition has three components. First, in order to conduct TDM, one should have a reasonable understanding of the main pharmacokinetic principles (discussed in the first part of this chapter). Second, blood concentration(s) of the drug monitored should be available. The third and most important component of the definition is clinical assessment such as patient characteristics, efficacy and toxicity of the drug and presence of drug interactions. Adjusting therapy solely according to drug concentrations, without incorporating clinical evaluation, overlooks the advantages offered by TDM.

Generally, therapeutic drug monitoring is meant to improve patient outcomes by increasing efficacy and minimizing toxicity. TDM is useful in those drugs with narrow reference range whose efficacy and toxicity are proportional to drug concentration. In other words, it is suitable for drugs that have well-established reference range, such as phenytoin and vancomycin. Additionally, TDM is useful in those drugs whose response cannot be assessed directly, and at the same time, there is a good correlation between drug concentration and response, such as in the case of some antiepileptic medications and antirejection drugs. Furthermore, TDM is beneficial in those drugs with unpredictable pharmacokinetics with wide interpatient variability, where administering the same dose to different individuals yields different drug concentrations and responses. For example, tacrolimus, an anti-rejection drug, has a wide inter-subject variability and its oral bioavailability ranges from 5% to 70%. Measuring tacrolimus levels would be helpful in monitoring therapy in those patients. TDM could also be used in cases of suspected toxicity in drugs that exhibit concentration-dependent adverse

Table 24.6 Concentration-dependent adverse reactions in select drugs

Drug	Concentration-dependent adverse reactions
Phenytoin	Nystagmus, dizziness, slurred speech, blurred vision, ataxia, depressed mental status, confusion, coma
Vancomycin	Nephrotoxicity
Lithium	Tremors, muscle weakness, delirium, seizures, renal failure
Valproic acid	Gastrointestinal upset, tremors, thrombocytopenia, central nervous system depression, increased liver enzymes, and serum ammonia
Digoxin	Nausea, vomiting, disorientation, bradycardia, hyperkalemia, arrhythmias
Tacrolimus	Neurotoxicity, nephrotoxicity
Theophylline	Nausea, vomiting, tremors, insomnia, arrhythmias, seizures, hyperthermia, brain damage
Voriconazole	Visual and auditory hallucinations

reactions and toxicity as in the examples depicted in Table 24.6.

In some situation, measuring drug concentrations is the only way to predict pharmacokinetic drug–drug interactions. Examples of those interactions include the interaction between phenytoin and valproic acid and the interaction between tacrolimus and clarithromycin. When valproic acid is added to a drug regimen containing phenytoin therapy, it displaces phenytoin from protein binding increasing its unbound fraction. This results in increased phenytoin clearance and V_d . However, with chronic concomitant therapy, valproic acid inhibits phenytoin metabolism, leading to another set of changes to phenytoin pharmacokinetics parameters. This complicated interaction on top of the nonlinear PK of phenytoin is an indication for monitoring the blood levels of phenytoin. Because valproic acid and phenytoin compete at the protein binding sites, measuring the free fraction of phenytoin, when available, could be more informative than the total level. Another example is clarithromycin interaction with tacrolimus. Clarithromycin, which is one of the most commonly prescribed antibiotics in the community, interferes with tacrolimus metabolism, leading to a significant increase in tacrolimus concentrations, which could lead to tacrolimus-induced nephrotoxicity in transplant patients [3]. With the help of TDM, tacrolimus level will need to be monitored when clarithromycin has to be given to the patient. Another indication for TDM is in the conditions where altered pharmacokinetics of the drugs is expected such as in patients with liver or renal disease, in pregnancy, and in older adults. For example, gentamicin half-life increases by 35-fold in patients with renal failure, making the use of TDM very useful in those situations. On the other hand, TDM is not useful in drugs with linear predictable pharmacokinetics and drugs with a wide therapeutic range. In addition, measuring drug levels is not needed in drugs where the pharmacological response is not correlated with drug concentration. Clopidogrel is an example where drug concentration is not correlated with the pharmacological response. The antiplatelet effect of clopidogrel persists beyond the presence of clop-

Table 24.7 Rationale of therapeutic drug monitoring (TDM)

TDM is useful	TDM is <i>not</i> useful
Drugs with a well-established reference range	Predictable pharmacokinetics
Drug response cannot be assessed directly and there is a good correlation between drug concentration and response	Pharmacological response not correlated with drug concentration
Drugs with narrow therapeutic window	Pharmacological response can be easily measured
Drugs with unpredictable pharmacokinetics	Wide therapeutic range
Suspected toxicity	
Medication adherence concerns	
Unpredictable drug interactions	
Disease-related altered pharmacokinetics	

idogrel and its active metabolite. Measuring the platelet function will be more useful than measuring clopidogrel concentration. Furthermore, TDM is not of value if the drug's pharmacological response can be easily measured, such as in the case of antihypertensives. Table 24.7 summarizes the rationale of therapeutic drug monitoring.

Exercise

You are the pharmacist in charge of the drug information center at a local hospital. The lab service manager contacted you to ask for your help. They are preparing a list of drug concentrations that might be needed for TDM. A preliminary list has been developed (Table 24.8), and he asks your help to review it.

Answer: Of the listed drugs in Table 24.8, TDM is of value in the following drugs: vancomycin, phenytoin, tobramycin, carbamazepine, tacrolimus, voriconazole, lithium, valproic acid, digoxin, and theophylline. Vancomycin trough level is a surrogate for the area under the concentration time curve to minimum inhibitory concentration ratio (AUC/MIC), an efficacy marker, and could be used to monitor its efficacy. Phenytoin exhibits a nonlinear pharmacokinetics profile and has a wide interpatient variability in pharmacokinetics, and TDM might be useful. For tobramycin, which is a concentration-dependent antimicrobial, TDM is useful for the aim of maximizing efficacy and minimizing adverse drug reactions. Carbamazepine induces its own metabolism and exhibits concentration-dependent adverse reactions. Tacrolimus, lithium, digoxin, and theophylline exhibit concentration-dependent adverse reactions, and TDM might be useful. In addition, tacrolimus has a wide interpatient variability in drug absorption, adding more benefits to its TDM. For valproic acid, it exhibits concentration-dependent saturable protein binding and concentration-dependent adverse reactions, and TDM might be beneficial. Lastly, voriconazole undergoes nonlinear pharmacokinetics, and its concentration might be related to its efficacy and adverse

Table 24.8 Therapeutic drug monitoring exercise: list of drugs

Vancomycin	Tobramycin	Valproic acid
Phenytoin	Carbamazepine	Lacosamide
Theophylline	Tacrolimus	Voriconazole
Valsartan	Amlodipine	Lithium
Clopidogrel	Vigabatrin	Digoxin; Ampicillin

reactions. On the other hand, TDM is not of value in the following drugs: ampicillin, lacosamide, clopidogrel, vigabatrin, valsartan, and amlodipine. Ampicillin has a wide therapeutic range and predictable kinetics. Similarly, lacosamide has predictable linear pharmacokinetics. Clopidogrel and vigabatrin blood concentrations do not correlate with the pharmacological response, and their levels are not useful. Lastly, valsartan and amlodipine have easily measurable drug responses, the blood pressure.

The utility of TDM is expected to grow substantially in the future as more evidence emerges on the impact of interpatient variability in drug pharmacokinetics and pharmacodynamics. Advances in pharmacogenomics, clinical analytics, and real-world data are highlighting that factors such as genetic polymorphisms, organ function, comorbidities, and drug–drug interactions can lead to significant differences in drug exposure and response among individuals. As this body of evidence expands, TDM may increasingly be applied not only to traditional narrow therapeutic range medications but also to drugs that have not historically required monitoring, especially in populations at higher risk of subtherapeutic effects or toxicity.

Approach to Therapeutic Drug Monitoring

Pharmacists play an important role in therapy assessments using therapeutic drug monitoring. The following are the general TDM steps whenever a patient gets started on a drug where we have the capability of measuring its concentration:

- *Assessment if the drug is indicated and if it is a reasonable option for the patient.* For instance, when a physician prescribes vancomycin for a patient, dose or order adjustments are only necessary after verifying that it is the suitable antimicrobial for the patient or if there are no other therapeutic alternatives available that do not require TDM.
- *Assessment of the appropriateness of initial dose of the drug.*
- *Determine if TDM of the selected drug is beneficial* (see previous section).
- *Assessment if a drug concentration monitoring is indicated.* This is a very important step. A drug level does not need to be measured just because we have the capability

of measuring it. Drug concentrations shall be used as a tool to assess patient's drug therapy rather than an ultimate goal to target (Remember: *treat the patient, not the level*). Measuring routine levels is not recommended. Drug levels are only measured if they are indicated. The following are the general indications for measuring a drug level:

- Lack of efficacy.
- Suspected drug toxicity.
- Medication adherence concerns.
- Presence of unpredictable drug interactions.
- Drugs with nonlinear pharmacokinetics or wide interpatient variability.
- Presence of a superimposing comorbidity that might complicate the drug's pharmacokinetic profile, e.g., pregnancy.
- To determine the patient's "therapeutic" drug concentration.
- *Determine the timing of the drug level withdrawal.* Drug levels are generally measured at steady state (3–5 drug half-lives). However, pre-steady-state levels can be beneficial in some situations to determine the adequacy of the dosage. With regard to the timing related to the dose, generally, trough levels are recommended. However, for some drugs, peak levels might be needed, such as in the case of conventional dosing of aminoglycosides. Always check drug-specific TDM information to determine the appropriate timing of the drug concentration.
- *Assessment of the drug level in the context of the patient's clinical picture.* This allows for determining if dose alteration is needed.

Clinical Pearls

- Pharmacokinetic concepts are essential in day-to-day clinical practice.
- Bioavailability, salt factor and rate of drug absorption help in determining the equivalent dosage in formulation conversions.
- Volume of distribution reflects drug distribution in the body and helps determine the loading dose of the drug.
- Total clearance is the sum of the drug's renal and non-renal clearance. Appreciation of the renal and non-renal components of drug clearance helps assess the need for dosage adjustments in renal and/or hepatic impairment.
- Knowledge of the drug half-life helps determine the time it takes for the drug to reach steady state and the time it takes to get out of the system.
- TDM is an important therapeutic tool in disease management. If used for the right drug, at the right time, and when needed, TDM will be a powerful tool to optimize medication therapy.

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Chapter Objectives

1. Understand the different types of liver diseases.
2. Interpret and understand the significance of liver chemistry tests.
3. Interpret and utilize the different scoring systems to assess hepatic function.
4. Assess drug-induced liver injury and differentiate between its different types.

Background

The liver is the largest organ of the body, and it performs a multitude of complex functions that helps in maintaining homeostasis and health. The liver receives its blood supply from two different sources. It receives oxygen-rich blood via the hepatic artery and receives nutrient-rich blood via the portal vein arising from the stomach, intestine, pancreas, and spleen. Hepatocytes constitute the majority of the cells in the liver, and they are involved in carrying out the different functions of the liver. These include synthesis of serum proteins (such as albumin and coagulation factors), regulation of metabolism (glucose, lipids, amino acids, and cholesterol), and biotransformation and detoxification of endogenous as well as exogenous compounds, including drugs [1]. This array of functions performed by the liver makes it more complicated to measure the function of the liver. There is no one test available that can provide an assessment of the liver's total functional capacity. Thus, the diagnosis and assessment of liver function and disease involves a detailed examination of the patient's clinical history, physical examination, imaging, as well as patterns of abnormalities in laboratory tests of liver injury and function. This chapter aims to help the pharmacist to have an assessment of the patient's liver function

and to investigate possible drug-induced causes of liver disease as well as manage drug therapy in patients with existing liver disease.

Liver Diseases

There are many causes of liver diseases, and they can manifest in different ways. They can be classified clinically into three broad categories [1]:

- *Hepatocellular*: This mainly involves damage to the hepatocytes. Examples of the common causes of hepatocellular injury include acute viral hepatitis and alcoholic liver disease.
- *Cholestatic*: This mainly involves inhibition of bile flow, which can result from intrahepatic or extrahepatic cholestasis. Intrahepatic cholestasis involves a dysfunction in the secretion of bile by the hepatocytes, as well as diseases involving the bile ducts within the liver. Extrahepatic cholestasis involves the obstruction of the bile ducts outside the liver, which can happen due to several reasons such as gall stones, primary sclerosing cholangitis, pancreatitis, and tumors [2].
- *Mixed*: It features both hepatocellular and cholestatic injury. This includes many drug-induced liver diseases and viral hepatitis.

Liver Chemistry Tests

Due to the multitude of physiological functions that the liver performs, there is no one single test that can assess total liver function. However, utilizing a panel of liver tests that can measure some of the functions carried out by the liver or detect liver damage can provide information on the presence of liver disease, type of liver disease, extent of liver damage, as well as response to treatment. At most healthcare facilities

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ties, liver chemistry tests can be ordered as a panel referred to as *LFT panel* or *LFTs* (liver function tests). This naming can be misleading, as not all the liver chemistry tests included actually measure the “liver function.” *LFTs* usually consist of a series of blood tests measuring aminotransferases, including alanine aminotransferase (ALT), aspartate aminotransferase (AST), and alkaline phosphatase (ALP), as well as bilirubin and albumin. Table 25.1 separates liver chemistry tests into four broad categories based on the hepatic process they assess.

It is important to note that an abnormal *LFT* usually means a value exceeding the upper reference limit, which can vary among laboratories. It is common practice to define the normal range as mean value ± 2 standard deviations of the values observed in the reference normal population. Thus, abnormal aminotransferase levels can be observed in the normal population, and patients with chronic liver diseases can sometimes present with normal aminotransferase levels [2–5].

Liver Synthetic Function Tests

One of the main functions of the liver is the biosynthesis of various proteins, notably albumin, and the clotting factors. The measurement of the levels of these proteins can provide an assessment of the biosynthetic function of the liver. However, these tests are usually not sensitive enough to detect lower levels of liver injury or dysfunction due to the liver's high functional reserve. This means that the liver can maintain normal levels of its biosynthetic function despite having significant damage. Inadequate synthetic function is usually observed in hepatic cirrhosis or massive liver injury. In these situations, these tests are very useful in determining prognosis and recovery.

Table 25.1 Categories of liver chemistry tests by the hepatic process [2]

Hepatic process	Liver chemistry test
Synthetic function	Albumin Prealbumin PT/INR
Cholestasis	Bilirubin ALP GGT 5'-nucleotidase
Hepatocellular injury	ALT AST
Detoxification	Ammonia

ALT alanine aminotransferase, *AST* aspartate aminotransferase, *ALP* alkaline phosphatase, *GGT* gamma-glutamyl transferase, *INR* international normalized ratio, *PT* prothrombin time

Albumin

Albumin is a major plasma protein that is synthesized exclusively by hepatocytes. It functions to maintain plasma oncotic pressure and binds and transports various molecules, including drugs, hormones, and waste products such as bilirubin. The albumin normal range is 40–50 g/L, and it has a relatively long serum half-life of ≈ 19 –21 days [6]. Therefore, the decline in serum albumin following the onset of liver damage is generally slow. Thus, albumin levels are generally reduced in patients with chronic cirrhosis but are usually within normal limits in patients with acute hepatic dysfunction due to acute viral hepatitis or drug-induced hepatic injury [2].

In addition, albumin levels can be reduced in other clinical scenarios irrespective of liver disease. These include malnutrition or malabsorption, chronic kidney disease, severe burns, increased blood volume, and systemic inflammation. Hospitalized patients who are acutely ill usually have low albumin levels due to a combination of systemic inflammation, malnutrition, and IV fluid administration. A low albumin level, together with no alterations in other liver tests or clinical evidence of liver disease, is usually indicative of a non-hepatic cause for hypoalbuminemia. On the other hand, in patients with known cirrhosis and decreased albumin levels, monitoring of albumin can help with assessing the prognosis of the case [3].

Another important aspect that is important for pharmacists to consider is that hypoalbuminemia will impact the pharmacokinetics of highly protein-bound drugs. Lower albumin levels will result in higher proportion of free drug which could potentially result in increased activity or increased side effects. A common example is phenytoin, where it is recommended to use the free phenytoin level rather than the total phenytoin level for therapeutic drug monitoring in patients with low albumin levels.

Prealbumin

Prealbumin is another plasma protein closely related to albumin. As compared to albumin, it has a shorter serum half-life of ≈ 2 days and is more sensitive to protein nutrition and less sensitive to hepatic function or fluid status [7]. It is mainly utilized clinically to assess the nutritional status of acutely ill hospitalized patients.

Prothrombin Time (PT)/International Normalized Ratio (INR)

All the blood-clotting factors are synthesized exclusively by the hepatocytes, with the exception of factor VIII, which is synthesized by the vascular endothelial cells. Compared to

albumin, the half-lives of the clotting factors are much shorter ranging from 6 h to 5 days, which makes measuring them a good indicator for hepatic synthetic function during acute hepatic dysfunction. However, the liver has a great synthetic reserve, and the levels of blood clotting factors will decrease only due to substantial hepatic impairment (>80% of the synthetic capability) [2].

Serum prothrombin time (PT) measures the activity of factors II, V, VII, and X, which are produced by the hepatocytes and require vitamin K for their activation. The international normalized ratio (INR) is a standardized index that standardizes PT, taking into account the differences in test reagents used at different laboratories. INR/PT values are elevated when there are reduced circulating levels of these factors or their activation is impaired. Thus, impaired hepatic synthesis, vitamin K deficiency, or using vitamin K antagonists (such as warfarin) will result in elevated PT/INR. Response to vitamin K administration can be used as a tool to examine the etiology of prolonged PT/INR. Elevated PT/INR due to vitamin K antagonists, malabsorption, or antibiotic-induced disturbance in gut flora will respond well to vitamin K, while vitamin K is not effective in the case of hepatic synthetic dysfunction.

Liver Tests That Reflect Cholestasis

These are tests that look at the excretory function of the liver. They include bilirubin, alkaline phosphatase (ALP), 5'-nucleotidase, and γ -glutamyl transpeptidase (GGT). These tests can help differentiate between hepatocellular and cholestatic liver diseases, but they cannot distinguish between intrahepatic and extrahepatic cholestasis. This is usually diagnosed by diagnostic imaging, as extrahepatic cholestasis is usually associated with dilation of the bile ducts.

Bilirubin

Bilirubin is the breakdown product of heme-containing proteins, most important of which is hemoglobin in red blood cells. Bilirubin is present in the blood as conjugated (direct) and unconjugated (indirect) bilirubin. Unconjugated bilirubin is lipophilic and is bound to albumin in the blood, while conjugated bilirubin is water-soluble and can be excreted in the bile as well as by the kidneys. Jaundice, also known as icterus, results from elevated bilirubin in the blood. It manifests as yellow discoloration of the skin and sclera of the eye [2].

In patients with elevated bilirubin, it is important to identify if only unconjugated bilirubin is elevated or both fractions are elevated. An isolated increase in unconjugated bilirubin (more than 70% unconjugated) is seen most commonly in hemolytic disorders or due to some genetic condi-

tions, such as Gilbert's syndrome, and is rarely due to liver disease [8]. Hemolytic disorders result in excessive destruction of red blood cells, leading to increased production of unconjugated bilirubin exceeding the capacity of the liver to conjugate and excrete bilirubin, leading to hyperbilirubinemia. On the other hand, conjugated hyperbilirubinemia (more than 50% conjugated bilirubin) usually results from liver or biliary tract disease. Elevation of conjugated bilirubin can happen in any type of liver disease, while isolated elevation in unconjugated bilirubin usually refers to a non-hepatic cause.

Bilirubin can also be measured in the urine. Unconjugated bilirubin is usually albumin-bound and is not cleared by the kidney, while conjugated bilirubin is water-soluble and can be filtered by the kidney [8]. A urine dipstick to test for bilirubin in the urine can be used as an alternative to the fractionation of serum bilirubin. The presence of bilirubinuria usually implies the presence of liver disease [8].

Alkaline Phosphatase (ALP), 5'-Nucleotidase, γ -Glutamyl Transpeptidase (GGT)

Alkaline phosphatase (ALP) is found in various body tissues, mainly in the liver, bone, small intestine, and placenta. ALP levels can be higher in children (up to three times the adult range) due to bone growth. Also, ALP can be elevated during late pregnancy as a result of increased placental ALP. Other than that, elevated ALP is usually correlated clinically with cholestatic liver disease. An increase in ALP levels of more than four times the normal range usually suggests a cholestatic disorder, while levels of three times normal or less can occur in any type of liver disease.

ALP can also be elevated due to non-hepatic causes, which include bone disorders (Paget disease, healing fractures, osteoporosis, rickets, osteomalacia, hypervitaminosis D, or vitamin D deficiency), diabetes mellitus, renal failure, sepsis, neoplasms, hyperparathyroidism, and hyperthyroidism [2]. One practical approach to identify if the liver is the source of the elevated ALP is to measure serum 5'-nucleotidase or GGT. If they are elevated, then it is highly likely that the elevated ALP is due to a liver disorder [8]. These enzymes are rarely elevated in non-hepatic disorders. In addition, GGT is usually elevated in alcoholic liver disease, where a GGT/ALP ratio >2.5 usually refers to alcohol abuse [2].

Liver Tests That Reflect Hepatocellular Injury

Aminotransferases (ALT and AST)

Aminotransferases (also known as transaminases) are sensitive indicators of hepatocyte injury and thus are useful in identifying acute hepatic diseases such as hepatitis [8]. They include aspartate aminotransferase (AST), also known as serum glutamic oxaloacetic transaminase (SGOT), as well as

alanine aminotransferase (ALT), also known as serum glutamic–pyruvic transaminase (SGPT). These enzymes are primarily located inside the hepatocytes and are released into the serum when the hepatocytes are damaged. Aminotransferase levels are very sensitive and are elevated even with minor hepatocyte injury. Small increases (up to 300 IU/L) are nonspecific and can occur with minor liver disorders, while marked elevations (more than 1000 IU/L) are usually a marker of extensive hepatocellular injury, which usually occurs due to viral hepatitis, ischemic hepatitis, or drug-induced liver injury [8]. However, aminotransferase levels provide no prognostic benefit in acute liver disease.

The ratio of AST to ALT can provide valuable diagnostic information. Alcoholic liver diseases are usually characterized by an AST:ALT ratio of more than 2. This is mainly due to lower ALT levels secondary to alcohol-induced deficiency of pyridoxal phosphate [8].

Tests Reflecting Detoxification

Ammonia

Ammonia is primarily produced in the body during normal protein metabolism as well as by bacterial catabolism of proteins in the colon. Ammonia is mainly detoxified in the liver (converted into urea) and the muscles (binds to glutamic acid to form glutamine). Therefore, patients with advanced liver disease or muscle wasting can develop hyperammonemia [9]. Ammonia possibly plays a role in the development of hepatic encephalopathy, which is a group of neuropsychiat-

ric symptoms ranging from changes in personality to coma. Increased ammonia in the blood, together with increased permeability of the blood–brain barrier to ammonia, commonly seen in patients with hepatic encephalopathy, results in ammonia crossing the blood–brain barrier and inducing brain edema [2, 9]. However, ammonia levels do not correlate well with hepatic encephalopathy in patients with chronic liver failure, and the diagnosis of hepatic encephalopathy relies more on the patient's history and clinical evaluation [2].

In addition, ammonia levels can be elevated as a result of medications (most commonly valproic acid), urea cycle disorders, Reye syndrome, and infections with ammonia-producing bacteria (such as mycoplasma) [2, 10].

Patterns of Liver Test Abnormalities

The liver has a complicated physiological role and performs a multitude of physiological functions. There is no one test that can provide an assessment of the overall liver function; however, each one of the tests mentioned in the previous sections assesses a certain aspect of the liver's function and status. Thus, it is important not to look at the results of any of the liver tests in isolation from the clinical picture, the results of other relevant tests, or the results of diagnostic imaging. By looking at the pattern of the different liver tests in tandem, one can determine the general category of liver disease. Table 25.2 depicts various patterns of liver tests associated with the different classes of liver diseases.

Table 25.2 Patterns of liver tests associated with the different classes of liver diseases [1]

Liver disorder	Bilirubin	Aminotransferases (ALT, AST)	ALP	Albumin	PT/INR
Acute hepatocellular disease (viral, drug-induced, or ischemic hepatitis)	Both fractions (conjugated and unconjugated) elevated Bilirubinuria	Both ALT and AST elevated (>1000 IU/L) ALT > AST	Normal or mild elevation (<3 times normal)	Normal	Usually normal Poor prognosis if elevated (more than 5 times normal) and not corrected by vitamin K
Chronic hepatocellular disease	Both fractions elevated Bilirubinuria	Mild elevation (<300 IU/L)	Normal or mild elevation (< 3 times normal)	Decreased	Prolonged Not corrected by vitamin K
Alcoholic hepatitis	Both fractions elevated Bilirubinuria	AST:ALT >2	Normal or mild elevation (<3 times normal)	Decreased.	Prolonged Not corrected by vitamin K
Hepatic cholestasis	Both fractions elevated Bilirubinuria	Normal or mild elevation (<300 IU/L)	Highly elevated (>4 times normal)	Normal unless chronic condition	Normal If prolonged will respond to vitamin K
Hemolytic disorders	Increased unconjugated fraction No Bilirubinuria	Normal	Normal	Normal	Normal

ALT alanine aminotransferase, AST aspartate aminotransferase, ALP alkaline phosphatase, INR international normalized ratio, PT prothrombin time

Other Liver Assessment Tests

Liver laboratory tests can provide a general category of liver disease. Additional tests are usually required to make the proper specific diagnosis and to assess the severity of disease. These include:

- *Liver ultrasound*: important in distinguishing between intrahepatic and extrahepatic cholestasis.
- *Percutaneous liver biopsy*: important for assessing hepatic fibrosis, which helps to identify the severity and stage of liver disease (early, advanced, precirrhotic, or cirrhotic).
- *Noninvasive tests to detect hepatic fibrosis*: these include transient elastography and magnetic resonance elastography [8].
- *Tests for viral hepatitis*: These include serologic tests for antibodies and molecular assays to detect viral genetic material. These tests, together with the clinical history, epidemiology, and patient risk factors, can help provide a correct diagnosis for viral hepatitis and distinguish between the different hepatitis viruses (Hepatitis A, B, C, D, or E) [2].

Scoring Systems of Liver Diseases

Liver cirrhosis is associated with high rates of mortality and morbidity, and the 1-year mortality rate ranges widely from 1% to 57% based on the stage of hepatic cirrhosis and the complications associated with it [11]. This highlights the importance of identifying patients at high risk. Liver biopsy is the most accurate way to assess the severity and stage of hepatic cirrhosis. However, cirrhosis can also be staged clinically by using the Child-Turcotte-Pugh classification (also referred to as Child-Pugh score or CTP score). Table 25.3 shows the details of CTP score criteria. CTP score is a good predictor of prognosis in hepatic cirrhosis and can provide guidance to identify patients eligible for liver transplant. In

Table 25.3 Child-Turcotte-Pugh (CTP) score [11]

	1	2	3
Encephalopathy	None	Grade I-II (or suppressed with medication)	Grade III-IV (or refractory)
Ascites	None	Mild	Moderate-refractory
Albumin (g/L)	>35	28–35	<28
Bilirubin (umol/L)	<34	34–50	>50
INR	<1.7	1.7–2.3	>2.3

Class A (Mild): ≤6; Class B (Moderate): 7–9; Class C (Severe): ≥10

Table 25.4 Three-month mortality based on MELD and CTP scores [13]

		3-month mortality %
MELD score	<9	1.9
	10–19	6.0
	20–29	19.6
	30–39	52.6
	>40	71.3
CTP score	<7–9	4.3
	10–12	11.2
	13–15	40.1

CTP Child-Turcotte-Pugh (CTP) score, MELD Model for end-stage liver disease

addition, it is usually used to guide drug dosing adjustment in patients with liver cirrhosis.

The model for end-stage liver disease (MELD) is another scoring system that has currently replaced the Child-Pugh score for identifying liver transplant candidates. It incorporates only three noninvasive variables: INR, serum bilirubin level, and serum creatinine concentration [11]. It uses a linear regression mathematical equation involving these three variables to provide a numerical value that correlates with mortality from liver disease [12]. The higher the MELD score, the higher the mortality and morbidity [13]. Table 25.4 depicts the correlation of mortality rates with CTP and MELD scores [13].

Drug-Induced Liver Injury

The liver plays a major role in detoxifying and eliminating drugs, and at the same time is susceptible to toxicity due to exposure to medications. Over one thousand medications, herbal supplements, illicit drugs, and chemicals can induce hepatotoxicity [14]. The incidence of drug-induced liver injury (DILI) has been increasing, and it is the main reason for post-marketing drug withdrawals [14]. Most cases of DILI recover completely with no residual damage or injury. However, the rare cases that progress into developing acute liver failure, chronic liver injury, cirrhosis, or vanishing bile duct syndrome (chronic cholestasis and loss of intrahepatic bile ducts) can result in death or requirement of liver transplantation [15].

Adults are generally more likely to develop DILI as compared to children, and women are at higher risk than men. Other factors that can increase the risk of DILI include obesity, malnutrition, pregnancy, concurrent medications, genetic susceptibility, and a history of drug reactions [16]. Preexisting liver diseases and other comorbidities do not affect the risk of developing DILI as much as they affect the ability of the patient to recover from DILI [17].

Types of DILI

DILI can be either due to direct toxic effects or can be idiosyncratic. Idiosyncratic drug-induced hepatotoxicity is generally more predominant, while only a few agents can induce direct hepatotoxicity (e.g., acetaminophen and methotrexate) [14].

Direct Toxic

Direct toxic drug-induced hepatotoxicity is generally predictable and dose-related and usually occurs within a short time period after exposure. In addition, direct hepatotoxins usually produce distinct morphological abnormalities that are reproducible and characteristic of each toxin [16, 18].

Idiosyncratic

On the other hand, idiosyncratic drug-induced hepatotoxicity is usually unpredictable and dose-independent and can occur within a few days up to 12 months after exposure. Unlike direct hepatotoxins, extrahepatic manifestations such as arthralgia, fever, rash, leukocytosis, and eosinophilia can occur in about one quarter of patients with idiosyncratic drug-induced hepatotoxicity [18].

Patterns of Liver Enzyme Changes in DILI

Similar to general liver diseases, DILI can be categorized based on the changes in liver enzymes into hepatocellular, cholestatic, or mixed disease [19].

Hepatocellular Injury

DILI due to hepatocellular injury is usually characterized by marked elevations in ALT and AST, which usually precede elevations in total bilirubin, and are associated with modest increase in ALP levels [16]. Hepatocellular injury is defined by the US Food and Drug Administration as an increase in ALT of at least three times above the upper limit of normal (ULN) together with a rise in total bilirubin of at least two times above ULN and $R > 5$ [19], where R is defined as in Eq. 25.1.

$$R = \frac{(\text{Measured ALT} / \text{ULN of ALT})}{\div (\text{Measured ALP} / \text{ULN of ALP})}. \quad (25.1)$$

Common examples of DILI due to hepatocellular injury are toxicity due to isoniazid and acetaminophen.

Cholestatic Injury

Cholestatic injury is characterized by an increase in ALP level that precedes or is more prominent than increases in ALT or AST [16]. It is defined as ALP >3 times above ULN

and total bilirubin above 2 times ULN, and R less than or equal to 2. This injury can be caused by medications like erythromycin, carbamazepine, or amoxicillin-clavulanic acid [19].

Mixed Injury

In this kind of injury, both ALT and ALP are elevated, and R ranges from 3 to 4. Medications such as phenytoin, phenobarbital, and sulfonamides could cause mixed injury [16, 19].

Diagnosis of DILI

The clinical symptoms and signs, as well as the patterns of liver test abnormalities in DILI, can be difficult to distinguish from other forms of liver diseases. There is no specific clinical, laboratory, or histological feature that can definitely diagnose DILI [20]. The diagnosis of DILI is more of a diagnosis of exclusion that relies on clinical judgment, potential of the agent in question to cause liver injury as compared to other causes of liver injury, and liver test abnormalities. It is important to consider the following six factors in DILI diagnosis [15]:

1. Time to onset.
2. Time to recovery.
3. Injury pattern (hepatocellular, cholestatic, or mixed).
4. Exclusion of other causes of liver injury.
5. The drug in question is reported to induce liver injury.
6. Response to reexposure.

Because of these multiple factors, the diagnosis of DILI can be complicated, and identifying causality can be debatable. A number of tools have been developed to standardize the assessment of causality [15]. These include:

- *Roussel Uclaf Causality Assessment Method (RUCAM) scale*: specifically developed for assessment of DILI, and includes liver-specific criteria, including risk factors and use of other hepatotoxic drugs [21]. Despite its complexity, the RUCAM scale remains the most widely used and studied tool to assess causality in DILI [20]. See Table 25.5 for details of the RUCAM scale.
- *Maria and Victorino (M & V) Scale*: a modification of the RUCAM scale meant to simplify it and to improve its usability [22].
- *Naranjo Probability Scale*: not specific to DILI and can be used to assess any potential drug-related adverse effect [23] (see Chap. 2).

Table 25.5 RUCAM Scale to assess causality for DILI

	Hepatocellular type		Cholestatic or mixed type		Assessment
1. Time to onset	Initial treatment	Subsequent treatment	Initial treatment	Subsequent treatment	Score
From the beginning of the drug:					
Suggestive	5–90 days	1–15 days	5–90 days	1–90 days	+2
Compatible	<5 or > 90 days	>15 days	<5 or > 90 days	>90 days	+1
From cessation of the drug:					
Compatible	≤15 days	≤15 days	≤30 days	≤30 days	+1
If reaction occurred before starting the drug or ≥ 15 days (hepatocellular) or ≥30 days (cholestatic) after stopping, the injury should be considered unrelated, and RUCAM cannot be calculated					
2. Course	Change in ALT between peak value and ULN		Change in ALP (or total bilirubin) between peak value and ULN		Score
After stopping the drug:					
Highly suggestive	Decrease ≥50% within 8 days		Not applicable		+3
Suggestive	Decrease ≥50% within 30 days		Decrease ≥50% within 180 days		+2
Compatible	Not applicable		Decrease <50% within 180 days		+1
Inconclusive	No information or decrease ≥50% after 30 days		Persistence or increase or no information		0
Against the role of the drug	Decrease <50% after 30 days OR recurrent increase		Not applicable		–2
If the drug is continued:					
Inconclusive	All situations		All situations		0
3. Risk factors					Score
Alcohol or pregnancy	Alcohol: Present		Alcohol or pregnancy: Present		+1
	Absent		Absent		0
Age	Age of the patient: ≥55 years				+1
	<55 years				0
4. Concomitant drugs					Score
None or no information or concomitant drug with an incompatible time to onset					0
Concomitant drug with suggestive or compatible time to onset					–1
Concomitant drug known to be hepatotoxic, with a suggestive time to onset					–2
Concomitant drug with clear evidence for its role (positive rechallenge or clear link to injury, and typical signature)					–3
5. Exclusion of other causes of liver injury					Score
Group I (6 causes):			All causes in groups I and II were ruled out		+2
Acute viral hepatitis due to HAV, HBV, or HCV, biliary obstruction, alcoholism, recent history of hypotension, shock, or ischemia			The 6 causes of group I were ruled out		+1
			Five or 4 causes of group I were ruled out		0
Group II:			Less than 4 causes of group 1 were ruled out		–2
Complications of underlying liver disease(s), or clinical features or serologic, and virologic tests indicating acute CMV, EBV, or HSV infection			Nondrug cause is highly probable		–3
6. Previous information on hepatotoxicity of the drug:					Score
Reaction labeled in the product characteristics					+2
Reaction published but unlabeled					+1
Reaction unknown					0
7. Response to readministration:					Score
Positive	Doubling of ALT with drug alone		Doubling of ALP (or bilirubin) with drug alone		+3
Compatible	Doubling of the ALT with the suspect drug combined with another drug that had been given at the time of onset of the initial injury		Doubling of the ALP (or bilirubin) with the suspect drug combined with another drug that had been given at the time of onset of the initial injury		+1
Negative	Increase of ALT but less than ULN with drug alone		Increase of ALP (or bilirubin) but less than ULN with drug alone		–2
Not done or not interpretable	Other situations		Other situations		0

Roussel Uclaf Causality Assessment Method (RUCAM) score <3 (unlikely), 4–5 (possible), 6–8 (probable), and >8 (highly probable). (Adapted from LiverTox.nih.gov [15])

Dosage Adjustment in Liver Diseases

As the liver has a wide range of physiological functions, liver impairment can have a significant impact on various aspects of drug pharmacokinetics. Unfortunately, there is no single laboratory test that can measure the overall liver function in the same way as creatinine clearance is used to assess renal function [24]. In patients with acute hepatitis, there is a mild or transient decrease in liver drug metabolism that usually does not necessitate dosage adjustment. On the other hand, patients with liver cirrhosis, the permanent loss of hepatocyte function necessitates dosage adjustment for hepatically metabolized drugs [24]. CTP score can be used to guide drug dosing in patients with liver cirrhosis. Some drug monographs will provide dosing guidance based on CTP scores. In general, for drugs that are primarily metabolized by the liver (>60%), it is appropriate to moderately decrease the initial dose (~25%), for a CTP score of 8–9, and to decrease the initial dose further for CTP score of 10 or greater (~50%) [24].

Oral drug absorption can also be impacted as hepatic dysfunction can result in decreased first-pass metabolism, resulting in increased bioavailability of oral drugs affected by first-pass metabolism [24]. In addition, decreased plasma albumin in patients with liver disease can significantly affect highly protein-bound drugs. The decreased albumin can result in an increased free fraction of the drug, which has to be taken into account during dose adjustments. It is also important to account for the increased volume of distribution in patients with significant ascites. Water-soluble medications, such as beta-lactams might require a higher loading dose to account for the increased volume.

For medications where there is no clear recommendation for drug dosing in hepatic dysfunction, the pharmacist can utilize their understanding of the drug's pharmacokinetics, and the availability of alternatives to guide their decision. These principles include:

- Change to a therapeutic alternative not affected by the liver function.
- Decrease initial dose, which can be guided by CTP score.
- Titrate dose slowly with close monitoring of therapeutic response and adverse effects.
- Close therapeutic drug monitoring, if available.

Clinical Pearls

- The liver performs a wide array of physiological functions, and there is no single test to assess its overall function.
- Assessment of patients with liver disease requires incorporating an understanding of the different liver chemistry

tests with the patients' overall clinical picture, and other investigations.

- Diagnosis of drug-induced liver disease is a diagnosis of exclusion. The pharmacist has to take many factors into consideration to assess causality.
- Hepatic dysfunction can significantly impair various aspects of drug pharmacokinetics, which necessitates close monitoring by the pharmacist.

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Rene R. Breault

Chapter Objectives

1. Describe basic kidney physiology and function.
2. Explain the definitions of acute kidney injury (AKI) and chronic kidney disease (CKD).
3. Describe the risk factors for both chronic kidney disease (CKD) and acute kidney injury (AKI).
4. Apply a systematic approach to assessing patients' kidney function.
5. Interpret the various laboratory tests used to assess kidney function.

Background

Kidney disease affects 1 in 10 Canadians with millions more at risk [1]. Worldwide, approximately 10% of the population is affected by chronic kidney disease (CKD) [2]. A significant proportion of individuals may go unrecognized and therefore not receive timely treatment to delay progression of their kidney dysfunction. Pharmacists are in an ideal position to be able to help screen and assess patients who may be at risk of CKD or AKI. Research has shown that targeted screening by pharmacists can help identify patients with CKD, which may have important implications for prevention and management of the disease [3]. Assessing kidney function is an important skill not only for determining the appropriateness of drug dosages but also to help with ongoing monitoring and management of kidney disease.

The kidneys are responsible for the following functions [4]:

1. Regulation of fluid volume, osmolarity, blood pressure, electrolyte concentrations, and acidity.

2. Excretion of metabolic end products and foreign substances such as urea, toxins, and drugs.
3. Synthesis of renin, erythropoietin, and calcitriol (vitamin D3).

The kidneys receive their blood supply from the renal arteries and receive about 20% of the cardiac output under normal resting conditions [5]. The functional unit of the kidney is the nephron, and each kidney contains approximately one million of these microscopic subunits, which work to filter the blood and produce urine [5]. Three exchange processes occur within the nephrons: (1) glomerular filtration, (2) tubular reabsorption, and (3) tubular secretion. When the kidneys begin to fail due to acute injury/toxicity or chronic disease, these exchange processes can become impaired, and if undetected or untreated, can progress to impede the regulatory, excretory, and synthetic functions of the kidney.

Approach to Assessing Renal Function

Assessment of renal function involves the following key steps:

1. Obtain patient clinical history (medical/medication history, signs, and symptoms) and demographic information to determine if they are at risk for renal dysfunction.
2. Obtain relevant laboratory data in order to assess markers of renal function.
3. Determine if the patient has acute kidney injury (AKI) or chronic kidney disease (CKD) according to guideline definitions.
4. Ensure appropriate therapies are in place to manage the renal dysfunction (AKI or CKD).
5. Discontinue any nephrotoxic drugs (AKI) and/or adjust drug dosages (CKD).
6. Provide ongoing monitoring and assessment of renal function and potential associated complications.

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Markers of Kidney Function

Glomerular filtration rate (GFR) cannot be measured directly. An ideal marker of kidney function should be at a stable concentration in plasma, physiologically inert, freely filtered at the glomerulus, not secreted, reabsorbed, synthesized, or metabolized in the kidney; and the amount secreted at the glomerulus is equal to the amount excreted in the urine.

Exogenous and Endogenous Markers

Table 26.1 lists the various exogenous and endogenous markers used to assess kidney function. These methods are generally used for research purposes and are typically not used in clinical practice except in circumstances when specific patient factors may render traditional methods of assessment, such as serum creatinine, unreliable or inaccurate. Their invasiveness and cost also prohibit their widespread use in everyday practice.

Serum creatinine (SCr) is the most commonly used marker to assess renal function. SCr is used to estimate glomerular filtration rate and to calculate creatinine clearance for the purpose of drug dosage adjustments in renal impairment. Serum creatinine can be used to assess patients with acute kidney injury and chronic kidney disease, which will be discussed later in the chapter. Proteinuria (protein in the urine) is also used to screen for and monitor progression of chronic kidney disease.

Serum Creatinine (Reference Range 50–110 $\mu\text{mol/L}$)

Creatinine is a nonprotein, nitrogenous, metabolic by-product of the muscles, which under normal physiologic conditions remains constant if muscle mass is not significantly changed. It is mostly eliminated by glomerular filtration, and at steady state, the rate of creatinine production equals its excretion. Although there is an inverse relationship between SCr and kidney function, it should not be the only method for evaluation of renal function, as there are a number of factors that can affect SCr concentrations (Table 26.2).

Creatinine clearance (CrCl) can be measured using a 24-h or timed urine collection to estimate a patient's renal function. This method is often difficult to implement and prone to errors in collection, but it can be useful in specific patients. It

Table 26.2 Factors affecting serum creatinine concentration (SCr)

Decreased SCr	Increased SCr
Paralysis, low activity level	Renal impairment
Elderly	Large dietary protein intake
Decreased muscle mass	Vigorous exercise
Cirrhosis	Increased muscle mass

is more common to use the measured SCr to calculate an estimate of a patient's renal function. The most utilized equation is the Cockcroft–Gault formula (Eq. 24.1) [6]. The formula uses the patient's age and body weight, in addition to the SCr:

$$\text{CrCl (ml/min)} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{\text{SCr } (\mu\text{mol/L})} \times 1.2 \text{ (males)} \quad (26.1)$$

There has been debate in recent years as to which weight to use in the formula, with weight-adjusted and nonweight-based formulas being developed in attempts to improve the accuracy of the equation. Using total body weight in an overweight individual may overestimate renal function and, conversely, underestimate renal function in an underweight person. Some clinicians use an adjusted or ideal body weight in these individuals in hopes that this may provide a more accurate estimate of creatinine clearance. It should also be noted that this formula should only be used in patients with a SCr at steady state, and it will not be of clinical value in patients with acute kidney injury or on dialysis. Creatinine clearance as calculated by the Cockcroft–Gault equation, has been used to determine dosages for many medications that are renally excreted, and there are many resources available to pharmacists to determine drug dosages based on renal function.

Equations to estimate GFR include the Modification of Diet in Renal Disease (MDRD) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formulas [7, 8]. These formulas are used to determine the staging of chronic kidney disease due to their improved predictive performance over the Cockcroft–Gault equation. It has also been suggested that they be incorporated into drug monographs to provide dosage recommendations for patients with renal dysfunction. Many laboratories now report an estimated GFR with the measured SCr using one of the aforementioned equations.

Both of the above equations are normalized to 1.73 m² body surface area. Online calculators are readily available for clinicians to utilize these formulas. As with the Cockcroft–Gault formula, they should only be used with stable chronic kidney disease, not in patients with acute kidney injury.

Table 26.1 Markers of kidney function

Exogenous markers	Endogenous markers
Inulin	Cystatin C
Iothalamate	
Cr-EDTA (ethylenediamine tetra-acetic)	

Blood Urea Nitrogen (BUN) (Reference Range 2.9–8.2 mmol/L)

BUN is the concentration of nitrogen (as urea) in the serum. Its serum concentration depends on urea production, which occurs in the liver, GFR, and tubular reabsorption. On its own, it cannot be used to predictably assess renal function. In conjunction with other laboratory data, it can be used to monitor hydration, renal function, and protein tolerance and catabolism. It can also be used to predict the risk of uremic syndrome in patients with severe renal failure. An elevated BUN can occur with high protein diets, upper GI bleeding, dehydration and/or volume depletion, or acute kidney injury. Usually, a low BUN does not have physiological consequences, but may be low in the presence of malnutrition or in individuals who have extensive liver damage.

Urine Protein (Proteinuria)

Healthy individuals normally excrete very small amounts of smaller molecular weight proteins in the urine (10–150 mg/day). In the presence of kidney damage, the glomerulus becomes more permeable to larger proteins, including albumin. Increased albumin excretion (albuminuria) can occur in the setting of diabetic nephropathy, glomerular disease, and hypertension. Table 26.3 describes the relationship of albuminuria and proteinuria to severity of renal dysfunction.

Other Urinalyses

Urinalysis to look for blood cells, cellular casts, specific gravity, and sodium may provide additional information as to presence of kidney injury. Table 26.4 summarizes the various urinalysis tests relevant to assessing kidney function.

Hematuria refers to the presence of red blood cells (RBC) in the urine. A few RBC may normally be seen in the urine under microscopy. Red blood cells of >3/high-powered field may be indicative of intrinsic renal failure, infection, or a consequence of damage from renal stones or trauma. Women of childbearing age may have some residual RBCs in the urine as a result of menstruation.

Pyuria refers to the presence of white blood cells (WBC) or leukocytes in the urine. Again, it may be normal to find a

Table 26.4 Urinalysis tests

Parameter	Reference range/normal findings	Comments
Red blood cells	1–3/high-powered field (HPF)	Persistent hematuria may be seen in glomerulonephritis, infection, and renal stones
White blood cells	0–2/HPF	May be present in conditions such as interstitial nephritis
Specific gravity	1.016–1.022	Correlates with kidneys' concentrating ability and will be increased or decreased depending on various physiological or disease processes
Urine sodium	Variable	Often used to assess volume status and acute kidney injury
%FE _{Na}	Variable	Used in assessing acute kidney injury

%FE_{Na} the fractional excretion of sodium

few white blood cells under microscopy. Large amounts of WBCs may indicate the presence of inflammation or infection in the urinary tract. They may also be seen in interstitial nephritis.

Casts are cylindrical masses of glycoproteins formed in the renal tubules. There are different types of casts depending on the types of cells they form around. For example, red cell casts form around RBCs, white cell casts around WBCs, or epithelial cell casts. There are also clear or hyaline casts, which are usually not an indication of a specific disease process. The presence of WBC casts typically indicates infection or inflammation, and RBC casts and epithelial cell casts usually are representative of significant damage to the kidney.

Specific gravity is an indication of the kidneys' ability to concentrate urine. Specific gravity (SG) is the ratio of the weight of a given fluid to the weight of an equal volume of distilled water. Patients with normal kidney function and normal fluid intake have an SG between approximately 1.016 and 1.022, but can dilute urine to approximately 1.001 and concentrate urine to approximately 1.035. A specific gravity of 1.010 indicates that the urinary osmolality is the same as the plasma (isosthenuric). Individuals who are volume-depleted, such as in prerenal failure, would present with a concentrated urine (SG ≥1.022), whereas patients with intrinsic renal failure would have kidneys that would be unable to dilute or concentrate urine, so SG would remain around 1.010.

Urine sodium concentration and %FE_{Na} are helpful indicators of a patient's volume status and can vary depending on clinical status and presence of kidney damage or other disease processes. The %FE_{Na} is the fraction of sodium that is filtered by the glomerulus and eventually excreted in the urine and may be useful in distinguishing types of acute kidney injury.

Table 26.3 Albuminuria/proteinuria and severity of renal impairment [9]

Measure	Normal-mild	Moderate	Severe
Albumin excretion rate (mg/24 h)	<30	30–300	>300
Protein excretion rate (mg/24 h)	<150	150–500	>500
Albumin/creatinine ratio (mg/mmol)	<3	3–30	>30
Protein/creatinine ratio (mg/mmol)	<15	15–50	>50
Urine dipstick	Negative to trace	Trace to +	+ or greater

Types of Kidney Dysfunction

Acute Kidney Injury

Acute kidney injury is defined by an abrupt decrease in kidney function that includes, but is not limited to acute renal failure. A patient is experiencing AKI when any of the following criteria are met [10]:

- Increase in serum creatinine by ≥ 26.5 $\mu\text{mol/L}$ within 48 h or.
- Increase in serum creatinine to ≥ 1.5 times baseline, which is known or presumed to have occurred within the prior 7 days or.
- Urine output < 0.5 ml/kg/h for 6 h.

The staging of AKI can be assessed based on serum creatinine and urine output criteria as outlined in Table 26.5 [10].

Patient Assessment

It is important to be able to identify patients who may be at risk of developing AKI. These include patients receiving potentially nephrotoxic drugs, the elderly, or those who may be experiencing or at risk of the following conditions:

Volume depletion (severe burns, blood loss, dehydration)	Renal artery stenosis
Cirrhosis with ascites	Renal artery or vein thrombosis
Heart failure	Glomerulonephritis
Cardiomyopathy	Benign prostatic hypertrophy
Hypotension/shock	Nephrolithiasis
	Chronic kidney disease

Patients at risk of AKI should have a baseline serum creatinine measured when their kidney function is stable, to assess for changes over time to determine if there is an acute deterioration in renal function. When no baseline value is

available, often repeated serum creatinine measurements can often be used to assess the rate of deterioration, or urine output can be measured. This information, along with the clinical history of the patient, can be used to determine if the individual meets the criteria for AKI.

Symptoms

Signs and symptoms of acute kidney injury may be nonspecific but can include changes in urinary habits, sudden weight gain, or flank pain. Other signs may include edema, orthostatic hypotension, colored or foamy urine, or hypertension.

Classification of AKI

AKI can be divided into three categories based on their different etiology, pathophysiology, and management: prerenal, intrarenal, and postrenal [11].

- *Prerenal injury* is a functional response to renal hypoperfusion and is not associated with a structural injury to the kidney itself.
- *Postrenal* AKI results from obstruction of the urinary collecting system. Obstruction may occur at the level of the bladder, urethra, ureters, or renal pelvis. This must occur in both kidneys in order to cause AKI.
- *Intrarenal (intrinsic)* AKI occurs due to an injury to a structural component of the kidney. There are four main types of intrarenal AKI based on the location of the injury: tubules, interstitium, glomerulus, and vascular capillaries.

Causes of each type of AKI are outlined in Table 26.6 [11]. Various laboratory parameters can be used to help confirm the type of AKI the patient is experiencing in conjunc-

Table 26.5 Staging of acute kidney injury (AKI) [10]

Stage	Serum creatinine	Urine output
1	1.5–1.9 times baseline or ≥ 26.5 $\mu\text{mol/L}$ increase	< 0.5 ml/kg/h for 6–12 h
2	2.0–2.9 times baseline	< 0.5 ml/kg/h for ≥ 12 h
3	3.0 times baseline or Increase in serum creatinine to ≥ 353.6 $\mu\text{mol/L}$ or Initiation of renal replacement therapy or, in patients < 18 years, decrease in eGFR to < 35 ml/min/1.73 m ²	< 0.3 ml/kg/h for ≥ 24 h or Anuria for ≥ 12 h

Table 26.6 Causes of acute kidney injury

Prerenal	Intrarenal	Postrenal
Hypovolemia	Ischemic causes	Nephrolithiasis
Hemorrhage	Hypotension	Blood clots
Cutaneous losses (burns, sweat)	Hypovolemic shock	Surgical injury
Gastrointestinal losses (diarrhea, vomiting)	Sepsis	Malignancy
Renal losses (diuresis)	Systemic lupus erythematosus	Neurogenic bladder
Decreased effective blood volume	Hemolytic-uremic syndrome	Prostate cancer
Heart failure	Renal artery thromboembolism	Benign prostatic hypertrophy
Cirrhosis	Multiple myeloma	Urethral strictures
Nephrotic syndrome	Ethylene glycol ingestion	Drug-induced
Intrarenal vasoconstriction	Drug-induced	
Hepatorenal syndrome		
Drug-induced		

Table 26.7 Differentiating types of AKI based on laboratory findings

Laboratory tests	Acute kidney injury type		
	Prerenal	Intrarenal	Postrenal
Urine sediment	Normal	Cast, cellular debris	Cellular debris
Specific gravity	↑	↓ or ↔	↔
Urinary RBC	None	2–4+	Variable
Urinary WBC	None	2–4+	1+
Urine sodium (mmol/L)	<20	>40	>40
FE _{Na} (%)	<1	>2	Variable
BUN	↑↑	↑	↑
Serum creatinine	↑↑	↑↑	↑↑

%FE_{Na} the fractional excretion of sodium, BUN blood urea nitrogen, RBC red blood cells, WBC white blood cells

tion with additional clinical information. The relevant laboratory parameters are summarized in Table 26.7.

Ongoing Assessment and Monitoring

Treatment goals of AKI include minimizing the degree of insult to the kidney, reducing extrarenal complications, and expediting the patients' recovery of renal function. General treatment includes hydration, discontinuing any nephrotoxic agents, and treatment of underlying conditions contributing to the injury. Diuretics, in particular loop diuretics like furosemide, are used if there is volume overload. Renal replacement therapy (dialysis) is used when correction of electrolyte imbalances is required, particularly with increased potassium, fluid overload, and when removal of uremic or other toxins is necessary.

Individuals should have daily or more frequent monitoring of their serum creatinine and urinary output to help assess recovery of renal function. Additional monitoring should include electrolytes, BUN, fluid status, and blood pressure. If kidney function recovers, patients should receive regular monitoring of their kidney function, as they are at an increased risk of developing chronic kidney disease throughout their lifetime.

Chronic Kidney Disease

Chronic kidney disease (CKD) is a progressive loss of kidney function over a period of months or years. More specifically, it can be defined as either kidney damage or a glomerular filtration rate (GFR) of <60 ml/min/1.73 m² for ≥3 months [9]. Classification of CKD is provided in Table 26.8.

Patient Assessment

Screening for CKD should be targeted for individuals at increased risk of developing CKD, including the following [12]:

Table 26.8 Classification of chronic kidney disease [9]

Stage	Description	GFR(ml/min/1.73 m ²)
1	Kidney damage with normal or ↑ GFR	≥90
2	Kidney damage with mild ↓ GFR	60–89
3	Moderate ↓ GFR	30–59
4	Severe ↓ GFR	15–29
5	Kidney failure/end-stage renal disease	<15 or dialysis

GFR glomerular filtration rate

- Hypertension.
- Diabetes mellitus.
- Family history of stage 5 CKD.
- Hereditary kidney disease.
- Previous history of acute kidney injury.
- Vascular disease.
- Multisystem disease with potential kidney involvement (e.g., systemic lupus erythematosus).
- Advanced age.

Individuals should also be assessed for current or prior use of potentially nephrotoxic drugs, which may cause AKI or contribute to progression of CKD (Table 26.9).

Screening for CKD should include a serum creatinine (and associated eGFR), a random urine albumin or protein/creatinine ratio, and urinalysis. An eGFR of <60 ml/min/1.73 m² for ≥3 months is diagnostic for CKD. A new finding of a reduced eGFR should prompt repeat testing to exclude causes of acute deterioration of eGFR, such as acute kidney injury.

Symptoms

At the early stages of CKD, patients are largely asymptomatic. In stages 3–5 CKD, patients may begin to experience general symptoms related to fatigue, edema, and decreased urine output. Cardiovascular symptoms may arise related to hypertension, heart failure, pericarditis, and atherosclerosis. With the accumulation of uremic toxins, individuals may experience nausea and vomiting, anorexia, bleeding, and pruritus. Neuromuscular symptoms include restless leg syndrome, muscle cramps, impaired cognition, and peripheral neuropathy. Anemia may also be present at the later stages of CKD due to the kidney's inability to produce erythropoietin. An imbalance of calcium and phosphate homeostasis and vitamin D metabolism may contribute to signs and symptoms related to renal bone disease, such as bone pain and increased risk of fractures.

Ongoing Assessment and Monitoring of Patients with CKD

Individuals with CKD are usually initiated on therapy to slow the progression of renal impairment with angiotensin

Table 26.9 Nephrotoxic drugs

Angiotensin receptor blockers	Antivirals (acyclovir, indinavir, tenofovir)
Angiotensin converting enzyme inhibitors	Radiographic contrast dyes
Antineoplastics (mitomycin, cisplatin, methotrexate)	Diuretics
Antimicrobials (aminoglycosides, amphotericin B, vancomycin, cephalosporins, penicillins, sulfonamides)	Lithium
	Nonsteroidal anti-inflammatory drugs (NSAIDs)

converting enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARBs). Within 2 weeks of initiation or dosage increases, patients should have their potassium and eGFR checked. ACEi and ARBs can cause a reversible reduction in eGFR when beginning therapy. General recommendations for assessment and management include [12]:

- Start at the lowest dose of ACEi or ARB and titrate up slowly to the maximum tolerated dose.
- Stop ACEi or ARB if the reduction in eGFR exceeds 25% from baseline.
- Recheck eGFR in 2–3 weeks if reduction in eGFR is 5–25% from baseline.
- Increases in serum potassium can be expected of up to 0.5 mmol/L.

Certain medications that are renally excreted or may potentially be nephrotoxic should be temporarily held in the setting of acute illnesses where patients are unable to maintain adequate fluid intake until they have recovered. In this clinical setting, continuing these medications may increase the risk of the patient experiencing acute kidney injury or adverse effects. The medications include sulfonylureas, ACEi/ARB, diuretics, metformin, and NSAIDs [12].

Individuals with CKD are also at increased risk for cardiac disease and should be assessed to ensure they are receiving appropriate therapy to reduce their overall cardiovascular risk. This includes assessing patients for initiation of statin and antiplatelet therapy. Statin therapy is generally recommended for all patients with CKD and diabetes unless there are contraindications [13]. In patients without diabetes, patients aged >50 years should receive a statin, and those <50 years should receive statin therapy if they have known cardiovascular disease or prior ischemic stroke, or if their estimated 10-year incidence of cardiovascular disease is >10% [14]. Low-dose aspirin therapy can be used for secondary prevention in patients with established cardiovascular disease where there are no contraindications [15].

Finally, all patients with CKD should have regular assessments of their drug therapy to determine if dosage adjustments are required. This should occur at regular intervals

whenever the patient has a new eGFR calculated or a new medication is initiated, including nonprescription therapies. A general approach to adjusting drug dosages is outlined as follows [16]:

- Obtain patient history, including relevant demographic and clinical history.
- Calculate creatinine clearance using an appropriate weight-adjusted or nonweight-based formula, compared to eGFR from lab results.
- Review current medications and determine which drugs may require dosage adjustment.
- Consult one or more drug-dosing references to determine an appropriate dosage.
- Monitor for response to the drug as well as for adverse effects.
- Revise regimen if required based on response and clinical status.

Clinical Pearls

- Pharmacists have an important role in the assessment and monitoring of both acute kidney injury and chronic kidney disease.
- A systematic assessment process includes evaluation of relevant markers of kidney function in conjunction with the patients' clinical history and presentation.
- Targeted screening and assessment of kidney function should occur in those patients at highest risk, including those with diabetes, hypertension, vascular disease, the elderly, and those receiving renally eliminated or potentially nephrotoxic drugs.

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Cecilia Lau

Chapter Objectives

1. Describe the core elements of approach to infectious disease assessment.
2. Describe the use of empiric, definitive, and prophylactic antimicrobial therapies.
3. Describe the use of an antibiogram.
4. Describe the approach to interpretation of culture results.

Background

Infectious diseases encompass illnesses associated with infection from pathogenic bacteria, fungi, protozoa and viruses. The epidemiology and incidence of specific diseases vary depending on extrinsic and intrinsic factors to the host. History and physical exam for signs and symptoms will often suggest a probable diagnosis. In selected patients, this diagnosis can be confirmed by obtaining specimens for culture, serology, polymerase chain reaction identification, microscopy, and pathology when indicated. Imaging and procedures may further assist with the diagnosis. Microbiological surveillance of common isolates in specific illnesses has allowed clinicians to identify the most common organisms associated with specific infections. This has also allowed accumulation of susceptibility patterns, presented in antibiograms. Additional consideration must also be given regarding host exposures to likely pathogens in the age of global travel when antimicrobial resistance patterns can differ. Globally, infectious syndromes may have similar pathogenesis, diagnostic workup and common pathogens, allowing empiric treatment. However, the prescribed antimicrobials should be tailored to the identified pathogen, its susceptibility,

pharmacokinetic/pharmacodynamics parameters, availability of an antimicrobial, and patient tolerability. Judicious prescribing of antimicrobials at every opportunity is required to decrease unnecessary use of broad-spectrum antimicrobials, which is associated with increased cost, resistance, and exposure of patient to adverse effects. The ultimate goal is to prolong the longevity of all antimicrobials in the era of increased global antimicrobial resistance.

In clinical practice, pharmacists are tasked to maintain competency in the management of different infectious syndromes, acquire knowledge on novel antimicrobials, and minimize the impact of inappropriate use of antimicrobials while ensuring the resolution of an infection. Pharmacist intervention encourages adherence to guideline-directed care in infectious disease, with specific tailoring to ensure antimicrobial indication, dosing regimen, and choice based on site of infection and duration of treatment are appropriate. As medication stewards, the role of the pharmacists has been specifically highlighted as champions for implementation of Antimicrobial Stewardship Program (ASP). Pharmacists working in all practice settings have an important role in ASP as they possess the unique skills and expertise to provide tailored advice based on disease and patient factors.

Approach to Infections with "I.N.F.E.C.T.I.O.N.S."

The collaborative approach to infections can be divided into core elements. Key pharmacists' roles for each of the elements or highest impact interventions are as listed in Table 27.1, which may vary depending on the practice setting.

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Table 27.1 Approach to infections

	Core element	Pharmacist role
I	Infection and/or indication for each antimicrobial prescribed must exist	Recognize if an infection or indications for each antimicrobial exist Intervene when an antimicrobial is not indicated Intervene when there is therapeutic duplication Intervene when additional antimicrobial is required
N	Not working alone in making therapeutic decisions	Collaboration with other healthcare team members Pharmacists are the medication experts and are frequently consulted to ensure optimal dosing regimen (drug, dose, route, duration) is prescribed for the indication and the sites involved
F	Foci of infection should be assessed for source control and impact on choice and duration of therapy	Anticipate impact of presence or absence of source control on duration of therapy
E	Empiric therapy should target common pathogenic organisms with consideration of additional risk factors	Intervene when empiric therapy is inappropriate for suspected infection
C	Culture to be done in clinically relevant specimens in selected patients	Interpret preliminary and final culture and sensitivity results and recommend modifications to empiric or definitive therapy
T	Timing of first antimicrobial exposure has implications	Intervene when delay in prescribing is appropriate. Intervene when therapy should not be delayed
I	Inquire into allergies, organ functions, genetic/metabolic abnormalities	Determine the appropriate drug choice and dosing regimen tailored to patient
O	Ongoing assessment and monitoring required	Determine if dose and interval of empiric therapy are appropriate based on indication and organ function
N	Narrowest spectrum antimicrobial should be used whenever possible and applicable	Determine if streamlining to narrowest spectrum antimicrobial is appropriate. Intervene as necessary
S	Seamless care	Pharmacists play an important role in counselling on the medications prescribed and to assess efficacy, tolerability, and ensure compliance pharmacists can maximize the future care of these patients by clear documentation of a patient's intolerance to an antimicrobial

Infection and/or Indication for Each Antimicrobial Prescribed Must Exist

Humans live in harmony with a plethora of microorganisms, some of which possess the capacity to cause disease. Microbial flora may be classified as residential flora, which

is found consistently at a site and promptly re-establishes its presence if disturbed or transient flora, which may colonize the host for a specific time period but does not permanently establish itself. Furthermore, microorganisms colonize different parts of the body based on tissue tropism as well as the presence of an ideal habitat. Thus, the normal organisms that live on the skin are appreciably different than what lives in different parts of our gastrointestinal tract based on factors such as oxygen and natural host defence with gastric acid. When host defenses are breached, some organisms that normally live in symbiosis with the host may cause disease. Acquisition of a pathogenic organism is also possible from exposure to communicable pathogens under the right circumstances, e.g., viruses responsible for Ebola, influenza. Some infections will also require additional vector for transmission to occur, such as the deposit of *Plasmodium* sporozoites into an individual while an infected *Anopheles* mosquito is taking a blood meal, transmitting malaria, which is endemic in certain parts of the world, limited by the natural habitat of the mosquito. Thus, causative pathogens of infectious diseases may be intrinsic to the host or extrinsically acquired.

Signs and Symptoms Suggestive of Infections

With the milieu of organisms living within us as well as continuous exposure to the outside world, familiarity with common signs and symptoms associated with infection will alert clinicians to the possibility of infection. General warning signs can include fever and chills, hypothermia, tachycardia, tachypnea, nausea, vomiting, diarrhea, decreased appetite, decreased urine output, or abnormal mentation. Some infections may have pathognomonic findings, but in general the differential diagnosis can be wide and varied when most patients present with such nonspecific signs and symptoms. Different infections can have different signs and symptoms as they originate from different parts of the body, and also largely reflect an individual's response. Specific localizing signs may be present on exam, such as swelling or erythema surrounding an infected joint or costovertebral pain in pyelonephritis. An examination of bloodwork, when available, may reveal abnormalities in white blood cell counts or a rise in inflammatory markers. Yet, none of these signs and symptoms is specific to infections only, as many noninfectious causes exist for the same presentation. To illustrate, a patient with severe medication allergy may present with generalized rash, high fever, leukocytosis driven by eosinophilia, and high inflammatory markers and does not signify that an infection exists; this is the body's adverse reaction to a medication. In addition, signs and symptoms are particularly difficult to interpret and may be absent at the extremes of the age spectrum or in immunocompromised patients.

Antimicrobial Indication Broadly Defined

Regardless whether an infection is present or not, there should be a clear indication for each antimicrobial prescribed as outlined in Fig. 27.1. Antimicrobials are utilized to prevent or treat infections. Thus, patients prescribed antimicrobials may or may not have an infection. Prophylaxis is used to prevent infections when patient, procedural or situational specific factors increase the risk for certain infections. These patients do not have an existing infection but are considered at risk. Treatment is categorized into empiric and definitive therapy. These patients have a suspected or documented infection and require effective therapy.

Primary prophylactic therapy is prescribed to prevent a first infection, e.g., Surgical prophylaxis, posttransplant opportunistic infection, and AIDS patients with low CD4 counts. The choice and duration of the required preventive therapy vary depending on each situation, duration of perceived risk and pathogens targeted. Surgical prophylaxis is of finite duration, usually ≤ 24 h postoperatively, whereas some posttransplant prophylaxis is continued lifelong. *Secondary prophylaxis* is prescribed to prevent the recurrence of a previously documented infection after a treatment course has been completed. Evidence-based guidelines exist for many of the afore-mentioned at risk patient populations and should be adhered to.

Empiric therapy is defined as the use of antimicrobials to provide coverage of the most likely pathogens in specific infections or based on likelihood of effectiveness when a pathogen is identified, but the result of the isolate's susceptibility is unknown at the time of prescribing. Although presumptions can be made based on known spectrum of activity of the different classes of antibiotics, other factors such as local resistance pattern will often define which antimicrobials are the preferred choices within a geographical location. Even within an institute, the susceptibility patterns can vary greatly among different wards and change over time based on patterns of antimicrobial usage.

Definitive therapy, also known as targeted therapy, is based on susceptibility data. The antimicrobial prescribed may be narrower or broader in its spectrum of activity than the empiric therapy. Whenever possible, the narrowest agents are preferred to minimize overall antibiotic exposure, which can have downstream effect on resistance profiles. When narrower spectrum antimicrobial is chosen, the perceived benefits of streamlining include improvement in resistance profile, decreased adverse events, decreased length of stay in hospital, cost savings to patient, and healthcare [1]. Hesitancy to modify a chosen regimen includes complex susceptibility pattern in a polymicrobial infection, inability to discern contamination, or colonization from pathogenic existence of the

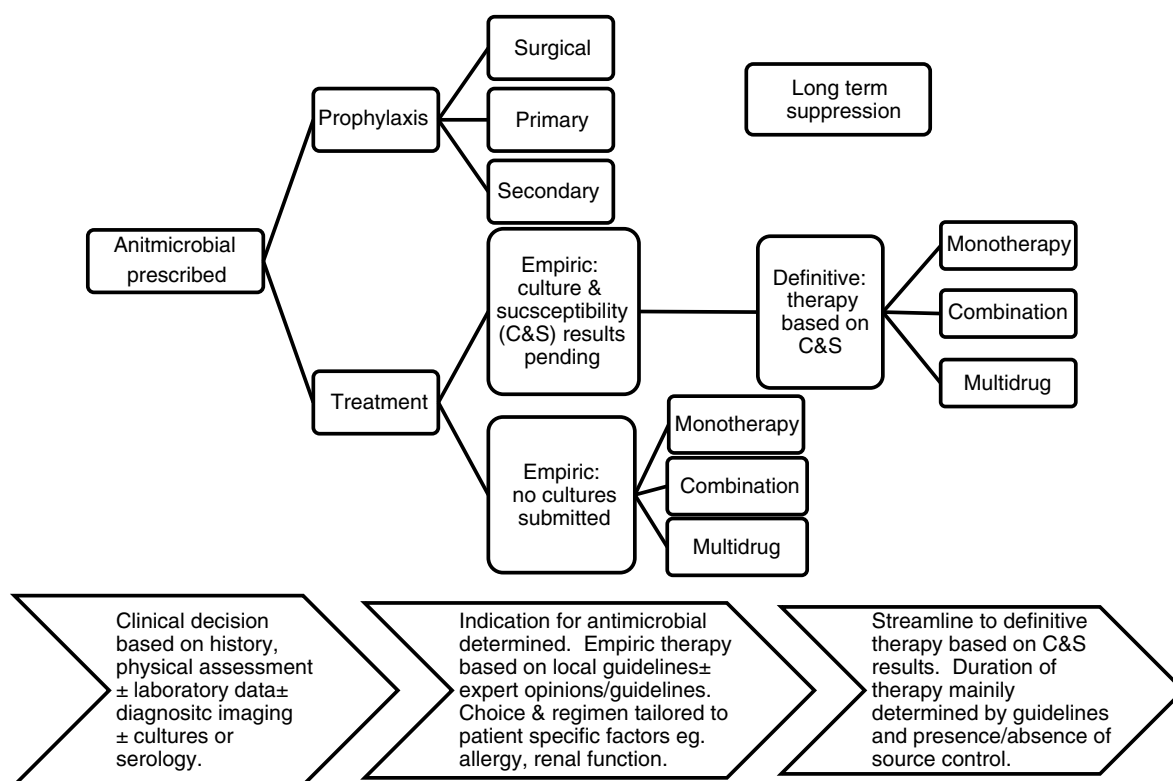


Fig. 27.1 Indications for the prescribed antimicrobial therapy. C&S, culture, and sensitivities

identified organisms, concerns regarding the reliability of antimicrobial influenced culture results, and the perception that the patient is improving with empiric, often broader spectrum, antimicrobial therapy.

Lifelong suppression therapy is an ill-defined hybrid of secondary prophylaxis and definitive therapy. Although an appropriate treatment course of an infection has been completed and no evidence-based guideline exists to support the continuation of treatment, some infections may be deemed incurable, and the attending specialists may choose to prescribe prolonged suppression therapy. Paucity of data makes determining the right dosing regimen difficult, where one prescriber may choose a lower dose while another choose a higher dose, with both based on experience. In one study on prolonged antibiotic use defined as 1 year and longer, 43/202 (21%) were prescribed long-term suppressive antimicrobial therapy for infected prosthetic devices (valves, cardiac devices, arthroplasty, spinal fixation), chronic osteomyelitis and vascular graft infections; 34/43 (79%) were prescribed by infectious disease specialists [2]. This highlights the importance of transparency in prescribing, especially regarding specific unique decisions made after careful deliberations between the patient and the prescriber.

Not Working Alone in Making Therapeutic Decisions

In patient care, clinical assessment, diagnostic investigations, and therapeutic decisions often occur as a continuous and overlapping process that involves many team members. Signs and symptoms elucidated by a clinician's history and physical exam will often identify a wide range of differential diagnosis, which may include infections. Clinicians should be cognizant of specific host-related factors that influence the likelihood of contracting an infection, such as a patient's overall immune status from acquired or congenital disease or iatrogenic immunodeficiency from splenectomy or immunosuppressive therapy. In susceptible individuals, exposure to animals (e.g., playful bites from a cat, affectionate licking of open wounds by a dog) can cause diseases ranging from cellulitis, bacteremia to osteomyelitis. Sick contacts are also important to note as general knowledge of transmission, incubation and timeline to disease manifestations helps narrow down potential infectious etiology. For the same reason, recent or remote travel history for work, leisure or missionary trips may reveal a risk of acquiring endemic infections specific to the geographic location. Altogether, a collaborative patient intake assessment can offer insight to foci of an

infection, probable causative pathogens and evaluation of actual or potential complications from the infection.

Currently, pharmacist training does not include in-depth history taking, physical exam, interpretation of many diagnostic investigations, all of which are crucial to the management of various infectious diseases. Interdisciplinary collaboration is encouraged whenever possible, as individual team members contribute their expertise to improve the quality of patient care. As a team, every patient should be assessed for severity of illness, potential source of infection, need for diagnostic investigations, while contemplating the choice of empiric antimicrobial therapy and the necessity of other pharmacologic and non-pharmacologic interventions. Common indications for additional tests ordered include:

1. Diagnosing or supporting the diagnosis.
2. Identifying the foci of an infection and assessing for possible complications.
3. Deciding if further culture for isolation of culpable organism(s) is possible or required.
4. Determining if an infection is amenable to source control.

Thus, patient care process in infectious diseases may often involve team members due to their specialist knowledge pertaining to their area.

Pharmacists are, however, the medication experts and are frequently consulted to ensure optimal dosing regimen (drug, dose, route, duration) is prescribed for the indication and the sites involved. The choice of the antimicrobial may be impacted by the distribution characteristics of an agent. For example, in meningitis, the chosen antimicrobials must have adequate blood-brain barrier penetration and also ideally bactericidal as central nervous system lacks many of the natural host defense mechanisms. In addition, dosing can be complex as individual antimicrobials may be impacted by organ function relevant to its clearance. Continuous assessment is required to ensure appropriate adjustments are made based on worsening or normalization of organ functions. Finally, antimicrobials can certainly have drug-drug or drug-disease interactions or contraindications, such as the substantial decrease of valproate acid levels in patients treated with carbapenems or exacerbation of myasthenia gravis from aminoglycoside use. For these reasons, antimicrobial therapy guidance from pharmacists plays an important role. In turn, pharmacists may rely on team members (most responsible physician, interventional radiologist, general surgeon, nurse clinician, and physician assistant) to confirm that improvement has been documented.

Foci of Infection Should Be Assessed for Source Control, Impact on Choice, and Duration of Therapy

Source Control

Strategies to remove the foci of infection are known as source control. It may involve minimally invasive procedures or invasive operations for the purpose of drainage, debridement, or resection. The decision of whether source control is needed or should be attempted depends on many factors and often requires assessments from various specialists. It may also depend on the severity of illness. For example, a patient with an acute abdomen in septic shock found to have a perforated viscus on imaging will benefit from immediate life-saving surgical intervention involving washout of fecal peritonitis, control of perforation with immediate or delayed of re-establishment of gut continuity. On the other hand, a clinically stable patient found to have contained ruptured appendix may be conservatively treated for days with antimicrobials while awaiting surgery. For catheter-associated bloodstream infections (CABSI), the removal of the infected line and insertion into a different site when the continued need for intravenous access exists, increases the success of medical therapy and significantly decreases the chance of reoccurrence, especially for the virulent and biofilm-loving pathogens such as *Staphylococcus aureus*. In contrast, when some less virulent coagulase-negative *Staphylococcus* species are isolated in CABSI, a reasonable approach is to retain the line and treat with antimicrobials in patients with difficult vascular access. Furthermore, situations may exist where the risks of a procedure outweigh the benefits of source control, e.g., multiple redo open heart valve replacements subjecting patient to surgery-related risks of complications, including renal failure, ischemic bowel, sternal wound infections, and heart block.

Impact on Choice of Therapy

For an antimicrobial to achieve a therapeutic effect, an adequate amount of the drug must be delivered to the site of infection. The ability to achieve or detect local concentrations at the site of infection is desirable, but its consequences are not certain. Generally, the local concentration of the antimicrobial should be at least equivalent to the MIC of the targeted organism, although it has been suggested that higher multiples may be more efficacious. There is evidence to suggest that even at subinhibitory concentrations, the antimicrobial may still exert positive influences on the host defenses against infections through altering bacterial morphology, adherence properties, enhanced phagocytosis and promotes intracellular killing of bacteria by leukocytes [3]. Clinical cure may still be possible with lower than desired concentration at the site of infection, but the approach to patient care should target utilizing therapy with the highest probability of optimal outcome.

Antimicrobial choice and dosing should be appropriate for the targeted site of infection. In addition to being proven effective for a given pathogen in an in vitro setting, antimicrobial prescribing must consider each agent's ability to reach the site of infection, dosing modification required to achieve effective concentration, and, if all else is equal, final selection may be based on the perceived benefit of the pharmacodynamics parameters associated with the antimicrobial. Experience and bodies of literature continue to accumulate regarding known difficult to penetrate sites of infection, such as in central nervous system, ophthalmic, bone, and prostate infections, which are reflected in many point-of-care references. Some examples where the site of infection is important to the choice and dose of antimicrobial prescribed include daptomycin use in lung infections, tigecycline use in bacteremias, and the choice of antimicrobials based on the foci of post-neurosurgery infections. Daptomycin, although a reasonable alternative for susceptible gram-positive infections, cannot be recommended in pneumonia as the drug is inhibited by the pulmonary surfactant, a primary component of the epithelial lung fluid lining the lung surfaces. Proposed mechanism of inhibition includes insertion of daptomycin into the lipid aggregates of the surfactant, thus sequestering and inhibiting the action of the antibiotic. Tigecycline has excellent distribution in the body, but its resultant low serum concentrations have been associated with breakthrough recurrence in the off-label treatment for bacteremia with susceptible pathogens. First-generation cephalosporins, e.g., cefazolin, are effective for most surgical site infections and is routinely recommended in guidelines. However, if the surgical infection involves a deep organ space where penetration of antimicrobials is known to be problematic, the choice changes to include coverage with antimicrobials that offers proven penetration to the site of infection. In post-neurosurgery with image-proven brain abscesses, for maximal penetration and potential pathogen coverage, antimicrobials are prescribed at highest dose appropriate to the patient's organ function may be meropenem 2 g IV every 8 h plus vancomycin IV dosed to target steady state trough of 15–20 mg/L. Conversely, if the post-neurosurgery surgical infection is limited to a superficial wound infection, cefazolin or other alternatives, including oral agents may be appropriate.

Impact on Duration of Therapy

Pharmacists play an instrumental role in assessment of many aspects of prescribing, including appropriate duration of therapy. Table 27.2 provides examples of duration of therapy recommended for common infectious syndromes, where anatomical sites, source control and other factors may impact the duration of therapy. In addition to source control, duration of therapy may be influenced by the site of infection, immune status of the patient, and the causative pathogen. Further tailoring is recommended based on patient response

Table 27.2 Examples of duration of antimicrobial therapy impacted by foci of infection, source control, and other factors

Site	Infections	Duration of therapy	Impact of foci identification and source control
Bacteremia	Foci removed	7 days vs 10–14 days	From clearance of bacteremia or removal of foci
Bone	Osteomyelitis	Acute → 42 days Chronic → treat till ESR normalizes, may be ≥90 days	Dependent on the causative pathogens and acuity of the infection. If source control involves amputation distal to the site of infection and no other foci of infection is identified, antimicrobial may be discontinued
Heart	Endocarditis— Native valve	<i>Staphylococcus aureus</i> Mitral/aortic valve, uncomplicated → 28–42 days Tricuspid valve, uncomplicated → 14 days <i>Streptococcus viridans</i> → 14–28 <i>Enterococcus</i> species → 28–42	Dependent on the causative pathogens as well as valve(s) involved. due to anatomy of the valve and small area for debridement, and lack of data, short course postoperative antibiotic cannot be suggested. A full course of treatment based on causative pathogen is required. in general, if the removed valve sent for culture demonstrates no growth, treatment may start from documented clearance of bacteremia, whichever is sooner
Joint	Septic arthritis	14–28 days	Arthroscopic or open debridement of joints is desirable to minimize damage to joint from cytokine-induced inflammatory reaction

ESR erythrocyte sedimentation rate

to therapy in specific infections. Pharmacists are unlikely to participate in source control, but our assessment of a patient should include inquiry into the feasibility of source control to anticipate its impact on antimicrobial treatment.

One must, however, be mindful that not all infections require source control, such as for uncomplicated urinary tract infections, pneumonia, or meningitis. The same indications with complicated circumstances may necessitate the removal of a long-standing indwelling Foley catheter, debridement of an empyema or replacement of a ventriculo-peritoneal shunt. In selected infections, if source control is incomplete or cannot be performed, a longer-than-usual duration may or may not be required. If the nidus of an infection has been removed and there is no concern for residual infection, antimicrobial therapy may be appropriately discontinued despite indication-specific, guideline-recommended duration of treatment. Suppose a patient diagnosed with image-proven left foot osteomyelitis (OM), which evolved from chronic bilateral diabetic foot ulcer, undergoes a left below-knee amputation, therapy for the left foot OM should be shortened from ≥6 weeks to ≤48 h post-operatively to decrease stump infection rate. However, the right foot remains, and when assessed clinically, requires appropriate wound care in addition to a short course of antimicrobial therapy. If this same patient presented with bacteremia, but workup is negative for disseminated or metastatic disease, the most appropriate therapy would be an antimicrobial that will effectively treat both the bacteremia and the diabetic foot ulcer, with duration determined by whichever indication requires the longer course.

Empiric Therapy Should Target Common Pathogenic Organisms with Consideration of Additional Risk Factors

Some common medically important bacteria are presented in Fig. 27.2 based on Gram stain cell wall morphology. Preliminary growth and Gram stain identification is labour-intensive. It requires interpretation and further biochemical tests to confirm the identification. Similarly, identification of viral, fungal, and protozoan pathogens must also follow specific protocols to identify the organism. Some organisms do not grow well in artificial media. Thus, other techniques such as nucleic acid testing, serology, and microscopy must be employed where applicable and available. Each organism is capable of causing a range of infections unique to it, which are often present as one manifestation, although concurrent manifestations in the same patient may be possible. Until more rapid and reliable testing is widely available, clinicians must continue to rely on existing knowledge on these pathogens and the specific infectious syndromes attributable to them.

Fortunately, widely available references exist to offer guidance to clinicians when initiating empiric therapy based on a suspected diagnosis. A list of references commonly used by practitioners involved in care of patients with infection is provided in Table 27.3. Adherence to local guidelines whenever possible can maximize the likelihood of prescribing the most appropriate therapy based on local susceptibility patterns of the suspected pathogens, availability of antimicrobials, and cost to patient and healthcare system.

Fig. 27.2 Common potentially medically important bacteria classified based on Gram stain morphology of the cell wall. *HECK-Yes comprised of clinically relevant bacteria, species specific, with moderate to high risk of inducible AmpC beta-lactamase expression (*Enterobacter cloacae*, *Citrobacter freundii*, *Klebsiella aerogenes*), commonly encountered and those with inducible chromosomal AmpC genes (*Hafnia alvei*, *Yersinia enterocolitica*) less commonly encountered

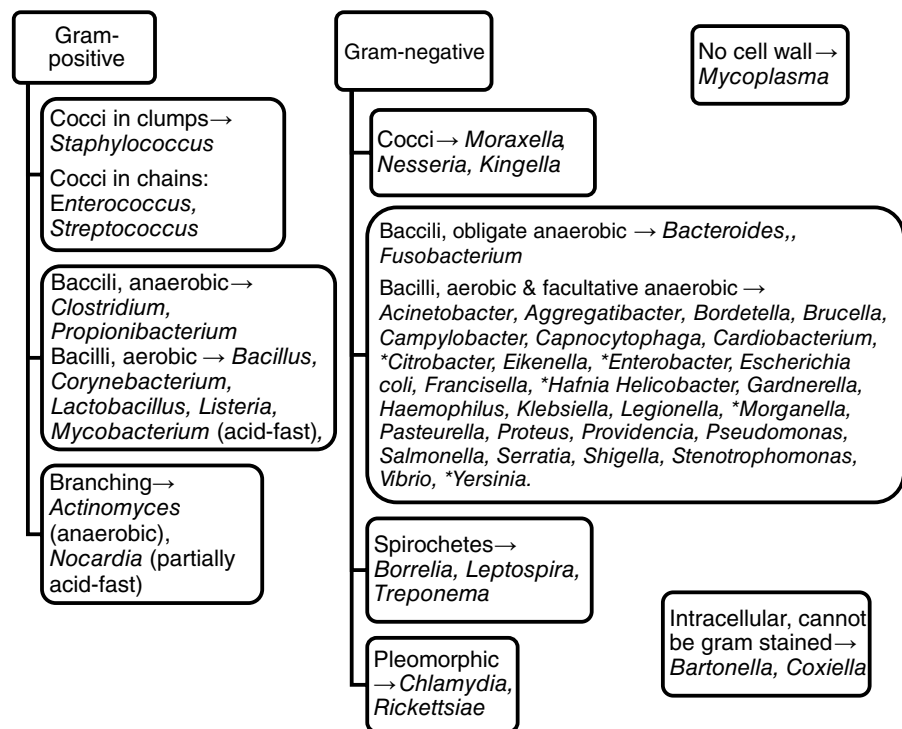


Table 27.3 Commonly used references in infectious disease patient care

Reference	Comments	Cost
Antimicrobial and treatment-focused bedside references		
Pros: Regularly updated, allows point-of-care access to succinct indications or organ-specific antimicrobial recommendations		
Cons: Requires navigation or search to obtain relevant information, must interpret with caution to applicability to local practice		
<i>Sanford guide collection</i>	Available as pocket books or individual APP purchase for electronic devices, relating to antimicrobial therapy, HIV/AIDS therapy, viral hepatitis therapy	\$
<i>John Hopkins guides</i>	Available as pocket books or in-APP purchases for electronic devices, relating to antibiotics, HIV	\$
<i>Bugs and drugs</i>	Available as individual APP purchase for electronic devices	\$
Infectious disease-focused websites with easy-to-interpret disease and treatment-related information		
Pros: Regularly updated; provides succinct summary on disease, pathogens, and treatment; information for the general public and health care professionals can be found		
Cons: Requires navigation or search to obtain relevant information; must interpret with caution to local practice		
<i>World Health Organization</i>	http://www.who.int/topics/infectious_diseases/en/	None
<i>European Centre for Disease Prevention and Control</i>	https://ecdc.europa.eu/en/home	None
<i>U.S. Department of Health & human services, Centers for Disease Control and Prevention</i>	https://www.cdc.gov/	None
<i>Infectious Diseases Society of America</i>	https://www.idsociety.org/Index.aspx	None
References for physician or specialist trainees practicing in infectious diseases		
Pros: Provides in-depth learning opportunities regarding specific pathogens and diseases		
Cons: New editions with cost implications, non-pharmacy practice targeted		
<i>Oxford handbook of infectious disease and microbiology</i>	Available as printed pocket book, eBook, or APP purchases for electronic devices	\$
<i>The AST handbook of transplant infections</i>	Available as printed book or eBook	\$
<i>Comprehensive review of infectious diseases</i>	Available as printed book	\$\$
<i>Mandell, Douglas, and Bennett's principles and practice of infectious diseases</i>	Available as 2-volume set printed book and eBook	
<i>Feigin and Cherry's textbook of Pediatric infectious diseases</i>	Available as 2-volume set printed book and eBook	

Where no reference exists, or may be incomplete, pharmacists can assist their team to make the best decisions by participating in evidence-based medicine review of available primary literature. Based on the suspected infectious diagnosis, patients should be prescribed an antimicrobial with dosing regimen tailored to host factors.

In an outpatient setting, clinicians may prescribe a course of empiric therapy based on symptom and physical assessment to treat a clinically well patient with no imaging, laboratory, serologic or microbiologic workup. There may be times when outpatient work-up should be considered. Suppose, an otherwise young healthy patient, with no recent travel or hospitalization, presents to her physician with dysuria, suprapubic pain, increased urinary frequency and urgency, but denies fever and flank pain. She may be diagnosed and prescribed treatment empirically for cystitis, for which *Escherichia coli* (*E. coli*) is known to be the causative pathogen in over 75% of cases. Common references recommend trimethoprim-sulfamethoxazole (TMP-SMX) or nitrofurantoin, but suggest local resistance patterns for *E. coli* should be confirmed to be <20% before prescribing TMP-SMX. After consulting a local antibiogram, which documents ~75% susceptibility to TMP-SMX, a prescription for nitrofurantoin 100 mg PO BID for 5 days is written. If this same patient is currently in the third trimester of a pregnancy, treatment should be culture-directed, and an end-of-treatment repeat culture should be done 1 week after the last dose of antibiotic. Some commonly used antimicrobials in the treatment of cystitis are not recommended due to risk to newborn, e.g., nitrofurantoin should be avoided due to risk of hemolytic anemia to the newborn.

Some patients with infections and significant comorbidities may require hospitalization for closer monitoring and management of exacerbations of chronic diseases. Hospitalization for a noninfectious cause should not result in an automatic escalation of antimicrobial therapy. These patients may still respond appropriately to the prescribed empiric course of antimicrobial based on an indication without further work-up. Unfortunately, it is possible to develop new infections while hospitalized and for these patients, the process of I.N.F.E.C.T.I.O.N.S. should be instituted for the new infection.

Culture Should Be Done in Clinically Relevant Specimens in Selected Patients

In some patients not improving on empiric therapy or if the patients present to the hospital with sepsis or septic shock, cultures may be required to ensure treatment is appropriate, as there are many noninfectious causes that may result in the same physiologic derangements. The criteria upon which sepsis and septic shock are diagnosed are listed in Table 27.4

Table 27.4 Sepsis and septic shock: the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3) [4]

Terminology	Sepsis-3 definition	Diagnostic criteria
Sepsis	Life-threatening organ dysfunction caused by dysregulated host response to infection	Infection <i>plus</i> measure of organ dysfunction identified with: 1. A raised sequential (sepsis-related) organ failure assessment (SOFA) score, with a change in score of ≥ 2 from baseline <i>or</i> : 2. Quick SOFA (qSOFA) score considered positive if patient has ≥ 2 of the following clinical criteria: Respiratory rate ≥ 22 bpm. Altered mentation, <i>or</i> . Systolic blood pressure of 100 mmHg or less.
Septic shock	Subset of sepsis with particularly profound circulatory, cellular, and metabolic abnormalities associated with substantially increased mortality	Vasopressor requirement to maintain a mean arterial pressure (MAP) of >65 mm hg <i>and</i> serum lactate level > 2 mmol/L in the absence of hypovolemia The presence of both criteria predicts hospital mortality in excess of 40%

[4]. Appropriate collection, processing, and interpretation of clinically relevant specimens allow confirmation of diagnosis and streamlining to definitive therapy. Such may be the case in an otherwise healthy individual who complains of several-day history of fever and chills, increased difficulty in breathing and fatigue that is on treatment with a respiratory fluoroquinolone, now presents with septic shock and specimens submitted for culture include blood, sputum, and urine. This patient has a radiographic evidence of pneumonia, while history and urinalysis are negative for signs and symptoms of infection of the urinary tract. Suppose, the urine grew yeast, blood grew *Streptococcus pneumoniae* and sputum grew *Streptococcus pneumoniae* and yeast. The yeast in urine is most likely insignificant, as it reflects a result of antibiotic pressure from the recent use of a fluoroquinolone. The yeast in sputum is considered normal flora in respiratory tract, but may be significant in immunocompromised patients, which this patient is not. Of the cultures submitted, the significant pathogen most compatible with the presenting symptoms and adjunctive test results is *Streptococcus pneumoniae*, which is known to be a respiratory pathogen.

In the absence of microbiologic sampling or where culture identifies presence of multiple organisms, a decision must be made based on the likelihood of an individual or group of organisms' pathogenicity pertaining to the specific infection as well as its local susceptibility pattern.

Timing of First Antimicrobial Exposure Has Implications

Mortality

Debates continue to exist as to the optimal time to first dose of antibiotics. It was previously demonstrated that administration of an antimicrobial effective for the isolated or suspected pathogens within the first hour of documented hypotension was associated with a survival rate of 79.9% [5]. Each hour of delay over the next 6 h was associated with an average decrease in survival of 7.6% [5]. Yet, as more studies are conducted, data from larger trials where antimicrobial is evaluated as part of sepsis bundles, sensitivity analysis does not currently support a specific association between mortality and time to first dose of antibiotics. Most frontline clinicians, however, would agree that empiric therapy for patients in septic shock should not be unnecessarily delayed, despite knowing that antimicrobial naïve patient will typically generate the highest yield of causative pathogen from appropriate specimens collected. Therefore, in hemodynamically stable patients, cultures should be collected prior to first dose of antimicrobial.

Interpretation of Culture Results

Cultures should be interpreted within the framework of recent antimicrobial use. When antimicrobial is given before a blood culture that subsequently showed no growth, clinicians must interpret the significance of the negative result. Empiric narrowing of antimicrobials can be performed with caution after careful consideration of the negative result, combined with an assessment of the most likely pathogen causing the presentation. The opposite is also true. A positive culture in a patient administered an antimicrobial usually effective against the organism may suggest difficult-to-eradicate pathogen due to its propensity to metastasize and seed in the body, causing many foci of infection that must be drained, debrided or resected. Alternately, it may suggest the need to optimize dosing regimens, reassess if the choice is appropriate for the site of infection or consider the possibility of resistance.

Inquire into Allergies, Organ Functions, and Genetic/Metabolic Abnormalities

Pharmacists are at a unique position to offer advice on antimicrobial therapy as the team member likely most knowledgeable in antimicrobials and their spectrum of activity. In practice, pharmacists are instrumental to ensure choice and dosing of medications are appropriate for patient-specific factors such as allergies and organ functions relevant to drug clearances.

Allergies and Intolerances

Some medical conditions, especially viral infections, can appear as exanthemata and in some cases, there's known association of high incidence of rash when an antibiotic (likely not indicated) is prescribed. For example, 70–100% of individuals with Epstein–Barr Virus mononucleosis-like syndrome treated with amoxicillin develop rash. For some patients, intolerances such as headache and gastrointestinal upset are reported as allergies. It is thus not surprising that up to 10% of the general population claims to have allergies to penicillin, with the actual frequency of anaphylactic type reactions to penicillin are closer to 0.01–0.05% [6]. Clinicians, including pharmacists, should elucidate the indication for the inciting antibiotic, timeline and evolution of any reaction, management for the reaction and if possible, a review of other antimicrobials given since. A patient with a true allergy to penicillin will be at risk of reaction to all penicillins. If a patient with reported penicillin allergy demonstrates tolerance to other penicillins, e.g., piperacillin-tazobactam, patient should be counselled and have allergy de-labeled.

Cross-Reactivity Among Beta-Lactam Antimicrobials

From several studies conducted in the 1970s, cross-reactivity between penicillins and cephalosporins has been reported to be 7.7–8.1% and thus majority of literature quotes a cross-reactivity of <10% between the groups, whereas reports of as high as 42% from early retrospective studies have been reported [7, 8]. More recent studies have demonstrated that the rate of cross-reactivity between the two groups is much lower than found in earlier studies (<1%) and appears to be driven by side-chain similarities between specific agents. Regardless of whether you can complete a detailed allergy assessment at the time of prescribing, it is important for pharmacists to be familiar with penicillins and cephalosporins with similar side chains as the presence of identical side chains is a factor contributing to cross-reactivity between penicillins and cephalosporins, in the range of 10% to 38% [9]. Cross-reactivity has been detected between ceftriaxone, cefotaxime, and cefepime, which share an identical side chain at the R1 position. However, it is overall hard to accurately predict the risk of cross-reactivity, as some reactions may be due to antibodies to the common beta-lactam ring. With this in mind, Table 27.5 provides groupings of beta-lactam antibiotics that share similar side chains [6]. With regard to carbapenems, cross-reactivity among penicillins and carbapenems is estimated to be in the range of 1–4% [6, 9].

Sulfonamide Allergy

Non-antimicrobial sulfonamide allergy does not have implications on the prescribing of sulfonamide antimicrobials,

Table 27.5 Beta-lactam antibiotics with similar side chains, where cross-reaction between agents within a group is possible

β-Lactams with common R1 side chains at C-7						
Group 1		Group 2		Group 3		
Penicillin		Amoxicillin	Cefprozil	Cefepime		Cefpirome
Cefoxitin		Ampicillin	Cephadrine	Cefetamet		Ceftizoxime
Cephalothin		Cefaclor	Cephalexin	Cefeteram		Ceftodoxime
Cephadrine		Cefadroxil	Ceftatizine	Cefotaxime		Ceftriaxone
β-Lactams with common R2 side chains at C-1						
Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7
Cefadroxil	Cefmetazole	Cefotaxime	Ceftibuten	Cefuroxime	Cefdinir	Aztreonam
Cephalexin	Cefoperazone	Cephalothin	Ceftizoxime	Cefoxitin	Cefixime	Ceftazidime
Cephadrine	Cefotetan	Cephapirin				Cefsulodin
	Cefamandole					

Adapted from Ref. [6]

e.g., TMP-SMX can be used in patients reporting allergy to sulfonamide diuretics. Most commonly, patients may report occurrence of maculopapular rash while on therapy. Where necessary, continuation of therapy to complete treatment may be tolerated. Alternatively, treatment may be held, and a desensitization protocol can be used to rechallenge these patients, often inducing tolerance [9].

Although less common than beta-lactam allergies, allergic reactions have been reported in all classes of antimicrobials. Mild cutaneous reaction, where there is an absence of exfoliation of skin or involvement of mucosal and internal organs, is commonly reported. Patients with severe cutaneous adverse reactions such as Stevens–Johnson syndrome/toxic epidermal necrosis or drug rash with eosinophilia and systemic symptoms (DRESS), in general, should not be rechallenged, and alternative therapy should be prescribed. Where no alternative exists, or where patient reports allergy to multiple classes of antimicrobials, referral to allergist may assist with defining the culprit.

Drug allergies and intolerances are significant barriers to prescribing the most efficacious therapy. With proper counselling, most patients will consent to being rechallenged with the inciting medications where mild reactions were reported or history of reaction is vague, e.g., childhood penicillin allergy where medical attention was not required to manage the reaction. Patient should be counselled to self-monitor and most ideally, be closely observed.

Organ Function Relevant to Antimicrobial Therapy

Many antimicrobials require dosage adjustments for decreased renal and hepatic functions. For many renally cleared antimicrobials, dosage adjustments are range-based. For patients whose estimation of renal function (Cockcroft and Gault equation) lies on the cusp, it is useful to assess the overall trend of creatinine. If the trend portends improving renal function, the more aggressive dosing may be chosen with the exception of dialysis-dependent patients whose creatinine is artificially influenced by the dose of dialysis

administered. Dialysis dosing guidelines should be consulted for recommended adjustments based on the mode of dialysis employed. If a non-dialyzed patient with a renal function at steady state is lying on the cusp of dosing adjustments, an assessment of the risk and benefit of potentially overdosing versus inadequately treated infection may be required. In all cases, adjustments may also be indication and organ-specific. In jurisdictions where pharmacists may automatically adjust prescriptions based on organ function, prescribers should be encouraged to clearly indicate when dosing regimen is tailored to a specific indication or fluctuating renal function to avoid dosing errors.

Although some antimicrobials are cleared via liver metabolism, the dosage adjustment guidance are lacking for most. It is important to recognize that some antimicrobials should be avoided or used with caution in liver disease. When no alternative exists, close monitoring for adverse effects from antimicrobial therapy should be done while on therapy.

Genetic/Metabolic Characteristics Relevant to Antimicrobial Therapy

Although uncommon, specific genetic, and metabolic characteristics can impact the choice and dosing of selected antimicrobials. Inappropriate choice and dosing can have serious implications. For example, prior to administering dapsone and primaquine, it is most ideal to check the glucose-6-phosphate dehydrogenase (G6PD) status of a patient as drug-induced acute hemolytic anemia can occur in individuals with G6PD deficiency. Dapsone is primarily used for treatment of leprosy, as an alternate in *Pneumocystis jirovecii* pneumonia (PCP) prophylaxis and also as part of second-line treatment for PCP. Consider checking G6PD status in patients deemed at risk of PCP, as patients may develop adverse effects while on TMP-SMX, the preferred first line for both treatment and prophylaxis, and an alternative is required for continuation of treatment. Primaquine is currently the only available anti-relapse therapy to cure the liver-stage infection and prevent relapse in malaria caused by *P. vivax* and *P. ovale*. It is also used as part of an alternative

treatment in PCP for patients intolerant of TMP-SMX and dapsone. Malaria-specific tailoring of dosing recommendations for G6PD deficiency exists, and the drug can be administered with close monitoring for the prevention of relapses of *P. vivax* and *P. ovale* infections. Some clinicians may loosely adhere to this recommendation as the trait of G6PD deficiency is highly prevalent, particularly in people of African, Asian, and Mediterranean descent; however, it can affect all races, although the severity of G6PD deficiency varies significantly among racial groups.

Consequences also exist for individuals with polymorphism in drug-metabolizing enzymes, in particular the cytochromes P450 (CYPs). This polymorphism is translated into risk differences concerning drugs metabolized by the highly polymorphic enzymes CYP2C9, CYP2C19, and CYP2D6. The resultant phenotypes include poor metabolizers, intermediate metabolizers, extensive metabolizers, and ultrarapid metabolizers. The most studied antimicrobial where polymorphism impacts the availability of active drug is voriconazole, where therapeutic drug monitoring of levels may be employed to guide dosage adjustments to ensure reasonable level is achieved. Discovery of the effects of genetic polymorphism on various medications, including antimicrobials, will continue to occur. Pharmacists should be aware of the impact and ameliorating strategies that can be utilized to enhance patient care.

Ongoing Assessment and Monitoring Required

Preliminary Results and Antibigram

When an organism is isolated from a specimen, it is important to assess whether empiric therapy remains appropriate based on known local susceptibility data. While awaiting the susceptibility results specific to a patient's isolate, the clinician can utilize the antibiogram to assess the probability of appropriate coverage with current therapy. Antibiogram offers an overall profile of antimicrobial susceptibility testing results of a specific microorganism to a panel of antimicrobials with reliable spectrum of activity in the literature. Based on the Clinical and Laboratory Standards Institute (CLSI) published guidelines, most North American and Canadian centers conduct annual compilation of data using only the first isolate per patient in the period analyzed and

include only organisms for which ≥ 30 isolates were tested. In addition to directing empiric antimicrobials while further susceptibility results are pending, it can also be used to detect and monitor trends in antimicrobial resistance as seen in Table 27.6. These data can be summarized at a ward, hospital or healthcare region level.

Although a useful tool, antibiograms should not be the sole deciding factor guiding therapy. The examples as shown in Table 27.6 would suggest that using vancomycin is likely 100% efficacious for the treatment of *Staphylococcus aureus* infections. Based on cumulative clinical evidence, however, it is well known that beta-lactams exhibit a more rapid bactericidal activity [10]. In *S. aureus* bacteremia (SAB), rapid achievement or eradication is desired to minimize the metastatic manifestations such as infective endocarditis, polyarticular septic arthritis, vertebral osteomyelitis with or without paraspinal abscesses, for which invasive source control may be required, such as valve replacement, joint debridement, debridement of vertebral joint or needle aspiration of abscesses. In patients with SAB, one acceptable approach may be to include concurrent use of a narrow-spectrum beta-lactam and vancomycin until susceptibility is available. On the other hand, the same antibiogram would allow clinicians to confidently prescribe oral treatment for a patient with MRSA or MSSA skin abscess, which required incision and drainage but is otherwise well and home-bound.

Clinical Monitoring

Patients should be assessed for clinical improvement or deterioration on a regular basis while on treatment. Changes in clinical status may be related to efficacy or toxicity. For example, clinical failure remains a possibility when an infection is caused by a pathogen capable of producing inducible resistance while on therapy, e.g., inducible AmpC beta-lactamase *Enterobacter cloacae* complex treated with a cephalosporin. A patient may also develop reactions to the antimicrobial, which may have physical manifestations such as rash and vasculitis.

Laboratory Monitoring

Ongoing assessment of bloodwork would also assist in monitoring for response and tolerance to therapy. Normalization of leukocytosis may suggest that the patient is responding to therapy. However, continual decrease of the white blood

Table 27.6 Example of an antibiogram

2017 Antibiogram	CEFAZOLIN	CLOXACILLIN	CLINDAMYCIN	TETRACYCLINE	TMP-SMX	VANCOMYCIN
<i>S. aureus</i> —MRSA only	N/A	N/A	90[89]	99[89]	98[89]	100[89]
<i>S. aureus</i> —ALL	75 [375]	75 [375]	83[357]	97[354]	99[354]	100[384]
2014 Antibiogram						
<i>S. aureus</i> —MRSA only	N/A	N/A	69[87]	93[90]	97[89]	100[95]
<i>S. aureus</i> —ALL	71 [332]	71 [332]	80[307]	96[300]	97[311]	100[328]

% of isolates susceptible [# isolates tested]; MRSA Methicillin-resistant *Staphylococcus aureus*, N/A not applicable

cells and development of neutropenia may suggest drug-associated adverse effects. An antimicrobial's adverse profile should be carefully reviewed and appropriate bloodwork ordered at regular intervals for assessment. Weekly or more frequent monitoring may be required if the patient is hospitalized for monitoring of infection or has fluctuating organ functions. For stable patients, on long-term therapy, monitoring may be decreased to $\geq$ monthly at the discretion of the follow-up physician. Table 27.7 provides a general guide on laboratory monitoring for the most commonly used antimicrobials, as well as unique or common adverse effects to consider as part of monitoring. Additional laboratory monitoring and increased frequency of monitoring may be required for concomitant risk factors.

Narrowest Spectrum Antimicrobial Should Be Used Whenever Possible and Applicable

Interpreting Patient-Specific Microbiologic Data

The ability to appropriately interpret microbiology data available empowers pharmacists to confidently assess indication as part of the patient care process. Astute examination of clinical signs and symptoms, microbiology data, combined with knowledge of the behavior inherent to an organism, enables practitioners to differentiate between colonization and true infection. An example for interpretation of blood cultures is shown in Table 27.8.

Interpretation of other specimens may follow a similar process in assuring the result is relevant and interpret its impact on therapy. In most patients, for specimens that are suitable for Gram stain, such as sputum or wound swabs, the presence or absence of white blood cells and bacteria may provide clues to the significance of subsequent growth. Suppose a patient is suspected to have developed a ventilator-associated pneumonia (VAP) from invasive ventilation required to support ruptured abdominal aortic aneurysm. A specimen was sent off for culture, and a broad-spectrum antimicrobial was started, e.g., piperacillin-tazobactam. The Gram stain from the endotracheal tube sample shows a high presence of polymorphs with predominance of a gram-negative bacilli with the final identification of *Stenotrophomonas maltophilia* (a gram-negative bacilli) and coagulase-negative *Staphylococcus* (a gram-positive cocci). This patient will require modification of therapy to adequately treat *Stenotrophomonas* VAP. The coagulase-negative *Staphylococcus* does not require treatment as 1. No predominant growth shown on Gram stain and 2. Unlikely to be a respiratory pathogen based on cumulative evidence. Exceptions will exist, such as for severely neutropenic patients from chemotherapy treatment for leukemia or immunosuppressed patients posttransplant, where the

absence of leukocytes does not exclude the possibility of an infection.

Streamlining to Definitive Therapy

When patients are assessed in follow-up, when the results of investigations, necessity and feasibility of source-control have been explored, and clinicians have the necessary information to finalize a therapeutic plan, streamlining to the narrowest spectrum definitive therapy should occur. This should be done in a timely manner to minimize unnecessary exposure to broader-than-necessary antimicrobials.

This is especially important in the context of ongoing synthesis of high-quality data that challenges commonly accepted practice, which may have based on expert opinions or past cumulative published experience. A case in point, for patients who are healthcare experienced, this may require reassurance that some deep-seated infections, such as infective endocarditis and osteomyelitis, may be initially treated with a set duration of parenteral antimicrobials with completion of the treatment using early oral switch based on results from randomized, controlled trials [11, 12]. Another area of evolving evidence that may surprise even health care professionals would be shorter duration of treatment for bacteremia (7 vs 10–14 days), sometimes combined with early oral switch dosed appropriately [13].

Seamless Care

The finalized plan should be made based on reasonable clinical improvement, along with a high confidence of obtaining the correct diagnosis with microbiologic data to support streamlining. This plan, including the choice, dosing regimen, duration of therapy and monitoring plan, should be clearly communicated to the patient.

Pharmacists play an important role in counselling on the medications prescribed and to assess efficacy, tolerability, and ensure compliance. Keeping up to date and recognizing appropriate scenarios where newly evolved evidence should not be applied to specific patients, infections, and microorganisms would increase our ability to do so. Additionally, pharmacists can maximize the future care of these patients by clear documentation of a patient's intolerance to an antimicrobial. It is especially important to document when desensitization or graded challenge was performed before institution of a course of treatment with an antimicrobial in a patient was known to be allergic to. Such information should be clearly communicated to patient as well as added to the patient's permanent medical records. As health care continues to work towards continuity of care, pharmacists' participation in seamless care will continue to evolve.

Table 27.7 Clinical and laboratory monitoring for commonly prescribed antimicrobials

	Laboratory monitoring (Frequency per week) ^a					Clinical monitoring for adverse effects
	CBC and Diff	Cr ± BUN	Liver enzymes	Electrolytes	Others	All patients should be counseled for self-monitoring of cutaneous, gastrointestinal adverse effects and anaphylactic/hypersensitivity reactions. Additional non-inclusive clinical monitoring listed for common or significant adverse effects
Antibiotics						
Aminoglycosides	1	2	1	–	–	Nephrotoxicity possible, especially with concurrent risk factors. Cochlear and vestibular toxicity may be nonreversible and cumulative. Neuromuscular blockade has been described. audiogram at regular interval recommended
Beta-lactams	1	1	1	1	–	Most commonly associated with allergy reactions. Rare hematologic toxicity but increases with prolonged use. Beware of agent-specific adverse effects, e.g., penicillins associated interstitial nephritis, myoclonic seizures with accumulation of inappropriately high dose
Fluoroquinolones	1	1	1	–	–	Rare: Hallucinations, psychosis, seizures, exacerbations of myasthenia gravis. Arthropathy with cartilage erosions maybe a concern in children; consider risk versus benefit. Tendinitis and rupture with concurrent risk factors. QT prolongation possible especially with concurrent risk factors
Macrolides	1	–	1	–	–	Acute psychosis possible with clarithromycin. Reversible cholestatic hepatitis and hearing loss with azithromycin possible. QT prolongation possible, especially with concurrent risk factors
Tetracyclines	1	1	1	–	–	Photosensitivity-related rash and hyperpigmentation possible. Teeth/bone discoloration
Miscellaneous antibiotics						
Chloramphenicol	1	1	1			Significant bone marrow suppression; fatal, aplastic anemia possible. Hemolytic anemia may be related to G6PD deficiency. Optic neuritis results in blindness possible
Clindamycin	1	1	1	–	–	Transient transaminitis possible
Daptomycin	1	1	1	–	CK weekly	Reversible skeletal muscle toxicity possible with >7 day use. Reversible eosinophilic pneumonia has been described with >10-day use
Linezolid	1	1	1	–	–	Reversible myelosuppression (thrombocytopenia, pancytopenia) usually with use >2 weeks. Peripheral neuropathy and optic neuropathy may be permanent. Fatal lactic acidosis reported
Metronidazole	1	1	1	–	–	Rare: Seizures, encephalopathy, cerebellar dysfunction, and peripheral neuropathy, usually reversible. Disulfiram reaction with co-ingestion of alcohol
Tigecycline	–	–	1	–	–	Significant dose related GI side effects. Transient transaminitis possible
TMP-SMX	1	1	–	1	–	Hyperkalemia, SJS, TEN, sweet’s syndrome. Drug-induced cholestasis and hepatitis possible
Vancomycin	1	1	–	–	–	Reversible nephrotoxicity and ototoxicity (vertigo, tinnitus, hearing loss) possible, especially with concurrent risk factors. Infusion-related reactions common including red man syndrome
Antifungals						
Amphotericin B	1	2	1	2	Mg twice weekly	Infusion related reactions may be minimized with reduced rate. Electrolyte abnormalities are common. Nephrotoxicity is common; reduced with lipid formulations and hydration
Azoles	1	1	1	–	–	Effect on QT interval, photosensitivity, and visual disturbances may be agent dependent. Potential for hepatic injury and rarely hepatic failure associated with all azoles

(continued)

Table 27.7 (continued)

	Laboratory monitoring (Frequency per week) ^a					Clinical monitoring for adverse effects
Echinocandins	–	–	1	–	–	Infrequent and minor side effects. Adhere to infusion time recommendation to minimize histamine release-related symptoms (pruritus, facial swelling, vasodilatation)
Antivirals						
Acyclovir, valacyclovir	1	1	1	–	Urinalysis weekly	Hydration and renally adjusted dosing to minimize accumulation associated neurotoxicity and nephrotoxicity
Cidofovir	1	1	1	1	Urinalysis weekly	Dose-related nephrotoxicity may be irreversible. Neutropenia is common
Foscarnet	1	2	1	2	Ca, Mg, Phos twice a week	Significant nephrotoxicity, usually reversible. Rarely: Hallucinations and seizures. Adhere to infusion rate limits to minimize metabolic abnormalities; may result in arrhythmias, tetany, seizures
Ganciclovir, valganciclovir	2	1	–	–	–	Significant reversible bone marrow suppression: Leukopenia, neutropenia, thrombocytopenia. Confusion, psychosis, seizures, and coma are possible. Nephrotoxicity

^aRepresents monitoring for 1. Therapeutic efficacy on leukocytosis 2. Hematologic toxicity 3. Renal or hepatic function for clearance of medication or potential toxicity. 4. Effects on electrolytes (potassium). *BUN* blood urea nitrogen, *Ca* calcium, *CBC and Diff* complete blood count with differential, *CK* creatine kinase, *CR* serum creatinine, *GI* gastrointestinal, *Mg* magnesium, *SJS* Stevens–Johnson syndrome, *TEN* toxic epidermal necrolysis

Table 27.8 Sample approach to interpretation of blood cultures: questions and implications in patient care

Is the culture antibiotic-influenced?
In the setting of recent or current use of antimicrobials: A negative blood culture may be representative of a false negative. A positive blood culture may indicate the need for source control, if applicable to a pathogen.
Is this collected from a line? If so, is there an accompanying peripheral blood culture?
If the patient has an existing central venous catheter and Growth from line specimen only → blood culture from line with a corresponding negative peripheral blood culture may suggest line infection. Indication and duration of antimicrobial is dependent on source control and the identified organism. Growth from line and peripheral specimen → differential time to positivity (dTTP) of ≥2 h where blood culture from the line grows faster may indicate catheter-associated bloodstream infection. A dTTP <2 h suggests bacteremia from other source. Growth from line specimen and no peripheral blood culture collected → repeat blood culture from line <i>and</i> peripheral site before administering antimicrobial. Positive blood cultures collected at time of newly inserted lines (arterial or venous) can represent true bacteremia from other source or possibly contamination at time of insertion. To give clarity, when possible, repeat blood culture from line(s) and peripheral site before administering antimicrobial. Polymicrobial growth in blood culture may be line-associated or may represent contamination. In the right setting, polymicrobial growth supports the diagnosis and severity of a true infection where multiple organisms may translocate into the bloodstream e.g. intra-abdominal sepsis.
Preliminary identification based on morphology is available. Considering the history, exam, and presentation, what organism is the result suggestive of? Is there a need to modify empiric therapy?
Medically important organisms can be distinguished by their unique morphology. Preliminary results are commonly reported as: Gram-positive cocci in clumps/clusters may represent → <i>staphylococcal</i> species. Assess the likelihood of the pathogen based on history, exam, and presentation. Consider adding vancomycin if coagulase-negative <i>staphylococcal</i> or MRSA infection is a possibility. Gram-positive cocci in chains may represent → <i>streptococcal</i> or <i>Enterococcal</i> species. Assess the likelihood of the pathogen based on history, exam, and presentation. Consider adding vancomycin if <i>Enterococcal</i> infection is a possibility. Gram-negative bacilli → assess the likelihood of the potential pathogens based on history, exam, bloodwork, and presentation, e.g., symptoms of urinary tract infection, abdominal pain, etc., broad-spectrum antimicrobials typically include adequate gram-negative coverage. In septic shock patient, consider adding a second agent for gram-negative coverage while awaiting final identification and susceptibility results.
How likely is this organism a contaminant based on assessment of existing risk factors and patients' response to therapy?
Contamination may be likely if peripheral or line culture is positive in one vial out of many nonantibiotic-influenced specimens collected. Interpretation will require knowledge of known specific pathogens virulence or status as a known colonizer on skin. In general, <i>Staphylococcus aureus</i> and yeast in blood should be viewed as true pathogens → repeat blood culture and modify empiric therapy to ensure adequate coverage. Coagulase-negative <i>staphylococcal</i> species may be pathogenic in appropriate settings, e.g., presence of prosthesis. However, it is the most common organism on human skin. If only one vial out of many is positive, the probability of contamination with this skin colonizer is high. This highlights the importance of repeating blood culture before starting therapy. It is important to note that one specific coagulase negative <i>staphylococcus</i> , <i>S. Lugdunensis</i> , may cause serious infections as usually seen for <i>Staphylococcus aureus</i> .

(continued)

Table 27.8 (continued)**The organism has been identified and the susceptibility result is pending. Is there a need to modify empiric therapy? Are there further non-pharmacologic concerns that should be discussed with the team?**

Modifications to empiric therapy can occur for selected organisms based on cumulative literature or information extracted from an antibiogram.

For example, a patient presents with acute abdominal pain, with several days history of fever and chills, increased fatigue, and worsening nausea and vomiting. Admission bloodwork was within normal parameters, and abdominal X-ray, chest X-ray (CXR), and CT abdomen were unremarkable. Patient was clinically stable, and blood cultures were collected. no antibiotics were started. overnight, therapy with meropenem was started for preliminary report of gram-negative bacilli growing in the submitted blood culture which is subsequently identified as *haemophilus influenzae*. You recall that *H. Influenzae* is highly susceptible to third-generation cephalosporins, and your institutional antibiogram confirms that in the previous year, 98% of isolates were susceptible to ceftriaxone and 68% to ampicillin. You recommended to your team to streamline patient to ceftriaxone at appropriate dose. Patient remains clinically stable. Your plan includes further narrowing to ampicillin if this strain is beta-lactamase negative. However, you also notice that blood cultures collected after doses of meropenem were administered is still positive for GNB. In discussion with the attending physician, the possibility of endocarditis was discussed, and echocardiogram was ordered. you also recall that your patient's past medical history includes documented hypogammaglobulinemia, which may predispose this patient to invasive *H. Influenzae* disease. You confirmed with local vaccination guidelines; your patient should receive one dose of *H. Influenzae* b vaccination based on age and risk factor; however, your institution's policy is for patient to obtain vaccination as outpatient. During team rounds, the patient was updated with regard to the change in antimicrobial choice and inquiries made into tolerance to the medications. You document your interventions, discussion, and recommendation for vaccination at a later date.

Susceptibility has been reported. Can the antimicrobial be further streamlined to a narrow spectrum antimicrobial?

Streamlining to the most effective therapy is safe even in patients who are critically ill when based on known pathogen and adequate source control where applicable.

For example, a patient presents with signs and symptoms worrisome for necrotizing fasciitis. Blood culture was collected, and the patient was immediately taken for debridement of the affected areas. Broad-spectrum antimicrobial coverage with imipenem and vancomycin were started preoperatively. Intraoperative confirmation of diagnosis of necrotizing fasciitis was made. Over the next 2 days, patient underwent repeat debridement and remains mechanically ventilated and dependent on vasopressor but is slowly being weaned. meanwhile, preliminary blood culture has shown gram-positive cocci (GPC) in chains, subsequently identified as *streptococcus pyogenes*. Intraoperative wound tissue from debridement also shows GPC in chains. There is no microbiologic evidence to support polymicrobial infection. Can this patient be streamlined to penicillin and clindamycin as per guidelines for the management of necrotizing fasciitis? Yes. *Streptococcus pyogenes* is exquisitely susceptible to penicillin with no reports of resistance globally. Clindamycin should be used concurrently to decrease the production of toxins. The slow clinical improvement is likely in response to the body's own response to the overwhelming infection and repeat iatrogenic trauma for source control. unless there is evidence of a new or concurrent infection requiring broad-spectrum antimicrobial, continuation of the unnecessarily broad coverage does not offer advantage over narrowed-spectrum definitive therapy. Additional factors to consider in the setting of invasive group A streptococcal infection includes prescribing chemoprophylaxis for household or close contacts to the index patient.

Do we have documented clearance of the bacteremia? Is it required?

Due to heterogeneity of the causative pathogens in bacteremia as well as the impact of source control in its role to remove foci of persistent bacteremia, it is difficult to conduct well-designed trials to study the necessity of document on clearance of bacteremia. Current practice is largely driven by observational, retrospective data as well as expert opinions.

Controversy exists regarding the importance of documented clearance of bacteremia. Subjecting patients to additional invasive blood draw and resource consumption must be weighed against the consequences of unrecognized sustained bacteremia suggestive of need for further investigations and treatment and may have implications on morbidity and mortality.

Repeat blood cultures to document sustained clearance are recommended in *S. Aureus* bacteremia, candidemia, or where culprit organism can cause endovascular infection e.g., *salmonella* spp.

What additional information should be considered while finalizing a plan on choice, dose, and duration?

Confirm that the chosen regimen is appropriate for indication, site of infection, organ function, and tolerance to antimicrobials.

Do the patient factors, culprit organism and the infectious syndrome, support early oral switch?

Is the planned duration of treatment reasonable based on guidelines and/or high-quality published randomized control trials?

Final choice may be influenced by cost to patient if payment required or disposition.

Clinical Pearls

- Encourage pre-antimicrobial microbiological sampling of suitable specimens when applicable to the practice setting and infectious diagnosis.
- Ensure empiric therapy is initiated for the suspected diagnosis based on common pathogens associated with specific infections, with consideration of patient-specific risk factors.
- Participate in the follow-up of microbiologic, serologic, radiographic, and other diagnostic investigations or procedures performed to assess the impact on therapy.
- Anticipate and make recommendations on most appropriate definitive therapy based on culture and susceptibility results whenever possible or narrow coverage based on most likely pathogenic organisms responsible when results are low-yield or when cultures are not performed.
- Participate in further reassessment and monitoring as required to ensure therapy is tolerated and resolution of the infection is achieved.
- Ensure the relevant team members and the patient are aware of the treatment and monitoring plans.
- Participate in allergy assessment and encourage de-labelling of allergies, when appropriate, or documentation of tolerance to antimicrobials for patients who report multiple allergies.

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Chapter Objectives

1. Describe the role of the pharmacist in the intensive care unit (ICU) and in the care of critically ill patients.
2. Describe the steps of critical care assessment, including collecting patient history, assessing the history of present illness, and conducting a review of systems.
3. Apply knowledge of routes of administration, intravenous compatibility, and pharmacokinetic changes in the critically ill to ensure effective and safe medication delivery to the patient.

Background

A critically ill patient is defined as a patient who has a life-threatening illness that often involves multisystem dysfunction and can result in morbidity and mortality. Generally, critically ill patients are cared for by an interprofessional team in the ICU. With the evolving role of pharmacists in clinical care, the presence of pharmacists in the critical care setting is becoming more common and presents a niche practice site for pharmacists. Pharmacist involvement in the care of the critically ill has been found to make a difference in patient outcomes [1]. For example, clinical pharmacy practice has been reported to result in improvements in fluid management and reduction in ventilator-associated pneumonia (VAP) occurrence, medication errors, and adverse drug reactions [1]. As a result, pharmacists' presence in the ICU is the standard of care in Canada, the USA and other countries, and evidence demonstrating the positive impact of pharmacy practice on ICU patient outcomes continues to emerge.

Critical care assessment incorporates aspects of a regular patient assessment, such as gathering a patient history, as well as specialty areas of assessment. Pharmacists in the ICU attend daily rounds with the interprofessional team where they can collect information on the patient's illness and status. An interprofessional team in the ICU usually consists of an intensivist, other specialists as required, a nursing team, a pharmacist, a respiratory therapist, a dietician, a physiotherapist, and a social worker. Critical care assessment starts with data collection. The information from daily rounds and the patient's chart can inform the pharmacist of the patient's history of present illness (HPI). Critically ill patients often have a more extensive HPI, including microbiology and laboratory results, as well as other investigations such as diagnostic imaging. Once a complete HPI is collected, pharmacists conduct a review of systems, which focuses on medication-related inquiries and illness progression or improvement. For a critically ill patient, there are additional topics of assessment that are unique to this population. Due to the nature of these patients' illnesses, dose adjustments are often required due to drug interactions and pharmacokinetic changes. Critically ill patients may also need a route of administration assessment due to difficulties swallowing, feeding tube insertion (e.g., nasogastric [NG] tubes), or lack of gastrointestinal access (e.g., nothing by mouth [NPO]). Intravenous (IV) lines are also common in this population, and an investigation into IV compatibility may be necessary. Due to the instability and characteristics of ICU patients' illnesses, there can be medical concerns, such as a patient's acid-base balance or fluid status that also need to be assessed.

A critically ill patient's medications can be split into three categories, including *disease-specific*, *hospital-specific*, and *patient-specific* medications, which can be remembered by the acronym (DHP). Separating the medications into three separate categories allows for organization and can help ensure the pharmacist does not miss any aspect of a critically ill patient's medication assessment. Patient-specific medications are those that the patient had been prescribed and was taking before admission. Pharmacists have a role in

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identifying these medications through medication reconciliation and assessing the need to continue or discontinue the drug while the patient is critically ill. Disease-specific medications are those that have been initiated to treat the patient's current illness. Therapeutic drug monitoring (TDM) may be necessary with some disease-specific medications with narrow reference ranges. Hospital-specific medications are those that are not used to treat the patient's illness but may be required due to the patient's presence in the hospital. Pharmacists need to assess if a patient requires such medications, which can include venous thromboembolism (VTE) and stress ulcer prophylaxis.

Pharmacists have an emerging role in the care of critically ill patients in the ICU and have been linked to improved clinical and patient outcomes. Critical care assessments need to be performed daily for each patient, as a patient's status can change quickly when critically ill. Knowledge of the specialty topics and specific skills required in critical care is necessary for any pharmacist pursuing a career in critical care.

Initial Assessment of a Critically Ill Patient

Past Medical History

Gathering the past medical history for the critically ill is quite similar to patients in the general population. This includes collecting the patient's demographics (age, sex, height, and weight), medical and medication histories, and allergies (see Chap. 1 for more details about patient history). In the ICU setting, patient information could be collected using the following sources:

- Patient or family: Patients in the ICU may be awake and able to provide and confirm their past medical history. However, given the nature of their critical illness, patients may be intubated or have an altered mental status or level of consciousness (LOC), hindering their ability to provide such information. Hence, pharmacists will need to rely on other sources. The patient's family members may provide a reliable source of information depending on the patient's disclosure of healthcare information to his/her family or the family's involvement in the patient's previous healthcare.
- Patient's chart.
- Provincial electronic health record (EHR), e.g., EPIC®, NetCare.
- Patient's healthcare providers in the community, e.g., family doctor.
- Patient's community pharmacy.

The patient's past medication history is essential information for the pharmacist. When collecting a patient's medication history, more steps are involved as the goal is to create the best possible medication history (BPMH), which is the first step in the medication reconciliation process. Medication reconciliation is the process of ensuring that the most accurate medication information is collected, and then each medication is reviewed and evaluated, and either continued, changed, or discontinued/held. Medication reconciliation also involves communication with all members of the interprofessional team, so that there is consistency amongst the information available to each discipline. When conducting a medication reconciliation, the list of medications should be completed using more than one source. At least two sources are consulted, as incorrect or incomplete information may be provided by any source. For example, on an EHR, it may show a medication that was dispensed by the pharmacy but was never actually taken by the patient, or the patient may have received medication samples from a physician, which would not be reported on the EHR. The BPMH will result in a list of drugs that the patient was taking before admission, which can be referred to as patient-specific medications.

Patient-Specific Medications

Once the pharmacist is aware of the patient-specific medications, an assessment should be made to determine the indication for each drug. This can prove difficult due to the inability of pharmacist to discuss with the patient. However, with the patient's past medical history, the pharmacist will likely be able to decipher the indication for most drugs. Due to the nature of the critical illness and possible hemodynamic instability of ICU patients, there are often drugs that can be temporarily discontinued while the patient is receiving this level of care. Examples of drugs that could be discontinued are bisphosphonates. The discontinuation of a bisphosphonate, while the patient is in the ICU, will not make a clinically significant difference in the patient's bone density but will decrease the number of drugs being administered to the patient. In addition, in the ICU, administering a drug can be more difficult due to the presence of enteral feeding tubes or a lack of gastrointestinal access. Continuing certain drugs can unnecessarily complicate patients' medication regimen and care, such as worrying about administering the bisphosphonate 30 min before food. Other medications may not be necessary for the patient to be taking in the ICU but require the pharmacist to contemplate the risks vs. benefits of discontinuation. For example, sudden discontinuation of antidepressants can cause withdrawal symptoms, and the pharmacist should consider if this discomfort to the patient is outweighed by pill burden and possible drug-drug interactions. Finally, some drugs should be discontinued because

they may cause harm to critically ill patients. For example, some patients in the ICU are at an increased risk of having acute kidney injury (AKI) due to factors such as renal hypoperfusion. Drugs that are nephrotoxic or may potentiate kidney injury may need to be discontinued. The pharmacist should still consider if discontinuing this drug will cause more harm than continuing it. Examples of medications that can cause kidney insult and may need to be held during an acute illness can be remembered by the acronym SADMANs. SADMANs stands for sulfonylureas, angiotensin-converting enzyme inhibitors (ACEIs), diuretics, metformin, angiotensin receptor blockers (ARBs), non-steroidal anti-inflammatory drugs (NSAIDs), and sodium-glucose linked transporter 2 (SGLT2) inhibitors. Note that this is just an example of some drugs that should be assessed for discontinuation. There are exceptions, as drugs such as diuretics are used in the ICU setting, while a drug such as metformin would rarely be continued in an unstable patient. As part of an interprofessional team providing care to critically ill patients, conducting assessments to continue or discontinue patient-specific medications is a role that pharmacists can take lead. A medication reconciliation to create a BPMH should be completed on admission, at any transition of care (e.g., when a patient is moved to another unit in the hospital), and at discharge as this minimizes the chance for medication errors and promotes seamless care.

History of Present Illness

After the pharmacist has collected the patient history, the next step is to determine the reason for admission and the history of present illness (HPI). Some common reasons for admission include sepsis, traumatic brain injury (TBI), shock, trauma, and respiratory failure. Once the HPI and the patient's current status are known, the pharmacist can begin to assess the patient's disease-specific medications for indication, efficacy, and adverse effects.

Disease-Specific Medications

Disease-specific medications are drugs that are used to treat the patient's current illness e.g. antimicrobials in patients with sepsis. These drugs will be individual to the patient and his/her illness. Pharmacists should refer to the disease-specific clinical guidelines or the best evidence available to inform their treatment decisions. In addition, there are classes of drugs that are commonly seen in the ICU such as those needed to treat the patient's condition and/or hemodynamic instability that critical care pharmacists need to be familiar with. These include sedatives, analgesics, vasopressors, and antimicrobials. Table 28.1 provides a summary of the common medications used in the ICU. Pharmacists

Table 28.1 Examples of common medications utilized in the intensive care unit

Drug class	Example
Vasopressors and inotropes	Epinephrine, norepinephrine, dobutamine, dopamine, vasopressin, phenylephrine, milrinone
Vasodilators	Sodium nitroprusside, labetalol, hydralazine, nicardipine, nitroglycerin, clevidipine
Sedatives and anesthetics	Propofol, midazolam, ketamine Dexmedetomidine
Opioid analgesics	Fentanyl, hydromorphone
Neuromuscular blockers	Succinylcholine; cisatracurium, rocuronium
Antiarrhythmic	Amiodarone, diltiazem, lidocaine

assess the suitability of the current patient-specific medications to ensure they are safe, effective, and indicated.

Hospital-Specific Medications

Hospital-specific medications are medications used to prevent complications that patients may experience due to the fact that they are in a hospital. Hospital-specific medications are not the drugs that the patient was previously on before admission and are not the ones that are used to currently treat the patient's reason for admission. These medications may often be overlooked due to the focus on disease-specific medications that are used to treat the patient's current illness. Therefore, pharmacists can have a significant role in assessing the patients' need for hospital-specific medications. Those medications should be assessed daily, as the need to continue these medications can change throughout the patient's hospital stay. Hospital-specific medications can be unintentionally and unnecessarily continued in some patients during their transition of care, and pharmacists can prevent this by completing a medication reconciliation at each care transition and at discharge.

Venous Thromboembolism (VTE) Prophylaxis

Pharmacists should determine for every patient in the ICU if VTE prophylaxis is appropriate. The pharmacist should assess the patient's risk of experiencing a VTE before deciding to initiate prophylaxis therapy. Critically ill patients require VTE prophylaxis more often than other patient populations due to factors related to their illness. VTE risk factors include the expected length of stay, degree of mobilization, surgery and type of surgery, and age. A longer length of stay increases the risk of a VTE. Patients who are immobile have a greater VTE risk due to the pooling of blood in their legs. Surgery is a risk factor due to immobilization during and after surgery, and it can result in damage to patients' vasculature. In addition, the length of surgery

Table 28.2 Recommended VTE prophylaxis dose in select anticoagulants

Drug class	Drug	Recommended prophylactic dose
Low molecular weight heparins	Dalteparin	5000 units subcutaneously once daily
	Enoxaparin	30 mg subcutaneously twice daily
	Tinzaparin	4500 units subcutaneously once daily
Unfractionated heparin	Heparin	5000 units subcutaneously every 12 h OR every 8 h

VTE venous thromboembolism. Note: The above dosing might vary based on patient-specific characteristics

correlates with a higher risk for development of VTE as the longer the surgery the longer the patient was immobilized for on the operating table. The type of surgery can also increase the risk for VTE. Surgeries with increased risk are those that require arteries and veins to be cut and repaired. Critically ill patients have multiple risk factors and often qualify for VTE prophylaxis. However, the pharmacist must also consider the patient's bleed risk and conduct a benefit vs. risk analysis to determine if pharmacological prophylaxis is appropriate. If the patient is unable to receive pharmacological VTE prophylaxis due to, for example, an active bleed, nonpharmacological VTE prophylaxis such as pneumatic compression stockings can be used. However, they are not as effective at VTE prevention compared to pharmacological therapy. Therefore, a reassessment of the patient's conditions should occur daily, as they should be switched to a pharmacological alternative as soon as it deems appropriate. Table 28.2 depicts examples of anticoagulants and their prophylaxis dosing. The choice of the agent and its dosing depends on patient-specific characteristics (e.g., weight, renal function) and the evidence supporting their use in certain patient populations.

Stress Ulcer Prophylaxis (SUP)

A stress ulcer is a gastric mucosal erosion that breaches the submucosa and puts the patient at a high risk for gastrointestinal bleeding. Stress ulcers generally result from ischemia secondary to reduced gastric blood flow, followed by reperfusion injury. Gastrointestinal bleeding can cause morbidity and at a minimum, a fourfold increase in ICU mortality [2]. Critically ill patients are at an increased risk of developing a stress ulcer due to the possible coexistence of multiple risk factors (Table 28.3). The patients at the highest risk for gastrointestinal bleeding are those with mechanical ventilation for >48 h and those with coagulopathy [2]. The need for stress ulcer prophylactic therapy should be assessed for each patient, and the presence of one or more risk factors may indicate the need to start therapy. Drugs used as prophylactic

Table 28.3 Risk factors for the development of a stress ulcer in the ICU

Mechanical ventilation >48 h	History of GI ulceration or bleeding within 1 year
Requirement of high-dose steroids	ICU stay $\geq$ 1 week
Coagulopathy	Major surgery lasting >4 h
Multiple trauma/traumatic brain injury	Acute lung injury
Acute kidney injury	Hepatic dysfunction
Sepsis	Hypotension

ICU intensive care unit, GI gastrointestinal

therapy include histamine-2 receptor blockers (H_2RBs) and proton pump inhibitors (PPIs). Patients who receive enteral nutrition may not require pharmacological SUP prophylaxis, especially the effectiveness of medications has not been clearly established [3, 4]. In addition, increasing the gastric pH secondary to SUP may increase the risk of ventilator-associated pneumonias and *Clostridioides difficile* infections [2]. Due to the potential risk of complications associated with prophylaxis, the pharmacist should assess the need for prophylaxis on a case-by-case basis and consider both the benefits and risks for each patient. In addition, the pharmacist should assess if prophylactic therapy requires continuation throughout the patient's stay in the ICU. SUP should be discontinued after the initial indication resolves, since extended use of acid suppressants is associated with certain risks and costs.

Bowel Routine

Constipation is one of the common complications in the ICU. Critically ill patients have an increased risk of developing constipation secondary to multiple risk factors related to their medications and their illnesses:

- Medication-induced constipation, e.g., sedatives, opioids.
- Reduced food intake secondary to poor appetite, reduced fibre intake, and/or reduced oral intake, e.g., NPO.
- Decreased mobility and/or surgery: Decreased mobility and surgery are also considered risk factors for constipation. Critical care patients are often confined to their beds due to their illnesses or due to immobilization during or after surgery.
- Metabolic disturbance, e.g., dehydration and renal failure.
- Presence of neurological disorders such as spinal cord injury.

Constipation in the ICU, if untreated, may result in multiple complications that could result in longer ICU stay and increased patient morbidity. These complications include

poor tolerance to feeds, poor nutrition, gastroparesis, and increased intra-abdominal pressure leading to decreased lung compliance in ventilated patients and difficulty weaning. A bowel routine is used in the hospital to prevent or treat constipation. All patients should be assessed if bowel routine is appropriate. The drugs most often prescribed are polyethylene glycol 3350 (PEG 3350), lactulose, sennosides, bisacodyl, or a combination. These drugs can be prescribed on an as-needed basis or as regularly scheduled dosing. Pharmacists should assess the continued need for these laxatives daily by consulting with the patient's nurse(s) or chart. They assess if the patient is having regular bowel movements, diarrhea, or constipation. Based on this assessment, the laxatives may be continued, discontinued, have a dose change, or be switched to an alternative regimen.

Daily Assessment

In contrast to other patient populations, critically ill patients' clinical status might change quickly throughout their ICU stay. As a result, following the initial assessment, ICU pharmacists should assess their patients daily to determine if a change in plans is needed. The daily assessment performed for each patient is quite comprehensive and involves a review of systems and review of patients' microbiological tests, laboratory tests, other investigations, and current medications. Daily assessment data are generally obtained from the daily multidisciplinary rounds, where the bedside nurse provides a detailed report of the patient's progress in a head-to-toe approach. As mentioned previously, other sources of information include the patient interview (if possible), the patient's chart, and the EHR. A review of systems is a methodical approach to assessing patients' current clinical status and is grouped into assessment by individual organ systems. Pharmacists may not be the healthcare professionals conducting the physical assessment on the patient, but they should be able to interpret the findings of the assessment and apply them to their patient care. Similar to the initial assessment, pharmacists should also conduct a review of the patient's medications (patient-specific, disease-specific, and hospital-specific) on a daily basis. The appropriateness of a drug therapy can change quickly in critically ill patients, and drugs may need to be initiated and discontinued or dose adjustments may be required. A generalized overview of a daily assessment of a critically ill patient by a pharmacist is presented in Table 28.4.

Review of Systems

The main organ systems that will be the focus of this chapter are the central nervous system, cardiovascular, respiratory, gastrointestinal, and genitourinary.

Table 28.4 Pharmacist's checklist of daily assessment of critically ill patients

Assessment category	Monitoring parameters
Central nervous system (CNS)	Level of consciousness (GCS) Presence of delirium Degree of sedation Pain control
Cardiovascular system (CV)	Blood pressure (SBP, DBP) Blood pressure target Heart rate and rhythm Presence of arrhythmias QTc interval $T_{\max}$ in the last 24 h Fluid status IV fluids administered Need for vasoactive agents (vasopressors and/or inotropes) Need for antihypertensive therapy Need for VTE prophylaxis
Respiratory system (RESP)	Respiratory sounds Respiratory rate ABGs Acid-base balance Respiratory secretions Infection (e.g., pneumonia) Presence of mechanical ventilation and ventilation parameters
Gastrointestinal system (GI)	Feeding status (e.g., DAT, oral intake, feeding through enteral feeding tube, NPO) Tube feed tolerance Any nausea or vomiting Need for stress ulcer prophylaxis Bowel sounds Bowel movements (BMs) Presence of constipation or diarrhea Need for bowel routine Liver function
Genitourinary system (GU)	Urine output Fluid balance Kidney function Infection (e.g., UTI) Acid-base balance
Microbiology (micro)	Risk of infection or presence of infection Cultures and sensitivities (C&S) Need for cultures Appropriateness of the antimicrobials Therapeutic drug monitoring (TDM)
Laboratory tests (labs)	Check relevant lab results regularly Check the trend of lab values (more informative than single values) Adjust drugs or drug dosages as appropriate
Other investigations	Check for new investigations daily Includes exploratory tests and diagnostic imaging
Medications	Disease-specific medications (D) Hospital-specific medications (H) Patient-specific medications (P) Need for dosage adjustments Assessment for the need for PRN medications Drug interactions Intravenous compatibility

ABGs arterial blood gases, DAT diet as tolerated, DBP diastolic blood pressure, DVT deep vein thrombosis, GCS Glasgow Coma Scale, MAP mean arterial pressure, NPO nothing by mouth, PRN when needed, SBP systolic blood pressure, UTI urinary tract infection

Central Nervous System (CNS)

Assessment of patients' level of consciousness (LOC) is essential in critical care assessment. The most common scale for determining a patient's level of consciousness that is performed at the bedside is the Glasgow Coma Scale (GCS) (see Chap. 3 "Physical Assessment for Pharmacists"). LOC might reflect the severity of the patient illness, degree of sedation, and/or adverse effects of medications. In addition, changes in LOC provide insight into patient progress. For example, sudden GCS changes in patients with subarachnoid hemorrhage (SAH) might indicate the development of delayed cerebral ischemia, a common complication following SAH. In addition, patients who are unconscious or comatose will have a feeding tube and will require an assessment of their drugs to determine if they can be administered through a feeding tube. More information regarding drugs and feeding tubes can be found under the section "[Routes of administration](#)".

In the ICU, patients are at risk of developing delirium due to their illness (e.g., infections, organ failure) and the medications they are getting. Delirium is defined as a disturbance in mental function and a reduced awareness of the environment. Delirium can present as confusion, disorientation, and a decreased level of alertness. Delirium is often transient, but in the ICU population, it is associated with long-term cognitive dysfunction, increased mortality, and prolonged ICU stay [5]. The presence of delirium can be determined with the use of the Confusion Assessment Method for the ICU (CAM-ICU) or with the use of the Intensive Care Delirium Screening Checklist (ICDSC). Pharmacists should be aware if a patient is experiencing delirium as it can be caused by drugs and will require an assessment to determine if a drug is an instigating factor. Some of the drugs that are associated with causing delirium include sedative hypnotics, such as benzodiazepines and narcotic analgesics. In patients experiencing delirium, assessment for alternative therapeutic options with less propensity in causing delirium is recommended. Delirium can be treated with the use of atypical antipsychotics such as quetiapine and haloperidol. Nonpharmacological modalities include promoting good sleep habits, ensuring the patient is well hydrated, and improving the patient's sensory input through devices such as hearing aids [5].

Sedation is another part of the CNS assessment in the ICU. Sedation of critically ill patients improves tube tolerance and mechanical ventilation optimization in intubated patients. In addition, it helps control agitation, reduces oxygen consumption, and reduces intracranial pressure (ICP). The Richmond Agitation-Sedation Scale (RASS) and the Sedation-Agitation Scale (SAS) can be used to assess a patient's degree of sedation. It is generally recommended that sedative drugs be used to maintain a light level of sedation (RASS of -2 or higher). Light sedation in this popula-

tion is associated with improved clinical outcomes, such as a shorter ICU length of stay [6]. Table 28.1 depicts examples of common sedatives in the ICU. Level of sedation required might need to be changed based on the patient condition or progress. For examples, TBI patients with elevated ICP might require higher level of sedation compared to general critically ill patients.

Another element in CNS review is pain and pain control. Sometimes pain assessment can be challenging in the critically ill as many patients are unable to report if they are experiencing pain. There are scales developed to help assess a patient who is unable to provide verbal information, such as the Behavioral Pain Scale (BPS) and the Critical-Care Pain Observation Tool (CPOT). Note that for these scales to be valid, the patient has to have intact motor function. Vital signs can also be used to assess for the presence of pain with an increase in blood pressure, heart rate, respiratory rate, and an increase in body temperature, signifying the patient may be in pain. However, vital signs should not be the sole aspect of pain assessment and should be used only with a validated pain assessment scale. Opioid analgesics such as morphine, hydromorphone, and fentanyl are considered the first-line therapy options in critically ill patients. However, nonopioid analgesics such as NSAIDs and acetaminophen should also be considered to decrease the amount of opioids being administered to the patient to decrease the risk of delirium or other adverse effects, such as constipation [6].

Cardiovascular System (CV)

It is important to closely and frequently monitor cardiovascular system function. It is useful to assess hemodynamic stability, disease severity, organ perfusion, response to therapies, adverse effects of medications, and/or need for additional therapies. The following values should be assessed regularly:

- Maximum temperature within the last 24 h ($T_{\max}$).
- Blood pressure (SBP and DBP) and mean arterial pressure (MAP). MAP is useful as it approximates organ perfusion. In the ICU, the target MAP for most patients is ≥ 65 mmHg [7]. Equation 28.1 depicts how patient's MAP is calculated.

$$\text{MAP} = \left(\frac{(2 \times \text{diastolic blood pressure})}{+ \text{systolic blood pressure}} \right) \div 3 \quad (28.1)$$

- Blood pressure target: BP target is disease-specific and will need to be determined on an individual basis.
- Central venous pressure (CVP): CVP is the blood pressure in the venae cava near the right atrium. It can be used in the ICU to monitor the patient's fluid status, as it reflects the amount of blood returning to the heart.

- Heart rate and rhythm: Patients are also continuously monitored with an electrocardiogram (ECG). This allows monitoring for heart rate and rhythm and allows detection of cardiac arrhythmias such as supraventricular tachycardia, atrial fibrillation, and ventricular tachycardia. In addition, QTc interval monitoring could be obtained, which is helpful in the presence of interacting drugs that cause QTc interval prolongation. Cardiac arrhythmias can be very serious and can cause mortality, and need to be addressed and treated immediately.
- Need for vasopressors (see Table 28.1) as in cases of shock (see below).
- Need for antihypertensives: In the ICU, patients can also experience a hypertensive crisis, which is defined as a systolic blood pressure ≥ 180 mmHg or a diastolic blood pressure ≥ 120 mmHg. Hypertensive emergencies are also associated with morbidity and mortality and are often treated with drugs such as sodium nitroprusside or labetalol (see Table 28.1).
- Some laboratory values and investigations (discussed later) can be interpreted in concert with the above assessment. For example, lactate can be used to identify if the patient is hypoxic, and a value >1 mmol/L is a sign that the patient is not getting a sufficient level of oxygen to the tissues.

Shock

Shock is the most common manifestation of circulatory dysfunction in the ICU and is a serious condition in which there is not enough blood flow to the other organs of the body. There are four types of shock:

- *Hypovolemic shock*, which is caused by insufficient blood volume due to loss of fluid from circulation.
- *Cardiogenic shock* is due to the inability of the heart to pump blood effectively and can be caused by conditions such as myocardial infarction or heart failure.
- *Obstructive shock* is due to obstruction of blood flow outside of the heart in conditions such as pulmonary embolism.
- *Distributive shock* is due to an abnormal distribution of blood flow due to vasodilation and can be caused by sepsis (septic shock), anaphylaxis, or spinal injuries (neurogenic shock).

Patients in shock often receive fluid resuscitation and inotropic agents and/or vasopressors. Inotropes are used to increase a patient's cardiac output, therefore increasing MAP and allowing the organs to maintain perfusion. Vasopressors cause vasoconstriction to increase the patient's MAP (see Table 28.1).

Respiratory System

Critically ill patients require close monitoring of their respiratory system as many require oxygen and potential mechanical ventilation. In daily rounds, generally, the respiratory therapist provides the progress report of the patient's respiratory status. Respiratory status includes respiratory rate, respiratory sounds, ventilation parameters (if applicable), quality, and quantity of respiratory secretions and arterial blood gases. These are also interpreted in concert with other laboratory values pertaining to acid-base balance (later discussed in this chapter), chest X-ray (CXR), and sputum cultures. Respiratory tract infections are also of concern in patients in the ICU and can be detected through the presence of respiratory distress (abnormal ABGs), hypotension, mental status changes, difficulty weaning from mechanical ventilation, and CXR. Respiratory secretions are often purulent, and a sputum sample should be cultured to determine the responsible pathogen.

Arterial Blood Gases

Laboratory tests that are used for monitoring of respiratory function include measuring arterial blood gases (ABGs). An ABG test measures:

- *Partial pressure of oxygen (PaO_2)*, which is a measure of the pressure of oxygen dissolved in the blood and how well the oxygen can move from the lungs into the blood. A low PaO_2 indicates the patient is not oxygenating properly and is hypoxemic.
- *Partial pressure of carbon dioxide ($PaCO_2$)*, which is a measure of the pressure of carbon dioxide dissolved in the blood and how well the carbon dioxide is moving out of the body.
- *pH* measures hydrogen ions (H^+) in a patient's blood. A low pH is an indicator of acidemia and a high pH is an indicator of alkalemia.
- *Bicarbonate (HCO_3^-)* is used as a buffer to counteract pH changes if the blood is too acidic or basic. A low HCO_3^- level indicates metabolic acidosis, and a high HCO_3^- level indicates metabolic alkalosis.
- *Oxygen saturation (SaO_2)* measures how much hemoglobin is carrying oxygen. SaO_2 can also be calculated with pulse oximetry.

Fraction of Inspired Oxygen (FiO_2)

FiO_2 is the fraction of oxygen in the volume of air inspired and is a measure of the percentage of oxygen participating in gas exchange in the lungs. FiO_2 is used in the calculation PaO_2/FiO_2 ratio and is used as a measure of oxygenation. PaO_2 should increase with increasing FiO_2 , and a decrease in the ratio indicates inadequate or decreased oxygen exchange

or hypoxemia. The ratio can be used to determine the amount of oxygen that will be delivered to the patient in the form of oxygen-enriched air. A normal $\text{PaO}_2/\text{FiO}_2$ is >400 mmHg, a $\text{PaO}_2/\text{FiO}_2$ ratio ≤ 300 mmHg indicates acute lung injury, and a $\text{PaO}_2/\text{FiO}_2 \leq 200$ mmHg is indicative of acute respiratory distress syndrome (ARDS) [8]. ARDS is a respiratory failure with rapid onset of lung inflammation and the inability of the lungs to effectively provide oxygen to the rest of the body or expel carbon dioxide out of the lungs [8]. Most patients with ARDS will require mechanical ventilation. Risk factors include pneumonia, sepsis, and severe trauma. CXR is also used to detect the presence of ARDS or monitor its progression.

Mechanical Ventilation

Mechanical ventilation is common in the critically ill patient population. Although pharmacists are not involved in determining the ventilation parameters, patients' current parameters can inform the pharmacist if the patient is doing better or worse. There are three main stages of mechanical ventilation. The first is the trigger, which is the signal that initiates the opening of the respiratory valve. The next is the limit which regulates gas flow into the lungs, and finally, cycling, which stops the inspiratory phase and leads to the opening of the expiratory valve. There are also three methods of cycling from the inspiratory to expiratory phase:

- Volume cycled—delivers a preset volume of gas into patients' lungs as the inspiratory phase begins, and once this volume is reached, the patient passively exhales. This is the most common type of ventilation cycling.
- Pressure cycled—gas flows into a patient's lungs when the inspiratory phase begins until a pressure-sensing mechanism reaches a preset level and the patient passively exhales.
- Time cycled—gas flows into a patient's lungs when the inspiratory phase begins until a timing mechanism reaches a preset duration and the patient passively exhales.

There are also different modes of mechanical ventilation and include control modes or support modes. In pressure control ventilation, the respiratory rate is set, and the gas is delivered nonspontaneously. This method is used primarily in patients that are not conscious or are under more sedation. Pressure support allows the patient to trigger the ventilation with a preset pressure assistance delivered with each breath to decrease the work of breathing. In this mode, the patient controls the respiratory rate, duration of inspiration, and tidal volume.

Pharmacists can monitor the status of patients with the use of ventilation parameters. There are four main parameters that the pharmacist can look at:

- Tidal volume: The volume of air inhaled or passively exhaled in a normal respiratory cycle. The tidal volume that is set is often dependent on patient characteristics such as lean body weight and their disease.
- Respiratory rate: The number of preset tidal volume breaths of the patient will be received from the ventilator. If the patient is on support ventilation, the patient can also spontaneously breathe above the set rate.
- FiO_2 : See above. An increase in this value signifies deterioration.
- Positive end expiratory pressure (PEEP): Maintains pressure in the lungs during exhalation phase and improves oxygenation by enhancing alveolar volume and increasing oxygen exchange through an increase in surface area.

Gastrointestinal System (GI)

Daily gastrointestinal (GI) assessment is essential for proper critical care assessment as it might influence the choice and dosage of drug regimens. The feeding status of every patient should be determined. While some patients might be awake and able to swallow and on diet as tolerated (DAT), others are getting nutrition through feeding tubes, e.g., nasogastric (NG) or orogastric (OG). See section "[Routes of administration](#)" for further details on how this might affect pharmacist's decisions on drug therapies. Altered drug absorption might also be present in critically ill patients secondary to multiple factors such as delayed gastric emptying, binding to feeds and vasopressor use. This should be taken into consideration when assessing the efficacy of orally administered medications. Other GI assessment considerations include the need for stress ulcer prophylaxis and bowel routine; tube feed tolerance; presence of GI symptoms such as nausea, vomiting, diarrhea, or constipation; and assessment of bowel sounds.

Bowel sounds are caused by the movement of the intestines as food is passed through the gastrointestinal tract and are detected with the use of a stethoscope. Reduced bowel sounds indicated that there is decreased intestinal activity, and this can indicate constipation. The absence of bowel sounds can indicate an ileus, which is a condition where there is a lack of intestinal movement, and this can lead to complications such as intestinal obstruction. There are times of the day when reduced bowel sounds are normal, such as during sleep. Hyperactive bowel sounds or an increase in intestinal activity can occur with diarrhea or directly after eating. The assessment of bowel sounds should also be accompanied by assessing patients' bowel movements.

Genitourinary System

Assessment of patients' renal function is essential for pharmacists as it helps guiding dosage adjustment of renally eliminated medications. In addition, it affects therapy deci-

sions in terms of the choice of non-nephrotoxic medications in patients with renal impairment. Renal function can be assessed through laboratory tests, e.g., serum creatinine and blood urea nitrogen, in conjunction with the patient's urine output. Normal urine output for an adult is 0.5–1 ml/kg/h. Critically ill patients are at an increased risk for the development of AKI, and the diagnosis of AKI relies on a decreased urine output and increased SCr and BUN. If AKI occurs, the pharmacist should assess a patient's drugs and discontinue nephrotoxic drugs, if possible. See Chap. 26 for further information regarding renal function assessment.

Fluid balance is also an important factor to monitor in critically ill patients. Hypovolemia can lead to low organ perfusion and decreased oxygenation to the tissues, and hypervolemia can lead to organ damage and conditions such as pulmonary edema. Maintaining a euvolemic fluid balance is required to prevent complications. Urine output can be used as an indicator of abnormal fluid balance either by reflecting polyuria or oliguria. Correcting an abnormal balance may include fluid administration in hypovolemia or the use of diuretics in hypervolemia. Abnormal fluid balance can also cause electrolyte disturbances, which can be dangerous and require correction. It can also cause acid-base imbalances, and this can be monitored through ABG tests. Refer to the section “[Acid-base balance](#)” for more information.

Patients in the ICU are also at an increased risk for urinary tract infections (UTIs) secondary to multiple risk factors such as urinary catheterization and immunosuppression and should be monitored. Some signs and symptoms of a UTI can include pyrexia, mental confusion or altered mental status, and dysuria. The urine should be cultured, and antimicrobial therapy might be required.

Microbiology

Sepsis is one of the main reasons for ICU admission [9]. In addition, critically ill patients are also at an increased risk for the development of nosocomial infections due to being immunocompromised or because of the multiple points of entry for bacteria. ICU patients often have catheters, IV lines, and could be on mechanical ventilation. Examples of those infections include VAP and catheter-associated urinary tract infections. Pharmacists have an important role in assuring that critically ill patients are prescribed appropriate antimicrobials with an appropriate dosage regimen throughout patients' ICU stay. Initiation of effective antimicrobials within the first hour after development of hypotension has been associated with increased survival in patients with septic shock [10]. Due to the severity of infections in critically ill patients and the need for quick initiation of antimicrobials, it is often not possible to wait for the results of cultures and empiric therapy is required. Once the initial culture is

back, pharmacists should step down the empiric therapy to the appropriate targeted antibiotic therapy to prevent unnecessary drug use. Continued empiric therapy puts the patient at an increased risk for adverse effects, drug interactions, and the development of antibiotic resistance. Pharmacists should keep in mind that in the ICU there can be changes in susceptibilities and resistance patterns and should always consult the site-specific antibiogram. Pharmacists should continue to follow-up with cultures and sensitivities daily and order cultures when appropriate. They should also be aware of the normal flora in culture media that are less likely to be pathogenic to avoid unnecessary use of antimicrobials. For example, the presence of *Candida* in sputum cultures generally does not need to be treated. TDM can also be required for some antimicrobials, for example vancomycin and aminoglycosides. Pharmacists have a role to identify which patients should receive TDM, monitor the drug levels throughout, and make appropriate dose adjustments based on the results. For more information on infectious diseases assessment, the reader can refer to Chap. 27 “[Infectious Disease Assessment](#).”

Labs

Critically ill patients are often hemodynamically unstable, mechanically ventilated, on multiple pharmacological and nonpharmacological interventions that might result in rapid changes in their status and organ function throughout their ICU stay. As a result, ICU patients have daily lab tests done or even multiple times throughout the day. Pharmacists should regularly check patients' lab values to monitor response to therapies, adverse drug reactions, and organ functions and to assess the need for further management. For example, a patient's renal function can change rapidly due to an AKI, and pharmacists must be aware of the patient's current kidney function so that they can dose the patient's medications appropriately. Pharmacists can also order or suggest additional lab tests to ensure that they get the information required to complete an informed assessment or to monitor the patient. Table 28.5 provides some of the lab tests that may be ordered in the ICU for monitoring critically ill patients. Abnormal values may indicate an underlying pathology and can lead to serious complications, and should always be investigated.

Other Investigations

In addition to patient history, the review of systems, lab and microbiology, other investigations such as diagnostic imaging and other exploratory tests could be done. Pharmacists are not required to have the skill to read the output of these tests (e.g., read a chest X-ray image); however, they need to

Table 28.5 Common lab tests that are ordered in the ICU

Category	Examples
Electrolytes	Sodium, potassium, chloride, calcium, magnesium, phosphate
Serum creatinine	
Blood urea nitrogen	
Complete blood count with differential	Hgb, RBC, HCT, MCV, MCHC, RDW WBC (total, neutrophils, lymphocytes, monocytes, eosinophils and basophils) Platelets
Arterial blood gases (ABG)	pH, PaCO ₂ , PaO ₂ , bicarbonate
Liver panel	ALT, AST, ALP Bilirubin Albumin
Coagulation	INR, aPTT
Glucose	
Other	Lactate, creatine kinase, troponin, C-reactive protein

aPTT activated partial thromboplastin time, *ICU* intensive care unit, *Hgb* hemoglobin, *RBC* red blood cell, *Hct* hematocrit, *MCV* mean corpuscular volume, *MCHC* mean corpuscular hemoglobin concentration, *PaCO₂* partial pressure of carbon dioxide, *PaO₂* partial pressure of oxygen, *RDW*, red blood cell distribution width, *WBC* white blood cell, *ALT* alanine transaminase, *AST* aspartate transaminase, *ALP* alkaline phosphatase, *INR* international normalized ratio

be able to interpret the findings of these tests (e.g., written report or interaction with another healthcare team member) in the context of patient progress and medication therapy monitoring. Examples of such investigations that are helpful in the initial and follow-up assessment of the critically ill include:

- **Electrocardiography (ECG):** An ECG records the electrical activity of the heart. It can detect arrhythmias, such as tachycardia or atrial fibrillation. In addition, the QTc interval could be recorded from the ECG output.
- **Chest X-ray (CXR):** CXR is generally ordered for intubated patients and also used to detect and monitor conditions such as pneumonia, pulmonary edema, and pneumothorax.
- **Computed tomography (CT):** CT scan can be used to image numerous parts of the body and can detect, for example, structural abnormalities or hemorrhages.
- **Echocardiogram (ECHO):** ECHO is often used to monitor heart function, such as calculating a patient's ejection fraction.

Knowledge of the results of these tests can help the pharmacist understand the patient's HPI, the course of the patient's illness, and inform drug-related decisions, recommendations, and monitoring.

Drug Interactions

Patients in the ICU are at an increased risk for drug interactions. Rates of drug-drug interactions have been reported to be more frequent in ICU patients compared to the general hospital population [1, 11]. The increased risk is attributed to polypharmacy and altered drugs' pharmacokinetics in the critically ill. Drug interactions are associated with longer ICU stays and can potentially have a negative impact on patients' outcomes. The pharmacist should always assess patients' medications for potential drug interactions in every practice setting, but extra diligence should be exhibited in critical care settings due to the increased risk.

Intravenous Compatibility

It is not uncommon for critically ill patients to have IV lines (peripheral and/or central). IV lines are utilized for the administration of fluids, drugs, and blood products. IV therapy offers a rapid method for delivering fluids to correct for dehydration and electrolyte imbalances. Drugs delivered through the intravenous route can have a more rapid therapeutic effect, have complete bioavailability, and provide an alternative method for delivering drugs when patients have no gastrointestinal access. There are occasions that multiple IV medications and fluids are, administered through a single IV line at the same time using a y-site. A y-site is a device that has two separate tubes that can both be used to deliver fluid through and they both connect to the primary IV line; therefore, there is an intermingling of fluids in the primary IV tubing. The pharmacist has a role in assuring that drugs and fluids that are administered simultaneously at a y-site are compatible. Incompatibility can also be dependent on the concentration of each drug, the diluent solution, and the material of the drug delivery device.

There are three types of incompatibilities: physical, chemical, and therapeutic. However, note that often physical or chemical incompatibilities can lead to therapeutic changes as well. Physical incompatibility is when two or more substances are combined, and a physical change occurs, such as a change in color and viscosity, or a precipitate is formed. Chemical incompatibility is a result of chemical interactions, such as decomposition or oxidation-reduction reactions, which lead to a change in the chemical properties of either substance. Therapeutic incompatibility refers to drug interactions. Implications of IV incompatibility include a potential for particulate emboli from precipitation or separation, patient harm due to toxicity, pH changes causing tissue irrita-

tion, or therapeutic failure. If there is an incompatibility, the pharmacist may recommend, for example, an alternative drug, a dosage change, a change in the solvent, or a separation of administration time. There are many resources that the pharmacist can use as a reference. References include Micromedex and UpToDate Lexidrug, which include Trissel's IV Compatibility [12]. More references on IV compatibility are available, and the pharmacist can choose a reference based on preference and availability.

Routes of Administration: Clinical Pearls

Critically ill patients may have dysphagia, have a feeding tube inserted, or lack gastrointestinal access. Pharmacists need to make sure that drugs are administered appropriately in ICU; the right drug in the right formulation and through the right route of administration.

- The most straightforward scenario is when the patient is able to receive oral medications. However, even if the patient can receive oral medications, it is important to look at the speech-language pathologist's report, if available, as there may be restrictions on receiving solids (e.g., tablets) vs. liquids.
- Some drugs have dosage regimen conversions among formulations (dose and/or frequency), and this should be accounted for. Examples of drugs that require dose conversions include carbamazepine and lithium. In addition, few drugs are not interchangeable between the tablet/capsule and liquid form such as posaconazole.
- It becomes more difficult if the patient has a feeding tube inserted as the patient is not able to swallow any medications. In this case, the first step is to assess if tablets can be crushed and capsules can be opened or if the drug is available in an alternative liquid formulation.
- Generally, sustained release (e.g., SR, ER, XL) and enteric coated tablets should not be crushed. If the XL drug also has an immediate release (IR) option available, the drug could be switched to IR and administered more frequently. Note that this is not an option for some drugs, and each drug needs to be assessed individually based on its properties. For example, patients on nifedipine XL cannot be switched to the immediate-release formulation due to the adverse effect profile of the latter. An alternative antihypertensive agent should be considered, e.g., amlodipine.
- The capsule may contain powder if it is an immediate-release formulation or beads if it is an extended-release formulation. It is important to note that the administration of the suspended contents of the capsule through the feeding tube might not be possible in all scenarios, and pharmacist should consult the pertinent product monograph for drug-specific details. For example, administering dabigatran without the capsule shell can increase the oral bioavailability from the normal range of 3–7% up to 75% [13]. This increase in bioavailability can put the patient at risk of adverse effects and a toxic concentration of the drug in the body; therefore, dabigatran should only be administered to patients who can take the capsule orally.
- Administration of the sustained-release beads of some drugs through the feeding tube could result in tube clogging as in the cases of venlafaxine and duloxetine.
- Drugs available in film-coated formulations are generally recommended not be crushed by the manufacturer. There are multiple reasons for film coating such as masking bitter taste, light and humidity protection, and ease for swallowing. Some of these reasons might not be relevant in tube-fed patients, and crushing film-coated tablets is possible. For example, a patient with a feeding tube will not taste the drug and bitter-tasting film-coated tablets could be crushed.
- Another option for patients with feeding tubes is to switch them to the liquid, injectable, transdermal, or rectal forms of the drug, if available, and account for any dose conversions. If switching to a liquid format, elixirs and suspensions are preferred over syrups as syrups can cause clumping in the tube. An example of a drug that cannot be given in its liquid form to a patient with a feeding tube is ciprofloxacin liquid as it will adhere to the tube and block it.
- Drugs administered through enteral feeding tubes might bind to feeds, and this might result in reduced bioavailability. Holding tube feeds before and after dosage administration is recommended. Phenytoin is an example of drug affected by this type of interaction.

There are references available that can be used as a resource to determine if a drug can be crushed, opened, etc. A reference for the use of drugs in patients with a feeding tube is: *Handbook of Drug Administration Via Enteral Feeding Tubes* [14]. More references on routes of administration are available and the pharmacist can choose a reference based on preference and availability.

Pharmacotherapy Variability in Critical Illness

Pharmacotherapy variability is defined as “the variability in drug response among and within individuals that is attributed to the inter and intra-individual differences in the action and disposition of drugs” [15]. Pharmacotherapy variability in critical illness can negatively affect patient outcomes, so

clinicians must monitor for these changes. Such variability results from altered drug pharmacokinetics and pharmacodynamics in critically ill patients.

Pharmacokinetics of drugs are often based on studies done on healthy volunteers or on population parameters. When a patient has an acute critical illness, multiple physiological changes take place due to the illness itself, and multiple pharmacological and other supportive therapies are required to treat the patient. Those physiological changes make this population different from healthy volunteers, and the pharmacokinetics of drugs can be altered. Altered pharmacokinetics can lead to therapy failure, an increase in adverse drug reactions, and longer hospital stays, leading to increased overall healthcare costs.

Common manifestations of critical illness which has an impact on drug pharmacokinetics include systemic inflammatory response, alteration in body fluid volume, hemodynamic instability, plasma pH changes, organ dysfunction, and shock-related redirection of the blood flow to the core organs. In critically ill patients, the bioavailability of drugs from peripheral routes can be altered. Redirection of blood flow to core organs leads to reduced absorption from subcutaneous, intramuscular, and transdermal routes. Enteral drug bioavailability is also altered due to a reduction in mesenteric perfusion, delayed gastric emptying, changes in gastric pH, and alterations in GI transporters. Drug distribution is also altered due to increased fluid volume, inflammation-related alteration in plasma proteins, and tissue perfusion. Furthermore, drug metabolism might also be affected secondary to changes in blood flow to the liver and the alteration in liver enzymes. Lastly, excretion of drugs is also affected based on preexisting renal dysfunction, altered blood flow to the kidneys, alteration in urinary pH, augmented renal clearance (ARC), and changes in drug transporters. Also, it is not uncommon for ICU patients to be on dialysis, further complicating drug pharmacokinetics (discussed below). Pharmacists need to be aware of the potential changes in pharmacokinetic properties that can be encountered in critically ill patients and ensure that the drugs will still be effective and safe in these patients.

In addition to pharmacokinetic changes, altered pharmacodynamics (drug action) are pronounced due to severe physiological disturbances in critical illness. Altered receptor expression, signal transduction, and end-organ responsiveness can significantly modify drug effects. Figure 28.1 depicts *Hanafy's Neuro-CPK Pharmacovariability Wheel*, which outlines the factors affecting pharmacotherapy variability in neurocritical care. These factors are also relevant to pharmacotherapy in critical illness more broadly [15].

Renal Replacement Therapy

AKI is more often observed in critically ill patients than in the general hospital population and is a predictor of mortality [16]. Causes of AKI in the ICU population can be attributed to multiple causes, such as the use of nephrotoxic medications, contrast media, and renal hypoperfusion secondary to shock, such as in sepsis or trauma.

Some patients with AKI will require renal replacement therapy (RRT) if they are at risk of or have developed volume overload or a significant solute imbalance, or toxicity. This is in addition to those with other indications for RRT and those with end-stage renal disease (ESRD) admitted to the ICU. The following are the common indications for RRT in the critically ill [17]:

- Diuretic-resistant pulmonary edema.
- Refractory hyperkalemia (>6 mmol/L).
- Refractory metabolic acidosis.
- Uremic complications—pericarditis, encephalopathy, bleeding.
- Rising plasma concentrations of urea or creatinine.
- Oliguria (<0.5 mL/kg/h).
- Dialyzable intoxications.

Three different types of RRT are used in the ICU: peritoneal dialysis (PD), intermittent hemodialysis (IHD), and continuous renal replacement therapy (CRRT) [17]. This is in addition to hybrid modes of RRT such as sustained low-efficiency dialysis (SLED).

Peritoneal dialysis (PD) uses a dialysate to absorb waste and fluid with use of the peritoneum as a natural filter. This type of dialysis is not commonly used in the ICU and generally provides minimal drug removal.

Hemodialysis uses a dialysis machine with artificial membrane filters that utilize a concentration gradient between patient's blood and the dialysate to remove waste and fluid. IHD is provided in short intervals, usually every day or every 2–3 days and can result in rapid clearance of drugs.

CRRT is a slow and continuous RRT modality that is used for critically ill patients who require RRT and are hemodynamically unstable. Depending on the mechanism of solute removal, CRRT is divided into three main types: continuous venovenous hemofiltration (CVVH), continuous venovenous hemodialysis (CVVHD) and continuous venovenous hemodiafiltration (CVVHDF).

CRRT removes wastes at a much slower and steadier rate than IHD (Fig. 28.2).

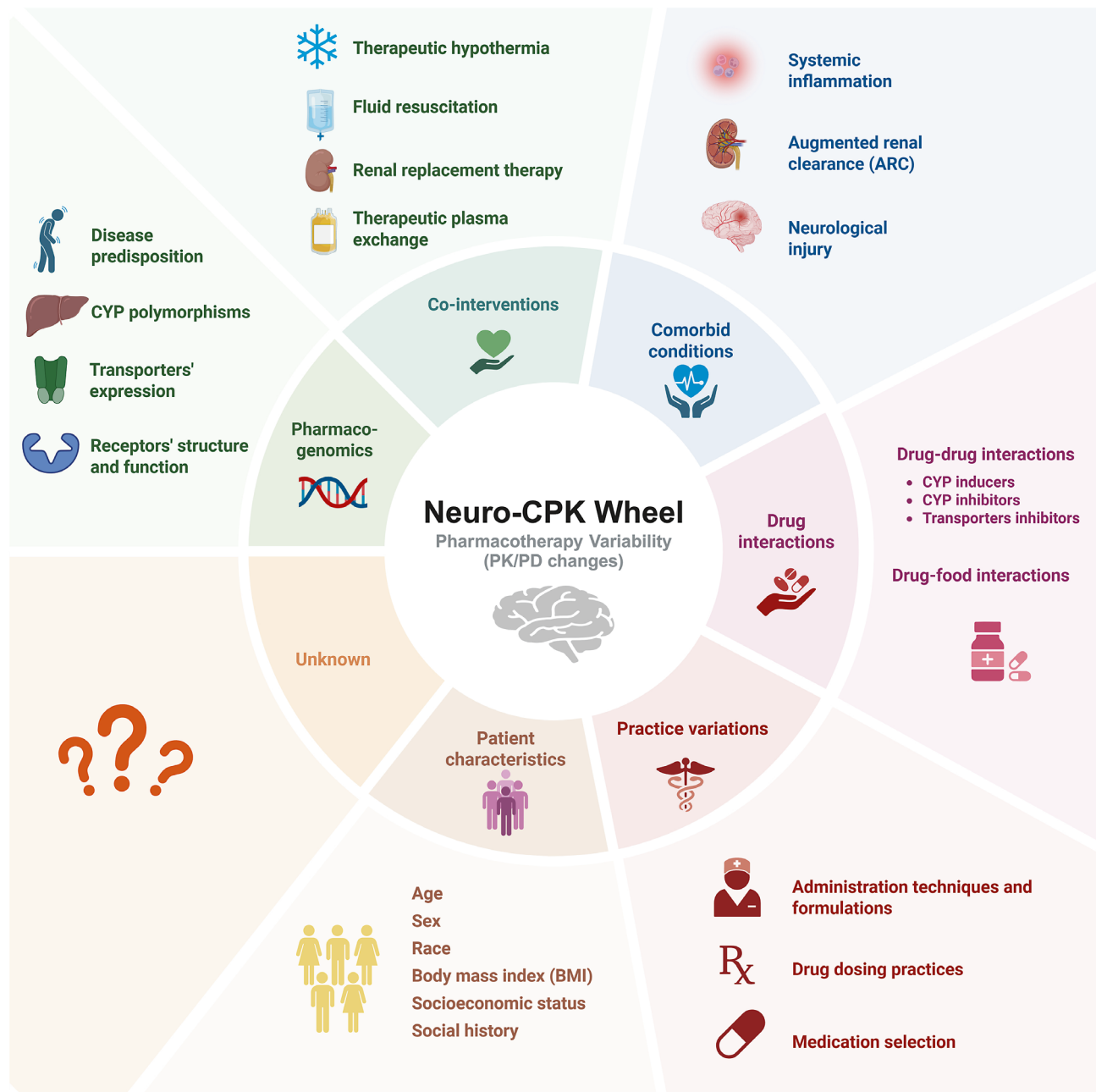


Fig. 28.1 Hanafy's Neuro-CPK pharmaco-variability Wheel illustrates the factors influencing the pharmacokinetics (PK) and pharmacodynamics (PD) of drugs in neurocritical care. Pharmacogenomics contributes to inter-individual variability in drug responses, affecting PK through mechanisms such as cytochrome P450 (CYP) polymorphisms and altered transporter expression, and PD through drug tolerance and structural/functional changes at the drug-target level. Co-interventions in neurocritical care, including renal replacement therapies, therapeutic hypothermia, fluid resuscitation, and therapeutic plasma exchange, can result in pharmacotherapy variability. Comorbid conditions like systemic inflammation, ARC, and neurological injury impact the body's drug response, leading to erratic and unpredictable drug levels. Practice variations between institutions and health care teams are critical factors often underemphasized. Different formula-

tions and administration techniques result in variable drug plasma levels. Drug dosing practices may also vary. Concomitant medication administration is prevalent among ICU patients, resulting in drug-drug interactions, with many drugs classified as CYP inducers or inhibitors. Drug-food interactions may also occur through adsorption, chelation, or complexation. Patient characteristics, including age, sex, race, and socioeconomic status, are important factors affecting drug concentrations in the body. Despite efforts to identify factors causing drug variability, gaps remain that are currently unexplainable. *Neuro-CPK* neurotherapeutics and Clinical Pharmacokinetic laboratory, *PK* pharmacokinetics, *PD* pharmacodynamics, *CYP* Cytochrome P450 enzymes. (Reprinted from Mahmoud SH et al., *Frontiers in Neurology* [15] under the terms of the Creative Commons Attribution License (CC BY))

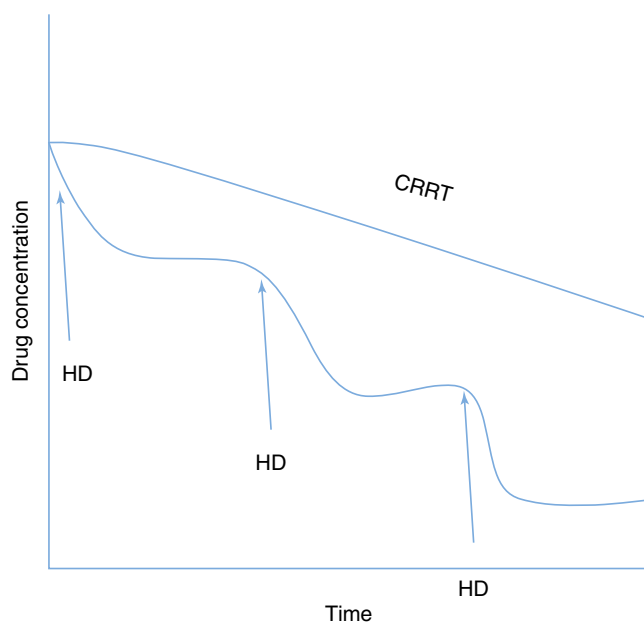


Fig. 28.2 CRRT results in continuous slow removal of solutes. IHD results in rapid and efficient removal of solutes in a short period of time and the drug clearance in-between runs depends on the patient's residual clearance. *CRRT* continuous renal replacement therapy, *HD* intermittent hemodialysis

Drug removal by RRT depends on the drug characteristics and the RRT modality. Drug-specific characteristics include molecular weight, protein binding, volume of distribution, and route of drug elimination [18, 19]. Drugs with a higher molecular weight and drugs that are highly protein bound are less likely to be removed. Drugs with a smaller volume of distribution will have greater dialyzability because they have a higher concentration in the plasma compared to drugs with high volumes of distribution. The volume of distribution of a drug has less impact on drug removal with CRRT compared to IHD because of the relatively slow and continuous process of CRRT, allowing drugs to equilibrate continuously with the vascular compartments. Pharmacists should also consider certain RRT factors, such as the dialysis membrane composition and pore size, the dialysate flow rate, and blood flow rates. This is in addition to the efficiency of the RRT mode on solute removal.

There are several drug dosing strategies in patients on RRT. The following are examples:

- In patients on PD, adjust dosing based on the patient's residual renal function.
- In patients on IHD, give the dose or a supplemental dose of the drug after the HD run.

- In patients on CRRT, it depends on the patient's residual renal clearance and nonrenal clearance. This is in addition to the factors described above. For example, if a drug is 100% renally eliminated and not protein bound, dose it similar to patients with creatinine clearance of 20–30 mL/min (CRRT clearance) assuming no residual renal function. For drugs that are highly protein bound and mainly hepatically metabolized, no dosage adjustment is needed assuming no protein binding alterations.

There are references available to help pharmacists with drug dosage adjustments in patients with decreased renal function or on dialysis. Some resources available include *Seyffart's Directory of Drug Dosage in Kidney Disease* and *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults* written by George Aronoff.

Augmented Renal Clearance

Although more attention is given to adjusting the doses of renally eliminated drugs in patients with renal impairment, little attention is given to whether patients exhibit an augmented renal clearance (ARC). ARC is the phenomenon of enhanced renal function in critically ill patients. It is defined as a CrCl of greater than 130 mL/min/1.73m². This phenomenon tends to occur in the following populations: age <50 years old, patients with a history of recent trauma, male sex, and those with lower critical illness severity scores [20, 21]. ARC has a significant impact on the dosing of renally eliminated drugs such as β -lactam antimicrobials and vancomycin. ARC might result in subtherapeutic concentrations of these drugs with subsequent therapy failure and possible worsened patient outcomes. The pathophysiology of ARC is largely unknown, but it is thought to be closely related to the vigorous sympathetic response attendant with severe critical illness, increased levels of acute phase proteins, alterations in vascular tone, cardiac output, and major organs blood flow, resulting in a hyperdynamic state and augmented glomerular filtration rate. This is in addition to the effects of administration of the pharmacotherapeutic strategies aimed at improving or maintaining organ perfusion (e.g., fluid resuscitation, vasopressors) [20]. Pharmacists need to be aware of ARC and monitor patients' renal function throughout their ICU stay. If ARC is detected, pharmacists must assess patient's medications for the presence of renally eliminated drugs and assess the need for dosage adjustment and/or medication changes. Figure 28.3 provides a stepwise process for assessing an ICU patient exhibiting ARC.

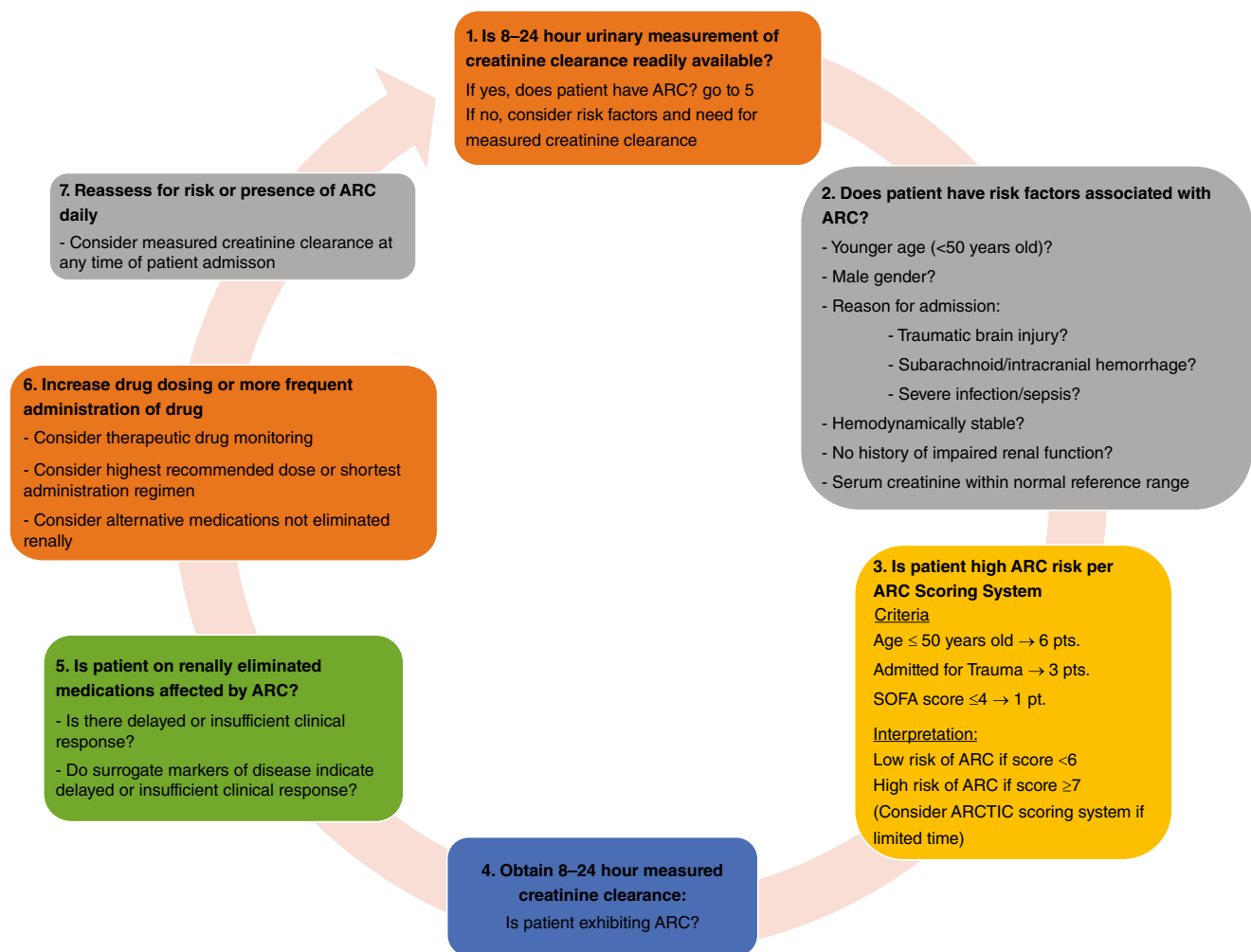


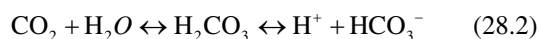
Fig. 28.3 Assessment of ICU patients for augmented renal clearance. (Reprinted from Mahmoud and Shen [20]). This is an open access article distributed under the Creative Commons Attribution License which permits unrestricted use, distribution, and reproduction in any medium,

provided the original work is properly cited (CC BY 4.0). *SOFA* sequential organ failure assessment, *ARCTIC* Augmented Renal Clearance in Trauma Intensive Care

Acid-Base Balance

Acid-base disorders are common complications in the ICU. Critical care pharmacists should be aware of these disorders and be able to assess ABG findings and identify potential drug-related causes.

Acid levels are determined by the concentration of H^+ ion in the body, and their concentration is tightly regulated within a normal pH range of 7.34–7.42 [22]. Acid in the body comes primarily from carbon dioxide (CO_2) as carbonic acid (H_2CO_3), which is balanced through the following equilibrium equation:



This equation can be used to help explain the few mechanisms the body uses to control the level of acid in the body.

There are three main mechanisms the body uses to maintain a normal body pH:

- The respiratory system controls the $PaCO_2$ and this regulates alveolar ventilation. The higher the level of H^+ , the more CO_2 is expired from the lungs, which decreases the concentration of acid in the body. This is a fast-acting mechanism of control.
- The renal system controls the level of HCO_3^- in the body and also can excrete other metabolic acids that the body produces, such as lactic acid, fatty acids, and ketone bodies. This is a slow mechanism of control and can take hours to days.
- Acute changes in pH can also be normalized through an acid buffer system with the use of HCO_3^- , sulfate (SO_4^{2-}), and hemoglobin.

In the healthy population, these mechanisms can control the pH and maintain it within the normal range. However, in patients with organ dysfunction and fluid imbalance, these mechanisms can fail or not be sufficient to correct for changes in balance. These patients can develop *acidemia*, blood pH is <7.34 , or *alkalemia*, a blood pH is >7.42 [23]. The development of acidemia occurs through the process known as *acidosis* and alkalemia through *alkalosis*. When conducting an acid-base assessment, the first step is to determine the pH of the blood to detect the presence of acidemia or alkalemia. Next, the concentration of HCO_3^- and the pCO_2 should be measured, as this can help determine if the cause of the abnormal pH is due to respiratory or metabolic processes.

There are four possible clinical manifestations of an acid-base disorder, and they include respiratory acidosis, respiratory alkalosis, metabolic acidosis, and metabolic alkalosis.

Respiratory acidosis occurs when there is an increase in the partial pressure of carbon dioxide (pCO_2) above the normal range and is defined as a pH <7.4 and pCO_2 of >40 mmHg [23]. The increase in pCO_2 is a sign that there is insufficient alveolar ventilation, and this can be caused by drug intoxication, CNS dysfunction, or respiratory obstruction or failure, such as with pneumonia, pulmonary edema, or chronic obstructive pulmonary disease (COPD). The signs and symptoms include headache, anxiety, tachycardia, hypotension, and arrhythmias and, if severe, can lead to coma. Acutely, the physiological compensation for respiratory acidosis is the use of the buffer system. The slower-acting renal system is the main compensatory mechanism for respiratory acidosis and increases the pH by excreting H^+ and retaining HCO_3^- , which can cause metabolic alkalosis. Treatment of respiratory acidosis includes identifying and treating the cause and continued monitoring of ABGs and respiratory rate.

Respiratory alkalosis occurs when there is a decrease in the pCO_2 below the normal value and is defined as a pH >7.4 and a $\text{pCO}_2 < 40$ mmHg [23]. The main cause of respiratory acidosis is hyperventilation; therefore, decreasing the pCO_2 . Causes of hyperventilation can include hypoxia or respiratory pathology, and examining the PaO_2 can help determine the underlying cause. Other causes of hyperventilation are CNS dysfunction, infection, drug-induced (e.g., nicotine), anxiety, pain, or can be due to overventilation that can occur in the ICU. Signs and symptoms are usually mild, such as dizziness, paresthesia, and confusion, but can become severe and lead to complications such as seizures. The body uses the buffer system to help compensate for alkalosis, and the renal system will retain H^+ and excrete HCO_3^- to lower the pH, which can cause metabolic acidosis. The treatment is similar to acidosis, with trying to identify and treat the cause and continuing to monitor ABGs and respiratory rate.

Metabolic acidosis occurs when there is a decrease in HCO_3^- and is defined as a pH <7.4 and $\text{HCO}_3^- < 22$ mmol/L

[23]. Signs and symptoms of metabolic acidosis include hyperventilation, hyperkalemia, insulin resistance, nausea and vomiting, and CNS effects such as confusion and coma. The respiratory system compensates for the low pH by decreasing the pCO_2 in the body through hyperventilation, and this mechanism can lead to respiratory alkalosis. The cause of metabolic acidosis can be determined by calculating the anion gap (AG), which is the difference in unmeasured anions and cations. The AG is calculated with the following equation and has a normal range of 8–14 mmol/L.

$$\text{AG} = [\text{Na}^+] - [\text{Cl}^-] - [\text{HCO}_3^-] \quad (28.3)$$

If the AG is normal, the cause of the acidosis is not due to an unmeasured cation or anion, and the primary cause is a loss of HCO_3^- . Causes of normal AG acidosis can be remembered with the acronym HARDUP, which stands for hyperalimentation, acetazolamide, renal tubular acidosis, diarrhea/dilution, ureteral diversion, and pancreatic fistula [24]. If the AG is >14 mmol/L, it is due to an excess of unmeasured cations or anions, and causes can be remembered with the acronym MUDPILES [24]. MUDPILES stands for methanol/metformin, uremia, diabetic ketoacidosis, propofol/paraldehyde, iron/isoniazid, lactate/linezolid, ethylene glycol, and salicylates/starvation [24]. Finally, a decreased AG, which is <8 mmol/L is uncommon and is usually associated with hypoalbuminemia [25]. Treatment of metabolic acidosis includes identifying the cause and treating it and potentially the administration of sodium bicarbonate.

Metabolic alkalosis occurs when there is an increase in HCO_3^- and is defined by a pH of >7.4 and $\text{HCO}_3^- > 28$ mmol/L [23]. Signs and symptoms include hypoventilation, hypokalemia, confusion, muscle weakness, tachycardia, and arrhythmias. The respiratory system compensates for the high pH by increasing the pCO_2 in the body through hypoventilation, and this mechanism can lead to respiratory acidosis. Causes include a loss of H^+ (e.g., from vomiting or diuretics), intake of excess alkali substances (e.g., excessive antacid use), or decreased renal function. Treatment includes identifying the cause and treating and correcting any fluid and electrolyte imbalances.

The changes that occur with ABGs in each acid-base disorder are presented in Table 28.6. Maintaining an acid-base balance to compensate for any abnormalities, and often the compensation for one disorder can lead to the development of another. For example, compensation for respiratory acidosis can lead to metabolic alkalosis. Due to the fast-acting mechanism of the respiratory system, respiratory acidosis or alkalosis reflects a current body dysfunction, while the slow-acting metabolic system could be caused by something that happened hours to days ago. The main treatment for these disorders is treating the underlying cause of the imbalance, and with treatment, the imbalance should correct itself.

Table 28.6 Expected changes in primary acid-base abnormalities^a

Disorder	pH	HCO ₃ ⁻	pCO ₂
Metabolic acidosis	↓	↓	↓
Metabolic alkalosis	↑	↑	↑
Acute respiratory acidosis	↓↓	↔	↑↑
Chronic respiratory acidosis	↓	↑↑	↑
Acute respiratory alkalosis	↑↑	↔	↓↓
Chronic respiratory alkalosis	↑	↓↓	↓

HCO₃⁻ bicarbonate, pCO₂ partial pressure of carbon dioxide

^aAdapted from Hubble [22], Copyright 2004, with permission from Elsevier

Conclusion

Pharmacists provide an essential clinical role in an interprofessional team and contribute to patient care, safety, and positive patient and clinical outcomes. Pharmacists working in critical care are becoming the standard of care and present an opportunity for pharmacists to work in a specialized practice. Critically ill patients require daily assessments and assessments on unique topics due to the complicated nature of their illnesses. The pharmacist needs to have the knowledge and the skills to provide care for these patients. Working in critical care can be challenging for a practitioner, but it can also be a practice site that provides learning opportunities and rewarding experiences.

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Assessment Considerations in Older Adults

29

Cheryl A. Sadowski

Chapter Objectives

1. Describe the process of comprehensive geriatric assessment.
2. Identify the domains of comprehensive geriatric assessment.
3. Describe practical tools that pharmacists can use in their assessment of older adults.

Background

Assessment of older adults, particularly those who are multimorbid, complex, or frail, requires a broader interprofessional assessment. This approach is referred to as comprehensive geriatric assessment and includes health and non-health domains. The assessment is completed with an interprofessional team, and the process focuses on functional abilities and values and goals of the patient, rather than cure or simply chronic disease management. The assessment has many domains, including cognition, function, activities of daily living, nutrition, social supports, and others. Pharmacists have an important role to play in contributing to the geriatric assessment through feedback on a variety of domains, and particularly through assessment of medications.

Definitions

Geriatrics is the health sciences specialty of caring for older adults. This specialty is sometimes referred to as a supraspecialty, versus a subspecialty, because it addresses not only the management of health conditions under internal medicine but also includes the overall health and wellness of older

adults and addresses health issues related to aging [1, 2]. The broader study of aging and older adults, using bio-psychosocial sciences, is the study of gerontology [3].

The current accepted terminology is “older adults” or “older people,” rather than terms that infer that these individuals have less value or are “senile” in old age [4]. Language is evolving in geriatrics to move away from terms that support negative stereotypes, such as the term “elderly”. In some clinical and research settings, terms such as young-old, middle-old, or old-old will be used to categorize the chronological age of this older age group, but this terminology is shifting to identify specific age ranges (e.g., age 65–74, 75–84, and 85 and older). The term “centenarian” refers to someone who has reached 100 years of age and “supercentenarians” as those reaching 110 years. These definitions are not just used to address demographic analyses through the national census but to consider the unique needs of specific age groups, to direct healthcare services, to design and analyze research meaningfully, and to communicate correctly to and about the patients one cares for.

Demographics

The 2016 census in Canada showed that for the first time in Canadian history, seniors outnumbered children aged 14 and under. In 2023, 18.9% of the population was 65y and older [5]. This increase in the older adult demographic is due to two factors: improved life expectancy and low fertility rates. Aging and an increasing number of older adults are a global phenomenon, with the age group of 60 years and older outnumbering those 5 years and younger since 2020 [6]. Aging is associated with increased health system use, and there have been calls for increased capacity for healthcare professionals and the system to be better prepared to care for this population [7].

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Epidemiology and Pharmacoepidemiology

Seniors use more medications than any other age group, with 51.2% of senior claimants in Canada using 1–4 different drug classes chronically, while 36.4% of seniors use five or more drug classes chronically. The most common medication class used by seniors in Canada is the HMG-CoA reductase inhibitors (statins) [8]. In addition to medications, over 60% of seniors use dietary supplements (e.g., vitamins) [9].

Seniors also use the healthcare system more than other age groups, with 37% of hospitalization episodes in Canada in those 65y and older. Seniors also have longer stays in hospital, on average 10.4 days versus 4.7 days for those less than 65y of age [10]. Seniors represent less 20% of the population but cost approximately half of health spending in Canada [11].

Disease and disability also increase with age. The most common chronic disease in older adults is hypertension, diagnosed in 66% of seniors in Canada [12]. However, if multimorbidity were a single diagnosis, it would be the most common condition, as an estimated 75% of seniors have co-existing conditions, although this may be as high as 93% [13, 14]. Another unique aspect of geriatric care is the concept of the “geriatric syndrome,” which is a multifactorial condition involving age-related risks and stressors that together overwhelm the patient’s system [15]. Syndromes are challenging to address due to their multifactorial nature, but there are resources to assess and manage common syndromes, such as falls or urinary incontinence.

Impact of Aging

Within the context of individualized patient care, it is important to consider the physiologic changes that occur with aging. These changes are part of normal aging and are not pathological (Table 29.1) [16]. These changes can subsequently affect pharmacokinetic parameters, with the least impact on drug absorption (Table 29.2) [17, 18]. Furthermore, changes in receptor density and affinity, post-receptor changes, or negative feedback can all lead to a change in how older adults respond to medications, which is described under “pharmacodynamics” [18–20]. In most cases, older adults are more sensitive to medications, and adverse events may occur (Table 29.3).

There are changes in how older adults present with illness. Often, seniors have vague complaints or do not follow the “textbook presentation” of a disease. This can be due to multimorbidity, with co-existing conditions affecting how another disease presents or progresses, or may be due to a

Table 29.1 Physiologic changes due to aging

Body system	Key changes
Body composition	↑ Fat ↓ Lean
Cardiovascular system	↓ Beta receptors ↓ Cardiac output
Pulmonary system	↓ Cilia ↓ Vital capacity ↓ Maximal breathing capacity ↑ Residual volume
Gastrointestinal system	↑ Gastric pH ↑ Gastric emptying time
Hepatic system	↓ Size ↓ Splanchnic blood flow
Renal system	↓ GFR ↓ Nephrons
Endocrine system	↓ Hormonal secretions
Nervous system	↓ Blood flow to CNS

CNS central nervous system, *GFR* glomerular filtration rate

Table 29.2 Changes in pharmacokinetic parameters due to aging

Parameter	Change
Absorption	Minimal change in GI absorption of passively absorbed medications; slight decrease in actively absorbed medications (e.g., B12, iron) IM absorption may be affected by blood flow (e.g., immobile/bedridden patients)
Distribution	Increased distribution of lipid-soluble medications, leading to longer half-life Decreased distribution of water-soluble medication, leading to higher maximum serum concentration C_{Max}
Metabolism	Slower first-pass effect Slower phase I metabolism (oxidation/reduction) Minimal changes in phase II metabolism (conjugation)
Excretion	Decreased renal elimination

GI gastrointestinal, *IM* intramuscular

physiologic decline with aging that affects the acuity and type of symptoms experienced [21–24].

The Imperative

There is a demographic imperative to address the care gap within the healthcare system by better preparing professionals to assess and manage care for older adults. Given the unique presentation of older adults, changes in pharmacokinetics and pharmacodynamics, and complexity in presentation, pharmacists require a deeper understanding of the interprofessional and comprehensive assessment processes required. The unique role of the pharmacist is essential in contributing to comprehensive geriatric assessment. The purpose of this chapter is to provide an introduction to the assessment of older adults for the pharmacist.

Table 29.3 Examples of pharmacodynamic alterations due to aging

System or function affected	Proposed mechanisms	Medication examples
Postural control	Fewer dopamine D2 receptors in striatum	Antipsychotics
Orthostasis	Blunting of β receptors response, receptor downregulation, changes in the vascular tree and autonomic nervous system	Blood pressure medications, TCAs, antipsychotics, diuretics
Thermoregulation	Poor temperature-regulating mechanisms (e.g., shivering, $\downarrow$ metabolic rate, $\downarrow$ vasoconstriction, $\downarrow$ thirst response, $\downarrow$ awareness of environmental or body changes in T, unable to tolerate extremes in temperature)	Medications affecting awareness, mobility, muscular activity, vasoconstrictor mechanisms CNS meds (e.g., phenothiazines, benzodiazepines, opioids, ethanol)
Higher cognitive function	Neuronal loss, receptor downregulation	Central anticholinergics, stimulants
Anticoagulation	Poor hepatic production of coagulation factors, dietary intake	Anticoagulants, thrombolytics
Tardive dyskinesia	Impaired or decreased dopamine-synthesizing neurons	Antipsychotics
Arrhythmias	Cardiac hypersensitivity	Antiarrhythmic medication

Comprehensive Geriatric Assessment

The use of comprehensive geriatric assessment (CGA) recognizes that a disease-based model for assessment and care does not work well for older adults who have multiple comorbidities in addition to complexities of aging, such as the lack of caregiver support, financial concerns, or difficulty navigating a complex healthcare system. CGA is an operational framework of integrated care and those with multimorbidity or frailty are best suited to CGA, versus implementing this process for all individuals age 65 years and older [25, 26]. The difference in CGA from regular patient assessment is that it includes non-health domains, focusing on functional capacity and quality of life, and includes an interprofessional team [27]. The core team is usually made up of a physician, nurse, and social worker, and additional assessment is provided by physiotherapists, occupational therapists, spiritual care professionals, psychologists, and, notably, pharmacists [28]. The process requires collaboration, but this may take place within the same setting, around the same time, or virtually and sequentially [28]. The role of the pharmacist primarily focuses on medication review, but an understanding of the other domains of assessment is essential to integrate appropriate interventions. Pharmacists may also have the training to become proficient

Table 29.4 Domains of comprehensive geriatric assessment

Domain	Example
Mental status	Psychosis, depression, anxiety
Cognitive status	Cognition, memory
Social status	Social supports Availability of spouse, family, friends
Values	Decisions regarding DNR, living will, power of attorney
Spirituality	Supports, beliefs Activity and socialization
Economic status	Ability to pay for care or treatments
Physical status	Review of disease states Review of medications
Functional status	Ability to do activities of daily living
Sensory impairment	Hearing, vision
Health maintenance/disease prevention	Primary, secondary, tertiary prevention Geriatric syndromes, falls, incontinence Nutrition Vaccinations
Risks	Abuse, fraud, self-neglect

DNR do not resuscitate

at other tests in other domains, which may augment the efficiency of the care team [29].

CGA is defined as “a multi-dimensional multi-disciplinary diagnostic process focused on assessing an older person’s medical, psychological, and functional capability to develop a coordinated and integrated plan for treatment and long-term follow-up focused on the individual’s needs” [25]. The initial assessment includes goal identification, a comprehensive care plan, who is responsible, and a timeline to review progress. The CGA process originated in hospital and demonstrated that older adults who underwent CGA, compared to usual hospital care, had less risk of being institutionalized, dying, or having a functional decline [30, 31]. There has been an expansion of CGA beyond the hospital setting, which can be conducted in clinics, in the patient’s home, long-term care, or community programs [32, 33]. Evidence is variable, but when CGA is focused on vulnerable older adults, there are some positive effects on frailty and mortality [33]. Currently, there is not a single, standardized set of tools, but the content below includes the most common resources for validated assessments under each domain. Table 29.4 depicts general CGA domains.

Interprofessional Assessment of Health

The pharmacist’s expertise in medications supports the primary role focusing on medication assessment. However, the pharmacist can participate in other components of CGA, learning to use the tools in a validated manner or in terms of interpreting the findings of the other team members in order to apply that learning to decisions about medications.

Physical Assessment

Pharmacists are familiar with vital signs and the routine measurement of pulse and blood pressure. The standard for a physical examination applies to older adults, but a few adjustments to focus on issues that are more prevalent in seniors [27]. One difference in measuring vital signs is to measure orthostatic blood pressure, which requires a sitting or supine blood pressure, followed by a repeated blood pressure measurement after the patient stands.

Prevention of disease should also be assessed, such as inquiring about vaccination status.

Mental Health

Cognitive impairment and neurocognitive disorders (dementia) occur almost exclusively in geriatrics. An assessment for cognitive functioning is a core component of CGA. The most common starting point is a screening test, such as the *Mini-Mental Status Exam* (MMSE), which provides some information about the patient's cognitive abilities but is not diagnostic for dementia [34]. Another tool that was first used to screen for mild cognitive impairment and is a more specific assessment of executive function is the *Montreal Cognitive Assessment* (MoCA) [35]. Both of these tools are relatively brief, about 5–10 min, have a series of healthcare professional scored questions, and result in a score out of 30 points. There are cut-points for normal versus abnormal, which can assist clinicians in identifying if further assessments are required. For example, a 26/30 or higher would be considered normal, but lower scores should be investigated further, for possible disease or a medication cause of impairment. There has been some controversy regarding copyright, cost, and access to some of these tests. Other open-access tools are sometimes used, such as the Saint Louis University Mental Status Exam (SLUMS), which is also a brief questionnaire assessing similar domains and is scored out of 30 points [36]. Given the diversity of the population, there are tools such as the Rowland Universal Dementia Assessment Scale (RUDAS) that was designed for culturally and linguistically diverse populations or those with lower levels of formal education [37]. The *clock drawing test* is a relatively quick test to administer, by observing the patient draw a clock, but it has a variety of proposed scoring methods [38]. The *2 trail making tests*, assess sequencing, prioritization, visuospatial skills, and handling multiple cognitive tasks. The first trail making test, includes sequential numbers or letters (A,B,C; 1,2,3), and the second includes alternating numbers and letters (A,1,B,2,C,3) [39]. In-depth cognitive assessments are conducted by occupational therapy or a neuropsychologist who can target specific domains with focused

validated tests [40]. The pharmacist can administer some of these tests if the appropriate training has been conducted, but should be cautioned that findings of low performance may be challenging to discuss with patients and also create a situation where further investigations and interventions should be addressed that may be beyond the scope of the practice of a pharmacist (e.g., driving ability, living independently). It is likely that another team member is administering these tests, and pharmacists should be reviewing these results as part of their assessment to determine if a patient may be experiencing cognitive impairment, which can then lead to assessment of possible medication adverse event (e.g., impaired cognition), and consideration of additional supports the patient may need in terms of drug information or medication management (e.g., home care, carer administering medications).

The assessment of mood is also part of a geriatric assessment. A screening tool that is commonly used is the *Geriatric Depression Scale Short Form* (GDS-SF) which has 15 questions, scored as yes/no by the patient [41, 42]. It is important to use a geriatric-focused questionnaire, as depression screening tools for younger patients often include questions about family demands or work, which may not be relevant for someone who is retired or who is not living with children. There is also a concern that patients with dementia will require an adapted tool that does not directly ask them questions that require abstract thinking. The *Cornell Scale* for Depression is such an instrument, which has the clinician interact with the patient and then the clinician fills out a scale regarding observations about the patient, such as behaviours or physical signs [43]. This type of observation takes more time than a straightforward questionnaire but is appropriate for a patient with dementia.

Function

A functional assessment is generally in the domain of rehabilitation therapists, but the pharmacist should be able to understand the impairment based on the assessments and scores. The two broad areas are the (basic) activities of daily living (ADL), or the self-care activities such as toileting, eating, and dressing, and the instrumental activities of daily living (IADL), which are activities needed to live independently, including taking medications, housework, and preparing meals. The best approach to assessing ADL and IADL is to observe the patient, which can be done in a simulated laboratory setting or in the patient's own home during a home visit. There are formal measures for ADL assessment, led by the *Katz Index of Independence in Activities of Daily Living*, which asks about or observes independence in six areas and scores the patient as independent (score of 1) or dependent (score of 0) [44]. The

IADL first-line tool is the *Lawton-Brody*, which also scores patients as 0 or 1 on their ability to perform eight different tasks [45]. There are a variety of other measures that a physiotherapist may use to assess function and disability, such as the *Functional Independence Measure* (FIM) which can also be combined with the *Functional Assessment Measure* (FAM), which scores the patient from 1 (total assistance) to 7 (fully independent) in six different domains, including cognition, communication, and sphincter control [46]. Pharmacists should particularly take note of functional measures, which can determine if the patient can independently take their medications or safely manage self-care activities.

Geriatric Syndromes

Geriatric syndromes are challenging to address because they are multifactorial and involve multiple organ systems [15]. Syndromes do not follow a disease trajectory, but they are associated with functional decline and disability [15].

Frailty

Frailty is a concept that is still evolving in terms of definition and criteria, but one method used in practice, research, and by organizations developing clinical practice guidelines is the Canadian Frailty Scale, a nine-item scale that provides assessment by image/picture, with stage 1 being fit and stage 9 being terminally ill [47]. Other methods of assessing frailty include counting the total number of deficits a person has [48], which can be done as an electronic frailty index as well. There are a variety of scales that include clinician and sometimes patient completion of some questions, such as the Edmonton Frail Scale, FRAIL, or Tilburg Frailty Indicator [49–51]. There is no universally used method of frailty assessment, but a tool that informs clinicians regarding functional decline can guide decision making regarding types of interventions and revision of goals of care.

Falls

Falls are also common in older adults, leading to fractures, soft tissue injuries, and decreased socialization through a fear of falling [52]. Inquiring about falls can be done by a variety of health professionals, including pharmacists who work in the community [53, 54]. The World Guidelines for Falls Prevention and Management for Older Adults recommends three key questions including the following: Have you fallen in the past year? Do you feel unsteady when standing or walking? Do you have worries about falling [55]? Falls are multifactorial and can include environmental, behavioral, medical, and medication risks. Determining if

Table 29.5 Medications associated with fall risk

Medication class	Example	Risk
Sedative hypnotics	Lorazepam, diazepam, zopiclone	High ^a OR = 1.3–1.6
Antidepressants (TCA, SSRI, other)	Amitriptyline, sertraline, citalopram	High ^a OR = 1.5–1.6
Antipsychotics	Quetiapine, haloperidol	High ^a OR = 1.3–1.7
Diuretics	Furosemide	Moderate OR = 1.1–1.4
Opioids	Morphine, hydromorphone	Low ^b OR = NS–1.3
NSAIDs	Ibuprofen, naproxen	Low ^b OR = NS

OR odds ratio, SSRI serotonin selective reuptake inhibitors, TCA tricyclic antidepressants

^aStudies consistently found to be statistically significant

^bStudies are inconsistent in statistical significance

the patient is at risk for medication-related falls is a priority for pharmacists. The medications with the strongest association for falls are psychotropics, including sedative hypnotics, antidepressants, and antipsychotics. These medications can increase the risk of falls by 50% from the baseline [56–58]. Other medications that can contribute to the fall risk include antiepileptic medications, opioids and diuretics, although the risk is less than psychotropics and may decrease over time (Table 29.5) [58–63]. Pharmacists can assess risk for falls through a medication review that is structured and personalized, and by prioritizing review and potential deprescribing of psychotropic medications, but also reviewing for undertreated conditions that can increase fall risk [64]. Tools to support a medication review include the STOPPFall to identify fall risk increasing drugs [65]. Pharmacists may conduct simple observations, or even validated measures, in the community pharmacy, but primarily they should be aware of how to interpret the results of basic measures. Some examples that may be used by a physiotherapist include the *Tinetti Balance and Gait Examination*, a series of assessments on gait and balance, scored from 0 to 2 [66]. A simple assessment is the *Timed Up and Go* (TUG), which requires the patient to be seated, rising from that position without using his/her arms, walking 3 m, turning around, and sitting again. The test is timed, and the patient should not take longer than 16 s [67]. Another simple test is the *Functional Reach Test*, where the patient stands close to a wall and reaches out his/her hands. The measurement should be at least 25 cm [68]. However, pharmacists should consider the layout and safety features of their pharmacy before administering this test, as a patient who is at risk of falls could fall and injure themselves during the activity. Concerns about geriatric patients who have gait or balance problems should be referred to a physiotherapist for further workup and in-depth assessments.

Lower Urinary Tract Symptoms

Older adults are more likely to experience lower urinary tract symptoms (LUTS) and urinary incontinence (UI). There is minimal research on the integration of continence care in community pharmacy; there is some evidence that this can be implemented from assessment through intervention and follow-up [69]. One tool to start the conversation and screen both males and females is the *International Consultation on Incontinence Questionnaire* (ICIQ), which includes four questions, starting with, “How often do you leak urine?” [70]. The *Bladder Self-Assessment Questionnaire* (B-SAQ) includes four questions similar to the ICIQ but also asks the patient to score how much of a bother those symptoms are [71]. There are also questionnaires specific to the type of LUTS, such as a *benign prostatic hyperplasia (BPH) questionnaire* for men, with seven questions on symptoms and four questions on bother, and an eight-question *overactive bladder questionnaire* (OAB-V8) for both men and women [72, 73]. After compiling the scores, the pharmacist can discuss concerns about LUTS or, in some cases, encourage the patient to see the family physician for further investigation and more invasive examinations (e.g., pelvic and/or rectal exam).

Nutrition

Assessing nutrition in older adults is important to prevent frailty and cachexia. In addition to changes with aging, such as reduced appetite or energy expenditure, or changes in taste and smell, there are often changes with dentition and socialization affecting eating patterns and functional abilities, and resources to purchase and prepare healthy meals [74]. Monitoring weight obtained by self-report should be verified by physically weighing the patient, but more complete anthropometric assessment (requiring the measurement of height), labwork (e.g., low albumin), or food diaries can be incorporated [74, 75]. Asking a question about their ability to swallow can inform about nutritional concerns, as well as the swallowing of medications, which can be done with the *EAT-10*, a ten-item questionnaire, scoring people from 0 (no problem) to 4 (severe problem) [76]. The *mini-nutrition assessment* (MNA) is commonly used as a screening tool in seniors, with scoring for food intake, weight loss, mobility, stress, neuropsychological problems, and body measurements, and has been shown to be superior in detecting malnutrition compared to other tools [77–79].

Values

To determine a patient’s priorities, a model of the 5Ms is supported by both the American and Canadian Geriatrics Societies [80]. This centers around asking a question, such as

“What matters most to you?” The other M’s include *memory/mind*, *mobility*, *medications*, and *multicomplexity*. Other tools have been developed to identify values, but may be termed “goals” or “preferences.” The development of validated tools is an area of research that is evolving, but is built on the foundation of being patient-directed rather than disease-directed [81]. This can help the clinician direct the assessment to delve deeper into some areas or provide guidance on the types of interventions that can be recommended or to expand the assessment and care planning to address other goals, such as life enjoyment [82].

Sensory Impairment

Senses are extremely important to assess as older adults experience changes in taste, vision, and hearing. In older adults, the loss of vision and hearing has dramatic effects on function and quality of life, and impaired taste can lead seniors to add excessive amounts of salt to food. Vision can be assessed by using a *Snellen chart* and *eye chart* that measures visual acuity. It is recommended that this is done at least every 5y but more frequently if there is exposure to treatments that impact eye health (e.g., steroids) or systemic disease (e.g., diabetes) [83]. Referral to an ophthalmologist for screening for diseases, such as glaucoma, could be considered [84].

Hearing is detected initially through observation during a conversation, such as noting when patients have difficulty in conversation or exhibit behaviours such as looking to a partner for support. Patients often do not report hearing loss partly because of stigma, but also because they may not be aware as presbycusis is a gradual loss of hearing. One recommendation is to simply ask the patient and caregiver/partner/family member if the patient is having difficulty hearing. Other recommended screening approaches can be administered by a health professional or even tested by family, and include such examples as a *whisper test*, where a series of digits or letters are read aloud in a whisper, about 1 m behind the patient, and they repeat what is heard [85, 86]. A validated questionnaire is the *Hearing Handicap Inventory*, which includes 25 questions that are scored out of 100 [87]. Formal assessment by an audiologist is appropriate to determine the type of hearing loss and to guide use of devices.

Social Supports

In order to provide appropriate recommendations for seniors, it is important to understand their circumstances and any potential harm they might be at risk of. In recent years, particularly during the COVID19 pandemic, the assessment of isolation and loneliness has been integrated into CGA and

includes a combination of measures, such as a friendship scale, loneliness scale, and interpersonal supports [88]. Life space, life engagement, and quality of life are not a core part of CGA but there is increasing development and use of tools for these important metrics that identify care needs and describe overall wellbeing for older adults. For social needs these areas are assessed by a social worker, nurse, or medical trainee using a checklist of questions. These questions include if the patient is living with someone, what supports the family, friends, or caregivers provide, if there are formal or informal caregivers involved, the type of housing he/she is residing in, the social network, environmental safety, or indicators of abuse [27]. Focused questions can also include the amount of financial resources the patient has available, how much informal or formal care is costing, or government funding accessed.

Another area of support that patients might describe is spiritual support. Religion and spirituality may play a role in determining choices related to medications (e.g., the use of animal or pork by-products), expectations for family, decisions about care, or the type of institutional supports the patient will accept. It is therefore necessary to ask about religion, spiritual beliefs, and the social, emotional, or psychological support that can be provided [89]. Formal spiritual care is a process that can be done by a religious leader, but may also be appropriate for other healthcare providers. The process starts with identifying spiritual needs and resources, and this can be triggered by findings from the CGA or part of a broader geriatrics team [90].

Medication Assessment in Older Adults

There are a number of generic tools that pharmacists can use for medication reviews. These include programs such as MedsCheck or other local tools. However, most of these tools are not designed for the unique multimorbid, complex, or frail older adult. Pharmacists should consider using tools that have been adapted or validated for seniors, including *TIMER* (Tool to Improve Medications in Elderly via Review), *NO TEARS*, *DITTO* (Drug Importance Tool for Treatment Optimization), the *Hierarchy of Utility*, or a ten-question self-administered tool that identifies medication risk in seniors (Table 29.6) [91–95]. These tools may improve the efficiency in identifying drug-related problems in seniors by providing a guiding framework and questions to ask, in the context of multimorbidity, grey areas of decision-making, and the potential for a high number of medications. The tools do not provide a score or direction that indicates an intervention is necessary, but they provide information for the pharmacist to consider in identifying polypharmacy, undertreatment, adherence, or management concerns. There is no evidence that one tool is superior to another.

Table 29.6 General medication assessment tools for geriatrics

Tool	Design	Categories and content included
TIMER	Structured two-page questionnaire with question prompts, checkboxes, tables, and references to guide the review	Medication insurance coverage Adherence Medication safety (adverse events, drug interactions) Therapeutic goals (related to disease states, common geriatric syndromes)
NO TEARS	Acronym to assist healthcare provider in reviewing each medication	Need and indication Open questions Tests and monitoring Evidence and guidelines Adverse events Risk reduction or prevention Simplification and switches
DITTO	A grid that is used once the importance and indication of medications are identified	The grid includes four levels of importance Vital, important, optional, not indicated Reasons for use include Cure, prevent complications, relieve symptoms
Hierarchy of utility	A sequence of ten steps to determine potentially inappropriate medications (PIM); six questions to determine the strength of indication; eight questions regarding the potential misuse or safety concerns	Example of questions included Potentially inappropriate medication <i>Accurately ascertain all medications used</i> <i>Define overall goals of care</i> Strength of indication (step 8 in the PIM assessment) <i>Does medication provide immediate relief of distressing symptoms?</i> Potential misuse <i>Is the medication associated with little benefit and high risk of toxicity in most older patients?</i>
Ten-question	A self-administered questionnaire with ten questions to be completed by the older adult	Example of questions included <i>Do you currently take five or more medications?</i> <i>Is it difficult for you to follow your medication regimen or do you sometimes choose not to?</i>

Medication Appropriateness

Medications are identified as inappropriate in seniors if they overall cause more harm than benefit, and there are safer alternatives available. There are two main types of tools—explicit and implicit, which are described below (Table 29.7) [96, 97]. Explicit tools include the ACOVE-3 criteria, The AGS Beers Criteria, STOPP/START, and others [98–101]. The AGS *Beers criteria* is updated every 3 years by the American Geriatrics Society and includes a list of potentially inappropriate medications (e.g., first-generation antihistamines), a list of medications to avoid in particular disease states (e.g., anticholinergic medications in delirium), medications to be used with caution (e.g., SSRI, which can cause SIADH), drug interactions, and medications requiring adjustment for renal function. The *STOPP criteria* were developed in Ireland for a context of practice in the European Union and also have periodic updates. They have many overlapping criteria with the AGS Beers Criteria, such as “do not initiate tricyclic antidepressants as first-line antidepressant therapy.” Other tools include the FORTA resource (based in Germany) that has a straightforward A,B,C,D ranking for medication risk [102], Choosing Wisely, and internationally adapted campaign to identify and reconsider potentially inappropriate medications as well as other forms of intervention [103], and many more from a variety of countries and settings. Many of these tools are written as statements without any scoring required, but they can be built into software to alert the clinician when a particular potentially inappropriate medication is being dispensed or used, or the tools are available through apps on mobile devices. They are designed

to be used as a guide for decision-making about prescribing, rather than mandatory criteria to follow without considering the patient in the decision-making. The *START* component of the STOPP/START criteria includes a list of medications that should be started, such as using bisphosphonates and vitamin D in a patient on long-term corticosteroids.

An implicit measure of medication appropriateness is the *Medication Appropriateness Index* (MAI) [104]. This tool is primarily used in research, due to the time it takes to administer ten questions about every medication the patient is taking. For an average regimen in seniors, it would take approximately 45 min to use. In addition, this tool does not identify medications that are missing. Despite these limitations, the questions are practical and can guide the pharmacist on decisions about the medications (e.g., Is the medication effective for the condition?).

Specific medications that are high risk include anticholinergic and sedative-hypnotic medications. One tool that combines assessment for both of these categories includes the Drug Burden Index, although this tool requires permission and a cost to use, it can provide a validated score to determine risk and guide decision making [105]. There are a variety of other tools that specifically focus on anticholinergic burden, but there is no consistent scoring or listing of the same medications on each tool [106].

Related to tools for appropriateness include tools to identify polypharmacy, and related resources to support deprescribing. For example, the ARMOR tool assists in evaluating polypharmacy by triggering the clinician to integrate other tools (e.g., The AGS Beers Criteria) as part of Assessment, Review for interactions and adverse drug events Minimize number of medications, Optimize for conditions and PK parameters, and Reassess for clinical, functional, and cognitive status [107]. Tools can also be built into an electronic health record to flag for volume or use of particular high-risk medications [108]. Regimen complexity overlaps with polypharmacy as number of medications can increase complexity, but tools that address this, such as the Medication Regimen Complexity Index (MRCI), also include medication directions (e.g. PRN versus scheduled), device complexity (e.g., oral versus injection or patch), and frequency of use [109]. The MRCI takes time to score, but reviewing the domains can assist pharmacists in identifying complexity for patients rather than simply counting the number of medications.

Functional Medication Management/ Medication Management Capacity

Medication management is challenging for many older adults due to the multistep process (e.g., ordering, obtaining, packaging/sorting, actuating, administering), the complexity

Table 29.7 Comparison of medication appropriateness tools

	Explicit	Implicit
Design	Criterion based	Judgment based
Development	Developed from published reviewers, consensus, expert opinions	The clinician uses information from the patient and published work
Clinical judgment	Applied with little or no clinical judgment	Clinician makes judgments about appropriateness
Focus	Usually drug or disease oriented	The patient
Benefit	Applied to databases Simple to apply	Most sensitive approach for the patient
Limitations	Do not take into account all factors that define high-quality healthcare for the individual	Time-consuming to provide care
	Do not address the burden of comorbid disease, patient preferences	
Example	The AGS Beers Criteria	Medication Appropriateness Index

of the regimen, and the physical and cognitive challenges of using multistep devices. Sometimes, disease can play a role, such as osteoarthritis or sensory impairment (e.g., low vision), which can affect how a senior can open blister packages, administer eye drops, or dial an insulin pen. In some cases, this is picked up as non-adherence, but these tools can assist in identifying why the patient is non-adherent, which is related to functional concerns. A review of 44 different tools indicates that there is growing interest in this area, yet few of these tools are integrated into standard CGA and pharmacy medication reviews [110]. These tools have been used primarily in research studies but could be used in clinical practice. One drawback is that many take more than 15 min to administer, which can be problematic in terms of a CGA. Some tools provide a standardized regimen to test the patient, rather than focusing on the patient's own regimen, which can make it difficult to apply the findings to make appropriate interventions for a specific patient. Two tools, the *Self-Administration of Medication* (SAM) and the *Drug Regimen Unassisted Grading Scale* (DRUGS), require less than 15 min to administer, assess the patient managing his/her own regimen, and identify if the patient requires additional supports, all of which may be useful to inform pharmacists and can be built into the medication review process [111, 112].

Clinical Pearls

- The comprehensive geriatric assessment is a broad assessment of health and non-health domains, which the pharmacist should be familiar with.
- The pharmacist should participate in geriatric assessment by contributing to medication assessment and engaging in dialogue about other domains with the interprofessional team.
- The pharmacist has an important role to play in assessing medications in older adults in all settings.
- There are a variety of validated tools to assess the appropriateness and functional management of medication that the pharmacist can incorporate into practice.

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Assessment Considerations in Pediatric Patients

30

Deonne Dersch-Mills

Chapter Objectives

1. Recognize challenges related to patient assessment that are unique to pediatrics and neonatology.
2. Describe the required components of a complete pharmacy assessment that are unique to pediatrics and neonatology.
3. Gain a basic understanding of the differences in pediatric assessment as it relates to vital signs and common laboratory parameters.

Background

Pediatric pharmacy technically includes the care of any patient under the age of 18 years, although practically, patients over the age of 15 can usually be cared for following similar dosing/monitoring practices as in adults. In Canada in 2023, children under 15 years comprised ~15% of the Canadian population (Statistics Canada, Population by sex and age group 2023), and accounted for 6.7% of hospital discharges (not including births in hospital) [1]. Approximately 50% of Canadian children receive one or more medications each year; this is up to 79% when only infants <1 year old are considered [2, 3]. Not all this medication use is acute; 19.8% of American children have taken any prescription medication in the last 30 days, 13.9% use that medication on a chronic basis [4]. However, polypharmacy is much less common than in adult patients. Around 20% of US children received at least one medication in the previous week, but less than 6% had used 2 or more medications in the previous week [5, 6]. The most common medications filled for children in Canadian community pharmacies in 2012 included antibiotics (amoxicillin, azithromycin and cefprozil most commonly), asthma medications (salbutamol, flutica-

sone, montelukast), and medications for attention deficit hyperactivity disorder (ADHD) (methylphenidate primarily) [2]. This mirrors the most common classes of medications dispensed to children in the United States in 2013–14: respiratory agents, psychotropic agents, and antibiotics [4].

Before moving forward, it is important for pharmacists to understand age-related terminology for children. Gestational age (GA) is the duration of time a newborn has been in utero (full term is considered ~40 weeks), while post-natal age (PNA) is the duration of time since birth (e.g. 2 weeks old). Corrected gestational age (CGA) is a term used commonly in infants born prematurely and refers to their gestational age plus post-natal age. For instance, a baby born at 30 weeks GA, with a PNA of 10 days, would have a CGA of 31 weeks and 3 days (commonly abbreviated as 31 + 3 weeks). A neonate (or newborn) is an infant <28 days PNA, or for premature infants, less than 44 weeks CGA. The term “infant” typically refers to a range of PNA from 28 days to 1 year. The term “toddler” typically refers to ages 1–3 years, and “school-age” refers to age from 4–9 years. “Adolescent” typically refers to age 10–19 years, though the true onset of adolescence is determined by the onset of puberty. These distinctions are important due to developmental changes that occur in pharmacokinetics (see more below).

There is a distinct lack of research on medication use in children, which adds its own challenges to the practice of pharmacy in children. Up to 90% of medication use in hospitalized children and 30% of medication use in ambulatory children is considered “off-label,” that is, not included in the product labeling/monograph, and thus not supported by the manufacturer [2]. Only 25% of medications on the market are approved for children, and this number is even less for infants [7]. This results in a lack of detailed pharmacokinetic, dosing, adverse effect, and efficacy data for appropriate medication management in children. As well, due to the lack of studied indications in children, there is little impetus for manufacturers to make formulations appropriate for children, and so formulation issues related to availability, palatability, excipients, or measurability are commonplace.

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Due to the above-mentioned factors, children are at higher risk of medication errors and have a higher risk of harm from medication errors, so the pharmacist plays an important role in caring for infants and children who use medications [8]. Approximately 8% of pediatric Emergency Department visits are medication-related, of which 2/3 are deemed preventable (examples include adverse drug reactions, sub-therapeutic doses, or non-adherence due to a variety of factors) [9]. There are an estimated 0.85 adverse drug reactions per 100 inpatient admissions in pediatrics, of which 21% would be considered preventable [10]. Children with medical complexity are at especially high risk due to multiple comorbidities, medications, caregivers and transitions in care settings [11]. Medication safety risks like this highlight the importance of a complete pharmacist assessment for children using medications.

In many ways, assessment of a pediatric patient is not different than assessment of an adult patient. Pharmacists should gather information on medical history, medication history, medical conditions, and any other relevant information, and then ensure that all medications are indicated, dosed appropriately, effective, safe, and adhered to. However, each of these steps contains nuances for infants and children. There may be additional steps within each component of the assessment, and there is a significantly smaller evidence base to support decisions and judgment calls made within each step of the assessment.

Pharmacokinetic Differences

One of the major nuances between pediatric and adult assessments is the need to consider pharmacokinetic ontogeny, or pharmacokinetic differences based on the age of the

child. While it is not surprising that there are pharmacokinetic differences between children and adults, there are also important differences between premature newborns, term newborns, infants, toddlers, school-age children and adolescents. Table 30.1 provides a summary of major pharmacokinetic considerations in infants and children. This variability in pharmacokinetics makes dosing medications in children especially challenging as there is no “standard” dose of a medication. Medication doses may vary based on CGA, PNA, as well as body weight or surface area. Most notable is the need to use neonatal (not pediatric) dosing in newborns as the first few months of life are characterized by rapidly changing organ function (renal and hepatic most importantly) as well as changes in drugs’ volume of distribution.

Typically, newborns are characterized by reduced renal and hepatic clearance at birth (especially with premature birth), which develops rapidly in the first month of life and reaches adults values at 6 months to 1 year of age. Childhood is typically characterized by excellent renal and hepatic clearance, even exceeding that of adults in some cases. Normal clearance values approach that of adults around adolescence. This is an oversimplification, given that different aspects of renal and hepatic clearance develop at different rates. Pharmacodynamic differences in infants and children are much less well understood, but more fulsome reviews of pharmacokinetic and pharmacodynamic differences in children are available [12–14]. By selecting doses recommended by age, the pharmacokinetic differences that occur throughout childhood are taken into account. All pharmacists should ensure that medication doses in children are age and/or weight appropriate by consulting neonatal and/or pediatric dosing references with each and every prescription.

Table 30.1 Summary of pharmacokinetic considerations throughout childhood [12–14]

	Neonates	Infants	Children	Adolescents
Absorption—Enteral	Changes in the degree of absorption of some medications due to elevated gastric pH and reduced pool of bile salts Changes in rate of absorption due to prolonged gastric emptying Consider frequent feeds (for medications requiring an empty stomach, this is challenging with q3-4h feeds) and frequent spit-ups/regurgitation (include counselling on what to do if baby spits up after taking medication)		Biggest factor relates to ability to take medication (related to taste and formulations)	Non-compliance becomes an increasingly important factor
Absorption—other routes	<u>Rectal</u> - consider frequent stooling patterns. First-pass metabolism may be bypassed depending on the depth of insertion (lower insertion will bypass first pass metabolism via lower and middle rectal veins, higher insertion still goes through first pass via upper rectal vein) <u>Percutaneous</u> - much higher in neonates (especially premature) due to immature keratinization (first 2 weeks of life), high body surface area: Volume ratio, and high blood flow to skin. Use caution with topical steroids (lowest potency possible).		Similar to adults	Similar to adults

(continued)

Table 30.1 (continued)

	Neonates	Infants	Children	Adolescents
Distribution	Increased volume of distribution for water-soluble medications. Approaches adult values around 6–9 months. Increased CNS permeability in newborn period and early infancy. Reduced protein binding of medications in neonatal period and early infancy (due to altered binding proteins as well as displacing substances such as bilirubin).		Similar to adults; binding proteins approach adult values at 3–5 years	Similar to adults
Metabolism	Reduced (in general) Phase I: CYP3A7 mature at birth, CYP2E1, CYP2D6 mature in around 2 weeks Phase II: Sulfation and methylation relatively mature at birth	Variable Phase I: CYP3A4, 2C9, 2C19, 1A2 mature ~6 mos-1 year Phase II: Glucuronidation mature 2 months-3 years; acetylation mature in 1–4 years (note other pathways may take over for immature systems e.g. sulfation of acetaminophen)	Increased (in general); typically exceeding adult values (i.e. frequently require higher mg/kg doses)	Similar to adults.
Elimination	Reduced at birth; both GFR and tubular secretion rapidly double in first 1–2 weeks of life.	Developing- GFR reaches ~50% adult values around 48 weeks PMA	Increased (in general) GFR reaches adult values before 2 years, tubular functions mature more slowly	Similar to adults

CYP Cytochrome P450

Steps in Assessment

Information Gathering

In pediatric practice, the gathering of patient history is often not done with the patient themselves. Until children are mature enough to provide this information themselves (note that this age varies depending on the child), parents and other caregivers are relied upon to provide an accurate history of the patient's illness(s) and medication(s). This provides an extra challenge to history gathering as the most appropriate person may not be available to provide this history, depending on the circumstances of the child's illness. Changing custody or living arrangements may impact a caregiver's ability to provide a history, or one parent may primarily manage the medical care of the child, and another caregiver might be the only one present at the time of pharmacists' information gathering. Additionally, information related to symptoms of illness will not be provided first-hand but rely on observations made by the caregiver. This introduces a source of potential inaccuracy or missing information that is not as commonly encountered when patients provide their own history.

Medical History

While medical histories of children tend to be shorter than that of adults, this is not always the case and many pediatric patients may have long medical histories before they reach school age. Gathering this information is not different than

gathering it for adult patients, through the use of a patient (caregiver) history corroborated by medical records.

An important difference to note is that in infants (especially those in the first 3 months of life), maternal pregnancy and delivery history is relevant and should be gathered wherever possible as well. Maternal conditions during pregnancy and the circumstances of delivery may be very relevant to an infant's medical care. For instance, a mother using methadone for a substance abuse condition may result in her infant developing signs and symptoms of withdrawal several days after discharge home. Likewise, whether or not a mother received intrapartum prophylactic antibiotics is directly related to the infant's risk of Group B *Streptococcus* infection in the newborn period. Very few medications are contraindicated in breastfeeding, but knowing if an infant is receiving even small amounts of medications via breastmilk may be relevant if concerns arise regarding a potential drug interaction or adverse effect.

Medication History

As with medical history, medication histories are often short in duration and length in children as compared to adults. Again, there are exceptions to this in children with complex medical needs. All relevant details collected for adult medical histories should be collected for pediatric patients, with some additional factors to consider.

Many pediatric medications do not come from the manufacturer in appropriate formulations for administration to children, and thus asking about the formulation is an impor-

tant step in pediatric medication history. Extemporaneously compounded liquids are especially risky and prone to confusion, as often caregivers know the volume they provide to the child but not the concentration. If there are multiple concentrations available due to variability in the recipe used by the pharmacy during compounding, this can lead to dose errors and subsequent sequelae (see Box 30.1 for example). If the actual product is not physically available and there are questions regarding the strength of a formulation or volume given, call the pharmacy that provided the previous fill and clarify what was provided. Medications in tablet form (except many sustained or controlled release formulations) may be crushed or dissolved for ease of administration and the appropriateness of this practice should be assessed by the pharmacist depending on the size of the tablet, ease of splitting, and the possibility of alternate dosing intervals (for example, a daily dose of ASA 20 mg can be delivered as a half tablet (40 mg) every other day instead of a quarter tablet (20 mg) every day). Some medications may be given in a “dissolve-a-dose” manner, whereby a tablet is crushed and suspended in a small amount of liquid, then the dose is measured from the resulting suspension. Certainly, accurate dosing in this situation relies on solubility and/or appropriate suspension of the tablet, and this should be considered when doing an assessment of the patient’s dose/response.

Box 30.1: Example of an Error Related to Formulation

There is no commercially available clonidine oral liquid on the market, and so it is commonly compounded by pharmacies for use in infants and children who cannot swallow tablets or who require very small doses.

Colin is a 10 kg child discharged from a hospital with a prescription for clonidine 50 micrograms PO q6h. His parents fill the prescription at the outpatient pharmacy within the hospital on their way home. The outpatient pharmacy provides them with 500 mL of 10 micrograms/mL compounded clonidine suspension with instructions to take 5 mL four times a day.

Colin’s parents go to their neighborhood pharmacy for the next refill. That pharmacy only has a recipe for clonidine 100 micrograms/mL oral liquid and so fills the prescription with that suspension (with instructions to give 0.5 mL four times a day). At home, Colin’s parents are used to giving 5 mL four times daily and so continue to do so. The next day Colin is acting lethargic and weak. His parents take him to the nearest emergency department and while they are reviewing the medications with the team, the pharmacist there discovers the error—a 10 times overdose of clonidine.

Even with appropriate commercially available formulations, administration of pediatric medications can be especially challenging. Literature shows that measurement errors are common in caregivers of children and so observing the caregivers measure (and administer if appropriate) the medications may provide insight into dosing appropriateness [15]. Considerations regarding the measurement device should also be made. Household teaspoons are not appropriate for measuring medications, and even some medication syringes do not have appropriate markings for accurate measurement of small doses. Ensuring an appropriate measuring device is an important part of a pharmacist’s assessment of dosing accuracy (see Fig. 30.1).

Likewise, how a caregiver gives the medication to the child may be relevant. Medications mixed in a small amount of food/milk immediately prior to administration is appropriate, but if it is mixed in a large amount that the child does not consume all of, the dose taken will not be accurate. Instructions to give medications on an “empty stomach” or to avoid dairy products are not likely to be followed in newborn infants who typically drink formula or breastmilk every 3–4 h. In these cases, the medication may be given with feeds for ease of administration, but a slightly higher dose may need to be used empirically, or the dose may be titrated up based on monitoring parameters (e.g., levothyroxine based on TSH). If this occurs, the administration conditions should just be kept consistent for the individual patient. Parents and caregivers may ask if it is appropriate to mix all a child’s medications together in one syringe/cup prior to administration. This practice should ideally be discouraged as it may contribute to inaccurate measurement, potential incompatibilities, and will present challenges with re-dosing or estimating the portion of dosages consumed if only part of the “mixture” is taken/spit up.



Fig. 30.1 Examples of inappropriate (3 on left) and appropriate measuring devices (3 on right)

Other things to consider in the medication history of infants and children include medication storage, timing, and palatability. Depending on the frequency of administration of a medication and the child's schedule (school, daycare, activities), medication may not always be stored in an ideal manner. The timing of medications may also be impacted by bedtimes. For instance, if a medication is meant to be given every 6 h around the clock, it may be appropriate to give it four times daily (QID) instead to accommodate uninterrupted sleep. Lastly, an assessment of the palatability of the medication formulation and the caregiver's report of how often the child takes/receives their full dose should be completed. These types of issues will be further explored in the section on adherence assessment.

Indication Assessment

Like in the adult setting, "Is this therapy indicated?" is an important question to be assessed by the pharmacist. Due to the relative lack of studies in pediatric patients, many medications do not have an official indication in pediatrics (i.e. pediatric indications and dosing are not included in the product monograph). Because of the lack of appropriate information in the product labeling, alternative data sources thus need to be accessed to assess the use of the medication in an infant or child. Pediatric medication information sources should be available and reviewed to aid in this part of the assessment, but primary literature may also have to be consulted more often than in adults.

Assessing whether an agent is "first-line" or "ideal" for a child includes special considerations that may not be necessary in adult patients. An agent may be selected not because it has the best data to support it, but because it is the only agent with pediatric dosing information, or because it is the only one with a formulation appropriate for the age group (or, it has the best-tasting formulation). A medication may be selected because it has a less frequent administration schedule, accommodating a child's school or activity schedule, or to limit the number of dosage times due to unpleasantness of administration (e.g., bad taste or painful injection). In this sense, assessment of whether a medication is the most appropriate choice for the patient is increasingly challenging.

Because of differences in pathophysiology or pharmacodynamics in pediatric patients as compared to adults, an "appropriate" or "ideal" medication choice in adults may not be so in pediatrics. Likely pathogens in bacterial infections differ between age groups, the causes of thrombosis or hypertension are very different in children, and systemic steroids can have very a different safety profile depending on the age of the child. For this reason, complete assessment of whether a medication is "indicated" for a child requires much more in-depth look into references than for a typical

adult patient. Pediatric or neonatal guidelines, reviews, studies, and other literature need to be consulted for this component of the assessment. One cannot rely on what is appropriate in adult patients to be appropriate in newborns, infants, or children. Of note, a list of potentially inappropriate medications for children has been developed and list includes medications that should either be avoided or used with caution in children because of either high risks or because safer alternatives are available [16].

Dose Assessment

As alluded to above, comprehensive dose assessment in pediatric patients requires several more steps than in adult patients. First, the pharmacist needs to determine how the medication dose is characterized: by gestational age (in newborns), post-natal age, by body weight, by body surface area, or some combination of the above.

Once that is determined, the pharmacist must gather the needed information to categorize the patient into the appropriate dosing range using a pediatric dosing reference. A common error is to use pediatric doses for neonates; therefore, the age of infants needs to be confirmed. In addition to selection of the appropriate dosing category, the pharmacist must also ensure that the appropriate dose for the indication in question is considered. Doses for medications can vary significantly for different indications. For example, the ASA dose for antiplatelet effects is 1–5 mg/kg/day, where anti-inflammatory doses range from 60–100 mg/kg/day.

The next step is another common source of error: calculations. Doses should never be estimated, and calculations should never be done in one's head. A calculator is an essential tool for pediatric pharmacists! Pediatric doses are typically listed in mg/kg/DAY divided qXh or can be listed as mg/kg/DOSE given qXh. This is a common source of error and must be carefully checked. While it is common to see pediatric doses that are higher than adult doses on a mg/kg basis because of pharmacokinetic differences noted above, it is important to note that adult maximum daily doses (total mg) should typically still be observed in pediatric patients. Box 30.2 provides an example.

Many medications that children use are on a short-term basis only (e.g., antibiotics); however with medications that are used on an ongoing basis, pharmacists need to ensure that doses are checked with each (re)fill. Children, and especially infants, grow quickly, and medications dosed on a mg/kg basis may need adjustment for growth. Each fill is an opportunity for the pharmacist to assess the patient's weight, need to continue a medication, its apparent effectiveness and the presence of any adverse effects as part of the decision as to whether to adjust the dose (or discontinue the medication altogether if appropriate).

Box 30.2: An Example of a Pediatric Dose Approaching an Adult Dose

Rex is a 12-year-old boy being treated for a pneumonia using amoxicillin. He weighs 40 kg.

His prescription states:

90 mg / kg / day = 1350 mg PO TID $\times$ 7 days

While 90 mg/kg is much higher than most adults would receive, this is an appropriate dose for pneumonia in children. However, the adult maximum daily dose of amoxicillin for pneumonia is 3000 mg, so Rex's dose should be adjusted to 1000 mg PO TID even though this will only be 67 mg/kg/day.

Lastly, the pharmacist needs to identify an appropriate formulation; one that allows the dose to be measured accurately, is (reasonably) palatable, and has suitable stability. As mentioned above, this may be a significant challenge on its own. As an additional challenge, newborns <44 weeks CGA should ideally not use medications containing preservatives (e.g., benzyl alcohol, propylene glycol) and children should ideally not use medications containing alcohol [16]. This is due to adverse effects reported with these additives (gasping syndrome, metabolic acidosis, hypoglycemia) [17, 18]. However, if an alternative, preservative-free and/or alcohol-free product is not available, the benefit of the medication should be weighed against the risks from a small dose of alcohol or preservative.

Monitoring: Efficacy and Safety

Symptom Assessment

In pediatrics, often patients are non-verbal or are not at a developmental stage where they can describe their symptoms, and so a caregiver's external assessment of the patients' symptoms must be relied upon. There is a greater focus on *observation* of signs and symptoms when determining a patients need or response to medications. While these observations have some degree of objectivity, many assessments done in this way may include some subjectivity. For this reason, other objective measures may be relied upon more frequently in younger children than in older children or adults. For instance, the objective presence of a fever and results of key lab values (e.g., white blood cells) can supplement a parent's subjective report of a child not improving in an assessment of a child's response to antibiotics. Pain scoring tools that include a measurement of heart rate, blood pressure, as well as an observation of the infants' behavior can add an objective component to a parent's assessment of their child's pain. There are many pediatric-specific assessment tools for a variety of medical conditions and pharmacists should seek

those out when needed. Table 30.2 includes some examples of condition-specific assessment tools for infants and children. It should be noted, however, that dismissing a parent's assessment of their child due to its subjectivity is not advised. Caregivers know typical behavior in their children, and slight changes in behavior (e.g., poor feeding, decreased energy, altered sleep) can be important signs of illness or adverse medication effects in children. Having said that, if subjective observations contradict objective measures, one should proceed with caution. When assessing a child's illness, there are some red flags that should elicit immediate referral to medical care. Table 30.3 depicts a list of the red flag symptoms that prompts referral in pediatric patients.

Physical Assessment

Physical assessment in pediatrics can be limited by both the pharmacists' knowledge of the nuances of pediatric physical exam and by the patient's tolerability of the exam itself. It is beyond the scope of this chapter to describe the differences

Table 30.2 Examples of pediatric-specific assessment tools [19–25]

Condition	Assessment tool
Pain	Neonatal pain, agitation and sedation scale (N-PASS) face, legs, activity, cry and Consolability scale (FLACC) Faces pain scale
Sedation	Neonatal pain, agitation and sedation scale (N-PASS) State behavior scale (SBS) COMFORT scale
Nausea	Pediatric nausea assessment tool (PeNAT)
Asthma	Childhood asthma control test (C-ACT) (asthma control) Pediatric respiratory assessment measure (PRAM) (asthma exacerbation)

Table 30.3 Red flag symptoms in pediatrics that prompts immediate referral to health care practitioner [26, 27]

System	Symptom
General appearance	Increasing lethargy/confusion
Vital signs	Rash with fever Fever in infant <3 months old Very fast or very slow heart rate (see table 28–4 for age-based normals) Elevated respiratory rate at rest (see table 28–4 for age-based normals)
Central nervous system	Change in/loss of consciousness Lack of response to pain Decreased tone/floppiness Seizure-like activity
Cardiovascular	Very fast or very slow heart rate (see table 28–4 for age-based normal values)
Respiratory	Very fast respiratory rate (see table 28–4 for age-based normal values) Labored or noisy breathing Irregular or absent respiration
Derm	Skin color changes- dusky/blue Very dry lips or mouth
Genitourinary	Severely reduced/lack of urine output

Table 30.4 Typical normal vital sign ranges, comments on physical exam in children [28–31]

Age group	Normal respiratory rate (breaths per minute)	Definition of tachypnea (breaths per minute)	Comments on pediatric respiratory exam
Newborn	34–50	>60	Respiratory rate is best measured when baby is settled or sleeping, not crying. Increased work of breathing can be indicated by nasal flaring, subcostal and intracostal retraction, head bob, or tracheal tug. Infants and children have narrower airways, and a small degree of inflammation can reduce the diameter significantly (eg. Croup). Infants less than a year of age are obligate nasal breathers, and nasal congestion can be a significant source of respiratory distress, in addition to being a cause of poor feeding. Noisy breathing including wheeze, stridor or grunting are signs of respiratory distress and should be referred.
Infant	25–40	>50	
Child 1–5 years	20–30	>40	
Child >5 years and adolescents	15–25	>30	
Age group	Normal heart rate when awake	Normal heart rate when sleeping	Comments on cardiac exam in pediatrics
Newborn	100–205	90–160	“Exercise” in infants is best represented by feeding, thus poor feeding/sweating while feeding may be a sign of cardiac compromise. Congenital heart defects are a common cause of cardiac compromise in infants- echocardiography is required to assess heart structure. It is important to assess a child’s heart rate at rest and not when they are crying/upset.
Infant	100–190	90–160	
Toddler (1–2 y)	100–140	80–120	
Preschooler (3–5 y)	80–120	65–100	
School-age (6–11 y)	75–120	60–90	
Adolescent	60–100	50–90	
Age group	Normal systolic/diastolic blood pressure (mmHg)	Definition of hypotension- systolic (mmHg)	Comments on pediatric blood pressure measurement in pediatrics
Neonates	70–85/35–55	<60	BP measurement requires pediatric-specific, appropriately sized-cuffs. Definitions of hyper- and hypo-tension require knowledge of the child’s age/gender based percentile of height and use of specific blood pressure tables. A quick way to estimate normal systolic blood pressure in a child: Median systolic blood pressure = 2(age) + 90 mmHg. The most common cause of hypertension in children are renal pathologies, which differs significantly from adults.
Infant	70–100/40–55	< 70	
Child up to 5 y	90–100/40–60	< 70 + (age in years × 2)	
Child >5 y to 11 y	100–110/60–75	< 70 + (age in years × 2)	
Adolescent	110–120/65–80	< 90	

in physical exam between children and adults, but vital sign assessment can be a simple addition to a pharmacists’ assessment in many cases, and the major differences between children and adults are summarized in Table 30.4.

Lab Values

As in adults, lab values are an important part of assessing both the efficacy and toxicities of medications in children. Table 30.5 outlines some of the most used laboratory values and how they differ between children and adults.

Monitoring renal function is especially important for pharmacists, both for assessing the need for dose adjustment in renal dysfunction, but also for monitoring for nephrotoxicity of medications. Because normal ranges of serum creatinine in children are relatively wide, serum creatinine values can double, reflecting acute kidney injury, but remain in the “normal” range. It is important therefore for pharmacists to follow trends in serum creatinine rather than just absolute values. In cases where the pharmacist needs to estimate renal function, the formula used in children differs from that used

in adults. The most accurate and commonly used formula is called the “Bedside Schwartz” formula (Eq. 30.1). Estimation of GFR in infants and especially in newborns is challenging as renal function continues to develop rapidly after birth, with most infants reaching full renal function around 6 months of age. Again, following trends in serum creatinine (expecting a downward trend in the first 1–2 weeks of life) is the most effective method of assessing renal function. Other factors such as urine output and hydration should also be considered in the overall assessment of renal function. Renal dysfunction in children is fortunately uncommon, and children with renal dysfunction are likely to be followed by a pediatric nephrology team.

“**Bedside Schwarz**” formula for estimating Glomerular Filtration Rate GFR in children >1 year of age [35]:

$$\text{GFR (ml / min / 1.73 m}^2\text{)} = \frac{36.2 \times \text{Height (cm)}}{\text{Serum Creatinine (}\mu\text{mol / L)}} \quad (30.1)$$

Table 30.5 Common lab values used for monitoring medications and how they differ in children [32–34]

Lab Value	Normal values	Comments
Hemoglobin	Birth 150–200 g/L Nadir (in term infants) = 90–110 g/L, occurs ~6–8 weeks of age Nadir (in preterm infants) = 60–80 g/L, occurs ~3–7 weeks of age Normalizes (adult values) ~ 6 months of age	It is normal for infants to have a drop in Hgb following birth, due to a switch from fetal to adult hemoglobin.
White blood cells	Neonate 9–30 cells/mm ³ Infant 6–18 cells/mm ³ Child 5–15 cells/mm ³ Adolescence ~ adult values	Newborns may have a decrease in white blood cells in response to infection as cells are used up and replaced at a slower rate.
Platelets	Same as adults	Newborns may have slightly lower platelets counts, with a lower limit of 150 x10 ⁹ /L.
Electrolytes	Same as adults	Newborns may have slightly lower sodium (some sodium wasting due to renal immaturity) and higher K (normal state of potassium retention) values than infants and children. Serum phosphate also tends to be higher in infants and children due to high needs during bone growth and high energy needs.
Creatinine	Newborns 27–88 µmol/L Infants 18–35 µmol/L Children 27–62 µmol/L Adolescents 44–88 µmol/L	Note that creatinine in the first few days of life often reflects maternal renal function and a downward trend is expected in first 2 weeks of life.

Because infectious diseases are commonplace in children, interpretation of bacterial cultures is worth mentioning. Sputum cultures are challenging as contamination with saliva is common. It is therefore necessary to note the presence of endothelial cells and/or nasopharyngeal flora in these cultures, as they may indicate a lack of appropriate specimen. Urinary samples, likewise, can be difficult to obtain appropriately without catheterization and thus have a higher likelihood of contamination by skin flora. Lastly, there is a higher rate of false negative blood cultures. Blood cultures in children are typically limited to 2 bottles (versus 4 in adults), and use smaller volumes of blood, resulting in a higher risk of false negative results. The reduced number of bottles also makes distinguishing between pathogenic growth and contamination more difficult. In adults, growth of skin flora in 1 out of 4 blood culture bottles is typically associated with contamination, however in children growth of the same

pathogen in 1 out of 2 bottles is less convincing as contamination. For these reasons, it is not uncommon to empirically treat a child for an infection despite negative cultures, purely based on signs and symptoms of infection. This represents a particular challenge for pharmacists attempting to tailor antibiotic therapy to infectious pathogens. Antibiotic therapy may have to be assessed based on “typical” pathogens rather than actual.

When considering “typical” pathogens, these vary by age and so pharmacists need to consider this when assessing the appropriateness of antibiotic therapy. Another major factor in the “likely” bacteria pathogens is immunization status, and pharmacists should be familiar with vaccination history in patients presenting with infectious diseases. For instance, unimmunized children are at much higher risk of infection with *Hemophilus influenza* type B (HiB), which can be a significant pathogen in both meningitis and pneumonia. Table 30.6 outlines common pathogens in common infections throughout infancy and childhood, assuming vaccination in accordance with local immunization programs.

Adherence Assessment

An assessment of adherence in children should include not only overt nonadherence (eg. the child dislikes the taste and spits it out, or the parent forgets to give it on the weekends) but also “covert” nonadherence (e.g., parents make measurement errors, or the product is used beyond its stability date). Asking to see the caregiver show the dose given as measured, and an examination of the medication bottles can assist with this. Questions regarding the appropriate storage of medication are also important (e.g., using coolers for refrigerated medications on long trips).

Measurability

Swallowing skills and variability in dosages often make the use of solid dosage forms impractical in young children. If the tablet is appropriate to split, and if a child’s dose can be rounded to accommodate a half (or even quarter) tablet, this may be the most practical approach. However, in younger children and for smaller doses, the use of oral liquids may be the only solution. Oral liquids should be measured in appropriately sized devices. Often, especially in infants, doses may be so small that accurate measurement becomes challenging. Doses less than 1 mL are ideally measured in 1 mL (or even 0.5 mL) oral syringes to maximize accuracy. Typically, manufacturer guidance dictates that doses <20% of the syringe size cannot be accurately measured (e.g. no less than 0.2 mL in a 1 mL syringe) although exceptions may be required when alternatives do not exist.

Occasionally, unique solutions to measurability issues need to be explored. Very small doses of some injectable products

Table 30.6 Common pathogens in common childhood infections [36–41]

Infectious Disease	Age Group	Common Pathogens
Meningitis	Newborns	Group B <i>Streptococcus</i> <i>E.coli</i> Other bacteria, including <i>Listeria monocytogenes</i>
	Infants 1–2 months	Group B <i>Streptococcus</i> Gram negative bacteria <i>Streptococcus pneumoniae</i> <i>Neisseria meningitidis</i>
	3 months–9 years	<i>Streptococcus pneumoniae</i> <i>Neisseria meningitidis</i> <i>Hemophilus influenzae b</i> (if unimmunized)
	10 years and up	<i>Neisseria meningitidis</i> <i>Streptococcus pneumoniae</i> <i>Hemophilus influenzae b</i> (if unimmunized)
Acute otitis media	All ages	Viruses (e.g. respiratory syncytial virus (RSV), parainfluenza, influenza, adenovirus, coronavirus) <i>Streptococcus pneumoniae</i> Non-typeable <i>Hemophilus influenza</i> <i>Moraxella catarrhalis</i> <i>Streptococcus pyogenes</i>
Pneumonia	Newborns	Group B <i>Streptococcus</i> Gram negative enteric bacteria <i>Ureaplasma urealyticum</i>
	Infants <3 months	<i>Chlamydia trachomatis</i> RSV Parainfluenza <i>Streptococcus pneumoniae</i> <i>Bordetella pertussis</i>
	Children ≤5 years	Respiratory viruses (e.g. RSV, parainfluenza, human metapneumovirus) <i>Streptococcus pneumoniae</i> <i>Mycoplasma pneumoniae</i> (note that <i>Staph aureus</i> can be a common cause of post-influenza pneumonia)
	Children >5 years and up	<i>Mycoplasma pneumoniae</i> <i>Chlamydia pneumoniae</i> <i>Streptococcus pneumoniae</i>
Urinary tract infections	All ages	<i>Escherichia coli</i> <i>Klebsiella</i> <i>Proteus</i> <i>Enterococcus</i> <i>Pseudomonas</i> <i>Enterobacter</i> <i>Citrobacter</i> <i>Serratia</i>

(e.g. enoxaparin doses <10 mg) may be most easily measured using an insulin syringe (e.g., 1 unit = 0.01 mL = 1 mg enoxaparin). Very small doses of oral medications only available in capsules, may require (very careful) preparation of powder papers, which involves weighing of powders from capsules and packaging in specially folded papers. Serial dilutions may be needed to attain measurable volumes of some liquids.

Whatever the situation, pharmacists must ensure that parents and caregivers using medications in children have an accurate method of measurement available as well as a practical approach to giving the medication.

Palatability

If a medication must be delivered to the child as an unpalatable oral liquid, there are some approaches to helping mask the taste of the medication. In infants, most medications can be mixed with a small amount of formula or breast milk to improve palatability. It is important to instruct parents or caregivers not to mix medications in a full volume of feed (i.e. a full bottle) as the complete dose may not be delivered if the full amount of feed is not taken. Older children may find that eating frozen treats prior to taking the medication may numb the tongue enough and reduce the bad taste. Similarly, allowing the child to choose a food or drink to wash the medication taste away may be helpful. Concentrated, sugary beverages or foods (honey, chocolate syrup) or other strongly flavored foods (cheeses) may help mask or remove the taste from the child's mouth.

Medication Administration

Administering oral medications to children can be a significant challenge in itself; pharmacy team members caring for children should be familiar with techniques to support caregivers and children in the medication administration process [42]. After measurement with an appropriate device, oral liquids can be given to infants mixed with a small amount of milk/formula through a bottle nipple, or for older children, they can be mixed in a *small amount* of palatable food or liquid before administration. Alternatively, oral liquids can simply be expelled at the back of the mouth followed with a drink of palatable liquid. Iron liquid can stain teeth, and so rinsing the mouth after administration or using a straw if the medication has been diluted in a liquid is advised (note that these precautions are unnecessary in infants without teeth, and less important in those without permanent teeth).

The ability to swallow oral solid dosage forms makes oral medication administration easier, and this can usually occur around age 5 or 6 years. If children struggle with pill swallowing, they can practice with candies of increasing sizes (e.g., start with sprinkles, slowly increasing the size of the candies up to the target tablet size; usually, small jellybeans are sufficiently sized for most medications). Children can place the medication/candy on the back of their tongue, tilt their head back, and have a large drink of liquid to facilitate this. In some cases, special training may help teach pill swallowing to children who require chronic use of oral medications best delivered as oral solid dosage forms [43].

Medications to be administered via feeding tubes (e.g., nasogastric tubes, gastro-jejunal tubes) can be administered without palatability concerns, however the location of the

medication absorption should be considered in the context of the intestinal administration site (e.g. ensure the medication can be absorbed if administered into the jejunum), and the tubes should be flushed with water afterwards to ensure complete delivery of the dose.

Adolescents

As a final comment on adherence assessment, it is important to consider adolescents specifically. Increasing independence with medication use and administration is often given to adolescents as they move towards adulthood, which requires a change in approach. Where previously assessment was conducted primarily with the caregivers on behalf of the child, the pharmacist should begin to include the adolescents in these conversations to build their health literacy and knowledge of their conditions/medications. Also of note,

adolescents often become more self-conscious of their medical conditions and associated medication use, and strategies to increase the discreteness of their medication use may be appreciated and improve adherence. For example, selection of a dry powder inhaler instead of a metered-dose inhaler with a spacer device or choosing a once or twice-a-day administered agent instead of one that has to be used at school. Considerations such as these should come into a pharmacist's assessment as their pediatric patients move towards adulthood.

Assessing medication use in children has many similarities to that in adults, and should be approached in a similarly systematic way, bearing slight nuances in mind. Table 30.7 summarizes the approach to pediatric assessment, highlights common sources of errors, and provides some tips for minimizing these errors.

Table 30.7 Pediatric assessment steps, common errors, and tips to avoid errors

Step in assessment	Common errors	Pediatric considerations
1. Information gathering.	Caregivers confusing mg with mL when stating dose. Unknown or incorrect concentration dispensed.	Try to take history from most appropriate caregiver. Include maternal history if <3 months of age. Include formulation, strength, and administration details in medication history.
2. Indication assessment.	Unfamiliarity with pediatric indications.	Be sure to use pediatric references. Consider "other" reasons for selection of agent-administration schedules, palatability, and availability of pediatric data.
3. Dosage assessment.	Mathematical errors. Using dose for wrong age group. mg/kg/day vs mg/kg/dose, Exceeding adult daily maximum dosage. Patients outgrowing doses of chronic medications.	Use neonatal doses for neonates, age-appropriate pediatric doses for infants and children. Check dose for specific indications. Double check math. Consider growth with medications used chronically. Keep adult maximum daily doses in mind.
4. Efficacy and safety assessment.	Unfamiliarity with pediatric "normal" values for physical exam or laboratory parameters.	Try to include objective and subjective measures. Physical exam, laboratory values, assessment of renal function, and typical pathogens may differ from adults.
5. Adherence assessment.	"Covert" non-adherence (parents inappropriately measuring medications or using compounded medications beyond expiry date)	Ensure caregivers have a practical approach to measurement and administration of the required doses. Assess for palatability issues. Observe measurement and/or administration of medications by caregivers. Help transition adolescents to adulthood and independent medication use.

Clinical Pearls

- Assessment of dose requires additional steps in children as compared to adults- categorization by age/weight, calculation of appropriate dose, and determination of accurate measurement processes are a few major steps that need to be included.
- Infants and children have variable pharmacokinetic and pharmacodynamic parameters, and thus dosages vary widely. The use of pediatric/neonatal references is essential for appropriate dosing and indication assessment by pharmacists.
- Finding appropriate formulations for children is a challenge for pharmacists. Careful assessment of doses/concentrations, measurability, and method of administration needs to be included in the overall medication assessment for infants and children.
- Interpretation of diagnostic tests and vital signs varies in children, and a basic understanding of these differences is important for pharmacists assessing children's medication therapy.

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Nese Yuksel

Chapter Objectives

1. Describe the goals of therapy and management strategy for menopause.
2. Assess women for menopausal symptoms and who may be considering menopausal hormone therapy.
3. Describe the goals of therapy and therapeutic options for women seeking hormonal contraceptives.
4. Assess women who are seeking hormonal contraception.

Assessment of the Menopausal Woman

Background

Menopause is part of the normal aging process and is defined as the cessation of menstruation due to the loss of ovarian function. The average age of menopause in North America is 51 years (45–55 years) [1]. The menopause transition or perimenopause is the time leading up to the menopause. The Stages of Reproductive Aging Workshop (STRAW+10) classification depicts the stages of menopause, which are classified according to changes in a woman's menstrual patterns and follicle stimulating hormone (FSH) (Fig. 31.1) [2]. Many women start having symptoms, with or without menstrual cycle changes, during the menopause transition. Hormone levels fluctuate and decline during the menopause transition, with symptoms starting up to 4–8 years before the last menses.

Menopause can be a natural process or induced through surgery (bilateral oophorectomy), chemotherapy, or radiation. Menopause before the age of 45 years is considered early, while premature menopause occurs before the age of 40, which is also referred to as premature ovarian insufficiency [3].

The early loss of estrogen before the natural age of menopause increases the risk of long-term health consequences such as cardiovascular disease (CVD), osteoporosis, cognition and dementia, and early mortality [3].

The most common menopausal symptoms are vasomotor (hot flashes, night sweats), genitourinary, sleep disturbances, mood, cognition, and muscle and joint pain (Table 31.1). Approximately 80% of women will have vasomotor symptoms (VMS), while 25% will be severe enough to affect quality of life [1]. Genitourinary syndrome of menopause (GSM) captures the range of urogenital symptoms associated with estrogen deficiency, including vaginal dryness, urinary incontinence, recurrent urinary tract infection, and dyspareunia. GSM affects more than 50% of post-menopausal women [4].

This chapter focuses on the assessment required for the management of menopausal symptoms. The period of midlife is also an opportune time to review long-term health risks such as CVD, osteoporosis, and breast cancer.

Diagnosis

Natural menopause is diagnosed when a woman has not had a menstrual period for 12 months without any other cause. With the loss of ovarian function, the levels of estrogen and progesterone fall, resulting in a subsequent increase in FSH levels. There are no specific diagnostic tests for menopause, but an elevated FSH level will help confirm the diagnosis when a woman has not had a menstrual period for 12 months or longer. In women who have had a hysterectomy before the cessation of ovarian function, hormone levels (estrogen, progesterone) and FSH are helpful in determining menopause status. Perimenopause is associated with fluctuating hormone levels, and FSH levels can be variable and erratic; these measurements should not be used for diagnosis.

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Menarche					FMP (0)					
Stage	-5	-4	-3b	-3a	-2	-1	+1 a	+1b	+1c	+2
Terminology	REPRODUCTIVE				MENOPAUSAL TRANSITION		POSTMENOPAUSE			
	Early	Peak	Late		Early	Late	Early			Late
					Perimenopause					
Duration	variable				variable	1-3 years	2 years (1+1)	3-6 years	Remaining lifespan	
PRINCIPAL CRITERIA										
Menstrual Cycle	Variable to regular	Regular	Regular	Subtle changes in Flow/ Length	Variable Length Persistent ≥7- day difference in length of consecutive cycles	Interval of amenorrhea of ≥60 days				
SUPPORTIVE CRITERIA										
Endocrine FSH AMH Inhibin B			Low Low	Variable* Low Low	↑Variable* Low Low	↑>25 IU/L** Low Low	↑Variable Low Low	Stabilizes Very Low Very Low		
Antral Follicle Count			Low	Low	Low	Low	Very Low	Very Low		
DESCRIPTIVE CHARACTERISTICS										
Symptoms						Vasomotor symptoms Likely	Vasomotor symptoms Most Likely			Increasing symptoms of urogenital atrophy

* Blood draw on cycle days 2-5 ↑ = elevated

** Approximate expected level based on assays using current international pituitary standard⁶⁷⁻⁶⁹

Fig. 31.1 Stages of reproductive aging workshop (STRAW+10). (Reprinted with permission from Wolters Kluwer Health, Inc. Harlow et al. [2])

Table 31.1 Menopause symptoms

Classification	Types of symptoms
Vasomotor	Hot flashes Night sweats
Genitourinary	Vaginal dryness Urge/stress incontinence Dyspareunia Frequent urinary tract infections
Sleep	Fragmented sleep Insomnia
Mood symptoms	Anxiety, irritability, depressive symptoms, mood swings
Musculoskeletal	Stiffness/soreness Muscle/joint pains
Memory/concentration	Memory changes Difficulty concentrating/focus Brain fog
Sexual	Low libido
Others	Fatigue Headaches/migraines Dry skin/eyes Heart palpitation

Treatment Approaches

Menopausal hormone therapy (MHT) is the most effective option for bothersome vasomotor symptoms [1]. MHT can also help improve fragmented sleep, mood issues, GSM symptoms, and prevent bone loss. MHT should be individualized to the woman after consideration of the woman's benefits and risks (Table 31.2). Many women worldwide discontinued MHT after the initial publication of the Women's Health Initiative (WHI) in 2002 showed an increased risk of coronary heart disease (CHD), stroke, and breast cancer with estrogen and progestogen [5]. However, these results have been recently questioned, as many outcomes in the study (such as CHD, breast cancer) were not statistically significant [6]. The estrogen-alone arm of the WHI study did not show the same risks of CHD or breast cancer [7]. Current professional guidelines agree that systemic MHT is safe and effective for initiating in women less than 60 years of age or less than 10 years post-menopause

Table 31.2 Benefits and risks of menopausal hormone therapy

Type of MHT regimen	Benefits	Risks	Considerations
Systemic estrogen alone	Relief of menopausal symptoms Prevention of bone loss and reduce osteoporotic fractures Improve quality of life	Venous thromboembolism (VTE) Cardiovascular disease (CVD) Stroke Gallstones Note: Increase in breast cancer was not seen in large randomized trials such as WHI estrogen therapy-alone arm	Age (≥ 60 years) or time since menopause (≥ 10 years postmenopause) are risk factors for VTE, stroke, and CVD when <i>initiating</i> MHT. The greatest risk of VTE is in the first 1–2 years. Transdermal estrogens at standard doses may have less risk of VTE compared to oral estrogen.
Systemic estrogen and progestogen	Same as estrogen alone	Venous thromboembolism (VTE) Cardiovascular disease (CVD) Stroke Breast cancer Gallstones	Same as estrogen alone Progesterone may have less risk of breast cancer compared to synthetic progestins; however, this data is primarily from observational data
Local vaginal estrogen therapy	Relief of GSM symptoms	Not the same risks as systemic estrogen therapy	Low systemic estrogen levels, as long as standard doses are used

CVD cardiovascular disease, GSM genitourinary syndrome of menopause, HT hormone therapy, VTE venous thromboembolism, WHI Women's Health Initiative

[1, 8, 9]. Menopausal hormone therapy also reduces osteoporotic fractures and is an option for the treatment of osteoporosis in women who have concurrent menopausal symptoms [10].

The current recommendations for symptom relief are to continue systemic MHT for as long as the women needs it for symptoms [1, 8, 9]. For early or premature menopause, MHT is recommended until the average age of menopause, not only for symptom relief but also for prevention of osteoporosis, CVD, and cognitive changes [1, 3]. In women with an intact uterus, estrogen is provided along with a progestogen to reduce the risk of endometrial cancer. Progestogen refers to both synthetic progestins and natural progesterone. Other options for MHT include products that do not require additional progestogen for endometrial protection, such as a tissue-selective estrogen complex (TSEC) and tibolone. A TSEC uses a selective estrogen receptor modulator (bazedoxifene) for endometrial protection in combination with conjugated estrogen. Tibolone, also known as a Selective Tissue Estrogenic Activity Regulator (STEAR), has 3 active metabolites with estrogenic, progestogenic, and androgenic activities [1]. Women who have had a hysterectomy can use estrogen alone.

The management of perimenopause may differ slightly. Options include combined hormonal contraception (if irregular or heavy bleeding or if contraception is required), menopausal hormone therapy, or estrogen in combination with a levonorgestrel–intrauterine system (LNG-IUS). The need for contraception during the perimenopause should also be considered [1].

Non-hormonal prescription medications for vasomotor symptoms include antidepressants (selective serotonin reuptake inhibitors (SSRIs) such as paroxetine, citalopram, and escitalopram, serotonin norepinephrine reuptake inhibitors (SNRIs) such as venlafaxine and desvenlafaxine), gabapenti-

noids (i.e., gabapentin, pregabalin), clonidine, and oxybutynin [1]. These medications are not as effective as MHT for vasomotor symptoms; however, they may be an option in women who have contraindications to or choose not to take MHT [11]. A new class of non-hormonal medications has come on the Canadian market called neurokinin (NK)-receptor antagonist. Fezolinetant, an NK-3 receptor antagonist is now available for moderate to severe vasomotor symptoms. Another agent, elizinetant, an NK-1,3 receptor antagonist, has recently received approval by Health Canada. The NK receptor antagonists target the kisspeptin, neurokinin B, and dynorphin (KNDy) neurons, which have been identified as the physiologic mechanism behind the cause of vasomotor symptoms. The loss of estrogen results in overactivity of the KNDy neuron, which leads to vasomotor symptoms [11].

For mild vasomotor symptoms, alternate approaches include phytoestrogens (e.g., soy-based products), natural health products (e.g., black cohosh), or non-pharmacologic approaches including lifestyle measures (e.g., weight loss, avoiding triggers) or complementary therapy including cognitive behavioral therapy (CBT), hypnosis, mindfulness, and acupuncture [1, 11]. Though studies have shown mixed results with phytoestrogens and acupuncture, there is some compelling evidence for CBT, mindfulness, and hypnosis [1, 11]. There is a lack of evidence to support many of the natural health products for vasomotor symptoms.

Local vaginal estrogen therapy is preferred for GSM only [12]. Vaginal estrogen therapy has very little systemic absorption and can be used in women with contraindications to systemic estrogen. In women with estrogen receptor-positive breast cancer, vaginal estrogen therapy can be considered if other non-hormonal therapies for GSM have failed and if the patient is not on aromatase inhibitors [13]. Other options for GSM include intravaginal DHEA (prasterone) and an oral selective estrogen receptor modulator called

ospemifene. Non-pharmacologic options for GSM include lubricants, moisturizers, vaginal dilators, and pelvic floor physiotherapy [12].

Patient Assessment

Pharmacists play an important role in assessing women for menopausal symptoms, discussing options, reviewing the risks and benefits for each of the options, including MHT, and managing the care of these patients. As with any patient care process, the initial assessment begins with a complete relevant history, including demographics, medical history, social history (smoking, alcohol use), family history (e.g., family history of breast cancer, cardiovascular disease), medications (prescription and non-prescription), and laboratory and physical exam findings. The following information should be collected specifically for the care of women suffering from menopausal symptoms.

Gynecologic History

The purpose of the gynecologic history is to capture the menopausal status of the woman (peri- or postmenopausal), to determine if the woman has had a hysterectomy or bilateral oophorectomy, and to identify if she is experiencing any issues with her menstrual periods or vaginal bleeding if the uterus is still intact. In a perimenopausal woman, it is important to capture menstrual regularity, cycle length, flow, or any other issues with her periods. Questions to ask include:

Are you still having menstrual periods?

If no:

Did you have a hysterectomy? When did you have your hysterectomy?

Have you had your ovaries removed (one or both)? When did you have the surgery?

When was your last menstrual period?

Are you having any vaginal spotting or bleeding (if uterus still intact)? Could there be other reasons for your menstrual period stopping (ie hormonal contraceptives, endometrial ablation)?

If yes, still having periods:

Are you having any changes in your menstrual patterns?

Are your periods regular or irregular?

How often do you get your periods?

Are your periods heavy or light? How long do the periods last?

Are your periods painful?

Do you get spotting or bleeding in between periods?

Consider the need for contraception if the last menstrual period was <2 years for woman less than 50 years or < 1 year if older than 50 [14].

Are you sexually active? Do you need contraception?

What contraception do you currently use?

Menopause Symptom History

A good symptom history is important in capturing the type of symptoms experienced, the severity, how bothersome the symptoms are to the woman and the impact on the woman's quality of life.

For each symptom, it is important to ask:

What symptoms are you having? (see Table 31.1 for symptoms).

How often are you having the symptoms?

How would you describe the severity of these symptoms (for example, on a scale of "mild to severe" or "not at all to extremely")?

When did the symptoms start? Are they worse at certain times of the day or in certain parts of your cycle?

How long do the symptoms last? A hot flash can last from 1–5 minutes. If longer than this, consider other possible causes of hot flashes.

Is there anything that makes the symptoms worse?

How are the symptoms affecting your quality of life?

For genitourinary symptoms, are you having pain with intercourse?

Medical History

A comprehensive personal medical history is important when assessing menopausal women for several reasons: [1] to identify diseases or drugs that may lead to symptoms overlapping with menopausal symptoms; [2] to identify long-term health consequences which may need preventative measures or treatment such as osteoporosis, hypertension, hyperlipidemia, or other cardiovascular diseases; and [3] to capture contraindications to MHT and other medications. Conditions or medications that may contribute to VMS should be identified initially and dealt with if appropriate prior to treatment (Table 31.3). Contraindications to MHT include [1]:

Table 31.3 Medical conditions and medications that can cause vasomotor symptoms

Medical conditions	Hyperthyroid Hyperparathyroid Carcinoid tumor Pheochromocytoma Lymphoma
Medications	Raloxifene GnRH agonists Aromatase inhibitors Nicotinic acid (niacin) Opiates Calcium channel blockers Chemotherapy Antipsychotics Some antidepressants (e.g., SSRI)

GnRH gonadotropin-releasing hormone, *SSRI* selective serotonin reuptake inhibitor

- Undiagnosed abnormal vaginal bleeding.
- Known, suspected or history of breast cancer or other estrogen-sensitive cancers.
- Active liver disease.
- Active or history of venous thromboembolism.
- Known thrombophilia.
- Coronary heart disease.
- Cerebrovascular accident.
- Pregnancy.

Capturing other comorbidities is important to identify if women have further risks for venous thromboembolism (VTE), cardiovascular disease (CVD), and stroke, as well as choosing among the different menopausal hormone therapy options. Comorbidities that are important to consider include [15]:

- Obesity.
- Smoking.
- Hypertension.
- High lipids.
- Diabetes.
- Other CVD risk.
- Gallstones.

Symptom Management

Many women will try a number of measures, such as lifestyle changes, non-pharmacologic options, or natural health products before seeking help from a health-care provider. Identifying if the woman has tried MHT or non-hormonal medications is also helpful. The following are questions to ask:

What have you tried to help your symptoms (including lifestyle factors, herbals, vitamins, other medications)?
Are you currently taking menopausal hormone therapy or have you tried in the past?
What has helped with your symptoms?
What was the name of the product? What dose did you use?
How long did you try it? Was it effective for you? Why did you stop?
Did you experience any side effects?

Additional Assessment Considerations

- An elevated FSH can help confirm menopause in postmenopausal women. In the perimenopause, FSH and estrogen levels fluctuate and are not diagnostic.
- Other laboratory testing includes a TSH level, as symptoms from low thyroid levels can overlap with menopause symptoms.

- Women who have undiagnosed abnormal uterine bleeding should be referred. Check to see if they have had a pelvic ultrasound and/or endometrial biopsy completed.
- Assessment of CVD risk includes checking blood pressure, lipids, and glucose.
- Check when their last pap smear and mammogram were completed and the results, as these should be completed according to local guidelines.
- Bone mineral density (BMD) measurement can be considered if the criteria for BMD testing are met (see Chap. "Osteoporosis").

Decision-Making with Menopausal Hormone Therapy

Menopause can be a confusing time for women, and decision-making regarding MHT may feel complex. Communicating in a balanced way about the benefits and risks of MHT is important for informed decision-making (see Table 31.2) [1, 8, 9]. Systemic MHT has a number of benefits, including relieving vasomotor symptoms, helping with sleep, mood, and improving quality of life. For GSM alone, vaginal estrogen therapy can be considered. See Table 31.4 for GSM medications available in Canada. Approximately 40% of women on systemic MHT continue to have GSM symptoms; therefore, vaginal estrogen therapy can be considered in combination with systemic MHT. Choosing among the systemic MHT products will be dependent on patient preference, as well as patient-specific factors. Estrogen is given continuously every day, and the progestogen is given continuously every day or cyclically (12–14 days every month). See Table 31.5 for MHT products in Canada and common doses. Transdermal estrogen does not have a high first-pass

Table 31.4 Medications for GSM in Canada

Formulation	Type of estrogen	Starting doses
Estrogen		
Vaginal cream	Conjugated estrogen	0.5 gm vaginally daily for 14 days, then 0.5 gm 2–3 times weekly
Vaginal tablet	17 β -Estradiol	10 mcg tablet vaginally daily for 14 days, then one tablet twice weekly
Vaginal ovule	17 β -Estradiol	4 or 10 mcg ovule daily for 14 days, then one tablet twice weekly
Vaginal cream	Estrone	0.5 gm vaginally daily for 14 days, then 0.5 gm 2–3 times weekly
Intravaginal DHEA		
Vaginal	Prasterone	6.5 mg ovule inserted daily
Selective estrogen receptor modulator (SERM)		
Oral	Ospemifene	60 mg tablet daily

GSM Genitourinary symptoms of menopause

Table 31.5 Systemic menopausal hormone therapy products in Canada

Formulation	Type of estrogen	Starting doses
Estrogen		
Oral	Conjugated estrogen (CE) 17 β -Estradiol	0.3–0.625 mg tablet daily 0.5–1 mg tablet daily
Transdermal patch	17 β -Estradiol	25–50 μ g patches twice weekly or once weekly (depending on product)
Transdermal gel	17 β -Estradiol (gel)	1–2 metered doses/actuation daily 0.5–1 mg packets daily (depending on product)
Progestogen		
Oral	Progesterone micronized	100 mg daily for continuous regimen 200 mg daily for 12–14 days every month for cyclic regimen
Oral	Medroxyprogesterone acetate	2.5 mg daily continuous regimen 5 mg daily for 12–14 days every month for cyclic regimen
Oral	Norethindrone acetate	5 mg daily
Combination of estrogen and progestogen products		
Oral	17 β -Estradiol/progesterone micronized	1 mg estradiol/100 mg progesterone
Oral	17 β -Estradiol/norethindrone acetate	1 mg estradiol/0.5 mg norethindrone tablet daily 0.5 mg estradiol/0.1 mg norethindrone tablet daily
Oral	17 β -Estradiol/drospirenone	1 mg estradiol/1 mg drospirenone tablet daily
Transdermal patch	17 β -Estradiol/norethindrone acetate	140/50 (50 μ g estradiol/140 μ g norethindrone) twice weekly 250/50 (50 μ g estradiol/250 μ g norethindrone) twice weekly
Tissue-selective estrogen complex (TSEC)		
Oral	Conjugated estrogen (CE)/bazedoxifene	0.45 mg CE/20 mg bazedoxifene tablet daily
Selective tissue estrogenic activity regulator (STEAR)		

effect through the liver, and at standard doses, may have less risk of VTE compared to oral products. Transdermal estrogen may be preferred over oral estrogen in women who are smokers, with high triglyceride levels, low libido, migraines, gall bladder disease, and shift workers [1].

Monitoring Menopausal Hormone Therapy

Women should be asked about improvements in symptoms, as well as breakthrough bleeding and adverse effects on follow-up. Common adverse effects with estrogen include

breakthrough bleeding, breast tenderness, water retention, and headache. Progestogen adverse effects include mood changes such as depression and anxiety, irritability, headache, breast tenderness, and water retention. Natural progesterone may cause drowsiness. Breakthrough bleeding, one of the most common reasons for early MHT discontinuation, can occur up to 6 months after starting MHT. Other common adverse effects will improve with time. Breakthrough bleeding in a postmenopausal woman that occurs after 1 year on MHT will require further investigation and should be referred. Adverse effects of estrogen that continue to be bothersome can be managed by changing to a lower dose or switching product formulations. Progestogen side effects can be handled by changing to a different progestogen, switching to cyclic if on continuous or using LNG-IUS instead. There is no time frame for MHT discontinuation in the post-menopausal woman; current guidelines recommend to continue MHT for the time that the woman needs it for symptom relief [1].

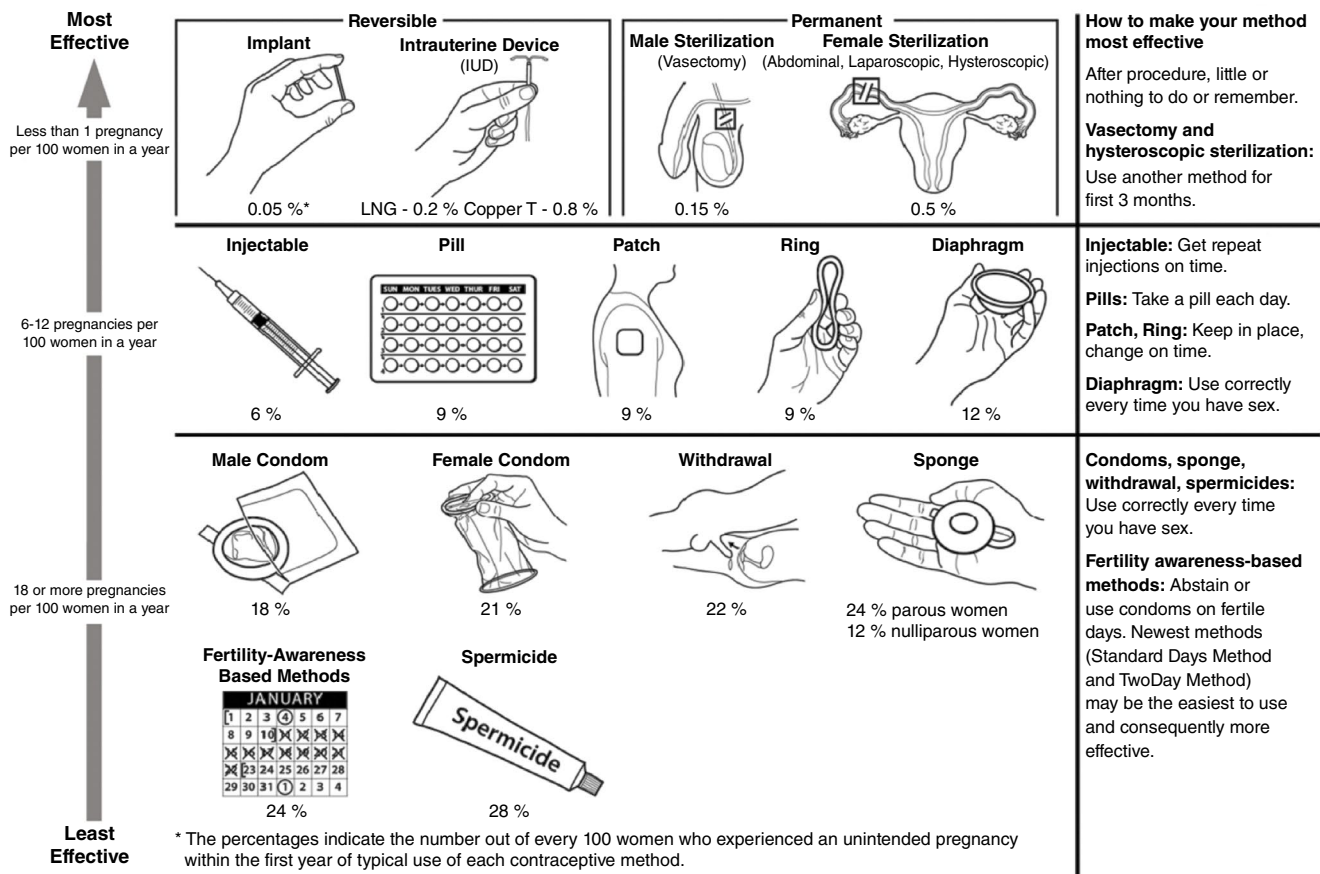
Assessment of Women for Hormonal Contraception

Background

It is estimated that 40% of all pregnancies in Canada are unintended, and nearly half are from women on some form of contraception [16, 17]. Unintended pregnancy results in adverse maternal and child health outcomes, as well as significant costs to the health-care system [17]. When helping women select among the contraceptive options, considerations include effectiveness of the contraceptive method (typical failure rates), ease of use, access, cost, and patient-specific factors. Tailoring the contraceptive method to the individual's needs is important for the woman's commitment to the chosen method and continued adherence. Adherence to contraceptives is poor, with 60% of women using incorrectly or inconsistently [18]. Long-acting reversible contraceptives (LARC) such as the intrauterine devices, including levonorgestrel intrauterine system (LNG-IUS) have the highest effectiveness and continuation rates as their use does not require regular action by the user [19]. Contraceptive options include both non-hormonal methods (barrier contraceptives) and hormonal contraception. Efficacy rates for the various contraceptive methods can be found in Fig. 31.2.

Hormonal Contraceptive Options

Hormonal contraceptives refer to both combined hormonal contraceptives and progestin-only contraceptives. Combined hormonal contraceptives (CHC) contain an estrogen (ethinyl estradiol or estetrol) and a progestin and are available as



CS 242797

CONDOMS SHOULD ALWAYS BE USED TO REDUCE THE RISK OF SEXUALLY TRANSMITTED INFECTIONS.

Other Methods of Contraception**Lactational Amenorrhea Method:** LAM is a highly effective, temporary method of contraception.**Emergency Contraception:** Emergency contraceptive pills or a copper IUD after unprotected intercourse substantially reduces risk of pregnancy.

Adapted from World Health Organization (WHO) Department of Reproductive Health and Research, Johns Hopkins Bloomberg School of Public Health/Center for Communication Programs (CCP). Knowledge for health project.

Family planning: a global handbook for providers (2011 update). Baltimore, MD; Geneva, Switzerland: CCP and WHO; 2011; and Trussell J. Contraceptive failure in the United States. Contraception 2011;83:397-404.



U.S. Department of
Health and Human Services
Centers for Disease
Control and Prevention

Fig. 31.2 Effectiveness of family planning methods (Source: CDC). Available at <https://www.cdc.gov/mmwr/volumes/65/rr/rr6503a1.htm> (Accessed August 14, 2025). The use of the material does not imply an endorsement by the Centers for Disease Control and Prevention (CDC), Agency for Toxic Substances and Disease Registry (ATDSR), Health

and Human Services (HHS) or the United States Government of any particular organization, service, or product and that any views expressed in the book do not necessarily represent the views of CDC, ATDSR, HHS or the United States Government. The material is otherwise available on the agency website for no charge

combined oral contraceptives (COC), transdermal patch, and vaginal ring. CHCs can be safely used in healthy women after careful consideration of contraindications. In addition to preventing pregnancy, CHCs have many non-contraceptive benefits and are used for a number of other conditions (Table 31.6).

Current guidelines recommend LARCs such as LNG-IUS and the contraceptive implant as a first-line contraceptive option for women [16]. The LNG-IUS is an intrauterine device that slowly releases a progestin (levonorgestrel) directly into the uterus. LNG-IUS is especially favorable as a LARC as it is as effective as permanent contraception. The duration of use is 8 years for the 52 mg LNG-IUS and 5 years for the 19.5 mg LNG-IUS. LNG-IUS also has other non-

contraceptive uses, including abnormal uterine bleeding, dysmenorrhea, endometriosis, and providing endometrial protection for women on estrogen alone [19]. The contraceptive implant contains the progestin etonorgestrel and is inserted under the skin of the upper arm for up to 3 years duration.

Progestin-only contraceptives include the progestin-only pill (POP) and depot injection (depot medroxyprogesterone acetate). Progestin-only contraceptives are options in women where estrogens may not be appropriate, such as smokers over age 35, postpartum (≤ 6 months), or breastfeeding [20].

This chapter focuses on the assessment of CHC and POP as these agents are some of the most commonly used methods of contraception by women. In Canada, the use of com-

bined oral contraceptives is second only to the use of condoms [21].

Patient Assessment of Hormonal Contraception

Pharmacists play an important role in assessing women for hormonal contraception, discussing options, and reviewing the risks and benefits for each option. As with any patient care process, the initial assessment begins with a complete relevant history, including demographics, medical history, social history (smoking, alcohol use), family history, medications (prescription and non-prescription), laboratory, and physical exam findings. The following information should be collected specifically for women seeking hormonal contraception.

Reasons for Hormonal Contraception

A woman may be interested in hormonal contraception for other reasons than contraception. Inquiring about other reasons that the woman may be considering hormonal contraception is important (see Table 31.6). Additionally, she may also not be aware of non-contraceptive benefits, which would be important to discuss when counseling.

If the woman is primarily interested in contraception, it may be an opportune time to discuss the benefits of long-acting reversible contraceptives such as LNG-IUS prior to the assessment. LNG-IUS or the contraceptive implant may be a good option to consider if the woman is not planning on getting pregnant in the next year and has no other contraindications. If the woman is interested in an LNG-IUS or the contraceptive implant, then refer her to her physician.

Table 31.6 Non-contraceptive benefits of CHC

Menstrual-related benefits
Decreased menstrual blood loss and anemia
Decreased dysmenorrhea
Reduced PMS symptoms
Others
Decreased acne
Improved hirsutism
Improved bone health
Perimenopausal symptoms
Prevention of ovarian and endometrial cancer

CHC combined hormonal contraceptives, PMS premenstrual syndrome

Menstrual History

A good history captures baseline information on a woman’s menstrual cycle. The possibility of pregnancy should be ruled out before considering a hormonal contraceptive. Any undiagnosed abnormal uterine bleeding should be identified and the woman referred to her physician.

- When was your last menstrual period?
- How often do you get your periods? Are they regular or irregular?
- Are your periods heavy? How long do they last?
- Do you get spotting or bleeding in between periods? Has this been assessed?
- Have you had unprotected intercourse since your last menstrual period? If there is a possibility that the woman is pregnant, recommend a pregnancy test and refer to physician.

Medical History

A comprehensive medical history is important to identify hormonal contraindications and to help choose among the different contraceptive options. See Table 31.7 for the World Health Organization’s absolute contraindications (Category 4) to CHC [22]. For POP, the only absolute contraindication is current breast cancer. Special attention is needed for capturing cardiovascular risk factors (smoking, obesity, high lipids, hypertension, diabetes, and history of CVD), breast cancer risk, and liver disease. Questions to ask during the history include:

Table 31.7 Absolute contraindications to CHC (WHO Medical Eligibility Criteria Category 4) [22]

Smokes 15 cigs/day or more and over 35 years of age
Cardiovascular disease
History of hypertension (not controlled)
History of stroke
History of migraines with aura
Diabetes with microvascular complications
Venous thromboembolism—Current or past
Thrombophilia
Breast cancer history—Current or past
Active or past liver disease
Given birth in the last 3 weeks
Breastfeeding <6 weeks postpartum
Rheumatic diseases such as lupus
Other active cancers/chemotherapy

CHC combined hormonal contraceptives, WHO World Health Organization

Do you smoke? CHC is contraindicated in women over the age of 35 years who smoke.

What is your weight? Measure the woman's weight to get a baseline. Determining weight/BMI is helpful in choosing among the hormonal contraceptives (see Decision Making). The risk of VTE is also increased in obese women.

Do you suffer from migraines? Are they associated with an aura? Migraines with aura are associated with a higher risk of stroke compared to women without migraines. CHC should not be used in women with migraines with aura.

Have you had a blood clot in the past? CHC is associated with an increased risk of VTE. A history of VTE is a contraindication to the use of CHC.

What medications are you currently taking? The current medication list will be important to screen for possible drug interactions with hormonal contraceptives. Medications of most concern for both CHC and POP are inducers of cytochrome P450 CYP3A4, including anticonvulsants, rifampin, some antiretrovirals, and the natural health product, St John's wort.

It will also be important to identify if the woman is recently postpartum, if she is less than 6 weeks postpartum and breastfeeding, she is not a candidate for CHC, while less than 6 months postpartum is a relative contraindication [22]. Progestin-only contraceptives should be considered in both of these scenarios.

Past Contraceptive Use

It is important to capture the types of contraceptives (both non-hormonal and hormonal methods) that have been used in the past. Time frame, duration of use, and experience with the contraceptive method, as well as any side effects experienced, will help in hormonal contraceptive decision-making.

What type of contraception are you currently using?

Have you been on hormonal contraception in the past? Which ones and for how long?

How satisfied were you with this method?

Did you have any side effects?

Physical Assessment

A blood pressure measurement should be performed at baseline prior to initiating CHC. If the blood pressure is $\geq 140/90$, the woman should be referred to their physician. A pap smear, pelvic examination, or STI testing is not needed prior to starting a hormonal contraceptive. These can be completed as part of routine healthy women's care, but are not necessary before starting hormonal contraception.

Decision-Making with Hormonal Contraceptives

In the absence of contraindications, deciding among the hormonal contraceptive options is dependent on the woman's preference, as well as previous experience with hormonal contraceptives. The CHCs vary with the type of estrogen (ethinyl estradiol, estetrol), doses of ethinyl estradiol (10–35 μg ethinyl estradiol), the types of progestins, dose regimens (monophasic, biphasic, triphasic), and the formulation (oral, transdermal patch, vaginal ring). Phasic formulations (e.g., triphasic) were formulated to decrease the exposure to progestins; however, it is unclear if they provide any advantage over monophasic products. The progestins have different progestogenic, androgenic, and estrogenic properties. First-generation (e.g., norethindrone) and second-generation progestins (e.g., levonorgestrel) may have lower VTE risk compared to other progestins; however, this is controversial. Current Canadian guidelines do not recommend preferential prescribing of the type of progestin in CHCs [23]. The classic CHC regimen is 21 days of the CHC, followed by 7 days of hormone-free intervals (21/7). Several products are now available with shortened hormone-free intervals (HFI) of 4 days (24/4) or with estrogen only during the HFI. Continuous (i.e., daily dosing with no HFI) or extended dosing of CHC (i.e., 7-day HFI every 3 months, 84/7) may be considered in women who have symptoms during the HFI (e.g., headaches, pelvic pain, endometriosis, PCOS) or do not want to have withdrawal bleed. If considering continuous or extended CHC regimens, use monophasic products or extended dosing-specific formulations.

The CHC patch or ring may be preferred in women who want the convenience of weekly application (patch is applied once a week for 3 weeks followed by 7-day HFI) or monthly administration (ring is inserted for 3 weeks followed by 7-day HFI). The CHC patch should be avoided in women over 90 kg. Most studies with oral contraceptives do not indicate a decrease in contraceptive efficacy with obesity, but this is not clear in all studies. As a decrease in effectiveness cannot be ruled out in women with BMI ≥ 30 and as obesity is associated with increased VTE risk, the LNG-IUS may be a consideration in obese women [23].

POPs may be considered in situations where estrogen is contraindicated, such as migraines with aura or smoking in women over the age of 35 years. They are also often used for postpartum and in women who are breastfeeding [20]. There are two types of POPs in Canada, including norethindrone 0.35 mg and drospirenone 4 mg. Norethindrone POP is used

daily with no hormone free interval. It is important to take the pill at the same time each day and if it is delayed more than 3 h then back-up contraception (for example, barrier methods) must be used for up to 48 h [20]. Drospirenone POP is taken daily for 24 days with a 4-day hormone-free interval. It does not have the same 3-h window of administration as norethindrone POP.

This is a good time to ask about the woman's familiarity with non-hormonal contraceptives (i.e., barrier methods) as well as to prepare for counseling on the prevention of sexually transmitted infections and recommendations for backup contraception for missed hormonal contraceptives.

Monitoring Hormonal Contraceptives

Ideally, it is best to have a follow-up 1–3 months after the woman has started the hormonal contraceptive. On follow-up, the woman should be asked about her satisfaction with the method, as well as breakthrough bleeding and adverse effects. Table 31.8 summarizes the adverse effects of CHC. POP would have the same adverse effects under the progestin-related column in the table. If using for other non-contraceptive benefits as well, inquire about symptom improvement. Inquire about her adherence and if there are any adherence issues. Blood pressure should be repeated. It is also a good time to assess if the woman has any changes in health status (i.e., new medications, new medical conditions).

Table 31.8 Adverse effects of CHC

Estrogen related	Estrogen deficiency	Progestin related	Progestin deficiency
Nausea	Early/midcycle BTB	Breast tenderness	Late BTB/spotting
Headaches	Hypomenorrhea	Fluid retention	Heavy menstrual flows
Breast tenderness	Menopausal symptoms (vasomotor, insomnia)	Bloating	Delayed menses
Fluid retention	Mood (irritability, depression)	Mood (anxiety, depression)	
Poor contact lens fit		Headache	
Chloasma		Appetite changes	

CHC combined hormonal contraceptives, BTB breakthrough bleeding

Clinical Pearls

Menopause

- Pharmacists play an important role in assessing women for menopausal concerns.
- Assessment of women for menopausal concerns includes gynecologic history to capture menopause status, experience of menopausal symptoms, impact on quality of life, and detailed medical history to capture contraindications or risks to medication options.
- Menopausal hormone therapy is a safe and effective option when initiating in women younger than 60 years of age or within the first 10 years of their last menstrual period and after careful consideration of benefits and risks.

Contraception

- Pharmacists play an important role in assessing women for contraceptive care.
- Assessment of women for hormonal contraceptives includes history of menstrual cycles, previous use and experience of contraceptives, and detailed medical history to determine if there are any contraindications or risks to using hormonal contraceptives.
- Hormonal contraceptives are safe to use in healthy women of reproductive age who do not have contraindications.

Gender Statement

We have used the word “women’s health” to refer to sex-specific conditions or health conditions that may impact women differently. We do acknowledge that not all people who identify as women are born biologically female nor do all people assigned female at birth will identify as woman.

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Chapter Objectives

1. Describe the classification of anemia.
2. Complete a patient assessment and interpret laboratory findings to determine the most likely cause of anemia.
3. Apply a monitoring and follow-up plan for patients initiated on treatment for anemia.

Background

Anemia is commonly encountered in clinical practice and is characterized by a decrease in hemoglobin (Hgb) or red blood cells (RBCs), resulting in reduced oxygen-carrying capacity of the body [1]. Anemia is defined by the World Health Organization as a Hgb <130 g/L in adult men, <120 g/L in non-pregnant adult women, and <110 g/L in pregnant women [2]. Severe anemia is a Hgb <80 g/L in adult men and non-pregnant women [2]. It is important to note that Hgb values vary based on sex as well as ethnicity. As examples, women typically have lower Hgb concentrations than men, and Black individuals have lower values as compared to White individuals [2].

Anemia can cause significant morbidity and mortality. Older adults with anemia have higher hospitalization and mortality rates, while anemia in children can impair cognitive and psychomotor development. Globally, anemia impacts approximately 25% of the population [3]. In the United States, estimates suggest almost 7% of the population has anemia, with age, sex, race, poverty income ratio, and birthplace associated with differences in anemia prevalence [4].

Given the prevalence of anemia, pharmacists play an important role in assessing patients with anemia, determin-

ing potential cause(s), and identifying the need for additional laboratory testing or referrals where appropriate. Pharmacists can also assist patients with treatment options, dietary recommendations, and managing drug-drug interactions with oral iron supplements.

Clinical Presentation

Signs and symptoms of anemia can vary considerably depending on factors such as the rate of development and overall health status of the patient [5]. In many cases, mild anemia is asymptomatic and may be found when ordering a complete blood count (CBC) as part of routine bloodwork or other investigations [1]. In patients that are otherwise healthy, signs or symptoms may not be obvious even at low Hgb concentrations if the anemia develops slowly over time [5]. In anemias that develop very rapidly, symptoms tend to be more pronounced. Because hemoglobin delivers oxygen, signs and symptoms are often related to lack of oxygen delivery including fatigue, weakness, dizziness, or shortness of breath. Cardiac signs or symptoms may include chest pain, palpitations, or tachycardia. On examination, patients may exhibit pallor of skin and/or mucous membranes. In the elderly, signs and symptoms of anemia may overlap with other conditions and anemia has been associated with increased falls, reduced cognition, and overall physical decline [6]. Depending on the cause of the anemia, patients may have additional symptoms related to underlying nutritional deficiencies such as neurologic symptoms as a result of vitamin B12 deficiency. General signs and symptoms of anemia are summarized in Table 32.1.

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Table 32.1 General signs and symptoms of anemia

Symptoms	Signs
Fatigue	Pale skin or mucous membranes
Weakness	Tachycardia
Dizziness or lightheadedness	Palpitations
Headache	Shortness of breath on exertion

Etiology

Anemia occurs when there is an imbalance in the production and destruction or loss of RBCs. There are three primary causes of anemia: blood loss, inadequate RBC production, and increased RBC destruction [5]. Blood loss can be caused by acute (e.g., trauma) or chronic (e.g., gastrointestinal ulcers) bleeding. Nutritional deficiencies (B12, folate, and iron), chronic kidney disease, thyroid disease, liver disease, bone marrow failure, and anemia due to chronic disease or inflammation can lead to inadequate RBC production. Increased destruction of RBCs may be caused by hereditary (e.g., sickle cell anemia, thalassemia) or acquired (e.g., immune hemolytic anemia) conditions.

Approach to the Assessment of Anemia

Anemia reflects an underlying disease or condition; therefore, it is important to conduct a thorough work-up to determine the cause in order to guide appropriate management. Laboratory tests should be evaluated in the context of the patient's history and physical examination to diagnose anemia. Laboratory tests also play an important role in assessing response to treatment.

History and Physical Examination

A detailed patient history may provide clues as to the cause of the anemia [1]. Specific questions to ask the patient may depend on the situation as well as laboratory tests that are available.

- *Signs of blood loss:* Consider blood loss from the gastrointestinal (GI) tract, genitourinary tract, or as a result of trauma. A menstrual history should be taken for women to rule out heavy menstrual bleeding.
- *Past medical history:* Chronic conditions associated with anemia include: rheumatoid arthritis, systemic lupus erythematosus, chronic kidney disease, congestive heart failure, liver disease, thyroid disease, hemolytic disorders, aplastic anemia, certain cancers (e.g., leukemia, lymphoma), infections (e.g., human immunodeficiency virus, tuberculosis, osteomyelitis), inflammatory bowel disease, celiac disease.

- *Surgeries or procedures:* Recent surgeries may cause anemia or result in secondary bleeding. Gastric bypass surgery can lead to reduced absorption of vitamins. A history of recent blood donation or frequent laboratory testing (e.g., in hospital settings) should also be considered.
- *Medications:* Drugs may cause or contribute to anemia through different mechanisms. For example, non-steroid anti-inflammatory drugs (NSAIDs), anticoagulants, and antiplatelet agents may promote bleeding and iron deficiency. Certain chemotherapy or antimalarial agents, zidovudine, trimethoprim, sulfasalazine, phenytoin, phenobarbital, metformin, and proton pump inhibitors may cause macrocytic anemias. Ribavirin, and less commonly select antibiotics, NSAIDs, and other agents can cause hemolytic anemia.
- *Diet:* Dietary history may suggest possible deficiencies such as B12, folate, or iron in the diet.
- *Family history:* This may be useful to identify potential inherited anemias, such as thalassemia or sickle cell disease.
- *Pregnancy:* Individuals that are pregnant have increased iron demands and are at a higher risk of iron deficiency anemia.

In addition to general signs of anemia on physical examination, there may be additional findings that are suggestive of specific causes [1]. For example, jaundice or scleral icterus may be a sign of hemolytic anemia. Patients with B12 deficiency may experience altered mental status or numbness and tingling in their hands and feet. Iron deficiency anemia can cause brittle nails and a smooth or swollen tongue.

Laboratory Tests

As anemia is defined as reduced oxygen carrying capacity with a reduction in red blood cells (RBCs) or hemoglobin (Hgb), the initial laboratory tests used for diagnosis of anemia include the complete blood count (CBC), which includes RBC and Hgb. Other tests may also include reticulocyte count, iron studies, vitamin B12, and folate levels [7]. A summary of these laboratory tests and what they measure is found in Table 32.2. A decreased Hgb or RBC count confirms that the patient has anemia. It is important to recognize that Hgb and Hct may decrease when plasma volume increases (fluid overload) and may increase when plasma volume decreases (dehydration) [8]. Evaluating trends in Hgb (chronic versus more recent onset) as well as absolute values in the context of the patient history is important.

After confirming low Hgb or RBC count, the next step is to assess lab values to help categorize the anemia. A common method to categorize the anemias is to look at mean corpuscular volume (MCV) to help narrow down potential

causes. An algorithm for assessing anemia based on MCV is found in Fig. 32.1. Anemias are commonly classified as microcytic (MCV < 82 fL), macrocytic (MCV > 100 fL), or normocytic (MCV 82–100 fL). MCV represents a measurement of the average size of RBCs and, therefore, can sometimes be misleading in patients with mixed anemias. For example, MCV may appear normal in patients who have both anemia of chronic disease and iron deficiency anemia. In some cases, the red cell distribution width (RDW) may provide additional information. A normal RDW indicates homogeneity in RBC size, whereas an increased RDW indicates variation in RBC size and can occur with iron deficiency and megaloblastic anemia [7]. A peripheral smear may be ordered to examine the size and shape of RBCs as

well as abnormal circulating cells. Digital peripheral blood smears are currently being explored and are likely to be available in the future to support anemia assessments [9].

In patients with microcytic anemia, iron studies are needed to differentiate the cause. A low serum ferritin (less than 30 µg/L), is usually the best indicator of iron deficiency as it reflects iron storage in the body. However, ferritin is an acute-phase protein and is therefore elevated by inflammation. In this situation, a ferritin level greater than 100 µg/L suggests iron deficiency is unlikely [10]. Other iron studies, including serum iron, total iron binding capacity (TIBC), and transferrin saturation, are often not that helpful in distinguishing iron deficiency anemia from anemia of chronic disease [8]. A trial of iron therapy may be necessary to confirm the diagnosis. RDW is often increased in nutritional deficiencies, including iron deficiency anemia. In iron deficiency anemia, smaller microcytic cells are formed. Table 32.3 compares laboratory test results in patients with iron deficiency anemia and anemia of chronic disease. In patients with ferritin within the reference range or elevated, further investigations are needed to determine the cause.

Normocytic anemia can have a number of different causes, and pharmacists should refer patients when appropriate [1]. It is important to correlate findings with the patient's history, physical exam, and appropriate tests. The reticulocyte count and/or a peripheral smear can be useful in differentiating potential causes of normocytic anemia. A high reticulocyte count suggests the bone marrow is functioning appropriately and has the building blocks to respond to the anemia, and thus, potential causes may include acute blood loss or hemolysis. In patients with normocytic anemia and a low reticulocyte count, potential causes may include bone marrow failure or chronic infection, inflammation, malignancy, or chronic kidney disease (Fig. 32.1). Evaluation of other blood cells (white blood cells and platelets), serum creatinine, liver tests, as well as consideration of the patient's history and physical examination, will aid in narrowing down the cause.

Assessing macrocytic anemias should include reviewing medications and substances, such as alcohol, to determine whether these may be implicated in causing the anemia [11]. Methotrexate, pemetrexed, zidovudine, chemotherapy drugs, and hydroxyurea are common causes of macrocytic anemia.

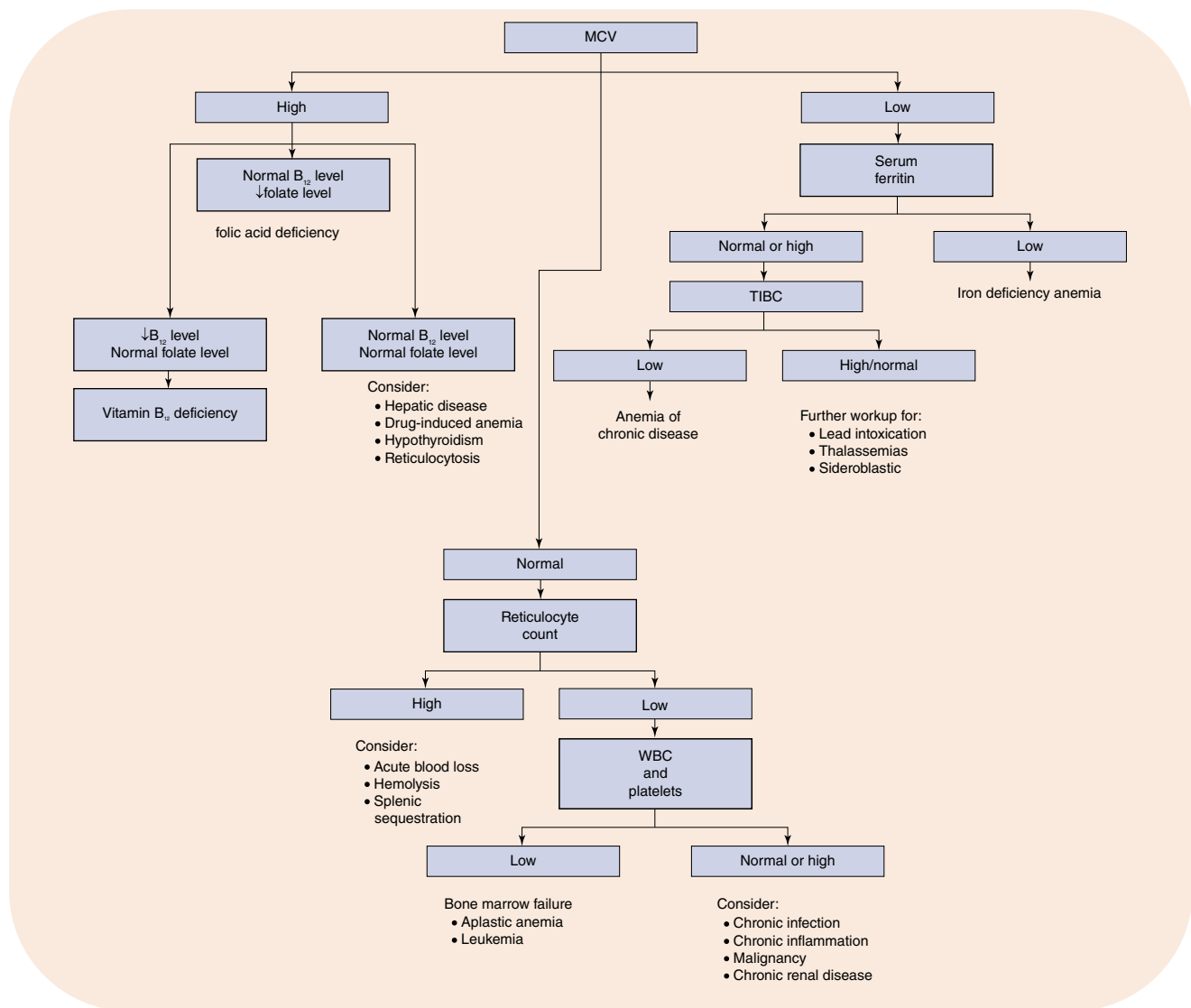
Table 32.2 Common laboratory tests used to diagnose anemias

Test	Reference range ^a	Description
<i>Complete blood count</i>		
Red blood cells (RBCs)	(Males) 4.3–6.0 × 10 ¹² /L (Females) 3.8–5.2 × 10 ¹² /L	–
Hemoglobin (Hgb)	(Males) 135–175 g/L (Females) 120–160 g/L	Amount of hemoglobin in a unit of blood
Hematocrit (Hct)	(Males) 0.40–0.52 (Females) 0.36–0.48	Percentage volume of RBCs in the blood
Mean corpuscular volume (MCV)	82–100 fL	Size of the average RBC
Mean corpuscular hemoglobin concentration (MCHC)	310–360 g/L	Average concentration of Hgb in the RBC
Red cell distribution width (RDW)	<16%	Measures variation in RBC volume
<i>Iron studies</i>		
Serum iron	8–35 µmol/L	Measures iron bound to transferrin
Total iron binding capacity (TIBC)	40–75 µmol/L	Measures iron-binding capacity of transferrin
Transferrin saturation index	Male: 0.12–0.60 Female: 0.10–0.55 Non-binary or prefer not to disclose: 0.10–0.60	Ratio of serum iron to TIBC expressed as a percent
Ferritin	(Males) 30–500 µg/L (Females) 120–300 µg/L	Indicator of iron body stores. Caution interpreting ferritin in the presence of inflammatory conditions or malignancy
<i>Others</i>		
Vitamin B12	>160 pmol/L	–
Folate-serum	>10.0 nmol/L	–
Reticulocyte count	20–120 × 10 ⁹ /L	Immature RBCs

^aExample adult reference ranges obtained from Alberta Precision Laboratories. Accessed November 13, 2024. Available at <https://www.albertahealthservices.ca/webapps/labservices/indexAPL.asp>

Table 32.3 Laboratory differentiation of iron deficiency anemia (IDA) and anemia of chronic disease (ACD)

Test	IDA	ACD	IDA + ACD
Mean corpuscular volume (MCV)	↓	Normal or ↓	Normal or ↓
Red cell distribution width (RDW)	↑	Normal	↑
Serum iron	↓	↓	↓
Total iron binding capacity (TIBC)	↑	Normal or ↓	Normal or ↓
Transferrin saturation	↓	Normal or ↓	↓
Ferritin	↓	Normal	Normal



Source: Joseph T. DiPiro, Gary C. yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach, 12e*
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Fig. 32.1 General algorithm for diagnosis of anemias based on laboratory data. (Reprinted with permission from McGraw-Hill Education, Chapter 122 Anemias, DiPiro JT, Yee GC, Haines ST, Nolin TD,

Ellingrod VL, Posey L. *DiPiro's Pharmacotherapy: A Pathophysiologic Approach, 12th Edition*; 2023)

To rule out nutritional deficiencies, vitamin B12 and serum folate levels should be ordered. Low serum folate levels suggest folate deficiency. Deficiency in vitamin B12 is caused by low dietary intake or, more frequently, poor absorption [1]. Long-term use of medications, including metformin, proton pump inhibitors, and histamine H2 antagonists, can also cause B12 deficiency [12]. A vitamin B12 level <160 pmol/L is suggestive of deficiency. However, vitamin B12 levels at the lower end of the reference range may be associated with clinical symptoms of B12 deficiency and would therefore require treatment.

Management and Follow-Up Assessment

Management of the anemia depends on the cause as well as the patient's clinical status. For example, in anemia of chronic disease, the anemia is often corrected by treating the underlying disease. Specific treatment of the various types of anemia is beyond the scope of this chapter. However, treatment and follow-up assessment of iron, folic acid, and vitamin B12 deficiency are briefly summarized below.

Once the diagnosis of iron deficiency anemia and the underlying cause has been determined, treatment is often ini-

tiated with oral iron supplements. Patients may be encouraged to increase dietary intake of foods rich in heme iron (e.g., animal meat, fish) or non-heme iron (e.g., lentils, tofu) [13]. Ascorbic acid (vitamin C) increases absorption of non-heme iron, whereas tannins found in tea and coffee can decrease absorption. For oral iron supplements, the usual target dose is 100–200 mg of elemental iron per day [14]. Common iron supplements, as well as their elemental iron content, are summarized in Table 32.4 [14]. The optimal site of absorption for iron is the duodenum; thus, iron in slow or delayed release formulations are available, but may not be well absorbed as they may bypass the optimal site of absorption, the duodenum [14]. For maximum absorption, oral iron should be administered on an empty stomach, 1 h before or 2 h after a meal. Administration of oral iron supplements with a glass of orange juice can improve absorption. Side effects commonly encountered with oral iron supplements are abdominal pain, nausea, constipation, diarrhea, metallic taste, and dark stools. If patients are having difficulty tolerating oral iron, management strategies may include starting with a lower dose of elemental iron and titrating slowly, switching to a preparation with a lower amount of elemental iron, alternate day dosing, or administering iron with small snacks or meals. Studies have also compared daily iron dosing and alternate day dosing as hepcidin levels in the body increase with iron supplementation and decrease iron absorption for approximately 24 hours. However, to achieve hemoglobin increases, both alternate-day and daily dosing strategies are reasonable strategies. For patients needing a rapid response, daily dosing may be preferred. Where tolerance is of concern, alternate-day dosing may be utilized [15]. There are a number of drug-drug interactions with iron supplements to be aware of, including antacids, proton pump inhibitors, H₂-antagonists, and tetracycline or doxycycline that can decrease iron absorption. In addition, iron can impact the absorption of levodopa, levothyroxine, fluoroquinolones, bisphosphonates, integrase strand transfer inhibitors, tetracycline, and doxycycline; therefore, administration

times should be separated [14]. To assess response to treatment, a CBC should be ordered approximately 4 weeks after starting therapy [16]. Hgb is expected to increase >10 g/L after 4 weeks of treatment [16]. Iron deficiency anemia usually corrects within 2–4 months of starting therapy if appropriate doses are used and the underlying cause is corrected. However, oral iron is recommended to be continued for about 3 months after Hgb normalizes in order to replenish iron stores. A serum ferritin should be ordered to confirm repletion of iron stores prior to discontinuing therapy.

Folate deficiency is treated with folic acid supplementation. The dose of folic acid used and duration depend on the cause of the deficiency; however, common doses for treatment are 1–5 mg orally per day [11]. Recommended intake of folic acid in adults through diet and supplements is 400 µg/day. The United States Preventive Services Task Force recommends women planning or capable of pregnancy take a daily supplement of 400–800 µg/day [17]. Foods that are rich in folic acid include green leafy vegetables, citrus fruits, and grains. It is important that vitamin B12 deficiency is ruled out prior to starting folic acid, as folate may correct the anemia but does not treat the neurologic manifestations of vitamin B12. CBC should be repeated at approximately 1 month to assess response to treatment. Usually, Hgb normalizes within 2 months. Folate level should be repeated in approximately 3–4 months.

B12 deficiency can be treated with oral or parenteral vitamin B12 [11]. The decision to use oral versus parenteral initially may depend on the cause of the anemia (e.g., malabsorption), the severity, and the presence of neurologic symptoms. High-dose oral vitamin B12 (1–2 mg/day) is as effective as intramuscular administration in terms of correcting the anemia and neurologic symptoms [18]. However, parenteral vitamin B12 may be preferred until B12 levels are corrected in those with neurologic symptoms as well as patients who are non-adherent to oral B12 therapy. There are a number of dosing schedules for parenteral vitamin B12 [19]. For patients with dietary deficiency, lower doses of vitamin B12 can be used (e.g., 250 µg/day) [19]. Patients with pernicious anemia or any B12 deficiency caused by malabsorption require lifelong therapy [11]. CBC and vitamin B12 levels should be repeated 1–2 months after initiation of therapy to ensure correction.

Table 32.4 Comparison of oral iron supplements

Iron supplement	Elemental iron	Usual maximum dose (adults)
Ferrous ascorbate, capsule	100 mg	One capsule a day
Ferrous gluconate 300 mg tablet	35 mg	Two tablets 3 times daily
Ferrous sulfate 300 mg tablet	60 mg	One tablet 3 times daily
Ferrous fumarate 300 mg tablet	100 mg	One tablet 2 times daily
Heme-iron polypeptide (e.g., Proferrin®) 11 mg tablet	11 mg as heme iron	One tablet 3 times daily
Polysaccharide-iron complex (e.g., Feramax®) 150 mg capsule	150 mg	One capsule once daily

Clinical Pearls

- Investigation of the underlying cause of the anemia is necessary.
- MCV categorizes anemia into microcytic, normocytic, and macrocytic.

- In mixed anemias, MCV may be misleading—assessing MCV along with RDW (and/or the peripheral smear) may be helpful to identify mixed anemias.
- Reticulocytes can help categorize anemias as hypo- or hyper-proliferative.
- Pharmacists play an important role in assessing patients with anemia and monitoring response to treatment and referring when appropriate.

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